
INDIAN RADIO LICENSING

Owning and Operating a 10 W Walkie-Talkie in India

Every lawful route to a 10 W hand-held radio, the ASOC examination, the question bank, what expedition groups actually use, and the listening and Citizen Band verdicts.

Compiled: 2026-08-25

Jurisdiction: India - DoT, WPC, Telecommunications Act 2023

Status: Research summary, not legal advice

Contents

About this document	8
How to read the evidence tags	8
A warning about the question bank	8
Contents	9
Part 1 — The executive answer: can one person legally own and use a 10 W walkie-talkie in India?	13
1.1 If you only read one page	14
1.2 The action plan, in brief	16
1.3 Why radio is licensed at all	17
1.4 The legal framework in plain language	18
1.5 The taxonomy of licences: five different things, often confused	23
1.6 Every lawful route to 10 W — the decision tree	28
1.7 Routes that do not work — stated bluntly	31
1.8 The three uncertainties a reader must not gloss over	35
1.9 What changed in the ninety days before this report	35
1.9A Adversarial fact-check log — 2026-08-23	36
1.10 Abbreviations used in this report	38
1.11 Status note	39
Part 2 — What you may do without a licence, and the separate law of equipment	40
2.1 If you only read one page of this part	40
2.2 The four gates	41
2.3 Where the power to de-license comes from, and what the 2023 Act changed	42
2.4 The complete catalogue of India's de-licensed bands	44
2.5 The three bands that can legally carry voice	48
2.6 The verdict on licence-free use, and on 10 W	51
2.7 Equipment gate 1 — Equipment Type Approval (ETA)	53

2.8 Equipment gate 2 — import	56
2.9 Equipment gate 3 — possession, and the death of the Dealer Possession Licence	60
2.10 Baofeng-class imported radios — the answer, gate by gate	64
2.11 Which radios sold in India are type-approved?	67
2.12 What to ask WPC in writing before you spend money	69
2.13 Summary of this part	70
2.14 Disclaimer for this part	71
Part 3 — The amateur radio route: grades, the ASOC examination, and the licence	71
3.1 A note on evidence tagging in this part	72
3.2 The chain of authority	72
3.3 One document, not two	74
3.4 The grades, and the authoritative power and band table	74
3.5 Eligibility	81
3.6 What an amateur licence does not permit — the restrictions that decide this question	81
3.7 The ASOC examination	83
3.8 After the exam	99
3.9 Step-by-step action plan with realistic timelines	109
3.10 Master document checklist	110
3.11 Common claims about the amateur route that are wrong	112
3.12 Questions to put to WPC in writing, before you buy a radio	114
3.13 What this part does not cover	115
3.14 Disclaimer for this part	115
Part 4 — The ASOC syllabus in detail, and how to study for it	115
4.1 Which syllabus is legally operative — and the honest uncertainty	116
4.2 Exam structure, weighting and pass marks	117
4.3 Section A — Radio Theory and Practice, topic by topic	120
4.4 Section B — National and ITU Radio Regulations, topic by topic	123
4.5 Part II — the Morse test (General grade only)	131
4.6 Errors in the mainstream study material — check every one of these	132

4.7 Official and free primary documents — assemble this folder first	134
4.8 Institutional courses — NIAR and ARSI	135
4.9 Regional clubs and training bodies	138
4.10 Books, videos, mock tests and Morse resources	142
4.11 A concrete study plan	149
4.12 What remains genuinely unresolved in this part	152
Part 5 — The question bank: what has actually been asked, and what could be asked	154
5.1 How to read this part	154
5.2 If you only read one page	155
Part 5A — DOCUMENTED QUESTIONS	156
A.1 Why there are no official past papers, and what exists instead	157
A.2 Source register — full provenance	157
A.3 S1 — the official WPC/DoT sample paper (100 questions)	158
A.4 S2 and S8 — the NIAR Study Manual question bank (230 questions)	168
A.5 S3 — DARTS "Sample ASOC Exam Question Paper" (100 questions)	185
A.6 S4 — DARTS "Amateur Radio Exam Question Paper Sample" (188 questions, with a published key)	194
A.7 S5 — historical government papers, Grade I / Grade II, pre-2010 descriptive era	209
A.8 S6 — SIARS, "Queries: ASOC Exam questions" (11 items, answered)	217
A.9 S7 — AARS/INDRA study book, Annexure-III descriptive sample paper (31 items)	218
A.10 What was looked for and could not be found — the honest negatives	220
A.11 Count and reliability assessment	222
Part 5B — PRACTICE QUESTIONS (constructed from the syllabus)	224
THESE ARE NOT PAST PAPERS	224
B.0 How Part 5B works	224
B.1 The paper's SECTION A — RADIO THEORY AND PRACTICE	227
B.2 GROUP C — INTERFERENCE AND ELECTROMAGNETIC COMPATIBILITY	241
B.3 GROUP D — SAFETY	243
B.4 The paper's SECTION B — RADIO REGULATIONS	245
B.5 Practice-question answer key at a glance	262

5.3 Critical uncertainties a reader must carry away from this part	263
5.4 Disclaimer for this part	265
Part 6 — Penalties, enforcement, risk, and appendices	265
6.1 Which law bites today, 2026-08-23	265
6.2 The penalty provisions, exactly	268
6.3 Search, seizure, inspection and arrest	278
6.4 Real enforcement — what has actually happened, with dates	282
6.5 Defences that do not work	288
6.6 Honest risk assessment of operating unlicensed	289
Appendix A — Glossary of abbreviations	291
A.1 Institutions and bodies	292
A.2 Statutes, rules and instruments	294
A.3 Licences, authorisations and certificates	294
A.4 Technical terms	295
A.5 Process and payment terms	296
Appendix B — Forms, fees and payment heads	297
B.1 The amateur radio route (the realistic individual route to 10 W)	297
B.2 Equipment and possession permissions	299
B.3 Captive and commercial routes (for completeness — closed to natural persons)	300
B.4 Forms — there is no numbered form	301
B.5 Bharatkosh payment heads	302
Appendix C — Key contacts	303
C.1 WPC headquarters and the amateur section	303
C.2 WPC field units — the Wireless Monitoring Organisation	304
C.3 NIAR — National Institute of Amateur Radio	305
C.4 ARSI — Amateur Radio Society of India	306
Appendix D — Sources, grouped by topic	307
D.1 Statutes	307
D.2 Commencement notifications	308

D.3 Subordinate rules	308
D.4 Licence-exemption notifications	310
D.5 WPC orders, office memoranda and FAQs	310
D.6 DoT and portal service pages (all fetched 2026-08-23)	313
D.7 Official press releases and audit reports	314
D.8 Judgments	315
D.9 Study resources, clubs and exam material	316
D.10 Secondary analysis and news	317
D.11 Sources that could not be fetched	318
Appendix E — Legal disclaimer	319
Part 7 — Choosing a radio: a worked example	320
7.1 If you only read one page	321
7.2 The three intended uses, judged separately	323
7.3 The UV-32 examined	331
7.4 The seller: who actually runs the India storefront	339
7.5 Better hardware for a licensed Indian amateur	347
7.6 What to actually do for convoy communication	349
7.7 The plan, the sequence, and what to confirm before spending money	354
7.8 Disclaimer for this part	359
Part 8 — What Indian expedition groups actually use	359
8.1 If you only read one page	361
8.2 The ranked practice picture, in detail	363
8.3 Bluetooth and mesh helmet intercoms	367
8.4 Radios in practice — who really uses them, and what they get wrong	380
8.5 The Citizen Band correction	389
8.6 Why community advice on this subject is unreliable	395
8.7 The recommended architecture for a mixed group	398
8.8 Disclaimer for this part	405
Part 9 — Listening, receiving, and the Citizen Band verdict	406

9.1 If you only read one page	407
9.2 Why no radio covers all frequencies	412
9.3 The widest hardware that exists	420
9.4 The legal asymmetry, and why it exists	431
9.5 Receive legality in India, question by question	434
9.6 The Citizen Band verdict	458
9.7 The correction to Parts 7 and 8	478
9.8 The recommended progression	484
9.9 Disclaimer for this part	495
Disclaimer	497

Owning and Operating a 10 W Walkie-Talkie in India

About this document

This report answers one practical question: **what must a private individual in India do to legally own and operate a 10 W hand-held two-way radio?** It covers every lawful route, the licence that each requires, the examination that stands in front of the realistic route, and what happens to people who skip all of it.

It was assembled from primary sources — the Department of Telecommunications, the Wireless Planning and Coordination Wing, the Saral Sanchar portal, gazette notifications, and the text of the governing Acts and Rules — supplemented by institute and club material where no official source exists.

How to read the evidence tags

Radio law is enforced with equipment seizure and prosecution, so every substantive claim in this document carries its evidential status inline:

Tag	Meaning
[VERIFIED]	Stated on a primary or official source that was fetched directly. The source URL and document date accompany the claim.
[REPORTED]	From a credible secondary source — an institute, an established club, a mainstream outlet, a vendor compliance page. The source is named.
[UNCERTAIN]	Could not be confirmed, sources disagree, or the source is undated. Treat as a question to put to WPC, not as settled law.

Never act on a [REPORTED] or [UNCERTAIN] claim as though it were settled. Where sources conflict, the report shows both and says which is more authoritative.

A warning about the question bank

Part 5 contains two kinds of question, and they must not be confused. **Section A** holds questions traceable to a real published source, each attributed. **Section B** holds practice questions constructed from the official syllabus — useful for study, but never presented as past papers.

Contents

- **Part 1 — The executive answer: can one person legally own and use a 10 W walkie-talkie in India?**
 - 1.1 If you only read one page
 - 1.2 The action plan, in brief
 - 1.3 Why radio is licensed at all
 - 1.4 The legal framework in plain language
 - 1.5 The taxonomy of licences: five different things, often confused
 - 1.6 Every lawful route to 10 W — the decision tree
 - 1.7 Routes that do not work — stated bluntly
 - 1.8 The three uncertainties a reader must not gloss over
 - 1.9 What changed in the ninety days before this report
 - 1.9A Adversarial fact-check log — 2026-08-23
 - 1.10 Abbreviations used in this report
 - 1.11 Status note
- **Part 2 — What you may do without a licence, and the separate law of equipment**
 - 2.1 If you only read one page of this part
 - 2.2 The four gates
 - 2.3 Where the power to de-license comes from, and what the 2023 Act changed
 - 2.4 The complete catalogue of India's de-licensed bands
 - 2.5 The three bands that can legally carry voice
 - 2.6 The verdict on licence-free use, and on 10 W
 - 2.7 Equipment gate 1 — Equipment Type Approval (ETA)
 - 2.8 Equipment gate 2 — import
 - 2.9 Equipment gate 3 — possession, and the death of the Dealer Possession Licence
 - 2.10 Baofeng-class imported radios — the answer, gate by gate
 - 2.11 Which radios sold in India are type-approved?
 - 2.12 What to ask WPC in writing before you spend money
 - 2.13 Summary of this part
 - 2.14 Disclaimer for this part

- Part 3 — The amateur radio route: grades, the ASOC examination, and the licence
 - 3.1 A note on evidence tagging in this part
 - 3.2 The chain of authority
 - 3.3 One document, not two
 - 3.4 The grades, and the authoritative power and band table
 - 3.5 Eligibility
 - 3.6 What an amateur licence does *not* permit — the restrictions that decide this question
 - 3.7 The ASOC examination
 - 3.8 After the exam
 - 3.9 Step-by-step action plan with realistic timelines
 - 3.10 Master document checklist
 - 3.11 Common claims about the amateur route that are wrong
 - 3.12 Questions to put to WPC in writing, before you buy a radio
 - 3.13 What this part does not cover
 - 3.14 Disclaimer for this part
- Part 4 — The ASOC syllabus in detail, and how to study for it
 - 4.1 Which syllabus is legally operative — and the honest uncertainty
 - 4.2 Exam structure, weighting and pass marks
 - 4.3 Section A — Radio Theory and Practice, topic by topic
 - 4.4 Section B — National and ITU Radio Regulations, topic by topic
 - 4.5 Part II — the Morse test (General grade only)
 - 4.6 Errors in the mainstream study material — check every one of these
 - 4.7 Official and free primary documents — assemble this folder first
 - 4.8 Institutional courses — NIAR and ARSI
 - 4.9 Regional clubs and training bodies
 - 4.10 Books, videos, mock tests and Morse resources
 - 4.11 A concrete study plan
 - 4.12 What remains genuinely unresolved in this part
- Part 5 — The question bank: what has actually been asked, and what could be asked
 - 5.1 How to read this part
 - 5.2 If you only read one page

- **Part 5A — DOCUMENTED QUESTIONS**
 - A.1 Why there are no official past papers, and what exists instead
 - A.2 Source register — full provenance
 - A.3 S1 — the official WPC/DoT sample paper (100 questions)
 - A.4 S2 and S8 — the NIAR Study Manual question bank (230 questions)
 - A.5 S3 — DARTS "Sample ASOC Exam Question Paper" (100 questions)
 - A.6 S4 — DARTS "Amateur Radio Exam Question Paper Sample" (188 questions, with a published key)
 - A.7 S5 — historical government papers, Grade I / Grade II, pre-2010 descriptive era
 - A.8 S6 — SIARS, "Queries: ASOC Exam questions" (11 items, answered)
 - A.9 S7 — AARS/INDRA study book, Annexure-III descriptive sample paper (31 items)
 - A.10 What was looked for and could not be found — the honest negatives
 - A.11 Count and reliability assessment
- **Part 5B — PRACTICE QUESTIONS (constructed from the syllabus)**
 - B.0 How Part 5B works
 - B.1 The paper's SECTION A — RADIO THEORY AND PRACTICE
 - B.2 GROUP C — INTERFERENCE AND ELECTROMAGNETIC COMPATIBILITY
 - B.3 GROUP D — SAFETY
 - B.4 The paper's SECTION B — RADIO REGULATIONS
 - B.5 Practice-question answer key at a glance
- **5.3 Critical uncertainties a reader must carry away from this part**
- **5.4 Disclaimer for this part**
- **Part 6 — Penalties, enforcement, risk, and appendices**
 - 6.1 Which law bites today, 2026-08-23
 - 6.2 The penalty provisions, exactly
 - 6.3 Search, seizure, inspection and arrest
 - 6.4 Real enforcement — what has actually happened, with dates
 - 6.5 Defences that do not work
 - 6.6 Honest risk assessment of operating unlicensed
- **Appendix A — Glossary of abbreviations**
 - A.1 Institutions and bodies
 - A.2 Statutes, rules and instruments
 - A.3 Licences, authorisations and certificates
 - A.4 Technical terms
 - A.5 Process and payment terms

- **Appendix B — Forms, fees and payment heads**
 - B.1 The amateur radio route (the realistic individual route to 10 W)
 - B.2 Equipment and possession permissions
 - B.3 Captive and commercial routes (for completeness — closed to natural persons)
 - B.4 Forms — there is no numbered form
 - B.5 Bharatkosh payment heads
- **Appendix C — Key contacts**
 - C.1 WPC headquarters and the amateur section
 - C.2 WPC field units — the Wireless Monitoring Organisation
 - C.3 NIAR — National Institute of Amateur Radio
 - C.4 ARSI — Amateur Radio Society of India
- **Appendix D — Sources, grouped by topic**
 - D.1 Statutes
 - D.2 Commencement notifications
 - D.3 Subordinate rules
 - D.4 Licence-exemption notifications
 - D.5 WPC orders, office memoranda and FAQs
 - D.6 DoT and portal service pages (all fetched 2026-08-23)
 - D.7 Official press releases and audit reports
 - D.8 Judgments
 - D.9 Study resources, clubs and exam material
 - D.10 Secondary analysis and news
 - D.11 Sources that could not be fetched
- **Appendix E — Legal disclaimer**
- **Part 7 — Choosing a radio: a worked example**
 - 7.1 If you only read one page
 - 7.2 The three intended uses, judged separately
 - 7.3 The UV-32 examined
 - 7.4 The seller: who actually runs the India storefront
 - 7.5 Better hardware for a licensed Indian amateur
 - 7.6 What to actually do for convoy communication
 - 7.7 The plan, the sequence, and what to confirm before spending money
 - 7.8 Disclaimer for this part

- **Part 8 — What Indian expedition groups actually use**
 - 8.1 If you only read one page
 - 8.2 The ranked practice picture, in detail
 - 8.3 Bluetooth and mesh helmet intercoms
 - 8.4 Radios in practice — who really uses them, and what they get wrong
 - 8.5 The Citizen Band correction
 - 8.6 Why community advice on this subject is unreliable
 - 8.7 The recommended architecture for a mixed group
 - 8.8 Disclaimer for this part
- **Part 9 — Listening, receiving, and the Citizen Band verdict**
 - 9.1 If you only read one page
 - 9.2 Why no radio covers all frequencies
 - 9.3 The widest hardware that exists
 - 9.4 The legal asymmetry, and why it exists
 - 9.5 Receive legality in India, question by question
 - 9.6 The Citizen Band verdict
 - 9.7 The correction to Parts 7 and 8
 - 9.8 The recommended progression
 - 9.9 Disclaimer for this part

Part 1 — The executive answer: can one person legally own and use a 10 W walkie-talkie in India?

Position as at 2026-08-23. Indian radio law changed three times in the ninety days before this date. Every figure below carries the date of the source it came from.

Evidence tags used throughout this report:

Tag	Meaning
[VERIFIED]	Stated on a primary or official source that was actually fetched — a gazette PDF, the bare text of an Act, a Department of Telecommunications (DoT) or Press Information Bureau (PIB) page. URL and document date given.
[REPORTED]	From a named credible secondary source, or an official document retrieved only from a non-government mirror.
[UNCERTAIN]	Could not be confirmed, sources disagree, or the source is undated.

Nothing tagged **[REPORTED]** or **[UNCERTAIN]** below should be treated as settled law.

1.1 If you only read one page

The one-sentence answer. Yes — one lawful route exists for a private individual in India to own and operate a hand-held radio at the 10 watt level, and it is the **Amateur Station Operator Certificate (ASOC)**, the Indian amateur radio ("ham") licence. Every other route is either capped far below 10 W, closed to natural persons, or both. The catch is that an amateur licence restricts *who you may talk to and about what* far more tightly than most readers expect.

Question	Answer	Confidence
Can a 10 W walkie-talkie ever be licence-free in India?	No. The highest licence-exempt power for any two-way voice radio is 5 W effective radiated power (ERP) on Citizen Band (CB), 26.957–27.283 MHz , and the highest licence-exempt hand-held ultra-high-frequency (UHF) allocation is PMR446 at 446.0–446.2 MHz, 500 milliwatt (mW) e.r.p. — one twentieth of 10 W. (The only other UHF exemption that permits voice, LPD433 at 434.04–434.79 MHz, is capped at 10 mW — see §1.5, Group E.)	[VERIFIED]
Is there any route to 10 W for one individual?	Yes, exactly one: an ASOC. An ASOC (Restricted) — the entry-level grade, no Morse test — authorises 10 watts maximum direct-current (DC) input power on 50–54 MHz, 144–146 MHz and 434–438 MHz. (The same grade separately carries ten high-frequency bands at 50 W DC input , voice emissions A3E/H3E/J3E/R3E only. The 10 W figure governs only 6 m, 2 m and 70 cm — the bands a hand-held actually uses.)	[VERIFIED] as to the table — Annexure-I of the 2010 WPC Office Memorandum was re-read page by page from the scan during the adversarial fact-check of 2026-08-23 and the figures above are exact; [UNCERTAIN] whether that 2010 table is still the operative specification (see §1.8)
Does "10 W" on the licence mean 10 W out of the antenna socket?	No. The limit is DC input power to the final stage of the transmitter , not radio-frequency (RF) output. A radio <i>advertised</i> as "10 W output" almost certainly draws more than 10 W DC and therefore exceeds the Restricted ceiling.	[VERIFIED] on the limit; the conversion arithmetic is engineering inference, [UNCERTAIN]
So which grade do I actually need?	If you want " the number on my licence is 10 W " → ASOC (Restricted) . If you want 10 watts of RF at the antenna → ASOC (General) (25 W DC input), which adds an 8 words-per-minute Morse test.	[VERIFIED] on both ceilings; the mapping between them is inference

Question	Answer	Confidence
What will it cost me in government fees?	₹1,100 or ₹2,100 total , once. ₹100 examination fee plus ₹1,000 for 20 years' validity or ₹2,000 for lifetime validity (defined as up to age 80).	[VERIFIED]
Can I use it to talk to my family, my staff, my security guards, or my trekking group?	No. Amateur radio may be used only to communicate with other authorised amateur stations , in plain language limited to messages of a technical nature , solely with a personal aim and without any pecuniary interest . Business communications and third-party messages are expressly forbidden.	[VERIFIED]
Can I get a commercial or captive ("CUG") walkie-talkie licence instead?	Not as an individual. Since 2026-06-23 the eligibility rule for captive telecommunication authorisations admits only companies and government bodies . A natural person is not eligible for any of them.	[VERIFIED]
Can I just buy a Baofeng-class radio and be careful?	No. Those radios cover 136–174 / 400–470 MHz or wider. Almost all of that is licensed spectrum. Unauthorised possession is a contravention carrying a civil penalty of up to ₹50,000 for a first instance and up to ₹2 lakh for each later instance (Third Schedule, item 1(a)). If — as appears to be the case — the Indian Wireless Telegraphy Act 1933 is still in force, s.6(1A) also puts up to three years' imprisonment on the table for possessing a <i>transmitter</i> without licence.	[VERIFIED] as to both provisions (Third Schedule read from the gazette PDF of Act 44 of 2023; s.6(1A) read from DoT's own copy of the 1933 Act). Two hedges added on fact-check 2026-08-23: (a) the ₹50,000 penalty is imposed by an Adjudicating Officer under Chapter VIII, and Chapter VIII has not been shown to be commenced — see §1.8 and Part 6 §6.1, so the civil track may not yet be operable; (b) the s.43 search-and-seize power is predicated on reason to believe an offence punishable under s.42 has been committed , and bare unauthorised possession of an ordinary walkie-talkie is a Third Schedule <i>civil</i> contravention, not an s.42 offence — so s.43 does not plainly reach it. The 1933 Act's own s.7 search power (offences under s.6) does. [UNCERTAIN] which power an officer would in practice invoke

The blunt version. If what you actually want is *a radio to coordinate a family trip, a shop, a wedding, an event crew or a security detail*, then 10 W is not lawfully available to you in India in 2026, and an amateur licence will not make it lawful — it will simply give you a licence you would immediately be breaching. Your lawful options in that case are a **500 mW PMR446 hand-held** (no licence, no exam, no fee), a **5 W ERP CB set**, or **renting handsets from a licensed Public Mobile Radio Trunking Service (PMRTS) operator**. If what you want is *to get into radio as a technical hobby and operate at 10 W*, the ASOC is a genuinely open, cheap and quick door.

1.2 The action plan, in brief

For the reader who has decided the amateur route fits. Each step is expanded in later parts of this report.

#	Step	Fee	Realistic elapsed time
1	Decide grade. Restricted (no Morse, 10 W DC input) or General (8 wpm Morse, 25 W DC input).	—	—
2	Register on the Saral Sanchar portal (saralsanchar.gov.in) as Individual (WPC Exams/Certificates) and apply on the Amateur/HAM module. Since March 2025 the exam application and the certificate application are a single step, with self-certified documents.	₹100 exam fee [VERIFIED]	Same day to file
3	Sit the examination at a Wireless Monitoring Station (WMS) of the Wireless Monitoring Organisation (WMO). One paper, two sections — radio theory and practice (Part A), and radio regulations (Part B). Pass mark: rule 5(5) of the 2024 Rules requires "a minimum of forty per cent of the total marks in respect of each category of ASOC" [VERIFIED — G.S.R. 675(E)], and DoT's WPC FAQ of 2024-12-05 states 40 % in each Part and 40 % overall for both grades. But two older official documents still in circulation — Annexure-II to the 2010 OM and the Saral Sanchar HAM FAQ — say 40 %/50 % for Restricted and 50 %/60 % for General. Assume the harder figures until WPC confirms; see Part 3 §3.7.12.	included	Date fixed after scrutiny; exams run "as and when there are sufficient number of candidates" [VERIFIED — Saral Sanchar HAM FAQ Q14]
4	Pay the certificate fee; the ASOC issues with your call sign printed on it (VU3 block for Restricted, VU2 for General). Payment must be made within two years of the result.	₹1,000 (20 years) or ₹2,000 (lifetime) [VERIFIED]	Weeks
5	Buy or import equipment. An ASOC holder needs no separate import authorisation for amateur radio equipment [VERIFIED — rule 7(3), gazette text read directly]. The parallel claim that an ASOC holder needs no radio-equipment-possession authorisation rests on rule 10(f) of the 2026 REPA Rules, which the 2026-08-23 fact-check could not re-verify — see §1.5, Group B. Treat it as [UNCERTAIN] and ask WPC.	—	—
6	Operate — within 144–146 MHz and 434–438 MHz (and 50–54 MHz), within the power ceiling, keeping a bound-book log and an equipment register, identifying by call sign at the start and end of each transmission and at least every ten minutes.	—	Ongoing

Total government outlay: ₹1,100–₹2,100. Total realistic elapsed time: roughly three to nine months, dominated by examination scheduling and result processing. The fees are [VERIFIED]; the elapsed-time range is a judgement drawn from the research, not a published service standard.

1.3 Why radio is licensed at all

A beginner reasonably asks why a 200-gram plastic object should need a government certificate when a mobile phone does not. Four reasons, and they are worth understanding because they explain every rule that follows.

1. Spectrum is a shared physical medium with no property lines. Two transmitters on the same frequency in the same place do not politely queue; they wreck each other's signal. Radio waves cross fences, districts and borders. The only way to keep a frequency usable is for one authority to decide who may transmit, where, on what, and how loudly. Indian law now says this in terms: section 4(1) of the Telecommunications Act, 2023 provides that "*The Central Government, being the owner of the spectrum on behalf of the people, shall assign the spectrum in accordance with this Act, and may notify a National Frequency Allocation Plan from time to time.*" [VERIFIED — Act No. 44 of 2023, <https://egazette.gov.in/WriteReadData/2023/250880.pdf>, 2023-12-24]

2. Some users of the spectrum are safety-of-life services. Aviation, maritime, railway, police, fire and disaster-response traffic share the same physical medium as hobby radio. Interference from a cheap unfiltered transmitter is not a nuisance to them; it is a hazard. This is why even the licence-free Indian bands are granted "*on non-interference, non-protection and shared and non-exclusive basis*", and why a licensed user's complaint can force an exempt user to relocate, reduce power, change antenna, or stop altogether. [VERIFIED — G.S.R. 1047(E), 2018-10-18, rule 4]

3. Enforcement is real and technical. The Wireless Monitoring Organisation, the field unit of the Wireless Planning and Coordination (WPC) Wing, has run direction-finding and monitoring stations since 1952 — "*one International Satellite Monitoring Earth Station (ISMES), five International Monitoring Stations (IMSSs), and 22 Wireless Monitoring Stations (WMSs)*" — tasked with "*resolving harmful interference ... adherence to licensing conditions and ensuring interference-free wireless environments*". [VERIFIED — PIB Release ID 1988428, 2023-12-19, <https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1988428®=48&lang=2>]

4. In India there is an explicit security dimension. Two-way radios that work off the public network are attractive to people who do not want to be on the public network. Following the April 2025 Pahalgam attack, the Central Consumer Protection Authority (CCPA) notified guidelines on **2025-05-27** regulating the online listing and sale of walkie-talkies, and on **2026-01-16** imposed penalties of **₹10 lakh each on Meesho, Flipkart, Amazon and Meta Platforms** and ₹1 lakh each on four smaller sellers, for listing radios without Equipment Type Approval or frequency disclosure. [VERIFIED — Press Information Bureau (PIB) Release ID 2132575, 2025-05-30, and PIB Release ID 2215261, 2026-01-16; see Part 2 §2.10.1 and Part 6 §6.4.7.

The link to the Pahalgam attack is [REPORTED] — MediaNama, 2025-06-03. The guidelines' own operative text could not be retrieved from consumeraffairs.nic.in or doca.gov.in, so clause-level detail is [UNCERTAIN].] The practical consequence for a buyer is that the cheap "licence-free, 5 km range" UHF hand-held is now an actively policed product category.

1.4 The legal framework in plain language

1.4.1 WHO REGULATES

The **Wireless Planning and Coordination (WPC) Wing** is not an independent regulator. It is an internal wing of the **Department of Telecommunications (DoT)**, Ministry of Communications, headed by the **Wireless Adviser to the Government of India**. It exercises the Central Government's powers; it has no separate legal personality. Its field and enforcement arm is the **Wireless Monitoring Organisation (WMO)**, and its regional licensing arms are the **Regional Licensing Offices (RLOs)** at New Delhi, Kolkata, Mumbai and in the South. [VERIFIED — PIB Release ID 1988428, 2023-12-19; <https://eservices.dot.gov.in/network-short-range-uhf-hand-held-radio>]

A practical note: the historic standalone WPC website wpc.dot.gov.in **no longer responds** (connection refused on port 443, re-tested 2026-08-23). Live WPC content is at dot.gov.in and, for services, at eservices.dot.gov.in. The transactional portals are saralsanchar.gov.in (the legacy portal, still the route for amateur radio applications) and eservices.dot.gov.in/authorisation-portal (the new portal for authorisations under the 2023 Act). [VERIFIED — direct testing, 2026-08-23]

Refinement added on fact-check 2026-08-23: the Saral Sanchar *home page* now returns **HTTP 301** to <https://eservices.dot.gov.in/saral-sanchar>; deep links such as saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf still serve content directly. The DoT landing page confirms that Saral Sanchar remains the counter for "**WPC Certificate of Proficiency (HAM, RTR, GMDSS)**", "**WPC Network and Non-Network Licenses**", "**WPC DPL/NDPL Licenses**", SACFA clearances and WPC import licences, and carries the notice that new applications under the Indian Telegraph Act 1885 have stopped "for all kind of Unified Licenses, Unified Licenses (VNO), Standalone Licenses, Registrations, Permissions". [VERIFIED — <https://eservices.dot.gov.in/saral-sanchar>, read 2026-08-23] Do not type saralsanchar.gov.in and conclude from the redirect that the amateur counter has moved; it has not.

1.4.2 THE THREE STATUTES, AND THE AWKWARD FACT THAT ALL THREE ARE LIVE

Statute	What it does	Status as at 2026-08-23
Indian Telegraph Act, 1885 (Act 13 of 1885)	Section 4 gives the Central Government the "exclusive privilege of establishing, maintaining and working telegraphs" and the power to licence others. "Telegraph" is defined widely enough to catch every radio transmitter. Section 20 punishes establishing, maintaining or working an unauthorised wireless telegraph with imprisonment up to three years, or fine (no ceiling stated for the wireless limb), or both , and s.20(3) allows forfeiture on conviction. Note s.20(2): a wireless-telegraph offence under this section is bailable and non-cognizable — materially softer than the 2023 Act's s.42(7).	Scheduled for repeal by s.60 of the 2023 Act, but that repeal appears not to have been commenced. Upgraded on fact-check 2026-08-23 from [REPORTED] to [VERIFIED]: the section texts were read on a gov.in host — https://wbconsumers.gov.in/writereaddata/ACT%20&%20RULES/Relevant%20Act%20&%20Rules/THE%20INDIAN%20TELEGRAPH%20ACT,%201885.htm (India Code still returns HTTP 403 and DoT's own PDF path returns 403/404)
Indian Wireless Telegraphy Act, 1933 (Act 17 of 1933)	"An Act to regulate the possession of wireless telegraphy apparatus." Section 3 bans possession without a licence; s.6(1A) punishes possessing a <i>wireless transmitter</i> without licence with up to three years' imprisonment; s.6(2) lets a court presume possession where apparatus is under a person's "ostensible charge" or in premises under his "effective control"; s.11 makes clear that a possession licence never authorises transmission.	Same — scheduled for repeal, apparently not yet repealed. [VERIFIED — DoT's own copy, https://eservices.dot.gov.in/sites/default/files/2024-11/The_Indian_Wireless_Act,_1933.pdf]

Statute	What it does	Status as at 2026-08-23
Telecommunications Act, 2023 (Act 44 of 2023)	Consolidates the law. Section 3(1) requires an authorisation to (a) provide telecommunication services, (b) establish or operate a telecommunication network, or (c) possess radio equipment . Section 4(2) requires a separate spectrum assignment to use spectrum. Section 47 governs amateur operator certification. Sections 42–43 create offences and search powers; the Third Schedule creates civil penalties.	In force in pieces. [VERIFIED]

Commencement, as far as it could be established:

Notification	Dated	Effective	Provisions commenced
S.O. 2408(E)	2024-06-21	2024-06-26	ss. 1, 2, 10–30, 42–44, 46, 47, 50–58, 61, 62
S.O. 2623(E)	2024-07-04	2024-07-05	ss. 6–8, 48, 59(b)
S.O. 3368(E)	2026-06-23	2026-06-23	s. 3(1) and s. 3(6)

All three [VERIFIED] from primary gazette PDFs; the most recent is at <https://egazette.gov.in/WriteReadData/2026/273762.pdf>.

One discrepancy found on fact-check 2026-08-23, on the "Dated" column only. The India Code footnotes to section 1 of the Act give the *notification* dates as **26 June 2024** for S.O. 2408(E) and **5 July 2024** for S.O. 2623(E) — i.e. the same day as the commencement each appoints — whereas the table above dates them 21 June and 4 July. **The effective dates (2024-06-26 and 2024-07-05) and the lists of provisions are the same on both accounts and are not in doubt.** The signing dates are [UNCERTAIN]; nothing in this report turns on them. The egazette portal itself could not be reached during the fact-check (TLS certificate chain error), so the 2026 notification could not be re-fetched independently — the S.O. 3368(E) entry rests on the earlier retrieval recorded in Part 2 §2.3 from <https://www.dot.gov.in/static/uploads/2026/06/35e1d307f3cbb6396b4c48802ed4b044.pdf>, which now returns HTTP 403 to an unauthenticated fetch. **Treat the S.O. 3368(E) row as [VERIFIED] on the original retrieval but currently uncheckable.**

Two consequences follow, and they matter:

- **Possession of radio equipment became an authorisation-requiring act on 2026-06-23**, under the 2023 Act itself. Before that date, the 2023 Act imposed no such requirement (the 1933 Act did). [VERIFIED]

- **Section 60 — the repeal of the 1885 and 1933 Acts — does not appear in any commencement notification that could be found.** Section 61 (which deems the old rules to continue) *was* commenced in 2024; section 60 was not. Strong circumstantial evidence supports this: DoT was still making de-licensing rules **under both old Acts** as late as 2026-06-11, and the July 2026 possession rules expressly speak of applications "made for grant of a license under the Indian Wireless Telegraphy Act, 1933 ... prior to the date of commencement of these rules". [VERIFIED as to those documents; [UNCERTAIN] as a definitive negative — no usable public gazette search was available.]

Do not tell yourself the old Acts are repealed. The honest formulation is: *they are repealed by section 60 of the 2023 Act, but section 60 has not been brought into force, so both Acts appear to remain operative.* This matters because the 1933 Act's three-year criminal penalty for possessing a wireless transmitter without licence is harsher than anything in the new Act's possession regime.

1.4.3 POSSESSION, OPERATION AND SPECTRUM ARE THREE DIFFERENT PERMISSIONS

This is the point on which most online advice about Indian walkie-talkies is simply wrong.

Legal act	Old law	New law	In force?	Exposure without permission
Possessing the radio	IWTA 1933 s.3 (licence to possess)	Telecom Act 2023 s.3(1)(c)	Yes, from 2026-06-23	Civil penalty up to ₹50,000 first instance, up to ₹2 lakh each subsequent instance (Third Schedule item 1(a)); plus, if the 1933 Act stands, up to 3 years' imprisonment under s.6(1A)
Establishing / operating a network	ITA 1885 s.4 (operating licence)	Telecom Act 2023 s.3(1)(b)	Yes, from 2026-06-23	s.42(1): up to 3 years and/or up to ₹2 crore, cognizable and non-bailable (s.42(7))
Using a frequency	ITA 1885 s.4 + a WPC frequency assignment	Telecom Act 2023 s.4(2)	Section 4 not found to be in force	No penalty in the new Act for bare unassigned use; s.20 of the 1885 Act would bite if that Act stands

[VERIFIED] for the provisions and penalties; [UNCERTAIN] for the "in force" entries that rest on a negative search.

Two nuances a beginner would not guess:

- **You do not have to switch the radio on.** Under the 1933 Act a court "may presume" possession where the apparatus is under your "ostensible charge" or in premises under your "effective control". A 10 W transceiver in your cupboard is possession. [VERIFIED]

- **A private pair of walkie-talkies is arguably not a "telecommunication network".** The 2023 Act defines that term as a system "*used or intended to be used for providing telecommunication services*" — which a purely personal radio pair is not. On the statutory text, the three-year offence in s.42(1) does not reach private walkie-talkie use, leaving the civil possession penalty as the main new-Act exposure. [VERIFIED on the text; whether a prosecutor or court would agree is [UNCERTAIN] — no case law was found.]

1.4.4 SUBORDINATE RULES CURRENTLY GOVERNING A HAND-HELD RADIO

Instrument	Gazette	Date	What it does
Telecommunications (Amateur Services) Rules, 2024	G.S.R. 675(E)	2024-10-29 (published 2024-10-30)	Creates ASOC (General) and ASOC (Restricted); supersedes the Indian Wireless Telegraphs (Amateur Service) Rules, 1978; abolishes the separate amateur station licence
Telecommunications (Authorisation for Captive Telecommunication Services) Rules, 2026	G.S.R. 511(E)	2026-06-23	Captive mobile radio trunking (CMRTS), captive non-public network (CNPN), captive very small aperture terminal (VSAT), captive general service. Individuals ineligible
Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026 ("REPA Rules")	G.S.R. 592(E)	2026-07-08; applications live 2026-08-06	Possession authorisation — for dealers and testers only ; rule 10 lists who needs none, including ASOC holders
Telecommunications (Terms and Conditions for Migration) Rules, 2026	G.S.R. 510(E)	2026-06-23	Migration of old licences to new authorisations
Use of Low Power and Very Low Power Short Range Radio Frequency Devices (Exemption from Licensing Requirement) Rules, 2018	G.S.R. 1047(E)	2018-10-18	The de-licensing rules; Table V is PMR446 at 446.0–446.2 MHz, 500 mW e.r.p.
Use of low power equipment in the Citizen Band 26.957–27.283 MHz (Exemption from Licensing Requirement) Rules, 2005	G.S.R. 533(E)	2005-08-12 (amended by G.S.R. 35(E), 2007-01-10)	CB at 5 W ERP , built-in antenna
WPC Office Memorandum No. L-14011/255/2004-AMT, Annexure-I	—	2010-08-13	The amateur band / emission / power table. Not a gazette instrument; an executive memorandum
WPC Order No. P-11014/34/2009-PP	—	2023-12-11	Spectrum charges (royalty and licence fee) for captive users

All [VERIFIED] as to identity and date. Note that the older de-licensing rules were made under the 1885 and 1933 Acts and survive the new Act by force of sections 3(4), 4(7) and 61 — nothing in the 2023 Act made a previously licence-free radio unlawful. [VERIFIED]

1.5 The taxonomy of licences: five different things, often confused

Indian radio permissions divide by **what the permission is about**, not by who wants it. Read the group headings before the rows; conflating these groups is the commonest and most expensive mistake.

GROUP A — PERMISSIONS ABOUT THE *PERSON* (OPERATOR COMPETENCE)

Permission	Legal basis	Fee	Validity	Open to an individual?
ASOC (Restricted) — entry-level amateur certificate, call sign block VU3	Telecom Act 2023 s.47; Amateur Services Rules 2024, rules 3–6	₹100 exam + ₹1,000 (20 yr) or ₹2,000 (lifetime, to age 80)	20 years or lifetime	Yes — age ≥12, Indian citizen
ASOC (General) — call sign block VU2; adds 8 wpm Morse	same	same	same	Yes
RTR (Radio Telephone Operator's Restricted certificate)	Telecom Act 2023 s.46; RTR Rules 2025	not captured	—	Aeronautical use only
GMDSS (Global Maritime Distress and Safety System) Commercial Radio Operator Certificate of Proficiency	Telecom Act 2023 s.46; GMDSS Rules 2024	not captured	—	Maritime use only

[VERIFIED] for the ASOC rows — G.S.R. 675(E), rules 4(1), 5(2), 6(1); <https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc>

Other ASOC charges, all [VERIFIED] from the gazette: duplicate certificate or change of station location **₹1,000** (rule 7(6)); late renewal within two years of expiry **₹1,000** late fee, with nothing considered after two years (rule 8); amateur society call sign **₹2,000** (rule 12); special-event call sign **₹200**, applied for at least 90 days ahead and valid 90 days (rule 13); reciprocal certificate for a foreign national **₹1,000** (rule 4(2)). The certificate is non-transferable and fees are non-refundable (rules 10 and 11).

GROUP B — PERMISSIONS ABOUT THE DEVICE (HARDWARE LEGALITY)

Permission	Legal basis	Who needs it	Fee	Status 2026-08-23
Equipment Type Approval (ETA)	De-licensing rules (e.g. G.S.R. 1047(E) rule 5(2))	Importer or Indian representative of any radio device sold for use in a de-licensed band	₹10,000 per application	Open; granted on self-declaration since 2024-09-09
WPC Import Licence	Notification No. 71 of 1953 under the Sea Customs Act 1878	Any entity importing a wireless transmitter/ receiver not covered by an undertaking	₹500 per licence, 1 year	Closed to new Telegraph Act applications; migrating to the possession authorisation from 2026-08-06. Individuals were never eligible
Dealer Possession Licence (DPL) / Non-Dealer Possession Licence (NDPL)	IWTA 1933; Possession Rules 1965	Dealers and non-dealer holders of wireless equipment	"nominal fees of Rs. 60"	Applications and renewals stopped from 2026-07-03
Radio Equipment Possession Authorisation (REPA)	Telecom Act 2023 s.3(1)(c); REPA Rules 2026	Dealers and test/demonstration importers only	₹1,000 application; ₹10,000 per year (dealers) or ₹2,000 per year, minimum ₹500 (test/demo)	Open from 2026-08-06
MTCTE (Mandatory Testing and Certification of Telecom Equipment)	Telecom Act 2023 s.19; Framework Rules 2025	Manufacturers/ importers of notified equipment	not captured	Open

[VERIFIED] for all rows except MTCTE, verified as to existence and legal basis only. Sources: <https://eservices.dot.gov.in/equipment-type-approval-eta> ; <https://eservices.dot.gov.in/import-license> ; <https://eservices.dot.gov.in/dealer-possession-license> ; <https://eservices.dot.gov.in/radio-equipment-possession-authorisation-services> ; G.S.R. 592(E).

Two things follow that a beginner would not guess:

- **An ordinary end user cannot apply for a possession authorisation even if he wants to.** Rule 4(1) of the REPA Rules restricts applicants to persons who intend to deal in radio equipment or to import it purely for testing or demonstration. An end user's possession must instead be covered by one of the gateways in **rule 10** — in practice an ASOC (rule 10(f)), a spectrum assignment (rule 10(e)), or a licence-exempt use (rules 10(c)/(d)). [VERIFIED on the original retrieval of G.S.R. 592(E)] **Fact-check caveat, 2026-08-23, and the reader should act on it.** The final 2026 REPA Rules could not be re-fetched from any gov.in URL during the fact-check. What was re-read is the **February 2025 draft** of the same rules, and in the draft the numbering and the substance both differ: the exemption provision is **rule 4**, not rule 10, and it exempts only (a) equipment exempted under s.3(3)/(4), (b) possession by an entity holding an authorisation under **s.3(1)(a) or (b)** plus a spectrum assignment (or a surviving Telegraph Act licence), and (c) equipment notified as needed by a user of a telecommunication service. **An ASOC is granted under s.47, not under s.3(1)(a) or (b), and an amateur holds no spectrum assignment** — so on the draft text an ASOC holder would *not* have been exempt. If the final rules added an express ASOC gateway, as this report's earlier retrieval records at rule 10(f), the position is as stated. If they did not, an ASOC holder's possession of amateur equipment rests on a different argument. **This is now [UNCERTAIN] and it is one of the two questions to put to WPC in writing before buying hardware** (the other is the power table, §1.8). The corresponding rows in §1.6 and the action plan in §1.2 carry the same caveat. The **import** exemption is unaffected — it comes from rule 7(3) of the Amateur Services Rules 2024, whose gazette text was read directly and is [VERIFIED].
- **ETA does not legalise anything for the operator.** ETA certifies that a *model* meets the technical limits of a *de-licensed band*. There is no ETA that makes a 10 W transceiver on amateur or land-mobile frequencies lawful for an unlicensed person to hold or use. [VERIFIED]

GROUP C — PERMISSIONS ABOUT THE TRANSMISSION (NETWORK AND SPECTRUM)

Permission	Legal basis	Scope	Cost	Open to an individual?
Legacy WPC "Network" licences — nine sub-categories, including Short Range UHF Hand Held Radio and Fixed/Land Mobile HF/VHF/UHF (below 806 MHz)	ITA 1885 s.4 + WPC frequency assignment	Private closed-group radio networks. Workflow: Letter of Intent (LoI) → Decision Letter (DL) → Wireless Operating Licence (WOL)	Royalty ₹2,000–₹75,000 per year by band, coverage and city category; plus licence fee ₹500 per base station and ₹250 per hand-held per year	DoT's page says "Any individual or an Indian company" — but see §1.7

Permission	Legal basis	Scope	Cost	Open to an individual?
Captive Mobile Radio Trunking Service (CMRTS) authorisation	Captive Rules 2026	Trunked captive land mobile through a repeater	₹10,000 processing + ₹20,000 initial guarantee; ₹300 per user terminal per year, minimum ₹5,000	No
Captive non-public network (CNP) / captive VSAT / captive general service	Captive Rules 2026	Private 5G-type networks; satellite; residual wireline and wireless captive networks	Captive general: nil fees. Captive VSAT: ₹7.5 lakh entry fee	No — and captive <i>general</i> is additionally barred to private limited companies
Public Mobile Radio Trunking Service (PMRTS) authorisation	Miscellaneous Services Rules 2026	Commercial trunking sold to third parties	₹10,000 processing + ₹20,000 entry + ₹20,000 guarantee	No — but you can be a <i>subscriber</i>
Spectrum assignment	Telecom Act 2023 s.4 (not found in force) or ITA 1885 s.4 + WPC Order 2023-12-11	The right to use a specific frequency	Royalty + licence fee as above	Only as an adjunct to one of the above

[VERIFIED] for the fee figures — WPC Order No. P-11014/34/2009-PP dated 2023-12-11; G.S.R. 511(E) Schedule and rule 38; <https://eservices.dot.gov.in/fees-charges-for-services>.

GROUP D — PERMISSION ABOUT THE SITE

SACFA — Standing Advisory Committee on Frequency Allocation clearance is a *siting* clearance for wireless installations: masts, towers, base-station antennas. Fees: **₹1,000 per general application**, ₹100 per small cell, ₹100 per additional antenna. Auto-cleared sites are approved instantly; scrutiny cases take up to 30 days. [VERIFIED — <https://eservices.dot.gov.in/sacfa-standing-advisory-committee-frequency-allocation-clearance>]

For an amateur, the requirement has a statutory hook that an earlier draft of this part omitted: **Annexure A, paragraph 4(3) of the Telecommunications (Amateur Services) Rules, 2024 provides that "The Standing Advisory Committee for Frequency Allocation (SACFA) clearance shall be required for an amateur station, as applicable."** [VERIFIED — G.S.R. 675(E), gazette text read directly, 2026-08-23] The words "as applicable" are then filled in by WPC's own FAQ: "SACFA clearance is required if the amateur station antenna is installed on a tower, building, or mast." [REPORTED — WPC FAQ No. P-14036/02/2023-COP, 2024-12-05, Q6; the scan was read in full during the fact-check, but it is hosted by ARSI rather than on a gov.in URL, so the tag stands] A hand-held with its stock rubber-duck antenna therefore raises no siting

question; a rooftop mast does. Note also 2010 OM Annexure-I note (ii), which subjects the whole band authorisation to "*site clearance as per the procedure prescribed by the Standing Advisory committee on Radio frequency allocation (SACFA) as applicable*". [VERIFIED — scan read 2026-08-23]

GROUP E — NO PERMISSION AT ALL (LICENCE-EXEMPT)

The complete licence-exempt power ladder for anything resembling a hand-held voice radio:

Band	Maximum power	Instrument	Usable as a voice hand-held?
26.957–27.283 MHz (CB)	5 W ERP , built-in antenna only	G.S.R. 533(E), 2005-08-12	Yes — but this is a high-frequency (HF) 11-metre band; a truly built-in antenna on a hand-held is electrically tiny and performance is poor
446.0–446.2 MHz (PMR446)	500 mW e.r.p. , integral antenna, hand-portable, peer-to-peer, no base station or repeater	G.S.R. 1047(E), 2018-10-18, Table V	Yes — this is the licence-free walkie-talkie in India
433.05–434.79 MHz	10 mW e.r.p. , ≤10 % duty cycle; voice allowed only in the 434.04–434.79 MHz sub-band , at ≤25 kHz with listen-before-talk and 1-minute maximum transmissions	G.S.R. 347(E), 2022-05-09	Technically yes; 10 mW gives a few hundred metres. A curiosity
169.4–169.5875 MHz	500 mW ERP	G.S.R. 1047(E), Table IV	No — assistive listening only
865–868 MHz	25 mW / 500 mW / 2 W e.r.p. by category	G.S.R. 853(E), 2021-12-10	No — radio-frequency identification (RFID) and telemetry, no voice allowance
2.4 GHz / 5 GHz / 6 GHz	4 W ERP and below	G.S.R. 45(E) 2005; G.S.R. 1048(E) 2018; 6 GHz rules 2026	No — wireless local-area networking

[VERIFIED] — the band and power figures were read from the gazette texts, retrieved from DoT's own compilation of subordinate legislation and from Internet Archive captures of DoT PDFs.

Note one source conflict here. Research file r01 could only reach the PMR446 rule through a secondary bare-act reproduction and tagged the 500 mW figure [REPORTED]; research files r02 and r09 both retrieved the gazette text itself (Gazette of India Extraordinary Part II §3(i), No. 753, 2018-10-18) and quote Table V verbatim. The primary retrieval governs: **500 mW e.r.p. at 446.0–446.2 MHz is [VERIFIED]**.

1.6 Every lawful route to 10 W — the decision tree

Work down this tree. It has only one branch that ends in a lawful 10 W hand-held for a private individual.

- Q1. Do you actually need 10 W, or do you need "a walkie-talkie that works"?
- └ 500 mW is enough (a few hundred metres to ~2 km, line of sight)
 - PMR446, ETA-approved, 446.0–446.2 MHz. No licence, no exam, no fee. Any person, any purpose. Available today. [VERIFIED]
 - └ You want more range and can accept 27 MHz HF and a built-in antenna
 - Citizen Band, 5 W ERP. No licence, no exam, no fee. Any purpose. This is the licence-free ceiling for two-way voice radio in India. [VERIFIED]
 - └ You specifically need 10 W → go to Q2.
- Q2. What will you use it FOR?
- └ Business, security, staff, events, a delivery fleet, coordinating a shop
 - Amateur radio is NOT available to you. "Communications of business" are expressly forbidden. [VERIFIED]
 - Captive/CUG authorisation requires a company or a government body. A natural person is not eligible. [VERIFIED]
 - Realistic options: subscribe to a PMRTS operator's service, or operate your employer's licensed radios, or incorporate a company and pursue a captive authorisation – and even then the plain "captive general" category is barred to private limited companies.
 - └ Talking to family, friends or a trekking group who are not licensed
 - NOT available. An amateur station may communicate only "with other authorised amateur stations". [VERIFIED]
 - Fall back to PMR446 at 500 mW.
 - └ Self-training, technical experimentation, and contacting other licensed amateurs → go to Q3.
- Q3. What does "10 W" mean to you?
- └ "The number written on my authorisation is 10 W"
 - ASOC (Restricted). 10 W maximum DC input to the final stage on 50–54, 144–146 and 434–438 MHz. No Morse test. ₹1,100–₹2,100. [VERIFIED]
 - └ "10 watts of RF coming out of the antenna socket"
 - A radio delivering 10 W RF draws well over 10 W DC, so it exceeds the Restricted ceiling. You need ASOC (General): 25 W DC input, plus an 8 words-per-minute Morse test. [VERIFIED on both ceilings; the DC-to-RF mapping is engineering inference, UNCERTAIN]

THE FULL ROUTE COMPARISON

#	Route	What you must hold	Bands	Power ceiling	Government cost	Effort	Realistic elapsed time	Purpos limits
0a	PMR446, licence-exempt	Nothing (device needs ETA)	446.0–446.2 MHz	500 mW e.r.p. , integral antenna, no repeater	₹0	Buy the radio	Same day	None. A person, purpos

#	Route	What you must hold	Bands	Power ceiling	Government cost	Effort	Realistic elapsed time	Purposes and limits
0b	Citizen Band, licence-exempt	Nothing (device needs ETA)	26.957–27.283 MHz	5 W ERP , built-in antenna	₹0	Buy the radio	Same day	None
0c	433 MHz low-power device	Nothing	433.05–434.79 MHz	10 mW e.r.p.	₹0	Buy the radio	Same day	Voice only in 433.05–434.04 MHz, at 10 kHz, 1-minute transmission
1	ASOC (Restricted)	ASOC, call sign VU3	50–54, 144–146, 434–438 MHz; plus ten HF bands at 50 W	10 W DC input (≈4–6.5 W RF, inferred)	₹1,100 or ₹2,100 once	Written exam, no Morse	~3–9 months	Self-training, technical investigation, contact with other licensed amateur operators. No business, no third-party messages.
2	ASOC (General)	ASOC, call sign VU2	Same, plus 5725–5840 MHz and amateur satellite; 400 W on HF	25 W DC input on VHF/UHF	₹1,100 or ₹2,100 once	Written exam plus 8 wpm Morse	Longer — Morse practice dominates	Identical to Restricted
3	Legacy WPC "Short Range UHF Hand Held Radio" licence	Wireless Operating Licence via LoI → DL	Probably the UHF Short Range Radio (USR) channels, 350.1625–350.5375 MHz	[UNCERTAIN] — no DoT document states one; one secondary source says 2 W	Royalty ≈₹2,000–3,000/yr + ₹250 per hand-held/yr, on the USR reading	Application, documents built for companies, frequency assignment	3–7 months if open at all	Personal business, security, emergency. DoT's own words.

#	Route	What you must hold	Bands	Power ceiling	Government cost	Effort	Realistic elapsed time	Purpose limits
4	Captive authorisation (CMRTS / CNPN / captive general)	Authorisation + separate spectrum assignment	Trunking bands per NFAP footnote IND 18	Per assignment	₹10,000 processing + ₹20,000 guarantee (CMRTS)	Corporate document set, auditor's certificate, board resolution	0.6–2.6 years observed for CMRTS	Captive only; can connect public network
5	Subscribe to a PMRTS operator	Nothing — the operator holds everything	811–824 / 856–869 MHz	The operator's	Commercial tariff	Low	Days to weeks	Whatever operator authorises
6	Operate an employer's licensed radios	Nothing personal, for land mobile	As assigned	As assigned	₹0 to you	Be employed there	—	The employer's stated purpose
7	Government / public protection and disaster relief	—	—	—	—	—	—	—

[VERIFIED] for the fee, band, power, purpose and eligibility cells except where marked [UNCERTAIN]. **Elapsed-time figures are judgement, not published service standards**, except the CMRTS range, which is [REPORTED] from the ITU-APT Foundation of India's documented sample of six actual grants (0.6 to 2.6 years).

Cost detail for the two routes that matter

	ASOC (Restricted)	ASOC (General)	Legacy USR-class network licence (if obtainable)
Examination fee	₹100 [VERIFIED]	₹100 [VERIFIED]	none
Certificate / licence fee	₹1,000 for 20 years, or ₹2,000 lifetime (to age 80) [VERIFIED]	same [VERIFIED]	Spectrum royalty, formula-based
Annual recurring cost	Nil [VERIFIED]	Nil [VERIFIED]	Royalty ≈ ₹2,000–₹3,000/yr for USR at minimum fixed-site rates, [UNCERTAIN] which minimum applies; plus ₹250 per hand-held per year and ₹500 per base station per year [VERIFIED]

	ASOC (Restricted)	ASOC (General)	Legacy USR-class network licence (if obtainable)
Possession authorisation	Not required — REPA rule 10(f) [VERIFIED on original retrieval; [UNCERTAIN] after the 2026-08-23 fact-check could not re-fetch the final rules — see §1.5, Group B]	same [UNCERTAIN]	Not required — REPA rule 10(a)/(e) [VERIFIED on original retrieval]
Import authorisation	Not required for amateur equipment — rule 7(3) [VERIFIED]	Not required [VERIFIED]	WPC import licence, ₹500 [VERIFIED]
SACFA	Only if the antenna goes on a tower, building or mast [REPORTED]	same	Required for the land-mobile and captive-trunking licences; not listed in DoT's published process for the hand-held service [VERIFIED that it is absent; [UNCERTAIN] that it is exempt]
Renewal	Apply ≥12 months before expiry; ₹1,000 late fee up to two years; nothing after two years [VERIFIED]	same	Five years at a time [VERIFIED]

A note on the draft that could change the last column: the **draft Telecommunications (Spectrum Assignment by Administrative Process) Rules, 2026** (G.S.R. 494(E), 2026-06-17, comments closed 2026-07-18) propose a flat **₹15,000 per frequency spot for five years** for UHF Short Range Radio, plus **₹250 per hand-held radio station per year**. These are the best current indication of what a lawful private VHF/UHF assignment would cost — but **they are draft figures and not yet law**, and no final version could be found as at 2026-08-23. [VERIFIED as to the draft text; **[UNCERTAIN]** as to whether it will be notified in this form]

1.7 Routes that do not work — stated bluntly

A 10 W radio can never be licence-exempt in India. Not on 446 MHz, where the cap is 500 mW — one twentieth. Not on 27 MHz, where the cap is 5 W ERP and the antenna must be built in. Not on 433 MHz, where the cap is 10 mW. The exemptions are *conditional on the power limit*: transmit above it and the exemption simply does not apply to you, and you fall back to needing an authorisation under s.3(1) and an assignment under s.4(2), which you do not have. [VERIFIED]

The captive / CUG route is shut to natural persons. Rule 5 of the Captive Rules 2026 admits only "a company" or a listed government body. A partnership firm, a sole proprietorship, a limited liability partnership, a society, a trust and a resident welfare association are all outside it too. And the residual "captive general service" category — the one that on its face covers a plain private wireless network — is barred even to private limited companies by the first proviso to rule 5. [VERIFIED — G.S.R. 511(E), 2026-06-23] There is a discretionary relaxation power in the second proviso, but it is exercised by the Central Government in the public interest, not on request.

Fact-check note, 2026-08-23 — the headline holds, one sub-claim is downgraded. The final 2026 rules could not be re-fetched from a gov.in URL during the fact-check (the DoT static-upload paths return HTTP 403 and the egazette portal presents a broken TLS chain). What was re-fetched is the **draft** of the same rules — G.S.R. 614(E), notified 2025-09-10, read in full from [https://thc.nic.in/Central Governmental Rules/Telecommunications \(Authorisation for Captive Telecommunication Services\) Rules, 2025.pdf](https://thc.nic.in/Central%20Governmental%20Rules/Telecommunications%20(Authorisation%20for%20Captive%20Telecommunication%20Services)%20Rules,%202025.pdf) — whose eligibility rule (numbered **rule 7(1)** in the draft, renumbered to rule 5 in the final version according to this report's earlier retrieval) reads: "*An applicant seeking authorisation under these rules shall be a company incorporated under the Companies Act, or a Ministry or Department of the Central or State Government, the Parliament of India, Legislative Assembly of a State of India, any Court of law, or any agency or autonomous organisation of the Central or State Government.*" **The central claim — that a natural person is not eligible for any captive authorisation — is therefore independently corroborated at draft stage and stands as [VERIFIED].** Two narrower points are **downgraded to [UNCERTAIN]** because they depend on provisos that do not exist in the draft text and could not be re-read in the final: (a) that the "*captive general service*" category is additionally barred to **private limited** companies by a first proviso; and (b) the exact gazette number **G.S.R. 511(E)** — a contemporaneous public post cites **G.S.R. 512(E)** for a rule set gazetted the same day (2026-06-23), and the sibling instruments in this family carry adjacent numbers, so a transposition cannot be excluded. Cite the rules by name and date, not by number, until you have the gazette in hand.

The individual walkie-talkie licence that used to exist is, at best, in limbo. Until mid-2026 DoT published a network licence category called "Short Range UHF Hand Held Radio", described in DoT's own words as being for "*owning and operating short-range Ultra High Frequency (UHF) Hand-Held Radios such as Walkie Talkies*" for "*personal, business, security, and emergency applications*", and open to "**Any individual** or an Indian company". [VERIFIED that the page says this — <https://eservices.dot.gov.in/network-short-range-uhf-hand-held-radio>] Then, from 2026-06-25, DoT stopped accepting new applications under the Indian Telegraph Act 1885. **Here the research files disagree, and the conflict must be stated openly:**

- Research file r01 reads the transition notice as having closed this counter outright.
- Research file r03 points out that the notice enumerates "*Unified Licenses, Unified Licenses (VNO), Standalone Licenses, Registrations, Permissions*" and **does not name WPC Network licences**, while the Saral Sanchar portal still advertises "WPC Network, Non Network and DPL/NDPL Licenses" among the things it issues.

r03's reading is the better-supported one, because it turns on the exact enumeration in the notice rather than on an inference from it. The honest position is therefore: *the notice is [VERIFIED]; whether it captures a WPC Network licence application is [UNCERTAIN]; nothing in the 2026 authorisation rules replaces this category for individuals; and a reader who wants this route must ask a Regional Licensing Office in writing whether a new application can be filed today, and on which portal.* Even if it can, the document checklist demands a Corporate Identity Number (CIN) or corporate address proof and an authorisation letter — a set built for

organisations — and one secondary source puts the USR power ceiling at 2 W, which would make the route incapable of delivering 10 W in any event. [REPORTED — HFUnderground wiki, undated; not confirmed by any DoT document]

An amateur licence will not legalise talking to your family or running your business. Three separate prohibitions bite, and they are cumulative:

1. Radio communications "*may be exchanged with other authorised amateur stations*". Your spouse, your driver and your shop assistant are not authorised amateur stations unless each holds an ASOC. [VERIFIED — Amateur Services Rules 2024, Annexure A, para 2(1)(a)]
2. "*Transmissions shall be made in plain language and limited to messages of a technical nature.*" [VERIFIED — Annexure A, para 2(1)(b)]
3. The holder "*shall not transmit messages containing ... advertisements, communications of business, statements on topics of political or industrial controversy, or third party messages*". [VERIFIED — Annexure A, para 2(2)(b)]

And section 47 of the Act itself now defines amateur services statutorily as being carried out "*solely with a personal aim and without any pecuniary interest*". [VERIFIED] Using a 10 W hand-held under an ASOC to coordinate a security detail, a wedding, a film shoot, a warehouse or a delivery fleet is business communication and is squarely prohibited. Whether Indian hams in practice converse socially, and are tolerated in doing so, is a separate question — the text is narrow, and the tolerance is unwritten. [VERIFIED on the text; [UNCERTAIN] on enforcement practice]

"It has an ETA sticker, so I can use it" is wrong. ETA is about the device's right to be sold into the Indian market for de-licensed-band use. It says nothing about your right to transmit. [VERIFIED]

"I'll import a Baofeng-class radio myself" does not work either. The WPC import licence is granted only to "*any entity registered in India and holding valid permission granted by DoT*". [VERIFIED — <https://eservices.dot.gov.in/import-license>, re-read 2026-08-23; the same page now also carries the notice that new applications under the Indian Telegraph Act 1885 have stopped and that Radio Equipment Possession Authorisation applications opened on the Authorisation Portal on 2026-08-06] **Hedge added on fact-check:** DoT does not say "individuals are ineligible" in those words. The conclusion that a private person cannot obtain this licence is an **inference** from two things — the eligibility formula ("entity registered in India" *and* "holding valid permission granted by DoT", which a private person by definition does not hold) and the document checklist, which demands a Corporate Identity Number or corporate address proof and an authorisation letter. The inference is strong, but it is an inference: [VERIFIED] as to the quoted eligibility text, [UNCERTAIN] as a categorical statement that no individual has ever been granted one. The single express exception is genuinely useful: "*An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment.*" [VERIFIED — Amateur Services Rules 2024, rule 7(3), gazette text read directly 2026-08-23] Whether a wideband 136–174 / 400–470 MHz consumer hand-held counts as "amateur radio equipment" for that purpose is [UNCERTAIN] — the 2024 Rules define the term circularly, as "*a radio equipment required for operating an amateur station*" (rule 2(1)(b)), which offers no help on a

radio that also covers non-amateur spectrum. Separately, in March 2026 the Directorate of Revenue Intelligence seized 11,060 Baofeng BF-888S sets at Nhava Sheva and two Mumbai importers were arrested under the Customs Act. [REPORTED — independently re-confirmed on fact-check 2026-08-23 across Press Trust of India copy carried by Communications Today, Telangana Today, Devdiscourse and Varindia; the reports also state that DoT has "blacklisted" the BF-888S, but **no gazette notification or published DoT list of blacklisted models could be located** — that specific assertion is [UNCERTAIN], see Part 2 §2.10]

Foreign radio services do not map onto India. FRS and GMRS (462/467 MHz in the United States) and MURS (151/154 MHz) have no Indian counterpart, and the US and European CB channel plans are not India's. Transmitting on them here is unauthorised use of spectrum. [Inference from the exhaustive Indian exemption list; [UNCERTAIN] only in the sense that this proves a negative]

Finally, "worst case they just confiscate it" understates the exposure — but the search power is narrower than an earlier draft of this part implied. Section 43 of the 2023 Act lets an authorised officer search any building, vehicle, vessel, aircraft or place and "*take possession*" of the equipment, without a conviction. [VERIFIED — s.43 read from the gazette PDF of Act 44 of 2023] **Correction made on fact-check 2026-08-23:** s.43 is *conditional*. It applies only where the officer "*has reason to believe that any unauthorised telecommunication network or telecommunication equipment or radio equipment in respect of which an offence punishable under section 42 has been committed is kept or concealed*". Bare unauthorised possession of an ordinary walkie-talkie is **not** an s.42 offence — it is a Third Schedule *civil* contravention (item 1(a)). The s.42(3) possession offences are narrow: jammers, SIM-box-class radio equipment "that can accommodate more than specified number of subscriber identity modules", and equipment knowingly using tampered telecommunication identifiers. So on the statutory text, **s.43 does not plainly reach a plain unlicensed hand-held**. What does reach it is s.7 of the Indian Wireless Telegraphy Act 1933 — a specially empowered officer may search "*any building, vessel or place*" for apparatus in respect of which an offence under s.6 has been committed and "*take possession thereof*" — which is one more reason the repeal question in §1.8 matters. [VERIFIED as to both texts; [UNCERTAIN] which power an officer would in practice invoke, and no case law on s.43 was found] Unauthorised possession attracts a civil penalty of up to ₹50,000 for a first instance and up to ₹2 lakh for each subsequent instance under the Third Schedule. [VERIFIED — Third Schedule, item 1(a), read verbatim from the gazette PDF] **But that penalty is imposed by an Adjudicating Officer under Chapter VIII, and Chapter VIII (ss. 31–41) has not been found in any commencement notification** — see §1.8 and Part 6 §6.1. Until it is commenced there may be no officer competent to impose it. [UNCERTAIN] If the Indian Wireless Telegraphy Act 1933 is indeed still in force — and the evidence says it is — s.6(1A)'s three-year imprisonment (or fine up to ₹1,000, or both) for possessing a wireless transmitter without licence also remains on the books, and unlike the civil track it needs no new machinery. [VERIFIED as to the text — DoT's own copy read directly 2026-08-23; [UNCERTAIN] as to whether s.60 of the 2023 Act has repealed it] Penalties and the real enforcement record are developed in a later part of this report.

1.8 The three uncertainties a reader must not gloss over

1. What "10 W" means. The ASOC power table's column heading is "**Max DC Input Power unless Otherwise specified**", and note (iv) states: "*DC input power is the total direct current power input to the final stage of the transmitter.*" [VERIFIED — WPC Office Memorandum No. L-14011/255/2004-AMT, 2010-08-13, Annexure-I] Consumer hand-helds are advertised by **RF output**. At power-amplifier efficiencies plausible for a VHF/UHF FM hand-held, 10 W of DC input yields roughly 4–6.5 W of RF output, and a set advertised as "10 W output" would draw roughly 15–25 W DC — over the Restricted ceiling, inside the General ceiling. **The two ceilings are [VERIFIED]; the conversion between them is engineering inference and is [UNCERTAIN].** Do not buy hardware on the strength of the arithmetic alone.

2. Whether the 2010 power table is still the operative specification. The Telecommunications (Amateur Services) Rules, 2024 contain **no frequencies and no wattages at all**. Rule 7(1) delegates the whole question: an ASOC holder "*shall operate amateur stations on such frequency bands and emissions and transmit power as specified by the Central Government in respect of each category of ASOC*", with the General holder permitted higher power "*as may be specified in his ASOC*". [VERIFIED] The most recent published table anyone could locate is Annexure-I to the 2010 WPC Office Memorandum, whose status WPC's own Central Public Information Officer confirmed in a formal Right to Information reply of 2020-12-28: "*the list of allocated bands and frequency for amateur radio operation in India may be seen at Annexure I of this Ministry's OM No. L-14011/255/2004-AMT dated 13/08/2010*". [VERIFIED — WPC letter No. P-14033/02/2020-COP] But that reply predates the 2024 Rules by four years, no post-2024 specification could be found, and wpc.dot.gov.in was unreachable throughout the research. **[UNCERTAIN] whether a newer specification exists.** If the issued certificate states different figures from Annexure-I, the certificate governs. **Ask WPC, in writing, for the current specification under rule 7(1) before buying equipment.** The published nodal contact is the Senior Deputy Wireless Adviser, 011-23036540, umang.srivastava@nic.in. [VERIFIED — <https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc>]

3. Whether the old Acts are repealed. Section 60 of the 2023 Act repeals the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933 — but section 60 does not appear in any commencement notification that could be found, and DoT was still making rules **under both old Acts** in June 2026. [VERIFIED as to those documents; [UNCERTAIN] as a definitive negative] The practical consequence is that the criminal-possession exposure under the 1933 Act probably still sits alongside the civil-penalty regime of the 2023 Act. A related gap: **Chapter VIII of the 2023 Act (sections 31–41), which creates the Adjudicating Officer and the civil-penalty machinery, also does not appear in any commencement notification** — yet the 2026 Rules repeatedly invoke section 32. [UNCERTAIN]

1.9 What changed in the ninety days before this report

A reader who consults advice written before mid-2026 will be reading superseded law.

Date	Change	Effect on a private radio owner
2026-06-17	Draft Telecommunications (Spectrum Assignment by Administrative Process) Rules, 2026 published (G.S.R. 494(E)); comments closed 2026-07-18	Would set the fee schedule for VHF/UHF land mobile. Not yet law
2026-06-23	S.O. 3368(E) brings s.3(1) and s.3(6) of the 2023 Act into force	Possessing radio equipment now requires an authorisation under the 2023 Act
2026-06-23	Captive, Miscellaneous, Principal Services and Migration Rules 2026 notified	The captive route becomes an authorisation regime — and closes to individuals
2026-06-25	Telecom eServices Authorisation Portal opens; new applications under the Indian Telegraph Act 1885 stop for Unified Licences, standalone licences, registrations and permissions	The legacy licensing window narrows; the position of WPC Network licences is [UNCERTAIN]
2026-07-03	DPL, NDPL, Manufacturing and Testing Licence applications and renewals stop	The old possession-licence route closes
2026-07-08	REPA Rules 2026 notified (G.S.R. 592(E))	New possession regime — dealers and testers only; ASOC holders exempt
2026-08-06	Radio Equipment Possession Authorisation applications go live on the portal	Operative from seventeen days before this report

All [VERIFIED] from gazette PDFs or DoT portal notices. **Two re-checked on 2026-08-23:** the 2026-08-06 REPA go-live is confirmed on DoT's own live pages, which state that "*Applications for new Radio Equipment Possession Authorisation under the Radio Equipment Possession Authorisation Rules, 2026, have been enabled on the Authorisation Portal with effect from 06.08.2026*"; and the 2026-06-25 stop-notice wording was re-read on the Saral Sanchar landing page and **does not name WPC Network licences**, which supports the §1.7 reading. The "ASOC holders exempt" cell in the 2026-07-08 row is downgraded to [UNCERTAIN] — see §1.5, Group B.

1.9A Adversarial fact-check log — 2026-08-23

The highest-stakes claims in this report were independently re-verified against primary sources fetched afresh, on the assumption that the draft contained errors. What that pass established:

Re-verified and confirmed, from the primary document itself:

- **The amateur power and band table.** Annexure-I to WPC OM No. L-14011/255/2004-AMT dated 13/08/2010 was read page by page from the scan (https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf). Restricted: ten HF bands at **50 watts** (A3E, H3E, J3E, R3E) and 50–54 / 144–146 / 434–438 MHz at **10 watts** (F1B, F2B, F3E, F3C). General: the same HF bands at **400 watts** with a wider emission set, and 50–54 / 144–146 / 434–438 / 5725–5840 MHz at **25 watts**. Column heading "Max DC Input Power unless Otherwise specified"; note (iv) "DC input power is the total direct current power input to the final stage of the transmitter"; para 11 reserves **VU2** for General and **VU3** for Restricted; para 12 supersedes the notification of 4 October 1985. Every figure in this report's tables matches the scan exactly.
- **Every fee and every rule number in the Amateur Services Rules 2024.** The gazette text (Gazette of India Extraordinary Part II §3(i) No. 617, 30 October 2024, notification dated 29 October 2024, **G.S.R. 675(E)**) was extracted and read: rule 4(1) age ≥ 12 and Indian citizenship; rule 4(2) reciprocal certificate **₹1,000**; rule 5(2) examination fee **₹100**; rule 5(4) Morse for General "as may be specified"; rule 5(5) pass at **forty per cent of the total marks**; rule 6(1) **₹1,000** for twenty years / **₹2,000** lifetime, lifetime defined as to age eighty; rule 6(2) two-year payment window; rule 7(3) no separate import authorisation; rule 7(6) duplicate or change of location **₹1,000**; rule 8(1) renewal twelve months ahead, **₹1,000** late fee, nothing after two years; rule 8(2) lifetime extended ten years at a time free; rule 10 no refund; rule 11 non-transferable; rule 12 amateur society **₹2,000** with four holders; rule 13 special call sign **₹200**. Annexure A paragraphs 2(1)(a), 2(1)(b), 2(2)(b), 4(3), 6 and 9(1)(b) all read as quoted in this report.
- **Penalties.** Third Schedule item 1(a) (₹50,000 first / ₹2 lakh subsequent), Second Schedule banding (₹1 lakh to ₹5 crore), s.42(1) (three years / ₹2 crore), s.42(3) (three years / ₹50 lakh), s.42(7) cognizable and non-bailable, s.43 search, s.47 and its Explanation, s.60(1) repeal and s.61 savings — all read verbatim from the gazette text of Act 44 of 2023. IWTA 1933 ss. 3, 6(1), 6(1A), 6(2), 6(3), 7, 8 and 11 read from DoT's own PDF. ITA 1885 ss. 3(1), 4 and 20 read from a gov.in host.
- **Licence-exempt figures.** PMR446 **446.0–446.2 MHz at 500 mW e.r.p.**, hand-portable, integral antenna, peer-to-peer, no base station or repeater, 6.25/12.5 kHz channels — Table V of **G.S.R. 1047(E) of 2018-10-18**. Citizen Band **26.957–27.283 MHz at 5 W ERP with a built-in antenna** — **G.S.R. 533(E) of 2005-08-12**, amended by **G.S.R. 35(E) of 2007-01-10**. Both numbers, both dates and both G.S.R. references were reproduced independently by more than one source. **No licence-exempt allocation anywhere in Indian law permits a 10 W hand-held.**
- **Exam mechanics.** Saral Sanchar's official HAM FAQ confirms **₹100 per grade and ₹200 for both grades in one application**, twenty-seven examination centres, Restricted 25 questions per Part over one hour and General 50 questions per Part over two hours, and the Part A exemption for electronics/telecom degree and diploma holders. WPC FAQ No. P-14036/02/2023-COP of 2024-12-05 was read in full and confirms 40 % per Part plus 40 % overall, 8 wpm Morse for General only, SACFA only for tower/building/mast antennas, and 90 days of portable operation.

- **Eligibility of the legacy walkie-talkie licence.** DoT's page for "Network – Short Range UHF Hand Held Radio" still states eligibility as individual citizens of India or Indian companies, and purposes "personal, business, security, and emergency" — and still states **no frequency band and no power limit**, confirming that the 2 W figure in this report is not DoT's.
- **wpc.dot.gov.in is dead** (connection refused, port 443) and **no post-2024 amateur band or power specification exists in any source that could be found.**

Corrected or hedged as a result (each change is marked at the point where it appears): the s.43 search power's s.42 predicate; the operability of the Third Schedule civil penalty given the Chapter VIII commencement gap; the ASOC exemption from the possession-authorisation regime; the exact gazette numbers of the June 2026 captive rules and the "private limited company" proviso; the "individuals are ineligible" reading of the import-licence page; the pass-mark conflict; the statutory hook for SACFA; the signing dates of the 2024 commencement notifications; and the upgrade of the Indian Telegraph Act 1885 section texts from [REPORTED] to [VERIFIED].

Could not be verified either way, and the reader must not assume: whether Annexure-I is still the operative specification under rule 7(1) of the 2024 Rules; whether section 60 of the 2023 Act has been commenced; whether Chapter VIII has been commenced; the text of the three June–July 2026 rule sets, none of which could be re-fetched from a government host; and whether any published DoT list of "blacklisted" radio models exists.

1.10 Abbreviations used in this report

Abbreviation	Expansion
ASOC	Amateur Station Operator Certificate
CB	Citizen Band (26.957–27.283 MHz in India)
CCPA	Central Consumer Protection Authority
CIN	Corporate Identity Number
CMRTS / CNPN	Captive Mobile Radio Trunking Service / Captive Non-Public Network
CUG	Closed User Group
DL / LoI / WOL	Decision Letter / Letter of Intent / Wireless Operating Licence
DoT	Department of Telecommunications, Ministry of Communications
DPL / NDPL	Dealer Possession Licence / Non-Dealer Possession Licence
e.r.p. / ERP	Effective radiated power (referenced to a half-wave dipole; includes antenna gain)
EIRP	Equivalent isotropically radiated power
ETA	Equipment Type Approval
FM	Frequency modulation (emission designator F3E)

Abbreviation	Expansion
GMDSS	Global Maritime Distress and Safety System
G.S.R. / S.O.	General Statutory Rule / Statutory Order — gazette notification prefixes
HF / VHF / UHF	High frequency (3–30 MHz) / very high frequency (30–300 MHz) / ultra high frequency (300–3000 MHz)
ITA 1885 / IWTA 1933	Indian Telegraph Act 1885 / Indian Wireless Telegraphy Act 1933
MTCTE	Mandatory Testing and Certification of Telecom Equipment
NFAP	National Frequency Allocation Plan
OM	Office Memorandum
PMR446	Personal Mobile Radio at 446 MHz
PMRTS	Public Mobile Radio Trunking Service
REPA	Radio Equipment Possession Authorisation
RLO	Regional Licensing Office of WPC
SACFA	Standing Advisory Committee on Frequency Allocation
USR	UHF Short Range Radio (350–351 MHz)
VSAT	Very Small Aperture Terminal (satellite)
WMO / WMS	Wireless Monitoring Organisation / Wireless Monitoring Station
WPC	Wireless Planning and Coordination Wing of DoT
wpm	Words per minute (Morse code speed)

1.11 Status note

This part is a research summary compiled on 2026-08-23. It is not legal advice. Indian radio law is mid-transition from the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933 to the Telecommunications Act 2023; three of the instruments relied on here took effect in June, July and August 2026; the amateur power table is a 2010 executive memorandum whose continued validity could not be independently confirmed; and the repeal of the two old Acts appears not to have been commenced. Confirm every figure with the WPC Wing of DoT, or with an established amateur radio club such as the Amateur Radio Society of India (ARSI) or the National Institute of Amateur Radio (NIAR), before spending money or transmitting.

Part 2 — What you may do without a licence, and the separate law of equipment

This part answers two questions that are constantly confused with each other:

1. **Operator legality** — may *you* transmit, and with what, without any licence at all?
2. **Equipment legality** — may that particular *radio* lawfully be type-approved, imported, sold to you, and held by you?

They are independent. You must satisfy both. A radio can be perfectly legal to own and completely illegal to key up; a transmission can be perfectly legal in principle and illegal because the hardware was never approved or lawfully imported.

Abbreviations used throughout, expanded on first use: **WPC** = Wireless Planning and Coordination Wing, the spectrum regulator inside the **DoT** = Department of Telecommunications, Ministry of Communications. **G.S.R.** = General Statutory Rules, the Gazette of India series in which subordinate rules are published; **S.O.** = Statutory Order, the series used for commencement and similar orders. **ETA** = Equipment Type Approval. **ERP** / **e.r.p.** = effective radiated power, referenced to a half-wave dipole; **EIRP** / **e.i.r.p.** = equivalent isotropically radiated power (2.15 dB higher for the same radiated power). **ASOC** = Amateur Station Operator Certificate. **PMR446** = Personal Mobile Radio in the 446 MHz band. **CB** = Citizen Band. **SRD** = short-range device.

2.1 If you only read one page of this part

Question	Answer	Evidence
Can <i>any</i> hand-held two-way radio be used in India with no licence at all?	Yes. A hand-portable radio with an integral antenna, working peer-to-peer in 446.0–446.2 MHz at not more than 500 mW e.r.p. , on 6.25 kHz or 12.5 kHz channels, and holding a WPC ETA, needs no licence from anybody, for any purpose.	[VERIFIED] — G.S.R. 1047(E), 2018-10-18, rule 3 and Table V
Is a second licence-free route available?	Citizen Band, 26.957–27.283 MHz, 5 W ERP with a built-in antenna (G.S.R. 533(E), 2005-08-12). Highest licence-free power in India for voice — but see the unresolved conflict in §2.5.2 before relying on it.	[VERIFIED] on the rule text; [UNCERTAIN] on the conflict
Can 10 W ever be licence-exempt in India?	No. Never, in any band, under any notification in force. The highest figure anywhere in the licence-exempt corpus for a two-way voice radio is 5 W ERP on CB; the highest UHF hand-held figure is 500 mW e.r.p. on PMR446.	[VERIFIED] — exhaustive enumeration of DoT's own list of de-licensing rules

Question	Answer	Evidence
Does licence-exemption also exempt the equipment?	No. Every exemption rule separately requires the model to be type approved . Licence-exempt is not approval-exempt.	[VERIFIED] — e.g. G.S.R. 1047(E) rule 5(2)
Is a Baofeng UV-5R-class radio (136–174 / 400–520 MHz, 4–8 W, detachable antenna) legal?	Not to import privately, not to buy from a compliant dealer, not to possess without an ASOC or a spectrum assignment, and not to operate outside amateur allocations. The brand is irrelevant; frequency, power, antenna and ETA are what matter — a Baofeng built to PMR446 limits with an Indian ETA is lawful for anyone.	[VERIFIED] on each gate; [REPORTED] that the BF-888S specifically is "blacklisted"
Does the ETA sticker let me transmit?	No. ETA certifies the <i>device</i> for <i>de-licensed band</i> operation only. It says nothing about your right to transmit, and no ETA exists for licensed-band equipment.	[VERIFIED] — WPC import compendium, 2022-07-06, §2.1–2.2

The one-line verdict. If what you want is *a walkie-talkie you can legally own and use today, with no exam, no fee and no paperwork*, buy an **ETA-approved PMR446 set locked to 446.0–446.2 MHz at ≤500 mW e.r.p.** If you want 10 W, no part of this section helps you: you must go to the amateur, captive or commercial routes covered elsewhere in this report.

2.2 The four gates

#	Gate	What it controls	Governing instrument	Who it binds
1	Equipment Type Approval (ETA)	the model is technically approved for a de-licensed band	rule 5(2) of the relevant de-licensing rules (e.g. G.S.R. 1047(E), 2018-10-18)	manufacturer / importer / authorised Indian representative
2	Import	the goods may lawfully cross the border	Notification No. 71-Customs, 1953-09-25 (S.R.O. 1815) under s.19, Sea Customs Act 1878; DGFT (Directorate General of Foreign Trade) import policy	whoever imports
3	Possession	you may lawfully hold the hardware	Telecommunications Act 2023, s.3(1)(c) + Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026, G.S.R. 592(E), 2026-07-08	whoever possesses
4	Operation / spectrum	you may lawfully transmit	Telecommunications Act 2023, s.4(2); ASOC, frequency assignment, or a de-licensing notification	whoever transmits

All four instruments were fetched and read by the underlying research. [VERIFIED]

Gate 4 is Part 1 and Parts 3–4 of this report. Gates 1–3 are §2.7 to §2.9 below. Gate 4 in its *licence-free* form is §2.3 to §2.6.

2.3 Where the power to de-license comes from, and what the 2023 Act changed

Every de-licensing notification in force today was made under the **Indian Telegraph Act, 1885** (sections 4 and 7) and the **Indian Wireless Telegraphy Act, 1933** (sections 4 and 10). The standard recital, verbatim from G.S.R. 1047(E):

"**G.S.R. 1047(E)**.—In exercise of the powers conferred by sections 4 and 7 of the Indian Telegraph Act, 1885 (13 of 1885) and sections 4 and 10 of the Indian Wireless Telegraphy Act, 1933 (17 of 1933), the Central Government hereby makes the following rules, namely:"

[**VERIFIED**] — Gazette of India, Extraordinary, Part II—Section 3—Sub-section (i), No. 753, 2018-10-18; PDF at [https://dot.gov.in/sites/default/files/License Exemption for SRD Device G_S_R_1047\(E\) dated 18th October, 2018_1.pdf](https://dot.gov.in/sites/default/files/License%20Exemption%20for%20SRD%20Device%20G_S_R_1047(E)%20dated%2018th%20October,%202018_1.pdf) (recovered from an Internet Archive capture of 2023-02-05; the live DoT path now returns HTTP 404).

Section 4 of the 1885 Act is the Government's exclusive privilege over telegraphs and its power to license; section 4 of the 1933 Act is the prohibition on *possessing* wireless apparatus without a licence. A de-licensing notification therefore cuts a hole in both the operating-licence requirement and the possession-licence requirement at once. [**VERIFIED**]

The Telecommunications Act, 2023 (Act 44 of 2023, assented 2023-12-24) does not disturb any of this. Section 3(4) continues pre-existing exemptions from authorisation; section 4(7) continues pre-existing spectrum exemptions; section 61 continues all rules made under the 1885 and 1933 Acts until superseded. [**VERIFIED**] on the statutory text. Nothing in the new Act makes a previously licence-free walkie-talkie unlawful.

Commencement — where the research files disagree, and which to believe.

Position	Source	Weight
Section 3(1) (authorisation, including to possess radio equipment) and 3(6) were commenced on 2026-06-23 by S.O. 3368(E) , gazette No. 3235 of 2026-06-23	quoted verbatim from the gazette PDF at https://www.dot.gov.in/static/uploads/2026/06/35e1d307f3cbb6396b4c48802ed4b044.pdf [VERIFIED]	Highest — the commencement notification itself

Position	Source	Weight
Section 60 (repeal of the 1885 and 1933 Acts) has not been shown to be commenced; WPC was still making rules under both old Acts on 2026-06-10 and 2026-06-11 (G.S.R. 466(E), G.S.R. 468(E)), and G.S.R. 468(E) rule 4(1) still says "Notwithstanding anything contained in the Indian Wireless Telegraphy (Possession) Rules, 1965 ..."	gazette recitals [VERIFIED] ; the conclusion that the old Acts survive is [UNCERTAIN] because it proves a negative	Strong
"Its parent Acts are repealed" — one research file writes as though the 1885 and 1933 Acts are already repealed	quotes s.60(1) text [VERIFIED] , but offers no commencement notification	Weakest on this point — the text of a repeal is not proof that the repeal is in force

Resolution for the reader: as at 2026-08-23, treat the 1885 and 1933 Acts as **still in force alongside** the Telecommunications Act 2023, with s.3(1) of the new Act operative since 2026-06-23. **[VERIFIED]** as to s.3(1); **[UNCERTAIN]** as to the repeal. The practical consequence is nil for exemptions — they survive either way by ss.3(4), 4(7) and 61 — but it matters for which penalty provisions bite, which is Part 6's problem.

What "licence-exempt" actually means. Every exemption is granted "on non-interference, non-protection and shared and non-exclusive basis". You get no protection at all, and a licensed user's complaint can put you off the air. Verbatim, G.S.R. 1047(E) rule 4(1):

"...where any person whom a license has been issued under the provisions of section 4 of the Act; and section 4 of the Indian Wireless Telegraphy Act, 1933 informs the Authority that his licensed system is getting harmful interference from any other radio communication system exempted under these rules, then such authority shall call upon the user of such unlicensed wireless equipment to take necessary steps to avoid interference by relocating the equipment, reducing the power and using special type of antennae, failing which such Authority shall recommend discontinuation of such wireless use."

"(2) The Authority shall give a reasonable opportunity to the user of wireless equipment before making recommendation of discontinuation of wireless use under sub-rule (1)"

[VERIFIED] — G.S.R. 1047(E), rule 4. The original 2005 CB rule was harsher — "shall discontinue its use forthwith" — until G.S.R. 35(E) of 2007-01-10 substituted the graduated remedy and added "Provided that, before such discontinuation, a reasonable opportunity to explain the circumstances shall be offered to such unlicensed user of wireless equipment." **[VERIFIED]**

2.4 The complete catalogue of India's de-licensed bands

Two official lists exist and they are not identical: DoT's consolidated compilation *"List of Gazette Notifications regarding De-licensing band, issued by WPC Wing, DoT"* (16 entries, contents to 2018) and WPC's *"List of sub-ordinate legislations (Rules) framed for license-exempt use of spectrum"* (17 entries, republished on the DoT site on 2025-07-24). Both are official; the newer WPC list omits two inductive-application entries that appear in the older compilation.

[VERIFIED] — both lists read.

2.4.1 MASTER TABLE

Band	Notification	Date	Subject	Power limit	Use for a hand-held voice radio
9–50 kHz	G.S.R. 83(E)	2014-02-11	very low power RF devices incl. RFID (radio frequency identification)	field-strength limited	None
50–200 kHz	G.S.R. 90(E)	2009-02-10	very low power RF devices incl. RFID	field-strength limited	None
9 kHz–30 MHz (inductive sub-bands)	G.S.R. 870(E)	2021-12-21	inductive applications	field-strength limited	None
302–351 kHz	G.S.R. 697(E)	2015-09-16	inductive applications	field-strength limited	None
302–435 kHz; 855–1050 kHz; 1.89–2.30 MHz	G.S.R. 996(E)	2018-10-05	inductive applications	field-strength limited	None
13.553–13.567 MHz	G.S.R. 884(E)	2010-11-04	very low power RF devices, indoor	field-strength limited	None
26.957–27.283 MHz (Citizen Band)	G.S.R. 533(E), amended by G.S.R. 35(E)	2005-08-12; amendment 2007-01-10	wireless equipment "intended to be used while in motion or during halts"	5 W ERP, built-in antenna	Yes — highest licence-free voice power in India. See §2.5.2
36–38 MHz	G.S.R. 696(E)	2015-09-16	wireless microphones	50 mW; audio channel bandwidth ≤200 kHz	No — one-way audio

Band	Notification	Date	Subject	Power limit	Use for a hand-held voice radio
335.7125–335.8375 MHz (6 spot frequencies)	G.S.R. 532(E), amended by G.S.R. 34(E)	2005-08-12; 2007-01-10	remote control of cranes	low power spot frequencies	None
402–405 MHz	G.S.R. 673(E)	2008-09-23	medical implants, cardiac monitoring	very low power (25 µW e.r.p. in the corresponding 2018 table)	None
433.05–434.79 MHz	G.S.R. 347(E) (supersedes G.S.R. 680(E) of 2012-09-12 and G.S.R. 698(E) of 2015-09-16)	2022-05-09 (gazette No. 332, 2022-05-11)	low power RF devices	1 mW / 10 mW e.r.p.; voice expressly allowed at 434.04–434.79 MHz	Technically yes, at 10 mW — a few hundred metres. See §2.5.3
Multiple SRD bands, incl. 446.0–446.2 MHz	G.S.R. 1047(E)	2018-10-18 (gazette No. 753)	low and very low power short-range devices, incl. "Personal Mobile Radio 446 MHz"	Table V: 500 mW e.r.p.; other tables 50 nW to 500 mW	Yes — this is the licence-free walkie-talkie. See §2.5.1
865–868 MHz	G.S.R. 853(E) (supersedes the 2005 RFID rules; earlier chain G.S.R. 168(E) 2005-03-11 → G.S.R. 37(E) 2007 → G.S.R. 564(E) 2008-07-30 at 1 W transmitter / 4 W ERP)	2021-12-10 (gazette No. 694, 2021-12-13)	short-range devices	25 mW / 500 mW / 25 mW / 2 W e.r.p. by category	None — RFID and telemetry, no voice allowance
2.4–2.4835 GHz	G.S.R. 45(E)	2005-01-28	low power equipment (W-LAN, Bluetooth)	1 W (30 dBm) transmitter, 4 W (36 dBm) ERP , spread ≥10 MHz, antenna within 5 m of rooftop else SACFA clearance	No in practice — see note below

Band	Notification	Date	Subject	Power limit	Use for a hand-held voice radio
5150–5250, 5250–5350, 5470–5725, 5725–5875 MHz	G.S.R. 1048(E)	notified 2018-10-18, published 2018-10-22 (gazette No. 754)	wireless access systems / RLAN	1 W conducted (5150–5250, 5725–5875); 250 mW (5250–5350, 5470–5725); 250 mW client devices	None
76–77 GHz	G.S.R. 699(E)	2015-09-16	short-range automotive radar	5 W (37 dBm) ERP	None — radar, not communications
1.6–10.6 GHz and above (ultra-wideband)	G.S.R. 1046(E)	2018-10-18	UWB devices	very low power spectral density	None

[VERIFIED] — band/notification/date entries from the two official lists; power figures read from the individual notification texts, except the field-strength and power-spectral-density entries, which were not transcribed and are described only by category.

2.4.2 ADDITIONS SINCE THE 2022 COMPILATION

Band	Notification	Date	Subject	Power	Tag
various (2018 SRD Rules amended: transport/traffic telematics excluded from rule 3; Table IX S.No. 14 changed from 24.15–24.25 GHz to 24.00–24.25 GHz)	G.S.R. 271(E)	2024-05-09	amendment to G.S.R. 1047(E)	—	[VERIFIED] as to the notification number and date — DoT's own copy at https://eservices.dot.gov.in/sites/default/files/circular-notifications/use-of-low-power-and-very-low-power-short-range-radio-frequency-devices-exemption-from-licensing-requirement-amendment-rules-2024.pdf . The substance of the amendment is [REPORTED] — Lexplosion summary. Table V (PMR446) does not appear to be touched, but confirm against the notification before relying on it. <i>(An earlier draft of this part recorded the number as unknown and the date as 2024-05-10; the gazette copy listed in Part 6 Appendix D.4 supersedes that.)</i>

Band	Notification	Date	Subject	Power	Tag
5925–6425 MHz (lower 6 GHz)	draft G.S.R. 316(E) 2025-05-16; final G.S.R. 47(E)	2026-01-20	low-power indoor / very-low-power outdoor RLAN	30 dBm e.i.r.p. indoor; 14 dBm e.i.r.p. outdoor	[REPORTED] — and contested : a second research thread found that as at NFAP-2025 (2025-12-30) the lower 6 GHz band had not been released, with industry bodies publicly complaining. Both accounts are [REPORTED] ; the conflict is unresolved and irrelevant to walkie-talkies
5875–5905 MHz	G.S.R. 466(E)	2026-06-10	C-V2X (cellular vehicle-to-everything) on-board units	33 dBm max e.i.r.p.; 23 dBm/MHz max power spectral density	[VERIFIED]
77–81 GHz	G.S.R. 468(E)	2026-06-11	short-range automotive radar	50 dBm average e.i.r.p.; 55 dBm peak	[VERIFIED] as to the notification and its recital; [REPORTED] as to the power figures

2.4.3 TWO WARNINGS ABOUT THE OFFICIAL LISTS

- DoT's own index contains typographical errors.** Entry 12 prints the PMR446 band as "446–462.2 MHz"; the notification says **446.0–446.2 MHz**. The index also prints "27.140-25.150 MHz" for what is plainly 27.150 MHz, and dates G.S.R. 1048(E) 18.10.2018 when the gazette carrying it is No. 754 of 2018-10-22 (signed 18 October, published 22 October). **Quote the notifications, never the index.** **[VERIFIED]** — both documents read side by side.
- The 2.4 GHz rule is drafted more widely than its description.** Rule 3 of G.S.R. 45(E) exempts "any wireless equipment" in 2.4–2.4835 GHz subject to a spread-spectrum condition ("in Spread of 10 MHz or higher"); DoT's index describes it as Wi-Fi and Bluetooth. **[VERIFIED]** on the rule text. Whether WPC would accept a 2.4 GHz *voice* two-way radio under it is **[UNCERTAIN]**; the 10 MHz spreading requirement excludes conventional narrowband FM, and no such radio is known to hold an Indian ETA. Do not build a plan on this.

There is no de-licensed allocation anywhere in the Indian VHF land-mobile range (136–174 MHz) or the general UHF land-mobile range (400–470 MHz) other than the 200 kHz PMR446 sliver and the 433 MHz LPD sliver. **[VERIFIED]** — exhaustive review of both official lists plus the post-2022 notifications.

2.5 The three bands that can legally carry voice

2.5.1 PMR446 — 446.0–446.2 MHz, 500 mW E.R.P. — THE PRACTICAL ANSWER

Table V of G.S.R. 1047(E), verbatim:

S.No.	Frequency range in MHz	Transmit power limit	Additional parameters	EN No.
1	446.0-446.2	500 mW e.r.p.	Channel spacing: 6.25 kHz and 12.5 kHz	EN 300 113-2, EN 301 166-2, EN 300 296-2

And the note beneath it — the legally decisive sentence, verbatim:

"Note: For the purpose of this Table, personal mobile radio 446 MHz device means hand portable radio with no base station or repeater use and uses integral antennas only in order to maximise sharing and minimise interference, and which operates in short range peer-to-peer mode and shall be used neither as a part of infrastructure network nor as a repeater;"

The rules define the power unit, verbatim:

"(c) 'effective radiated power (in a given direction)' or e.r.p. means the product of the power supplied to the antenna and its gain relative to a half-wave dipole in a given direction;"

[VERIFIED] — all three quotations from the G.S.R. 1047(E) gazette text; the notification is signed "[No. R-11017/04/2018-PP] MAKRAND PATHAK, Assistant Wireless Adviser".

Every condition, unpacked:

Condition	Requirement
Band	446.0–446.2 MHz only (200 kHz total)
Power	≤500 mW e.r.p. (dipole-referenced; ≈0.82 W e.i.r.p.)
Channel spacing	6.25 kHz and 12.5 kHz — both the 16-channel analogue and 32-channel digital plans are inside the rule
Form factor	hand-portable only
Antenna	integral only — no detachable or external antenna
Topology	short-range peer-to-peer; no base station, no repeater, not part of an infrastructure network
Standards	ETSI (European Telecommunications Standards Institute) EN 300 113-2, EN 301 166-2, EN 300 296-2
Type approval	required (rule 5(2))
Who may use it	"any person" — no restriction by nationality, age, purpose, or personal versus commercial use

Is European PMR446 legal in India? Yes. India adopted the European band edges, the European power limit, both European channel plans, the European topology restrictions, and cites the ETSI harmonised standards by number. [VERIFIED] This is the single most commonly mis-stated fact in Indian radio law. Two caveats: a European-market handset is only lawful here if **that model** holds an Indian ETA [VERIFIED] — rule 5(2); and anything the European Union permits above 500 mW, or any repeater or base configuration, is outside the Indian exemption [VERIFIED] .

Common mis-statements this report corrects:

Mis-statement (and where it appears)	Reality
"The 446 MHz licence-free limit is 0.5 W EIRP" (several vendor pages)	The gazette says 500 mW e.r.p. , dipole-referenced. EIRP would be ≈ 0.82 W. [VERIFIED]
"The band is 446.0–446.1 MHz" (etmytalk.com vendor page)	446.0–446.2 MHz. [VERIFIED]
"G.S.R. 1047(E) was published on 2018-10-08" (thejeshgn.com wiki)	The gazette masthead reads 2018-10-18 , Gazette No. 753. [VERIFIED]
"The de-licensed band is 446–462.2 MHz" (DoT's own index, entry 12)	Typo in the index. [VERIFIED]
"No walkie-talkie can be used in India without a licence"	False since 2018-10-18. DoT itself calls compliant PMR446 sets "license free walkie-talkie". [VERIFIED] / [REPORTED]

2.5.2 CITIZEN BAND — 26.957–27.283 MHz, 5 W ERP — THE HIGHEST LICENCE-FREE VOICE POWER, WITH A CAVEAT

Rules title: *"Use of low power equipment in the Citizen band 26.957 – 27.283 MHz (Exemption from Licensing Requirement) Rules, 2005"*, G.S.R. 533(E), 2005-08-12, signed "[No. R-11014/04/2005-LR] ASHOK KUMAR, Jt. Wireless Advisor." Rule 3, verbatim:

"3. Use of wireless equipment in the band 26.957 – 27.283 MHz.- Notwithstanding anything contained in any law for the time being in force, no licence shall be required by any person to establish, maintain, work, possess or deal in any wireless equipment intended to be used while in motion or during halts, on non-interference, non-protection and shared (non-exclusive) basis, in the frequency band 26.957 – 27.283 MHz with 5 Watt Effective Radiated Power and built-in antenna"

and rule 2(b), verbatim:

"(b) 'Effective Radiated Power' includes the gain of the antenna, if any;"

[VERIFIED] — retrieved three times independently: WPC's own published copy at http://www.wpc.dot.gov.in/DocFiles/CB_Radio_Delicensing.doc (Internet Archive capture 2006-11-27, document properties "WPC WING"); DoT's scanned gazette PDF *"Delicensing in 335 MHz band*

[GSR 532 (E)] and 26_957-27_283 MHz band [GSR 533 (E)]" read as page images; and the DoT eServices compilation of subordinate legislation. G.S.R. 35(E) of 2007-01-10 amends only rule 4 (interference) and **does not change the 5 W figure.** [VERIFIED]

A conflict between the research inputs, resolved. One research file lists the 2005 CB rules under WPC's index heading "Very low power equipment in the Citizen Band" and records the power limit as [UNCERTAIN] because it did not read the rule's text. Two other files read the operative rule directly and both record **5 W ERP**. The operative rule governs; an index heading does not. **Treat 5 W ERP as [VERIFIED].**

Four caveats you must not gloss over:

1. **The 5 W is ERP, including antenna gain — not transmitter output.** A 27 MHz hand-held with a short helical antenna is very inefficient, so a radio whose conducted output exceeds 5 W could in principle radiate under 5 W ERP. **Do not rely on this.** [UNCERTAIN] — no WPC test method, guidance or enforcement precedent was found, and DoT's own 2021 enforcement circular describes its monitoring arm measuring "transmit power" (§2.6.3), which suggests the conducted figure is what gets policed in practice.
2. **"Built-in antenna" is a hard condition.** A CB set with an SO-239 socket or a mag-mount whip falls outside the exemption on its face. [VERIFIED]
3. **The 5 W figure is in tension with the 2018 SRD Rules.** G.S.R. 1047(E) Table IX row 2 permits only **10 mW e.r.p.** across 26957–27283 kHz for a "non-specific short range device". G.S.R. 1047(E) contains **no supersession clause at all** (unlike G.S.R. 1048(E), 853(E) and 347(E), which each expressly supersede named rules), and both official lists continue to carry G.S.R. 533(E) and G.S.R. 1047(E) as live. [VERIFIED] on the absence of the supersession clause and on both entries appearing. [UNCERTAIN] — the two instruments are formally inconsistent for a 27 MHz device and no WPC clarification was found. **Put this to WPC in writing before buying a 5 W CB set.**
4. **Type approval under the CB rules is unclear.** Unlike every other exemption rule examined, the 2005 CB Rules as read contain **no equipment or type-approval rule** (rules 1–4 only: short title, definitions, exemption, interference). Whether ETA is nonetheless required for a CB set — via the import/EXIM regime or a rule made elsewhere — could not be established. [UNCERTAIN]

Finally, the physics: 27 MHz is the 11-metre band. Skip propagation, high noise, and an electrically tiny "built-in" antenna mean a 5 W CB hand-held is not a drop-in substitute for a UHF walkie-talkie.

2.5.3 LPD433 — 434.04–434.79 MHz, 10 MW E.R.P., VOICE EXPRESSLY ALLOWED

G.S.R. 347(E), 2022-05-09 (gazette No. 332, 2022-05-11), *"The Use of Low Power Radio Frequency Devices in the frequency band 433.05 to 434.79 MHz (Exemption from License) Rules, 2022"*, superseding the 2012 and 2015 rules. Table, verbatim:

Sl. No.	Frequency range in MHz	Transmit/ Radiated Power limit	Spectrum access and mitigation	Modulation / max occupied bandwidth
1	433.05-434.79	10 mW e.r.p.	≤10% duty cycle	not specified
2	433.05-434.79	1 mW e.r.p.	Note 1	power density limited to -13 dBm/10 kHz for wideband modulation >250 kHz
3	434.04-434.79	10 mW e.r.p.	Note 1	≤25 kHz

"Note 1: Voice applications are allowed with a maximum bandwidth of 25 kHz, with a spectrum access technique such as LBT or equivalent and a maximum transmit period of 1 minute for each transmission. Other audio/ video applications are excluded."

[VERIFIED] — G.S.R. 347(E). "LBT" is listen-before-talk. This is the European LPD433 allocation, and India expressly permits voice on it — but 10 mW is one fiftieth of PMR446 and one five-hundredth of CB. Range is a few hundred metres. It is a curiosity, not an option. The ubiquitous cheap "433 MHz walkie-talkie" lives here, and almost none of them respects 10 mW.

2.6 The verdict on licence-free use, and on 10 W

2.6.1 THE COMPLETE LICENCE-EXEMPT POWER LADDER

Rank	Power	Band	Instrument	Device type
1=	5 W (37 dBm) ERP	76–77 GHz	G.S.R. 699(E)	automotive radar — not communications
1=	5 W ERP	26.957–27.283 MHz	G.S.R. 533(E)	Citizen Band two-way voice, built-in antenna
3	4 W ERP (1 W transmitter)	2.4–2.4835 GHz	G.S.R. 45(E)	spread spectrum, ≥10 MHz spreading
4	2 W e.r.p.	865–868 MHz	G.S.R. 853(E) Table IV	RFID interrogator, ≤4 s per transmission
5	1 W (30 dBm) conducted	5150–5250, 5725–5875 MHz	G.S.R. 1048(E)	RLAN access point
6=	500 mW e.r.p.	446.0–446.2 MHz	G.S.R. 1047(E) Table V	PMR446 hand-portable two-way radio
6=	500 mW e.r.p.	169.4–169.5875 MHz	G.S.R. 1047(E) Tables IV, IX	assistive listening, metering
6=	500 mW e.r.p.	865–868 MHz	G.S.R. 853(E) Table II	tracking and tracing

Rank	Power	Band	Instrument	Device type
6=	500 mW e.i.r.p.	2446–2454 MHz	G.S.R. 1047(E) Table VII	RFID
10	250 mW conducted	5250–5350, 5470–5725 MHz	G.S.R. 1048(E)	RLAN
11	100 mW	several	various	model control, telematics, 61 GHz
12	50 mW	36–38 MHz	G.S.R. 696(E)	wireless microphone
13	25 mW	865–868 MHz; 2400–2483.5 MHz	G.S.R. 853(E); G.S.R. 1047(E) Table VI	SRD
14	10 mW	433.05–434.79 MHz; 26957–27283 kHz	G.S.R. 347(E); G.S.R. 1047(E) Table IX	LPD (voice allowed at 434.04–434.79); non-specific SRD
...	down to 50 nW	87.5–108 MHz	G.S.R. 1047(E) Table III	personal FM transmitter

[VERIFIED]

2.6.2 THE VERDICT

10 W is never licence-exempt in India, for any two-way radio, in any band, under any notification in force. [VERIFIED] — this is a negative established by exhaustive enumeration: both of DoT's own official lists of de-licensing rules were read, every notification capable of bearing on voice communications was read, and the maximum figure in each was identified. Nothing reaches 10 W for a communications device.

Three corollaries the reader must internalise:

1. A **10 W UHF walkie-talkie** — the Baofeng UV-5R/UV-9R class and every "10 W long-range" radio sold online — is outside every Indian exemption on **two** independent grounds: power *and* frequency. Even at reduced power settings such radios cover 136–174 MHz and 400–470 MHz, which are licensed land-mobile ranges. [VERIFIED]
2. **Turning a 10 W radio down to 0.5 W does not bring it inside the PMR446 exemption.** The exemption also requires an integral antenna, hand-portable form, no repeater capability, operation confined to 446.0–446.2 MHz, and — decisively — an **ETA for that model**. A frequency-agile radio with a detachable antenna fails on every limb. [VERIFIED]
3. Because ERP includes antenna gain, a *nominal* 10 W radio might in theory radiate under a 5 W ERP CB ceiling. This is legally untested in India and enforcement appears to measure transmit power. [UNCERTAIN] — **do not treat this as a route.**

2.6.3 THE 2021 CRACKDOWN — PROOF THAT THIS IS ENFORCED

DoT published a circular titled "*Public Notice in regard to mis-use of 446 MHz PMR Radio*", dated 2021-12-28. The DoT page exists at <https://www.dot.gov.in/spectrummanagement/public-notice-regard-mis-use-446-mhz-pmr-radio28122021> [VERIFIED] as to the URL and title, but its body could

not be rendered (HTTP 403 to automated fetch; a JavaScript-only shell to plain HTTP clients) and no Internet Archive capture exists. The text below is transcribed by *Communications Today*, 2021-12-31 — **[REPORTED]** , and **[UNCERTAIN]** as to whether the transcription is complete:

"Such PMR devices mean hand-held portable radio with no base station or repeater use and operates in peer-to-peer mode. However, such license-exempt PMR devices (popularly known as license free walkie-talkie, or 446 MHz license-free radio etc.) require type approval from the WPC Wing, DoT"

"During a recent spectrum monitoring drive undertaken by the Wireless Monitoring Organisation (WMO) DoT observed that few of such PMR devices (Smart Talky- M/s Vertel, Aspera, Beta 8-9 LF, PCOM etc.) though type approved, but are being operated with higher (than permissible) transmit power of around 5 Watt, in violation with the relevant rules/ Gazette Notification."

"That, the certificate will not be valid in case any change in the above parameters and not conforming to the relevant Gazette Notification; and That, use of such equipment in case not conforming to above notification will require a specific wireless operating license, as applicable from this Ministry."

"Finding of non-compliance during such monitoring may lead to revocation of the ETA certificate issued and initiation of other legal proceedings as per provision under Section 20 of the Indian Telegraph Act, 1885 and Section 6 of the Indian Wireless Telegraphy Act, 1933 & other relevant provisions"

Four lessons. (a) The Wireless Monitoring Organisation physically goes looking for over-powered "licence-free" radios; this is not a dead letter. (b) Named radios **sold in India as licence-free** were found transmitting at roughly **5 W — ten times the legal 500 mW** — while holding genuine ETAs. Ask any seller for the ETA number and check it (§2.11). (c) In DoT's own words, a non-conforming device "will require a specific wireless operating license": there is no middle ground. (d) DoT itself uses the phrase "license free walkie-talkie", which settles whether such a thing exists in India.

2.7 Equipment gate 1 — Equipment Type Approval (ETA)

2.7.1 WHAT ETA IS, AND THE CRUCIAL LIMIT ON IT

DoT's own service page, verbatim:

"Equipment Type Approval (ETA) is a certification issued by the Wireless Planning & Coordination (WPC) Wing of the Department of Telecommunications (DoT), Government of India. It is required for the import, sale, and use of wireless communication devices that operate in de-licensed frequency bands or involve radio frequency (RF) transmission. Equipment Type Approval (ETA) is permitted on self-declaration."

[VERIFIED] — <https://eservices.dot.gov.in/equipment-type-approval-eta> , fetched 2026-08-23. Note that the trailing words "or involve radio frequency (RF) transmission" read as if ETA covers all RF devices. It does not. WPC's own compendium is authoritative, §2.1 verbatim:

"Central government, by framing rules under the Indian Telegraph Act (ITA) 1885 and Indian Wireless Telegraphy Act (IWTA) 1933, allows license exempt use of certain frequency bands subject to compliance with the technical parameters mentioned in the respective notifications; however, equipment complying with such notifications need to be type approved and accordingly Equipment Type approval (ETA) certificate is issued to such compliant RF devices."

and §2.2:

"For license-band RF devices: - Generally, an 'operating' license issued under the Indian Telegraph Act, 1885 is required to establish, maintain and work any wireless equipment in India. However, for 'Dealers' of wireless equipment, a Dealer Possession License (DPL) is required which is issued under the Indian Wireless Telegraphy Act, 1933."

[VERIFIED] — WPC *Compendium of Orders/Circulars/Guidelines ... in regard to Import licensing requirement*, F. No. R-11018/02/2017-PP, 2022-07-06, signed Ashim Dutta, Dy. Wireless Adviser.

Therefore: ETA exists only for licence-exempt bands. A 10 W VHF/UHF hand-held is licensed-band equipment. **No ETA exists for it and none can be obtained.** Its legality flows from the operator's authorisation, not from any type-approval certificate. This is the point that most commercial "WPC consultant" websites blur, and it cuts both ways: a customs officer or seller who demands an ETA for amateur equipment is asking for a document that cannot exist.

[VERIFIED]

The statutory hook for PMR446 is rule 5 of G.S.R. 1047(E), verbatim:

"5. **Equipment.**—(1) The equipment shall comply with the respective EN number for effective use of spectrum and to avoid harmful interference. (2) The wireless equipment shall be type approved and designed and constructed in such a manner that the bandwidth of emission and other parameters shall conform to the limits specified in rule 3 and the application for obtaining equipment type approval shall be made to the Central Government in the format given in Annexure. (3) The safety related requirements shall be as per the International or National standards such as ITU/ETSI/ANSI/BIS/ICNIRP for the respective devices and frequency bands."

[VERIFIED] The same requirement appears in G.S.R. 45(E) rule 6, G.S.R. 1048(E) rule 7, G.S.R. 853(E) rule 5, G.S.R. 696(E)/698(E) rule 5(b) and G.S.R. 466(E). [VERIFIED]

There is no numbered form. The prescribed format is simply the "*ANNEXURE — APPLICATION FOR EQUIPMENT TYPE APPROVAL (Refer sub-rule (2) of rule 5)*" to G.S.R. 1047(E), with Section A (applicant) and Section B (transmitter details: frequency range, preset switchable channels, number of voice/data/TV channels). [VERIFIED] In practice the application is made through the Saral Sanchar portal, not on paper.

2.7.2 SELF-DECLARATION IS NOW THE DEFAULT

WPC Office Memorandum No. **R-11017/01/2018-PP(part-1)** dated 2024-09-09, signed Amit Kumar Singh, Assistant Wireless Adviser, verbatim:

"With the objective to promote the Ease of Doing Business in the telecom sector, the competent authority has decided that, henceforth, all the applications of Equipment Type Approval (ETA) for wireless equipment devices, which have been license-exempt through various Gazette notifications issued from time to time, shall be granted to the applicant on a self-declaration basis."

"3. These ETAs shall be granted for ensuring compliance with RF regulations. Therefore, ETA holders must obtain No Objection Certificate or other clearances (if applicable) from the Directorate General of Foreign Trade (DGFT) before importing the equipment."

[VERIFIED] — PDF at [https://eservices.dot.gov.in/sites/default/files/2024-11/Issuance%20of%20Equipment%20Type%20Approval%20\(ETA\)%20forLicense-Exempt.pdf](https://eservices.dot.gov.in/sites/default/files/2024-11/Issuance%20of%20Equipment%20Type%20Approval%20(ETA)%20forLicense-Exempt.pdf). CBIC (Central Board of Indirect Taxes and Customs) circulated it to all Customs formations by **Instruction No. 24/2024-Customs dated 2024-10-22**, which also records that "Earlier Instructions no. 16/2022-Customs dated 21.07.2022 stands modified to that extent." [VERIFIED]

The older two-track structure — self-declaration ETA for devices in the DGFT "FREE" category, scrutiny-based ETA (with an accredited-laboratory RF test report and a manufacturer agreement) for "Restricted" category devices — is set out in the 2022 compendium §2.1.

[VERIFIED] The 2024 memorandum says *all* licence-exempt ETA now goes on self-declaration, yet DoT's live page still lists four excluded categories and the scrutiny guidelines remain published. The safest statement: **self-declaration is the default; a residual scrutiny path survives at least for the four excluded categories.** [UNCERTAIN] as to whether scrutiny formally survives for anything else.

2.7.3 ETA AT A GLANCE

Item	Value	Evidence
Issuing body	WPC Wing, DoT	[VERIFIED]
Default route since 2024-09-09	self-declaration	[VERIFIED] — OM R-11017/01/2018-PP(part-1)
Portal	saralsanchar.gov.in	[VERIFIED]
Fee	Rs. 10,000 per application	[VERIFIED] — DoT eServices ETA page and WPC's "Guidelines for filling Scrutiny based ETA Application": "Payment of Rs.10000/- per application is to be made through Saral Sanchar portal." Note a conflict: a parallel research thread could not confirm any fee and recorded consultancy claims that the self-declaration route carries no government fee [UNCERTAIN]. DoT's own page is the better source; verify on Saral Sanchar before budgeting
Fee basis	per application; industry describes it as per model	[VERIFIED] / [REPORTED]
Validity	"Lifetime unless revoked by WPC Wing"	[VERIFIED]

Item	Value	Evidence
Documents	RF test report; authorisation from manufacturer (for an Indian representative); technical literature	[VERIFIED]
Excluded from self-declaration	radar, jamming devices, drones, satellite equipment	[VERIFIED]
Waived entirely	test / prototype / sample / pre-launch units, up to 1000 units per model , on a self-declaration to Customs	[VERIFIED] — O.M. ETA-WPC/Policy/2018-19, 2019-02-26
Who applies	the manufacturer or its authorised Indian representative / importer — never the end user	[VERIFIED] — Annexure to G.S.R. 1047(E), 1048(E), 853(E); 2019 OM
Typical elapsed time	4–6 weeks end to end for a first product, of which lab testing 1–3 weeks	[REPORTED] — PCN Global India, 2026-07-06

An individual end-user never applies for an ETA. Your job is to buy a model that already has one.

2.8 Equipment gate 2 — import

2.8.1 THE ROOT PROHIBITION DATES FROM 1953

WPC compendium §1.1, verbatim: *"Under Notification No. 71 of 1953 issued under the Sea Custom Act 1878, a license is required from Ministry of Communications for import of any wireless transmitter/receiver/ transceiver."* [VERIFIED]

The gazette text itself (S.R.O. 1815, 1953-09-25), verbatim:

"the Central Government hereby restricts with effect from the 15th October 1953, the bringing into India of apparatus for wireless telegraphs as follows:— (a) in the case of transmitting apparatus and of receiving apparatus incorporated in a single unit with transmitting apparatus, to persons who have been licensed to import such apparatus by the Ministry of Communications in Form I, annexed hereto; (b) in other cases, to persons who apply to a customs officer to import the same in Form II, annexed hereto"

[VERIFIED] — reproduced as Annex-1 to the 2022 compendium. The two forms are **Form I** (licence to import transmitting apparatus) and **Form II** (application to import receiving apparatus, expressly "NOT to be used for Transmitting Apparatus").

This 1953 restriction is why a transmitter arriving at an Indian port without a WPC import licence or a recognised exemption is treated as restricted goods and is liable to seizure.

2.8.2 THE WPC IMPORT LICENCE

Item	Value (verbatim from DoT eServices, fetched 2026-08-23)
Purpose	"The Import License is required for any entity that intends to bring wireless devices or equipment—such as antennas, transceivers, and satellite phones—into India."
Who can apply	"Import License can be granted to any entity registered in India and holding valid permission granted by DoT"
Fee	"Fee for grant of Import license is Rs. 500 per import license."
Validity	one year

[VERIFIED] The document checklist presupposes a corporate applicant: service licence, **copy of DPL** (Dealer Possession Licence), Corporate Identity Number (CIN) / Limited Liability Partnership Identification Number (LLPIN) / Permanent Account Number (PAN), identity proof of authorised person, purchase order and proforma invoice, technical literature, WPC frequency authorisation/assignment letters, ETA copies, security undertakings. **[VERIFIED]** **There is no realistic individual pathway here.**

A sample import licence reproduced in the compendium (RLO Mumbai, 2022-05-30) carries the cleanest official statement that import legality is not operating legality, verbatim:

"This is only a clearance from technical angle and does not pertain to administrative and/or financial aspects of import. Please note that the grant of import license does not imply permission to establish, maintain and operate a Wireless Telegraph equipment. A separate license will be required for the usages of the equipment."

[VERIFIED]

2.8.3 WHAT DOES NOT NEED AN IMPORT LICENCE

All from the 2022 compendium, **[VERIFIED]** :

Case	Relief
Licence-exempt-band devices in the DGFT "FREE" category	a system-generated Import Undertaking from Saral Sanchar suffices; no import licence
Test / prototype / sample / pre-launch units	ETA waived up to 1000 units per model; undertaking to Customs (O.M. 2019-02-26)
Standalone RF antennas, cables, connectors, batteries	no WPC import licence or NOC needed — but an RF amplifier does need one
Non-RF products (audio transmitters, optical fibre equipment)	outside WPC's remit entirely

Case	Relief
Domestic OEMs (original equipment manufacturers) in the Domestic Tariff Area	may import transceivers and radio modules against a valid DPL on an undertaking to Customs (O.M. R-11018/06/2019-PP, 2019-07-26)
Telecom Service Providers	self-declaration import certificate, QR-verifiable, auto-downloaded from Saral Sanchar (O.M. R-11017/02/2021-PP, 2021-10-21)
Experimental / technology-trial / manufacturing-and-testing / demonstration licensees	import permission is built into the operating licence; licence fee Rs. 5,000 in each category (O.M. R-14016/01/2019-NT(Pt.), 2019-07-23)
An ASOC holder importing amateur radio equipment	no separate import authorisation — rule 7(3), Telecommunications (Amateur Services) Rules, 2024, G.S.R. 675(E), 2024-10-29: <i>"An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment."</i> [VERIFIED]

The last row is the single most valuable practical privilege attached to an amateur licence, and it is developed in the amateur part of this report. Note its scope limit: rule 2(1)(b) defines "amateur radio equipment" as *"a radio equipment required for operating an amateur station"*. Whether a wideband 136–174 / 400–470 MHz consumer hand-held qualifies is [UNCERTAIN].

2.8.4 THE 2026 GAP IN THE IMPORT CHAIN — AN UNRESOLVED PROBLEM

DoT stopped accepting Dealer Possession Licence applications and renewals from 2026-07-03 (§2.9), yet the eServices Import Licence page still lists "Copy of DPL" among the required documents and still quotes Rs. 500 for one year. The research files differ:

Reading	Source	Tag
The import licence has migrated to the Authorisation Portal along with everything else from 2026-08-06	one research file's status table	plausible but inferential; treat as [UNCERTAIN]
DoT has published nothing explaining how an import licence is obtained after 2026-07-03, nor whether the "import for sale" limb of rule 4(1)(a) of the 2026 Possession Rules now subsumes it	a second research file, after searching	[UNCERTAIN]

This is the weakest link in the whole equipment chain. Anyone planning a commercial import should put the question to WPC in writing before shipping anything.

2.8.5 CUSTOMS CLASSIFICATION AND DUTY — NOT SETTLED

Claim	Tag
Amateur radio equipment is HSN (Harmonised System of Nomenclature) code 8525 60 00, Customs Notification 24/2005 Sl. No. 16, basic customs duty NIL, Goods and Services Tax (GST) 18%, import policy FREE	[REPORTED] — Devadas K.M. VU2DH, <i>Import of Amateur Radio Equipment: 2023-24</i> , circulated by NIAR. The guide's own rates were stated valid only to 2024-03-31, HS 2022 restructured heading 8525, and the notification could not be fetched. [UNCERTAIN]
Importing by post or courier invites classification under HSN 9804 as a personal/household item at 38.5% duty	[REPORTED] — same source
Import of second-hand amateur equipment is restricted and needs a Chartered Engineer certificate	[REPORTED] — same source; not verified against DGFT policy
ITC(HS) code for walkie-talkies generally	secondary sources give 8517 62 90, 8525 60 00, 8529 10 and 9804 inconsistently. [UNCERTAIN] — check the live ITC(HS) schedule at dgft.gov.in

2.8.6 WHAT ACTUALLY HAPPENS TO RADIOS AT THE BORDER

- Import of a wireless transmitter without a WPC import licence contravenes the 1953 restriction. [VERIFIED]
- In March 2026 the Directorate of Revenue Intelligence (DRI) seized **11,060 Baofeng BF-888S sets** worth about Rs 2.5 crore at Nhava Sheva, within a larger consignment; **two importers were arrested under the Customs Act**. Press copy: *"The owners of the two Mumbai firms which had imported these good without due permissions are a father-son duo, and both were arrested under the Customs Act."* [REPORTED] — PTI via Communications Today and Deccan Herald, 2026-03-22. Reports state the seized sets operate in 400–480 MHz and that "licence-free operation is allowed only within the 446.0–446.2 MHz band". [REPORTED] — Free Press Journal, 2026-03-22.
- The Saral Sanchar import undertaking that an importer signs includes, verbatim: *"We clearly understand that we are liable to be prosecuted under Indian law, in case of any wrong declaration."* [VERIFIED]
- **Import offences are prosecuted under the Customs Act 1962, which the Telecommunications Act 2023 does not touch.** That regime is materially harsher than the civil-penalty scheme discussed in Part 6. [REPORTED] on the practice; the exact Customs Act machinery is [UNCERTAIN] — sections 111, 112 and 125 could not be fetched from indiacode.nic.in, legislative.gov.in or cbic.gov.in, and **this report deliberately does not cite them.**

- Practitioner account, [REPORTED] : *"Normally, though its not mentioned in import policy or Customs tariff, a customs officer insist for Equipment Type Approval which is not available with any HAM, invariably it is detained ... Prominent brands like Icom, Yaesu, Kenwood, Alinco are already having ETA approval and normally not insisted."* — VU2DH. The claim that those brands hold Indian ETAs is questionable, given that ETA is a de-licensed-band instrument, and is [UNCERTAIN] .
- Traveller account, [REPORTED] : *"Customs may seize the device. In serious cases involving restricted frequencies, you could face fines or legal action."* — indiabaggagerules.com, 2026-05-01.
- **Whether an individual may lawfully carry a compliant European PMR446 handset into India in baggage and use it** — given the ETA requirement and DGFT policy — is not settled by anything in the sources. [UNCERTAIN] The RF parameters would be lawful; the model's lack of an Indian ETA would not be.
- For a sense of the personal cost of getting this wrong, Part 6 §6.4.6 covers *Frederick Joseph Iseman v State NCT of Delhi* (Delhi High Court, 2024-11-12): a satellite phone found in a transit passenger's baggage on 2018-12-30, never switched on and with a flat battery, produced a seized passport and criminal proceedings that were still live in the High Court almost six years later. [VERIFIED] — order text at <https://indiankanoon.org/doc/72044028/> .

2.9 Equipment gate 3 — possession, and the death of the Dealer Possession Licence

2.9.1 THE OLD REGIME IS CLOSED

DoT notice, verbatim from <https://eservices.dot.gov.in/dealer-possession-license> , fetched 2026-08-23:

"NOTE: This is to inform that the acceptance of all applications including renewals under Indian Telegraph Act 1885 has been stopped for Dealer Possession License (DPL)/ Non-Dealer Possession License (NDPL) and Manufacturing Licence & Testing License. Accordingly, the Saral Sanchar Portal and National Single Window System (NSWS) will not accept any new applications for the same from 03.07.2026 onwards"

and:

"All applicants intending to obtain Radio Equipment Possession Authorisation under the provisions of the Telecommunication Act, 2023 may apply online through the Authorisation Portal by logging in to <https://eservices.dot.gov.in/authorisation-portal> with effect from 06.08.2026."

[VERIFIED]

For reference, the DPL as DoT described it: issued *"to authorized dealers of wireless equipment to possess, stock, and sell radio frequency devices"*, open to *"individuals or a company registered under Companies Act 2013"*, fee *"nominal fees of Rs. 60"*, validity **1 year**, process including **police**

verification. [VERIFIED] (Several compliance firms state 5-year validity; DoT's own page says 1 year — prefer DoT. [REPORTED] versus [VERIFIED] .) NDPL was the sibling instrument for a non-dealer holder of wireless equipment; DoT never published a separate description, so its precise scope is [UNCERTAIN] .

Practical consequence: almost every third-party guide to DPL, NDPL and WPC import licences now describes a dead regime. Distrust anything dated before mid-2026.

2.9.2 THE NEW REGIME — RADIO EQUIPMENT POSSESSION AUTHORISATION RULES, 2026

G.S.R. 592(E), Gazette of India Extraordinary Part II §3(i) No. 534, **2026-07-08**, F. No. 24-03/2025-UBB, signed Devendra Kumar Rai, Jt. Secy., made under s.56(1) and s.56(2)(a) read with **s.3(1)(c)** of the Telecommunications Act 2023, preceded by draft G.S.R. 154(E) of 2025-02-27 (which was styled "...Rules, 2025"). Applications live on the Authorisation Portal from **2026-08-06**.

[VERIFIED]

Rule 4(1) — the only two purposes for which an authorisation exists, verbatim:

"A person who meets the eligibility criteria under sub-rule (2) may apply under sub-rule (3) to obtain authorisation to possess radio equipment if he intends to— (a) deal with such radio equipment through manufacture, purchase or import for sale, let for hire, repair or testing or demonstration thereof; or (b) import or purchase or hire radio equipment only for the purposes of testing or demonstration."

[VERIFIED] **There is no category for "I want to own a radio."** Eligibility under rule 4(2) is limited to companies and LLPs, persons holding a trade licence/permit/registration, government bodies, and existing 1965-Rules licence holders. [VERIFIED]

Fees and duration:

Item	Amount	Rule
Application fee (non-refundable)	Rs. 1,000	4(3)(b)
Authorisation fee — dealer type (rule 4(1)(a))	Rs. 10,000 per annum	5(2)(a)(i)
Authorisation fee — testing/ demonstration only (rule 4(1)(b))	Rs. 2,000 per annum or pro rata, minimum Rs. 500	5(2)(a)(ii)
Modification of an authorisation	Rs. 1,000 non-refundable	5(4)
Duration — dealer type	at least one year, up to five years	6(a)
Duration — testing/demonstration	up to twelve months	6(b)
Renewal	apply not later than one month before expiry; late applications for sufficient cause on payment of late fees	7

[VERIFIED] Note the scale of the change: the old DPL cost Rs. 60; the successor costs Rs. 10,000 per annum. Casual possession authorisations have been priced out of existence.

Rule 8(1) — the sales restriction that governs whether a shop may sell you a radio, verbatim:

"An authorised entity that holds authorisation for possession of radio equipment as referred to in clause (a) of sub-rule (1) of rule 4 shall give possession thereof only to a person as referred to in rule 10 or who holds an authorisation granted under these rules."

[VERIFIED] This is the modern successor to the 1965 rule that a dealer may not sell a complete wireless set to an unlicensed person.

Rule 10 — who needs no authorisation at all, verbatim and complete:

"An authorisation under these rules shall not be required if a person— (a) holds a licence, registration, permission, by whatever name called, under the Indian Telegraph Act, 1885 (13 of 1885) or the Indian Wireless Telegraphy Act, 1933 (17 of 1933) and continues to operate under the terms and conditions of such licence, registration or permission under sub-section (6) of section 3 of the Act; (b) holds an authorisation under any other rule under the Act, the terms and conditions of which do not require it to obtain authorisation under clause (c) of sub-section (1) of section 3 of the Act; (c) holds an exemption under sub-section (3) of section 3 of the Act and an exemption from the requirement of assignment under sub-section (6) of section 4 of the Act; (d) holds an exemption, in respect of possession of radio equipment, as referred to in sub-section (4) of section 3 or sub-section (7) of section 4 of the Act; (e) has spectrum assigned to it under sub-sections (4), (8) or (9) of section 4 of the Act; **(f) holds an Amateur Station Operator Certificate in accordance with the Telecommunications (Amateur Services) Rules, 2024;** (g) is in possession of radio equipment that can accommodate not more than four subscriber identity modules; or (h) is a user who holds proof of subscription to a telecommunication service provided by an authorised entity.

Explanation.—For the removal of doubts, it is clarified that a person as referred to in this rule shall not manufacture, sell or let for hire the radio equipment in its possession."

[VERIFIED]

Other obligations worth knowing if you ever hold an authorisation: rule 5(3) records the **address at which the equipment must be kept**; rule 9(1) requires secure premises, an inventory, facilitation of physical verification, and *"forthwith report any instance of loss, theft, loss of possession or suspected misuse"*; rule 9(2) draws a line at **100 milliwatt** — below it, testing needs only non-interference and no protection; above it, the entity must hold a spectrum assignment; rule 12 sets disposal deadlines (two to three months) when an authorisation ends; rule 13 allows restrictions near the international borders, the Line of Control and the Line of Actual Control, and inspection without notice where the Government considers immediate action necessary; rule 14 makes breach a section 32 breach under the Act. **[VERIFIED]**

2.9.3 DO YOU, PERSONALLY, NEED A POSSESSION AUTHORISATION?

Your situation	Authorisation needed?	Basis
You own a compliant PMR446 set (446.0–446.2 MHz, ≤500 mW e.r.p., integral antenna, ETA)	No — the licence-exemption covers possession	rule 10(c)/(d) read with G.S.R. 1047(E) and Act ss.3(4)/4(7). [VERIFIED] on the rule text; the chain of cross-references is inferential, so [UNCERTAIN] in form — although nobody has ever needed a possession licence for a de-licensed-band device in India
You own a CB set within G.S.R. 533(E)	No , on the same reasoning	as above, plus the §2.5.2 caveats
You hold an ASOC	Probably no — but this is no longer a settled point	rule 10(f). Downgraded to [UNCERTAIN] on the adversarial fact-check of 2026-08-23: the final 2026 Rules could not be re-fetched from any government host, and the February 2025 draft of the same rules put the exemptions in rule 4 , not rule 10, with no amateur gateway — it exempted only equipment exempted under s.3(3)/(4), possession by a holder of an authorisation under s.3(1)(a)/(b) plus a spectrum assignment or surviving Telegraph Act licence, and equipment notified as needed by a user of a telecommunication service. An ASOC is granted under s.47 and carries no spectrum assignment. Ask WPC in writing before relying on this
You hold a frequency assignment (captive/CUG or commercial)	No	rule 10(e) [VERIFIED]
You are a private individual who simply wants to own a 10 W UHF hand-held, with no ASOC and no assignment	You cannot get one. The Rules provide no such applicant class, and you fall outside rule 4(2) eligibility unless you hold a trade licence	rules 4(1), 4(2) [VERIFIED] ; the conclusion that no lawful route remains is a reading of the Rules, [UNCERTAIN] as a matter of DoT practice
You run a shop that wants to sell walkie-talkies	Yes — dealer type: Rs. 1,000 application + Rs. 10,000 per annum, 1–5 years, security vetting possible	rules 4(1)(a), 5(1), 5(2)(a)(i), 6(a) [VERIFIED]

One drafting tension to flag, because it affects home-built equipment: the rule 10 Explanation says a rule-10 person "*shall not manufacture, sell or let for hire the radio equipment in its possession*", while the 2024 Amateur Rules expressly contemplate self-made apparatus (their

statutory equipment register has a column for *"in case assembled by Authorised entity write self-made"*). The better reading is that "manufacture" here means manufacture for trade, and that the specific amateur rules prevail over the general possession rules — but the drafting is genuinely awkward and this is [UNCERTAIN] . Put it to WPC if you intend to homebrew.

2.10 Baofeng-class imported radios — the answer, gate by gate

"Baofeng UV-5R class" here means a dual-band 136–174 / 400–520 MHz hand-held at roughly 4–8 W with a detachable antenna and open front-panel frequency entry. The same analysis applies to Quansheng, Retevis, TIDRADIO and every clone.

Gate	Verdict	Basis
Buy in India?	A compliant dealer must refuse you: rule 8(1) lets an authorised dealer give possession only to a rule-10 person or another authorisation holder. A private buyer with no ASOC and no assignment is neither.	[VERIFIED] — rule 8(1), 2026 Possession Rules
Own?	With an ASOC : on the earlier retrieval of the 2026 Possession Rules, rule 10(f) exempts you from any possession authorisation — but that exemption is [UNCERTAIN] after the 2026-08-23 fact-check (see §2.9.3). Without an ASOC or an assignment: you are possessing radio equipment without authorisation or exemption — Third Schedule item 1(a) of the Act, civil penalty up to Rs. 50,000 first offence, up to Rs. 2 lakh for each subsequent instance ; and, if the Indian Wireless Telegraphy Act 1933 still stands, s.6(1A) adds imprisonment up to three years , or fine up to Rs. 1,000, or both.	Penalty text [VERIFIED] — Third Schedule of Act 44 of 2023 and s.6(1A) of the 1933 Act both re-read from primary copies on 2026-08-23. The rule 10(f) exemption is [UNCERTAIN] . Whether the Third Schedule can be adjudicated today is a commencement question dealt with in Part 6 §6.1 and is [UNCERTAIN]

Gate	Verdict	Basis
Import?	As a private individual by post, courier or baggage: no — the 1953 Notification restricts bringing transmitting apparatus into India to persons licensed to import, and individuals are not eligible for the WPC import licence. As a licensed amateur importing amateur radio equipment : no separate import authorisation needed (rule 7(3), 2024 Amateur Rules) — subject to the scope question at §2.8.3. Commercially without permissions: the 2026 Nhava Sheva consignment was seized and the importers arrested.	[VERIFIED] / [VERIFIED] / [REPORTED]
Operate?	Only on frequencies you are authorised to use. An ASOC (Restricted) holder may use such hardware only within 50–54, 144–146 and 434–438 MHz and within the power ceiling of that grade. Outside those allocations, and for anyone without an ASOC, no. Note that a stock UV-5R covers 136–174 and 400–470 MHz — the great majority of which is unlawful for an Indian amateur.	[VERIFIED]

The "blacklist". Press reporting says DoT has blacklisted the BF-888S specifically, quoting PTI: *"The Baofeng BF-888S walkie-talkies have been blacklisted by the Department of Telecommunications (DoT) for operating beyond permissible frequencies as they can be used for unauthorised communication, posing a threat to national security. These devices require specific licences."* [REPORTED] — PTI via Communications Today and Telangana Today, 2026-03-22. **No gazette notification, DoT circular or published list naming specific models could be located**, including on a fresh search for this report. [UNCERTAIN] — ask WPC directly whether such a list exists and where it is published. Do not assume that a model's absence from an unpublished list means anything.

The brand is not the issue. Baofeng also markets a licence-free line for India; one research thread records a **Baofeng GT-1** advertised as a "license-free PMR 446 walkie talkie ... IN: 446–446.20 MHz" [REPORTED], but the product page could not be fetched and **no ETA number could be confirmed** [UNCERTAIN]. A Baofeng built to 446.0–446.2 MHz at ≤500 mW e.r.p. with an integral antenna and a genuine Indian ETA is lawful for anyone to buy and use. A UV-5R is not. **Frequency, power, antenna and ETA decide the question; the badge on the case decides nothing.**

Why cheap sets attract attention. Industry sources make the technical case that low-cost radios emit *"spurious transmissions on other frequencies"* which can land on licensed channels including police allocations, and that *"cheap/spurious handsets are unlikely to have any*

manufacturer warranty." [REPORTED] — BRUTFORCE, updated 2022-11-01, and Chierda, 2021-12-06. Both are vendor sources with a commercial interest: treat the technical point as sound and the legal characterisation as marketing.

2.10.1 AVAILABILITY ONLINE IS NOT EVIDENCE OF LEGALITY

The **CCPA** (Central Consumer Protection Authority, Department of Consumer Affairs) notified the *Guidelines for the Prevention and Regulation of Illegal Listing and Sale of Radio Equipment including Walkie Talkies on E-Commerce Platforms, 2025* on **2025-05-27** (PIB release 2132575, 2025-05-30), after examining **16,970** non-compliant listings and issuing 13 notices. CCPA's stated concern, verbatim from the PIB release:

"It was observed that Walkie-talkies are being sold on e-commerce platforms without mandatory and clear disclosures regarding the requirement of a wireless operating license or compliance with applicable laws. ... The omission of details such as frequency range, licensing obligations under the Indian Telegraph Act, 1885, or the Wireless Telegraphy Act, 1933, and the Use of Low Power, Very Low Power Short Range Radio Frequency Devices (Exemption from Licensing Requirement) Rules, 2018 and the potential legal consequences of unauthorized use, misleads consumers into believing that the devices are freely operable by the general public."

What the guidelines require, verbatim from the same release:

"Mandates that only authorized and compliant walkie-talkie devices operating on permitted frequencies are listed for sale on online platforms. Product listings to specify frequency ranges and other technical parameters and include proof of regulatory approval (Equipment Type Approval). Requires e-commerce entities to undertake due diligence and verify regulatory compliance, including licensing where applicable and listings lacking frequency information or necessary certification should be taken down. Prohibits misleading advertisements or product descriptions that may misinform consumers about the legal usage of such devices. Sellers should ensure that equipment listed for sale do not operate on frequencies which are not exempted from requirement of frequency assignment and authorization by DOT ..."

[VERIFIED] — PIB release 2132575. The full operative text of the guidelines could not be obtained: consumeraffairs.nic.in was unreachable and dca.gov.in returned "Access denied" for the PDF at https://consumeraffairs.nic.in/sites/default/files/file-uploads/latestnews/Guidelines_27052025.pdf. [UNCERTAIN] as to clause-level detail.

Enforcement to date:

Date	Event	Tag
2025-05-09	CCPA issues 13 notices to marketplaces including Amazon, Flipkart, Meesho, IndiaMART, TradeIndia	[VERIFIED] — PIB release 2127931
2025-05-27 / 30	Guidelines notified; PIB release	[VERIFIED]

Date	Event	Tag
2026-01-16	Final orders: Rs 10 lakh each on Meesho, Meta (Facebook Marketplace), Flipkart and Amazon; Rs 1 lakh each on Chimiya, JioMart, Talk Pro and MaskMan Toys — Rs 44 lakh total	[REPORTED] — Outlook Business 2026-01-16; Mint 2026-01-20; Storyboard18 2026-02-11
2026-01/02	Volumes cited: Flipkart 65,931 units sold with frequency blank or outside the exempt range; Amazon 2,602 units and 467 non-compliant listings; Meesho 2,209 units by a single seller; JioMart 58 units; 710 Facebook Marketplace listings delisted. One seller was found offering imported 400–470 MHz sets, some marketed as "license-free" or as toys with claimed 30 km range	[REPORTED]
reported 2026-02-12	TradeIndia fined Rs 10 lakh over uncertified walkie-talkie sales	[REPORTED] — Livemint, 2026-02-12. The date of the CCPA order itself is not stated in the report, so only the reporting date is given here [UNCERTAIN]
2026-03-21/22	DRI seizure of 11,060 BF-888S sets at Nhava Sheva; two arrests under the Customs Act	[REPORTED]

CCPA rejected the platforms' intermediary defence: *"intermediary protections are conditional upon the exercise of effective due diligence"*. [REPORTED] — Outlook Business. Meta has reportedly challenged CCPA's jurisdiction over Facebook Marketplace listings. [REPORTED]

Read this as the regulator's clearest recent statement of the law: the listing being taken down and the seller fined is the accurate signal, not the fact that you can add the item to a cart.

2.11 Which radios sold in India are type-approved?

There is no published list of approved models. What can be established is the structure of the legitimate market.

Class	What it is	ETA?	Who may buy	Examples
A — Licence-free PMR446 sets	446.0–446.2 MHz, ≤500 mW e.r.p., integral antenna, no repeater or base use	Required (rule 5(2), G.S.R. 1047(E)) [VERIFIED]	anyone	TalkPRO (X-10, LF446, H250A, 4GL), BRUTFORCE, Baofeng GT-1, Motorola Talkabout T82 in PMR446 configuration — [REPORTED] from vendor sites; no individual ETA number could be confirmed [UNCERTAIN]
B — Licensed-band professional radios	VHF 136–174 MHz, UHF 403–470 MHz, microwave	None exists and none is needed	only a rule-10 person or an authorisation holder (rule 8(1)) [VERIFIED] ; the buyer needs a frequency assignment under s.4(2) [VERIFIED]	Motorola DP series, Hytera, Kenwood TK series, Icom — [REPORTED]
C — Amateur transceivers	HF/VHF/UHF amateur allocations	None — amateur bands are licensed spectrum	ASOC holders	Icom, Yaesu, Kenwood, Alinco — [REPORTED]

An Indian licence-free vendor states the Class A test plainly, verbatim: *"License-Free walkie talkies operating on 446–446.2 MHz band with a Maximum effective radiated power (ERP) of 0.5W are classified as License Free"* and *"It is mandatory for License-Free Walkie Talkies to have Equipment Type Approval Certificate (ETA) from the WPC."* [REPORTED] — TalkPRO FAQs. A vendor for Class B states the converse: *"Customer cannot buy or operate Licensed radios without prior approval from the WPC."* [REPORTED]

2.11.1 HOW TO VERIFY A SPECIFIC MODEL YOURSELF

Step	Action	Note
1	Ask the seller for the ETA number for that exact make and model	A refusal or a vague answer is your answer
2	Look it up on Saral Sanchar: https://saralsanchar.gov.in/wpc_eta_report.php → "List of Issued ETA (Self-Declaration)", searching by ETA No. or by Equipment Make/Model	[VERIFIED] — page fetched. The search is CAPTCHA-protected and cannot be scripted; do it by hand
3	Expect the record under "Other"	The equipment-category dropdown has no "walkie-talkie", "two-way radio" or "PMR" entry [VERIFIED]

Step	Action	Note
4	Cross-check the frequency range on the ETA against 446.0–446.2 MHz and the power against 500 mW e.r.p.	An ETA whose range reads "400–470 MHz" is not a licence-free radio
5	Confirm the antenna is genuinely integral and that the set has no repeater or base mode	Table V note [VERIFIED]

Because the register is CAPTCHA-gated, **no verified list of currently approved walkie-talkie models exists in this report.** **[UNCERTAIN]** — a human must search the register for the specific models under consideration.

2.11.2 RED FLAGS IN A LISTING

Red flag	Why it matters
No frequency range stated	Exactly the omission CCPA penalised [VERIFIED]
Range stated as "400–470 MHz", "UHF", or "136–174 / 400–470 MHz"	Licensed spectrum — a Class B radio being mis-sold as licence-free [VERIFIED]
"10 W", "8 W", "long range 5–30 km" plus "licence-free"	Mutually contradictory: the licence-free ceiling is 500 mW e.r.p. [VERIFIED]
Detachable/SMA antenna on a "licence-free" set	Breaks the integral-antenna condition even at 500 mW [VERIFIED]
Repeater or base-station mode advertised	Outside the Table V definition [VERIFIED]
No ETA number, or an ETA that will not verify on Saral Sanchar	The model may be unapproved, or approved for something else
Marketed as a toy	A pattern CCPA specifically found [REPORTED]
An ETA number is quoted but the radio transmits at ~5 W	Precisely what DoT's Wireless Monitoring Organisation found in 2021; the ETA can be revoked and the <i>user</i> then needs an operating licence [REPORTED]

2.12 What to ask WPC in writing before you spend money

Six things this part could not settle. Each is a legitimate question for the WPC Wing, DoT, or for a recognised amateur club:

#	Question	Why it matters	Tag
1	Does the 5 W ERP Citizen Band allowance in G.S.R. 533(E) survive alongside the 10 mW e.r.p. figure in G.S.R. 1047(E) Table IX row 2 for a 27 MHz hand-held?	Decides whether the highest licence-free voice power in India is really available	[UNCERTAIN]

#	Question	Why it matters	Tag
2	Is ETA required for a Citizen Band set, given that G.S.R. 533(E) contains no equipment rule?	Decides whether a CB purchase is complete	[UNCERTAIN]
3	How does one obtain a WPC import licence after 2026-07-03, now that the DPL it requires no longer exists?	The weakest link in the import chain	[UNCERTAIN]
4	Is there a published DoT list of blacklisted or prohibited radio models, and where?	Press asserts the BF-888S is blacklisted; no published instrument was found	[UNCERTAIN]
5	Which specific walkie-talkie models currently hold a valid Indian ETA?	The register is CAPTCHA-gated and cannot be enumerated here	[UNCERTAIN]
6	May an individual bring a compliant European PMR446 handset into India as baggage and use it, given the ETA requirement and DGFT policy?	The RF parameters are lawful; the model approval is not	[UNCERTAIN]

Two further items are recorded for completeness: the operative text of the 2024 amendment to the 2018 SRD Rules — **G.S.R. 271(E) dated 2024-05-09**, whose number and date are [VERIFIED] from DoT's own copy but whose clause-level effect on Table V is [REPORTED] only, so confirm that Table V was untouched before relying categorically on the PMR446 figures — and whether the scrutiny-based ETA route formally survives the 2024-09-09 memorandum ([UNCERTAIN]).

2.13 Summary of this part

- 1. A licence-free walkie-talkie exists in India, and DoT itself uses that phrase.** It is a PMR446 set: 446.0–446.2 MHz, ≤500 mW e.r.p., integral antenna, peer-to-peer only, 6.25 or 12.5 kHz channels, holding a WPC ETA. Any person may buy and use one for any purpose, with no fee, no exam and no paperwork. [VERIFIED]
- 2. The highest licence-free voice power in India is 5 W ERP on Citizen Band**, with a built-in antenna, under G.S.R. 533(E) of 2005-08-12 — subject to an unresolved conflict with the 2018 SRD Rules and to the physics of 27 MHz. [VERIFIED] on the rule; [UNCERTAIN] on the conflict.
- 3. Nothing in the licence-exempt corpus reaches 10 W for a two-way radio.** Not in any band, not under any notification in force, not in 2026. [VERIFIED]
- 4. Licence-exempt is not approval-exempt.** Every exemption requires the model to hold an ETA, and ETA is available *only* for de-licensed bands — never for a 10 W VHF/UHF hand-held. [VERIFIED]
- 5. Equipment law has moved sharply in 2024–2026.** ETA went to self-declaration on 2024-09-09; the Dealer Possession Licence died on 2026-07-03; the Radio Equipment Possession Authorisation Rules, 2026 opened on 2026-08-06 with fees 167 times the old DPL and **no applicant category for a private owner.** [VERIFIED]

6. **For an individual, the only two clean answers are: buy an ETA-approved PMR446 set, or get an ASOC.** An ASOC certainly solves **import** — rule 7(3) of the Amateur Services Rules 2024, gazette text read directly, **[VERIFIED]**. Whether it also solves **possession** (recorded here as rule 10(f) of the 2026 Possession Rules) could not be re-verified on the 2026-08-23 fact-check and is **[UNCERTAIN]** — see §2.9.3. Everything in between — a UV-5R bought online, a European handset in a suitcase, a "10 W licence-free" listing — is a gate failure somewhere in the chain, and the enforcement record of 2025–2026 shows the gates being checked. **[VERIFIED]** / **[REPORTED]**

7. **Independent re-verification of the two licence-exempt figures the reader will act on, 2026-08-23.** Table V of G.S.R. 1047(E) — **446.0–446.2 MHz, 500 mW e.r.p.**, 6.25 kHz and 12.5 kHz channel spacing, hand-portable, integral antennas only, peer-to-peer, no base station and no repeater, EN 300 113-2 / EN 301 166-2 / EN 300 296-2 — was reproduced independently from a bare-act text of the 2018 Rules and matches this report word for word. The Citizen Band figures — **G.S.R. 533(E) of 12 August 2005**, band **26.957–27.283 MHz, 5 watt effective radiated power, built-in antenna**, amended by **G.S.R. 35(E) of 10 January 2007** which touches only the interference rule — were confirmed from two further independent reproductions. **No source found anywhere in the licence-exempt corpus permits a hand-held above 5 W ERP, and none permits any UHF hand-held above 500 mW e.r.p.** The answer to the reader's question is therefore firm: **a 10 W walkie-talkie cannot be licence-exempt in India.** **[VERIFIED]**

2.14 Disclaimer for this part

This part is a research summary compiled on 2026-08-23. **It is not legal advice**, and nothing in it authorises the import, possession, sale or operation of any radio. Equipment law moved sharply in 2024–2026: Equipment Type Approval went to self-declaration on 2024-09-09, the Dealer Possession Licence closed on 2026-07-03, and the Radio Equipment Possession Authorisation Rules, 2026 opened on 2026-08-06. Six questions this part could not settle are listed at §2.12, and the import chain after 2026-07-03 (§2.8.4) is the weakest link of all. Confirm the current position with the WPC Wing of DoT, or with a licensed amateur radio club such as ARSI or NIAR, before buying, importing or transmitting. Claims tagged **[REPORTED]** rest on named secondary sources and may be wrong; claims tagged **[UNCERTAIN]** could not be confirmed at all.

Part 3 — The amateur radio route: grades, the ASOC examination, and the licence

This is the only route by which one private individual in India can lawfully hold and operate a 10 W hand-held radio. Everything in this part therefore deserves close reading. It is also the part of the report where the law changed most recently and most completely: the rules that

governed Indian amateur radio from 1978 until 2024 were swept away on 2024-10-30, and most of what is published online — including several amateur-club websites refreshed in 2025 — still describes the old regime.

Abbreviations used throughout, expanded on first use and collected here for convenience: **ASOC** (Amateur Station Operator Certificate), **DoT** (Department of Telecommunications), **WPC** (Wireless Planning and Coordination Wing of DoT), **WMO** (Wireless Monitoring Organisation), **WMS/IMS** (Wireless Monitoring Station / International Monitoring Station), **RHQ** (Regional Headquarters of the WMO), **G.S.R.** (General Statutory Rule, the numbering used for subordinate legislation in the Gazette of India), **OM** (Office Memorandum), **SACFA** (Standing Advisory Committee for Frequency Allocation), **ITU** (International Telecommunication Union), **NFAP** (National Frequency Allocation Plan), **NIAR** (National Institute of Amateur Radio, Hyderabad), **ARSI** (Amateur Radio Society of India), **wpm** (words per minute), **MCQ** (multiple-choice question).

3.1 A note on evidence tagging in this part

Three DoT documents that are central to this part — the 2010 syllabus and power memorandum, the December 2024 FAQ, and the February 2025 process memorandum — could only be retrieved as signed, numbered scans hosted by ARSI, not from a gov.in address. The underlying research files disagree on how to tag them: r04, r06 and r12 treat them as **[VERIFIED]** because they are official DoT documents read directly; r05 downgrades them to **[REPORTED — DoT document, ARSI-hosted scan]** because they were not fetched from a government domain.

This part follows the stricter convention where the distinction could matter to a reader's legal exposure, and says so explicitly at those points. Treat these three documents as near-primary: signed, numbered, internally consistent, and in two cases addressed to WPC with a request to upload them to the DoT website — but not independently confirmed on a government server.

3.2 The chain of authority

Instrument	Date	Status	Tag
Telecommunications Act, 2023 (44 of 2023)	assented 2023-12-24	Parent Act. Section 47 is the rule-making power for amateur services	[VERIFIED]
Telecommunications (Amateur Services) Rules, 2024 — G.S.R. 675(E)	notified 2024-10-29; gazetted 2024-10-30	In force. Supersedes the Indian Wireless Telegraphs (Amateur Service) Rules, 1978	[VERIFIED]
WPC FAQ No. P-14036/02/2023-COP	2024-12-05	Departmental clarification: pass criteria, Morse speed, SACFA scope, portable operation	[REPORTED — DoT document, ARSI-hosted scan]

Instrument	Date	Status	Tag
WPC OM No. P-14036/03/2023-COP	2025-02-28	"Modified Amateur/HAM module": one-step application, self-certification, QR-coded certificate	[REPORTED — DoT document, ARSI-hosted scan]
WPC OM No. L-14011/255/2004-AMT, Annexure-I	2010-08-13	The operative frequency / emission / power table	[VERIFIED] as to content; [UNCERTAIN] as to continued legal force — see §3.4.5
WPC OM No. L-14011/255/2004-AMT, Annexure-II	2010-08-13	The published syllabus and examination scheme	[REPORTED — DoT document, ARSI-hosted scan]
WPC OM No. R-11014/03/2015-PP	2016-03-10	Discontinued Intelligence Bureau / Ministry of Home Affairs security clearance for amateur applicants	[VERIFIED]
Indian Wireless Telegraphs (Amateur Service) Rules, 1978 (as amended 2005, 2009)	1978; amendments effective 2010-04-01	Superseded 2024-10-30	[VERIFIED]

Sources: gazette text at <https://eservices.dot.gov.in/sites/default/files/circular-notifications/amateur-services-rules-2024.pdf> (Gazette of India, Extraordinary, Part II, Section 3, Sub-section (i), No. 617, 2024-10-30, CG-DL-E-01112024-258404, file No. 24-09/2024-UBB, signed Devendra Kumar Rai, Joint Secretary); 2010 OM at https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf ; 2024 FAQ at <https://arsi.info/wp-content/uploads/2024/12/Frequently-Asked-Questions-on-the-Telecommunication-Amateur-Service-Rules-2024.pdf> ; 2016 security-clearance OM at https://arsi.info/wp-content/uploads/2020/11/Security_Clearance-OM-R-11014032015-PP.pdf . All fetched 2026-08-23.

The supersession clause, verbatim **[VERIFIED]**:

"Now, therefore, in exercise of the powers conferred by section 47 read with clause (zi) to sub-section (2) of section 56 of the Telecommunications Act, 2023 (44 of 2023), and **in supersession of the Indian Wireless Telegraphs (Amateur Service) Rules, 1978**, except as respects things done or omitted to be done before such supersession and without overriding the terms and conditions of existing arrangements under those rules till the date of expiry of such arrangements, the Central Government hereby makes the following rules, namely:—"

Note the saving: certificates and licences already issued under the 1978 Rules continue to their stated expiry. **[VERIFIED]**

One numbering trap. DoT's own December 2024 FAQ cites the rules as "GSR 657(E) dated 29.10.2024". The gazette itself reads **G.S.R. 675(E)** in both the Hindi and English texts. The FAQ contains a digit transposition; cite 675(E). **[VERIFIED — both documents read]**

3.3 One document, not two

Under the 1978 regime an Indian amateur held two separate things: an Amateur Station Operator's Certificate, won by examination, and an *Amateur Wireless Telegraph Station Licence*, which authorised the station. **The station licence no longer exists.** Since 2024-10-30 the ASOC is itself the operating authorisation and carries the call sign. **[VERIFIED]**

Rule 3, verbatim:

"3. Scope. – (1) No person shall install or operate an amateur station except under and in accordance with the terms and conditions of an ASOC granted under these rules. (2) There shall be the following two categories of ASOC, which may be granted by the Central Government under sub-rule (1) of rule 6, namely,- (a) Amateur Station Operator Certificate- General, or ASOC (General); and (b) Amateur Station Operator Certificate- Restricted, or ASOC (Restricted)."

Rule 6(1), verbatim:

"The Central Government shall, upon successful completion of the examination under rule 5, issue any of the following categories of certificate, as applied for, **along with unique call signs for amateur station operators**, subject to payment of fees as specified below,-"

[VERIFIED] Any guide that tells you to "pass the ASOC exam, then apply separately for the station licence" is describing the pre-2024 world.

3.4 The grades, and the authoritative power and band table

3.4.1 THE 2024 RULES CONTAIN NO FREQUENCIES AND NO WATTAGES

This is the single most consequential structural fact in this part. The words "watt", "MHz" and "kHz" do not appear anywhere in the Telecommunications (Amateur Services) Rules, 2024.

[VERIFIED] The Rules delegate the entire question:

"7. General conditions applicable to an amateur station operator. – (1) An ASOC holder shall operate amateur stations on such frequency bands and emissions and transmit power **as specified by the Central Government** in respect of each category of ASOC, and the holder of ASOC (General) shall be permitted use of **more emission types and higher power** as may be specified in his ASOC, than the holder of ASOC (Restricted)." — Rule 7(1) **[VERIFIED]**

"3. Frequencies, emissions and power. - The amateur station shall be operated on frequencies that are within the frequency bands authorised to respective categories of certificates and on such classes of emissions and power **as specified by the Central Government.**" — Annexure A, paragraph 3 **[VERIFIED]**

3.4.2 THE OPERATIVE TABLE: WPC OM OF 2010-08-13, ANNEXURE-I

The specification the Rules point to has not been republished since 2010. WPC's covering memorandum says so in terms:

"8. The frequency, emission and power authorized to the Amateurs in India have been revised as per **Annexure-I**. These are issued in the light of the NFAP 2008, which is already available on the WPC website."

"12. This O.M shall supercede the 'Frequency bands, Emission and Power' that was notified by WPC Wing through its notification dated 4th October 1985." — OM L-14011/255/2004-AMT, 2010-08-13, paragraphs 8 and 12 **[VERIFIED — read from the scan]**

WPC's Central Public Information Officer (CPIO) reconfirmed Annexure-I in a formal reply under the Right to Information (RTI) Act, 2005:

"2. It is informed that the list of allocated bands and frequency for amateur radio operation in India may be seen at **Annexure I of this Ministry's OM No. L-14011/255/2004-AMT dated 13/08/2010**. Relevant pages of the above OM has been enclosed." — WPC letter No. P-14033/02/2020-COP dated 2020-12-28, signed M. Revathi, Senior Deputy Wireless Advisor (C) and CPIO **[VERIFIED — reproduced at page 33 of NIAR's rules compilation, <http://www.niar.org/downloads/Amateur-Radio-Rules.pdf>]**

Annexure-I, transcribed exactly. The column heading is "Max DC Input Power unless Otherwise specified".

ASOC (RESTRICTED) — call sign block VU3

Frequency band	Emission	Max DC input power	Remarks
1820–1860* kHz	A3E, H3E, J3E, R3E	50 watts	Old "Grade II" licensees are also authorized to use A1A emission in these bands
3500–3700* kHz	A3E, H3E, J3E, R3E	50 watts	"
3890–3900* kHz	A3E, H3E, J3E, R3E	50 watts	"
7000–7100 kHz	A3E, H3E, J3E, R3E	50 watts	"
7100–7200 kHz	A3E, H3E, J3E, R3E	50 watts	"
14000–14350 kHz	A3E, H3E, J3E, R3E	50 watts	"
18068–18168\$ kHz	A3E, H3E, J3E, R3E	50 watts	"

Frequency band	Emission	Max DC input power	Remarks
21000–21450 kHz	A3E, H3E, J3E, R3E	50 watts	"
24890–24990\$ kHz	A3E, H3E, J3E, R3E	50 watts	"
28000–29700 kHz	A3E, H3E, J3E, R3E	50 watts	"
50–54 MHz (6 m)	F1B, F2B, F3E, F3C	10 watts	
144–146 MHz (2 m)	F1B, F2B, F3E, F3C	10 watts	Old "Grade II" licensees are also authorized to use A1A and A2A emissions in these bands
434–438@ MHz (70 cm)	F1B, F2B, F3E, F3C	10 watts	"

ASOC (GENERAL) — call sign block VU2

Frequency band	Emission	Max DC input power
1820–1860* kHz	A1A, A2A, A3E, H3E, J3E, R3E, F1B, F2B, F3E, F3C, A3C, A3F	400 watts
3500–3700* kHz	"	400 watts
3890–3900* kHz	"	400 watts
7000–7100 kHz	"	400 watts
7100–7200 kHz	"	400 watts
14000–14350 kHz	"	400 watts
18068–18168\$ kHz	"	400 watts
21000–21450 kHz	"	400 watts
24890–24990\$ kHz	"	400 watts
28000–29700 kHz	"	400 watts
50–54 MHz (6 m)	F1B, F2B, F3E, F3C	25 watts
144–146 MHz (2 m)	F1B, F2B, F3E, F3C	25 watts
434–438@ MHz (70 cm)	F1B, F2B, F3E, F3C	25 watts
5725–5840@ MHz (5 cm)	F1B, F2B, F3E, F3C	25 watts

Footnotes, verbatim:

"* On primary shared basis as per the relevant provision of Radio Regulations. \$ Authorization is on non-interference and non protection basis. @ On secondary basis as per the relevant provisions of Radio Regulations."

Notes, verbatim:

"(i) All the allocation subject to the relevant provisions of the Radio Regulations. Amateur satellite service is permitted for General Grade in the appropriate sub bands in accordance with Radio Regulations and in those cases the maximum output RF power (eirp) is 30 dBw.

(ii) The above authorization is subject to site clearance as per the procedure prescribed by the Standing Advisory committee on Radio frequency allocation (SACFA) as applicable.

(iii) For A3F emission, the transmission shall be restricted to call-sign of the station, location and other particular of the station. They shall be limited to point to point test transmission employing a standard interlace and scanning with a bandwidth not more than 4 KHz.

(iv) **DC input power is the total direct current power input to the final stage of the transmitter."**

[VERIFIED — all of the above read directly from the scanned Annexure-I; the same Annexure-I is enclosed with the WPC RTI reply of 2020-12-28 and reproduced in NIAR's compilation.]

Independently re-verified, 2026-08-23. As part of an adversarial fact-check of this report, the twelve-page scan at https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf was fetched again and every page was read afresh, without reference to the transcription above. **Both tables above match the scan exactly** — every frequency band, every emission designator, every wattage, the three footnote symbols (*, \$, @) and all four notes (i)–(iv). Also confirmed from the same document: the OM number and date (No. L-14011/255/2004-AMT, dated 13/8/2010, signed Munesh Kumar, Assistant Wireless Advisor); paragraph 8 (*"The frequency, emission and power authorized to the Amateurs in India have been revised as per Annexure-I. These are issued in the light of the NFAP 2008"*); paragraph 11 (VU2 reserved for General, VU3 for Restricted); paragraph 12 (supersedes the frequency, emission and power notified on 4 October 1985); and the Annexure-II syllabus including **"There will be NO Morse test for restricted grade"** and the 8 wpm Morse receiving and sending tests for General. The fee-migration table on page 1 of the same OM independently confirms the ₹100 examination fee and the ₹1,000 / ₹2,000 licence fees, and confirms that the ₹200 change-of-location, ₹100 duplicate and ₹100-per-half-year late-renewal figures are the **2010** figures, superseded by the 2024 Rules. **The single most consequential number in this report — 10 W DC input on 2 m and 70 cm for ASOC (Restricted) — is confirmed.** The doubt recorded in §3.4.5 is *not* about what Annexure-I says; it is only about whether Annexure-I is still the operative specification under rule 7(1) of the 2024 Rules. That doubt is unresolved and the fact-check found nothing to resolve it: no post-2024 specification exists in any source that could be located, and wpc.dot.gov.in still refuses connections on port 443.

Headline summary:

Grade	Ten HF bands (1820 kHz – 29700 kHz)	6 m / 2 m / 70 cm	5 cm (5725–5840 MHz)	Morse	Call sign
ASOC (Restricted)	50 W DC input	10 W DC input	not authorised	none	VU3 / VU31–VU39
ASOC (General)	400 W DC input	25 W DC input	25 W DC input	8 wpm send and receive	VU2 / VU21–VU29

3.4.3 EMISSION DESIGNATORS, DECODED

The codes come from ITU Radio Regulations Appendix 1 and are themselves examinable.

Code	Meaning
A1A	Morse telegraphy, keyed unmodulated carrier (CW)
A2A	Morse telegraphy, amplitude-modulated tone
A3E	Amplitude modulation (AM), double sideband, telephony
H3E	AM single sideband, full carrier
R3E	AM single sideband, reduced carrier
J3E	Single sideband, suppressed carrier (ordinary voice SSB)
F1B	Frequency-shift keying telegraphy (radioteletype, digital)
F2B	Frequency modulation of an audio tone, telegraphy
F3E	Frequency modulation (FM) telephony — what a walkie-talkie transmits
F3C	FM facsimile
A3C	AM facsimile
A3F	AM television

[VERIFIED — ITU Radio Regulations Appendix 1 designators; the Indian syllabus expressly examines "Designation of Emission — Appendix I"]

Critical restriction: on 144–146 MHz and 434–438 MHz, **both grades are confined to F1B, F2B, F3E and F3C — FM and digital only. Single sideband and AM (J3E, A3E) are not authorised on VHF/UHF in India for either grade.** An ordinary FM hand-held is F3E and is squarely inside the authorised set. **[VERIFIED]**

Whether the digital voice modes (DMR, D-STAR, C4FM) fall inside F1B/F2B/F3E/F3C is **[UNCERTAIN]** — digital voice is normally designated F7W or F1E/F1D, none of which appears in Annexure-I.

3.4.4 THE TRAP: 10 W MEANS DC INPUT, NOT RF OUTPUT

Annexure-I's column heading is "Max DC Input Power unless Otherwise specified" and note (iv) defines it as "the total direct current power input to the final stage of the transmitter".

[VERIFIED]

Consumer hand-helds are advertised by **RF output**. The two numbers are not the same. A transmitter drawing 10 W of direct current into its power amplifier delivers materially less than 10 W of radio-frequency power out: at power-amplifier efficiencies plausible for a VHF/UHF FM hand-held, roughly 40 to 65 per cent, 10 W DC in yields about 4 to 6.5 W RF out. **The 10 W DC-input ceiling is [VERIFIED]; the efficiency range and the resulting RF figure are engineering inference, not law — [UNCERTAIN].**

The practical consequence:

- A hand-held sold as "**5 W output**" typically draws 8–12 W DC — at or slightly over the Restricted ceiling.
- A hand-held sold as "**10 W output**" must draw roughly 15–25 W DC — **over the Restricted ceiling, inside the General ceiling of 25 W.**

So the honest answer to "I want a 10 W walkie-talkie, legally" splits in two:

- If "10 W" means *the number my licence names* → **ASOC (Restricted) is sufficient.**
- If "10 W" means *10 watts of RF at the antenna socket* → **you need ASOC (General),** and therefore the 8 wpm Morse test.

How WPC actually measures or enforces "DC input to the final stage" on a modern solid-state hand-held, where no manufacturer publishes that figure, is **[UNCERTAIN]**. This is a good question to put to WPC in writing.

3.4.5 THE WEAKEST LINK IN THE CHAIN — READ THIS BEFORE RELYING ON THE TABLE

The 2010 OM was issued under the 1978 Rules. Those Rules were superseded on 2024-10-30, and the 2024 Rules contain **no express saving clause for executive instruments** made under them. No post-2024 frequency, emission or power specification could be found on DoT's document index, its media library, the eServices ASOC pages, the December 2024 FAQ, or ARSI's WPC-documents page. The WPC website wpc.dot.gov.in was unreachable throughout the research (connection timeout on port 443) and the Internet Archive holds no substantive captures of it after 2019.

The practical position is almost certainly that Annexure-I continues to govern — WPC's own 2020 RTI reply points to it and nothing has replaced it — but a strict reading leaves the authorising instrument technically unanchored. **[UNCERTAIN]**

Two further wrinkles follow from the same delegation:

- Rule 7(1) says a General holder gets higher power "as may be specified **in his ASOC**", implying the certificate itself may carry the band and power endorsements. No specimen of a post-2024 QR-coded ASOC was available to inspect. **If an issued certificate states figures different from Annexure-I, the certificate governs. [UNCERTAIN]**
- **Action for the reader:** before buying equipment, ask WPC in writing to confirm the current authorised bands, emissions and transmit power for ASOC (Restricted) and ASOC (General), citing Rule 7(1). Contact details are in §3.12.

3.4.6 BAND-EDGE DISCREPANCIES (FLAGGED, NONE AFFECT A 2 M / 70 CM HAND-HELD)

Band	2010 WPC OM Annexure-I (authoritative)	NIAR booklet p.20	NIAR standalone sheet	Wikipedia
160 m	1820–1860 kHz	1820–1825 kHz	1800–1825 kHz	1800–1825 kHz
40 m	7000–7100 + 7100–7200 kHz	7000–7200 kHz	7000–7200 kHz	7000–7200 kHz
30 m	absent for both grades	10100–10150 kHz shown for both	same	shown for General only
6 m	50–54 MHz	50–52 MHz	50–52 MHz	50–54 MHz

[UNCERTAIN] — the 2010 OM is the authoritative instrument and was read directly from the scan, so 1820–1860 and 50–54 are the better figures. Two live oddities:

- **30 m (10100–10150 kHz) does not appear in Annexure-I at all**, for either grade, although NIAR's widely-copied table shows it and NFAP designates amateur there. Both Annexure-I pages were checked; this is a genuine gap, not a scanning artefact. Confirm with WPC if 30 m matters.
- NFAP-2022's India column splits 6 m at 50–52 MHz with a parenthetical noting that the ITU Radio Regulations allocate 50–54, while the 2010 OM authorises 50–54 to licensees.

[VERIFIED — NFAP 2022, 47–68 MHz table, India column]

A separate trap worth stating: **an NFAP allocation is not a licensee authorisation**. NFAP-2022 allocates the amateur service (secondary) in India at 135.7–137.8 kHz (2200 m) and 472–479 kHz (630 m) **[VERIFIED]**, but neither band appears in the WPC authorisation table. On the plain reading of Rule 7(1), an Indian amateur may not transmit there. **[UNCERTAIN — no document expressly forbids it; it is simply not authorised.]**

Finally, note that **70 cm is secondary in India**. NFAP-2022's India column for 432–438 MHz lists RADIOLOCATION, FIXED and MOBILE in capitals (primary) and "Amateur" in lower case (secondary). You must accept interference and must not cause it. 144–146 MHz, by contrast, is AMATEUR / AMATEUR-SATELLITE in capitals — primary. **[VERIFIED]**

3.5 Eligibility

Rule 4(1), verbatim **[VERIFIED]**:

"4. Eligibility conditions for obtaining an Amateur Station Operator Certificate. – (1) The ASOC (General) or the ASOC (Restricted), as the case may be, shall be granted under sub-rule (1) of rule 6 to any person, **being a citizen of India**, who,- (a) **is not less than twelve years of age**; and (b) qualifies the examination specified under rule 5 for the grant of an ASOC:"

Requirement	Position under the 2024 Rules	Tag
Citizenship	Indian citizen (narrow reciprocal route for foreigners)	[VERIFIED]
Minimum age	12. The old 18-year default with a 12–18 exception is abolished	[VERIFIED]
Maximum age	None for holding. "Lifetime" validity is defined as running only to age 80	[VERIFIED]
Educational qualification	None required	[VERIFIED — absence in the Rules]
Medical test	None	[VERIFIED — absence in the Rules]
Police verification / security clearance	None. Discontinued for amateur applicants in 2016 — see §3.8.1	[VERIFIED]
Parent/guardian undertaking	Required for applicants aged 12–18, in the prescribed format	[VERIFIED — Saral Sanchar FAQ Q2]

Non-citizens. Rule 4(1) proviso and 4(2) allow a foreign national holding an equivalent certificate from a recognised country to apply, subject to administrative clearance of the Ministry of External Affairs, on payment of **one thousand rupees**. Rule 6(3) limits such a certificate to the duration of the visa or twelve months, whichever is less. **[VERIFIED]** Rule 4(3) says the Government "may specify the list of countries" whose certificates are recognised; **no such published list was found.** **[UNCERTAIN]**

3.6 What an amateur licence does *not* permit — the restrictions that decide this question

For a reader whose real goal is a walkie-talkie, this section matters more than the power table.

3.6.1 THE PURPOSE LIMITATION

Rule 2(1)(c), verbatim **[VERIFIED]**:

"'amateur services' means radio communication services for the purpose of **self-training, intercommunication and technical investigations** carried out by an amateur, that is, by a duly authorised person interested in radio technique, **solely with a personal aim and without any pecuniary interest**;"

3.6.2 MESSAGES — ANNEXURE A, PARAGRAPH 2, VERBATIM

[VERIFIED]

"2. Messages. - (1) (a) Radio communications may be exchanged with **other authorised amateur stations**. The amateur stations shall not communicate with amateur stations of countries whose administrations have notified the International Telecommunication Convention of their objection to such radio communications;

(b) Transmissions shall be made in **plain language and limited to messages of a technical nature**.

(2) The authorised entity shall **not transmit messages containing**, - (a) reproduction of broadcast programmes, tape recordings or transmissions of entertainment value or music; (b) false or misleading calls, signals, news, advertisements, **communications of business**, statements on topics of political or industrial controversy, or **third party messages**; or (c) use of indecent, obscene, offensive language, or signals.

(3) In case of failure of normal telecommunication facilities, an ASOC holder maybe permitted by the Central Government or State Government to handle third party messages pertaining to natural or manmade calamities, **originating from and addressed to a competent civil authority namely, District Magistrates or Deputy Commissioners or Collectors of the district or any other officer authorised by them.**"

3.6.3 THE FOUR HARD PROHIBITIONS

1. **No business use of any kind.** "Communications of business" is expressly barred and the definition requires "without any pecuniary interest". An ASOC cannot lawfully be used to run a warehouse, a delivery fleet, a shop, a construction site, an event company, a film shoot, a wedding, a housekeeping team or a security detail. [VERIFIED]
2. **Only licensed amateurs may be at the other end.** Paragraph 2(1)(a) permits exchanges with "other authorised amateur stations" only, and 2(2)(b) bans third-party messages. **A licensed amateur may not lawfully talk to an unlicensed spouse, child, colleague or driver on a second handset.** Every party to the contact needs their own ASOC. This is the point on which most prospective buyers of a pair of walkie-talkies are mistaken. [VERIFIED]
3. **No broadcasting or entertainment.** No music, no relaying broadcast programmes, no transmission to an unspecified audience. [VERIFIED]
4. **Third-party traffic only in the narrow disaster case** described in paragraph 2(3): normal telecom must have failed, the traffic must concern a calamity, it must originate from or be addressed to a District Magistrate / Deputy Commissioner / Collector or an officer they authorise, and the Central or State Government must permit it. [VERIFIED]

A fifth point, less obvious and worth flagging honestly: the 2024 drafting is **stricter** than the old conditions. The 1978-era Annexure I paragraph II(1)(b) permitted "messages of a technical nature relating to tests **and to remarks of personal character** (excluding business affairs or transactions)". The 2024 text drops the personal-remarks clause entirely and limits transmissions to "messages of a technical nature". **[VERIFIED — texts compared]** Read literally, ordinary conversation between two licensed amateurs is now outside the permitted content. In practice Indian amateurs converse socially and this is universally tolerated, but no WPC clarification on the point was found. **[UNCERTAIN]**

Even if every family member obtains their own ASOC, the correspondent problem disappears but paragraph 2(1)(b) and 2(2)(b) still constrain content, and using the amateur bands as a household intercom is not "self-training, intercommunication and technical investigations". **[VERIFIED on the text; enforcement practice UNCERTAIN]**

3.6.4 ONE AFFIRMATIVE DUTY

"(7) The ASOC holder **shall provide radio communication services** in amateur services frequency band, at the request of the Central Government, or a State Government, or any officer specially authorised in this behalf ... **during natural calamities and disasters, without any fees or charge.**"
— Rule 7(7) **[VERIFIED]**

3.7 The ASOC examination

3.7.1 STATUTORY BASIS

Rule 5, verbatim **[VERIFIED]**:

"5. Amateur station operator examination. – (1) Any person satisfying the eligibility criteria set forth under rule 4 ... may make an application for appearing in the amateur station operator examination for obtaining the ASOC, to the Central Government, in the form specified for this purpose in the portal, **one month before the date of such examination.**

(2) An application under sub-rule (1) shall be accompanied by **examination fee of one hundred rupees.**

(3) The Central Government shall publish the syllabus, place, manner, date and time for the examination ... including the date of announcement of the results of such examination.

(4) The **language of such examination shall be English**, and examination for ASOC (General) shall also require such **proficiency in Morse Code** as may be specified by the Central Government.

(5) The successful completion of the examination under this rule shall require a **minimum of forty per cent of the total marks** in respect of each category of ASOC."

Four points follow directly. The examination is **in English only** — there is no Hindi or regional-language paper. **Morse survives in the statute** for General grade. The Rules specify almost nothing about format; everything else is delegated to what the Government "publishes" under Rule 5(3), and that published material is thin and partly stale. And the statutory pass floor is a flat **40 per cent of total marks**.

There is a structural oddity in Rule 5(1): you must apply "one month before the date of such examination", but no examination date exists until after your application has been scrutinised (§3.7.7). In practice the one-month figure works as a lead time, not a deadline you can compute. **[UNCERTAIN as to how WPC reconciles this]**

3.7.2 WHERE YOU APPLY — SARAL SANCHAR, AND NOWHERE ELSE

- **There is no offline route.** All amateur applications have been online only since **2020-08-01**, under WPC OM File No. Project/ASMS/02/2019 (Part-1) dated 2020-07-27. **[REPORTED — NIAR, <https://www.qsl.net/vu2jos/Exam.pdf> , 2025-03-21, quoting the OM]** A three-month transitional window for hard copies closed at the end of 2020. As of 2026-08 there is no paper route; if you cannot use the portal you need help, not a form.
- The portal is **Saral Sanchar**, <https://saralsanchar.gov.in> , notified under Rule 15. **[VERIFIED]**
- **As of 2026-08-23, <https://saralsanchar.gov.in/> 302-redirects to <https://eservices.dot.gov.in/saral-sanchar> ; the legacy login remains reachable at <https://saralsanchar.gov.in/LoginHome.php> .** **[VERIFIED — observed 2026-08-23]**
- **ASOC applications are NOT on the new DoT Authorisation Portal.** That portal handles service authorisations, licence migration and radio-equipment possession. DoT's ASOC service page, last updated 2026-08-20, still says "Submit application saral sanchar portal". **[VERIFIED — <https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc>]** DoT has been migrating other WPC services to the Authorisation Portal through 2026, so **re-check this before applying. [UNCERTAIN — volatile]**

3.7.3 THERE IS NO NUMBERED FORM

Rule 5(1) and every other application provision in the 2024 Rules say only "in the form specified for this purpose in the portal". **[VERIFIED — re-confirmed on 2026-08-23 by reading the whole of G.S.R. 675(E): the words "in the form specified for this purpose in the portal" (or "in the form as may be specified") appear in rules 4(2), 5(1), 7(8), 8(1), 8(2), 12(1), 12(2), 12(4) and 13(1), and the Rules contain no form number, no annexed proforma and no schedule of forms. Annexure A is the terms and conditions, and the only tabular proforma anywhere in the instrument is the equipment register in Annexure A paragraph 7(4), which is unnumbered.]** The old numbered paper forms belong to the superseded 1978 Rules. **If a website, coaching centre or agent quotes you a "Form No." for the ASOC exam or certificate, it is quoting a superseded instrument.** What you will actually encounter:

Artefact	Exact label you will see	Tag
Exam application (web form)	"Application for Amateur Exam"	[REPORTED — NIAR screenshots, 2025-03-21]
Exam application ID	EXAM + year + serial, e.g. EXAM20250200019	[REPORTED — NIAR]
Certificate application (web form)	"APPLICATION FOR WPC (Amateur Station Operator's Certificate)", ID pattern HAM<YYYYMMDD>0000NN	[REPORTED — NIAR, License.pdf, 2025-03-01]
Admit card	"e-ADMIT CARD / FOR AMATEUR STATION OPERATING CERTIFICATE EXAMINATION- 2025"	[REPORTED — NIAR model admit card]
Nationality proforma A	"SELF DECLARATION OF CHARACTER AND NATIONALITY (in case valid Indian passport is available)"	[REPORTED — DoT document, ARSI-hosted scan, OM 2025-02-28 enclosure]
Nationality proforma B	"NATIONALITY CERTIFICATE (in case valid Indian passport is not available)"	[REPORTED — DoT document, ARSI-hosted scan]
Minor's undertaking	"ASOC EXAMINATION UNDERTAKING (To be issued by Parent/Guardian/Head of Institution of candidates who are between 12 to 18 years of age)"	[REPORTED — DARTS archive of the pre-2020 form set; the current proforma could not be located on any gov.in URL — [UNCERTAIN]]

Obsolete forms you will find online and must not use: "ANNEXURE II — Application from an individual for a licence to Establish, Maintain and Work an Amateur Wireless Telegraph Station in India"; "APPENDIX III — NATIONALITY CERTIFICATE"; the four-copy personal-particulars sheet asking for height, complexion and identification marks; and the letter requesting Section A exemption addressed to the Officer in Charge of a Wireless Monitoring Station. The old demand-draft payment to "PAO (HQ), DOT" is also dead — Bharatkosh has replaced it entirely. [REPORTED — DARTS archive, 2018, <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Form.pdf>]

Proforma B, verbatim, because the attesting-officer list is narrow [REPORTED — DoT document, ARSI-hosted scan]:

"Certified that I have known Shri. / Smt / Kum son / daughter of Shri..... for the last..... years and that to the best of my knowledge and belief he/she bears a good moral character. He/ She is ofnationality. He/ She is not related to me.

Note- This certificate should be from one of the officers listed below:- 1. Gazetted officers of Central or State Governments. 2. Members of Parliament or State Legislatures. 3. Sub-Divisional Magistrate/ Officer or Tahsildars or Naib/Deputy Tehsildars authorised to exercise magisterial powers."

Note a 2025 change: the pre-2025 version of that list also allowed a school or college principal for a bona fide student. **That option is not in the 2025 proforma. [REPORTED — DARTS archive vs. the 2025 OM]**

3.7.4 THE 2025 PROCESS CHANGE — ONE APPLICATION, SELF-CERTIFICATION

WPC OM No. P-14036/03/2023-COP dated **2025-02-28**, "Launch of modified Amateur/HAM module on Saral Sanchar portal of DoT", signed Sachin Kumar, Assistant Wireless Adviser, verbatim **[REPORTED — DoT document, ARSI-hosted scan]:**

2.1 Applicant need to submit a single application for appearing in ASO Certificate examination and for grant of ASO Certificate for passed applicants.

2.2 Requirement of attested documents has been done away with. Applicant now need only to submit self-certified documents and produce the original documents on the day of examination. This verification is a one-time requirement. If an applicant is unable to produce the original documents at the time of the examination, ASO Certificate will be granted only after production of original document at the concerned office.

2.3 The Call Sign, as selected by the applicant on the portal, will be system-generated - based on availability.

2.4 Requirement of attested Nationality Certificate has also been simplified. Applicant now need to submit either of the following: i) Copy of valid Indian Passport and signed self-declaration as per attached proforma; or ii) Attested nationality and character certificate as per attached proforma.

2.5 Applicant's photograph will be digitally captured on the day of the examination.

2.6 ASO Certificate will be QR code authenticated.

1. From March 2025, applicant need to file online application on the modified Amateur/HAM module of Saral Sanchar Portal at <https://saralsanchar.gov.in>

Caveat on "one-step". The user manual enclosed with that same OM, dated 2025-03-01, still contains a separate "Section-7 ASO Certificate Application" that you complete *after* the result is published, with its own fee payment. "One-step" appears to mean one continuous tracked journey inside one module, not literally one submission. **[UNCERTAIN — verify on the live portal]**

3.7.5 APPLYING, STEP BY STEP, WITH THE ACTUAL FIELD NAMES

Registration (once). **[VERIFIED — DoT user manual] / [REPORTED — NIAR walkthrough]**

1. Go to the portal and click **Registration**.
2. Registration Type: **"Individual (WPC Exams/ Certificates)"**. The DoT manual notes: "Only Individual (WPC Exams/ Certificates) or Individual/ Proprietor/ Partnership Firm can apply for Amateur License." **[VERIFIED]**
3. Have ready a **PAN** (Permanent Account Number; a minor without one uses a parent's or guardian's), a mobile phone for a one-time password by SMS, and a logged-in email account for an email one-time password. **[REPORTED — NIAR]**

4. Fill contact details, solve the CAPTCHA, validate **two separate one-time passwords**, submit.
5. Credentials arrive by email; first login forces a password change plus another mobile one-time password.
6. On the dashboard, click **APPLY NOW** on the **WPC** tab.

Exam application. Path: **WPC "Apply now"** → **Apply for Examinations** → **Amateur Station Exam** → **Go.** [VERIFIED — DoT manual] The 2025 module has five steps [REPORTED — DoT document, ARSI-hosted scan]:

Step	Tab	Contents
1	Applicant Details	Name; Upload Photograph ; Upload Signature ; father/mother/husband name; date of birth + proof upload; place of birth; occupation; nationality; nationality and character certificate (branches on Indian resident / Indian resident with passport / Indian non-resident)
2	Other Details	Identity proof; present and permanent address details and proofs. "In case of Indian applicant at least one of ID proof or Nationality certificate is mandatory."
3	Exam Details	"Category of License Applied" (Amateur General Grade / Amateur Restricted Grade); " Nodal Office for Examination " (your exam centre); "Is Exempted from Part A" (Yes/No); "Reason for Exemption"; proof-of-exemption upload; " Fee to be Paid: 100 "
4	Location Details	Station address same as present address (Y/N); station address; pin code; district; state; latitude (8°4' to 37°6' N) and longitude (68°7' to 97°25' E) in degrees/minutes/seconds; station address proof (rent agreement or owner's consent)
5	Declaration	"I ACCEPT THE ABOVE CONDITIONS"

Then **Save Draft** → **Validate Application** → **Submit**.

Payment. "After Submitting the application, Applicant is redirected to pay the fees. Fees is prepopulated. Applicant has to click Pay Now to get redirected to Bharatkosh Portal for Payment." The dashboard warns: "Please make payment through Saral Sanchar integrated BharatKosh portal ... Any other mode of payment will not be considered." [REPORTED — DoT document, ARSI-hosted scan]

The step people miss — upload the signed application. Verbatim:

"Applicant has to download the PDF of the Application by clicking the Download Button. Take a printout this pdf Manually sign on the print and then scan and upload to final submit the application or If user has PDF editor, they can embed the signature in the PDF and upload."

[REPORTED — DoT document, ARSI-hosted scan] Until you do this, your application is not with WPC.

Scrutiny. The application goes to the WPC/WMO nodal officer. If something is wrong it is reverted to your dashboard under "Previous Applications" with a "Modify Application" button, and only the flagged serial numbers are editable. Correct, re-download, re-sign, re-upload.

[REPORTED — DoT document, ARSI-hosted scan]

3.7.6 FEES

Item	Fee	Basis	Tag
Examination, Restricted	₹100	Rule 5(2)	[VERIFIED]
Examination, General	₹100	Rule 5(2)	[VERIFIED]
Examination, both grades in one sitting	₹200	Saral Sanchar official FAQ Q3	[VERIFIED]
ASOC, 20-year validity (either grade)	₹1,000	Rule 6(1)	[VERIFIED]
ASOC, lifetime validity (to age 80, either grade)	₹2,000	Rule 6(1)	[VERIFIED]
Total entry cost, Restricted	₹1,100 (20 yr) or ₹2,100 (lifetime)	—	[VERIFIED]

Official fee table, verbatim from the Saral Sanchar FAQ Q3 (https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf) **[VERIFIED]**:

S. No.	Category	Examination Fee
1	Restricted	Rs. 100/-
2	General	Rs 100/-
3	General & Restricted	Rs 200/-

NIAR's practical advice: **"Those wanting to appear for General Grade are advised to appear in both grades"** — pay ₹200 so that a Morse failure still leaves you with a Restricted pass.

[REPORTED — NIAR, 2025-03-21] This is directly supported by the official FAQ Q10

[VERIFIED]: "If a candidate passes in Part I and II, he/she will be issued a GENERAL Licence. If a candidate passes only in Part I of the examination he/she will be issued a RESTRICTED Licence."

One conflict to note: the official FAQ says one application can cover both grades for ₹200, but the 2025 portal screenshots show a single-choice dropdown with "Fee to be Paid: 100". **[UNCERTAIN — ask WPC or the helpdesk whether two applications are now needed.]**

The ₹100 examination fee has been stable since **2010-04-01** (it was ₹10–₹25 before that) and was re-enacted verbatim in the 2024 Rules. It is unlikely to be stale. **[VERIFIED]**

Bharatkosh account heads (the Non-Tax Receipt Portal, <https://bharatkosh.gov.in> , now embedded in the Saral Sanchar flow). You should not normally need these, but they resolve payment failures. **[VERIFIED — Saral Sanchar FAQ Q24 and Q25]**

	Examination fee	Certificate / licence fee
Purpose head	"Amateur Exam conducted by Wireless Monitoring Stations and RHQs"	"Amateur License"
Payment type	"Exam fee"	"License fee" / "License renewal fee"
Function head	127500103050100 — Proficiency Certificate	127500103050100 — Proficiency Certificate
Pay and Accounts Office	regional , matching your exam centre (below)	077188 — Controller of Communication Accounts, PAO HQ
Drawing and Disbursing Officer	regional DDO CCA	205155 — DDO, SO(Cash), New Delhi

Regional HQ	PAO code	DDO code
Western RHQ, Mumbai	077148 — CCA Maharashtra	201562
Northern RHQ, New Delhi	077177 — CCA New Delhi	201540
Southern RHQ, Chennai	077153 — CCA Tamil Nadu	201579
Eastern RHQ, Kolkata	077155 — CCA West Bengal	201589

3.7.7 DOCUMENTS, PHOTOGRAPH AND SIGNATURE

Uploaded with the application [REPORTED — NIAR, 2025-03-21, <https://www.qsl.net/vu2jos/Exam.pdf>]:

#	Document	Specification as published by NIAR (2025-03-21)
a	Photograph	Scanned colour passport-size, JPG, 300 × 300 pixels
b	Signature	Scanned, JPG, 300 × 300 pixels, 150 kB
c	Date-of-birth proof	Birth certificate / SSC certificate / class 10 marks memo; must show father's name; self-certified; PDF under 1 MB
d	Present address proof	Aadhaar / election card / driving licence / ration card / electricity bill / bank passbook / telephone bill / rent agreement; self-certified; PDF under 1 MB
e	Permanent address proof	Same list; self-certified; PDF under 1 MB
f	Nationality certificate	(i) passport copy, self-certified, plus the signed self-declaration; or (ii) the attested proforma. "The name, designation and address of attesting officer must be clearly readable"
g	Part A exemption proof (if claiming)	Degree in engineering with electronics/telecom, B.Sc./M.Sc. with electronics/telecom, or diploma in engineering with electronics/telecom; self-certified; PDF under 1 MB

#	Document	Specification as published by NIAR (2025-03-21)
h	Station latitude and longitude	Entered in degrees, minutes, seconds
i	Station address proof	Rent agreement or owner's consent, if the station is not at your present address [REPORTED — DoT document, ARSI-hosted scan]
j	Parent/guardian undertaking	Only for applicants aged 12–18 [VERIFIED]

Attestation. Since 2025-02-28, documents are **self-certified**; gazetted-officer attestation is abolished for the exam and certificate route. The one exception is the **nationality certificate** where you have no passport — that still requires attestation by a gazetted officer, a Member of Parliament or a State Legislator, or an SDM / Tahsildar exercising magisterial powers.

[REPORTED — DoT document, ARSI-hosted scan] Most club guides, including NIAR's older ones, still say everything must be attested; they predate the OM.

Photograph and signature sizes — four official specifications exist and they conflict:

Source	Date	Photograph	Signature	Documents
WPC "Instructions for filing online applications"	c. 2020	JPEG, under 50 kB	JPEG, under 50 kB	PDF, under 50 kB each
DoT "ASOC Manual" user manual (gov.in)	2020-05-11	under 250 kB	under 150 kB	PDF under 1 MB
NIAR guide	2020-08-12	JPEG, under 50 kB	JPEG, under 50 kB	—
NIAR guide	2025-03-21	JPG, 300 × 300 px	JPG, 300 × 300 px, 150 kB	PDF under 1 MB

Only the second row is on a gov.in URL **[VERIFIED]**; the others are **[REPORTED]**.

Practical guidance: prepare a colour passport-style JPG at 300 × 300 pixels, under 150 kB, and a signature JPG at the same size. That satisfies every published specification except the strictest 50 kB one; if the portal rejects it, re-compress under 50 kB. **Trust the limit the portal displays on the day over anything in this table.**

The 2025 photograph puzzle. OM paragraph 2.5 says the photograph "will be digitally captured on the day of the examination", yet the NIAR walkthrough published three weeks later still shows an "Upload Photograph" control, and the model admit card carries a photo box. Either you upload one for the admit card and a fresh one is captured for the certificate, or the OM describes a change not yet live. **[UNCERTAIN — prepare a photograph regardless.]**

Originals to carry to the exam hall. Official FAQ Q16, verbatim **[VERIFIED]**:

"The Admit Card is to be produced for verification at the time of the test. At least one original valid Photo Identification Card (such as Aadhaar Card, Driving Licence, Passport, PAN Card, Voter ID,) should also be presented. Date of Birth Proof, Address proof, Proof for Exemption in Part A and Undertaking certificate (For students between ages 12 and 18) may also be produced to the examiner for verification."

The admit card is stricter, requiring the identity card to be original, unexpired, not damaged or smudged, and requiring originals of everything uploaded. If an original is unavailable you are **not barred from the exam**, but the certificate is withheld until you present it at the WMS/IMS or RHQ office. **[REPORTED — NIAR model admit card; DoT OM ¶2.2]**

3.7.8 EXAM CENTRES

The examination is conducted by the **Wireless Monitoring Organisation** field units — 27 Wireless Monitoring Stations and International Monitoring Stations. Official list, verbatim from the Saral Sanchar FAQ Q4: "Examination may be conducted at any of the Wireless Monitoring Station/International Monitoring Station in India." **[VERIFIED]**

#	Centre	#	Centre	#	Centre
1	Ahmedabad	10	Gorakhpur	19	New Delhi
2	Ajmer	11	Hyderabad	20	Patna
3	Bengaluru	12	Jalandhar	21	Raipur
4	Bhopal	13	Jammu	22	Ranchi
5	Bhubaneswar	14	Kolkata	23	Shillong
6	Chennai	15	Lucknow	24	Siliguri
7	Dehradun	16	Mangalore	25	Thiruvananthapuram
8	Dibrugarh	17	Mumbai	26	Vijayawada
9	Goa	18	Nagpur	27	Vishakapatnam

Regional mapping (it determines your Bharatkosh payment head) **[VERIFIED — same FAQ, Q24]:**

Regional HQ	Centres served
Western RHQ, Mumbai	Ahmedabad, Bhopal, Goa, Mumbai, Nagpur, Raipur
Northern RHQ, New Delhi	Ajmer, Dehradun, Gorakhpur, Jalandhar, Jammu, Lucknow, New Delhi
Southern RHQ, Chennai	Bengaluru, Chennai, Hyderabad, Mangalore, Thiruvananthapuram, Vijayawada, Vishakhapatnam

Regional HQ	Centres served
Eastern RHQ, Kolkata	Bhubaneswar, Dibrugarh, Kolkata, Patna, Ranchi, Shillong, Siliguri

Street addresses and telephone numbers for all 27 are compiled by NIAR at

<http://www.niar.org/downloads/ham-downloads/List-of-Wireless-Monitoring-Stations.pdf>

[REPORTED] — but that list is undated and four entries carry no address. **Confirm the address on your own admit card. [UNCERTAIN]**

Sitting the exam in a city with no monitoring station. Official FAQ Q15, verbatim **[VERIFIED]:**

"For examination venues in cities other than where Wireless Monitoring Stations/ International Monitoring Stations are situated, there should be a minimum of 20 candidates and the centre should be either a School/University/Education Centre/Amateur Radio Society or Club. The application may be submitted to the nearest monitoring station."

This is how clubs bring an examiner to a college or a hamfest. **One individual cannot trigger it.**

Two choices lock at submission. Official FAQ Q7, Q8 and Q9 **[VERIFIED]:** "no change will be allowed after submission of application" for the **exam centre**, and "No change in category will be permitted after submission of the application" for the **grade**. Choose carefully.

3.7.9 HOW DATES ARE NOTIFIED — THERE IS NO EXAM CALENDAR

Unlike the aeronautical RTR(A) certificate — the Radio Telephone Operator's Restricted Certificate of Proficiency (Aeronautical) — for which DoT publishes an annual schedule, **ASOC exam dates are not published in advance. [REPORTED]** The mechanism is per-candidate:

Official FAQ Q14, verbatim **[VERIFIED]:**

"The Date, Venue and Time of the examination will be intimated to the candidate by the respective Monitoring Station. The examination will be conducted as and when there are sufficient number of candidates."

DoT user manual, verbatim **[VERIFIED]:**

"Once the application is scrutinized successfully then hall ticket will be issued and applicant will be notified of the exam date and venue through SMS. If scrutiny is not successful then applicant has to modify and resubmit it."

Download path: **Apply for Examinations → Amateur → Current Application → Action → Download Admit Card. [REPORTED — NIAR, 2025]**

Frequency. NIAR says the exam runs "almost every month" at the monitoring stations, and its 2024-11-09 FAQ says "every month when there are sufficient numbers of candidates".

[REPORTED] Older accounts describe monthly exams in Delhi, Mumbai, Kolkata and Chennai and quarterly elsewhere. **[REPORTED]** The official FAQ deliberately says only "as and when there are sufficient number of candidates". **[VERIFIED]** **Do not plan around a fixed date.**

Exam-hall rules, from the admit card's instructions to candidates **[REPORTED — NIAR model admit card, 2025-03-21]:**

"1. Print this Admit Card on an A4 size paper. Admit Card is valid only if the candidate's photograph image is legibly printed. ... 4. Mobile phones or any other electronic gadgets such as calculators, watches are NOT ALLOWED inside the test centre. 5. Candidates should report at the test venue 30 minutes prior to the start of the test. No candidate will be permitted after the commencement of the test. 6. The Admit Card is provisional subject to the condition that the candidate has fulfilled the eligibility conditions like age, academic qualification (in case of exemption from Part A) etc. as mentioned in the application. 7. This Admit card is valid only for the test date and time as specified above."

Note item 4: **no calculators and no watches.** The syllabus contains Ohm's law and resonance arithmetic; expect to do it by hand. Note item 6: **the admit card is provisional** — if your originals do not support your claims, including the Part A exemption, the pass can be undone.

3.7.10 EXAM FORMAT AND MARKING

The detailed scheme is **Annexure II to WPC OM No. L-14011/255/2004-AMT dated 2010-08-13**, which was issued under the now-superseded 1978 Rules. No replacement has been published, although Rule 5(3) requires one and the admit card links to one. DoT's own December 2024 FAQ still uses Annexure II's "Part A / Part B" vocabulary, which strongly suggests it remains operative in substance. **[UNCERTAIN as to formal legal status; [REPORTED] as to content.]**

Annexure II, verbatim **[REPORTED — DoT document, ARSI-hosted scan]:**

"1. The examination shall consist of the following two parts:

PART I - Written Test — It shall comprise of one paper containing two sections as under:

Section A: Radio Theory and Practice

Note- Applicants holding degree in Engineering/Science or Diploma in Engineering and having studied electronics or telecommunications shall be exempted from appearing in Section A of Part-I of the test.

Section B: National and International Telecommunication Union (ITU) Radio Regulations applicable to the operation of amateur station and those relating to the working of station generally.

PART II – Morse (Only for General Certificate) — Morse reception and sending (8 wpm)"

One written paper, two sections, plus a practical Morse test for General only. There is no second written paper and no interview.

Question counts, verbatim from the official Saral Sanchar FAQ Q5 **[VERIFIED]**:

"The examination consists of two parts; Part A: Basic Electronics and Part B: radio Regulations. The Restricted Grade examination consists of **25 Questions each in Part A and Part B** and is for **one hour**. The General Grade Examination consists of **50 Questions each in Part A and Part B** and is for **two hours**."

Marks per question. The research files disagree here, and the disagreement is resolvable. r05 tagged marks-per-question **[UNCERTAIN]** because no current official document states it. r11 resolved it from the official WPC sample question paper — "Sample question paper of the examination for award of Amateur Station Operator's Licence (General and Restricted Grade)", Government of India, Ministry of Communications and IT, DoT, 7 pages, 100 questions, originally at <http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf> and now reachable only through the Internet Archive (earliest capture 2016-08-04) and club mirrors. Its instruction block reads, verbatim **[VERIFIED — archived government document]**:

"Note: 1. All questions are of Multiple Choice type. 2. Maximum marks 100. 3. Restricted grade candidates have to answer the first 25 questions(each question carries **Two** marks) from each section and the time allotted for it is **One** hour. 4. General grade candidates have to answer all the 50 questions(each question carries **One** mark) from each section and the time allotted for it is **Two** hours."

r11 is the better-evidenced source here because it read the paper itself. The paper is undated on its face and is an archived copy of a page that no longer resolves, so treat the marks-per-question figures as well-supported but not currently republished by DoT.

Resulting format table:

	ASOC (Restricted)	ASOC (General)
Written papers	1	1
Sections	2 — Part A / Section A (Radio Theory and Practice); Part B / Section B (Radio Regulations)	Same 2
Questions attempted	Q1–Q25 of each section = 50	Q1–Q50 of each section = 100
Marks per question	2	1
Section weighting	50 : 50	50 : 50
Maximum marks	100	100
Format	MCQ, four options	MCQ, four options
Duration	1 hour	2 hours

	ASOC (Restricted)	ASOC (General)
Morse	None	8 wpm, sending and receiving
Language	English	English (plus Morse)

The "four options" detail is **[REPORTED]** — it appears in the Indian Institute of Hams' reproduction of the WPC FAQ ("Objective questions (with four choice)") and is stated flatly by NIAR and by the pre-2020 WPC application form.

Negative marking: no source, official or secondary, says anything either way. **[UNCERTAIN]**

One secondary source to disregard: hamradio-arcad.com (post dated 2020-08-13) says "There are 100 questions ... Each question carries 1-mark each. The duration of the written test is of one hour." That combines the General question count with the Restricted duration and is almost certainly an error. **[REPORTED — prefer the official FAQ.]**

3.7.11 THE PART A EXEMPTION — A LARGE SHORTCUT FOR AN ENGINEER

Official FAQ Q12, verbatim **[VERIFIED]**:

"Candidates having the following Qualifications are exempted from appearing in Part A of the Examination (i) B. Tech/BE Degree in Electronics or Telecommunication (ii) B.Sc/M.Sc with Electronics or Telecommunication (iii) Diploma in Engineering with Electronics or Telecommunication"

Annexure II words it more broadly: "degree in Engineering/Science or Diploma in Engineering and having studied electronics or telecommunications". **[REPORTED — DoT document, ARSI-hosted scan]** The portal implements it as a Yes/No radio button plus a "Reason for Exemption" dropdown and a proof upload, and the admit card prints "Exempted from part A: Yes/No".

Two cautions.

- **The arithmetic of the exemption is unclear.** If Part A is dropped, does the section-wise minimum apply only to Part B, and is the total then out of 50 rather than 100? No source says. **[UNCERTAIN — put this to WPC in writing before relying on the exemption.]**
- Whether the exemption survives into any post-2024 syllabus is **[UNCERTAIN]**; the exemption is in the 2010 Annexure II and in the currently published Saral Sanchar FAQ, both of which predate the 2024 Rules.

3.7.12 PASS MARKS — A GENUINE CONFLICT BETWEEN OFFICIAL DOCUMENTS

Four official or quasi-official statements exist:

#	Source	Date	Restricted	General
A	WPC OM L-14011/255/2004-AMT, Annexure II [REPORTED — DoT document, ARSI-hosted scan; the scan was re-read page by page on 2026-08-23 and the quoted words are exact]	2010-08-13	"at least 40 % in each section and 50 % in aggregate"	"at least 50 % in each section and 60 % in aggregate"
B	Saral Sanchar official FAQ, Q6 [VERIFIED — live on saralsanchar.gov.in]	PDF created 2020-08-24	40 % / 50 %	50 % / 60 %, plus Morse
C	Telecommunications (Amateur Services) Rules, 2024, Rule 5(5) [VERIFIED]	2024-10-29	"a minimum of forty per cent of the total marks "	same
D	DoT WPC FAQ No. P-14036/02/2023-COP, Q8 [REPORTED — DoT document, ARSI-hosted scan; both pages were re-read on 2026-08-23 and the table below is a faithful reproduction]	2024-12-05	40 % in each part, 40 % overall	40 % in each part, 40 % overall , plus 8 wpm Morse

Statement D, verbatim, as a table in the FAQ:

ASO Certificate Category	Written Examination	Total Pass Percentage	Morse Proficiency Requirements
General ASO Certificate	Minimum 40% in both Part A and Part B of the written exam	40% overall	Minimum 8 words per minute in both transmission and reception
Restricted ASO Certificate	Minimum 40% in both Part A and Part B of the written exam	40% overall	Not required

Signed "(Rahul Sondhara) Engineer(COP)", addressed to "Sr. DWA (IT&SACFA), WPC Wing (with request to arrange to upload on DOT website and Saral Sanchar Portal of DOT)."

Which governs. Statements A and B describe the pre-2024 regime and are mutually consistent; B is still live on a government portal, but its PDF was created four years before the 2024 Rules and is stale rather than authoritative. **Statement C is the binding law** — a floor of 40 per cent of total marks, with no section-wise minimum. **Statement D is the Department's own interpretation of C**, issued five weeks after the Rules came into force; it applies C's 40 per cent to the aggregate for both grades and adds a 40 per cent per-part minimum, which C does not require but which the Department may specify under Rule 5(3).

Best reading as at 2026-08: 40 per cent in Part A, 40 per cent in Part B, and 40 per cent overall, for both grades, plus 8 wpm Morse for General. This remains [UNCERTAIN] only because the older, stricter FAQ is still published on the Government's own portal and a candidate cannot rule out an examiner applying it.

Practical advice: aim for 60 per cent or better in each part. That clears every version of the rule, and the extra preparation is trivial.

Disregard one further secondary figure: several sources give General as "50 % per section, 55 % overall". The primary 2010 OM says 60 per cent aggregate. **[REPORTED — prefer the OM.]**

3.7.13 MORSE CODE — STILL TESTED, CONTRARY TO COMMON BELIEF

Morse has not been abolished in India. The ITU removed the international Morse requirement for high-frequency amateur access at the 2003 World Radiocommunication Conference, and many administrations dropped it entirely — the United Kingdom in 2003, the United States in 2007. India did not. It reduced the speed and confined the test to one grade, and re-enacted the requirement in 2024.

- **The statute.** Rule 5(4): examination for ASOC (General) "shall also require such proficiency in Morse Code as may be specified by the Central Government". **[VERIFIED]**
- **The speed.** The DoT WPC FAQ of 2024-12-05 sets it at "Minimum 8 words per minute in both transmission and reception" for General, and "Not required" for Restricted. **[REPORTED — DoT document, ARSI-hosted scan]**
- Annexure II states flatly: "There will be **no Morse test for Restricted Grade.**" **[REPORTED — DoT document, ARSI-hosted scan]**

History, for context:

Period	Grades	Morse
1978 Rules as framed	Advanced, Grade I, Grade II, Grade II (Restricted), Short Wave Listener	Advanced and Grade I: 12 wpm; Grade II: 5 wpm; Restricted: none [REPORTED]
Amendment Rules 2005 and 2009, effective 2010-04-01	General, Restricted	General 8 wpm ; Restricted none [REPORTED — DoT document, ARSI-hosted scan]
Telecommunications (Amateur Services) Rules, 2024, in force 2024-10-30	ASOC (General), ASOC (Restricted)	General 8 wpm per the 2024-12-05 FAQ; Restricted none [VERIFIED] + [REPORTED]

The test procedure, verbatim from Annexure II **[REPORTED — DoT document, ARSI-hosted scan]**:

"Morse receiving: (Speed: 8 words per minute) The test piece will consist of a plain language passage of 200 characters which may comprise of letters, figures. Test piece may also contain the following punctuations ie full stop; comma; semi colon; break-sign; hyphen and question mark. The average words shall contain five characters and each figure and punctuation will be counted as two characters. The test will be for five consecutive minutes at a speed of 8 words per minute. A short practice piece of one minute shall be sent at the prescribed speed before the start of the actual test. **Candidates will not be allowed more than one attempt in Morse reception and sending test.** The test may be written in ink or pencil but must be legible. Over-writing will be treated as error. ...

More than 5 errors will disqualify a candidate. However if a candidate receives without any error in any part of the passage continuously for one minute duration will be declared successful in the Morse reception test.

Morse Sending (Speed: 8 words per minute) ... Candidates are required to send for five consecutive minutes at a speed not less than 8 words per minute ...

Note- A candidate shall have to pass both receiving and sending simultaneously."

Four operational points: **one attempt per sitting**; the error rule is generous, because a single clean error-free minute anywhere in the passage passes you outright; **you must pass sending and receiving in the same sitting** and cannot bank one; and overwriting counts as an error, so handwriting matters.

Whether the 8 wpm test is currently being administered at every centre, and whether any change is proposed, is **[UNCERTAIN]**.

3.7.14 RESULTS, FAILURE AND RE-EXAMINATION

- **Results.** Rule 5(3) requires the Government to publish "the date of announcement of the results". **[VERIFIED]** In practice the result appears on your portal dashboard: "After examination, examiner will update the exam result and that can be seen on the same tab. If the applicant has passed, then he can apply to the ASO Certificate. If he is failed, he has to reapply for the exam." **[REPORTED — DoT document, ARSI-hosted scan]** **No published turnaround time exists. [UNCERTAIN]**
- **If you fail, you simply apply again** — a fresh application on the portal with a fresh ₹100 fee. Rule 5(2) attaches the fee to every application; there is no re-sit concession. **[VERIFIED by implication from Rule 5(2)]**
- **No limit on attempts** appears anywhere in the 2024 Rules. **[VERIFIED — absence of any such provision in the full text]**
- **Document verification is one-time**, per OM paragraph 2.2, so a repeat candidate whose originals were already checked should not need to repeat that step. **[REPORTED — DoT document, ARSI-hosted scan]** Whether the portal short-cuts the uploads for a repeat candidate is **[UNCERTAIN]**.
- **Partial passes are not banked between sittings**, other than the General/Restricted split within a single sitting described in §3.7.6. Nothing suggests a passed Part A or a passed Morse test carries forward to a later attempt. **[UNCERTAIN]**
- **The "one-month waiting period" claim is not established.** Several search summaries assert that a failed candidate must wait a month before re-appearing. It is not in the 2024 Rules, the 1978 Rules, the Saral Sanchar FAQ, the 2024-12-05 FAQ, or either DoT user manual. It appears to be a garbled restatement of Rule 5(1)'s one-month application lead time. **[UNCERTAIN — not established.]** In practice the binding constraint is when your monitoring station next has enough candidates, which is likely to be a month or more anyway.

- **Malpractice.** The 1978 Rules contained an express power to debar a candidate for impersonation, fabricated documents or unfair means. **There is no equivalent provision in the 2024 Rules.** They deal with false information at the certificate stage instead: Rule 9(c) permits suspension or cancellation where the holder "wilfully furnished incorrect or false information to the Central Government", after a hearing. **[VERIFIED]** Whether WPC retains an administrative debarment power for exam malpractice is **[UNCERTAIN]**.

3.8 After the exam

3.8.1 POLICE VERIFICATION — THE HONEST POSITION

Dozens of pages, including club pages refreshed in 2025, say that after passing you must clear "Local Police and Central Home Department verification" and that this takes about twelve months. **[REPORTED — e.g. <https://www.nbars.in/asoc> ; Wikipedia]** Wikipedia footnotes that sentence to a **2007 Times of India article**. It is nineteen-year-old journalism describing a paper-era process.

What the Government actually decided, verbatim **[VERIFIED]** — Department of Telecommunications, WPC Wing, **Office Memorandum No. R-11014/03/2015-PP dated 2016-03-10**, subject "Security clearance in respect of (i) Wireless Telegraph license (ii) Radio Amateur license", signed D. Jha, Senior Deputy Wireless Adviser:

"The security clearance for the different categories of wireless licenses was used to be taken from the Ministry of Home Affairs. The matter was reviewed in the Ministry of Home Affairs and it has been decided to discontinue security clearance by the IB/MHA for the following cases:

- a) Application of firm/individual for grant of Wireless Telegraph license in non-disturbed areas. However, security verification in respect of the applicants located in J&K, North-Eastern States and LWE affected areas will continue to be referred to the intelligence Bureau. A list of LWE affected areas is enclosed at Annexure-I.
- b) Application of Indian nationals/foreigners for grant of Radio Amateur license.
 1. The above orders will be applicable with immediate effect including the cases pending with the Intelligence Bureau/MHA."

("IB" is the Intelligence Bureau, "MHA" the Ministry of Home Affairs, "LWE" left-wing extremism; Annexure-I lists 106 LWE-affected districts.)

Read carefully: the geographic carve-back for Jammu and Kashmir, the North-Eastern States and LWE districts is attached to clause **(a)**, ordinary wireless telegraph licences. Clause **(b)**, radio amateur licences, carries **no geographic qualification at all**. On the face of the document, amateur applicants anywhere in India are outside the clearance loop. **[VERIFIED as to the text]**

Corroboration by silence **[VERIFIED]**: the 2024 Rules contain no verification, vetting or clearance provision; the December 2024 FAQ does not mention one; the February 2025 OM and the March 2025 user manual describe every screen and contain no verification step; and DoT's ASOC service page lists only "Photo Id card; Application through saral sanchar portal".

What replaced it: self-certification, an originals check on exam day, a signed declaration ("I hereby certify that there are no criminal proceedings pending against me and I have not been convicted by any court of law ... I am aware that it is a criminal offence to furnish any false information"), and a revocation warning printed on the certificate itself.

Honest caveat. Do not read this as a guarantee that no enquiry will ever occur — the Government can always make one — but do not build your timeline around a verification step that the current documents do not contain. Whether any state police force still runs an informal enquiry in Jammu and Kashmir, the North-East or LWE districts for *amateur* applicants is **[UNCERTAIN]**; if you live in one of those areas, ask WPC.

3.8.2 CLAIMING THE CERTIFICATE

After the result appears **[REPORTED — DoT document, ARSI-hosted scan; corroborated by NIAR's [License.pdf](#), 2025-03-01]**:

1. **Apply for Certificates → Amateur (Apply Now) → "Exam conducted through this portal" → Get Data → "Apply amateur License" → Save Draft.** Your earlier data is pulled forward and an application ID is generated.
2. **Step 1: Applicant Information** — pre-filled.
3. **Step 2: Call Sign** — see §3.8.3.
4. **Step 3: Payment Details** — choose 20-year or lifetime validity; the portal computes and displays the exact "Valid till" date.
5. **Step 4: Declaration**, then Validate and Submit.
6. **Pay ₹1,000 or ₹2,000** through the Saral Sanchar-integrated Bharatkosh page.
7. **Download the PDF, sign it, upload it** ("Upload Signed Application and Final Submit").
8. "Once the applicant validates and submits the form successfully, Certificate will be uploaded by WPC Nodal User." Download it from **Apply for Certificates → Amateur → Current Application → Action → Download Certificate.**

What arrives: a single PDF titled "Amateur Radio Station Operator Certificate", QR-code authenticated, digitally signed by the Assistant Wireless Adviser (COP), showing name, parent/spouse name, date of birth, address, category, call sign, certificate number, "Valid Upto", and the **location of the station with latitude and longitude**. It includes a tear-off pocket card.

[REPORTED — NIAR sample of a migrated certificate dated 2025-12-02, <https://qsl.net/vu2jos/Extension.pdf>]

A naming inconsistency to expect: NIAR's March 2025 guide still reproduces an older "Amateur Wireless Station License" template citing the 1978 Rules, while the December 2025 sample shows the new form. **[UNCERTAIN]** which template a brand-new applicant receives today; the substance is the same either way.

3.8.3 CALL SIGN ALLOCATION

You choose your own suffix, subject to availability. OM 2025-02-28 ¶2.3: "The Call Sign, as selected by the applicant on the portal, will be system-generated - based on availability."

[REPORTED — DoT document, ARSI-hosted scan] The DoT user manual of 2025-03-01 explains the mechanism:

"Applicant can choose from the available list of call signs, available call sign are auto list as per entered character. After the submit Call sign is locked for 24 hours or till the fee payment done whichever is earlier."

The screen shows a fixed **VU** prefix box, a grade-digit box (2 for General, 3 for Restricted) and a free box for the remainder. **You type characters, the portal lists what is free, you pick one, and you have 24 hours to pay before it is released. Pay promptly if you care about the suffix.** The older advice to "give three choices and hope" describes the pre-2025 module and is superseded. **[VERIFIED as to the manual's text]**

Series	Who / what	Status
VU2 + 2 or 3 letters	General grade (formerly Advanced Grade and Grade I)	Legacy; valid; largely exhausted
VU3 + 3 letters	Restricted grade (formerly Grade II and Grade II Restricted)	Legacy; valid
VU21–VU29 + 2 letters	General grade, new issues from 2025-06-25	Current
VU31–VU39 + 2 letters	Restricted grade, new issues from 2025-06-25	Current
VU4	Andaman and Nicobar Islands	Expedition / special
VU7	Lakshadweep	Expedition / special
AT..., AU...	Special event call signs	Temporary, 90 days

The 2025-06-25 change is **[REPORTED — NIAR, https://qsl.net/vu2jos/New_Bio.pdf and [New_Calls.htm](#) ; independently reported by Amateur Radio Newsline, 2025-07-31, quoting Jose Jacob VU2JOS]. No WPC circular announcing the new blocks could be located. [UNCERTAIN]** Existing VU2 and VU3 call signs remain valid.

Two practical notes. One source describes the new Restricted format as two digits followed by **three** letters; NIAR's list of actually-issued call signs (VU32AB, VU33AC, VU33EE, VU33NG, VU35RJ, VU37JD and so on) is uniformly **two** letters, and NIAR is the better source because it is compiled from real issues. **[REPORTED — conflict resolved in NIAR's favour]** And NIAR reports that "Echolink and LOTW are having problems to accommodate these new prefixes" — if Logbook of the World or EchoLink matter to you, expect to raise a support ticket. **[REPORTED]**

Once issued, the call sign carries an obligation. Annexure A ¶9(1)(b) **[VERIFIED]**:

"The call sign endorsed in the certificate shall be sent for identification **at the beginning and at the end of each period of transmission. When the period of transmission exceeds ten minutes the call sign shall be repeated.** ASOC holder shall not make transmission without identification or with false identification."

3.8.4 THE TWO-YEAR CLIFF

Rule 6(2), verbatim **[VERIFIED]**:

"(2) On successful completion of the examination by the applicant, payment for the grant of ASOC of either category shall be made **within a period of two years from the date of declaration of results** of the examination under rule 5, failing which no certificate shall be granted to such applicant: Provided that such applicant, who becomes ineligible for grant of ASOC may apply for reappearing in the examination referred to in rule 5 for grant of the said certificate."

Pass the exam, then let two years lapse without paying, and you sit the exam again. There is no fee-free grace period and no extension provision.

3.8.5 VALIDITY, AND WHETHER "LIFETIME" IS WORTH ₹1,000 MORE

Two options only, chosen when you pay: **20 years** or **lifetime**, where lifetime is legally capped at your 80th birthday (Rule 6(1) Explanation). **[VERIFIED]**

How the portal computes it, verbatim from the DoT user manual of 2025-03-01 **[VERIFIED]**:

"Validity will be displayed according to the birth date of the applicant and certificate issuing date. If 20 years is selected, the validity will be 20 years from the issuing date or 80th birthday of the applicant, whichever is smaller. If lifetime is selected, the validity will be 80 years from the issuing date or 80th birthday of the applicant, whichever is smaller."

Arithmetic (analysis, not law): lifetime costs ₹2,000 against ₹1,000 for 20 years. It is worth it if you are more than 20 years short of 80 — that is, if you are under 60 — because it removes a renewal cycle. At 61 or older the two options converge on your 80th birthday and lifetime is strictly worse value.

After 80 the certificate does not die. Rule 8(2): "An ASOC with lifetime validity may be extended for **ten years at a time, without any additional fees**, upon specific request made by the ASOC holder." **[VERIFIED]** Note the wording is confined to lifetime certificates; whether a 20-year ASOC that runs past 80 gets the free extension is **[UNCERTAIN]**.

3.8.6 TIMELINES — WHAT TO ACTUALLY EXPECT

There is no published service-level standard. Neither the Rules, the office memoranda, the DoT eServices service page, nor any citizen's charter commits to a number. **[UNCERTAIN]**

Claim	Source	Assessment
"Up to 12 months" / "on average 12 months"	Wikipedia, NBARS, VU22DX, several club pages	[REPORTED] Traces to a 2007 newspaper article describing the pre-2016, pre-portal, MHA-clearance process. Disregard.
52 days from exam day to licence in hand	NIAR biography of S. Gopinath VU33GN: "appeared for the Ham exam on 1 August 2025 and got the license on 22 Sept 2025"	[REPORTED] The one clean, dated, recent data point found.
"The whole process takes several days / weeks"	NIAR migration guide	[REPORTED] Migration is a simpler workflow; a lower bound, not comparable.

Where the delay actually sits:

1. **Waiting for an exam slot — the dominant and least predictable component.** The examination runs "as and when there are sufficient number of candidates". Weeks in Bengaluru, Hyderabad, Mumbai or Delhi; potentially months in a thin region. This is not part of "after the exam" but it dominates the real-world wait.
2. **Application scrutiny before the exam** — days to a few weeks. **[UNCERTAIN as to duration]**
3. **Result declaration** — no published interval. **[UNCERTAIN]**
4. **Certificate application, payment and issuance** — the VU33GN data point plus the absence of any verification step suggest weeks, not months.

Planning estimate, labelled as an estimate and not a commitment: budget roughly **two to four months from passing the exam to holding the certificate**, and treat the wait for an exam slot as a separate and larger unknown. Realistically, **three to six months from deciding to apply to holding a call sign** is a fair expectation in a city with a monitoring station, longer elsewhere.

3.8.7 RENEWAL

Rule 8, verbatim **[VERIFIED]**:

"8. Renewal of Amateur Station Operator Certificate. – (1) An ASOC holder may apply for renewal of his ASOC by making an application in the form specified for this purpose in the portal, **at least twelve months prior to the date of expiry** of such certificate, along with payment of fees as specified in sub-rule (1) of rule 6:

Provided that any application for renewal after the expiry of such time period and **up to two years after the date of such expiry**, may be made upon payment of **late fees of one thousand rupees**:

Provided further that **no application for renewal shall be considered if received after two years from the date of expiry** of the ASOC.

(2) An ASOC with lifetime validity may be extended for ten years at a time, without any additional fees, upon specific request made by the ASOC holder in the form specified for this purpose."

A real drafting ambiguity. The first proviso applies the late fee to an application made "after the expiry of **such time period**". That could mean (a) the twelve-months-before-expiry deadline — in which case anyone applying with eleven months to go owes ₹1,000 — or (b) the validity of the certificate itself. NIAR reads it as (b), and (b) is the only reading that makes the "up to two years after the date of such expiry" clause coherent. But the text is ambiguous. **[UNCERTAIN — apply at least twelve months early and avoid the question entirely.]**

What changed in 2024: the late-renewal window shrank from **five years to two**, and the surcharge changed from **₹100 per half-year to a flat ₹1,000**. **[VERIFIED]** And **the 40-contacts-a-year renewal condition is gone** — Rule 14(1)(b) of the 2009 Rules required the licensee to certify contacts "on at least 40 occasions per year"; no equivalent appears in the 2024 Rules. **[VERIFIED]**

Missing the two-year window has consequences the Rules do not spell out. Combining Rule 6(2)'s proviso with Rule 3(1)'s prohibition, the practical result is that you would have to sit the examination again. **[UNCERTAIN — inference, not stated]** There is also a hard equipment consequence: under the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026 (G.S.R. 592(E), 2026-07-08), Rule 10(f) exempts an ASOC holder from needing a possession authorisation, and Rule 12 requires a person who "ceases to be such person" to dispose of the radio equipment **"Within two months of such cessation."** Letting the ASOC lapse starts a two-month disposal clock on the radio itself.

Fact-check downgrade, 2026-08-23. The 2026 REPA Rules could not be re-fetched from any gov.in host during the adversarial fact-check (DoT static-upload paths return HTTP 403, the egazette portal presents a broken TLS chain, and the rules are not indexed by the bare-act aggregators). What could be re-read is the **February 2025 draft** of the same rules. In the draft, the exemption provision is **rule 4, not rule 10**, and it does **not** mention ASOC holders: it exempts equipment exempted under s.3(3)/(4) of the Act, possession by an entity holding an authorisation under **s.3(1)(a) or (b)** together with a spectrum assignment or a surviving Telegraph Act licence, and equipment notified as required by a user of a telecommunication service. An ASOC is granted under **s.47**, and an amateur holds no spectrum assignment, so on the draft text an ASOC holder would not have been exempt. Either the final rules added an express amateur gateway — which is what this report's earlier retrieval records at rule 10(f) — or they did not. **The rule 10(f) exemption and the rule 12 two-month disposal clock are therefore both downgraded to [UNCERTAIN].** Ask WPC in writing,

alongside the power table question in §3.4.5, whether an ASOC holder requires a Radio Equipment Possession Authorisation. The **import** exemption is unaffected: it comes from rule 7(3) of the Amateur Services Rules 2024, whose gazette text was read directly and is **[VERIFIED]**.

3.8.8 COMPLETE FEE SCHEDULE UNDER THE 2024 RULES

#	Item	Fee	Rule	Tag
1	Examination, per grade (₹200 for both)	₹100	5(2)	[VERIFIED]
2	ASOC, 20-year validity	₹1,000	6(1)	[VERIFIED]
3	ASOC, lifetime (to age 80)	₹2,000	6(1)	[VERIFIED]
4	Renewal, 20 years	₹1,000	8(1) with 6(1)	[VERIFIED]
5	Renewal, lifetime	₹2,000	8(1) with 6(1)	[VERIFIED]
6	Extension of a lifetime ASOC past age 80	Nil , 10 years at a time	8(2)	[VERIFIED]
7	Late-renewal surcharge (up to 2 years after expiry)	₹1,000	8(1), first proviso	[VERIFIED]
8	Renewal more than 2 years after expiry	Not possible	8(1), second proviso	[VERIFIED]
9	Permanent change of station location	₹1,000	7(6)	[VERIFIED]
10	Duplicate ASOC	₹1,000	7(6)	[VERIFIED]
11	Special (event) call sign	₹200	13(1)	[VERIFIED]
12	Amateur society (club station), minimum 4 ASOC holders	₹2,000	12(1)	[VERIFIED]
13	Reciprocal ASOC for a foreign national	₹1,000	4(2)	[VERIFIED]
14	SACFA clearance (only if the antenna is on a tower, building or mast)	₹1,000	not in the Amateur Rules	[REPORTED — NIAR fee sheet and SACFA guide]
15	Refund on suspension, cancellation or surrender	None	10, 11(2)	[VERIFIED]

Obsolete fee figures still circulating. Change of location was ₹200 and a duplicate ₹100 under the 2010 schedule; both are now **₹1,000**. Late renewal was ₹100 per half-year; it is now a flat ₹1,000. There was also a ₹200 "mobile endorsement"; the 2024 Rules have no such concept. The WPC-authored FAQ redistributed by several clubs, and the "Beginner's Corner" page of the club

that styles itself TARS, still print the old figures. **[UNCERTAIN — the expansion of "TARS" could not be confirmed from the club's own pages.] Where a FAQ and the Rules conflict, the Rules win. [VERIFIED]**

The 2024 Rules did not raise the core fees. ₹100 exam, ₹1,000 for 20 years and ₹2,000 for lifetime are the same figures as under the previous regime. What changed is the minimum age (18 → 12), the supersession of the 1978 Rules, the disappearance of the published band and power annexure, and the digital-by-design portal obligation. **[VERIFIED]**

Every fee in the table above was re-verified on 2026-08-23 against the gazette text of G.S.R. 675(E) (Gazette of India Extraordinary Part II §3(i) No. 617, 30 October 2024, notification dated 29 October 2024), extracted and read rule by rule: rule 4(2) ₹1,000 reciprocal; rule 5(2) ₹100 examination; rule 6(1) ₹1,000 for twenty years and ₹2,000 for lifetime, "lifetime" defined by the Explanation as "till the ASOC holder attains the age of eighty years"; rule 7(6) ₹1,000 for a duplicate or a change of station location; rule 8(1) ₹1,000 late fee, with the second proviso barring any renewal application received more than two years after expiry; rule 8(2) lifetime extension of ten years at a time "without any additional fees"; rule 12(1) ₹2,000 for an amateur society of at least four ASOC holders; rule 13(1) ₹200 for a special event call sign. **No discrepancy was found.** The ₹200-for-both-grades examination fee is not in the Rules — it comes from Q3 of the official Saral Sanchar HAM FAQ, which was also re-read on 2026-08-23 and does state a three-row table of Restricted ₹100 / General ₹100 / "General & Restricted" ₹200. That FAQ predates the 2024 Rules (it still speaks of an "Amateur Wireless Telegraph Station Licence" and gives the superseded ₹200 change-of-location and ₹100 duplicate figures), so the ₹200 concession is **[VERIFIED]** as published but **[UNCERTAIN]** as to whether it survives rule 5(2)'s per-application wording. Budget ₹200 and be prepared to be told ₹100 per application.

One further re-verification. The obsolete-fee paragraph above is confirmed directly from page 1 of the 2010 OM, which tabulates the pre-2010 and post-2010 figures side by side: examination fee "Rs.100/- for each grade", licence fee "Rs.1000/- for 20 years and Rs. 2000/- for lifetime for each grade", change of location Rs.200, mobile endorsement Rs.200, duplicate licence Rs.100, and late renewal "Rs.100/- for every half year or part thereof".

3.8.9 POST-ISSUE OBLIGATIONS AND THE PRACTICAL OPERATING ENVELOPE

Topic	Position	Tag
Operating away from home	Permitted for up to 90 days at a location other than the certificate address, outside government-restricted areas, for amateur activities only, with the equipment restored afterwards. Beyond 90 days at one place, file a change of location and pay ₹1,000.	[REPORTED — WPC FAQ 2024-12-05, Q3 and Q4, DoT document, ARSI-hosted scan]
Carrying the radio while travelling	Not prohibited, except in government-restricted areas.	[REPORTED — same FAQ, Q5]

Topic	Position	Tag
Standing policy	"Amateur Radio Stations [new as well as existing] are allowed to operate anywhere in India except those locations restricted by Government of India from time to time."	[VERIFIED — WPC OM No. P-11015/01/2019-COP dated 2019-06-07]
SACFA clearance	Required only "if the amateur station antenna is installed on a tower, building, or mast". A hand-held with its stock antenna does not trigger it; a rooftop mast does. Applied for online.	[REPORTED — WPC FAQ 2024-12-05, Q6 and Q7]
Station log	Mandatory, in a bound book (not loose-leaf) : date and time of each transmission, summary, experiments, call signs worked, opening and closing times, and for portable or mobile operation "the particulars of temporary location". Times in Indian Standard Time; no gaps; entries made at the time; preserved one year from the last entry. The Rules do not authorise an electronic log.	[VERIFIED — Annexure A ¶6]; whether WPC accepts electronic logs in practice is [UNCERTAIN]
Equipment register	Make, model and type, serial number, source (or "self-made"), date of receipt or assembly, seller's receipt and certificate numbers, details of any sale or transfer.	[VERIFIED — Annexure A ¶7(4)]
Frequency-measuring equipment	The holder "shall have at the authorised amateur station, a reliable frequency measuring equipment". An anachronism for a synthesised hand-held, but still in the Rules.	[VERIFIED — Annexure A ¶4(2)]; enforcement [UNCERTAIN]
Identification and procedure	Call sign at the start and end of each transmission, repeated every ten minutes; calls limited to three repetitions; test signals not more than 30 seconds; no unmodulated carrier.	[VERIFIED — Annexure A ¶9]
Secrecy	A message you are not entitled to receive must not be disclosed to anyone other than an authorised officer or a court.	[VERIFIED — Annexure A ¶8]
Inspection	Produce the certificate, the log, other records and the equipment itself to an authorised officer.	[VERIFIED — Rule 14, Annexure A ¶10]
Import	"An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment." Whether this extends to a wideband consumer hand-held covering 136–174 and 400–470 MHz is [UNCERTAIN].	[VERIFIED — Rule 7(3)]

Topic	Position	Tag
Possession	ASOC holders are exempt from the Radio Equipment Possession Authorisation regime.	[UNCERTAIN] — recorded as Rule 10(f) of the REPA Rules 2026 on the original retrieval, but the final rules could not be re-fetched on 2026-08-23 and the February 2025 draft contained no amateur gateway. See §3.8.7. Confirm with WPC
Non-transferable; surrender	The ASOC cannot be transferred; it may be surrendered at any time with no refund.	[VERIFIED — Rule 11]
Suspension or cancellation	For breach of the ASOC terms, of the radio-equipment rules or the Act, of the ITU Convention or Radio Regulations, or for wilfully furnishing false information — after "a reasonable opportunity of making a representation and being heard". No refund.	[VERIFIED — Rules 9 and 10]
Club stations	A group of minimum four ASOC holders , fee ₹2,000, one named custodian; valid 20 years or co-terminus with the custodian's ASOC; changes notified within 30 days; lapses if fewer than four original applicants remain.	[VERIFIED — Rule 12]
Special event call signs	₹200, applied for at least 90 days before the event, valid 90 days from commencement. In practice applied for by email to awacop-dot@gov.in , not through the portal, with a SACFA clearance copy enclosed and a written report to WPC within a week of the event.	[VERIFIED — Rule 13]; process [REPORTED — NIAR, 2025-09-17]
Ships	An ASOC holder may be permitted to operate from an Indian-registered ship on specific request through the portal.	[VERIFIED — Rule 7(8)]
Duplicate after loss	₹1,000. NIAR's 2020 guidance required a scanned police First Information Report for a lost licence; the post-2024 guidance is silent.	[VERIFIED as to fee]; FIR requirement [UNCERTAIN]

Migration of old paper certificates does not apply to a new applicant, but you will encounter it. Holders of certificates issued manually before the Saral Sanchar module went live must migrate on the portal, free of charge, to receive a QR-authenticated certificate. The deadline was first 2025-09-15, extended to 2025-12-15 by OM dated 2025-10-09, and then to **2026-06-15** by OM No. L-14011/01/2023-COP dated 2026-01-05. **[VERIFIED]** That date has now passed and no further extension was found. **[UNCERTAIN]** Note that r04 records only the original 2025-09-15 deadline; r06 located both extension memoranda and is the better source here.

3.9 Step-by-step action plan with realistic timelines

Timings are estimates and are labelled as such. Only the statutory deadlines (marked "statutory") are law.

Step	What you do	Realistic elapsed time	Cost
0	Decide the grade. Restricted if the 10 W DC-input figure is what you need and you want the shortest path. General if you want 10 W of RF output, HF power, satellite access or the 5 cm band (5725–5840 MHz) — and are willing to learn Morse at 8 wpm.	—	—
1	Confirm the power table with WPC in writing (§3.4.5). Do this before buying any radio.	1–4 weeks for a reply; do not block on it	₹0
2	Assemble documents (§3.10) and prepare the photograph and signature files. Obtain the attested nationality certificate now if you have no passport — this is the slowest single document.	1–2 weeks	₹0
3	Study. Independent estimates run from "a few weeks" (NIAR) to "four to eight weeks of steady preparation" (NBARS) to a three-month Sunday course (IACDM/West Bengal Radio Club). If you hold a qualifying electronics degree or diploma, claim the Part A exemption and study only Radio Regulations. [REPORTED] For General, start Morse practice immediately: 8 wpm typically takes 6–10 weeks of daily practice from zero.	4–8 weeks, in parallel with steps 4–6	₹0 on the free study route
4	Register on the portal as "Individual (WPC Exams/Certificates)". Have your PAN, mobile and email ready.	Same day	₹0
5	File the exam application — five steps, then Save Draft, Validate, Submit. Grade and exam centre lock at submission. Consider entering both grades if aiming for General.	1–2 hours	—
6	Pay the exam fee via the integrated Bharatkosh flow, then download, sign, scan and upload the signed application PDF. Nothing reaches WPC until you do.	Same day	₹100 (or ₹200 for both grades)
7	Scrutiny. Watch the dashboard; if reverted, only the flagged serial numbers are editable.	Days to a few weeks [UNCERTAIN]	₹0
8	Admit card issued with venue, date and time; SMS notification; download from the same tab. Rule 5(1) contemplates applying one month before the exam (statutory), but in practice the wait is until the station has enough candidates.	Weeks to months — the dominant unknown	₹0

Step	What you do	Realistic elapsed time	Cost
9	Sit the exam. Report 30 minutes early. Carry the printed admit card, an original photo identity card, and originals of everything you self-certified. No phone, no calculator, no watch. Your photograph may be captured at the centre.	1 hour (Restricted) or 2 hours + Morse (General)	₹0
10	Result appears on the dashboard. No published turnaround. [UNCERTAIN]	Weeks [UNCERTAIN]	₹0
11	Apply for the certificate , pick your call sign from the live availability list, choose 20-year or lifetime validity. The call sign locks for 24 hours or until you pay.	1 hour	—
12	Pay the certificate fee (statutory deadline: within two years of the result), then download, sign and upload the signed PDF.	Same day	₹1,000 or ₹2,000
13	Certificate issued — QR-authenticated PDF with your call sign and station location. One documented 2025 case ran 52 days from exam day to licence in hand. [REPORTED]	4–10 weeks after the exam (estimate)	₹0
14	Before transmitting: start the bound-book log and the equipment register; check that your radio can be constrained to 144–146 MHz and 434–438 MHz; apply for SACFA clearance only if you are putting the antenna on a tower, building or mast (₹1,000).	—	₹0 or ₹1,000
15	Renewal (statutory): apply at least twelve months before expiry. ₹1,000 late fee up to two years after expiry; nothing is renewable after two years.	—	₹1,000 / ₹2,000

Total statutory outlay for a Restricted ASOC: ₹1,100 (₹100 exam + ₹1,000 for 20 years) **or ₹2,100** for lifetime validity. **[VERIFIED]** Free study material exists, so ₹1,100 can be the genuine all-in government cost.

Total realistic elapsed time: three to six months from deciding to apply to holding a call sign, in a city with a Wireless Monitoring Station; longer in a thin region. The twelve-month figure that dominates search results is pre-2016 and should be disregarded.

3.10 Master document checklist

#	Document	Needed for	Format	Notes
1	Photograph	Exam application	Colour passport-style JPG, 300 × 300 px, under 150 kB	Specifications conflict (§3.7.7). A fresh photograph may also be captured at the centre.
2	Signature image	Exam application	JPG, 300 × 300 px, under 150 kB	White background preferred

#	Document	Needed for	Format	Notes
3	Date-of-birth proof	Exam application; original at the exam	Self-certified PDF under 1 MB	Birth certificate, SSC certificate or class 10 marks memo; must show father's name
4	Present address proof	Exam application; original at the exam	Self-certified PDF under 1 MB	Aadhaar, voter ID, driving licence, ration card, electricity bill, bank passbook, telephone bill or rent agreement
5	Permanent address proof	Exam application; original at the exam	Self-certified PDF under 1 MB	Same list
6	Photo identity card	Original, at the exam hall	Original, unexpired, undamaged	Aadhaar, driving licence, passport, PAN card or voter ID
7	Nationality: passport copy plus signed self-declaration	Exam application	Self-certified PDF	Proforma A, from the 2025-02-28 OM
7-alt	Nationality certificate, attested	Exam application, if you have no passport	PDF under 1 MB	Proforma B. Attester must be a gazetted officer, MP or State Legislator, or SDM/Tahsildar with magisterial powers. Name, designation and address must be legible.
8	Part A exemption proof	Exam application, if claiming	Self-certified PDF under 1 MB	B.Tech/BE, B.Sc./M.Sc. or diploma with electronics or telecommunications. Carry the original and the marks memo.
9	Station address proof	Exam application, step 4	PDF	Rent agreement or owner's consent, if the station is not at your present address
10	Station latitude and longitude	Exam application, step 4	Degrees, minutes, seconds	Ranges accepted: 8°4'–37°6' N, 68°7'–97°25' E
11	Parent or guardian undertaking	Exam application, if aged 12–18	Prescribed proforma	The current proforma could not be located on a gov.in URL [UNCERTAIN] — ask the monitoring station
12	PAN	Portal registration	—	A minor without one may use a parent's or guardian's

#	Document	Needed for	Format	Notes
13	Signed application PDF	Both the exam and the certificate stages	Printed, signed, scanned — or signature embedded	The most commonly missed step. Without it the application does not reach WPC.
14	Admit card	Exam hall	Printed on A4	Valid only if the photograph prints legibly

3.11 Common claims about the amateur route that are wrong

Common claim	Reality	Tag
"The 1978 Rules govern Indian amateur radio."	Superseded on 2024-10-30 by the Telecommunications (Amateur Services) Rules, 2024.	[VERIFIED]
"You need an ASOC <i>and</i> a separate station licence."	One document since 2024. The ASOC carries the call sign and the station location.	[VERIFIED]
"General grade needs 50 % per section and 60 % aggregate."	40 % per part and 40 % overall for both grades, per Rule 5(5) and WPC's own December 2024 FAQ — but a stale FAQ with the old figures is still live on saralsanchar.gov.in.	[VERIFIED] statute; [UNCERTAIN] which the examiner applies
"The minimum age is 18."	Flat minimum of 12 since 2024; no guardian certificate required by the Rules, though the portal requires an undertaking for 12–18.	[VERIFIED]
"Restricted grade is VHF-only at 10 W."	Restricted also gets ten HF bands at 50 W . The 10 W applies only to 6 m, 2 m and 70 cm.	[VERIFIED]
"Restricted grade gets 100 W."	50 W on HF, 10 W on VHF/UHF. The 100 W figure circulating in search results is wrong.	[VERIFIED]
"India still issues a Short Wave Listener licence."	Abolished on 2010-04-01. Receiving needs no licence today. ARSI's public page is out of date.	[VERIFIED]
"Morse was abolished in India."	It was not. Rule 5(4) retains it for General; the December 2024 FAQ fixes it at 8 wpm sending and receiving.	[VERIFIED] statute; [REPORTED] speed
"You must renew with proof of 40 contacts per year."	Dropped in 2024.	[VERIFIED]
"You must clear police / Home Ministry verification."	Discontinued for amateur applicants by WPC OM R-11014/03/2015-PP dated 2016-03-10, and absent from every official document issued since.	[VERIFIED]
"It takes about 12 months."	Traces to a 2007 newspaper article about the paper-era process. One documented 2025 case ran 52 days from exam to licence.	[REPORTED]

Common claim	Reality	Tag
"You submit three call sign choices and hope."	Since March 2025 the portal shows live availability; you pick, and the choice locks for 24 hours or until payment.	[VERIFIED]
"Indian hams are VU2 or VU3."	New issues since 2025-06-25 are VU21–VU29 (General) and VU31–VU39 (Restricted).	[REPORTED]
"All documents must be attested by a gazetted officer."	Abolished for the exam and certificate route on 2025-02-28; self-certification plus originals at the exam. Only the no-passport nationality certificate still needs attestation.	[REPORTED — DoT document, ARSI-hosted scan]
"Change of address costs ₹200 and a duplicate ₹100."	Both are ₹1,000 under Rule 7(6).	[VERIFIED]
"Late renewal is ₹100 per half-year and you have five years."	Flat ₹1,000 and a hard two-year cut-off.	[VERIFIED]
"You may only transmit from your registered address."	Up to 90 days elsewhere, outside restricted areas, with no permission and no fee. The old ₹200 mobile endorsement no longer exists.	[REPORTED — WPC FAQ 2024-12-05]
"SACFA clearance is needed for every ham station."	Only if the antenna is on a tower, building or mast.	[REPORTED — same FAQ]
"SSB works on 2 m in India."	VHF/UHF emissions are limited to F1B, F2B, F3E and F3C. No SSB, no AM.	[VERIFIED]
"A ham licence lets you talk to your family on a second handset."	No. Only other authorised amateur stations; third-party messages are banned.	[VERIFIED]
"10 W means 10 W of RF output."	Annexure-I specifies DC input to the final stage.	[VERIFIED]
"The 2024 Rules contain the band and power table."	They contain no frequencies and no wattages at all.	[VERIFIED]
"Lifetime means for life."	It means until you turn 80, then free ten-year extensions on request.	[VERIFIED]
"You apply for a ham licence on the new DoT Authorisation Portal."	No. ASOC remains on the Saral Sanchar Amateur/HAM module as at 2026-08-23.	[VERIFIED]
"wpc.dot.gov.in is the WPC website."	It is unreachable. WPC content now lives on dot.gov.in and eservices.dot.gov.in.	[VERIFIED as to unreachability]

3.12 Questions to put to WPC in writing, before you buy a radio

Address: **Assistant Wireless Adviser (COP), WPC Wing, Department of Telecommunications, 6th Floor, Sanchar Bhawan, 20 Ashoka Road, New Delhi – 110001.**
 Email **awacop-dot@gov.in**. Telephone **011-2303 6181 / 011-2303 6534 / 011-2303 6539**. DoT eServices nodal officer for ASOC: **Senior Deputy Wireless Advisor, 011-2303 6540, umang.srivastava@nic.in**. Saral Sanchar helpdesk: **011-2335 0020 / 011-2335 0025**, Monday to Friday 09:30–17:30. **[VERIFIED]** NIAR is the most reliable unofficial help desk: National Institute of Amateur Radio, Raj Bhavan Road, Hyderabad 500082, +91 40 2331 0287, **info@niar.org**. **[REPORTED]**

1. **The power table.** What frequency bands, emission classes and maximum transmit power are currently specified under Rule 7(1) for ASOC (Restricted) and ASOC (General)? Does Annexure-I to OM No. L-14011/255/2004-AMT dated 2010-08-13 remain in force after the supersession of the 1978 Rules? (§3.4.5)
2. **What the certificate itself says.** Does an issued ASOC print the authorised bands, emissions and power? If so, what does a Restricted certificate state?
3. **DC input versus RF output.** How is "DC input power to the final stage" measured or assessed for a modern solid-state hand-held? Is a radio rated at 5 W RF output compliant with the 10 W Restricted limit? (§3.4.4)
4. **Pass marks.** Which applies today — the 2024-12-05 FAQ (40 % per part, 40 % overall, both grades) or the FAQ still hosted at saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf (Restricted 40 %/50 %, General 50 %/60 %)? Will the stale FAQ be withdrawn? (§3.7.12)
5. **Part A exemption arithmetic.** If exempted from Part A, is the pass computed on Part B alone, and out of what maximum? (§3.7.11)
6. **Negative marking,** and the marks-per-question value for each grade. (§3.7.10)
7. **Re-examination.** Any minimum interval before re-applying after a failure? Any cap on attempts? Does the one-time document verification carry over? (§3.7.14)
8. **Combined application.** Can one application still cover both grades for ₹200, or must two be filed? (§3.7.6)
9. **The current syllabus.** What is the URL of the syllabus linked from the e-Admit Card? Has the 2010 Annexure II been reissued to cite the 2024 Rules and the current edition of the ITU Radio Regulations instead of the 1978 Rules and the 2008 edition?
10. **Exam calendar and results.** Will dates be published in advance under Rule 5(3), as they are for the aeronautical RTR(A) certificate? How long after the exam are results published?
11. **Digital voice.** Are DMR, D-STAR and C4FM within the emissions authorised on 144–146 MHz and 434–438 MHz? (§3.4.3)
12. **30 m.** Is 10100–10150 kHz authorised to either grade? It is absent from Annexure-I but appears in NIAR's widely-copied table. (§3.4.6)
13. **Local enquiry.** If you live in Jammu and Kashmir, the North-Eastern States or an LWE-affected district: does any local verification still apply to an amateur applicant? (§3.8.1)

14. **Portal.** Will the Amateur/HAM module remain on Saral Sanchar, or move to the DoT Authorisation Portal? (§3.7.2)

3.13 What this part does not cover

The legality of the *equipment* is a separate question from the legality of the *operator*. Equipment Type Approval, import rules, dealer possession licences, the status of cheap imported wideband hand-helds, penalties, enforcement, the full syllabus, and study resources are dealt with in other parts of this report. Two cross-references matter enough to repeat here:

- **An ASOC holder needs no separate import authorisation** — Rule 7(3), read directly from the gazette text of G.S.R. 675(E) on 2026-08-23. **[VERIFIED]** The parallel claim that an ASOC holder needs **no radio-equipment possession authorisation** (recorded as Rule 10(f) of the REPA Rules 2026) could not be re-verified on fact-check and is **[UNCERTAIN]** — see §3.8.7. Do not present it to anybody as settled until WPC confirms it in writing.
- **A stock imported hand-held transmits far outside the Indian amateur allocations** — typically 136–174 MHz and 400–470 MHz against permitted ranges of 144–146 MHz and 434–438 MHz. Holding an ASOC does not make transmission outside those two windows lawful. **[VERIFIED]**

3.14 Disclaimer for this part

This is a research summary, not legal advice. Indian spectrum regulation is changing quickly: the governing rules for amateur services were replaced in October 2024, the application module was rebuilt in March 2025, new call sign blocks began issuing in June 2025, and a new equipment-possession framework took effect in July 2026. Several figures here — notably the pass marks, the photograph specifications, and above all **the power and band table itself** — are stated differently by different official documents, or rest on an office memorandum issued under rules that have since been superseded. Confirm the current position with the WPC Wing (contacts in §3.12) or with a licensed amateur radio club before sitting the examination, purchasing equipment, or transmitting.

Part 4 — The ASOC syllabus in detail, and how to study for it

This part assumes you have accepted the conclusion of Parts 1–3: the only realistic route by which one private individual in India can lawfully operate a 10 W handheld transceiver is the **Amateur Station Operator's Certificate (ASOC)**, granted by the **Wireless Planning and Coordination (WPC) Wing** of the **Department of Telecommunications (DoT)** under the **Telecommunications (Amateur Services) Rules, 2024**. Everything below is about passing that examination.

Two grades exist, and which one you sit changes how much you have to study:

Grade	Call-sign block	VHF/UHF power ceiling	Morse required?	Study consequence
ASOC (Restricted)	VU3	10 W DC input on 50–54, 144–146, 434–438 MHz	No	Written paper only
ASOC (General)	VU2	25 W DC input on the same bands, 400 W on HF	Yes — 8 words per minute (wpm), send and receive	Written paper plus a Morse practical, adding roughly 6–10 weeks

[VERIFIED — power and band figures from WPC Office Memorandum (OM) No.

L-14011/255/2004-AMT dated 2010-08-13, Annexure-I, mirrored at

https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf ; call-sign blocks from paragraph 11 of the same OM. Morse requirement

[VERIFIED] — Rule 5(4), Telecommunications (Amateur Services) Rules, 2024, G.S.R. 675(E) dated 2024-10-29, and DoT WPC FAQ No. P-14036/02/2023-COP dated 2024-12-05.]

"DC input" means the total direct-current power fed to the final stage of the transmitter, not radiated radio-frequency (RF) output. Part 1 §1.8 and Part 3 §3.4.4 of this report explain why that distinction decides whether Restricted grade genuinely delivers the 10 W you want. For study purposes the important point is simply this: **both grades sit the same written paper**, so the syllabus below is what you must learn either way.

4.1 Which syllabus is legally operative — and the honest uncertainty

The one detailed published syllabus dates from 2010. It is **Annexure II to WPC Office Memorandum No. L-14011/255/2004-AMT dated 2010-08-13**, headed "SYLLABUS AND THE DETAILS OF EXAMINATIONS FOR THE AWARD OF AMATEUR STATION OPERATOR'S LICENCE (Restricted) and (General)". [VERIFIED — scanned original read page by page; archived at

https://web.archive.org/web/20100827005753/http://wpc.dot.gov.in/DocFiles/amateur_new_guidelines.pdf , byte-identical mirror at https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf]

Paragraph 10 of that OM is the enabling sentence:

"The revised syllabus for the Amateur is placed at Annexure II of this OM. The exams as per the new pattern shall be conducted by the various Wireless Monitoring Stations."

[VERIFIED — OM 2010-08-13, para 10]

Rule 5(3) of the 2024 Rules obliges the Government to publish a syllabus on its portal, and it appears not to have done so. Rule 5(3) and rule 15 both require publication of the syllabus. No syllabus document could be found on eservices.dot.gov.in , saralsanchar.gov.in ,

dot.gov.in, or in the Amateur Radio Society of India's curated list of WPC documents.

[UNCERTAIN — this is absence of evidence, not evidence of absence. Ask the WPC Certificate of Proficiency (COP) Section directly: awacop-dot@gov.in, 011-2303 6181.]

The WPC website itself is down. wpc.dot.gov.in timed out from multiple network paths throughout the research (tested repeatedly on 2026-08-23). The official syllabus and the official sample paper survive only in the Internet Archive and on club mirrors. [VERIFIED that it is unreachable; UNCERTAIN whether it has been permanently retired.]

The best available updated restatement is not a government document. Jose Jacob VU2JOS of the National Institute of Amateur Radio (NIAR) publishes a four-page syllabus PDF, created 2024-11-13, at <https://qsl.net/vu2jos/Syllabus.pdf>. It reproduces the 2010 syllabus but replaces the two superseded national instruments in Section B with the Telecommunications Act, 2023 and the Telecommunications (Amateur Services) Rules, 2024, and it carries the corrected 40 % pass note. [REPORTED — hosted on a personal qsl.net site, not a [.gov.in](https://dot.gov.in) domain.]

Conflict between the research inputs, surfaced. One research pass treats the VU2JOS PDF as "the current syllabus to study from"; another treats the 2010 Annexure II as the operative instrument and the VU2JOS PDF as a helpful secondary restatement. **The second reading is the more defensible one:** a personal website cannot amend a government syllabus. The practical resolution is that the two documents differ only in Section B's list of national instruments and in the pass note, and on both of those points the *law itself* (the 2024 Rules) tells you what to learn. Study the 2010 Annexure II for scope, and the 2024 Rules for the national-regulations content.

There is one further discrepancy between them, and it matters: the VU2JOS and NIAR versions print **QRO** in the examined Q-code list, whereas a 220 dpi read of the original 2010 scan shows unambiguously **QRQ**. See section 4.6, row 1. [VERIFIED]

4.2 Exam structure, weighting and pass marks

4.2.1 STRUCTURE

"1. The examination shall consist of the following two parts: **PART I - Written Test ... Section A:** Radio Theory and Practice ... **Section B:** National and International Telecommunication Union (ITU) Radio Regulations applicable to the operation of amateur station and those relating to the working of station generally. **PART II – Morse (Only for General Certificate)** — Morse reception and sending (8 wpm)"

[VERIFIED — OM 2010-08-13, Annexure II, page 1]

Naming drift to be aware of: the 2010 syllabus and the sample paper say **Section A / Section B**; DoT's own FAQs say **Part A / Part B**. They are the same two sections. [VERIFIED]

4.2.2 WEIGHTING — THE DEFINITIVE TABLE

Both grades sit the *same* 100-question paper. Restricted candidates answer only the first 25 questions of each section, at double marks.

	Section A (Radio Theory and Practice)	Section B (Radio Regulations)	Total	Time	Morse
ASOC (Restricted)	Q1–Q25, 2 marks each = 50 marks	Q1–Q25, 2 marks each = 50 marks	100	1 hour	None
ASOC (General)	Q1–Q50, 1 mark each = 50 marks	Q1–Q50, 1 mark each = 50 marks	100	2 hours	8 wpm send and receive

[VERIFIED — official WPC/DoT "Sample question paper of the examination for award of Amateur Station Operator's Licence (General and Restricted Grade)", instruction block, archived at <https://web.archive.org/web/20160804195151/http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf> (the typo "Saple" is in the Government's own filename); byte-identical live mirror at <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-1.pdf>. Independently corroborated by the DoT/Saral Sanchar FAQ Q5, https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf.]

All questions are **multiple choice (MCQ)**, four options each, with the answer written into a box at the right margin. [VERIFIED — the scanned paper]

The weighting is exactly 50:50 between radio theory and regulations at both grades. There is no published sub-weighting inside a section — no stated number of questions per syllabus topic. [VERIFIED by absence in every official document examined; the distribution of the sample paper itself is the only guide.]

Ignore the claim, which circulates on the IIT Bombay Ham Radio Club page and on hamradio.coursify.me, that the paper may be "one, two, or three hours as per the Grade", or that Restricted grade "permits only reception". Both are wrong: three hours belonged to the pre-2010 Advanced grade, and Restricted grade certainly transmits. [REPORTED — inaccurate; <https://www.aero.iitb.ac.in/hrc/hamexam.php>]

4.2.3 PASS MARKS — A LIVE CONFLICT BETWEEN OFFICIAL DOCUMENTS, RESOLVED

Source	Date	Restricted	General
OM L-14011/255/2004-AMT, Annexure II	2010-08-13	40 % each section, 50 % aggregate	50 % each section, 60 % aggregate
DoT/Saral Sanchar FAQ_for_HAM.pdf (still live)	PDF created 2020-08-24	40 % each section, 50 % aggregate	50 % each section, 60 % aggregate
Telecommunications (Amateur Services) Rules, 2024, rule 5(5)	2024-10-29	40 % of total marks	40 % of total marks

Source	Date	Restricted	General
DoT WPC FAQ No. P-14036/02/2023-COP, Q8	2024-12-05	40 % in both Part A and Part B; 40 % overall	40 % in both Part A and Part B; 40 % overall; Morse minimum 8 wpm both ways

Rule 5(5), verbatim:

"(5) The successful completion of the examination under this rule shall require a minimum of forty per cent of the total marks in respect of each category of ASOC."

[VERIFIED — Telecommunications (Amateur Services) Rules, 2024]

Which governs. The 2024 Rules are subordinate legislation made under section 47 of the Telecommunications Act, 2023; the 2010 OM is an executive instruction. Rule 5(5) prevails, and the December 2024 FAQ is the Department's own reading of it. **A candidate today needs 40 % in Part A, 40 % in Part B and 40 % overall, at either grade.** [VERIFIED — FAQ hosted on [https://www.saralsanchar.gov.in/circular/HAM/FAQ_on_Telecommunication_\(Amateur%20Service\)_Rules_2024.pdf](https://www.saralsanchar.gov.in/circular/HAM/FAQ_on_Telecommunication_(Amateur%20Service)_Rules_2024.pdf) and indexed at <https://dot.gov.in/spectrummanagement/frequently-asked-questions-telecommunication-amateur-service-rules-2024> . One research pass could only obtain this FAQ as an image-only scan and left it unread; another read it in full. The reading above is the one supported by the text.]

Residual risk, stated plainly. The older, stricter FAQ is *still served* from the Government's own [saralsanchar.gov.in](https://www.saralsanchar.gov.in) domain, and NIAR, the South India Amateur Radio Society (SIARS), DARTS (the club at [darts.org.in](https://www.darts.org.in) in the Delhi National Capital Region — sources differ on what the abbreviation expands to, so it is used unexpanded throughout this report; see Part 6, Appendix A.5) and RIRCOM (the International Radio Communicators Club, Rajapalayam) all still print the old thresholds. A candidate cannot rule out an examiner applying the old figures. [UNCERTAIN]

Practical instruction: aim for 60 % or better in each section. That clears every version of the rule that anyone might apply, and the extra preparation is trivial.

4.2.4 THE SECTION A EXEMPTION — THE SINGLE BIGGEST SHORTCUT

"**Note-** Applicants holding degree in Engineering/Science or Diploma in Engineering and having studied electronics or telecommunications shall be exempted from appearing in Section A of Part-I of the test."

[VERIFIED — OM 2010-08-13, Annexure II §1]

DoT's FAQ Q12 spells out the accepted qualifications:

"(i) B. Tech/BE Degree in Electronics or Telecommunication (ii) B.Sc/M.Sc with Electronics or Telecommunication (iii) Diploma in Engineering with Electronics or Telecommunication"

[VERIFIED — https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf , Q12. Proof must be produced to the examiner on the day, per Q16 of the same FAQ.]

The application portal implements this as a Yes/No control plus a reason dropdown plus a document upload, and the admit card prints "Exempted from part A: Yes/No". [REPORTED — DoT user manual, ARSI-hosted scan]

If you qualify, your entire written syllabus collapses to Section B. That changes the study plan completely — see the exempted variant in section 4.11.

One unresolved question you should put to WPC in writing before relying on the exemption: if Part A is dropped, is the 40 % minimum applied only to Part B, and is the total then out of 50 rather than 100? No source addresses this. [UNCERTAIN]

4.3 Section A — Radio Theory and Practice, topic by topic

The syllabus text is reproduced faithfully; the commentary is this report's, based on the level actually tested in the official sample paper.

4.3.1 RESTRICTED CORE (EXAMINED AT BOTH GRADES) — 8 HEADINGS, 20 ENUMERATED SUB-TOPICS

#	Syllabus heading and items (as published)	What is actually tested, and to what depth	Primary study source
(i)	Elementary Electricity and Magnetism — 1. Passive devices (resistors, inductors, transformers, capacitors) and active devices (diodes, transistors). 2. Kirchhoff's current and voltage laws, simple applications. 3. Conductors and insulators, properties, units of circuit elements, Ohm's Law. 4. Conductance; definition of self and mutual inductance. 5. Power and energy: definition, units, simple applications. 6. Permanent magnets and electromagnets.	Definitions, units, one-step arithmetic. Ohm's Law; $P = VI = I^2R = V^2/R$; series/parallel resistance and capacitance; resistor colour code; mutual inductance $M = k\sqrt{L_1L_2}$; batteries in series and parallel. No calculus.	NIAR Study Manual ch. 3; ASOC Tutorials (India) SD-series
(ii)	Elementary Theory of Alternating Currents — 1. Sinusoidal quantities: peak, instantaneous, RMS (root mean square), average values. 2. Phase, reactance, impedance, power factor. 3. Series and parallel R-L-C circuits, resonance, coupled circuits. 4. Rectifiers, voltage regulation and smoothing circuits.	$RMS \approx 0.707 \times \text{peak}$; average $\approx 0.637 \times \text{peak}$; $X_L = 2\pi fL$, $X_C = 1/2\pi fC$; resonance $f = 1/(2\pi\sqrt{LC})$; behaviour of series vs parallel resonant circuits.	as above

#	Syllabus heading and items (as published)	What is actually tested, and to what depth	Primary study source
(iii)	Elementary theory of Semiconductor Devices — 1. Diodes and transistors: properties, use for amplifiers, oscillators, detectors and frequency changers.	Which device does what; common-emitter amplifier inverts phase; doping of NPN vs PNP.	as above
(iv)	Radio Receivers — 1. Principles and operation of TRF (tuned radio frequency) and superheterodyne receivers. 2. CW (continuous wave, i.e. Morse) reception. 3. Receiver characteristics: sensitivity, selectivity, fidelity, adjacent-channel and image interference, AVC (automatic volume control) and squelch circuits, signal-to-noise ratio.	The superhet block diagram is the single most-examined diagram in Section A. Standard intermediate frequency (IF) 455 kHz; local-oscillator arithmetic; BFO (beat frequency oscillator) is what makes CW audible; image frequency = signal $\pm 2 \times$ IF.	NIAR ch. 3; ASOC Tutorials SD-31 to SD-34
(v)	Transmitter — 1. Principles and operation of low-power transmitters, crystal oscillators, stability of oscillators. 2. Basic knowledge of semiconductor-based transmitter construction.	Master-oscillator/power-amplifier chain; why a crystal is more stable than an LC oscillator; Hartley vs Colpitts identification by tank circuit.	NIAR ch. 3; ASOC Tutorials SD-27
(vi)	Radio Wave Propagation — 1. Basic knowledge of the electromagnetic spectrum. 2. Wavelength, frequency, frequency bands. 3. Nature and propagation of radio waves; ground and sky waves; space waves; skip distance; skip zone; fading.	λ (metres) = 300 / f (MHz) — expect this arithmetic. Which band names cover which frequencies (see the ITU Article 2 table in section 4.4). Skip distance vs skip zone is a favourite distractor pair.	NIAR ch. 3
(vii)	Aerials — Common types of transmitting and receiving aerials.	Half-wave dipole length $\approx 143/f$ (MHz) metres; Yagi element roles (reflector, driven element, director); vertical vs horizontal polarisation.	NIAR ch. 3
(viii)	Frequency Measurement — Measurement of frequency and use of simple frequency meters.	Absorption wavemeter, heterodyne frequency meter, digital counter. Also links to Annexure A rule 4 of the 2024 Rules, which <i>requires</i> a reliable frequency-measuring device at the station.	NIAR ch. 3 + 2024 Rules Annexure A

[All syllabus text VERIFIED — OM 2010-08-13, Annexure II, pages 2–3]

The official sample paper also asks a handful of items not itemised above but plainly inside "radio theory and practice": decibels (dB, dBm) and gain, transformer volt-ampere rating, batteries in series and parallel, and the resistor colour code. [VERIFIED — sample paper]

4.3.2 GENERAL-GRADE ADDITIONS — 8 HEADINGS, 15 FURTHER SUB-TOPICS

#	Syllabus heading and items (as published)	Notes
(i)	Principles of Communications — analog and digital communication; need for modulation; amplitude, frequency and pulse modulation; demodulation.	Why modulate at all (antenna size, multiplexing); AM vs FM bandwidth; pulse-code modulation as the digital case.
(ii)	Alternating current — transformer construction; transformer losses; transformer as a matching device.	Turns ratio, impedance transformation by the square of the turns ratio; copper and iron (hysteresis, eddy-current) losses.
(iii)	Semiconductor devices and Transistors — elementary principles of conduction and construction; symbols and biasing methods.	Symbol recognition is examined directly.
(iv)	Power Supplies — half-wave and full-wave rectifiers; bridge rectifier; smoothing and regulating circuits.	Ripple frequency for half-wave vs full-wave; zener as a shunt regulator.
(vii)	Transmitters and Receivers — elementary principles of transmission and reception of facsimile and television signals.	Vestigial sideband for TV video; facsimile scanning.
(viii)	Propagation — characteristics of ionosphere and troposphere; properties of ionospheric layers; critical frequency and day/night frequencies.	D, E, F1, F2 layers; which layer disappears at night; critical frequency and maximum usable frequency (MUF); why 40 m works at night and 20 m by day.
(ix)	Aerials — principles of radiation; aerials for different frequency bands including microwave.	Parabolic dish gain, waveguides, standing wave ratio (SWR) and matching.
(x)	Space Communications — elementary principles of communication via satellites.	Uplink/downlink, transponder, the amateur-satellite service and OSCAR series. Indian amateur satellite operation is capped at 30 dBW equivalent isotropically radiated power (e.i.r.p.). [VERIFIED — OM 2010, Annexure-I, Note (i)]

[All syllabus text VERIFIED — OM 2010-08-13, Annexure II, pages 4–6]

Two things you must know about this list.

1. **The numbering gap (i)–(iv), then (vii)–(x), is in the original.** There are no items (v) or (vi). It is a drafting artefact: the 2010 list was produced by editing the older Advanced-grade list and deleting items. Nothing is missing from your syllabus. [VERIFIED — comparison of the 2010 OM with the pre-2010 syllabus at <https://indianhams.com/downloads/syllabus.pdf>]

2. **Three topics were deleted in 2010 and should not be examined:** motors and generators; moving-coil and moving-iron measuring instruments; and thermionic valves. Frequency measurement survives, but only as Restricted item (viii). Any study material that teaches you valve theory (triodes, thermionic emission, characteristic curves) is working from the **pre-2010** syllabus. [VERIFIED]

Do not study from <https://indianhams.com/downloads/syllabus.pdf> **or** <https://www.qsl.net/vu2msy/EXAM.htm> . Both still serve the pre-2010 four-grade syllabus (Grade II, Grade I, Advanced), with valves, 12 wpm Morse and 300-character passages. [VERIFIED — page content]

4.4 Section B — National and ITU Radio Regulations, topic by topic

Section B is half the paper for everyone and, for an exempted engineer, the *entire* written paper. It is also the part where the mainstream study material is most out of date. Budget your effort accordingly.

4.4.1 THE NATIONAL INSTRUMENTS — ONE IS DEAD, ONE IS ALIVE, ONE IS NEW

The syllabus says:

"(a) **Knowledge of** :- (i) the Indian Wireless Telegraph Rules, 1973. (ii) the Indian Wireless Telegraphs (Amateur Service) Rules, 1978 and amendments."

[VERIFIED — OM 2010-08-13, Annexure II]

Instrument named	Status today	What to do
Indian Wireless Telegraph Rules, 1973	Still in force , saved by section 61 of the Telecommunications Act, 2023 [VERIFIED]	Examinable. Caution: no authoritative full text could be located online. [UNCERTAIN — a real gap]
Indian Wireless Telegraphs (Amateur Service) Rules, 1978	Superseded on 2024-10-29/30 by the 2024 Rules [VERIFIED — preamble to G.S.R. 675(E)]	Do not learn it as live law. Read it only to see what changed.
Telecommunications (Amateur Services) Rules, 2024 , and its Annexure A	In force [VERIFIED]	This is the text to learn. It replaces the 1978 Rules in the syllabus.
Telecommunications Act, 2023	In force [VERIFIED]	Learn the framework: sections 47 (rule-making), 53 (digital delivery), 60 (repeals), 61 (savings).

Gazette citation note: the gazette itself reads **G.S.R. 675(E)** dated 2024-10-29 (published 2024-10-30). DoT's own December 2024 FAQ cites "GSR 657(E)" — a digit transposition in the FAQ. Cite 675(E). [VERIFIED]

What to learn from the 2024 Rules: grades and scope (rule 3), eligibility (rule 4: Indian citizen, minimum 12 years of age), the examination (rule 5), fees (rules 5(2) and 6), validity and renewal, suspension and cancellation, societies and special call signs, inspection, and the digital-portal rule (rule 15).

What to learn from Annexure A — this is the operating-procedure core and is heavily examinable:

Annexure A rule	Content you must know
1	The station may be used to receive Standard Frequency and Time Signal Service transmissions.
2	Messages. Plain language only; technical messages only; contacts only with authorised amateur stations. Forbidden: broadcast/music retransmission; false or misleading calls; advertisements; business communications; political or industrial controversy; third-party messages; indecent, obscene or offensive language.
2(3)	Third-party messages are permitted in a failure of normal telecommunications, if authorised, and only to or from a District Magistrate, Deputy Commissioner, Collector or officer authorised by them.
4	Frequency control. No energy radiated outside authorised bands; a reliable frequency-measuring device at the station, checked on every frequency change; SACFA (Standing Advisory Committee on Frequency Allocation) clearance as applicable. Note that the 2010 OM's own Annexure-I writes the name as "Standing Advisory committee on Radio frequency allocation"; the abbreviation is the same body.
5	Non-interference. Discontinue or restrict transmission on request; damped waves prohibited.
6	Log book. Bound book, not loose-leaf; date and time of every transmission; stations worked, times, emission type; opening and closing times; experiments and tests; temporary locations. "All times in the log shall be stated in the Indian Standard Time." No gaps; entries made contemporaneously; preserve for one year from the last entry.
7(4)	Equipment register — a nine-column register (make/model/type/serial, from whom received or "self-made", date received or assembled, seller's receipt and certificate numbers, to whom sold, date of sale, purchaser's certificate particulars, remarks).
8	Secrecy of correspondence. A message you were not entitled to receive must not be disclosed, copied or used, except to an officer authorised by the Central Government or a court.
9(1)	Listen before transmitting; send the call sign at the beginning and end of each transmission period, and repeat it whenever the period exceeds ten minutes; no unidentified or falsely identified transmission; avoid prolonged calls; use the phonetic alphabet and figure code when spelling out.
9(2)	Call and reply. Called station's call sign not more than three times; then DE (Morse) or "This is" (telephony); then the calling station's call sign not more than three times. A call may be sent three times at two-minute intervals; thereafter wait ten minutes, listening. General call: CQ (Morse) or "Hello all stations"/CQ (telephony).
9(3)	End of transmission AR (Morse) / "Over" (telephony). End of work VA (Morse) / "OUT" (telephony).

Annexure A rule	Content you must know
9(4)	Tests. Not more than 30 seconds , composed of a series of VVV followed by the call sign (or figures 1, 2, 3, 4... in the figure code on telephony). Longer tests require an artificial aerial. No unmodulated carrier without intelligible modulation.
10	Inspection. Any authorised officer may inspect, examine or test the apparatus; produce the certificate and the log.

[All VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A]

Note one divergence from the study manuals: NIAR teaches the general call as "CQ CQ CQ DE VU2JOS VU2JOS VU2JOS AR K". The rule text does not include the "AR K" ending; that is convention. [REPORTED — NIAR Study Manual; the rule text is VERIFIED]

4.4.2 THE ITU MATERIAL — A CLOSED, VERIFIABLE LIST

The syllabus specifies exactly ten ITU Radio Regulations (RR) references. Every one was checked against the **ITU Radio Regulations, Edition of 2024**, and all ten still map correctly, despite the syllabus citing the 2008 edition.

Syllabus item	Syllabus cites	ITU RR 2024 actual	Match
Designation of Emission	Appendix 1	Appendix 1 (REV.WRC-19) — Classification of emissions and necessary bandwidths	Yes
Phonetic alphabets and figure code	Appendix 14	Appendix 14 (REV.WRC-23)	Yes
Nomenclature of the Frequency and Wavelength	Article 2	Article 2 — Nomenclature	Yes
Frequency allocation for Amateur Services	Article 5	Article 5 — Frequency allocations	Yes
Interference, measures against interference and tests	Article 15	Article 15 — Interferences	Yes
Identification of Stations	Article 19	Article 19	Yes
Distress Signal, Call and Message Transmissions	Articles 30, 31, 32, 33	Art 30 general provisions; Art 31 GMDSS frequencies; Art 32 distress procedures; Art 33 urgency and safety procedures	Yes
Urgency Signal, Call and Message Transmissions	Articles 30, 31, 32, 33	as above	Yes
Amateur Station	Article 25	Article 25 — Amateur services	Yes
Call Sign series allocated to India	Appendix 42	Appendix 42 (REV.WRC-19)	Yes

[VERIFIED — ITU RR 2024 Vol. 1 (Articles) <https://search.itu.int/history/HistoryDigitalCollectionDocLibrary/1.49.48.en.101.pdf> and Vol. 2 (Appendices) ... [1.49.48.en.102.pdf](https://search.itu.int/history/HistoryDigitalCollectionDocLibrary/1.49.48.en.102.pdf) . GMDSS = Global Maritime Distress and Safety System.]

What to extract from each, in order of exam value:

Article 25 (Amateur services) — the single most examinable ITU text. Know: international amateur contacts are permitted unless an administration objects (25.1); transmissions limited to matters incidental to the amateur service and remarks of a personal character (25.2); **no encoding to obscure meaning** (25.2A); third-party international traffic only in emergencies or disaster relief (25.3); administrations decide whether to require Morse (25.5 — **this is why India can still require it**); administrations verify operational and technical qualifications (25.6); **maximum power is fixed by the administration** (25.7); call sign to be transmitted at short intervals (25.9). [VERIFIED — RR 2024 Vol. 1, pp. 323–324]

Article 2 (Nomenclature) — memorise the band table:

Band No.	Symbol	Frequency range	Metric subdivision
4	VLF	3–30 kHz	Myriametric
5	LF	30–300 kHz	Kilometric
6	MF	300–3000 kHz	Hectometric
7	HF	3–30 MHz	Decametric
8	VHF	30–300 MHz	Metric
9	UHF	300–3000 MHz	Decimetric
10	SHF	3–30 GHz	Centimetric
11	EHF	30–300 GHz	Millimetric
12	—	300–3000 GHz	Decimillimetric

Also RR 2.6: UTC (Coordinated Universal Time) is used in international radiocommunication and presented as a four-digit group 0000–2359. [VERIFIED — RR 2024 Vol. 1, pp. 25–26. Note the NIAR manual omits band 12.]

Appendix 1 (Emission designators) — three symbols plus a four-character bandwidth prefix. First symbol = modulation of the main carrier (N, A, H, R, J, B, C, F, G, D, P, K, L, M, Q, V, W, X); second = nature of the modulating signal (0, 1, 2, 3, 7, 8, 9, X); third = type of information (N, A, B, C, D, E, F, W, X). Bandwidth notation uses a letter in place of the decimal point: H = Hz, K = kHz, M = MHz, G = GHz — so 2.4 kHz is **2K40** and 12.5 kHz is **12K5**, and **the bandwidth goes before the class**, giving **3K00A3E**, never **A3E3K00**. [VERIFIED — RR 2024 Vol. 2, Appendix 1]

The emissions Indian amateurs may use, which you should be able to decode on sight:

Description	Designator
Morse telegraphy	A1A
Modulated CW Morse	A2A
Facsimile (AM)	A3C
AM voice	A3E
Fast-scan television	A3F
Radioteletype, frequency-shift keying	F1B
Radioteletype, audio frequency-shift keying	F2B
Facsimile (FM)	F3C
FM voice (narrowband)	F3E
Single sideband, full carrier	H3E
Single sideband, suppressed carrier	J3E
Single sideband, reduced carrier	R3E

[Emission list VERIFIED — OM 2010, Annexure-I; the symbol-by-symbol decoding as taught is REPORTED — NIAR Study Manual p.128]

Appendix 14 (Phonetic alphabet and figure code) — the 26 letter words (Alfa, Bravo, Charlie, Delta, Echo, Foxtrot, Golf, Hotel, India, Juliett, Kilo, Lima, Mike, November, Oscar, Papa, Quebec, Romeo, Sierra, Tango, Uniform, Victor, Whiskey, X-ray, Yankee, Zulu) **and the figure code, which candidates routinely neglect:** Nadazero, Unaone, Bissotwo, Terrathree, Kartefour, Pantafive, Soxisix, Setteseven, Oktoeight, Novenine, plus Decimal and Stop. [VERIFIED — RR 2024 Vol. 2, Appendix 14. ITU spells the figure words as single words; NIAR and the WPC sample paper split them into two ("Okto Eight"), and NIAR omits Decimal and Stop. Accept both spellings.]

Appendix 42 (Call sign series allocated to India) — ATA–AWZ, VTA–VWZ, 8TA–8YZ.

[VERIFIED — RR 2024 Vol. 2, Appendix 42.] **Trap:** the official WPC sample paper offers "8TA – 8WZ" as an option, which is wrong; the ITU allocation ends at 8YZ. [VERIFIED]

Article 19 (Identification of stations) — amateur call signs consist of one or two characters plus a single digit plus a suffix of not more than four characters, the last of which is a letter (19.68); the prohibition on the digits 0 and 1 does not apply to amateur stations (19.69). Indian practice on top: VU2 for General, VU3 for Restricted, suffix not more than three letters, and no suffix that duplicates a Q-code or a distress signal. [ITU provisions VERIFIED — RR 2024 Vol. 1, p. 270; VU2/VU3 split VERIFIED — OM 2010 para 11; the three-letter suffix limit REPORTED — NIAR Study Manual p.122]

Articles 30–33 (Distress, urgency, safety) —

Signal type	Morse	Voice	Priority
Distress	SOS three times, then DE, then call sign three times	MAYDAY three times, then "This is", then call sign three times	Absolute priority over everything
Urgency	XXX three times	PAN PAN	Second
Safety	TTT three times	SECURITE	Third
Test	Three series of VVV plus call sign	Figures 1, 2, 3, 4, 5 in the figure code plus call sign	Maximum 30 seconds

Distress frequencies taught for the exam: 500 kHz (radiotelegraphy), 2182 kHz (radiotelephony), 156.800 MHz (VHF channel 16). Distress content: call sign, exact location, nature of distress, assistance required, any other useful information. Any station hearing a distress call must immediately stop transmitting anything that could interfere, keep listening and render help if possible. [REPORTED — NIAR Study Manual p.124; the underlying provisions are ITU RR Articles 30–33 and the Indian 30-second test rule is VERIFIED from the 2024 Rules Annexure A rule 9(4)]

4.4.3 Q-CODES — THE EXACT EXAMINED SET OF 30

"QRA, QRG, QRH, QRI, QRL, QRM, QRN, **QRQ**, QRS, QRT, QRU, QRV, QRW, QRX, QRZ, QSA, QSB, QSL, QSO, QSU, QSV, QSW, QSX, QSY, QSZ, QTC, QTH, QTR, and QUM."

[VERIFIED — OM 2010-08-13, Annexure II, read from a 220 dpi render of the scan]

Learn all 30 as question-and-answer pairs. The high-frequency ones in past material: QRL (are you busy / frequency in use), QRM (man-made interference), QRN (atmospheric noise), QRT (stop sending), QRV (ready), QRX (wait / will call again), QRZ (who is calling me), QSB (fading), QSL (confirm receipt), QSO (contact), QSY (change frequency), QTH (position), QTR (correct time). QRK and QSA both carry a 1–5 scale (readability and strength respectively); QRI and QRM and QRN also use graded scales. [Meanings REPORTED — NIAR Study Manual pp. 122–123; the list itself VERIFIED]

The QRQ/QRO problem — learn both. The official 2010 syllabus (and the pre-2010 syllabus before it) lists **QRQ** ("shall I send faster?"). NIAR, SIARS, DARTS and the VU2JOS 2024 PDF all print **QRO** ("shall I increase power?") in its place, and NIAR's Q-code table then defines QRO and omits QRQ entirely. The syllabus is the authority, but the mass-market study material is what question-setters may have absorbed. **Memorise both.** [VERIFIED — official list; REPORTED — NIAR <http://www.niar.org/downloads/ham-downloads/Study-Manual.pdf> p.11, SIARS <https://www.siars.org.in/amateur-radio/licensing/syllabus/>, DARTS <https://darts.org.in/exam-syllabus-2/>]

Note what is **absent** from the examined list, despite being taught: QRP, QRB, QSP, QST, QTA, QSK and the numeric codes 73 and 88. [VERIFIED]

4.4.4 ABBREVIATIONS AND PROSIGNS — THE EXACT EXAMINED SET OF 17

Abbreviation	Meaning	Abbreviation	Meaning
AA	All after	K	Go ahead / invitation to transmit
AB	All before	NIL	I have nothing for you
AR	End of transmission of a message	OK	All correct
AS	Please wait / stand by	R	Roger — received fully
C	Yes / affirmative	TU	Thank you
CFM	Confirm	VA	End of work (spoken "Victor Alfa")
CL	Closing down	WA	Word after
CQ	General call to all stations	WB	Word before
DE	"This is" / "from"		

[List VERIFIED — OM 2010-08-13; meanings REPORTED — NIAR Study Manual p.121]

Also learn BK, 73, 55, OM and YL even though they are not in the official 17. **BK** (break — a third station entering between two others) is asked in the official WPC sample paper, Section B Q40, despite being absent from the syllabus list. [VERIFIED both facts]

4.4.5 SECTION B MATERIAL THE SYLLABUS DOES NOT ITEMISE BUT THE EXAM TESTS

Topic	What to know
RST reporting system	R = readability 1–5, S = signal strength 1–9, T = tone 1–9. Voice reports use RS only; Morse uses RST. S9 = "extremely strong", S8 = "strong" — NIAR's own exercise answer gets this wrong. [REPORTED — NIAR p.125]
Indian authorised bands, emissions and power	The 2010 OM Annexure-I table (see section 4.4.6). The 2024 Rules contain no band table; rule 7(1) delegates it. [VERIFIED]
Times-of-operation codes	HX = no specific working hours (amateur radio is HX); H24 = continuous; HJ = sunrise to sunset; HN = sunset to sunrise. [REPORTED — NIAR p.132]
IST and UTC	IST = UTC + 5:30; logs in IST; international convention in UTC, four digits. [REPORTED / VERIFIED at RR 2.6]
Standard Frequency and Time Signal Service	Syllabus item (c). India's station is ATA , operated by the National Physical Laboratory (NPL), New Delhi; standard frequencies 5.000, 10.000 and 15.000 MHz. [REPORTED — NIAR p.132; the permission to receive them is VERIFIED at Annexure A rule 1]
Repeaters	A station that simultaneously retransmits another on a different channel; standard Indian VHF shift 600 kHz . [REPORTED — NIAR p.129]

Topic	What to know
QSL cards and QSL bureaux	Confirmation of contact; the NIAR bureau.
Infringement categories	Unauthorised frequency / unauthorised emission / unauthorised period / unauthorised call sign. Asked in the official sample paper (Section B Q50). No primary source could be located — it is not in ITU RR Appendix 9. Probably Wireless Monitoring Organisation practice. [UNCERTAIN — but learn the four categories anyway, because they are examined]
Morse timing	Dot = 1 unit; dash = 3 units; space between elements = 1 unit; between letters = 3 units; between words = 5 units per the Indian material and the official sample paper , although the international standard is 7. Answer five in the exam. Error signal = eight dots. [VERIFIED — WPC sample paper Section B Q9 and NIAR p.139; the contradictory "seven dots" is on NIAR p.137]

4.4.6 THE INDIAN BAND, EMISSION AND POWER TABLE (MEMORISE THE VHF/UHF ROWS)

Band	Frequencies	Restricted: emissions	Restricted: max DC input	General: emissions	General: max DC input
160 m	1820–1860 kHz *	A3E, H3E, J3E, R3E	50 W	A1A, A2A, A3E, H3E, J3E, R3E, F1B, F2B, F3E, F3C, A3C, A3F	400 W
80 m	3500–3700 kHz *	same	50 W	same	400 W
75 m	3890–3900 kHz *	same	50 W	same	400 W
40 m	7000–7100 and 7100–7200 kHz	same	50 W	same	400 W
20 m	14000–14350 kHz	same	50 W	same	400 W
17 m	18068–18168 kHz \$	same	50 W	same	400 W
15 m	21000–21450 kHz	same	50 W	same	400 W
12 m	24890–24990 kHz \$	same	50 W	same	400 W
10 m	28000–29700 kHz	same	50 W	same	400 W
6 m	50–54 MHz	F1B, F2B, F3E, F3C	10 W	F1B, F2B, F3E, F3C	25 W
2 m	144–146 MHz	F1B, F2B, F3E, F3C	10 W	F1B, F2B, F3E, F3C	25 W
70 cm	434–438 MHz @	F1B, F2B, F3E, F3C	10 W	F1B, F2B, F3E, F3C	25 W

Band	Frequencies	Restricted: emissions	Restricted: max DC input	General: emissions	General: max DC input
5 cm	5725–5840 MHz @	not authorised	—	F1B, F2B, F3E, F3C	25 W

* primary shared basis; \$ non-interference, non-protection basis; @ secondary basis. "DC input power is the total direct current power input to the final stage of the transmitter." [VERIFIED — OM 2010-08-13, Annexure-I, pages 4–5]

Ten HF entries, not eleven — 30 m is absent. The nine HF rows above correspond to **ten** Annexure-I entries, because Annexure-I lists 40 m as two separate rows (7000–7100 kHz and 7100–7200 kHz); that is the "ten HF bands at 50 W" figure used in Parts 1, 3 and 5. An earlier draft of this table also carried a **30 m (10100–10150 kHz)** row. It has been removed, because a direct page-by-page read of the Annexure-I scan shows that **10100–10150 kHz does not appear in Annexure-I at all, for either grade**, even though NIAR's widely-copied table shows it and NFAP designates the amateur service there. Part 3 §3.4.6 records the same finding from the scan, and Part 5 §B.0.5 deliberately declines to make 30 m either a correct or an incorrect answer in its practice bank. Both Annexure-I pages were checked and this is a genuine gap, not a scanning artefact. [VERIFIED — the absence, from the 2010 OM scan; [UNCERTAIN] — whether 30 m is authorised by some instrument not located. Do not transmit on 30 m until WPC confirms in writing, citing rule 7(1); and if an examiner offers 30 m as an option, be aware that the mass-market study material treats it as authorised.]

A discrepancy you must know about. NIAR's current <http://www.niar.org/downloads/Authorised-Frequency-Bands.pdf> prints **1800–1825 kHz** for 160 m and **50–52 MHz** for 6 m, and shows 40 m as one 7000–7200 kHz block — all different from the 2010 OM. [VERIFIED that the two documents differ; UNCERTAIN which an examiner would mark correct. The safer assumption for exam purposes is the 2010 OM, because it is the only government instrument that states amateur power limits and it predates the NIAR revision. For *operating* purposes, treat this as an open question and ask WPC.]

The IARU (International Amateur Radio Union) Region 3 band plan is not law and is not in the syllabus. It is a voluntary agreement on mode segments. Do not spend exam preparation time on it. [REPORTED — <https://www.iaru.org/wp-content/uploads/2020/01/R3-004-IARU-Region-3-Bandplan-rev.2.pdf>]

4.5 Part II — the Morse test (General grade only)

"Morse receiving: (Speed: 8 words per minute) The test piece will consist of a plain language passage of **200 characters** which may comprise of letters, figures. Test piece may also contain the following punctuations ie full stop; comma; semi colon; break-sign; hyphen and question mark. The average words shall contain five characters and each figure and punctuation will be counted as two characters. The test will be for **five consecutive minutes** at a speed of 8 words per minute. A short

practice piece of one minute shall be sent at the prescribed speed before the start of the actual test. Candidates will not be allowed more than one attempt in Morse reception and sending test. ... Overwriting will be treated as error. ... **More than 5 errors will disqualify a candidate.** However if a candidate receives without any error in any part of the passage continuously for one minute duration will be declared successful in the Morse reception test."

"**Morse Sending (Speed: 8 words per minute)** ... Candidates are required to send for five consecutive minutes at a speed not less than 8 words per minute."

"**Note-** A candidate shall have to pass **both receiving and sending simultaneously.**"

[VERIFIED — OM 2010-08-13, Annexure II, pages 6–7]

Arithmetic: 8 wpm × 5 minutes × 5 characters per word = 200 characters. Figures and punctuation count as **two** characters each.

Morse has not been abolished in India. Rule 5(4) of the 2024 Rules retains it for ASOC (General), and the DoT FAQ of 2024-12-05 sets it at 8 wpm in both transmission and reception. [VERIFIED] Whether every Wireless Monitoring Station is actually administering it in practice could not be confirmed. [UNCERTAIN]

If you do not need HF and do not need 25 W, Restricted grade avoids this entirely and is several weeks shorter.

4.6 Errors in the mainstream study material — check every one of these

This table is arguably the most valuable page in this part. Each row is a real error in a widely used Indian study source that could cost you marks or, worse, lead you to operate unlawfully.

#	Error	Where it appears	The correct position
1	QRO listed in the examined Q-code set	NIAR Study Manual p.11; SIARS; DARTS; VU2JOS 2024 syllabus PDF	The official list has QRQ . Learn both. [VERIFIED]
2	"The range of VHF band is 300 to 3000 MHz"	NIAR Study Manual, Model Q&A Section-B Q44	VHF is 30–300 MHz ; 300–3000 MHz is UHF. [VERIFIED]
3	"Signal Strength 9 means: Strong signals"	NIAR exercise answer 11, p.134	RST S9 = extremely strong ; S8 = strong. NIAR's own table on p.125 says so. [VERIFIED — internal contradiction]
4	Word spacing given as seven dots (p.137) and five dots (p.139)	NIAR Study Manual	The exam expects five dots . [VERIFIED — WPC sample paper Section B Q9]
5	Call-sign series option "8TA – 8WZ"	The official WPC sample paper itself, Section B Q24(c)	ITU Appendix 42: 8TA–8YZ . [VERIFIED]

#	Error	Where it appears	The correct position
6	Emission designator written class-first, e.g. "A3E3K00"	The official WPC sample paper, Section B Q31	Bandwidth first: 3K00A3E . [VERIFIED — RR Appendix 1 §2(3)]
7	Pass marks "50 % each section, 60 % aggregate" for General	NIAR, SIARS, DARTS, RIRCOM, and DoT's own still-live FAQ_for_HAM.pdf	40 % / 40 % / 40 % since the 2024 Rules. [VERIFIED]
8	"Restricted Grade ... permits only reception"	IIT Bombay Ham Radio Club page	Restricted grade transmits: 50 W HF, 10 W VHF/UHF DC input. [VERIFIED]
9	"One, two, or three hours duration as per the Grade"	IIT Bombay Ham Radio Club; hamradio.coursify.me	Two grades only: 1 hour and 2 hours. [VERIFIED]
10	Change-of-location fee ₹200, duplicate ₹100	NIAR p.132; DoT FAQ_for_HAM.pdf Q21/Q22	2024 Rules rule 7(6): ₹1,000 for a duplicate or a change of location. [VERIFIED]
11	Section B(a)(ii) cites the "Indian Wireless Telegraphs (Amateur Service) Rules, 1978"	The official 2010 syllabus itself	Superseded by the 2024 Rules. [VERIFIED]
12	Phonetic figure words split in two, and "Decimal"/"Stop" missing	NIAR p.121	ITU Appendix 14 uses single words and includes Decimal and Stop. [VERIFIED]
13	"Passes in Section A → Restricted; passes in A and B → General"	RIRCOM FAQ	Garbled. The grades are applied for separately; the General/Restricted distinction is the written paper plus Morse, not one section versus two. [UNCERTAIN — no clean official statement]
14	160 m as 1800–1825 kHz; 6 m as 50–52 MHz	NIAR Authorised-Frequency-Bands.pdf	The 2010 OM says 1820–1860 kHz and 50–54 MHz. [VERIFIED that they differ]
15	Postal application with demand draft, self-addressed envelope, gazetted-officer character certificate	ARCAD how-to-become-a-ham page	Applications have been online only since 2020-08-01 . Do not follow it. [REPORTED]

4.7 Official and free primary documents — assemble this folder first

Document	URL	Date	Status
Telecommunications (Amateur Services) Rules, 2024, G.S.R. 675(E)	https://eservices.dot.gov.in/sites/default/files/circular-notifications/ASOC%20Service%20rules%2024.pdf (mirror: https://arsi.info/wp-content/uploads/2024/11/Gazette-Notification-Final-Amateur-Services-Rules-Oct-2024.pdf)	notified 2024-10-29	[VERIFIED] — the single most important document
DoT WPC FAQ on the 2024 Rules (pass marks, Morse)	https://www.saralsanchar.gov.in/circular/HAM/FAQ_on_Telecommunication_(Amateur%20Service)_Rules_2024.pdf	2024-12-05	[VERIFIED]
WPC OM L-14011/255/2004-AMT — Annexure I bands/ power, Annexure II syllabus	https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-255-2004-AMT-13082010.pdf	2010-08-13	[VERIFIED] — origin server unreachable
Official WPC sample question paper (100 MCQs, no answer key)	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-1.pdf (archived original: https://web.archive.org/web/20160804195151if_/http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf)	archived 2016-08-04; paper undated	[VERIFIED] — image-only scan, no text layer
DoT/Saral Sanchar FAQ on the exam (logistics, centres, Section A exemption)	https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf	pre-2024	[VERIFIED] — its pass-mark answer Q6 is stale
Telecommunications Act, 2023	https://eservices.dot.gov.in/sites/default/files/circular-notifications/Telecommunications-Act-2023.pdf	2023	[VERIFIED]
ITU Radio Regulations 2024, Vol. 1 (Articles)	https://search.itu.int/history/HistoryDigitalCollectionDocLibrary/1.49.48.en.101.pdf	2024	[VERIFIED]
ITU Radio Regulations 2024, Vol. 2 (Appendices)	https://search.itu.int/history/HistoryDigitalCollectionDocLibrary/1.49.48.en.102.pdf	2024	[VERIFIED]
National Frequency Allocation Plan 2022	https://arsi.info/wp-content/uploads/2024/04/NFAP-2022.pdf	2022	[VERIFIED] — reference only, not examined line by line
VU2JOS updated syllabus restatement	https://qsl.net/vu2jos/Syllabus.pdf	PDF created 2024-11-13	[REPORTED] — useful, not authoritative

Document	URL	Date	Status
ASOC service page (fees, eligibility, contacts)	https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc	"Last updated 20-08-2026"	[VERIFIED]

There is no government coaching. WPC states plainly: *"No, WPC does not provide any training for the examination."* [VERIFIED — WPC FAQ text, reproduced at <http://www.niar.org/downloads/Frequently-Asked-Questions.pdf> , PDF created 2020-08-24; restated in the FAQ dated 2024-11-09 at <https://qsl.net/vu2jos/FAQ.pdf>] Every course, class and book below is produced by a non-government body.

4.8 Institutional courses — NIAR and ARSI

4.8.1 NIAR — NATIONAL INSTITUTE OF AMATEUR RADIO, HYDERABAD

NIAR is the de-facto national training body and the only organisation publishing a purpose-written ASOC textbook.

Item	Detail
Address	6-3-1092/93, Raj Bhavan Road, Somajiguda, Hyderabad 500082, Telangana
Telephone	+91 40 2331 0287, +91 40 2331 0787
Email	info@niar.org
Website	http://www.niar.org
Classes	<i>"Classes for Amateur Radio is conducted at NIAR regularly."</i> — that one sentence is the entirety of what NIAR publishes about its classes. [REPORTED]
Course duration	<i>"It takes just a few weeks training to become eligible for the examination."</i> [REPORTED — http://www.niar.org/how-to-be-a-ham.html]
Course fee	Not published anywhere. [UNCERTAIN — telephone +91 40 2331 0287 or email info@niar.org]
Course calendar	Not published. NIAR's activity log suggests institutional/group batches are the normal mode rather than an open priced course. [REPORTED]
Free awareness programme	Introductory session for schools and colleges — outreach, not exam coaching. Arrange via info@niar.org . [REPORTED]

Website health warning: niar.org pages carry a 2014 copyright, several pages return HTTP 500, and HTTPS is broken (the TLS certificate does not cover niar.org or www.niar.org), so browsers will warn. Fetch over plain HTTP. **Confirm every NIAR price by telephone before paying.** [VERIFIED — site behaviour on 2026-08-23]

Printed study material (from <http://www.niar.org/training-material.html> , page **Last-Modified** 2024-07-23, prices themselves undated) [REPORTED]:

Item	NIAR member (₹)	Non-member (₹)	Postage (₹)
Study Manual	140	150	50
Exam Application Form (obsolete — applications are online)	20	25	20
Morse Code Practice Book	40	50	50
Morse Key	190	200	—
Morse Code Practice Oscillator	290	300	—

Payment by demand draft only, in favour of "National Institute of Amateur Radio", payable at Hyderabad. A printed Study Manual delivered anywhere in India therefore costs ₹200 for a non-member; a full Morse kit (book, key, oscillator) is ₹550 plus postage. [REPORTED]

The same manuals are free as PDFs — this is the important point:

Title	URL	Pages	PDF date
Study Manual ... (Restricted and General Grade)	http://www.niar.org/downloads/Study-Manual-General-Grade-22.pdf	147	created 2021-12-06, modified 2022-01-31
Study Manual ... (Restricted Grade)	http://www.niar.org/downloads/Study-Manual-Restricted-Grade-22.pdf	117	same
Study Manual (older edition)	http://www.niar.org/downloads/ham-downloads/Study-Manual.pdf	158	created 2016-05-05

[VERIFIED — all three downloaded and read]

Contents of the 2022 General-Grade manual: FAQs (p.5), Syllabus (p.10), Radio Theory and Practice Restricted (p.14), additional General topics (p.89), Radio Regulations (p.115), Introduction to Morse Code (p.134), Model Question Paper (p.138).

Known defects of the NIAR manual, and how to work around them: its Regulations chapter is written to the 1978 Rules; its fee table is pre-2024; and it carries the errors listed at section 4.6 rows 1–4, 7, 10, 12 and 14. **Use the manual for Section A theory and for Morse. Do not use its Regulations chapter as your primary Section B text — use the 2024 Rules themselves.**

Other free NIAR downloads worth having: the WPC FAQ (<http://www.niar.org/downloads/Frequently-Asked-Questions.pdf>), 2020-08-24, still the fullest statement of exam logistics), the Wireless Monitoring Station address list (<http://www.niar.org/downloads/Wireless-Monitoring-Stations-in-India.pdf>), the online application walkthrough (http://www.niar.org/downloads/ham-downloads/online_guide.pdf) and the VHF/UHF repeater list (<http://www.niar.org/downloads/VHF-UHF-Repeaters.pdf>).

The NIAR Android app has been removed from Google Play ([com.richtree.richcrm.niar](https://play.google.com/store/apps/details?id=com.richtree.richcrm.niar) returns HTTP 404). [VERIFIED — checked 2026-08-23]

NIAR membership, relevant only because it discounts materials: General ₹200 (renewal ₹150), Graded ₹1,000 (renewal ₹500), Life ₹5,000, Donor ₹5,001 and above. [REPORTED — <http://www.niar.org/membership.html>] Joining purely to save ₹10 on the manual is not worth it; join for the club station, library and QSL bureau.

4.8.2 ARSI — AMATEUR RADIO SOCIETY OF INDIA, BANGALORE

ARSI is India's IARU member society, not a coaching body. It runs no ASOC class and publishes no ASOC textbook. [REPORTED — <https://arsi.info/>]

Item	Detail
Address	635, 6th Main Road, next to Bank of India, Hanumanthnagar, Bangalore 560019
President	Ramesh Kumar VU2LU (elected at the 2025-08 AGM)
Website / contact	https://arsi.info/ ; https://arsi.info/contact-us/

Membership category	Fee	Eligibility
Patron Member	₹25,000	Any individual, firm or company
Life Member	₹4,000	—
Annual Member	₹250 + ₹50 joining fee	Holders of a valid amateur radio licence only
5-year Annual	₹1,100	Licence holders
10-year Annual	₹2,000	Licence holders
Institutional Life Member	₹10,000	Licensed clubs

[REPORTED — <https://arsi.info/arsi-membership/> , page undated]

Important: ordinary ARSI membership requires an existing licence, so a pre-exam candidate cannot join as an Annual Member. ARSI is a post-exam organisation for you.

What ARSI does contribute to your preparation:

1. **The best-curated legal reading list in India** — <https://arsi.info/important-wpc-documents/> collects the 2024 FAQ, the 2024 gazette, the Telecommunications Act 2023, NFAP 2011/2018/2022, the 2010 OM, the 2019 OM permitting operation anywhere in India, and the 2016 OM removing Ministry of Home Affairs security clearance. It also maintains an ITU Radio Regulations 2024 page at <https://arsi.info/itu-2024/> . [REPORTED — ARSI, but the documents are government originals]
2. **A structured CW (Morse) training video series** — see section 4.10.

3. The affiliated-club network — twelve clubs listed at <https://arsi.info/affiliated-clubs/>: VU2IIH Indian Institute of Hams (Bangalore), VU2ARC Bangalore Amateur Radio Club, VU2BK Mount Abu, VU2BPA Mumbai Amateur Radio Institute, VU2GIN Kolkata, VU2RDO Mangalore, VU2ECB DARES Mumbai, VU2LCI Lions Club Bangalore, VU2ISL Dhruva Space Hyderabad, VU2THR Tripura, VU2CWN CROWN Mumbai, MARS Chennai. This is **not** a complete map — DARTS, PHARC, WBRC, NBARS and ARCAD are all absent from it. [REPORTED]

No working website could be found for VU2IIH, the Indian Institute of Hams, Bangalore, despite it being ARSI's first affiliate. [UNCERTAIN — approach through ARSI or the Bangalore VHF nets.]

4.9 Regional clubs and training bodies

Ranked by how concrete their teaching offer is. **Fees are unpublished for almost all of them** — that is a finding, not an omission.

Body and city	What they actually offer	Cost	Contact
IACDM / West Bengal Radio Club (WBRC), Kolkata — Indian Academy of Communication and Disaster Management	The most structured course found: "Amateur Radio Communication (HAM Radio Training)", 3 months , Sundays 14:00–16:00 (about 24 contact hours), in person	Not published [UNCERTAIN]	9432919009; iacdmofficial@gmail.com ; vu2mqt@gmail.com ; 17 Railway Park, Sodepur, Kolkata 700110; https://iacdm.in/en-iacdm/ ; https://wbrc.in
OSCAR India / AISMIT, Kolkata (All India School of Management and Information Technology)	"Explore amateur radio and STEM with free online classes" — pan-India online sessions, evidenced by their video back-catalogue	Free	+91 90511 56526; aismitofficial@gmail.com ; https://aismit.org/ (their oscar-india.html page 404s, so the class calendar is unreadable) [UNCERTAIN]

Body and city	What they actually offer	Cost	Contact
DARTS, Delhi National Capital Region (the abbreviation's expansion could not be confirmed — see Part 6, Appendix A.5)	No instructor-led classes. The best free self-test material in India: three downloadable sample papers and five interactive quizzes (section 4.10)	Free	https://darts.org.in/ — note its syllabus page is stale (1973/1978 rules, old pass marks)
NBARS — North Bengal Amateur Radio Society, Siliguri	<i>"NBARS helps aspirants in North Bengal at every one of these steps, free of cost."</i> Informal coaching, mentor pairing, a private set of question papers for registered students, coordinated travel to the exam centre	Free	+91 9832366946; https://www.nbars.in/asoc
PHARC — Pune HAMs Amateur Radio Club (VU2PHQ), Pune	Exam facilitation and coordination, monthly meets (first Sunday), a VHF net, and an ASOC Exam Guide page with current fee figures	Not published [UNCERTAIN]	vu2phq@gmail.com ; http://punehams.in/asoc-exam-guide

Body and city	What they actually offer	Cost	Contact
ARCAD — Amateur Radio Club, Aniruddha's Academy of Disaster Management, Mumbai	Lists classroom trainings, workshops, antenna workshops, a camp and fox hunts, but publishes no dates, duration or fees ; offers a free self-assessment quiz	Not published [UNCERTAIN]	022 26057054 / 56; 601 Link Apartment, Old Khari Village, Khar (West), Mumbai 400052; https://hamradio-arcad.com/ — its "how to become a ham" page describes an obsolete postal procedure; ignore it
Academy of Ham Radio, Vijayawada (Ramesh Babu Arja)	<i>"For the last 38 years ... providing instruction to over 200 groups both online and offline"</i> ; 504 videos	Not published [UNCERTAIN]	https://www.youtube.com/@rdmramesh
RIRCOM / International Radio Communicators Club VU2IRD, Rajapalayam, Tamil Nadu	15-video Section B tutorial playlist, long-form Q&A videos, in-person model exams	Free (videos)	https://rircom.in/resources/faq/ ; https://www.youtube.com/@internationalradiocommunic9014 — spoken language (Tamil or English) not confirmed [UNCERTAIN]
ActiveHams / AARS-KL (ActiveHams Amateur Radio Society, Kerala) / LinuxClub, Kerala (Biju Gopi Thilaka VU2HBI)	Free resource hub plus three free Coursify courses, including "ASOC HAM Radio Exam India" — <i>"Live and Interactive Video Training Sessions"</i> , English with Malayalam video resources	Free	https://www.aars.in/ ; https://hamradio.coursify.me/courses/asoc-ham-radio-exam-india/more_info ; Telegram group linked from aars.in

Body and city	What they actually offer	Cost	Contact
Ham Radio India / Scout Library, Kerala	Claims "classes for ASOC ... conducted by WPC DoT" and application help, but only 13 public videos, mostly 2020–2022	Free	https://www.youtube.com/@HamRadioIndia ; Instagram @hamradio.india
Ham Radio Club, IIT Bombay	Student club; satellite-tracking sessions, not an ASOC course. Its exam page contains two significant errors (section 4.6 rows 8–9)	—	https://www.aero.iitb.ac.in/hrc/hamexam.php

Why clubs run classes at all — and why this can help you. The WPC FAQ permits an examination to be held away from a Wireless Monitoring Station, at a "*School/University/Education Centre/Amateur Radio Society or Club*", if **at least 20 candidates** are gathered. [VERIFIED — WPC FAQ] If your city has no monitoring station, finding a club that can muster twenty candidates is the practical route to a local sitting. The 27 standing centres are Ahmedabad, Ajmer, Bengaluru, Bhopal, Bhubaneswar, Chennai, Dehradun, Dibrugarh, Goa, Gorakhpur, Hyderabad, Jalandhar, Jammu, Kolkata, Lucknow, Mangalore, Mumbai, Nagpur, New Delhi, Patna, Raipur, Ranchi, Shillong, Siliguri, Thiruvananthapuram, Vijayawada and Vishakapatnam. [VERIFIED — WPC FAQ]

Coverage caveat, stated honestly. The research behind this section was conducted largely without a working general web search engine; findings came from direct page fetches, link-following from known hubs, the Internet Archive, the YouTube channel API and Google Play. Treat every statement of the form "no X was found" as "no X was found by these means", not as proof that X does not exist. [Method limitation, disclosed]

4.10 Books, videos, mock tests and Morse resources

4.10.1 BOOKS

Book	Why	Cost
NIAR Study Manual (General or Restricted grade)	The only purpose-written Indian ASOC textbook. Best for Section A theory and Morse. Regulations chapter is stale.	Free PDF ; ₹200 printed and posted, non-member
Biju Gopi Thilaka VU2HBI, Study Manual for Amateur Station Operator's Licence Examination — INDIA	~198–200 pages, and it appends three sample papers including a text transcription of the scanned official WPC paper — the easiest way to actually read the Government's own questions	Free — https://www.aars.in/ham-exam-book.pdf (2021 edition, PDF created 2021-06-10); 2019 edition at https://www.docdroid.net/VyBGBJd/asoc-exam-study-book-2020-ham-radio-india-study-book-by-biju-gt-pdf . Regulations content predates the 2024 Rules
V. K. Mehta, Principles of Electronics (S. Chand)	Cited by NIAR as a source for its own theory chapters; cheap, ubiquitous in Indian bookshops, pitched at exactly the Section A level	Typically ₹400–700 [UNCERTAIN — price not verified; no bookseller could be reached]
ARRL Handbook	Cited by NIAR but far beyond the syllabus . Buy it after you are licensed, not before.	Expensive
ON4UN/ON4WW, Ethics and Operating Procedures for the Radio Amateur	Not examined, but it is the best short guide to actually operating well	Free — https://www.aars.in/Ethics-and-Procedure-for-Radio-Amateur.pdf
William Pierpont N0HFF, The Art and Skill of Radio-Telegraphy	The standard free text on learning Morse	Free — https://www.aars.in/Radio-Telegraphy-bible.pdf

4.10.2 YOUTUBE CHANNELS AND VIDEO COURSES

All free. Metadata read on 2026-08-23. Language attributions are [REPORTED] — assessed from titles and channel descriptions, not by watching.

Tier 1 — structured courses on the Indian syllabus

Channel	URL	Language	What it covers	Note
ASOC Tutorials (India)	https://www.youtube.com/@ASOCIndia	English (one Hindi special on Morse key technique)	The most complete free ASOC course in existence: 51 videos in three series — SD-01 to SD-34+ (Section A theory), "Radio Regulations P1–P5 of 5" (Section B, 16–18 min each), and MC-01 to MC-10 "Learn Morse Code in 10 Days" (42 min to 1 h 24 min each)	644 subscribers. Budget 25–40 hours . Most posted 2020–2022, so the Section B parts predate the 2024 Rules — cross-check against the gazette
OSCAR India	https://www.youtube.com/@oscarindia	English and Hindi	225 videos. Playlists: ASOC Question Answer, Rules and Regulations, Decode Morse Code, Saral Sanchar Registration, Electronics in One Minute	4.9K subscribers — the largest Indian ASOC channel. Its Rules playlist contains " <i>Reforms in new Telecommunications (Amateur Services) Rules, 2024</i> " (posted 2025-01-10), one of the very few Indian study videos covering the 2024 Rules

Channel	URL	Language	What it covers	Note
Bharadwaj VU2NFK	https://www.youtube.com/playlist?list=PLAcwhL3Dop0wJ0niyQFupTCrVT4JomGLW	English	Short, well-structured: ASOC #0 introduction, #1 Section A syllabus, #2–#3 basic electricity, and #A "Rules and Regulations for Ham Radio" (49:42)	Posted 2019; about 2 hours total
RIRCOM VU2IRD	https://www.youtube.com/playlist?list=PLBBn24LAHG5v03meJ5T_80KRstGoYDH0c	English titles; spoken language unconfirmed [UNCERTAIN]	15-video "Tutorials" playlist walking Section B topic by topic: phonetics and numerals, Q-codes, call signs, frequency classification, emission codes, repeaters and satellites, QSL card, log book, equipment register, time signals, documents and fees	Plus long-form Section A and Section B Q&A videos (18–33 min each)

Tier 2 — Hindi and Hinglish

Channel	URL	Note
Radio Tarang VU2RT	https://www.youtube.com/@RadioTarangVU2RT	Hinglish. "Ham Radio Theory Classes zero to Hero" — semiconductors, diodes/transistors/KVL/KCL, AC theory. The freshest Indian ASOC content found (May–June 2026) , but only three theory videos so far
The Amateur Station Operator's Certificate (ASOC)	https://www.youtube.com/@ASOCEXAM	Hindi. "ASOC Exam Syllabus and Introduction to Ham Radio in India", 45:08 , posted 2026-08-10 — the longest Hindi syllabus video found. Brand new; too early to assess
HiTech BABAji	https://www.youtube.com/playlist?list=PLEJ_8N-Lfj4f__9mU8Il0i-jzr10er0JY	Hindi/English, 10 ASOC videos including ITU regulations, emission designation, licence stages, Bharatkosh payment, and "Provisions of Amateur Radio in Telecom Act 2023"
Tutorials Guru	https://www.youtube.com/@TutorialsGuru	Hindi. A 40-minute overview livestream; only three videos in the amateur radio playlist
Pravinder Sheoran	https://www.youtube.com/@pravindersheoran	Hindi. Single popular 6:54 explainer on getting the licence

Tier 3 — Malayalam

Channel	URL	Note
Ham Radio India	https://www.youtube.com/@HamRadioIndia	"Full syllabus for Ham Radio ASOC examination India — Malayalam" (17:02); also a fact-check video on whether a Baofeng BF-888s needs a licence
Scout Library	https://www.youtube.com/playlist?list=PLV9CG6mZ6ALkPqmtzezDh0Mz0JJQeeruG	9-video Ham Radio playlist: online exam application, hall-ticket download, radio programming
SM and SM Mentors	https://www.youtube.com/@SMSMENTORS	"How to get Amateur (HAM) Radio License in India" (11:50)

Tier 4 — useful but not a course

Calling CQ (https://www.youtube.com/@cq_calling , English, all 2026 — includes "ASOC Exam Application Step by Step" posted 2026-08-04, **the most current application walkthrough found**); ARSI VU2ZH (<https://www.youtube.com/@ARSIVU2ZH> , webinars plus the CW course); West Bengal Radio Club (<https://www.youtube.com/@WestBengalRadioClub> , Bengali/English); Ham Radio Club IIT Bombay; Jagdish Pandya VU2JGI (English/Gujarati, form submission process); Arekepudi Suresh Kumar (<https://www.youtube.com/@vuham> , Saral Sanchar registration); Tamiltronics (<https://www.youtube.com/@tamiltronics9663>).

Language coverage — the honest picture [REPORTED]:

Language	ASOC-specific video coverage
English	Good — a full course (ASOC Tutorials India) plus OSCAR India, VU2NFK, RIRCOM
Hindi	Thin but improving — OSCAR India, HiTech BABAji, Radio Tarang, @ASOCExam
Malayalam	Reasonable for syllabus overview and application; no full theory course
Tamil	Essentially none — one 4-minute human-interest video
Bengali	None found as video; NBARS's Bengali written manual is still unpublished [UNCERTAIN]
Telugu, Marathi, Gujarati, Kannada, Odia, Punjabi, Assamese	None found

Because rule 5(4) makes the **examination English-only** [VERIFIED], a candidate whose technical English is weak is best served by a Hindi or Malayalam explainer **plus** the English NIAR manual, not by either alone.

4.10.3 MOCK TESTS AND QUESTION BANKS

Downloadable sample papers (free PDFs):

Paper	URL	Pages	Provenance
Official WPC/DoT sample paper , 50 + 50 MCQs	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-1.pdf	7	Government [VERIFIED]. Image-only scan; no answer key was ever published
DARTS "Amateur Radio Exam Question Paper Sample" (paper 2)	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-2.pdf	19	PDF created 2014-02-24. Club-compiled practice, not a past paper. Roughly half is lifted from the US FCC question pool and several key answers are wrong. Use with caution [REPORTED]
DARTS "Sample ASOC Exam Question Paper" (paper 3)	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-3.pdf (mirror https://www.aars.in/ASOC-Exam-Sample-Question-Paper-3.pdf)	15	PDF created 2014-05-14. Club material, but its rubric matches the official paper [REPORTED]

Interactive online quizzes with instant scoring (free):

Quiz	Page
DARTS — Test Paper: Resistance	https://darts.org.in/test-paper-1/

Quiz	Page
DARTS — Test Paper: Capacitance	https://darts.org.in/test-paper-capacitance/
DARTS — Test Paper-1, Regulations Section B (30 questions)	https://darts.org.in/test-paper-1-regulation/
DARTS — Test Paper-2, Regulations Section B	https://darts.org.in/test-paper-2-regulation/
DARTS — Test Paper-3, Regulations Section B	https://darts.org.in/test-paper-3-regulation-sec-b/
ARCAD — "HAM Radio self-assessment Test by ARCAD" (~14 items)	https://hamradio-arcad.com/post/ham-practice-test
NIAR — about 50 solved Section B model questions	NIAR General-Grade manual, pp. 138–147

[REPORTED — darts.org.in and hamradio-arcad.com; DARTS quizzes are Google Forms that show the correct answer at the bottom of each response]

The apps finding, and the warning attached to it. A Google Play search on 2026-08-23 (India storefront) for "ASOC exam", "amateur station operator certificate India", "ham radio india license exam" and "ASOC ham exam India mock test" returned **no app targeting the Indian ASOC syllabus**. Everything returned targets the US FCC, Canadian, European or Bulgarian exams. [REPORTED — search results] The NIAR app has been removed. [VERIFIED]

Do not prepare with an FCC Technician app. Section B is entirely India- and ITU-specific, and even Section A differs in emphasis. Worse, a US app will teach you US band plans and US power limits — exactly the confusion that can get an Indian operator prosecuted.

A note on "past papers". No actual past paper from the current post-2010 MCQ regime has ever been published, by WPC or anyone else; candidates do not get their papers back. The official sample paper is the only artefact carrying the Government of India masthead, and it has no answer key. Anything advertised online as "ASOC previous year questions" is either that sample paper, a club compilation, or a remix of the two. [VERIFIED / UNCERTAIN — a private leaked paper cannot be disproved, only shown to be unindexed.] Part 5 of this report separates documented questions from constructed practice questions for exactly this reason.

4.10.4 MORSE (CW) RESOURCES — GENERAL GRADE ONLY

The bar: 8 wpm sending and receiving, five minutes, 5 errors maximum, both halves passed in the same sitting.

Resource	Type	Cost	URL
ARSI CW Training series — 15+ parts, Koch-style by letter groups (A/U/V/4, then N/D/B/6, ...); parts 1–12 posted 2024, parts 13–15 in 2025-09	YouTube playlist	Free	https://www.youtube.com/playlist?list=PLLzU02LrrhrzhL5aofid1W578FQ6k07YD

Resource	Type	Cost	URL
ASOC Tutorials (India) — "Learn Morse Code in 10 Days" MC-01 to MC-10, plus "Are you ready for Morse code receiving Examination?"	YouTube series	Free	https://www.youtube.com/@ASOCIndia
OSCAR India — "Decode Morse Code"	YouTube playlist	Free	https://www.youtube.com/playlist?list=PLR5acT8eNNJ_Xrl9PwbMTNjLhImFMiQWx
AARS Koch-method trainer	Browser app	Free	https://www.aars.in/koch.html
AARS online Morse key (sending practice)	Browser app	Free	https://www.aars.in/keying.html
AARS Morse player and Morse game	Browser apps	Free	https://www.aars.in/morse-player.html ; https://www.aars.in/simple-morse.html
IZ2UUF Morse Koch CW	Android app	Free	https://play.google.com/store/apps/details?id=net.iz2uuf.cwkoch
Coursify "Interactive Morse Code Training" (VU2HBI, VU2DPN)	Live online course	Free	https://hamradio.coursify.me/
NIAR Morse Code Practice Book / Key / Oscillator	Book and hardware	₹50 / ₹200 / ₹300 non-member; plus postage	Order from NIAR, demand draft only
NIAR manual ch. 5, "Introduction to Morse Code"	PDF chapter	Free	http://www.niar.org/downloads/Study-Manual-General-Grade-22.pdf pp. 134–139

[All REPORTED except the NIAR prices, which are REPORTED from an undated page, and the free status of the web tools, which was confirmed reachable on 2026-08-23]

NIAR's own links page recommends a software-download aggregator for Morse practice; that is not a first-party source and should not be followed.

4.10.5 TOTAL COST

Statutory and unavoidable [VERIFIED — G.S.R. 675(E), rules 5(2) and 6(1); the "both grades ₹200" figure from the WPC FAQ]:

Item	Cost
Examination fee, one grade	₹100
Examination fee, both grades in one sitting	₹200
ASOC certificate, 20-year validity	₹1,000

Item	Cost
ASOC certificate, lifetime (to age 80)	₹2,000
Duplicate certificate, or permanent change of location	₹1,000

A genuinely zero-rupee study route exists: the free NIAR PDF manual, the 2024 Rules gazette, the official sample paper, the ASOC Tutorials (India) video course and the DARTS quizzes. Total out-of-pocket for a Restricted-grade candidate is then **₹100 examination + ₹1,000 certificate = ₹1,100**, plus the radio.

4.11 A concrete study plan

This plan is this report's own synthesis, not any organisation's advice. The only sourced duration estimates available are NIAR's *"just a few weeks training"* [REPORTED] and NBARS's *"most candidates are ready in four to eight weeks of steady preparation"* [REPORTED]. The plan below sits inside that range for the written paper, and adds a longer parallel track for Morse.

4.11.1 CHOOSE YOUR VARIANT FIRST

Variant	Who	Written scope	Duration
A	Engineer or science graduate/diplomate in electronics or telecommunications, going for Restricted	Section B only (exemption claimed)	2–3 weeks , part time
B	No qualifying degree, going for Restricted	Sections A and B	6 weeks , part time
C	Going for General (25 W, HF, satellites)	Sections A and B plus 8 wpm Morse	6 weeks written, running in parallel with 8–12 weeks of daily Morse

Morse is the long pole. If you start General-grade preparation today, the Morse track — not the theory — sets your exam date. This report's estimate of 6–10 weeks from zero to 8 wpm at 20–30 minutes of daily practice is a synthesis, not a sourced figure; treat 12 weeks as a safer plan.

4.11.2 THE SIX-WEEK PLAN (VARIANT B; VARIANT A DOES WEEKS 3–6 ONLY; VARIANT C ADDS THE MORSE COLUMN THROUGHOUT)

Week	Section A (theory)	Section B (regulations)	Drills	Morse (General only)
0 (about 4 hours)	—	Read the Telecommunications (Amateur Services) Rules, 2024 gazette end to end, including Annexure A . It is short. Print it. Then read the 2010 Annexure II syllabus.	Sit the official WPC sample paper cold. You will score badly; that is the point. It calibrates the depth required, which is far shallower than the NIAR manual's bulk suggests.	Install the AARS Koch trainer or IZ2UUF. Learn the first letter group.
1	NIAR manual ch. 3: electricity and magnetism; AC theory. Back up with ASOC Tutorials SD-series for anything that does not land.	ITU Article 2 band nomenclature; Article 25 amateur services; the Indian band/power/emission table (section 4.4.6).	DARTS quizzes: Resistance, Capacitance.	20–30 minutes daily. Groups 1–2.
2	Semiconductors; radio receivers (spend real time on the superheterodyne block diagram); transmitters and oscillators.	ITU Appendix 1 emission designators — drill until you can decode A1A, F3E, J3E and write 3K00A3E correctly. Appendix 14 phonetics including the figure code .	Re-attempt the Section A half of the sample paper.	Groups 3–4.
3	Propagation; aerials; frequency measurement. (Variant A starts here.)	The 30 Q-codes and 17 abbreviations, plus BK and 73. Learn both QRQ and QRO .	DARTS Regulation Test Paper-1 (30 questions).	Groups 5–6. Start the ARSI CW series for structure.
4	<i>General grade only:</i> the additional topics — modulation, transformers, power supplies, facsimile and television, ionosphere and MUF, microwave aerials, satellites. <i>Restricted:</i> revise weak areas.	Annexure A operating procedure: messages permitted and forbidden, third-party traffic, log book (IST, one year), equipment register, secrecy, call and reply, AR/VA, the 30-second test rule, the ten-minute call-sign rule.	DARTS Regulation Test Papers 2 and 3.	Full alphabet. Begin sending practice on the AARS online key.

Week	Section A (theory)	Section B (regulations)	Drills	Morse (General only)
5	Weak-area revision using ASOC Tutorials videos.	Distress, urgency and safety (SOS/MAYDAY, XXX/PAN PAN, TTT/SECURITE, priorities, 500 kHz, 2182 kHz); RST; HX/H24/HJ/HN; standard frequency and time signals (ATA/NPL); repeaters and the 600 kHz shift; call-sign structure and Appendix 42; infringement categories.	The ~50 solved model questions at the back of the NIAR manual; the ARCAD self-assessment.	Push receiving speed to 8 wpm. Time yourself on five-minute passages.
6	—	Read the errors table at section 4.6 again and check every one of your memorised facts against it.	Re-sit the official sample paper under timed conditions (1 hour for Restricted, 2 hours for General). Then DARTS papers 2 and 3, discounting their US-derived items.	Full mock: five minutes receive, five minutes send, 5 errors maximum, both in one sitting.

4.11.3 SEQUENCING THE APPLICATION AGAINST THE STUDY PLAN

When	Action
End of week 1	Register on the DoT portal and start the application. Rule 5(1) requires the application "one month before the date of such examination" [VERIFIED], but no exam date exists until your application has been scrutinised, so treat the month as a lead time, not a computable deadline. [UNCERTAIN how WPC reconciles this]
Same time	Pay the ₹100 examination fee (₹200 if sitting both grades). Upload the Section A exemption proof if you are claiming it.
Ongoing	There is no published ASOC examination calendar. The FAQ says only: <i>"The Date, Venue and Time of the examination will be intimated to the candidate by the respective Monitoring Station. The examination will be conducted as and when there are sufficient number of candidates."</i> [VERIFIED] NIAR's 2025 guidance says "almost every month" [REPORTED]. Do not plan around a fixed date.
On notification	Download the e-admit card from the portal. Report 30 minutes early. No mobile phones, calculators or watches in the hall. [REPORTED — DoT/NIAR admit-card instructions]
On the day	Carry originals, including the degree or diploma if you claimed the Section A exemption. [VERIFIED — FAQ Q16]

When	Action
If you fail	Simply reapply, with a fresh ₹100 fee. No attempt limit appears anywhere in the 2024 Rules. The widely repeated claim of a compulsory one-month waiting period could not be traced to any source and is probably a garbled restatement of the rule 5(1) lead time. [UNCERTAIN — not established]

Portal warning. DoT's eServices site carries a notice, last updated 2025-08-08, directing applicants for authorisations under the Telecommunications Act, 2023 to <https://eservices.dot.gov.in/authorisation-portal> from 2026-06-25, and stating that Saral Sanchar will not accept new applications for those categories. **Whether ASOC is inside that list is unclear** — the notice enumerates service authorisations and the ASOC page itself still routes to Saral Sanchar. [VERIFIED that the notice exists; UNCERTAIN whether it captures ASOC.] **Every "how to apply on Saral Sanchar" video and PDF in this part may therefore be procedurally stale. Check the live portal before following any walkthrough.**

4.11.4 A FOUR-ITEM PRE-EXAM CHECKLIST

Check	Why
Can you state the pass mark you are being held to?	40 % in each part and 40 % overall is the current position, but the old 50 %/60 % figures are still published on a government domain. Aim for 60 %+ . [VERIFIED / UNCERTAIN]
Have you learned Section B from the 2024 Rules , not from a manual's regulations chapter?	Every mainstream Indian study text still teaches the superseded 1978 Rules. [VERIFIED]
Do you know both QRQ and QRO , and both the five-dot and seven-dot word-spacing answers?	The syllabus and the study material disagree on both. Answer QRQ if the option exists; answer five dots . [VERIFIED]
Have you attempted the official WPC sample paper at least twice?	It is the only artefact carrying the Government's masthead, and it is free. [VERIFIED]

4.12 What remains genuinely unresolved in this part

#	Open question	Who to ask
1	Has WPC published a syllabus under rule 5(3) of the 2024 Rules? None was found on any government domain. The 2010 Annexure II appears to remain operative by default. [UNCERTAIN]	WPC COP Section: awacop-dot@gov.in , 011-2303 6181; or Sr. Deputy Wireless Advisor, 011-23036540
2	Is wpc.dot.gov.in permanently retired? Unreachable throughout the research. If it returns, re-check the syllabus and sample-paper files for newer versions. [UNCERTAIN]	WPC helpline 011-2303 6534 / 011-2303 6539

#	Open question	Who to ask
3	Is the sample paper still representative? It is undated, first archived 2016-08-04, and cites the 1978 Rules. Whether the live paper has been updated for the 2024 Rules is unknown. [UNCERTAIN]	WPC
4	If Section A is exempted, is the 40 % applied to Part B alone, and is the total out of 50? No source addresses it. [UNCERTAIN]	WPC, in writing, before relying on the exemption
5	Which band figures would an examiner mark correct — the 2010 OM's 1820–1860 kHz and 50–54 MHz, or NIAR's 1800–1825 kHz and 50–52 MHz? And is 30 m (10100–10150 kHz) authorised at all? It is absent from Annexure-I for both grades but present in NIAR's table (§4.4.6). [UNCERTAIN]	WPC
6	What is NIAR's course fee, batch length and calendar? Published nowhere. [UNCERTAIN]	+91 40 2331 0287 / info@niar.org
7	What do IACDM (Kolkata), ARCAD (Mumbai), the Academy of Ham Radio (Vijayawada) and PHARC (Pune) charge? All unpublished. [UNCERTAIN]	The contacts in section 4.9
8	Has NBARS published its Bengali/Hindi/English ASOC manual? Still "compiling" as of 2026-08-23. It would be the first of its kind. [UNCERTAIN]	https://www.nbars.in/asoc
9	Is Morse actually being administered at every Wireless Monitoring Station? The rules and the December 2024 FAQ require it for General grade; no recent candidate account confirming it in practice was found. [UNCERTAIN]	Your chosen monitoring station, before you commit to General grade
10	Where is an authoritative full text of the Indian Wireless Telegraph Rules, 1973? It is one of only two national instruments the syllabus names, and no reliable copy could be located. [UNCERTAIN — a real gap in preparation]	WPC
11	What is the source of the four "infringement" categories (unauthorised frequency / emission / period / call sign)? Not in ITU RR Appendix 9; probably Wireless Monitoring Organisation practice. Examined nonetheless. [UNCERTAIN]	WPC

This part is a research summary compiled on 2026-08-23, not legal advice, and not an endorsement of any course, club or seller. Fees, syllabi, portals and course offerings change; several prices above are taken from undated or infrequently updated web pages and should be confirmed by telephone before payment. Confirm the current syllabus, examination pattern and pass marks with the WPC Wing, DoT (011-2303 6534 / 011-2303 6539; awacop-dot@gov.in) or with a licensed amateur radio club before relying on any figure here.

Part 5 — The question bank: what has actually been asked, and what could be asked

A naming warning before anything else. This part has two halves, and so does the examination itself, and they are not the same thing. In this part:

Label used here	Meaning
Part 5A — Documented questions	Real questions traceable to a published document. Nothing invented.
Part 5B — Practice questions	Questions constructed from the published syllabus. Not past papers.
"the paper's Section A"	The examination's own first half — Radio Theory and Practice , 50 marks
"the paper's Section B"	The examination's own second half — Radio Regulations , 50 marks

Abbreviations used throughout and expanded here on first use: **ASOC** — Amateur Station Operator's Certificate (called a "Licence" in older material; the Telecommunications (Amateur Services) Rules, 2024 call it a Certificate). **WPC** — the Wireless Planning and Coordination Wing of the **DoT**, the Department of Telecommunications, Ministry of Communications, Government of India. **MCQ** — multiple-choice question. **ITU** — International Telecommunication Union; **RR** — its Radio Regulations. **NIAR** — National Institute of Amateur Radio, Hyderabad. **DARTS** — the amateur radio club at darts.org.in, based in the Delhi National Capital Region; published sources expand the abbreviation inconsistently and no expansion could be confirmed, so it is left unexpanded here **[UNCERTAIN]**. **SIARS** — South India Amateur Radio Society, Chennai. **AARS** — ActiveHams Amateur Radio Society, Kerala; **INDRA** — the LinuxClub imprint that republished its study book. **ARSI** — Amateur Radio Society of India. **FCC** — the United States Federal Communications Commission. **OM** — Office Memorandum. **G.S.R.** — General Statutory Rule, the numbering used for subordinate legislation in the Gazette of India.

5.1 How to read this part

The request behind this report was for "the questions asked till now, plus the questions that could be asked". Those are two different things, and blending them would be the single most damaging thing this document could do to a candidate. So they are kept rigorously apart:

- **Part 5A** contains only questions transcribed from a real published document that was fetched and read during the research. Each block states the source, the source type, the document date and whether the source printed an answer key. Where an answer is supplied that the source did not print, it is marked as a derivation, not as an official key.
- **Part 5B** contains questions written from scratch against the published syllabus. Not one of them is a reproduction or a recollection of a question that has appeared in an ASOC examination. The whole of Part 5B is labelled as constructed practice.

Evidence tags carry through from the research exactly as they arrived. **[VERIFIED]** means a primary or official source was fetched — a gazette notification, the bare text of an Act, a WPC or DoT document, the ITU Radio Regulations. **[REPORTED]** means a named credible secondary source — NIAR, ARSI, an established club, a study book. **[UNCERTAIN]** means it could not be confirmed, the sources disagree, or the document is undated. No hedge has been upgraded into a certainty anywhere in this part.

5.2 If you only read one page

Genuine, traceable question count: 774 question items, from 7 distinct published documents, catalogued below as 8 source blocks (S1–S8, because the NIAR Study Manual supplies two). That splits into **670 objective items** (multiple-choice or single-answer short questions) and **104 descriptive or essay items** from the pre-2010 papers. [REPORTED — compiled 2026-08-23 from the seven documents listed at A.2, each fetched and read]

#	Source	Type	Questions	Answer key published?	Reliability
S1	Official WPC/DoT "Sample question paper ... Amateur Station Operator's Licence (General and Restricted Grade)"	Official government paper	100	No	Highest. The only artefact in existence carrying the Government of India masthead and the real exam rubric.
S2	NIAR Study Manual — 8 "Exercise" blocks plus "Model Questions & Answers"	Institute question bank	218	Partly (75 of 218)	High for regulations, mixed for theory; contains at least two demonstrable factual errors.
S3	DARTS "ASOC Exam Sample Question Paper 3"	Club-published practice paper	100	No	Medium. Correct rubric, Indian content, but no stated provenance.
S4	DARTS "Amateur Radio Exam Question Paper Sample" (paper 2)	Club-published practice paper	188	Yes, full key	Medium-low. Roughly half lifted verbatim from the US FCC pool; at least seventeen key answers wrong.
S5	brainstorms.in / VU2SWX — 7 Grade I / Grade II past papers , three carrying real 1990 exam dates	Historical government papers , collected by a candidate	7 papers = 114 items (41 MCQ + 73 descriptive)	No	High as historical artefacts; obsolete as a format guide.

#	Source	Type	Questions	Answer key published?	Reliability
S6	SIARS "Queries: ASOC Exam questions"	Club question-and-answer page	11	Yes, with reasoning	Medium. A club officer's answers, not WPC's.
S7	AARS/INDRA free study book, Annexure-III sample paper	Book-published descriptive paper	31	No	Medium. Pre-2010 descriptive format; useful as history.
S8	NIAR Study Manual "Model Type Questions" block — the same document as S2	Institute question bank	12 (these 12 plus S2's 218 make the 230 counted at A.4)	No	Medium.

Three honest negatives, and they matter more than the count:

- 1. There is no published, official, answered question bank for the ASOC.** WPC/DoT has released exactly one artefact — the undated seven-page sample paper (S1) — and it prints **no answer key**. [VERIFIED — the scanned paper has empty [] answer boxes at the right margin; <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-1.pdf>, fetched 2026-08-23]
- 2. No actual past paper from the current post-2010 MCQ regime has ever been published,** by WPC or by anyone else. Candidates do not get their papers back. Everything circulating as an "ASOC question paper" is either S1, a club compilation, or a re-mix of the two. [UNCERTAIN — it cannot be proved that no leaked paper exists somewhere private; only that none is publicly indexed.]
- 3. Candidate recollections are, in retrievable form, essentially nonexistent.** Zero blog posts, forum threads or Quora answers were found that reproduce a remembered ASOC question verbatim. The single most promising lead, a Reddit post describing one candidate's Restricted-grade experience, could not be retrieved at all. See A.10. **Treat any claim of "these are the questions that have been asked" with suspicion.**

What to do with that. The 50 questions that make up the Restricted-grade portion of the official sample paper (S1, the paper's Section A Q1–Q25 and Section B Q1–Q25, reproduced in full at A.3) are the single highest-value hour of study available anywhere in India. Everything else in Part 5A is supporting drill, and Part 5B exists because the documented haul, honestly counted, does not cover the syllabus.

Part 5A — DOCUMENTED QUESTIONS

Every question in this half is transcribed from a real published document or web page that was fetched and read during the research, on 2026-08-23 unless a different date is stated. Nothing here is invented. Constructed questions are in Part 5B and are deliberately absent from Part 5A.

A.1 Why there are no official past papers, and what exists instead

WPC does not publish past ASOC papers. It publishes one undated sample paper, and it publishes no answer key for it. The examination is set and conducted by the DoT monitoring stations, candidates do not retain their question papers, and no right-to-information disclosure of a past paper surfaced during this research. [VERIFIED — the WPC and Saral Sanchar official pages describe format, marks and pass criteria but publish no questions beyond S1; see the process research at [r05-asoc-exam-process.md](#)] [UNCERTAIN — whether any past paper has been released to an individual applicant privately]

The consequence for a candidate is precise and worth stating plainly: **the phrase "questions asked till now" has no honest answer for the post-2010 ASOC.** What exists is one government sample paper, one institute question bank, two club compilations, one club question-and-answer page, one book appendix, and one collection of seven genuine government papers from an examination format that was abolished in 2010 — seven documents in all. All of them are below.

A methodological warning about circular sourcing. Four of the apparently "different" question banks in circulation are largely the same material re-labelled: S1 is reprinted (badly transcribed) inside the AARS/INDRA book; DARTS paper 2's regulations items 36–60 are lifted from S5's 1990s Grade I/II blocks; the SIARS list is drawn from S1 and S4; and the AARS book reprints DARTS papers 2 and 3 as its own appendices. **Do not mistake repetition for corroboration.** When two sources agree, check first whether one simply copied the other. [REPORTED — established by comparing the documents directly]

A.2 Source register — full provenance

ID	Document	URL	Date evidence
S1	"Sample question paper of the examination for award of Amateur Station Operator's Licence (General and Restricted Grade)", Government of India, Ministry of Communications & IT, Department of Telecommunications. 7 pp, image-only scan.	Original (now returns 404): http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf · Archived: https://web.archive.org/web/20160804195151if_/http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf · Live byte-identical mirror: https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-1.pdf	Undated on its face. Earliest Internet Archive capture 2016-08-04 .
S2 / S8	NIAR, <i>Study Manual for Amateur Station Operator's Licence Examination</i> (Restricted and General), 158 pp	http://www.niar.org/downloads/ham-downloads/Study-Manual.pdf	Undated in the text. NIAR is the DoT-recognised national training institute.

ID	Document	URL	Date evidence
S3	"SAMPLE ASOC EXAM QUESTION PAPER", 15 pp	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-3.pdf · mirror https://www.aars.in/ASOC-Exam-Sample-Question-Paper-3.pdf	PDF creation date 2014-05-14 ; uploaded to the DARTS site 2018-02.
S4	"AMATEUR RADIO EXAM QUESTION PAPER SAMPLE", 19 pp	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-2.pdf	PDF creation date 2014-02-24 .
S5	question_paper_for_ham_radio_exam.zip → folder QPAPER/ containing Grade 1&2.TXT , Grade 2-1.TXT ... Grade 2-6.TXT	http://brainstorms.in/pdf/question_paper_for_ham_radio_exam.zip — linked from https://brainstorms.in/?p=247	Files timestamped 2003-05-12 ; blog post dated 2009-01-16 ; three papers carry printed sitting dates 1990-11-12 , 1990-12-10 and 1990-12-26 .
S6	"Queries: ASOC Exam questions"	https://www.siars.org.in/asoc_exam_qp1/	Page posted 2019-01-13 .
S7	Biju Gopi Thilaka VU2HBI, <i>Study Manual for Amateur Station Operator's Licence Examination — INDIA</i> , INDRA/LinuxClub 2021 edition, 198 pp, Annexure-III	https://www.aars.in/ham-exam-book.pdf	PDF creation date 2021-06-10 .

Provenance note on S5, which is important. The blog author, **Sunil TT, VU2SWX**, writes: "*I am posting some sample papers that I collected when I wrote the examination.*" The files themselves carry the masthead "**GOVERNMENT OF INDIA / MINISTRY OF COMMUNICATIONS / MONITORING ORGANIZATION**" and three carry real sitting dates. These are therefore the closest thing that exists in public to genuine *past papers* — but they belong to the pre-2010 **Grade I / Grade II descriptive** regime, which no longer exists. They are evidence of what was once asked, not of what will be asked. [REPORTED — <https://brainstorms.in/?p=247>, fetched 2026-08-23]

Method note on S1. The official sample paper is an image-only scan with no text layer. It was rendered to PNG at 200 dots per inch and transcribed by reading the images directly. Where the photocopy is genuinely ambiguous, the transcription says so rather than guessing.

A.3 S1 — the official WPC/DoT sample paper (100 questions)

Source type: official government paper. This is the single authoritative artefact, and the only one with a Government of India masthead. No answer key exists.

Every "→" line below is a **derivation with justification**, not a government answer. Where the paper itself is defective — and it is, in at least six places — the defect is flagged rather than smoothed over.

The paper's own rubric, transcribed verbatim from the scan:

Note: 1. All questions are of Multiple Choice type. 2. Maximum marks 100. 3. Restricted grade candidates have to answer the first 25 questions (each question carries Two marks) from each section and the time allotted for it is One hour. 4. General grade candidates have to answer all the 50 questions (each question carries One mark) from each section and the time allotted for it is Two hours.

[VERIFIED — transcribed from the scan of S1]

Practical consequence for a Restricted-grade candidate: only **the paper's Section A Q1–Q25** and **Section B Q1–Q25** are yours. Those 50 questions are the highest-value study material in this entire report.

A.3.1 THE PAPER'S SECTION-A (RADIO THEORY AND PRACTICE), Q1–Q50

Restricted grade: answer **Q1–Q25 only**.

A1. The Transformer works on the principle of a) Self Inductance b) Mutual Inductance c) both a & b d) None → **(b)**. A transformer transfers energy because flux produced by the primary links the secondary — which is the definition of mutual inductance.

A2. The Frequency of oscillation of a series RLC resonant circuit is a) $f=1/2\pi\sqrt{LC}$ b) $f=1/\pi\sqrt{LC}$ c) $f=2\pi\sqrt{LC}$ d) None → **(a)**. Resonance is where $X_L = X_C$, that is $2\pi fL = 1/(2\pi fC)$, giving $f = 1/(2\pi\sqrt{LC})$.

A3. The Capacitor is a a) Passive device b) Active device c) both a & b d) None → **(a)**. A capacitor stores energy but cannot supply net power gain.

A4. The frequency of a sinusoidal signal $v(s)=10\sin 100t$ volts is a) $100/2\pi$ c/s b) 50 c/s c) 100 c/s d) $100t$ c/s → **(a)**. $\omega = 100$ rad/s, and $f = \omega/2\pi = 100/2\pi$ cycles per second (approximately 15.9 Hz).

A5. The resistance value of an Insulator is a) Zero b) Very High c) Less than unity d) None → **(b)**.

A6. The Conductance is proportional to a) Resistance b) Inverse of Resistance c) Reactance c) Inverse of Reactance → **(b)**. $G = 1/R$. (Note that the paper mislabels the fourth option "c" instead of "d" — a defect in the government document itself.)

A7. The power can be expressed as a) $V=IR$ b) $P=VI$ c) $P=I^2R$ d) both b & c → **(d)**. Both $P = VI$ and $P = I^2R$ are valid; $V = IR$ is Ohm's law, not a power formula.

A8. The Law behind electromagnetism is a) Newton's law b) Faradays law c) Kirchoffs law d) keplers law → **(b)**. Faraday's law of electromagnetic induction.

A9. As per Kirchhoff's voltage law, in a closed circuit a) All the branch voltages meeting at a node should be zero b) All the branch currents meeting at a node should be zero c) The sum of all the voltage drops is equal to applied voltage d) None of the above → **(c)**. Option (b) is Kirchhoff's *current* law — the standard distractor.

A10. The Impedance of a resonant circuit is a) Zero b) Equal to resistance c) Unity d) Equal to reactance → **(b)**. At resonance the reactances cancel and only the resistive part remains. *(Strictly true of a series resonant circuit; a parallel resonant circuit shows maximum, dynamic resistance. The paper does not distinguish.)*

A11. The Bridge rectifier is a a) Full-wave type b) Half-wave type c) both a & b d) None of the above → **(a)**.

A12. A Diode can be used as a) Rectifier b) Detector c) Switch d) All the above → **(d)**.

A13. An Oscillator is based on a) Positive feedback b) Negative feedback c) No feedback d) both a & b → **(a)**. The Barkhausen criterion: loop gain of at least 1 with zero phase shift, that is, positive feedback.

A14. The output signal of an CE amplifier will be a) 45 deg out of phase b) 90deg out of phase c) 180 deg out of phase d) 270 deg out of phase → **(c)**. The common-emitter stage is the inverting configuration.

A15. In the process of modulation in communication system a) Signal frequency is more than carrier frequency b) Signal frequency is less than carrier frequency c) Signal frequency will be directly transmitted d) Signal frequency equals carrier frequency → **(b)**.

A16. The typical value of IF in a Superhetrodyne receiver is a) 450KHz b) 455KHz c) 500KHz d) 405KHz → **(b) 455 kHz**, the universal amplitude-modulation broadcast intermediate frequency. *(The AARS book's transcription of this paper prints option (a) as "430KHz"; the scan clearly reads 450KHz. Either way the answer is 455 kHz.)*

A17. The output signal of a detector in a communication receiver is in a) Audio Frequency range b) Radio Frequency range c) Intermediate Frequency range d) None of the above → **(a)**. Detection, or demodulation, recovers the audio.

A18. In a Radio receiver, the sensitivity will be governed by a) RF b) Detector c) AF d) None of above → **(a)**. Sensitivity is set by the noise figure and gain of the front-end radio-frequency stage.

A19. The noise in a Radio receiver can be expressed as a) S/N ratio b) N/S ratio c) Level of reference signal d) None of above → **(a)**.

A20. The following type of Oscillator will be more stable a) RC Oscillator b) Hartley Oscillator c) Colpitts Oscillator d) Crystal Oscillator → **(d)**. Quartz has a very high Q factor and a low temperature coefficient.

A21. The length of a dipole for a signal frequency of 300MHz is a) 10 mtrs b) 1 mtr c) 0.1 mtr d) 1 cm → **(b) 1 metre — as the examiner intends**, taking "length of a dipole" to mean one wavelength ($\lambda = 300/300 \text{ MHz} = 1 \text{ m}$). **[UNCERTAIN — the physically correct half-wave dipole is 0.5 m, which is not offered.]** Mark (b); do not over-think it in the hall.

A22. The frequency 7100 KHz can be transmitted by a) Tropospheric propagation b) Ionospheric propagation c) Ground wave propagation d) Surface wave propagation → **(b)**. 7.1 MHz is in the high-frequency range; long-haul HF works by ionospheric, or sky-wave, refraction.

A23. The most suitable antenna for receiving VHF range is a) Dipole b) Log periodic c) Yaagi d) None of the above → **(c) Yagi**, the answer the Indian study material consistently gives, on grounds of gain and directivity in the very-high-frequency band. **[UNCERTAIN — a plain dipole also receives VHF perfectly well; this question rewards the textbook answer, not the most defensible one.]**

A24. The skip distance is the distance between a) Transmitter and Receiver b) Transmitter and Ionospheric layer c) Receiver and Ionospheric layer d) Between two Ionospheric layers → **(a)**. Skip distance is the shortest ground distance from the transmitter at which a sky wave of a given frequency returns to earth — transmitter to receiver.

— end of the Restricted-grade portion of the paper's Section A —

A25. In A1A transmission a) Amplitude of the carrier is modulated b) Frequency of the carrier is modulated c) Both a & b d) None of above → **(a)**. A1A = amplitude modulation (A), single channel with no sub-carrier (1), telegraphy for aural reception (A) — that is, on-off keying of the carrier amplitude.

A26. The colour bands of a $4.7\text{K}\Omega \pm 5\%$ resistor are a) Yellow, Violet, Orange and Gold b) Yellow, Violet, Red and Gold c) Yellow, Violet, Orange and Silver d) Yellow, Violet, Red and Silver → **(b)**. Yellow = 4, Violet = 7, Red = $\times 100$, giving 4700Ω ; Gold = $\pm 5\%$.

A27. A signal(f_m) is amplitude modulated with a carrier(f_c), the frequency value of Lower Side Band is a) $f_m - f_c$ b) $f_m + f_c$ c) $f_c - f_m$ a) $f_c + f_m$ → **(c)**. The lower sideband is $f_c - f_m$. (The paper labels the last option "a" twice — another defect in the government document.)

A28. The doping concentrations of Emitter, Base and Collector in a NPN transistor is a) Heavy, Moderate and Light b) Heavy, Light and Moderate c) Moderate, Light and Heavy d) Moderate, Heavy and Light → **(b) Heavy, Light and Moderate**. Standard bipolar junction transistor construction: the emitter is heavily doped so that it injects carriers, the base is very lightly doped and thin so that few carriers recombine, the collector is moderately doped and physically largest. **[UNCERTAIN — several Indian coaching keys circulate answer (a). (b) is the textbook-correct one; if the examiner's key says (a), the question is defective, not you.]**

A29. The wavelength of a signal $v(s) = 5\cos 6\pi \times 10^6 t$ is a) 100m b) 50m c) 100cm d) 50cm → **(a) 100 m**. $\omega = 6\pi \times 10^6$ gives $f = 3 \times 10^6 \text{ Hz} = 3 \text{ MHz}$, so $\lambda = 300/3 = 100 \text{ m}$.

A30. The Mutual Inductance(M) between two coils of Self Inductance L_1 and L_2 respectively can be expressed as a) $M = \sqrt{L_1 L_2}$ b) $M = 1/\sqrt{L_1 L_2}$ c) $M = k\sqrt{L_1 L_2}$ d) $M = k/\sqrt{L_1 L_2}$ → **(c)**. k is the coefficient of coupling, with $0 \leq k \leq 1$.

A31. For a Step-down Transformer, the relation between the number of turns in primary (N_p) and the number of turns in secondary (N_s) is a) $N_s > N_p$ b) $N_s < N_p$ c) $N_s = N_p$ d) None of the above → **(b)**. Stepping the voltage down requires fewer secondary turns.

A32. The layer that reflects higher end frequencies in HF range during the day time is a) F_1 layer b) F_2 layer c) E layer d) D layer → **(b) F_2** . The F_2 layer has the highest electron density and returns the highest HF frequencies.

A33. The power conversion efficiency of a Full-wave rectifier is a) 40.6% b) 81.2% c) 100% d) Zero → **(b) 81.2 %**. Half-wave is 40.6 % — option (a) is the paired distractor.

A34. The length of dipole, reflector and director of Yaagi Antenna in terms of wave length are a) 0.55λ , 0.5λ and 0.45λ b) 0.45λ , 0.5λ and 0.55λ c) 0.5λ , 0.55λ and 0.45λ d) 0.5λ , 0.45λ and 0.55λ → **(c)**. The driven element is about 0.5λ ; the reflector is *longer*, about 0.55λ ; the director is *shorter*, about 0.45λ . Note the order the question asks for: dipole, reflector, director.

A35. The side-band system adopted for Television transmission a) Independent Wideband b) Vestigial Wideband c) Suppressed Sideband d) All the above → **(b) Vestigial Sideband**. (*The scan reads "Vestigial Sideband"; the AARS transcription mangles it to "Vestigial Wideband".*) Vestigial sideband conserves bandwidth while preserving the low video frequencies.

A36. One of the limitations of usage of BJT at Micro-wave frequencies is a) Size b) Transit-time c) Power d) all the above → **(b) Transit time**. Carrier transit time across the base becomes comparable to the radio-frequency period.

A37. The oscillator that makes a communication receiver capable of receiving morse code is a) Voltage Controlled Oscillator b) Local Oscillator c) Beat Frequency Oscillator d) Phase Locked Loop → **(c) the beat frequency oscillator**. An unmodulated keyed carrier is inaudible without a beat note.

A38. The circuit that enables the radio receiver output to remains cutoff when carrier is absent a) AGC b) Squelch c) Delayed AGC d) Noise Limiter → **(b) Squelch**.

A39. The category of pulse modulation that comes under digital systems is a) Pulse Code Modulation b) Pulse Amplitude Modulation c) Pulse Time Modulation d) All the above → **(a) pulse code modulation**. Pulse amplitude and pulse time modulation keep an analogue parameter; pulse code modulation quantises to a digital code.

A40. The highest frequency that is reflected by an Ionosphere layer is a) Maximum Usable Frequency b) Critical Frequency c) Carrier Frequency d) All the above → **(b) critical frequency**, the highest frequency returned at **vertical** incidence. [UNCERTAIN — for an oblique path the corresponding quantity is the maximum usable frequency, option (a). The question does not state the incidence angle. Indian study material teaches (b).]

A41. The device that can be used as a Voltage Regulator is a) Gun diode b) Zener Diode c) Pin Diode d) both a & c → **(b)**.

A42. The Reactance of a Capacitor increases when the frequency is a) increased b) decreased c) zero d) remains constant → **(b)**. $X_C = 1/(2\pi fC)$.

- A43.** The elements in the tank circuit of a Colpitts Oscillator are a) L and C b) R and C c) R and L d) R, L and C → **(a).** One inductor with a capacitive divider.
- A44.** The circuit that reduces ripples from the output of a Rectifier is a) Limiter b) Discriminator c) Blanker d) Filter → **(d).**
- A45.** Express the gain of an amplifier in dBm, when the output power of it is 100mW. a) 10 dBm b) 20 dBm c) 40 dBm d) 0 dBm → **(b) 20 dBm.** $\text{dBm} = 10 \cdot \log_{10}(P/1 \text{ mW}) = 10 \cdot \log_{10}(100) = 20$. *(The question is sloppily worded — dBm is an absolute power level, not a "gain".)*
- A46.** The output of an Amplifier with 10 dB gain and an input of 1mW is a) 10 mW b) 100 mW c) 1 mW d) 0 mW → **(a) 10 mW.** Plus 10 dB is a tenfold power increase.
- A47.** The Transformers are rated in a) Watts b) Volts c) Amperes d) Volt-Amperes → **(d) volt-amperes.** Core and copper losses depend on volts and amperes independently of the load power factor.
- A48.** Batteries are connected in parallel to get more a) Current capacity b) Voltage capacity c) both a & b d) None of the above → **(a).** Parallel connection adds capacity and current at the same voltage; series connection adds voltage.
- A49.** A Superhetrodyne receiver with local oscillator frequency range 3.455 to 30.455 MHz is capable of receiving the input signal range is a) 88-108 MHz b) 3-30 MHz c) 565-1600 KHz d) 2-17 MHz → **(b) 3–30 MHz.** With the standard 455 kHz intermediate frequency and high-side injection, $f_{\text{signal}} = f_{\text{LO}} - 0.455 \text{ MHz}$, giving 3.000–30.000 MHz.
- A50.** The Capability of a receiver to receive weakest possible signal is its a) Selectivity b) Sensitivity c) Fidelity d) All the above → **(b) Sensitivity.**

A.3.2 THE PAPER'S SECTION-B (RADIO REGULATIONS), Q1–Q50

Restricted grade: answer **Q1–Q25 only**.

- B1.** The value of bandwidth shown as 3K50 is a) 3050 Hz b) 3500 Hz c) 3500 KHz d) 3050 KHz → **(b) 3500 Hz.** ITU Radio Regulations Appendix 1: the letter occupies the position of the decimal point, and K means kilohertz, so $3\text{K}50 = 3.50 \text{ kHz} = 3500 \text{ Hz}$. [VERIFIED — RR Appendix 1]
- B2.** The first character in class of Emission signifies about a) Nature of signal(s) modulating the main carrier b) Type of modulation of the main carrier c) Type of Information to be transmitted d) Details of multiplexing the signal(s) → **(b).** First symbol = type of modulation; second = nature of the modulating signal; third = type of information.
- B3.** In the Morse code, the Test signal contains the characters of a) CQ CQ CQ b) V V V c) RT RT RT d) NON. → **(b) VVV.**
- B4.** The range of VHF band is a) 3 to 30 MHz b) 30 to 300 MHz c) 300 to 3000 MHz d) 0.3 to 3 MHz → **(b) 30–300 MHz. Study warning:** the NIAR manual's own model answer number 44 says "300 to 3000 MHz", which is **wrong** — that is the ultra-high-frequency range. Trust this paper. [VERIFIED — ITU RR Article 2 table; the NIAR error is documented at A.4]

B5. The time difference between IST and UTC is a) 05.30 Hrs. b) 05.00 Hrs. c) 05.15 Hrs. d) 05.45 Hrs. → **(a).** Indian Standard Time = Coordinated Universal Time + 5:30.

B6. The characters in the RST system stands for a) Readability, Signal and Test b) Readability, Signal strength and Test c) Readability, Signal strength and Tone d) Readability, Signal loss and Tone → **(c).**

B7. The VHF Frequency range that is authorised to Amateurs is a) 140 – 146 MHz b) 144 – 146 MHz c) 140 – 144 MHz d) 146 – 148 MHz → **(b) 144–146 MHz**, India's 2 metre allocation. (*Note that it is not the 144–148 MHz of ITU Region 2.*)

B8. The 'Single Side Band with Suppressed Carrier' is designated as a) H3E b) R3E c) J3E d) A3E → **(c) J3E.** H3E = full carrier, R3E = reduced carrier, J3E = suppressed carrier, A3E = double-sideband amplitude modulation.

B9. The space between two words in Morse code is a) a dot b) a dash c) a dot and a dash d) five dots → **(d) five dots — as this paper and the Indian study material teach. [UNCERTAIN and important: the international standard word space is seven units, not five. The NIAR manual itself prints seven on page 137 and five on page 139. For the ASOC hall, answer five.]**

B10. The Q Code for 'Are you busy?' is a) QRM b) QRL c) QSA d) QRN → **(b) QRL?**

B11. The distress frequency on Voice (Radio Telephony) is a) 2128 KHz b) 2182 KHz c) 1282 KHz d) 1228 KHz → **(b) 2182 kHz.**

B12. The characters that a Safety Signal contains in Morse code are a) XXX b) MAY DAY c) PAN PAN d) SSS → **[DEFECTIVE QUESTION.]** The international **safety** signal in Morse is **TTT**, which is **not offered**. XXX is the *urgency* signal — and this very paper says so at B38; MAYDAY and PAN PAN are voice signals, not Morse. The examiner's intended key is almost certainly **(a) XXX**. **[UNCERTAIN — mark (a), but know the question is wrong.]**

B13. The Phonetic used to represent digit '8' is a) Octa Eight b) Okta Eight c) Okto Eight d) Octo Eight → **(c) Okto Eight.** The ITU numeral phonetics are Nadazero, Unaone, Bissotwo, Terrathree, Kartefour, Pantafive, Soxisix, Setteseven, **Oktoeight**, Novenine. [VERIFIED — ITU RR Appendix 14]

B14. The Answer or Advice for the Q-Code 'QTH' is a) My exact location is b) What is your exact location? c) My correct time is Hrs. d) None of the above → **(a).** A Q-code *without* a question mark is the answer form; *with* a question mark, the question form.

B15. The written confirmation of a contact, exchanged between Amateurs is a) QSA3 b) QSL NR53 c) QST? d) QRX 1100 → **(b).** QSL means "I confirm receipt"; the QSL card is the written confirmation.

B16. The abbreviation used for 'All Before' is a) AA b) AB c) AR d) AS → **(a) AA.** (AR = end of transmission, AS = wait or stand by.)

B17. The calling Amateur's call sign in 'VU2DX DE VU2DJ' is a) VU2DX b) VU2DJ c) DE d) VU2 → **(b) VU2DJ.** "DE" means "from"; the station named *after* DE is the caller.

B18. The maximum characters that an Amateur Call Sign contains is a) Four b) Five c) Six d) Three → **(c) Six.** An Indian call sign such as VU2ABC is six characters — prefix VU, a digit, and up to three suffix letters. **[UNCERTAIN — the question does not say whether it means Indian call signs or ITU call signs generally; six matches the Indian format.]**

B19. The frequency range in 21 MHz band that is authorised to Amateurs is a) 21000 – 21350 KHz b) 21000 – 21450 KHz c) 21100 – 21150 KHz c) 21000 – 21400 KHz → **(b) 21000–21450 kHz.** *(The paper again mislabels the last option "c".)*

B20. The Emission that is used to sent Morse code by on/off keying the Unmodulated carrier in CW Transmission is a) A1A b) A2A c) A3A d) A5C → **(a) A1A.**

B21. The Q-code for 'The signal strength of your signals are Good' a) QSA 5 b) QSA 4 c) QSA 1 d) QSA 3. → **(b) QSA 4.** The QSA scale runs 1 barely perceptible, 2 weak, 3 fairly good, **4 good**, 5 very good.

B22. The character that represents the Morse code '– – · –' is a) Y b) Z c) C d) Q → **(d) Q.** (Y = – · – –, Z = – – ·, C = – · – ·, Q = – – · –.)

B23. The Amateur Licence will be issued in India by a) Wireless Monitoring Organisation b) Wireless Planning and Coordination Wing c) Telecomm Regulatory Authority of India d) Bharat Sanchar Nigam Limited → **(b) the WPC Wing of DoT.** The Wireless Monitoring Organisation *monitors*; the Telecom Regulatory Authority of India regulates tariffs and service quality and issues no licences of this kind.

B24. The Call sign blocks allotted to India are a) ATA – AWZ b) VTA – VWZ c) 8TA – 8WZ d) a & b e) a , b & c → **(e) a, b & c — as the paper intends. But note the typographical error: ITU Appendix 42 allots India ATA–AWZ, VTA–VWZ and 8TA–8YZ, not 8TA–8WZ. [VERIFIED against ITU Appendix 42; the error is the government paper's own]**

B25. The UTC stands for a) Universal Time for Coordination b) Universal Coordinated Time c) United States Telecomm Community d) Universal Telecommunication Centre → **(b).** *(The strictly correct English expansion is "Coordinated Universal Time"; (b) is the intended option.)*

— end of the Restricted-grade portion of the paper's Section B —

B26. The Emission 'C3F' stands for a) Double Side band Transmission b) Single side band Transmission c) Vestigial side band Transmission d) Suppressed side band Transmission → **(c).** C = vestigial sideband; 3 = single channel containing analogue information; F = television.

B27. The number of characters in a 8 wpm in 5 minutes passage should be a) 240 b) 200 c) 160 d) 400 → **(b) 200.** 8 words per minute × 5 characters per word × 5 minutes = 200. This matches the official Morse test specification — a 200-character passage at 8 words per minute over five minutes.

B28. The standard shift between transmitting and receiving frequency for Amateur Radio in VHF band is a) 500 KHz b) 600 KHz d) 1000 KHz d) 1200 KHz → **(b) 600 kHz.** *(Options mislabelled again.)*

B29. The equivalent time in hours of 1730(IST) in UTC is a) 1200Z b) 1230Z c) 0000Z d) 0530Z → **(a) 1200Z.** 17:30 – 5:30 = 12:00.

B30. The Q-code for 'I will call you again at 0400 hrs in the evening' is a) QRX0400 b) QRX1600 c) QRX4 d) QRX0400Z → **(b) QRX1600.** Four o'clock in the afternoon is 1600 on the 24-hour clock used on air.

B31. The Emission stands for an AM Broadcast with 3 KHz bandwidth is a) A3E3000K b) A3E3K00 c) A3E0K300 d) A3E0H30 → **(b) — as the paper intends. [DEFECT: none of the four options is correctly formed.]** ITU Radio Regulations Appendix 1 §2(3) puts the necessary bandwidth **before** the class of emission, so a 3 kHz amplitude-modulated broadcast is **3K00A3E**. [VERIFIED against RR Appendix 1] Choose (b) in the hall; know the real form.

B32. The distress frequency 156M800 Hz falls in the range of a) HF band b) UHF band c) Microwave band d) VHF band → **(d) VHF.** 156M800 means 156.800 MHz, marine VHF Channel 16, inside 30–300 MHz.

B33. The character for Morse code '· – – ·' is a) Full stop b) Comma c) Question mark d) Hyphen → **(c) question mark.** (· – – · = ?)

B34. In a Morse code transmission what will be the duration of a dot, when the duration of a dash is 30mSec. a) 90mSec b) 10mSec c) 1mSec d) 9mSec → **(b) 10 ms.** A dash is three dot-lengths.

B35. The Abbreviation for 'I have nothing for you' is a) QRU b) NIL c) NFU d) None of the above → **(b) NIL.** (QRU is the Q-code with the same meaning; the question asks for the abbreviation, which is NIL.)

B36. Frequencies those are authorised to use during the 0800 to 2000 Hrs. of the day will be indicated as a) H24 b) HN c) HJ d) HX → **(c) HJ,** meaning daytime hours only. (HN = night only, HX = intermittent, H24 = continuous.)

B37. The Phonetic used for alphabet 'N' is a) Norvey b) November c) Neighbor d) Night → **(b) November.**

B38. The urgent messages in a Morse code are indicated by a) PAN PAN b) XXX c) T T T d) V V V → **(b) XXX.** (PAN PAN is the voice urgency signal; TTT is the Morse safety signal; VVV is a test signal.)

B39. The Emission that indicates a FM Broadcast station is a) F1A b) F3A c) F3E d) F3C → **(c) F3E.** F = frequency modulation, 3 = single analogue channel, E = telephony.

B40. What are the letters required to be sent for a third station to enter between two stations in a Morse code transmission are a) BR b) BK c) BREAK d) BN → **(b) BK.** **Study note:** BK is asked here but does **not** appear in the official syllabus's closed list of seventeen abbreviations — learn it anyway. [VERIFIED — the syllabus list is at [r11-syllabus.md](#) §6.9]

B41. The word used in Voice which is equivalent to the word 'DE' transmitted in a Morse code transmission is a) FROM b) THIS IS c) CALLING d) All of the above → **(b) THIS IS.**

B42. The letters those are required to be sent in the time of Distress in Voice transmission are a) SOS b) MAY DAY c) PAN PAN d) SECURTIE → **(b) MAYDAY.** (SOS is Morse distress; PAN PAN is voice urgency; SÉCURITÉ is voice safety.)

B43. The Q-code for Test Signal is a) QRK b) QSU c) QSV d) QUM → **(c) QSV.**

B44. In abbreviation 73, refers to a) thanks b) welcome c) best regards d) none of these → **(c) best regards.**

B45. The frequencies those are designated with 'HX' can be used a) during the day time b) during the night time c) intermittently d) 24 Hrs. of a day → **(c) intermittently.**

B46. The suffix that is required to be sent along with the call sign for a Mobile Amateur Station is a) MOBILE b) MO c) MX d) None of the above → **(b) MO.**

B47. What will be the speed in wpm, when a message being transmitted contains 60 characters in a minute a) 5 wpm b) 8 wpm c) 10 wpm d) 12 wpm → **(d) 12 words per minute.** 60 characters ÷ 5 characters per word = 12.

B48. The ITU stands for a) International Trade Union b) Indian Trade Unions c) Indian Telecommunication Union d) International Telecommunication Union → **(d).**

B49. In RST system for Morse code transmission a) R and S need to be reported b) R and T need to be reported c) T and S need to be reported d) R, S and T are necessary → **(d).** The tone digit is reported for continuous-wave (Morse) work; it is omitted, leaving R and S only, for voice.

B50. The type of Infringement to be sent upon using incorrect emission is a) Un Authorised Frequency b) Un Authorised Period c) Un Authorised Emission d) Un Authorised Call Sign → **(c).**

A.3.3 DEFECTS FOUND IN THE OFFICIAL PAPER

Six internal defects were identified in S1. They are listed together because a candidate who spots one in the hall should not lose composure over it.

Item	Defect
A6, A27, B19, B28	Duplicate option letters — the fourth option is labelled with a letter already used.
B12	The safety signal TTT is not among the offered options, so no option is correct.
B24	India's third call-sign block is printed as 8TA–8WZ; ITU Appendix 42 says 8TA–8YZ.
B31	The emission designator is written class-first ("A3E3K00"); RR Appendix 1 requires bandwidth first ("3K00A3E").

[VERIFIED — each defect identified by reading the scan of S1 against the ITU Radio Regulations, Edition of 2024]

A.4 S2 and S8 — the NIAR Study Manual question bank (230 questions)

Source type: institute question bank. Document: NIAR, *Study Manual for Amateur Station Operator's Licence Examination*, 158 pp, <http://www.niar.org/downloads/ham-downloads/Study-Manual.pdf>, fetched 2026-08-23, undated in the text. NIAR is the DoT-recognised national training institute and runs many of the examination-preparation courses, so this is the most authoritative *non-government* bank available. [REPORTED]

Health warning on this source. The manual contains at least two demonstrable errors that would cost marks if learned uncritically:

1. Model answer **B-44** below states "The range of VHF band is: **300 to 3000 MHz**". That is the ultra-high-frequency range. VHF is 30–300 MHz — and the official paper offers 30–300 MHz as an option at S1 B4.
2. Word spacing in Morse is given as **seven** dots on page 137 and **five** dots on page 139. The examination expects five, per S1 B9.

Both are flagged in line below. The manual prints answers for 75 of its 230 items and leaves the rest unanswered. Answers marked **[NIAR]** are the manual's own; answers marked **[derived]** are this research's.

A.4.1 NIAR "MODEL QUESTIONS & ANSWERS", THE PAPER'S SECTION B (RADIO REGULATIONS) — 50 ITEMS WITH ANSWERS

Manual pages 156–157. Printed as question-and-answer, not as MCQ. Topic: **regulations**.

#	Question	Answer [NIAR]	Comment
B-1	The phonetic used for the digit '1' is	Una One	ITU numeral phonetics.
B-2	The Frequency range that is authorised to Amateurs in UHF band is	434–438 MHz	India's 70 cm amateur allocation.
B-3	The third symbol in class of an Emission signifies about	Type of Information to be transmitted	Consistent with S1 B2.
B-4	The call sign of an amateur station must be sent	At the beginning and end of each exchange of communications, and at least every 10 minutes while in communications	
B-5	The Q Code for 'Are you busy?' is	QRL?	Matches S1 B10.
B-6	What is simplex operation?	Transmitting and receiving on the same frequency	

#	Question	Answer [NIAR]	Comment
B-7	The message "PAN PAN PAN VU2NRO DE VU2JOS" is of type	Urgency signal	
B-8	What will be the time in UTC when the time in IST is 1000 Hrs	0430 Hrs	10:00 – 5:30.
B-9	If you contact another station and your signal is strong and perfectly readable, you should	Reduce your transmitter power output to the minimum needed to maintain contact	The minimum-necessary-power rule.
B-10	The radiation in and reception from unnecessary directions can be minimized by using	Directional antennas	
B-11	You are adjusting an antenna matching unit using an SWR bridge. You should adjust for	Minimum reflected power	
B-12	The transmission of characters "VVV VVV VVV DE VU2NRO" denotes the following type of signal	Test	Matches S1 B3.
B-13	The Amateur Licence is issued in India by	Wireless Planning & Coordination Wing	Matches S1 B23.
B-14	The Q-code for 'I will call you again at 0400 hrs in the evening' is	QRX 1600	Matches S1 B30.
B-15	Emissions shall be designated according to their	Necessary bandwidth and Classification	ITU RR Appendix 1.
B-16	You must keep the following document at your amateur station	Your Amateur Operator Licence	
B-17	The letters those are required to be sent in the time of Distress in Voice transmission are	MAY DAY	Matches S1 B42.
B-18	If you hear distress traffic and are unable to render assistance, you should	Maintain watch until you are certain that assistance is forthcoming	

#	Question	Answer [NIAR]	Comment
B-19	To make your call sign better understood when using voice transmissions, what should you do?	Use Standard International Phonetics for each letter of your call sign	
B-20	The first character in class of Emission signifies about	Type of modulation of the main carrier	Matches S1 B2.
B-21	The ITU stands for	International Telecommunication Union	Matches S1 B48.
B-22	The characters in the RST system stands for	Readability, Signal Strength and Tone	Matches S1 B6.
B-23	The international distress & calling frequency in Voice transmission is	2182 KHz	Matches S1 B11.
B-24	The Q-code for "Does my frequency vary"	QRH?	
B-25	The frequencies those are designated with 'HX' can be used	intermittently	Matches S1 B45.
B-26	UTC stands for	Universal Coordinated Time	Matches S1 B25.
B-27	The letters that are generally used for the first letters in Indian Amateur Radio call signs is	VU	
B-28	The Abbreviation for 'I have nothing for you' is	NIL	Matches S1 B35.
B-29	The suffix that is required to be sent along with the call sign for a Mobile Amateur Station is	MO	Matches S1 B46.
B-30	Frequencies those are authorised to use during the 0800 to 2000 Hrs. of the day will be indicated as	HJ	Matches S1 B36.
B-31	The meaning of Roger is	Received fully	
B-32	The type of Infringement to be sent upon using incorrect emission is	Unauthorized Emission	Matches S1 B50.

#	Question	Answer [NIAR]	Comment
B-33	The Q Code for 'Are you ready?' is	QRV?	
B-34	The organization responsible for the International Radio Regulations is	International Telecommunication Union	
B-35	The distress frequency 156M800 Hz falls in the range of	VHF band	Matches S1 B32.
B-36	The Standard Time & frequency station in India is operated by	National Physical Laboratory	The NPL runs station ATA on 10 MHz.
B-37	The Q-code for Test Signal is	QSV	Matches S1 B43.
B-38	The minimum age for a person to become a radio amateur in India is	12 years	Consistent with rule 4(1)(a) of the 2024 Rules. [VERIFIED]
B-39	In abbreviation 73 refers to	Best regards	Matches S1 B44.
B-40	Identification of a station primarily can be done by	Call sign	
B-41	The time difference between IST and UTC is	5.30 hrs	Matches S1 B5.
B-42	The Phonetic used for alphabet 'N' is	November	Matches S1 B37.
B-43	The 20 Meter Amateur Radio frequencies allotted to India is	14000 to 14350 kHz	
B-44	The range of VHF band is	<i>"300 to 3000 MHz"</i>	WRONG — that is UHF. VHF is 30–300 MHz. The official paper (S1 B4) offers 30–300 MHz. Do not learn NIAR's answer. [VERIFIED — ITU RR Article 2]
B-45	Can Radio Amateurs broadcast news	No	Broadcasting is forbidden to the amateur service.
B-46	Test signals cannot be sent for more than	30 seconds	Consistent with Annexure A rule 9(4)(a) of the 2024 Rules. [VERIFIED]
B-47	What is QSL Card	Written confirmation of a Contact	Matches S1 B15.

#	Question	Answer [NIAR]	Comment
B-48	The call sign blocks allotted for Amateurs in India is	ATA-AWZ, VTA-VWZ & 8TA-8YZ	Note that NIAR gets 8TA-8YZ right where the official paper (S1 B24) prints 8TA-8WZ.
B-49	The abbreviation for "Closing Down" is	CL	
B-50	The meaning of CQ is	General call for all stations	

A.4.2 NIAR SHORT-ANSWER EXERCISE, RADIO REGULATIONS — 25 ITEMS WITH ANSWERS

Manual page 134. Topic: **regulations**.

#	Question	Answer [NIAR]	Comment
R-1	What does the first symbol indicate in emission	Type of modulation of main carrier	
R-2	What is the meaning of QRM	I am being interfered with	
R-3	What is the meaning of QSL?	Can you acknowledge receipt?	<i>QSL?</i> is the question form.
R-4	What is the meaning of QRZ?	Who is calling me?	
R-5	How long can a Test signal be transmitted for?	30 seconds	
R-6	What does R stand for in RST Code?	Readability	
R-7	When is the suffix MO used for?	For Mobile operation	
R-8	Which type of signal has the highest priority?	Distress Signal	Distress, then urgency, then safety, then routine.
R-9	What is the Phonetics for 5?	Panta Five	
R-10	What is the meaning of Roger?	Received fully	
R-11	Signal Strength 9 means	Strong signals	Contradicts NIAR's own RST table on p.125, which says S9 is "extremely strong" and S8 is "strong". The table is right. [VERIFIED — internal contradiction]

#	Question	Answer [NIAR]	Comment
R-12	Ultra High Frequency is	300 to 3000 MHz	Correct — and note that it contradicts NIAR's own B-44.
R-13	What is the standard shift between transmitting and receiving frequency for Amateur Radio VHF Repeater	600 kHz	Matches S1 B28.
R-14	What is the minimum age for foreign nationals to apply for reciprocal licence	18 years	
R-15	Who operates Standard Time & Frequency station in India	National Physical Laboratory	
R-16	What is the Q code for sending test signal	QSV	
R-17	Amateur Radio licence can be renewed for	20 years or Life long	[UNCERTAIN — pre-2024 position. Verify against the Telecommunications (Amateur Services) Rules, 2024 before relying on it.]
R-18	What is the International Distress Frequency in Radio telephony	2182 kHz	
R-19	What is the speed of Morse code if 200 letters are sent in one minute?	40 wpm	$200 \div 5 = 40$.
R-20	What is the Morse Code alphabet denoted by: · · - ·	F	
R-21	What is the space duration between letters in Morse Code?	3 dits	
R-22	Mobile permission is issued for	90 days	[UNCERTAIN — pre-2024 figure.]
R-23	Amateur Radio Messages must be in	Plain language	Codes and ciphers are forbidden.
R-24	Power permitted for Restricted grade on 144-146 MHz is	10 watts	Directly relevant to the 10 W objective of this report. Corroborated by WPC OM L-14011/255/2004-AMT dated 2010-08-13, Annexure-I. [VERIFIED]
R-25	The fee for change of address is	Rs. 200	[UNCERTAIN — pre-2024 fee. Rule 7(6) of the 2024 Rules sets ₹1,000 for a change of location or a duplicate.] [VERIFIED — 2024 Rules r.7(6)]

A.4.3 NIAR EXERCISE, RADIO REGULATIONS MCQ, ITEMS 1–12 (MANUAL PP. 119–120)

No answer key printed; answers below are **[derived]**. Topic: **regulations**.

#	Question and options	Answer [derived]
RM-1	Who issues Amateur Radio Licence in India? a) WPC wing of Ministry of Communications & IT b) Department of Posts c) Telephone Department d) State Government	a
RM-2	What is the minimum age to become a radio amateur in India? a) No age limit b) 18 years c) 12 years d) 21 years	c
RM-3	How many types of Amateur Radio licences are there in India? a) Five b) Two c) One d) Three	b (Restricted and General)
RM-4	An Amateur Station is one which is a) Operated by the holder of a General Amateur Operator certificate of Competency on the amateur radio bands b) Owned and operated by a person who is not engaged professionally in radio communications c) Used exclusively to provide two-way communication in connection with activities of amateur sporting organisation d) Used primarily for emergency communications during floods, earthquakes and similar disasters	b
RM-5	When are third party messages permitted? a) Always b) Upon failure of normal telecommunication facilities and upon request from the Government c) Never d) When there are natural calamities	b
RM-6	What is the speed of Morse Code exam for General Grade ASOC exam? a) 20 WPM b) 12 WPM c) 8 WPM d) 5 WPM	c
RM-7	Are Radio Amateurs in India permitted to broadcast news? a) No b) Yes c) Occasionally d) Once per day	a
RM-8	Which one of the following is not true regarding issue of reciprocal licence? a) The applicant must have Amateur licence issued in his country b) The applicant must be above 20 years c) The licence shall be initially granted for one year or for the validity of the visa, whichever is earlier d) The country of which the applicant is a citizen should grant reciprocal licence to Indian nationals	b (the stated minimum is 18, per NIAR R-14)
RM-9	The full form of ITU is a) Indian Telecom University b) Indian Telecommunications Union c) International Telecommunications Union d) International Television Union	c
RM-10	The organization responsible for the International Radio Regulations is a) European Radio communications Office b) United Nations c) International Telecommunication Union d) European Telecommunication Standards Institute	c

#	Question and options	Answer [derived]
RM-11	(Amateur Service definition) ... a) a private radio service for personal gain and public benefit b) a public radio service used for public service Communications c) a radio communication service for the purpose of self training, intercommunication and technical investigation d) a private radio service intended only for emergency communications	c — this is the legal definition, and it is why a 10 W walkie-talkie used for family chat is not an amateur-radio use
RM-12	An Amateur Station is a station a) in the public radio service b) using radio communications for a commercial purpose c) using equipment for training new radio communications operators d) in the Amateur Service	d

Gap in the source: the manual's numbering then jumps from 12 to 25. Items 13–24 are absent from the published PDF. [VERIFIED — the extracted text was checked; there is no page carrying them]

A.4.4 NIAR EXERCISE, Q-CODES, PHONETICS AND PROCEDURE SIGNALS, ITEMS 25–34 PLUS A STRAY "11" (MANUAL PP. 125–126)

No answer key printed; answers **[derived]**. Topic: **regulations**.

#	Question and options	Answer [derived]
RM-25	What is the Phonetics for the alphabet R? a) Romeo b) Rome c) Romania d) Royal	a
RM-26	What is the normal prefix for Restricted Grade Amateur Radio Licence in India? a) VU2 b) VU3 c) VU5 d) VU9	b — VU3 is the Restricted-grade prefix; VU2 is General
RM-27	What is the Q Code for "My exact location is"? a) QRL b) QSL c) QRA d) QTH	d
RM-28	What is the meaning of QRZ? a) What is the exact time? b) What is your name? c) Who is calling me? d) When will we meet again?	c
RM-29	The Distress signal in Morse Code is: a) XXX b) SOS c) TTT d) V V V	b
RM-30	Pan Pan means: a) Urgency Signal b) Test Signal c) Weather warning d) Normal message	a
RM-31	What is meaning of CL in Morse code? a) clearing down b) cloudy c) see you later d) closing down	d
RM-32	If in Morse code three series of v v v are sent what does it represent? a) I am busy b) Test Signal c) Ending Signal d) General call	b
RM-33	Which type of signal has the highest priority? a) Urgent Signal b) Test Signal c) Distress Signal d) Weather warning	c
RM-34	What is the meaning of CQ? a) General call for all stations b) I have nothing for you c) I am busy d) I am closing down	a

#	Question and options	Answer [derived]
RM-11b	The Morse code signal SOS is sent by a station a) with an urgent message b) in grave and imminent danger and requiring immediate assistance c) making a report about a shipping hazard d) sending important weather information	b — <i>(printed out of sequence as "11" in the manual)</i>

A.4.5 NIAR EXERCISE, BANDS, POWER, EMISSIONS AND LOG-KEEPING, ITEMS 1–10 (MANUAL P. 133)

No answer key printed; answers **[derived]**. Topic: **regulations**.

#	Question and options	Answer [derived]
RB-1	What is the 40 Meter Band allocation for Amateur Radio License in India? a) 14.000 to 14.350 MHz b) 7.000 to 7.100 MHz c) 7.000 to 7.200 MHz d) 7.100 to 7.200 MHz	c [UNCERTAIN — Indian sources differ between 7.000–7.100 and 7.000–7.200 MHz depending on vintage; check the current authorised table]
RB-2	How much power is permitted on HF for Restricted Grade Amateur Radio Licence in India? a) 50 watts b) 100 watts c) 400 watts d) 25 watts	a — 50 W on HF for the Restricted grade [VERIFIED — 2010 OM, Annexure-I]
RB-3	By what name in general are the Amateur Radio Satellites known as: a) INSAT b) INTELSAT c) IRIDIUM d) OSCAR	d
RB-4	What is the meaning of DE in Morse code? a) How are you? b) This is c) Who are you? d) Distant station	b
RB-5	What type of emission is A1A? a) Morse Code b) SSB c) FM d) AM	a
RB-6	What is emission for SSB, suppressed carrier: a) A1A b) F3E c) J3E d) A3E	c
RB-7	When it is 10.30 pm IST, What will UTC time be? a) 1000 Hrs b) 1500 Hrs c) 0000 Hrs d) 1700 Hrs	d (22:30 – 5:30 = 17:00)
RB-8	The frequency band of 30 to 300 MHz is: a) HF b) UHF c) VHF d) SHF	c
RB-9	How long must a log book be preserved? a) One year b) One year from date of last entry c) Two years d) Permanently	b [VERIFIED — 2024 Rules, Annexure A r.6(4)]
RB-10	The Time in the log book must be in: a) IST b) UTC c) GMT d) Local Time	Answer as printed in the study material: b (UTC). But see the conflict note immediately below — the current Rules say IST.

A live conflict you must know about — log timekeeping. Every Indian study manual, including NIAR here, teaches that log times are kept in UTC, and the error table at A.6.3 below marks the DARTS key of "IST" as wrong on exactly that basis. **However, Annexure A rule 6(2) of the Telecommunications (Amateur Services) Rules, 2024 states verbatim: "All times in the log shall be stated in the Indian Standard Time."** [VERIFIED — G.S.R. 675(E) dated 2024-10-29, Annexure A r.6(2), quoted in full in the licence-conditions research] The Rules are subordinate legislation made under section 47 of the Telecommunications Act, 2023 and are far more authoritative than any club manual. **For your actual log, use IST.** For the examination, be aware that an examiner working from pre-2024 material may expect UTC — and that ITU RR 2.6 does require UTC for international radiocommunication activities generally, which is where the study material's habit comes from. [VERIFIED — ITU RR 2024, 2.6] **[UNCERTAIN — which answer a 2026 examiner marks as correct.]**

A.4.6 NIAR EXERCISE, BASIC ELECTRICITY AND CIRCUITS — 50 ITEMS (MANUAL PP. 31–35)

No answer key printed. Topic: **radio theory**. Eleven of the fifty refer to circuit diagrams that are images in the PDF and cannot be transcribed; they are listed for completeness but cannot be answered from text alone.

#	Question and options	Answer [derived]
T1-1	Resistivity of a wire depends on (A) length (B) material (C) cross section area (D) none of the above	B
T1-2	Resistance of a wire is r ohms. The wire is stretched to double its length, then its resistance in ohms is (A) $r/2$ (B) $4r$ (C) $2r$ (D) $r/4$	B (length doubles, area halves)
T1-3	Ampere second could be the unit of (A) power (B) conductance (C) energy (D) charge	D ($1 \text{ A}\cdot\text{s} = 1 \text{ coulomb}$)
T1-4	Conductance is expressed in terms of (A) ohm/m (B) m/ohm (C) mho/m (D) mho	D
T1-5	We have three resistances of values 2Ω , 3Ω and 6Ω . Which combination gives an effective resistance of 4Ω ? (A) all three in parallel (B) 2Ω in series with $3 \Omega \parallel 6 \Omega$ (C) 3Ω in series with $2 \Omega \parallel 6 \Omega$ (D) 6Ω in series with $2 \Omega \parallel 3 \Omega$	B ($3 \parallel 6 = 2$; $2+2 = 4$)
T1-6	Three equal resistors in series dissipate 10 W. Power dissipated when connected in parallel across the same emf? (A) 10 W (B) 30 W (C) 90 W (D) 270 W	C (resistance falls ninefold)
T1-7	Current I in the figure is (A) 1.5 A (B) 0.5 A (C) 3.5 A (D) 2.5 A	<i>figure not extractable</i>
T1-8	Four identical resistors first in parallel then in series. Ratio of the first combination to the second is (A) $1/16$ (B) $1/4$ (C) 4 (D) 16	A ($R/4 \div 4R$)
T1-9	The unit of electrical conductivity is (A) mho/metre (B) mho/sq. m (C) ohm/metre (D) ohm/sq. m	A

#	Question and options	Answer [derived]
T1-10	Which bulb will have the least resistance? (A) 220 V 60 W (B) 220 V 100 W (C) 115 V 60 W (D) 115 V 100 W	D ($R = V^2/P$)
T1-11	Ratio of the resistance of a 100 W 220 V lamp to a 100 W 110 V lamp (A) 4 (B) 2 (C) 1/2 (D) 1/4	A
T1-12	Two 1 k Ω , ½ W resistors in series. Combined value and wattage (A) 2 k Ω ½ W (B) 2 k Ω 1 W (C) 2 k Ω 2 W (D) 1 k Ω ½ W	B
T1-13	Three 3 Ω resistors form a triangle. Resistance between any two corners? (A) 3/4 Ω (B) 3 Ω (C) 2 Ω (D) 4/3 Ω	C (3 6)
T1-14	Five resistances connected as in figure — equivalent resistance A–B (A) 35 Ω (B) 25 Ω (C) 15 Ω (D) 5 Ω	<i>figure not extractable</i>
T1-15	How many different combinations may be obtained with three resistors each of resistance R? (A) 3 (B) 4 (C) 5 (D) 6	B
T1-16	Ohm's law is not applicable to (A) DC circuits (B) high currents (C) small resistors (D) semi-conductors	D
T1-17	Five resistances connected as shown to a 40 V supply. Voltage between P and Q (A) 40 V (B) 22.5 V (C) 20 V (D) 17.5 V	<i>figure not extractable</i>
T1-18	The current in the 5 ohm resistor will be (A) 2.1 A (B) 2.7 A (C) 3.0 A (D) 3.5 A	<i>figure not extractable</i>
T1-19	Least current will flow through (A) 25 Ω (B) 18 Ω (C) 10 Ω (D) 5 Ω	<i>figure not extractable</i>
T1-20	Total power loss in the circuit is (A) 10 W (B) 50.2 W (C) 205 W (D) 410 W	<i>figure not extractable</i>
T1-21	A resistance of 5 Ω is drawn so that its length doubles. Its resistance will now be (A) 5 Ω (B) 7.5 Ω (C) 10 Ω (D) 20 Ω	D
T1-22	Specific resistance of a substance is measured in (A) ohms (B) mhos (C) ohm-m (D) cm/ohm	C
T1-23	A wire of resistance R has its length and cross-section both doubled. Its resistance becomes (A) 4R (B) 2R (C) R (D) R/4	C
T1-24	Ohm's law is not applicable in all the following cases except (A) Electrolytes (B) Arc lamps (C) Insulators (D) Vacuum radio valves	C
T1-25	The element of an electric heater is made of (A) copper (B) steel (C) carbon (D) nichrome	D
T1-26	Two aluminium conductors of equal length; one has four times the cross-section. If the smaller has 100 Ω , the other has (A) 400 Ω (B) 100 Ω (C) 50 Ω (D) 25 Ω	D
T1-27	Variable resistors are (A) Wire wound (B) Thin film (C) Thick film (D) All of the above	D

#	Question and options	Answer [derived]
T1-28	Temperature coefficient of resistance is expressed in terms of (A) ohms/ohm·°C (B) mhos/ohm·°C (C) mhos/°C (D) ohms/°C	A
T1-29	Which material is expected to have least resistivity? (A) Copper (B) Lead (C) Mercury (D) Zinc	A
T1-30	A fuse is always installed in a circuit in (A) Series (B) Parallel	A
T1-31	The rating of fuse wire is expressed in terms of (A) Ohms (B) Mhos (C) Amperes (D) Watts	C
T1-32	Which of the following material is not used as fuse material? (A) Silver (B) Copper (C) Aluminium (D) Carbon	D
T1-33	The voltage drop across the 9 ohm resistor will be (A) 18 V (B) 12 V (C) 9 V (D) 6 V	<i>figure not extractable</i>
T1-34	The voltage drop will be least in which resistor? (A) 2 Ω (B) 3 Ω (C) 6 Ω (D) 3 Ω and 2 Ω	<i>figure not extractable</i>
T1-35	The current through the 6 ohm resistor will be (A) 1 A (B) 2 A (C) 3 A (D) 4 A	<i>figure not extractable</i>
T1-36	The ratio of power dissipated in circuit ADB to that in ABC (A) 1 (B) 1.5 (C) 0.67 (D) 2	<i>figure not extractable</i>
T1-37	Total power dissipated in the circuit is (A) 80 W (B) 100 W (C) 120 W (D) 150 W	<i>figure not extractable</i>
T1-38	The current carrying capacity of the fuse material depends on (A) cross-sectional area (B) length (C) material (D) all of the above	D
T1-39	Two 1 kΩ, ½ W resistors in series. Combined value and wattage (A) 2 kΩ ½ W (B) 2 kΩ 1 W (C) 2 kΩ 2 W (D) 1 kΩ ½ W	B (duplicate of T1-12)
T1-40	Ohm's law is not applicable in all the following cases except (A) Electrolysis (B) Arc lamps (C) Insulators (D) Vacuum radio valves	C (duplicate of T1-24)
T1-41	In which circuit will the voltage source produce the most current? (A) 10 V across 10 Ω (B) 10 V across two 10 Ω in series (C) 10 V across two 10 Ω in parallel (D) 1000 V across a megohm	C (5 Ω gives 2 A)
T1-42	The unit of conductivity is (A) mho/m (B) mho/sq. m (C) ohm/m (D) ohm/sq. m	A
T1-43	For the circuit shown, the reading in ammeter A will be (A) 2 A (B) 0.5 A (C) 0.4 A (D) 0.2 A	<i>figure not extractable</i>
T1-44	In the colour code for resistances black represents the number (A) 0 (B) 1 (C) 2 (D) 3	A
T1-45	In the colour code white represents the number (A) 9 (B) 8 (C) 6 (D) 5	A
T1-46	In the colour code number 3 is represented by (A) blue (B) orange (C) grey (D) violet	B

#	Question and options	Answer [derived]
T1-47	The condition for the validity under Ohm's law is (A) Temperature at positive end more than negative end (B) Current proportional to size of resistance (C) Resistance must be wire wound (D) Resistance must be uniform	D
T1-48	$R_1 = 47\text{ k}\Omega$ 1 W and $R_2 = 0\ \Omega$ 1 W in parallel. Combined value (A) $47\text{ k}\Omega$ 1 W (B) $47\text{ k}\Omega$ 2 W (C) $47\text{ k}\Omega$ $\frac{1}{2}$ W (D) $0\text{ k}\Omega$ 1 W	D (a $0\ \Omega$ branch shorts the pair)
T1-49	A rheostat differs from a potentiometer in that (A) rheostat has large number of turns (B) rheostat offers larger number of tapings (C) rheostat has lower wattage rating (D) rheostat has higher wattage rating	D
T1-50	Which consists of two plates separated by a dielectric and can store a charge? 1) Inductor 2) Capacitor 3) Transistor 4) Relay	2

A.4.7 NIAR EXERCISE, SEMICONDUCTOR DEVICES — 30 ITEMS (MANUAL PP. 59–62)

No answer key printed. Topic: **radio theory**.

#	Question and options	Answer [derived]
T2-1	A semi-conductor is (A) one which conducts only half of applied voltage (B) a material whose conductivity is between that of a conductor and an insulator (C) alternate layers of conducting material and insulator (D) conductivity the average of metal and insulator	B
T2-2	A p-n junction is forward biased when (a) positive pole to p and negative pole to n (b) positive to n and negative to p (c) positive to n and p (d) mechanical force applied forward	a
T2-3	At absolute zero, Si acts as? (a) non-metal (b) metal (c) insulator (d) none of these	c
T2-4	When N-type semiconductor is heated? (a) electrons increase, holes decrease (b) holes increase, electrons decrease (c) both remain same (d) both increase equally	d
T2-5	Radio waves of constant amplitude can be generated with (a) FET (b) filter (c) rectifier (d) oscillator	d
T2-6	The depletion layer in the P-N junction region is caused by? (a) drift of holes (b) diffusion of charge carriers (c) migration of impurity ions (d) drift of electrons	b
T2-7	To use a transistor as an amplifier (a) emitter-base forward biased and base-collector reverse biased (b) no bias required (c) both forward (d) both reverse	a
T2-8	For amplification by a triode, the signal is given to (a) the cathode (b) the grid (c) the glass envelope (d) the anode	b
T2-9	Copper and germanium cooled from room temperature to 80 K, then (a) resistance of each increases (b) copper decreases (c) copper increases, germanium decreases (d) copper decreases, germanium increases	d
T2-10	Diamond is very hard because? (a) covalent solid (b) large cohesive energy (c) high melting point (d) insoluble in all solvents	a

#	Question and options	Answer [derived]
T2-11	The part of the transistor heavily doped to produce large numbers of majority carriers is (a) emitter (b) base (c) collector (d) any of the above	a
T2-12	An Oscillator is nothing but an amplifier with (a) positive feedback (b) negative feedback (c) large gain (d) no feedback	a
T2-13	When a P-N junction diode is reverse biased the current across the junction is mainly due to (a) diffusion of charge (b) drift charges (c) depends on material (d) both drift and diffusion	b
T2-14	Which, added as an impurity into silicon, produces n-type semiconductor (a) Phosphorous (b) Aluminium (c) Magnesium (d) both b and c	a
T2-15	When an n-p-n transistor is used as an amplifier then? (a) electrons flow emitter to collector (b) holes flow emitter to collector (c) electrons flow collector to emitter (d) electrons flow battery to emitter	a
T2-16	When arsenic is added as an impurity to silicon, the resulting material is? (a) n-type semiconductor (b) p-type semiconductor (c) n-type conductor (d) Insulator	a
T2-17	To obtain a p-type germanium semiconductor, it must be doped with? (a) arsenic (b) antimony (c) indium (d) phosphorus	c
T2-18	A device passes current one way; reversing the battery drops current to almost zero. The device may be (a) A p-n junction (b) An intrinsic semiconductor (c) A p-type semiconductor (d) An n-type semiconductor	a
T2-19	The cause of the potential barrier in a p-n diode is? (a) depletion of positive charges near the junction (b) concentration of positive charges near the junction (c) depletion of negative charges (d) concentration of positive and negative charges near the junction	d
T2-20	In forward bias, the width of the potential barrier in a p-n junction diode? (a) increases (b) decreases (c) remains constant (d) first increases then decreases	b
T2-21	A depletion layer consists of? (a) electrons (b) protons (c) mobile ions (d) immobile ions	d
T2-22	Which of the following when added acts as an impurity into silicon to produce n-type semiconductor? (a) P (b) Al (c) B (d) Mg	a
T2-23	In a junction diode, the holes are due to (a) protons (b) extra electrons (c) neutrons (d) missing electrons	d
T2-24	The intrinsic semiconductor becomes an insulator at (a) 0 °C (b) 0 K (c) 300 K (d) -100 °C	b
T2-25	In a p-n junction (a) potentials of p and n become higher alternately (b) p side is at higher potential than n (c) n side is at higher potential than p (d) both at same potential	c

#	Question and options	Answer [derived]
T2-26	If a full wave rectifier operates from 50 Hz mains, the fundamental frequency in the ripple will be (a) 100 Hz (b) 25 Hz (c) 50 Hz (d) 70.7 Hz	a
T2-27	An n-p-n transistor conducts when? (a) both collector and emitter negative w.r.t. base (b) both positive w.r.t. base (c) collector positive and emitter negative w.r.t. base (d) collector positive and emitter at same potential as base	c
T2-28	Barrier potential of a p-n junction diode does not depend on? (a) doping density (b) diode design (c) temperature (d) forward bias	b
T2-29	Reverse bias applied to a junction diode (a) increases the minority carrier current (b) lowers the potential barrier (c) raises the potential barrier (d) increases the majority carrier current	c
T2-30	In semiconductors at room temperature (a) conduction band completely empty (b) valence band partially empty and conduction band partially filled (c) valence band completely filled and conduction band partially filled (d) valence band completely filled	b

A.4.8 NIAR EXERCISE, MODULATION, BANDS AND ANTENNAS (GENERAL-GRADE ADDITIONAL TOPICS) — 30 ITEMS (MANUAL PP. 82–85)

No answer key printed. Topic: **radio theory**.

#	Question and options	Answer [derived]
T3-1	What does AM mean? a) Angelo Marconi b) Anno median c) Amplitude modulation d) Amperes	c
T3-2	What frequency range is the High Frequency band? a) 3-30 KHz b) 30-300 KHz c) 3 to 30 MHz d) 30 to 300 MHz	c
T3-3	Sometimes computers are connected to a UPS system. What does UPS mean? a) United Parcel Service b) Uniform Product Support c) Under Paneling Storage d) Uninterruptible Power Supply	d
T3-4	A given signal's second harmonic is twice the given signal's _____ frequency? a) Fourier b) Foundation c) Fundamental d) Field	c
T3-5	Voltage is sometimes referred to as EMF or Electromotive...? a) Field b) Factor c) Flash d) Force	d
T3-6	"FET" is a type of transistor; its full name is _____ Effect Transistor? a) Field b) Factor c) Flash d) Force	a
T3-7	When measuring a small-signal amplifier one might be concerned with its "Noise..."? a) Fundamental b) Fall c) Force d) Figure	d
T3-8	Which of the following will have least wavelength a) VHF b) EHF c) HF d) UHF	b
T3-9	30 to 300 MHz frequency is categorized as: a) MF b) VHF c) UHF d) SHF	b

#	Question and options	Answer [derived]
T3-10	For a frequency of 30 MHz the wavelength will be: a) 10 m b) 1 m c) 0.1 m d) 0.01 m	a
T3-11	The range of audio frequency is: a) 20 kHz to 15 MHz b) 20 kHz to 150 MHz c) 20 Hz to 15 MHz d) 20 Hz to 15 kHz	d
T3-12	The unit of electric field intensity E is: a) Volts/metre b) Ampere/metre c) Coulombs/metre d) Henry/metre	a
T3-13	The current distribution on a half wave dipole is: a) uniform b) sinusoidal c) triangular d) complex	b
T3-14	For very high frequencies earth acts as a: a) conductor b) dielectric c) resistor d) None of these	b
T3-15	A wavelength of 15 mm could be expected in: a) VHF b) HF c) EHF d) SHF	c (15 mm corresponds to 20 GHz, near the SHF/ EHF boundary; NIAR intends EHF)
T3-16	The gain of an antenna: a) varies inversely as wavelength b) varies inversely as square of wavelength c) is independent of wavelength d) varies directly as wavelength	b
T3-17	Yagi antenna is used for: a) television b) radar c) medium wave broadcasting d) None of these	a
T3-18	Which of the following will increase the antenna radiation efficiency? a) Use of larger section of conductor b) providing insulation on conductor c) top loading of antenna d) any of the above	a
T3-19	The director in a Yagi antenna: a) is longer than the radiating element b) is shorter than the radiating element c) can be longer or shorter d) does not exist	b
T3-20	Antenna used in radar is: a) dipoles b) yagi antenna c) parabolic dishes d) None of these	c
T3-21	Television receiver antenna is usually: a) rhombic b) yagi c) parabolic dishes d) turnstile	b
T3-22	Which of the following antennas gives circular polarization: a) yagi-uda b) parabolic c) dipole d) helical	d
T3-23	Which of the following statements is false? a) A grounded $\lambda/4$ antenna is one half the length of a $\lambda/2$ dipole b) A vertical antenna transmits in all directions in a circle in the horizontal plane c) A grounded vertical antenna transmits ground waves and sky waves d) A counterpoise must be a good insulator	d
T3-24	Which of the following can make the antenna electrically longer? a) Series capacitor b) capacitive top loading c) vertical polarisation d) Circular polarisation	b

#	Question and options	Answer [derived]
T3-25	A unipole is also known as: a) omnidirectional radiator b) unidirectional radiator c) line radiator d) None of the above	a
T3-26	Radiation intensity does not depend upon: a) the antenna direction b) distance from the radiator c) both (a) and (b) d) None of the above	b
T3-27	The ratio of power radiated in the desired direction to the power radiated in the opposite direction is known as: a) transmission efficiency b) front to back ratio c) loss coefficient d) None of the above	b
T3-28	In case of antenna, decreasing Q: a) increases bandwidth b) decreases bandwidth c) independent of bandwidth d) None of the above	a
T3-29	The directional pattern of loop antennas: a) depends on shape of the loop b) is independent of shape of the loop c) depends on number of turns of loop d) None of the above	b
T3-30	The ratio of radiation intensity in a particular direction to average radiated power is called a) directive gain b) power gain c) directivity d) DIRP	a

A.4.9 S8 — NIAR "MODEL TYPE QUESTIONS", 12 ITEMS (MANUAL P. 86)

No answer key printed. Topic: **radio theory**.

#	Question and options	Answer [derived]
M-1	The following Amplifier is not used as audio amplifier because of distortion. a) Class A b) Class B d) Class C d) Both A & B (<i>option labels are mis-typed in the manual</i>)	Class C
M-2	It is desired to receive an amplitude modulated signal of frequency 2000 Hz. If the IF of the receiver is 465 KHz the LO frequency should be a) 2465 KHz b) 2000 KHz c) 465 KHz d) None	a [UNCERTAIN — the question is garbled; it works only if "2000 Hz" is read as "2000 kHz", giving $2000 + 465 = 2465$ kHz]
M-3	The range of MF is a) 30 to 300 KHz b) 300 to 3000 KHz c) 3 to 30 KHz d) None	b
M-4	The polarization of an electromagnetic wave is defined by its a) Propagation b) The E field c) The H field d) The transmitting antenna	b
M-5	The half wave dipole aerial has a theoretical impedance of a) 650 Ohms b) 720 Ohms c) 800 Ohms d) None	Treat as defective — the textbook figure for a free-space half-wave dipole is 73Ω, and none of the offered options is right.
M-6	A half wave antenna resonant at 7100 KHz is approximately the following metres long a) 20 Metres b) 40 Metres c) 80 Metres d) 160 Metres	a ($\lambda \approx 42$ m, so $\lambda/2 \approx 21$ m)

#	Question and options	Answer [derived]
M-7	As temperature increases, the resistance of a metallic conductor: a) Decreases b) Increases c) remains Constant d) Becomes negative	b
M-8	Impedance is a combination of a) Reactance with reluctance b) Resistance with conductance c) Resistance with reactance d) Reactance with radiation	c
M-9	An inductor and a capacitor are connected in series. At resonant frequency impedance is a) Minimum b) Maximum c) Totally resistive d) Reactance with radiation	a (and it is also purely resistive)
M-10	The major charge carriers of an NPN transistor are a) Holes b) Electrons c) Ions d) None	b
M-11	A diode in proper biasing condition converts a) AC to DC b) DC to AC c) AC to pulsating DC d) None	c
M-12	One millivolt is equal to a) 0.01 volt b) 0.0001 volt c) 0.001 volt d) None	c

A.5 S3 — DARTS "Sample ASOC Exam Question Paper" (100 questions)

Source type: club-published practice paper. Document: "SAMPLE ASOC EXAM QUESTION PAPER", 15 pp, PDF created 2014-05-14, <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-3.pdf> (mirrored at <https://www.aars.in/ASOC-Exam-Sample-Question-Paper-3.pdf>). Published by DARTS, the amateur radio club at darts.org.in in the Delhi National Capital Region. [REPORTED]

Why it is worth taking seriously. Its rubric reproduces the official one almost word for word — *"There will be 2 sections in the question paper. Each section contains 50 questions... Candidates appearing for Restricted grade should answer only first 25 questions from each of the section."* — and the content is Indian, not imported. **Why to be careful:** DARTS states no provenance whatsoever. It is *not* a past paper and must not be described as one. **No answer key is printed.**

Answers below are **[derived]**; an asterisk marks a question considered defective or genuinely ambiguous, explained after the table. Items 1–50 are **radio theory**; items 51–100 are **regulations**.

#	Question and options (verbatim)	Answer [derived]
1	Parameter that determines lowest signal reception capability of a receiver is a) Sensitivity b) Fidelity c) Selectivity d) Squelch	a

#	Question and options (verbatim)	Answer [derived]
2	TRF stands for a) Turning radio frequency b) Turning rod frequency c) Tuned radio feed d) Tuned radio frequency	d
3	If a super hetero dyne radio receiver is tuned to 1255 KHz & the IF frequency is 455KHz, then the frequency of the local oscillator of the receiver is a) 1255 KHz b) 800 KHz c) 1710 KHz d) 455 KHz	c
4	Kirchhoff's voltage Law states that a) The sum of voltages in a loop is zero b) The algebraic sum of voltages in a loop is zero c) The sum of currents at a junction is zero d) The algebraic sum of currents at a junction is zero	b
5	Among all these materials, which one has poor conductivity? a) Copper b) Aluminum c) Silicon d) Silver	c
6	A magnet does not attract which one in the following a) Nickel b) Aluminum c) Iron d) Cobalt	b
7	The unit of resistance is a) Mho b) Volts c) Ampere d) Ohm	d
8	The expanded form for 'LDR' is a) Light Emitting Display b) Low Dependent Reading c) Light Emitting Diode d) Light Dependent Resistor	d
9	The three leads of a common transistor are a) Base Collector Case b) Emitter Collector Bias c) Collector Base Omitter d) Collector Base Emitter	d
10	If R is resistance, I is current and V is voltage, then Ohms law states that a) $R=V/I$ b) $I=R/V$ c) $V=R/I$ d) $V/(R/I)=1$	a
11	If two capacitors are placed in series, is the final capacitance will be: a) Higher than the sum of the two capacitors b) Lower than the least of the two capacitors c) The sum of the two capacitors d) Cannot be determined the given data.	b
12	Which is the unit of energy in electronics? a) Ohm b) Ampere c) Joules d) Watt	c
13	The high frequency band (HF) is a) 3KHz to 30KHz b) 3MHz to 30MHz c) 30KHz to 300KHz d) 30MHz to 300MHz	b

#	Question and options (verbatim)	Answer [derived]
14	Among the following which is not a directional antenna a) Whip antenna b) Yagi-Uda antenna c) Loop antenna d) Dish antenna	a
15	The unit of current is a) Volts b) Ampere c) Watt d) Hertz	b
16	If a small value of inductance is connected in parallel with a large value, the combined inductance will be a) The same b) Higher c) Lower d) Cannot be determined with the given data.	c
17	The equivalent capacitance when three capacitors of 150 micro farad are connected in series is a) 450 micro farad b) 300 micro farad c) 150 micro farad d) 50 micro farad	d
18	A resistor with color bands: Red-Violet-Orange-Silver, has the value of a) 23 kilo Ohm, 5% tolerance b) 27 kilo Ohm, 5% tolerance c) 2.7 kilo Ohm, 10% tolerance d) 27 kilo Ohm, 10% tolerance	d
19	The abbreviation of RMS voltage is a) Railway Mail Service voltage b) Real Mean Square voltage c) Root Mean Square voltage d) Root Mean Service voltage	c
20	If the peak voltage of a sinusoidal ac supply is 150 volts, the RMS value is a) Less than peak value b) More than peak value c) Equal to peak value d) Not related to peak value of the voltage	a
21	A capacitor and inductor in series is called a a) Tuned Circuit b) Timing Circuit c) Delay Circuit d) Schmitt Circuit	a
22	A domestic inverter a) Converts ac to ac b) Converts ac to dc c) Converts dc to ac d) Converts dc to dc	c
23	The current in a circuit is 25mA, then the equivalent value is a) 0.00025 Ampere b) 0.0025 Ampere c) 0.025 Ampere d) 0.25 Ampere	c
24	If 2 mA current flows through a 100 Kohm resistor, then the voltage across it is a) 2 volts b) 20 volts c) 200 volts d) 0.2 volts	c
25	The tolerance bands with colors gold, silver & brown represent a) 5%, 10%, 2% b) 5%, 10%, 1% c) 10%, 5%, 1% d) 10%, 5%, 2%	b

#	Question and options (verbatim)	Answer [derived]
26	India uses which encoding technique for commercial TV broadcast a) NTSC b) SECAM c) PAL d) None of the above	c
27	The video bandwidth in a TV transmission in India is a) 455 KHz b) 10.7 MHz c) 4.2 MHz d) 5.5 MHz	d*
28	Sky wave propagation utilizes a) Troposphere b) Stratosphere c) Ionosphere d) Exosphere	c
29	Ionosphere does not have _ layer. a) C b) D c) E d) F	a
30	Which layer of ionosphere disappears during night time? a) C b) D c) E d) F	b
31	In dealing with sky/ground wave propagation, the area of silence where no signals are received is the a) dead zone b) skip distance c) skip zone d) propagation distance	c
32	Frequencies lesser than the critical frequency are a) passed into space b) the most desirable c) returned to earth d) refracted by the F2 layer	c
33	For mobile communications, the up-link frequency is ____ down link frequency. a) Less than b) More than c) Equal to d) Less or more than the	b
34	All India Radio (AIR) does not broadcast a) Amplitude Modulation transmission b) Frequency Modulation transmission c) Pulse Modulation transmission d) Both a and b	c
35	A diode can be used in the following stages of a radio receiver a) Detector b) Mixer c) Rectification d) All of the above	d
36	Geo stationary orbit is ___ km away from the earth a) 3100 b) 32786 c) 36786 d) 39000	c*
37	Which antenna is used for direction finding purpose? a) Loop antenna b) Active Whip antenna c) Whip antenna d) Dipole antenna	a
38	A transformer converts a) AC voltage into DC voltage b) DC voltage into AC voltage c) AC voltage into different AC voltage d) DC voltage into different DC voltage	c
39	A matching transformer (10:1 turns ratio) is used to match 200 ohms into ___ ohms. a) 2 b) 20 c) 200 d) 5	a

#	Question and options (verbatim)	Answer [derived]
40	The following are transformer losses a) Hysteresis losses b) Eddy current losses c) Copper losses d) All of the above	d
41	Digital modulation techniques are a) Amplitude modulation b) Frequency modulation c) Phase shift keying d) Phase modulation	c
42	The following idle component dissipates energy a) Capacitor b) Inductor c) Resistor d) All of the above	c
43	The most common stage used for amplifying an electrical signal is a) Common Base b) Common collector c) Common emitter d) Emitter follower	c
44	The biasing of a transistor is to a) To avoid thermal run away b) To provide stable operation c) To have a stable quiescent point d) All of the above	d
45	The ideal value of ripple factor should be a) 0 b) 1 c) 0.707 d) 1.414	a
46	The full wave rectifier employs a) 1 diode b) 2 diodes c) 3 diodes d) 4 diodes	b*
47	The following is used as a Voltage smoothing device a) Resistor b) Capacitor c) Inductor d) Both a and b	b
48	The following modulation is used for audio transmission in TV broadcast a) Amplitude modulation b) Frequency modulation c) Pulse modulation d) None of the above	b
49	NICAM technique is used in TV broadcast a) For transmitting video b) For transmitting mono aural audio c) For transmitting stereo audio d) For transmitting timing signals	c
50	Conventional Facsimile machine sends __ data over __ media a) Analog, air b) Digital, air c) Analog, telephone d) Digital, telephone	c
51	Under Indian Wireless Telegraph rules, 1973, the minimum age to work on a radio transmitting apparatus is ____ years a) 12 b) 15 c) 18 d) 21	a
52	An amateur is forbidden to transmit a) communications of business b) transmissions of entertainment value or music c) advertisements d) all the above	d

#	Question and options (verbatim)	Answer [derived]
53	Log containing chronological record of all transmissions should be preserved for a) 1 day b) 1 month c) 6 months d) 12 months	d
54	Log containing chronological record of all transmissions should contain a) Date and time of all transmissions b) Call signs intercepted c) Summary of communications d) All of the above	d
55	Which of the following should be given top priority? a) Safety signal b) Routine communication signal c) Distress signal d) Urgency signal	c
56	The following is not an amateur call sign of India a) VU2XYZ b) VU3ABC c) VU44AB d) VU3RS	c
57	The standard time & frequency signal in India is a) 5 MHz b) 10 MHz c) 20 MHz d) 100 MHz	b
58	The standard time & frequency signal in India call sign is a) VU2 b) VU3 c) ATA d) ATU	d
59	The abbreviation for end of message in Morse code is a) AA b) AR c) ED d) EN	b
60	QTR stands for? a) What is the correct time? b) What is your position in latitude and longitude? c) Thank you for sending d) What is Time and Hour?	a
61	The standard emission designation consists of a) 8 letters/numerals b) 3 letters/numerals c) 6 letters/numerals d) 2 letters/numerals	b*
62	Metric waves means a) 3-30 MHz b) 3-30 KHz c) 30-300 KHz d) 30-300 MHz	d
63	The standard bandwidth designation consists of a) 5 letters/numerals b) 3 letters/numerals c) 4 letters/numerals d) 2 letters/numerals	c
64	Which should be used as a suffix in call sign for mobile amateur stations a) ME b) MB c) MO d) ML	c
65	The Q code for "What is the correct time" is a) QCT b) QTM c) QRT d) QTR	d
66	The Q code for "Are my signals fading?" is a) QSB b) QAM c) QSI d) QSF	a
67	The Q code QRQ represents a) Shall I stop slowing? b) Shall I increase power? c) Shall I send faster? d) Shall I decrease power?	c

#	Question and options (verbatim)	Answer [derived]
68	The Phonetic alphabet for L is a) Lilly b) Lima c) Limca d) Lisa	b
69	The Phonetic for '3' is a) Terra Three b) Tele Three c) Tango Three d) Tango Tree	a
70	The expanded form for AA in Morse code is a) End of line b) End of message c) End of transmission d) End of signal	a
71	<i>(printed as "11") — a numbering typo in the source</i> The expanded form for TU in Morse code is a) Thank u b) Thank you c) Thank everyone d) None of the above	b
72	The expanded form for CQ in Morse code is a) Calling you b) Calling any station c) Calling Quebec d) Calling none	b
73	The abbreviation for 'stand by' in Morse code is a) SY b) AS c) SB d) AD	b
74	The Q code for "What is the name of your station?" is a) QNS b) QRA c) QYS d) QNA	b
75	The Q code for "Are you ready?" is a) QRY b) QRV c) QAR d) QRE	b
76	The expanded form of SOS is a) Save our selves b) Save one self c) Save one souls d) Save our souls	d
77	The distress signal in radio telephony is a) Help, Help, Help b) May day, May day, May day c) Save, Save, Save d) Save our souls	b
78	The phonetic for alphabet 'Y' is a) Yardly b) Yankee c) Yan d) Yarn	b
79	The phonetic for '9' is a) New Nine b) Nove Nine c) Novel Nine d) Nine	b
80	Which one of this is not a standard frequency signal a) 15 MHz b) 10 MHz c) 5 MHz d) 1 MHz	d
81	The renewal fee for General Grade Amateur license under new rules is a) Rs 1000 b) Rs 2000 c) Rs. 100 d) Rs.200	[UNCERTAIN — a 2014 question about the then-"new" rules. Under the 2024 Rules the ASOC fee is ₹1,000 for twenty years and ₹2,000 for lifetime, identically for both grades.] [VERIFIED — 2024 Rules r.6(1)]
82	The fee for mobile endorsement under new rules is a) Rs 100 b) Rs. 200 c) Rs. 1000 d) Rs. 2000	[UNCERTAIN — pre-2024 fee question; do not memorise a figure from a 2014 club paper]

#	Question and options (verbatim)	Answer [derived]
83	The maximum DC power permitted under new rules for Restricted grade amateurs under 144-146 MHz band is a) 10 watts b) 25 watts c) 50 watts d) 100 watts	a [VERIFIED — 2010 OM, Annexure-I]
84	The maximum DC power permitted under new rules for General grade amateurs under 7000-7100 KHz band is a) 50 watts b) 100 watts c) 200 watts d) 400 watts	d [VERIFIED — 2010 OM, Annexure-I]
85	The permitted emission under new rules for Restricted grade amateurs under 144-146 MHz band is a) A3E b) J3E c) F3E d) A1A	c [VERIFIED — 2010 OM, Annexure-I]
86	The first symbol in the designation for classification of emission represents a) Nature of signals modulating the main carrier b) Type of modulation of the main carrier c) Type of information to be transmitted d) None of the above	b
87	Emission designation of commercial FM broadcast transmission in India is a) ME b) J3E c) F3E d) R3E	c
88	Conventional Morse code transmission can be represented by a) J3E b) F3E c) ME d) A1A	d
89	The second symbol in the designation of emission represents a) Nature of signals modulating the main carrier b) Type of modulation of the main carrier c) Type of information to be transmitted d) None of the above	a
90	The following frequency band is permitted for general grade amateurs only a) 1820 - 1860 KHz b) 3500 - 3700 KHz c) 5725 - 5840 MHz d) 21000 - 21450 KHz	defective — see note
91	The fee for change of location new rules is a) Rs. 100 b) Rs. 200 c) Rs. 500 d) Rs. 1000	(d) ₹1,000 is the current figure under 2024 Rules r.7(6); at the time this paper was written the answer was ₹200. [VERIFIED — r.7(6)]
92	The maximum DC power permitted under new rules for General grade amateurs under 144-146 MHz band is a) 10 watts b) 25 watts c) 50 watts d) 100 watts	b — 25 W [VERIFIED — 2010 OM, Annexure-I, General Grade table]
93	The standard designation for a bandwidth of 4.8 KHz is a) 4800 Hz b) 4K8 c) 4K80 d) 4800	c
94	A bandwidth of 402 M represents a) 4000 MHz b) 402 MHz c) 4.2 MHz d) 420 MHz	b

#	Question and options (verbatim)	Answer [derived]
95	The emission C3F denotes a) Single side band analog audio emission b) Common analog radio emission c) Vestigial sideband analog TV emission d) Vestigial sideband digital TV emission	c
96	The expanded form for RST is a) Readability signal tone b) Readability strength tone c) Readability strength testing d) Readability signal strength tone	d*
97	The phonetic for 'W' is a) Wine b) White c) Whiskey d) Wheel	c
98	The UHF band extends from a) 3-30 KHz b) 30-300 MHz c) 3-30 MHz d) 300-3000 MHz	d
99	Which of the frequency bands was not allocated for Amateur service a) 7000 - 7100 KHz b) 7100 - 7200 KHz c) 10120 - 10240 KHz d) 28000 - 29700 KHz	defective — see note
100	'WPC' stands for a) Wireless Planning and Coordination b) Wireless Placement cell c) Wireless Planning cell d) Wireless Placement and Coordination	a

Notes on the flagged items:

- **Q27, video bandwidth in Indian television.** The intended answer is 5.5 MHz, but that figure is the sound-to-vision carrier spacing in CCIR System B/G. The *video* bandwidth is 5 MHz. Defective.
- **Q36, geostationary orbit distance.** The real figure is 35,786 km; the paper offers "36786". Choose the nearest option, (c).
- **Q46, the full-wave rectifier.** Two diodes in the centre-tapped configuration, four in a bridge. The paper wants (b), yet the official paper at S1 A11 establishes that a bridge rectifier is also full-wave. Ambiguous.
- **Q61, length of the standard emission designation.** The answer given here is (b), three letters/numerals, being the three-symbol class of emission. **SIARS publishes (c), six, for the identical question (see A.8).** Under ITU RR Appendix 1 a full designation is four bandwidth characters *plus* three class symbols, making seven. **[UNCERTAIN — the sources conflict; ask WPC.]**
- **Q82 and Q91, fees. [UNCERTAIN.]** These are pre-2024 fee questions. The current figures are in the fee table of the post-examination part of this report.
- **Q90, band permitted to General grade only. Defective.** Under the regime then in force, 1820–1860 kHz, 3500–3700 kHz *and* 5725–5840 MHz were all General-grade-only, so three of the four options are correct. Only 21000–21450 kHz, option (d), was open to the Restricted grade.

- **Q96, RST expanded.** Both (b) "Readability strength tone" and (d) "Readability signal strength tone" express the right thing. The official paper's wording at S1 B6 is "Readability, Signal strength and Tone", so (d).
- **Q99, band not allocated to the amateur service. Defective.** 10120–10240 kHz, option (c), is not an amateur band either — the 30 m allocation is 10100–10150 kHz — and 7100–7200 kHz has been amateur in ITU Region 3 since 2009. **[UNCERTAIN.]**

A.6 S4 — DARTS "Amateur Radio Exam Question Paper Sample" (188 questions, with a published key)

Source type: club-published practice paper. Document: 19 pp, PDF created 2014-02-24, source word-processor file titled "EXAM QUESTION PAPER". <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Sample-Question-Paper-2.pdf> [REPORTED]

Read this before you use it. This is the largest answered bank in circulation and the one most often passed around in Indian amateur-radio messaging groups — and it is **the least trustworthy source in this part:**

1. **Roughly half of it is lifted verbatim from the United States FCC Technician and General question pools**, with "FCC" search-and-replaced by "WPC". The tell-tales are unmistakable: azimuthal projection maps, 20 m data segments, "repeater trustee", "1500 Watts PEP", "ITU Regions 1 and 3 must yield to Region 2". None of that is Indian law.
2. **The published answer key contains at least seventeen demonstrable errors**, tabulated at A.6.3. Two of them — minimum age and log timekeeping — are on points the real examination does test.
3. It has **no stated provenance whatsoever**.

Use it for drill volume, not for law. Where it disagrees with S1 (the official paper) or with the Rules, S1 and the Rules win.

In the tables below, **Key** is **the answer DARTS published**, not a corrected answer. Rows whose key is believed wrong are listed separately at A.6.3.

A.6.1 BLOCK 1 — "RULES & REGULATIONS: OPERATING PROCEDURES", 60 ITEMS (TOPIC: REGULATIONS)

#	Question and options (verbatim)	Key
1	Which emission mode must be used to obtain assistance during a disaster? a) Only SSB b) Only SSB and CW c) Any mode d) Only CW	c
2	What should you do if a CW station sends "QRS" when using Morse code? a) Send slower b) Change frequency c) Increase your power d) Repeat everything twice	a

#	Question and options (verbatim)	Key
3	What is the recommended way to break into a conversation when using phone? a) Say "QRZ" several times followed by your call sign b) Say your call sign during a break between transmissions from the other stations c) Say "Break" "Break" "Break" and wait for a response d) Say "CQ" followed by the call sign of either station	c
4	Which of the following 20 meter band segments is most often used for most data transmissions? a) 14.000 – 14.050 MHz b) 14.070 - 14.100 MHz c) 14.150 - 14.225 MHz d) 14.275 - 14.350 MHz	b
5	What action should be taken if the frequency on which a net normally meets is in use just before the net begins? a) Reduce your output power and start the net as usual b) Increase your power output so that net participants will be able to hear you c) Ask the stations if the net may use the frequency, or move the net to a nearby clear frequency if necessary d) Cancel the net for that day	c
6	Which of the following is an advantage when using single sideband as compared to other voice modes on the HF amateur bands? a) Very high fidelity voice modulation b) Less bandwidth used and high power efficiency c) Ease of tuning on receive d) Less subject to static crashes (atmospherics)	b
7	What is an azimuthal projection map? a) A world map projection centered on the North Pole b) A world map projection centered on a particular location c) A world map that shows the angle at which an amateur satellite crosses the equator d) A world map that shows the number of degrees longitude that an amateur satellite appears to move westward at the equator with each orbit	b
8	How do you call another station on a repeater if you know the station's call sign? a) Say "break, break" then say the station's call sign b) Say the station's call sign then identify your own station c) Say "CQ" three times then the other station's call sign d) Wait for the station to call "CQ" then answer it	b
9	The frequency of 40 Meter band in MHz is a) 14 - 14.350 b) 7 - 7.2 c) 21 - 21.450 d) 15 - 15.400	b
10	Which sideband is commonly used in the VHF and UHF bands? a) Upper Side Band b) Lower side band c) Vestigial side band d) Double side band	d
11	When are you prohibited from helping a station in distress? a) When that station is not transmitting on amateur frequencies b) When the station in distress offers no call sign c) You are never prohibited from helping any station in distress d) When the station is not another amateur station	a
12	What is a practical way to avoid harmful interference when calling CQ using Morse code or CW? a) Send the letter "V" 12 times and then listen for a response b) Keep your CQ to less than 2 minutes in length to avoid interference with contacts already in progress c) Send "QRL? de" followed by your call sign and listen for a response d) Call CQ at low power first; if there is no indication of interference then increase power as necessary	d
13	What does it mean when a CW operator sends "KN" at the end of a transmission? a) Listening for novice stations b) Operating full break-in c) Listening only for a specific station or stations d) Closing station now	c

#	Question and options (verbatim)	Key
14	Who is accountable if a repeater station inadvertently retransmits communications that violate WPC rules? a) The repeater trustee b) The repeater control operator c) The transmitting station d) All of these answers are correct	c
15	Which of the following statements is true of the single sideband (SSB) voice mode? a) Only one sideband and the carrier are transmitted; the other sideband is suppressed b) Only one sideband is transmitted; the other sideband and carrier are suppressed c) SSB voice transmissions have higher average power than any other mode d) SSB is the only mode that is authorized on the 160, 75 and 40 meter amateur bands	b
16	Which layer of ionosphere disappears during night time? a) F b) E c) D d) C	c
17	When sending CW, what does a "C" mean when added to the RST report? a) Chirpy or unstable signal b) Report was read from S meter reading rather than estimated c) 100 percent copy d) Key clicks	a
18	How do you call another station on a repeater if you know the station's call sign? a) Say "break, break" then say the station's call sign b) Say the station's call sign then identify your own station c) Say "CQ" three times then the other station's call sign d) Wait for the station to call "CQ" then answer it	b
19	What does the Q signal "QSL" mean when operating CW? a) We have already confirmed by card b) I acknowledge receipt c) We have worked before d) Send slower	b
20	What is the first thing you should do if you are communicating with another amateur station and hear a station in distress break in? a) Continue your communication because you were on frequency first b) Acknowledge the station in distress and determine what assistance may be needed c) Change to a different frequency d) Immediately cease all transmissions	b
21	How do you indicate you are looking for any station with which to make contact? a) CQ followed by your call sign b) RST followed by your call sign c) QST followed by your call sign d) SK followed by your call sign	a
22	What should you transmit when responding to a call of CQ? a) Your own CQ followed by the other station's call sign b) Your call sign followed by the other station's call sign c) The other station's call sign followed by your call sign d) A signal report followed by your call sign	c
23	What must an amateur do when making a transmission to test equipment or antennas? a) Properly identify the station b) Make test transmissions only after 10:00 PM local time c) Notify the WPC of the test transmission d) State the purpose of the test during the test procedure	a
24	What is the meaning of the procedural signal "CQ"? a) Call on the quarter hour b) New antenna is being tested (no station should answer) c) Only the called station should transmit d) Calling any station	d
25	What brief statement is often used in place of "CQ" to indicate that you are listening for calls on a repeater? a) Say "Hello test" followed by your call sign b) Say your call sign c) Say the repeater call sign followed by your call sign d) Say the letters "QSY" followed by your call sign	b

#	Question and options (verbatim)	Key
26	Why should you use the International Telecommunication Union (ITU) phonetic alphabet when identifying your station? a) The words are internationally recognized substitutes for letters b) There is no advantage c) The words have been chosen to represent amateur radio terms d) It preserves traditions begun in the early days of amateur radio	a
27	Who is in charge of the repeater frequency band plan in your local area? a) The local WPC monitoring office b) Only WPC HO New Delhi c) The recognized frequency coordination body d) Amateur Radio society of India	b
28	What is the main purpose of repeater coordination? a) To reduce interference and promote proper use of spectrum b) To coordinate as many repeaters as possible in a small area c) To coordinate all possible frequencies available for repeater use d) To promote and encourage use of simplex frequencies	b
29	Which of these statements is true about legal power levels on the amateur bands? a) Always use the maximum power allowed to ensure that you complete the contact b) An amateur may use no more than 200 Watts PEP to make an amateur contact c) An amateur may use up to 1500 Watts PEP on any amateur frequency d) An amateur must use the minimum transmitter power necessary to carry out the desired communication	d
30	What is the proper way to break into a conversation between two stations that are using the frequency? a) Say your call sign between their transmissions b) Wait for them to finish and then call CQ c) Say "Break-break" between their transmissions d) Call one of the operators on the telephone to interrupt the conversation	c
31	Amateurs are forbidden to transmit about a) Equipments b) weather c) Antennas d) Third party messages	d
32	Standard time and frequency is transmitted on a) 7050 KHz b) 14050 KHz c) 21050 KHz d) 10000 KHz	d
33	What is considered to be proper repeater operating practice? a) Monitor before transmitting and keep transmissions short b) Identify legally c) Use the minimum amount of transmitter power necessary d) All of these answers are correct	a
34	What rule applies if two amateur stations want to use the same frequency? a) The station operator with a Restricted Grade license must yield the frequency to a General Grade licensee b) The station operator with a lower power output must yield the frequency to the station with a higher power output c) No frequency will be assigned for the exclusive use of any station and neither has priority d) Station operators in ITU Regions 1 and 3 must yield the frequency to stations in ITU Region 2	c
35	What should you do if you hear a newly licensed operator that is having trouble with their station? a) Tell them to get off the air until they learn how to operate properly b) Report them to the WPC HO c) Contact them and offer to help with the problem d) Move to another frequency	c
36	A3E indicates a) SSB b) AM-DSB voice c) FM Voice d) FSK	b
37	Line of sight propagation is the mode of communication in a) LF b) HF c) MF d) VHF	d
38	The wavelength of 300 MHz in Meters is a) 1 b) .1 c) 1.1 d) 0.01	a

#	Question and options (verbatim)	Key
39	Squelch control is used to eliminate a) static interference b) electrical disturbance c) receiver noise d) unwanted carrier	c
40	4th harmonic of 2.5 MHz is a) 10 MHz b) 15 MHz c) 8 MHz d) 7.5 MHz	a
41	The UHF range is a) 30 to 300 KHz b) 300 to 3000 KHz c) 3 to 30 GHz d) 300 to 3000 MHz	d
42	Indian amateurs can communicate with other amateurs in a) All countries b) Countries permitted by ITU c) Countries permitted by Indian Government d) Countries permitted by Indian Amateur society	a
43	Restricted grade amateurs can communicate on 7 MHz in a) A1 b) A3 c) F3 d) A3E	d
44	Lady amateurs are known as a) XL b) XYL c) YL d) LY	c
45	Minimum age to become an amateur is a) 18 years b) 14 years c) 12 years d) 16 years	a
46	All timing in the Log book should be in a) IST b) UTC c) GMT d) Local time	a
47	Amateurs should preserve their log for a period of a) 6 months b) 1 year c) 2 years d) 9 months from the date of the last entry	b
48	Q code to indicate time is a) QRG b) QRX c) QTR d) QSA	c
49	Test signal shall not be continued more than a) 30 seconds b) 1 minute c) 2 minutes d) 3 minutes	a
50	In India the standard time signal is broadcast by a) ISRO b) WPC c) OCS d) NPL	d
51	PANPAN transmitted thrice indicates a) Distress b) Emergency c) Urgency d) Distress and emergency	c
52	The broadcast of music is allowed in amateur service a) on request b) when channel is free c) never d) only for testing	c
53	The abbreviation VA means a) End of transmission b) End of message c) End of working d) End of schedule	c
54	SWLs are permitted to transmit in the frequency band of a) 7-7.1 MHz b) 3.89-3.9 MHz c) 144-146 MHz d) None of these	d
55	The amateur license is renewed by a) P & T b) Ministry of communication c) Monitoring stations d) None of these	b
56	FM Broadcasting station emission is a) A1E b) A3E c) J3E d) F3E	d
57	Q code to indicate the location of a station is a) QTL b) QTH c) QTN d) None of these	b
58	Amateur station on a ship can contact another amateur on land on a frequency authorized to a) the ship b) amateur stations c) by the ministry of communication d) ships calling frequency	b
59	SOS transmitted three times indicates a) urgency b) distress c) safety d) none of these	b
60	All timings in logbook should be in a) IST b) GMT c) UTC d) Local time	a

A.6.2 BLOCK 2 — "BASIC ELECTRONICS", 128 ITEMS (TOPIC: RADIO THEORY)

#	Question and options (verbatim)	Key
1	What is the name of a current that flows only in one direction? a) An alternating current b) A direct current c) A normal current d) A smooth current	c
2	What is the standard unit of frequency? a) The megacycle b) The Hertz c) One thousand cycles per second d) The electromagnetic force	b
3	How much voltage does an automobile battery usually supply? a) About 12 volts b) About 30 volts c) About 120 volts d) About 240 volts	a
4	What is the name of a current that reverses direction on a regular basis? a) An alternating current b) A direct current c) A circular current d) A vertical current	b
5	What is the term used to describe opposition to current flow in ordinary conductors such as wires? a) Inductance b) Resistance c) Counter EMF d) Magnetism	b
6	What instrument is used to measure the flow of current in an electrical circuit? a) Frequency meter b) SWR meter c) Ammeter d) Voltmeter	c
7	What instrument is used to measure Electromotive Force (EMF) between two points such as the poles of a battery? a) Magnetometer b) Voltmeter c) Ammeter d) Ohmmeter	b
8	What is the name for the distance a radio wave travels during one complete cycle? a) Wave speed b) Waveform c) Wavelength d) Wave spread	c
9	What term describes the number of times that an alternating current flows back and forth per second? a) Pulse rate b) Speed c) Wavelength d) Frequency	d
10	What does 50 hertz (Hz) mean? a) 5000 cycles per second b) 50 cycles per second c) 5000 meters per second d) 50 meters per second	b
11	Electromagnetic waves that oscillate more than 20,000 times per second as they travel through space are generally referred to as what? a) Gravity waves b) Sound waves c) Radio waves d) Gamma radiation	c
12	How fast does a radio wave travel through space? a) At the speed of light b) At the speed of sound c) Its speed is inversely proportional to its wavelength d) Its speed increases as the frequency increases	a
13	How does the wavelength of a radio wave relate to its frequency? a) The wavelength gets longer as the frequency increases b) The wavelength gets shorter as the frequency increases c) There is no relationship between wavelength and frequency d) The wavelength depends on the bandwidth of the signal	b
14	What is the formula for converting frequency to wavelength in meters? a) Wavelength in meters equals frequency in Hertz multiplied by 300 b) Wavelength in meters equals frequency in Hertz divided by 300 c) Wavelength in meters equals frequency in megahertz divided by 300 d) Wavelength in meters equals 300 divided by frequency in megahertz	d
15	What are sound waves in the range between 300 and 3000 Hertz called? a) Test signals b) Ultrasonic waves c) Voice frequencies d) Radio frequencies	c

#	Question and options (verbatim)	Key
16	What property of a radio wave is often used to identify the different bands amateur radio operators use? a) The physical length of the wave b) The magnetic intensity of the wave c) The time it takes for the wave to travel one mile d) The voltage standing wave ratio of the wave	a
17	What is the frequency range of the 2 meter band in the India? a) 144 to 146 MHz b) 222 to 225 MHz c) 434 to 438 MHz d) 50 to 54 MHz	a
18	What is used to convert radio signals into sounds we can hear? a) Transmitter b) Receiver c) Microphone d) Antenna	b
19	What is used to convert sounds from our voice into radio signals? a) Transmitter b) Receiver c) Speaker d) Antenna	a
20	What two devices are combined into one unit in a transceiver? a) Receiver, transmitter b) Receiver, transformer c) Receiver, transistor d) Transmitter, deceiver	a
21	What device is used to convert the alternating current from a wall outlet into low-voltage direct current? a) Inverter b) Compressor c) Power Supply d) Demodulator	c
22	What device is used to increase the output of a 10 watt radio to 100 watts? a) Amplifier b) Power supply c) Antenna d) Attenuator	a
23	Which of the battery types listed below offers the longest life when used with a hand-held radio, assuming each battery is the same physical size? a) Lead-acid b) Alkaline c) Nickel-cadmium d) Lithium-ion	d
24	What is the nominal voltage per cell of a fully charged nickel-cadmium battery? a) 1.0 volts b) 1.2 volts c) 1.5 volts d) 2.2 volts	b
25	What battery type on this list is not designed to be re-charged? a) Nickel-cadmium b) Carbon-zinc c) Lead-acid d) Lithium-ion	b
26	What is required to keep rechargeable batteries in good condition and ready for emergencies? a) They must be inspected for physical damage and replaced if necessary b) They should be stored in a cool and dry location c) They must be given a maintenance recharge at least every 6 months d) All of these answers are correct	d
27	What is the best way to get the most amount of energy from a battery? a) Draw current from the battery as rapidly as possible b) Draw current from the battery at the slowest rate needed c) Reverse the leads when the battery reaches the 1/2 charge level d) Charge the battery as frequently as possible	b
28	What formula is used to calculate current in a circuit? a) $I = E \times R$ b) $I = E / R$ c) $I = E + R$ d) $I = E - R$	b
29	What formula is used to calculate voltage in a circuit? a) $E = I \times R$ b) $E = I / R$ c) $E = I + R$ d) $E = I - R$	a
30	What formula is used to calculate resistance in a circuit? a) $R = E \times I$ b) $R = E / I$ c) $R = E + I$ d) $R = E - I$	b
31	What is the resistance of a circuit when a current of 3 amperes flows through a resistor connected to 90 volts? a) 3 ohms b) 30 ohms c) 93 ohms d) 270 ohms	b

#	Question and options (verbatim)	Key
32	What is the resistance in a circuit where the applied voltage is 12 volts and the current flow is 1.5 amperes? a) 18 ohms b) 0.125 ohms c) 8 ohms d) 13.5 ohms	c
33	What is the current flow in a circuit with an applied voltage of 120 volts and a resistance of 80 ohms? a) 9600 amperes b) 200 amperes c) 0.667 amperes d) 1.5 amperes	d
34	What is the voltage across the resistor if a current of 0.5 amperes flows through a 2 ohm resistor? a) 1 volt b) 0.25 volts c) 2.5 volts d) 1.5 volts	a
35	What is the voltage across the resistor if a current of 1 ampere flows through a 10 ohm resistor? a) 10 volts b) 1 volt c) 11 volts d) 9 volts	a
36	What is the voltage across the resistor if a current of 2 amperes flows through a 10 ohm resistor? a) 20 volts b) 0.2 volts c) 12 volts d) 8 volts	a
37	What is the current flowing through a 100 ohm resistor connected across 200 volts? a) 20,000 amperes b) 0.5 amperes c) 2 amperes d) 100 amperes	c
38	What is the current flowing through a 24 ohm resistor connected across 240 volts? a) 24,000 amperes b) 0.1 amperes c) 10 amperes d) 216 amperes	c
39	What is the formula used to calculate electrical power in a DC circuit? a) $P = E \times I$ b) $P = E / I$ c) $P = E - I$ d) $P = E + I$	a
40	How much power is represented by a voltage of 13.8 volts DC and a current of 10 amperes? a) 138 watts b) 0.7 watts c) 23.8 watts d) 3.8 watts	a
41	How much power is being used in a circuit when the voltage is 120 volts DC and the current is 2.5 amperes? a) 1440 watts b) 300 watts c) 48 watts d) 30 watts	b
42	How can you determine how many watts are being drawn by your transceiver when you are transmitting? a) Measure the DC voltage and divide it by 60 Hz b) Check the fuse in the power leads to see what size it is c) Look in the Radio Amateur's Handbook d) Measure the DC voltage at the transceiver and multiply by the current drawn when you transmit	d
43	How many amperes are flowing in a circuit when the applied voltage is 120 volts DC and the load is 1200 watts? a) 20 amperes b) 10 amperes c) 120 amperes d) 5 amperes	b
44	How many milliamperes is the same as 1.5 amperes? a) 15 milliamperes b) 150 milliamperes c) 1500 milliamperes d) 15000 milliamperes	c
45	What is another way to specify the frequency of a radio signal that is oscillating at 1,500,000 Hertz? a) 1500 kHz b) 1500 MHz c) 15 GHz d) 150 kHz	a
46	How many volts are equal to one kilovolt? a) One one-thousandth of a volt b) one hundred volts c) one thousand volts d) one million volts	c
47	How many volts are equal to one microvolt? a) one one-millionth of a volt b) one million volts c) one thousand kilovolts d) one one-thousandth of a volt	a
48	How many watts does a hand-held transceiver put out if the output power is 500 milliwatts? a) 0.02 watts b) 0.5 watts c) 5 watts d) 50 watts	b

#	Question and options (verbatim)	Key
49	What will happen to the resistance if the temperature of a carbon resistor is increased? a) It will increase by 20% for every 10 degrees centigrade b) It will stay the same c) It will change depending on the resistor's temperature coefficient rating d) It will become time dependent	c
50	What type of capacitor is often used in power-supply circuits to filter the rectified AC? a) Disc ceramic b) Vacuum variable c) Mica d) Electrolytic	d
51	Which of the following is the primary advantage of ceramic capacitors? a) Tight tolerance b) High stability c) High capacitance for given volume d) Comparatively low cost	d
52	Which of the following is an advantage of an electrolytic capacitor? a) Tight tolerance b) Non-polarized c) High capacitance for given volume d) Inexpensive RF capacitor	c
53	Which of the following is one effect of lead inductance in a capacitor used at VHF and above? a) Effective capacitance may be reduced b) Voltage rating may be reduced c) ESR may be reduced d) The polarity of the capacitor might become reversed	a
54	What is the main disadvantage of using a conventional wire-wound resistor in a resonant circuit? a) The resistor's tolerance value would not be adequate for such a circuit b) The resistor's inductance could detune the circuit c) The resistor could overheat d) The resistor's internal capacitance would detune the circuit	b
55	What is an advantage of using a ferrite core with a toroidal inductor? a) Large values of inductance may be obtained b) The magnetic properties of the core may be optimized for a specific range of frequencies c) Most of the magnetic field is contained in the core d) All of these choices are correct	d
56	How should two solenoid inductors be placed so as to minimize their mutual inductance? a) In line with their winding axis b) With their winding axes parallel to each other c) With their winding axes at right angles to each other d) Within the same shielded enclosure	c
57	Why might it be important to minimize the mutual inductance between two inductors? a) To increase the energy transfer between both circuits b) To reduce or eliminate unwanted coupling c) To reduce conducted emissions d) To increase the self-resonant frequency of both inductors	b
58	What is an effect of inter-turn capacitance in an inductor? a) The magnetic field may become inverted b) The inductor may become self resonant at some frequencies c) The permeability will increase d) The voltage rating may be exceeded	b
59	What is the common name for a capacitor connected across a transformer secondary that is used to absorb transient voltage spikes? a) Clipper capacitor b) Trimmer capacitor c) Feedback capacitor d) Suppressor capacitor	d
60	What is the common name for an inductor used to help smooth the DC output from the rectifier in a conventional power supply? a) Back EMF choke b) Repulsion coil c) Charging inductor d) Filter choke	d
61	What type of component is a thermistor? a) A resistor that is resistant to changes in value with temperature variations b) A device having a controlled change in resistance with temperature variations c) A special type of transistor for use at very cold temperatures d) A capacitor that changes value with temperature	b

#	Question and options (verbatim)	Key
62	What is the peak-inverse-voltage rating of a rectifier? a) The maximum voltage the rectifier will handle in the conducting direction b) 1.4 times the AC frequency c) The maximum voltage the rectifier will handle in the non-conducting direction d) 2.8 times the AC frequency	c
63	What are the two major ratings that must not be exceeded for silicon-diode rectifiers? a) Peak inverse voltage; average forward current b) Average power; average voltage c) Capacitive reactance; avalanche voltage d) Peak load impedance; peak voltage	a
64	What is the approximate junction threshold voltage of a germanium diode? a) 0.1 volt b) 0.3 volts c) 0.7 volts d) 1.0 volts	b
65	When two or more diodes are connected in parallel to increase current handling capacity, what is the purpose of the resistor connected in series with each diode? a) The resistors ensure the thermal stability of the power supply b) The resistors regulate the power supply output voltage c) The resistors ensure that one diode doesn't carry most of the current d) The resistors act as swamping resistors in the circuit	c
66	What is the approximate junction threshold voltage of a silicon diode? a) 0.1 volt b) 0.3 volts c) 0.7 volts d) 1.0 volts	c
67	Which of the following is an advantage of using a Schottky diode in an RF switching circuit as compared to a standard silicon diode? a) Lower capacitance b) Lower inductance c) Longer switching times d) Higher breakdown voltage	a
68	What are the stable operating points for a bipolar transistor that is used as a switch in a logic circuit? a) Its saturation and cut-off regions b) Its active region (between the cut-off and saturation regions) c) Between its peak and valley current points d) Between its enhancement and depletion modes	a
69	Why is it often necessary to insulate the case of a large power transistor? a) To increase the beta of the transistor b) To improve the power dissipation capability c) To reduce stray capacitance d) To avoid shorting the collector or drain voltage to ground	d
70	Which of the following describes the construction of a MOSFET? a) The gate is formed by a back-biased junction b) The gate is separated from the channel with a thin insulating layer c) The source is separated from the drain by a thin insulating layer d) The source is formed by depositing metal on silicon	b
71	Which element of a triode vacuum tube is used to regulate the flow of electrons between cathode and plate? a) Control grid b) Heater c) Screen Grid d) Suppressor grid	a
72	Which of the following solid state devices is most like a vacuum tube in its general characteristics? a) A bipolar transistor b) An FET c) A Tunnel diode d) A varistor	b
73	What is the primary purpose of a screen grid in a vacuum tube? a) To reduce grid-to-plate capacitance b) To increase efficiency c) To increase the high frequency response d) To decrease plate resistance	a
74	What is an advantage of the low internal resistance of Nickel Cadmium batteries? a) Long life b) High discharge current c) High voltage d) Rapid recharge	b
75	What is the minimum allowable discharge voltage for maximum life of a standard 12 volt lead acid battery? a) 6 volts b) 8.5 volts c) 10.5 volts d) 12 volts	c

#	Question and options (verbatim)	Key
76	When is it acceptable to recharge a carbon-zinc primary cell? a) As long as the voltage has not been allowed to drop below 1.0 volt b) When the cell is kept warm during the recharging period c) When a constant current charger is used d) Never	d
77	Which of the following is a rechargeable battery? a) Carbon-zinc b) Silver oxide c) Nickel Metal Hydride d) Mercury	c
78	What is impedance? a) The electric charge stored by a capacitor b) The inverse of resistance c) The opposition to the flow of current in an AC circuit d) The force of repulsion between two similar electric fields	c
79	What is reactance? a) Opposition to the flow of direct current caused by resistance b) Opposition to the flow of alternating current caused by capacitance or inductance c) A property of ideal resistors in AC circuits d) A large spark produced at switch contacts when an inductor is de-energized	b
80	Which of the following causes opposition to the flow of alternating current in an inductor? a) Conductance b) Reluctance c) Admittance d) Reactance	d
81	Which of the following causes opposition to the flow of alternating current in a capacitor? a) Conductance b) Reluctance c) Reactance d) Admittance	c
82	How does a coil react to AC? a) As the frequency of the applied AC increases, the reactance decreases b) As the amplitude of the applied AC increases, the reactance increases c) As the amplitude of the applied AC increases, the reactance decreases d) As the frequency of the applied AC increases, the reactance increases	d
83	How does a capacitor react to AC? a) As the frequency of the applied AC increases, the reactance decreases b) As the frequency of the applied AC increases, the reactance increases c) As the amplitude of the applied AC increases, the reactance increases d) As the amplitude of the applied AC increases, the reactance decreases	a
84	What happens when the impedance of an electrical load is equal to the internal impedance of the power source? a) The source delivers minimum power to the load b) The electrical load is shorted c) No current can flow through the circuit d) The source can deliver maximum power to the load	d
85	Why is impedance matching important? a) So the source can deliver maximum power to the load b) So the load will draw minimum power from the source c) To ensure that there is less resistance than reactance in the circuit d) To ensure that the resistance and reactance in the circuit are equal	a
86	What unit is used to measure reactance? a) Farad b) Ohm c) Ampere d) Siemens	b
87	What unit is used to measure impedance? a) Volt b) Ohm c) Ampere d) Watt	b
88	Why should core saturation of a conventional impedance matching transformer be avoided? a) Harmonics and distortion could result b) Magnetic flux would increase with frequency c) RF susceptance would increase d) Temporary changes of the core permeability could result	a
89	What is one reason to use an impedance matching transformer? a) To reduce power dissipation in the transmitter b) To maximize the transfer of power c) To minimize SWR at the antenna d) To minimize SWR in the transmission line	b

#	Question and options (verbatim)	Key
90	Which of the following devices can be used for impedance matching at radio frequencies? a) A transformer b) A Pi-network c) A length of transmission line d) All of these choices are correct	d
91	Which of the following describes one method of impedance matching between two AC circuits? a) Insert an LC network between the two circuits b) Reduce the power output of the first circuit c) Increase the power output of the first circuit d) Insert a circulator between the two circuits	a
92	A two-times increase or decrease in power results in a change of how many dB? a) 2 dB b) 3 dB c) 6 dB d) 12 dB	b
93	How does the total current relate to the individual currents in each branch of a parallel circuit? a) It equals the average of each branch current b) It decreases as more parallel branches are added to the circuit c) It equals the sum of the currents through each branch d) It is the sum of the reciprocal of each individual voltage drop	c
94	Capacitance is measured in a) Amperes b) watt c) Farad d) Coulombs	c
95	Frequency is a) cycles per second b) Kilo cycles per second c) Cycles per minute d) cycles per hour	a
96	Resonant frequency in a tuned circuit is equal to a) $1/2\pi LC$ b) $1/2\pi\sqrt{LC}$ c) $2\pi\sqrt{LC}$ d) $2\pi\sqrt{L+C}$	b
97	Power dissipated in a 400 Ohm resistor at 1 Amp is a) 40 Watts b) 400 KW c) 4 Watts d) 400 Watts	d
98	Zener diode is used for a) Rectification b) Voltage regulation c) Current regulation d) Switching	b
99	Plate current in a diode flows only when the plate is a) Negative with respect to cathode b) positive with respect to cathode c) when plate is at a lower voltage than cathode d) both at same potential	d
100	3 resistors of 2, 3 & 4 ohms are connected in series. The voltage across the circuit is 9 V; the current drawn by the 3 ohm resistor is a) 1.5 Amps b) 27 Amps c) 1 Amp d) 3 Amps	c
101	A superheterodyne receiver is tuned to 555 KHz and its local oscillator at 1010 KHz. The image frequency will be a) 1565 KHz b) 455 KHz c) 1465 KHz d) none of these	a
102	3 to 30 MHz band is known as a) MF b) LF c) VHF d) HF	d
103	The wavelength of a broadcast station at 1000 KHz is a) 30 Meters b) 300 Meters c) 0.3 Meters d) none of these	b
104	The core of power supply transformer is laminated to a) decrease impedance b) increase impedance c) decrease eddy current losses d) none of these	c
105	In a resonant circuit a) $X_L = X_C$ b) $X_L > X_C$ c) $X_L < X_C$ d) none of these	a
106	When frequency of a carrier is varied according to modulation the result is a) frequency modulation b) amplitude modulation c) product detection d) none of these	a
107	The phase relationship between the input and output of a common emitter circuit in degrees is a) 90 b) 180 c) 270 d) 0	d
108	The quartz crystal oscillator is known for its a) linearity b) stability c) high output d) flexibility	b

#	Question and options (verbatim)	Key
109	The effect of inter electrode capacitance of a triode is more predominant at a) HF b) VHF c) LF d) none of these	d
110	Which of the following is a characteristic of a liquid crystal display? a) It requires ambient or back lighting b) It offers a wide dynamic range c) It has a wide viewing angle d) All of these choices are correct	d
111	What is meant by the term MMIC? a) Multi Megabyte Integrated Circuit b) Monolithic Microwave Integrated Circuit c) Military-specification Manufactured Integrated Circuit d) Mode Modulated Integrated Circuit	b
112	What is a microprocessor? a) A low powered analog signal processor used as a microwave detector b) A miniature computer on a single integrated circuit chip c) A microwave detector, amplifier, and local oscillator on a chip d) A low voltage amplifier used in a microwave transmitter modulator stage	b
113	What safety feature does a power-supply bleeder resistor provide? a) It acts as a fuse for excess voltage b) It discharges the filter capacitors c) It removes shock hazards from the induction coils d) It eliminates ground-loop current	b
114	What components are used in a power-supply filter network? a) Diodes b) Transformers and transistors c) Quartz crystals d) Capacitors and inductors	d
115	What should be the minimum peak-inverse-voltage rating of the rectifier in a full-wave power supply? a) One-quarter the normal output voltage b) Half the normal output voltage c) Double the normal peak output voltage d) Equal to the normal output voltage	c
116	What should be the approximate minimum peak-inverse-voltage rating of the rectifier in a half-wave power supply? a) One-half the normal peak output voltage b) Half the normal output voltage c) Equal to the normal output voltage d) Two times the normal peak output voltage	d
117	What should be the impedance of a low-pass filter as compared to the impedance of the transmission line into which it is inserted? a) Substantially higher b) About the same c) Substantially lower d) Twice the transmission line impedance	b
118	What is an advantage of a crystal controlled transmitter? a) Stable output frequency b) Excellent modulation clarity c) Ease of switching between bands d) Ease of changing frequency	a
119	What type of receiver is suitable for CW and SSB reception but does not require a mixer stage or an IF amplifier? a) A super-regenerative receiver b) A TRF receiver c) A super-heterodyne receiver d) A direct conversion receiver	d
120	What type of circuit is used in many FM receivers to convert signals coming from the IF amplifier to audio? a) Product detector b) Phase inverter c) Mixer d) Discriminator	d
121	What portion of the AC cycle is converted to DC by a half-wave rectifier? a) 90 degrees b) 180 degrees c) 270 degrees d) 360 degrees	b
122	What portion of the AC cycle is converted to DC by a full-wave rectifier? a) 90 degrees b) 180 degrees c) 270 degrees d) 360 degrees	d

#	Question and options (verbatim)	Key
123	What is the output waveform of an unfiltered full-wave rectifier connected to a resistive load? a) A series of DC pulses at twice the frequency of the AC input b) A series of DC pulses at the same frequency as the AC input c) A sine wave at half the frequency of the AC input d) A steady DC voltage	a
124	Which of the following is a characteristic of a Class A amplifier? a) Low standby power b) High Efficiency c) No need for bias d) Low distortion	d
125	For which of the following modes is a Class C power stage appropriate for amplifying a modulated signal? a) SSB b) CW c) AM d) All of these answers are correct	b
126	Which of the following is an advantage of a Class C amplifier? a) High efficiency b) Linear operation c) No need for tuned circuits d) All of these answers are correct	a
127	How is the efficiency of an RF power amplifier determined? a) Divide the DC input power by the DC output power b) Divide the RF output power by the DC input power c) Multiply the RF input power by the reciprocal of the RF output power d) Add the RF input power to the DC output power	b
128	Which of the following describes a linear amplifier? a) Any RF power amplifier used in conjunction with an amateur transceiver b) An amplifier whose output preserves the input waveform c) A Class C high efficiency amplifier d) An amplifier used as a frequency multiplier	b

A.6.3 ERRORS IN THE DARTS PAPER-2 ANSWER KEY — DO NOT LEARN THESE

Item	DARTS key	Correct answer	Why it matters
Regulations Q3 — recommended way to break into a phone conversation	c ("Break Break Break")	b — say your call sign during a break between transmissions	SIARS publicly corrected exactly this item: "Break" is for emergencies only. It also contradicts this paper's own Q30.
Regulations Q10 — sideband commonly used on VHF/UHF	d (double sideband)	a — upper sideband	Basic operating practice.
Regulations Q11 — when you are prohibited from helping a station in distress	a	c — you are never prohibited	A safety-of-life point.
Regulations Q12 — avoiding harmful interference when calling CQ on CW	d	c — send "QRL? de " and listen	
Regulations Q30 — proper way to break into a conversation	c ("Break-break")	a — say your call sign between their transmissions	Directly contradicts the key given for Q3.

Item	DARTS key	Correct answer	Why it matters
Regulations Q31 — what amateurs are forbidden to transmit	d (third-party messages)	[UNCERTAIN — third-party traffic is <i>permitted</i> in India on failure of normal telecommunication facilities and on Government request. The key is at best misleading.] [VERIFIED — 2024 Rules, Annexure A r.2(3)]	
Regulations Q33 — proper repeater operating practice	a (monitor first, keep it short)	d — all of these answers are correct	SIARS also answers (d).
Regulations Q42 — Indian amateurs may communicate with amateurs in	a (all countries)	c — countries permitted by the Indian Government	The Rules allow the Government to bar communication with specified countries. (<i>SIARS also answers (a) — both are believed wrong.</i>) [UNCERTAIN]
Regulations Q45 — minimum age to become an amateur	a (18 years)	c — 12 years	Examinable, and the key is flatly wrong. Confirmed by rule 4(1)(a) of the 2024 Rules, by NIAR RM-2 and B-38, and by the DoT eServices ASOC page. [VERIFIED]
Regulations Q46 and Q60 — log timekeeping	a (IST) both times	The study material says UTC; the 2024 Rules say IST.	See the conflict note at A.4.5. Under Annexure A rule 6(2) of the 2024 Rules the DARTS key of IST is actually correct for the log itself , even though the club consensus and every study manual say UTC. [VERIFIED — Annexure A r.6(2)] [UNCERTAIN — which an examiner will mark correct]
Electronics Q1 — a current that flows only in one direction	c ("a normal current")	b — a direct current	The key appears offset by one for the first few items.
Electronics Q4 — a current that reverses direction regularly	b (direct current)	a — alternating current	Same offset.
Electronics Q99 — plate current in a diode flows only when the plate is	d (both at same potential)	b — positive with respect to the cathode	

Item	DARTS key	Correct answer	Why it matters
Electronics Q101 — superhet tuned to 555 kHz, LO 1010 kHz, image frequency	a (1565 kHz)	c — 1465 kHz (image = LO + IF = 1010 + 455)	
Electronics Q107 — phase relationship in a common-emitter stage	d (0°)	b — 180°	Contradicts the official paper at S1 A14.
Electronics Q109 — inter-electrode capacitance of a triode is most predominant at	d (none of these)	b — VHF	
Electronics Q110 — characteristic of a liquid crystal display	d (all of these)	a — it requires ambient or back lighting	Matches the US source item this was copied from.

A.7 S5 — historical government papers, Grade I / Grade II, pre-2010 descriptive era

Source type: historical government question papers, republished by a candidate. Where:

`question_paper_for_ham_radio_exam.zip` → folder `QPAPER/`, seven plain-text files, all timestamped 2003-05-12, linked from "Sample question papers for ASOC Examination" on *TT's Jottings — Blog of VU2SWX* (Sunil TT), post dated 2009-01-16, <https://brainstorms.in/?p=247> · file http://brainstorms.in/pdf/question_paper_for_ham_radio_exam.zip [REPORTED — all seven files downloaded and read on 2026-08-23]

Every file carries the masthead **GOVERNMENT OF INDIA / MINISTRY OF COMMUNICATIONS / MONITORING ORGANISATION**, and three carry printed sitting dates. **These are the closest thing to genuine past papers that exists in public.**

Their limitation is decisive: they belong to the Grade I / Grade II descriptive regime that was replaced in 2010 by the two-grade (General / Restricted) 100-question MCQ paper. The *topics* carried over almost unchanged; the *format* did not. Study them for coverage, not for format.

A.7.1 PAPER 1 — "GRADE 1&2" (COMBINED GRADE I / GRADE II PAPER), UNDATED

Rubric: "NOTE: Question No 1 from each section is compulsory. Grade I answer any 5 of the remaining questions – 2 Hours. Grade II answer any 2 of the remaining questions – 1 Hour."

SECTION A — Q.1, choose the correct answer (any 10) — topic: radio theory. Answers [derived]:

#	Question and options	Answer [derived]
i	A3E indicates a) SSB b) AM-DSB voice c) FM Voice d) FSK	b
ii	Capacitance is measured in a) Amperes b) watt c) Farad d) Coulombs	c
iii	Frequency is a) cycles per second b) Kilo cycles per second c) Cycles per minute d) cycles per hour	a
iv	Line of sight propagation is the mode of communication in a) LF b) HF c) MF d) VHF	d
v	The wavelength of 300 MHz in Meters is a) 1 b) .1 c) 1.1 d) 0.01	a
vi	4th harmonic of 2.5 MHz is a) 10 MHz b) 15 MHz c) 8 MHz d) 7.5 MHz	a
vii	Squelch control is used to eliminate a) static interference b) electrical disturbance c) receiver noise d) unwanted carrier	c
viii	Resonant frequency in a tuned circuit is equal to a) $1/2\pi LC$ b) $1/2\pi\sqrt{LC}$ c) $2\pi\sqrt{LC}$ d) $2\pi\sqrt{(L+C)}$	b
ix	Power dissipated in a 400 Ohm resistor at 1 Amp is a) 40 Watts b) 400 KW c) 4 Watts d) 400 Watts	d
x	The frequency of 40 Meter band in MHz is a) 14 – 14.350 b) 7 – 7.1 c) 21 – 21.450 d) 15 – 15.400	b
xi	Zener diode is used for a) Rectification b) Voltage regulation c) Current regulation d) Switching	b
xii	The UHF range is a) 30 to 300 KHz b) 300 to 3000 KHz c) 3 to 30 GHz d) 300 to 3000 MHz	d

SECTION A — descriptive questions (radio theory):

- **Q.2** Write short notes on any three: a) Skip distance b) AVC c) eddy currents d) Antenna matching
- **Q.3** Draw the circuit of a full wave bridge rectifier with smoothing circuit and explain its function.
- **Q.4** a) Explain briefly FM and its advantages over AM. b) Explain briefly dipole aerial.
- **Q.5** Draw the block diagram of a superheterodyne receiver and explain the function of each stage briefly.
- **Q.6** Explain briefly primary cell and secondary cell and how do they differ?
- **Q.7** State the colour code of resistors.

SECTION B — Q.1, choose the correct answer (any 10) — topic: regulations:

#	Question and options	Answer [derived]
i	Indian amateurs can communicate with other amateurs in a) All countries b) Countries permitted by ITU c) Countries permitted by Indian Government d) Countries permitted by Indian Amateur society	c
ii	Grade II amateurs can communicate on 7 MHz in a) A1 b) A3 c) F3 d) A3A	[UNCERTAIN — depends on the emission table then in force]
iii	Lady amateurs are known as a) XL b) XYL c) YL d) LY	c (XYL is a ham's wife)
iv	Minimum age to become an amateur is a) 18 years b) 14 years c) 12 years d) 16 years	c
v	All timing in the Log book should be in a) IST b) UTC c) GMT d) Local time	b — as the pre-2010 material taught; the 2024 Rules now require IST [VERIFIED — Annexure A r.6(2)]
vi	Amateurs should preserve their log for a period of a) 6 months b) 1 year c) 2 years d) 9 months from the date of the last entry	b (one year from the date of the last entry)
vii	Q code to indicate time is a) QRG b) QRX c) QTR d) QSA	c
viii	Test signal shall not be continued more than a) 30 seconds b) 1 minute c) 2 minutes d) 3 minutes	a
ix	In India the standard time signal is broadcast by a) ISRO b) WPC c) OCS d) NPL	d
x	PANPAN transmitted thrice indicates a) Distress b) Emergency c) Urgency d) Distress and emergency	c
xi	The broadcast of music is allowed in amateur service a) on request b) when channel is free c) never d) only for testing	c
xii	The abbreviation VA means a) End of transmission b) End of message c) End of working d) End of schedule	c

SECTION B — descriptive questions (regulations):

- **Q.2** State the conditions under which an amateur station can transmit third party messages.
- **Q.3** Write the meaning of four of the following: i) QRA ii) QRL iii) QRX 1600 iv) QRM v) QSY
- **Q.4** What are the international distress frequencies, and write a specimen form of a distress message in A3.
- **Q.5** a) Explain the need for a call sign. How are the call signs for amateurs in India formed? b) Who allots call signs for amateurs in India?
- **Q.6** (*log-keeping exercise*) On May 27 1986, Tuesday, presume you had a contact with Sri Vidyaprakash of Coimbatore on 14025 KHz signing VU2DVP at 2130 IST with good signal strength, tone and readability. Convey your particulars to him before closing down. Show the above contact in proper log form.

- **Q.7** What types of messages are the amateurs forbidden to transmit?

A.7.2 PAPER 2 — "GRADE 2-1", GRADE II, ONE HOUR, 100 MARKS, UNDATED

SECTION A — Q.1, choose the correct answer, any ten (compulsory) — radio theory:

#	Question and options	Answer [derived]
i	Plate current in a diode flows only when the plate is a) Negative w.r.t. cathode b) positive w.r.t. cathode c) at a lower voltage than cathode d) both at same potential	b
ii	Zener Diodes are used for a) Rectification b) Detection c) voltage regulation d) none of these	c
iii	3 resistors of 2, 3 & 4 ohms in series across 9 V; the current drawn by the 3 ohm resistor is a) 1.5 A b) 27 A c) 1 A d) 3 A	c ($9\text{ V} \div 9\ \Omega$)
iv	A superheterodyne receiver tuned to 555 KHz, local oscillator at 1010 KHz. The image frequency will be a) 1565 KHz b) 455 KHz c) 1465 KHz d) none of these	c (LO + IF)
v	3 to 30 MHz band is known as a) MF b) LF c) VHF d) HF	d
vi	The wavelength of a broadcast station at 1000 KHz is a) 30 Meters b) 300 Meters c) 0.3 Meters d) none of these	b
vii	The core of a power supply transformer is laminated to a) decrease impedance b) increase impedance c) decrease eddy current losses d) none of these	c
viii	In a resonant circuit a) $X_L = X_C$ b) $X_L > X_C$ c) $X_L < X_C$ d) none of these	a
ix	When the frequency of a carrier is varied according to modulation the result is a) frequency modulation b) amplitude modulation c) product detection d) none of these	a
x	The phase relationship between the input and output of a common emitter circuit in degrees is a) 90 b) 180 c) 270 d) 0	b
xi	The quartz crystal oscillator is known for its a) linearity b) stability c) high output d) flexibility	b
xii	The effect of interelectrode capacitance of a triode is more predominant at a) HF b) VHF c) LF d) none of these	b

SECTION A — descriptive (radio theory):

- **Q.2** Draw the block diagram of a superheterodyne receiver and explain briefly the function of each stage.
- **Q.3** Draw the circuit of a full wave rectifier and explain briefly its working.
- **Q.4** Write short notes on any three of the following: a) demodulation b) AVC c) Microphones d) Harmonic generators
- **Q.5** a) What are the colour codes for 100 Ohms, 470 kOhms and 4.7 MOhms with 5% tolerance? b) Is it possible to use a Ground plane antenna on HF communication? Justify your answer.

SECTION B — Q.1, choose the correct answer, any ten (compulsory) — regulations. The scanned text preserves only items i–iii and xi–xii legibly:

#	Question and options	Answer [derived]
i	SWLs are permitted to transmit in the frequency band of a) 7-7.1 MHz b) 3.89-3.9 MHz c) 144-146 MHz d) none of these	d (a Short Wave Listener holds no transmitting authority)
ii	The amateur license is renewed by a) P & T b) Ministry of communication c) monitoring stations d) none of these	b
iii	A3E emission is a) DSB b) SSB c) CW d) FSK	a
xi	Standard time and frequency is transmitted on a) 7050 KHz b) 14050 KHz c) 21050 KHz d) 15000 KHz	[UNCERTAIN — the NPL's station ATA transmits on 10 MHz; none of the four is right as printed. Defective.]
xii	Test signals are not to be transmitted for more than a) 10 minutes b) 5 minutes c) 1 minute d) none of these	c or d — the rule is 30 seconds, which is not offered. Defective.

SECTION B — descriptive (regulations):

- **Q.2** What are the frequency bands, power and emissions permitted to Gr II amateurs?
- **Q.3a** Give Q codes for the following expressions, answer any four: i) Your frequency varies ii) What is the name of the station? iii) Please change frequency to 14050 KHz iv) Who is calling me? v) Please increase power
- **Q.3b** Give the meaning of any four: i) QRS? ii) QRX 2000 iii) RST 599 iv) QRG 7051 KHz v) DE
- **Q.4** Write short notes on any three: a) Test signals b) secrecy of correspondence c) Amateur service d) Amateur Log Book
- **Q.5a** What do you understand by the term "inspection"?
- **Q.5b** What is the license fee for grade II and restricted amateurs? What is the surcharge for late renewal?
- **Q.5c** Explain the procedure to obtain a duplicate license and who issues a duplicate license?

A.7.3 PAPER 3 — "GRADE 2-2", GRADE II, ONE HOUR, 100 MARKS, UNDATED (FULLY DESCRIPTIVE)

Rubric: *"Answer any 3 questions from each section."*

SECTION A (radio theory):

- **Q.1** Answer in brief any two: (a) What is the difference between VFO and BFO (b) What are the advantages and disadvantages of FM over AM transmission (c) What is the difference between intermediate and image frequency
- **Q.2** (a) What is the peak value of mains voltage of 220 V AC? (b) Define Ohm's law. (c) What do you mean by the directivity of an antenna?

- **Q.3** (a) What is the time constant of a 2 μF capacitor and 250 $\text{k}\Omega$ resistor? (b) Define standing wave ratio.
- **Q.4** Write short notes on any three: (a) Skip zone (b) Squelch control (c) Zener diode (d) Loudspeaker
- **Q.5** Draw the block diagram of a transmitter and explain briefly the function of each stage.

SECTION B (regulations):

- **Q.1** Write short notes on any three: (a) Secrecy of correspondence (b) test signals (c) QSL cards (d) General call to all stations (F3E)
- **Q.2** (a) What types of messages can be exchanged among amateur stations? (b) What precautions are taken to avoid interference?
- **Q.3** (a) State the frequencies, band and power authorised to restricted amateurs. (b) What documents are to be kept in an amateur station?
- **Q.4** (a) How is a duplicate copy of the amateur license obtained? (b) What are the penalties for violation of the Indian Wireless Telegraph (Amateur Service) Rules 1978?
- **Q.5** Draw the specimen form of the Equipment register with two entries **or** describe six important conditions under which the amateur station equipment is to be kept.

A.7.4 PAPER 4 — "GRADE 2-3", GRADE II RESTRICTED, ONE HOUR, 100 MARKS, UNDATED

Rubric: *"Solve any three from each section."*

SECTION A (radio theory):

- **Q.1** Draw the block diagram of a superhet communication receiver and explain briefly the function of each stage.
- **Q.2** Write short notes on any three: (a) G.P. Antenna (b) FM (c) Capacitor microphone (d) skip distance
- **Q.3** (a) Define Ohm's law. (b) What series dropping resistor is required if five 6.3 V and five 12.6 V filaments are connected to 110 V and draw a current of 205 mA?
- **Q.4** (a) What is the value of the resistor with the following colours: (i) Black red white gold (ii) red green yellow silver (iii) orange brown red gold (b) What is a crystal oscillator and why is it preferred to an LC oscillator?
- **Q.5** Draw the schematic diagram of a full wave rectifier with smoothing circuits and explain how it works.

SECTION B (regulations):

- **Q.1(a)** Write the meaning of any four: (i) DE (ii) QRL IMI (iii) QRM 4 (iv) QRX 1600 (v) QSY 7020
- **Q.1(b)** Give the Q code or abbreviation used for any four: (i) send slow (ii) what is your location (iii) received correctly (iv) weather (v) calling any station
- **Q.2** What is the importance of a call sign? How are call signs of Indian amateurs formed?

- **Q.3** (a) What types of messages are permitted to be passed by Amateurs? (b) State the procedure to make a general call to all stations in radio telephony.
- **Q.4** Write short notes on any four: (a) QSL cards (b) Distress (c) test signals (d) Secrecy of correspondence (e) Amateur service (f) categories of license
- **Q.5** State the frequency, power and emissions authorized to restricted grade amateurs.

A.7.5 PAPER 5 — "GRADE 2-4", GRADE II RESTRICTED, SAT ON 1990-11-12, ONE HOUR, 100 MARKS

This paper carries a real examination date. Rubric: *"Answer any 3 questions from each section."*

SECTION A (radio theory):

- **Q.1** State Ohm's Law. What is the effective value of capacitance of two capacitors of 10 pF connected in parallel? What is the colour code of a resistance of 20 k Ω 10% tolerance?
- **Q.2** Draw the block diagram of a Communication Receiver. What is the necessity of a BFO?
- **Q.3** What do you know about skip distance and fading? What is the difference in mode of propagation between HF and VHF?
- **Q.4** Draw the block diagram of a low power transmitter for transmitting speech.
- **Q.5** Describe the aerial you are going to use for transmitting on 2 metres and 7 MHz.

SECTION B (regulations):

- **Q.1** Draw the specimen form of Log you are supposed to maintain.
- **Q.2** (i) Expand the following: QRM? QRU QTC2 VA AR (ii) Write the phonetics of K X Y M 2 6.
- **Q.3** What are the documents to be kept in an Amateur Station?
- **Q.4** Write briefly: (i) Precautions against interference (ii) Secrecy of correspondence (iii) Call and reply procedure.

A.7.6 PAPER 6 — "GRADE 2-5", GRADE II RESTRICTED, SAT ON 1990-12-10, ONE HOUR, 100 MARKS

SECTION A (radio theory):

- **Q.1** Answer any five: (a) Draw the symbol of a PNP and NPN transistor (b) What is the formula for determining the frequency in a resonant circuit? (c) What is the value of a resistor having red red orange colour combination? (d) What is the total capacitance of 3, 4 and 5 μ F capacitors in parallel? (e) What is the wavelength of a frequency of 150 MHz? (f) Define selectivity of a receiver.
- **Q.2** Draw the block diagram of a Superhet Receiver. Explain briefly the functions of each stage.
- **Q.3** What are the advantages and disadvantages of FM over AM?
- **Q.4** Write short notes on any three: (a) Crystal oscillator (b) Microphone (c) signal to noise ratio (d) AVC (e) Ground plane antenna

- **Q.5** (a) Describe a dipole antenna. (b) What are the functions of a filter circuit and a zener diode in a rectifier circuit?

SECTION B (regulations):

- **Q.1** Answer any five: (a) What is the meaning of F3E emission? (b) What is meant by UR RST 599? (c) At what intervals should a transmitting station identify itself? (d) What is the retention period of an amateur logbook? (e) What is meant by CQ DX? (f) What are the phonetics for VU3ABC?
- **Q.2** What are the frequency bands, emission and power authorized to restricted grade amateurs?
- **Q.3** (a) How to give a general call to all amateurs in radio telephony? (b) What is secrecy of correspondence?
- **Q.4** (a) What types of messages are permitted between amateur stations? (b) What are test signals? What is the procedure for the transmission of test signals?
- **Q.5** Write short notes on any three: (a) Amateur station (b) Penalties for breach of amateur service rules (c) Harmful interference (d) Equipment register

A.7.7 PAPER 7 — "GRADE 2-6", GRADE II RESTRICTED, SAT ON 1990-12-26, ONE HOUR, 100 MARKS

SECTION A (radio theory):

- **Q.1** What is the difference between a full wave and half wave rectifier? Draw a diagram of either of them.
- **Q.2** (a) Define Ohm's law. (b) Explain the colour code for resistance. How will a resistance of 46,000 Ohms 5% tolerance be colour coded?
- **Q.3** (a) Explain briefly SSB transmission and its advantages over DSB. (b) An ideal transformer has primary turns of 4000 and secondary 800. When primary voltage is 230 Volts, what is the secondary voltage?
- **Q.4** (a) What do you understand by Selectivity, Sensitivity and Fidelity as applied to a communication receiver? (*the source repeats "Selectivity" — evidently a typographical error for Sensitivity*)
- **Q.5** Draw a block diagram of your intended transmitter and explain briefly the function of each stage.

SECTION B (regulations):

- **Q.1** Answer any four: (a) Define amateur station. (b) What is the frequency band for restricted amateur service? (c) What is the Q code for "who is calling me?" (d) What is the emission code for amplitude modulated radio-telephony? (e) What are phonetics for JA5CQS?
- **Q.2** Explain the procedure by which an amateur station establishes contact with another amateur station on radiotelephony.

- **Q.3** What are the different columns of the log book maintained by an amateur? Why is it necessary to maintain a log?
- **Q.4** (a) What types of messages can be exchanged among amateur stations? (b) What precautions would you take for making test signals?
- **Q.5** (a) What is secrecy of correspondence? (b) How is a duplicate of an amateur license obtained?

A.7.8 WHAT THESE SEVEN PAPERS TELL YOU THAT NOTHING ELSE DOES

The recurring, near-guaranteed topics across a decade of real sittings were: Ohm's law and the resistor colour code; the superheterodyne block diagram; the full-wave rectifier and its smoothing; FM versus AM; skip distance and fading; dipole and ground-plane antennas; the log book (its columns, its timekeeping, its retention); Q-codes and abbreviations; phonetics for a given call sign; distress and urgency procedure; third-party messages; secrecy of correspondence; test signals; and the bands, power and emissions permitted to the restricted grade. **Every one of those themes survives into the modern MCQ paper.** [REPORTED — derived by reading all seven papers]

A.8 S6 — SIARS, "Queries: ASOC Exam questions" (11 items, answered)

Source type: club question-and-answer page. South India Amateur Radio Society, Chennai. Page posted **2019-01-13**; https://www.siars.org.in/asoc_exam_qp1/ [REPORTED]

Provenance, stated on the page itself: the questions were **downloaded from the internet by Ajaykumar VU, an ASOC aspirant from Kerala**, who sent them to SIARS asking for clarification. This is therefore *not* an independent question source — most of the items trace back to S1 or S4. Its value lies in the **answers with reasoning**, given by a club that trains candidates.

#	Question	SIARS answer	Note
1	The calling Amateur's call sign in 'VU2DX DE VU2DJ'?	VU2DJ	Same as S1 B17.
2	Standard shift between transmitting and receiving frequency for Amateur Radio in VHF band?	600 kHz	Same as S1 B28.
3	Recommended way to break into a conversation when using phone? a) Say "QRZ" several times followed by your call sign b) Say your call sign during a break between transmissions c) Say "Break" "Break" "Break" and wait for a response d) Say "CQ" followed by the call sign of either station	b — SIARS explains: give your call sign between overs; "Break" is for emergencies only	Corrects the DARTS paper-2 key of (c).

#	Question	SIARS answer	Note
4	Which 20 metre band segment is most often used for data transmissions? a) 14.000–14.050 b) 14.070–14.100 c) 14.150–14.225 d) 14.275–14.350 MHz	b	Copied from the US pool, but still correct as band-plan practice.
5	Proper repeater operating practice? a) Monitor before transmitting; keep transmissions short b) Identify legally c) Use minimum transmitter power necessary d) All of these answers are correct	d	Corrects the DARTS paper-2 key of (a).
6	Indian amateurs can communicate with other amateurs in: a) All countries b) Countries permitted by ITU c) Countries permitted by Indian Government d) Countries permitted by Indian Amateur society	a	[UNCERTAIN — the legally correct answer is believed to be (c); the Government may prohibit specified countries. Two club sources give (a). Ask WPC.]
7	Standard emission designation consists of: a) 8 b) 3 c) 6 d) 2 letters/numerals	c (6)	Conflicts with the three-symbol class of emission taught elsewhere; see the note on S3 Q61. [UNCERTAIN]
8	Abbreviation for 'stand by' in Morse code: a) SY b) AS c) SB d) AD	AS	
9	Renewal fee for General Grade Amateur licence: a) Rs 1000 b) Rs 2000 c) Rs 100 d) Rs 200	Rs 2000	[UNCERTAIN — a 2019 figure; the fee structure changed under the 2024 Rules.]
10	Call sign blocks allotted to India: a) ATA–AWZ b) VTA–VWZ c) 8TA–8WZ d) a & b e) a, b & c	e	Repeats the same 8TA–8WZ typographical error as the official paper; ITU Appendix 42 says 8TA–8YZ.
11	Speed in words per minute when a message contains 60 characters in a minute?	12 wpm (assuming 5 characters per word)	Same as S1 B47.

A.9 S7 — AARS/INDRA study book, Annexure-III descriptive sample paper (31 items)

Source type: book-published sample paper, pre-2010 descriptive format. Biju Gopi Thilaka VU2HBI, *Study Manual for Amateur Station Operator's Licence Examination — INDIA*, INDRA / LinuxClub edition, 198 pp, PDF created 2021-06-10, <https://www.aars.in/ham-exam-book.pdf> — **Annexure-III, "A Sample Question Paper for Amateur Radio Licencing Examination"**. [REPORTED — downloaded and read 2026-08-23]

(The same book also reprints S1, S3 and S4 as text; its transcription of S1 is a poor character recognition and should not be preferred over the scan. Annexure-III below is the one item unique to the book.)

SECTION-A — Radio Theory and Practice

1. What is Ohm's law? A resistor has the following colour bands: RED, RED, ORANGE and GOLD respectively. What is the value of this resistor?
2. State whether TRUE or FALSE and correct the FALSE statements: (a) When two resistors are connected in parallel, their resistance increases. (*False — it decreases*) (b) A capacitor allows DC current to pass through it. (*False — it blocks steady DC*) (c) When two capacitors are connected in parallel, their capacitance decreases. (*False — capacitances add*) (d) In a P-N junction diode, flow of current is possible only when the negative pole of the battery is connected to the p-type semiconductor and positive pole to n-type. (*False — that is reverse bias*) (e) Centre-tapped transformer secondary is not essential for a bridge rectifier circuit. (*True*) (f) In a transistor amplifier circuit, the emitter is always reverse biased and collector is always forward biased. (*False — the reverse*) (g) In a PNP transistor, the collector is applied with a negative voltage. (*True*) (h) SWR value is more than one for a matched system. (*False — a matched system is 1:1*) (i) The fourth band in a resistor is silver coloured. The tolerance of this resistor is 10%. (*True*) (j) Two capacitors 5 μF and 10 μF are connected in parallel. The 5 μF has a voltage rating of 10 V and the 10 μF a rating of 6 V. We can apply 9 volts to this circuit without damaging the 10 μF capacitor. (*False — in parallel both see 9 V, exceeding the 6 V rating*)
3. Draw the schematic diagram of the transmitter you intend to build for ham radio operation and explain the function of each stage in brief.
4. Explain the working function of the antenna you intend to use for ham radio transmissions.
5. Draw the block diagram of a super-heterodyne receiver. What are the superiorities of this type of receiver over the TRF receivers?
6. Write short notes on: Image frequency in a super-heterodyne receiver; Ionosphere; Single Side Band transmission; Percentage of modulation; Selectivity and sensitivity of receiver; Colpitts oscillator; Skip distance; Differences between fadeout and fading; A.G.C.; B.F.O.

(The italicised answers above are [derived]; the book prints none.)

SECTION-B — Rules & Regulations

1. What is a distress message? What action would you take if you receive a distress message?
2. Explain the meaning of the following: QRX 10 minutes, QSY 14.155 MHz, QRU?, QRV, QRL, KN, VA, AS, QSB, QRS.
3. Describe the procedure of establishing a contact with another amateur station using radiotelephony and radiotelegraphy.
4. What do you mean by "Amateur Radio Service"?

5. State whether TRUE or FALSE. Correct the FALSE statements: (a) In India, a Grade-I licence holder is allowed to use 400 watts of power in certain sub-bands. (b) In India, a Grade-II licence holder can directly use radiotelephony after getting his licence. (c) In India, ham radio operators are allowed to use the frequency band 7000 kHz to 7300 kHz. (d) In India, an amateur radio operator can handle third party messages. (e) Amateur radio operators are allowed to use secret language for communication. (*False — plain language only*) (f) 4S7VK is a call-sign of USA. (*False — 4S is Sri Lanka*) (g) An amateur radio operator should maintain the secrecy of communication. (*True*) (h) An amateur radio operator is allowed to communicate with the Military during emergency. (i) A message passed on by an amateur radio operator to a third party by taking money is called "QSP". (*False — accepting payment is prohibited; QSP means "I will relay to ..."*) (j) Third party messages can be handled by an amateur radio operator during a disaster.
6. What are the amateur radio bands of frequencies? Can a Restricted Grade Licence holder use High Frequencies (HF) for communication?
7. What is the condition to get a "Voice endorsement" by a Grade-II licence holder?

[UNCERTAIN on items 5(a)–(d), (h) and (j): these turn on the Rules in force. Several refer to the abolished Grade I / Grade II scheme. Answer them against the Telecommunications (Amateur Services) Rules, 2024 and the current bands table — not from this book.]

A.10 What was looked for and could not be found — the honest negatives

A.10.1 CANDIDATE RECOLLECTIONS: ESSENTIALLY NONE EXIST IN RETRIEVABLE FORM

Lead	Outcome
Reddit r/HamRadio, "My Ham Radio ASOC Exam and License Application Experience (Restricted Grade, India)" — https://www.reddit.com/r/HamRadio/comments/1rwts7m/	Could not be retrieved. Reddit returned HTTP 403 to every route tried (www, old.reddit, .json, .rss) and four third-party mirrors were 403, 502 or dead. All that survives is the search-engine snippet, which describes the <i>process</i> , not any question. [UNCERTAIN — the post may or may not contain remembered questions. This is the single highest-value gap in this part and is worth one manual look in a browser.]
Reddit r/amateurradio, "Gave the ASOC exam but no idea about the results" — https://www.reddit.com/r/amateurradio/comments/177r2ul/	Snippet concerns waiting for results, not content. Not retrievable.
Reddit r/amateurradio wiki, "gettingstartedindia"	Covers pass marks and process; no questions.
Reddit r/indianripoff, "HAM Radio License Exams: General and Restricted Grade"	Format description only.

Lead	Outcome
Blogs, Quora, forums	Zero posts found that reproduce a remembered ASOC question. Several search phrasings tried.
Messaging-group and social-media material reposted publicly	None found in an indexed, citable form.

A.10.2 OTHER SOURCES EXAMINED AND REJECTED OR UNAVAILABLE

Source	Why it is not in Part 5A
https://www.aahrs.in/asoc-exam-sample-questions (Active Amateur Ham Radio Society, Kerala)	The page renders only navigation; no question content in the served HTML. [UNCERTAIN — content may be behind the site's scripting.]
DARTS five self-scoring quizzes (Resistance, Capacitance, Regulation Section-B × 3), linked from https://darts.org.in/application-forms/	Live, free and self-scoring — but the items sit inside Google Forms, not published as text, so they cannot be transcribed or cited. Recommendation: take them; they cannot be quoted here.
ARCAD "HAM Radio self-assessment Test" — https://hamradio-arcad.com/post/ham-practice-test	Same problem: an embedded form of roughly 14 items. Not transcribable.
https://asoc.itshamradio.com/ ("ASOC Prep India")	A practice-test site. Its items are constructed practice , not documented past questions, so they would belong in Part 5B, not here.
https://www.qsl.net/vu2ukr/ham_radio_india.htm	Fetches; site map only. The advertised "Sample Model question papers" are not on the page. Last updated 2009-10-07.
WPC, Saral Sanchar and DoT eServices official pages and FAQs	Fetches. They describe format, marks and pass criteria. They publish no questions beyond S1.
NIAR "training material" page — http://www.niar.org/training-material.html	Points to the Study Manual already used as S2.
Mobile apps	No app targeting the Indian ASOC syllabus was found on Google Play (searched 2026-08-23, India storefront). Everything returned targets another country's examination. The former NIAR app is gone. Do not use a US FCC Technician app to prepare — the paper's Section B is entirely India- and ITU-specific. [REPORTED — Google Play search results]

A.11 Count and reliability assessment

A.11.1 COUNT BY SOURCE TYPE

Source type	Sources	Objective items	Descriptive items	Total	Answered by the source
Official government paper	S1	100	0	100	0
Institute question bank (NIAR)	S2 + S8	230	0	230	75
Club-published question banks	S3, S4	288	0	288	188
Historical government papers (pre-2010 Grade I/II, republished by a candidate)	S5 (7 papers)	41	73	114	0
Club question-and-answer page	S6	11	0	11	11
Book-published sample paper	S7	0	31	31	0
Candidate recollections of the modern examination	—	0	0	0	—
Total	7 documents (8 source blocks)	670	104	774	274

Sub-parts of a descriptive question — for example the ten true/false statements inside one numbered question — are counted as separate items, because each is independently answerable.

Deduplication warning. These 774 items are **not** 774 distinct questions. S4's regulations items 36–60 duplicate S5's 1990s blocks; S6 duplicates S1 and S4; the AARS book reprints S1, S3 and S4. After removing the obvious overlaps the estimate is roughly **600 substantively distinct questions**, of which only the 100 in S1 have any government provenance. [UNCERTAIN — an exact deduplicated count would require item-by-item matching that was not performed]

A.11.2 RELIABILITY, SOURCE BY SOURCE

S1 — official WPC/DoT sample paper. Reliability: highest, with caveats. It is the only document with government provenance, and its rubric is independently corroborated by the DoT/Saral Sanchar FAQ. But it is undated — the earliest archive capture is 2016-08-04 — so it certainly predates the Telecommunications (Amateur Services) Rules, 2024; it carries **no answer key**; and it contains at least six internal defects, tabulated at A.3.3. Treat every one of its 100 questions as *representative of style and scope*, and treat none of them as a statement of current law.

S2/S8 — NIAR Study Manual. Reliability: high for regulations, medium for theory. NIAR is the recognised national institute and its regulations answers align with S1 almost everywhere. Against that: it prints a plainly wrong VHF range at B-44, contradicts itself on Morse word

spacing and on RST signal strength 9, has a gap in its own numbering (regulations items 13–24 absent), quotes pre-2024 fees and validity periods, and leaves 155 of its 230 items unanswered. A good bank; verify the law elsewhere.

S3 — DARTS sample paper 3. Reliability: medium. Indian in content, correct in rubric, plausible in difficulty — but anonymous, undated except by file metadata (2014), unanswered, and containing at least five defective items. Useful drill; not evidence of anything.

S4 — DARTS sample paper 2. Reliability: medium-low; the answer key is actively dangerous. The only large *answered* bank, which is exactly why its errors matter. Roughly half is US FCC pool material with "WPC" substituted, and seventeen key answers were identified as wrong or misleading — including minimum age (it says 18; the answer is 12). Use it only with the error table at A.6.3 open beside it.

S5 — the 1990s Grade I/II papers. Reliability: high as artefacts, zero as a format guide. Three carry printed examination dates and all seven carry a government masthead, so as *historical past papers* they are the most genuine items in this part. They are also the only source showing what was actually set, repeatedly, over years. But the descriptive format was abolished in 2010, several of their answers depend on superseded Rules, and two of their MCQ items are internally defective. Read them for topic frequency, never for format.

S6 — SIARS. Reliability: medium; valuable chiefly as a corrective. A named club, a dated page and reasoned answers — and it independently corrects two of DARTS paper 2's bad keys. But it is derivative, it repeats the 8TA–8WZ typographical error, its fee figure is from 2019, and on at least two items its answer is believed wrong.

S7 — AARS/INDRA book, Annexure-III. Reliability: medium, historical. A named author with a call sign, a dated file and a coherent descriptive paper — but it belongs to the pre-2010 regime, several true/false items depend on the abolished Grade I / Grade II distinction, and no answers are printed.

Candidate recollections. Reliability: not assessable — none obtained. This is the most important finding in Part 5A. Anyone who tells you "these are the questions that have been asked" is, on the public evidence, either quoting S1 or repeating a club compilation.

A.11.3 WHAT TO ACTUALLY STUDY, IN ORDER

Priority	Material	Where	Why
1	S1, the paper's Section A Q1–Q25 and Section B Q1–Q25	A.3	The official paper, Restricted-grade portion. 50 questions. The highest-value hour of study available anywhere.
2	NIAR Model Q&A, the paper's Section B, all 50, plus the 25-item short-answer set	A.4.1, A.4.2	Answered, regulations-focused, institute-published.
3	S1, the remaining 50 questions	A.3	For margin, and mandatory if going for General grade.

Priority	Material	Where	Why
4	S3, DARTS paper 3	A.5	100 Indian-flavoured items for drill.
5	S5's recurring topic list	A.7.8	The themes that recurred across real sittings.
6	S4	A.6	Only with the error table at A.6.3 open beside it.
7	Part 5B below	Part 5B	Constructed practice, covering syllabus areas the documented haul does not reach.

Part 5B — PRACTICE QUESTIONS (constructed from the syllabus)

THESE ARE NOT PAST PAPERS

Every question in Part 5B was written from scratch, from the published syllabus and the primary regulatory sources. Not one of them is a reproduction of, or a recollection of, a question that has actually appeared in an ASOC examination. No official past papers exist in the public domain, and no official answer key has ever been published for the one official sample paper that does exist. Treat Part 5B as a **study aid**, not as a prediction of the paper.

Written against: the official syllabus — WPC Office Memorandum No. L-14011/255/2004-AMT dated 2010-08-13, Annexure II — and the Telecommunications (Amateur Services) Rules, 2024 (G.S.R. 675(E) dated 2024-10-29, Gazette of India Extraordinary No. 617 of 2024-10-30), cross-checked against the licence-exemption notifications, the penalties research and the recent-changes research. **Question count: 185.**

B.0 How Part 5B works

B.0.1 EVIDENCE TAGGING INSIDE A QUESTION BANK

Tagging every line of a question bank would make it unreadable, so the convention is:

- **Every explanation that states an Indian legal or regulatory fact carries an inline tag.** These are the questions that could actively mislead a reader into breaking the law, so each is tagged individually and cross-checked.
- **Explanations that state ordinary physics or radio engineering** — Ohm's law, Carson's rule, how a superheterodyne receiver behaves — are untagged. They are textbook facts, not claims about Indian law. Where the *Indian examination's own convention* diverges from international engineering practice, and there are at least three such places, the divergence is tagged explicitly and listed in the traps table below.
- **ITU Radio Regulations claims** carry **[VERIFIED – RR 2024 ...]** where the provision was read directly.

B.0.2 DIFFICULTY LABELS

easy — a single recall step, or one substitution into one formula; roughly the level of the official sample paper. **medium** — two steps, a unit conversion, or a fact that is commonly got wrong.

hard — multi-step arithmetic, a distinction between two similar concepts, or a regulatory point where the popular study material is wrong and the primary source disagrees with it.

On the evidence of the official sample paper the real examination is overwhelmingly *easy* by this scale. The medium and hard questions are here because understanding beats recall, not because the examiner is expected to ask them.

B.0.3 FIVE TRAPS BUILT INTO THIS BANK DELIBERATELY

These are places where the mass-market Indian study material is wrong, or where the examination's own convention differs from international practice. Each is the subject of at least one question below.

#	Trap	Position taken here
1	Pass mark	40 % in each part and 40 % overall, both grades. The 50 % / 60 % figures printed by NIAR, SIARS, DARTS, RIRCOM and by DoT's own stale FAQ_for_HAM.pdf were superseded by rule 5(5) of the 2024 Rules. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, r.5(5); DoT WPC FAQ No. P-14036/02/2023-COP dated 2024-12-05, Q8]
2	QRQ versus QRO	The official examined list contains QRQ (shall I send faster). NIAR, SIARS and DARTS all print QRO instead — a shared transcription error. Learn both. [VERIFIED — OM L-14011/255/2004-AMT dated 2010-08-13, Annexure II]
3	Morse word spacing	The Indian examination material uses five units between words; the international standard is seven. Answer five in the examination. [VERIFIED — official WPC sample paper, the paper's Section B Q9; NIAR Study Manual p.139 — and note NIAR's own p.137 says seven]
4	Power limits	Restricted = 50 W HF / 10 W VHF and UHF ; General = 400 W HF / 25 W VHF and UHF . Both figures are DC input to the final stage , not RF output. The "Restricted grade gets 100 W" claim circulating online is wrong. [VERIFIED — OM 2010-08-13, Annexure-I, note (iv)]
5	RST signal strength 9	S9 = extremely strong ; S8 = strong. NIAR's model-answer key contradicts NIAR's own RST table on this. [VERIFIED — internal contradiction in the NIAR Study Manual, pp.125 versus 134]

B.0.4 DISTRIBUTION OF THE 185 QUESTIONS

Weighting follows the real paper, which is **50 marks for the paper's Section A (Radio Theory and Practice) and 50 marks for its Section B (Radio Regulations)** at both grades. [VERIFIED — official WPC sample question paper instruction block; corroborated by the DoT/Saral Sanchar [FAQ_for_HAM.pdf](#) Q5]

Group	Topic	Questions	Numbers
A1	Elementary electricity and magnetism	13	Q1–Q13
A2	Alternating current, resonance and filters	13	Q14–Q26

Group	Topic	Questions	Numbers
A3	Semiconductor devices, power supplies and rectifiers	10	Q27–Q36
A4	Transmitters, receivers and modulation	15	Q37–Q51
A5	Propagation and the electromagnetic spectrum	10	Q52–Q61
A6	Antennas and transmission lines	10	Q62–Q71
A7	Decibels and power calculations	7	Q72–Q78
A8	Measurement and test equipment	7	Q79–Q85
C	Interference and electromagnetic compatibility	10	Q86–Q95
D	Safety — radio-frequency exposure, mains, antennas	9	Q96–Q104
B1	Indian legal framework, grades, eligibility, fees	15	Q105–Q119
B2	Authorised bands, emissions and power	12	Q120–Q131
B3	Licence conditions, station records and procedure	13	Q132–Q144
B4	Permitted and prohibited transmissions	8	Q145–Q152
B5	Penalties and enforcement	8	Q153–Q160
B6	ITU Radio Regulations	12	Q161–Q172
B7	International operating practice	13	Q173–Q185
	Total	185	

Roughly 85 questions cover the paper's Section A material and 81 cover its Section B material, plus 10 on interference, which straddles both, and 9 on safety.

Safety is not an itemised topic in the 2010 syllabus. Neither radio-frequency exposure nor mains safety appears anywhere in Annexure II. Those nine questions are included because the reader is about to own a transmitter; not because the examiner is expected to ask them. [VERIFIED — the full text of Annexure II was checked] Likewise **filters and transmission lines** are not named in the syllabus; they sit under "coupled circuits" and "aerials" respectively and are fair game, but a candidate short of time should study the named topics first.

B.0.5 TWO THINGS THIS BANK DELIBERATELY DOES NOT DO

1. **It does not resolve the 30 metre band question.** The bands research records that **10100–10150 kHz does not appear in Annexure-I of the 2010 OM at all**, for either grade, even though NIAR's widely-copied table shows it and the syllabus research reproduces NIAR's version. The two research files disagree, and the bands file states that the Annexure-I pages were checked directly, which makes it the more authoritative of the two. **No question in this bank makes 30 m either a correct or an incorrect answer.** [UNCERTAIN — ask WPC in writing to confirm whether 10100–10150 kHz is authorised]

2. **It does not test the IARU Region 3 band plan as if it were law.** The International Amateur Radio Union band plan is a voluntary agreement, it is not in the syllabus, and Q185 exists to make that point. [REPORTED — IARU Region 3 Interim Band Plan]

B.1 The paper's SECTION A — RADIO THEORY AND PRACTICE

50 marks of the paper. Candidates holding a degree in Engineering or Science, or a Diploma in Engineering, having studied electronics or telecommunications, are exempt from the whole of the paper's Section A. [VERIFIED — OM 2010-08-13, Annexure II §1 Note; DoT/Saral Sanchar [FAQ_for_HAM.pdf](#) Q12] An exempt candidate can skip everything down to group C and still be fully prepared.

A1. ELEMENTARY ELECTRICITY AND MAGNETISM

Syllabus: Annexure II §2.1 Section A (i) — passive and active devices, Kirchhoff's laws, conductors and insulators, Ohm's law, conductance, self and mutual inductance, power and energy, permanent magnets and electromagnets.

Q1 easy A resistance of 4 ohms is connected across a 12 V battery. What current flows? (a) 0.33 A (b) 3 A (c) 48 A (d) 8 A **Answer: (b).** Ohm's law, $I = V/R = 12/4 = 3$ A. Option (a) is R/V rather than V/R — the single most common slip in the whole paper. Always write the formula down before substituting.

Q2 easy The same 4 ohm resistor on 12 V dissipates how much power? (a) 3 W (b) 16 W (c) 36 W (d) 48 W **Answer: (c).** $P = V^2/R = 144/4 = 36$ W. You can also get there as $P = VI = 12 \times 3 = 36$ W, or $P = I^2R = 9 \times 4 = 36$ W. Knowing all three forms lets you answer whichever two quantities the question gives you.

Q3 easy Three 6 ohm resistors are connected in parallel. The total resistance is: (a) 18 ohms (b) 6 ohms (c) 2 ohms (d) 0.5 ohm **Answer: (c).** For n equal resistors in parallel, $R_{\text{total}} = R/n = 6/3 = 2$ ohms. A useful sanity check: a parallel combination is always **smaller than the smallest** resistor in the group, so any answer of 6 ohms or more must be wrong.

Q4 easy Kirchhoff's current law states that: (a) the sum of voltages around a closed loop is zero (b) the algebraic sum of currents entering and leaving a junction is zero (c) current is proportional to voltage (d) power equals voltage times current **Answer: (b).** The current law is a statement of charge conservation at a node — whatever flows in must flow out. Option (a) is Kirchhoff's *voltage* law, which is conservation of energy around a loop. Expect the examiner to offer both and rely on you knowing which is which.

Q5 medium Two capacitors of 10 microfarad each are connected in series. The resulting capacitance is: (a) 20 μF (b) 10 μF (c) 5 μF (d) 0.05 μF **Answer: (c).** Capacitors combine the opposite way round to resistors — in series, $1/C = 1/C_1 + 1/C_2$, so two equal capacitors in series give half the value. In parallel they simply add, giving 20 μF . Remembering "capacitors are backwards" saves you every time.

Q6 easy The unit of conductance is the: (a) ohm (b) siemens (c) henry (d) farad **Answer: (b).** Conductance G is the reciprocal of resistance, $G = 1/R$, and its SI unit is the siemens (S), historically called the mho. The henry is inductance and the farad is capacitance.

Q7 medium Two coils of $100\ \mu\text{H}$ each are magnetically coupled with a coefficient of coupling $k = 0.5$. The mutual inductance is: (a) $25\ \mu\text{H}$ (b) $50\ \mu\text{H}$ (c) $100\ \mu\text{H}$ (d) $200\ \mu\text{H}$ **Answer: (b).** $M = k\sqrt{L_1 \times L_2} = 0.5 \times \sqrt{(100 \times 100)} = 50\ \mu\text{H}$. The coupling coefficient runs from 0 (no shared flux) to 1 (all flux shared); a coil wound on a common ferrite core approaches $k = 1$, while two air-spaced coils in a radio-frequency tank circuit might be $k = 0.05$.

Q8 easy The resistance of a conductor depends on: (a) its length only (b) its cross-sectional area only (c) its length, cross-sectional area and the resistivity of the material (d) the current flowing through it **Answer: (c).** $R = \rho l/A$, where ρ is the specific resistance, or resistivity, of the material. Resistance is a property of the conductor, not of the current — which is precisely what makes Ohm's law linear for ordinary metals.

Q9 medium A resistor is marked brown, black, red, gold. Its value is: (a) 100 ohms $\pm 5\%$ (b) 1 kilohm $\pm 5\%$ (c) 10 kilohms $\pm 5\%$ (d) 1 kilohm $\pm 10\%$ **Answer: (b).** Brown = 1, black = 0, red = multiplier $\times 100$, so $10 \times 100 = 1000$ ohms. Gold in the fourth band means $\pm 5\%$ tolerance; silver would be $\pm 10\%$. Note that gold and silver mean *multiplier* $\div 10$ and $\div 100$ when they appear in the third band, but *tolerance* in the fourth.

Q10 easy Four 1.5 V dry cells are connected in series. The resulting voltage and capacity are: (a) 6 V, same capacity as one cell (b) 1.5 V, four times the capacity (c) 6 V, four times the capacity (d) 1.5 V, same capacity as one cell **Answer: (a).** Series connection adds voltages, but the same current flows through every cell, so the ampere-hour capacity is that of one cell. Parallel connection does the opposite: the voltage stays at 1.5 V and the capacity multiplies. This is exactly why a handheld transceiver's battery pack is a series stack — it needs the voltage.

Q11 easy An electromagnet differs from a permanent magnet in that: (a) it is stronger (b) its magnetism can be switched on and off, and its strength varied, by controlling the current (c) it has only one pole (d) it works only with alternating current **Answer: (b).** An electromagnet is a coil around a ferromagnetic core; the field exists only while current flows and is proportional to the ampere-turns. That controllability is why relays, loudspeakers and rotary machines use electromagnets. A magnet with only one pole has never been observed.

Q12 medium The energy consumed by a 100 W soldering iron left on for 3 hours is: (a) 300 W (b) 0.3 kWh (c) 0.3 kW (d) 300 J **Answer: (b).** Energy = power $\times$ time = $100\ \text{W} \times 3\ \text{h} = 300\ \text{Wh} = 0.3$ kilowatt-hour. Watch the units in the distractors: watts and kilowatts are *power*, a rate, not energy. 300 J would be 100 W for 3 seconds, not 3 hours.

Q13 hard A 12 V battery with an internal resistance of 0.5 ohm supplies a 5.5 ohm load. The terminal voltage of the battery on load is: (a) 12 V (b) 11 V (c) 1 V (d) 6 V **Answer: (b).** Total circuit resistance = $0.5 + 5.5 = 6$ ohms, so $I = 12/6 = 2\ \text{A}$. The internal resistance drops $I \times r = 2 \times 0.5 = 1\ \text{V}$, leaving 11 V at the terminals. This is why a handheld's transmit voltage sags the moment you key up: a 5 A transmit current through even 0.2 ohm of internal resistance and connector loss costs you a volt.

A2. ALTERNATING CURRENT, RESONANCE AND FILTERS

Syllabus: Annexure II §2.1 Section A (ii) — sinusoidal quantities, phase, reactance, impedance, power factor, series and parallel circuits, resonance, coupled circuits; and §2.2 (ii) for the General grade — transformer construction, losses, and the transformer as a matching device.

Q14 easy The RMS value of a sine wave whose peak value is 100 V is approximately: (a) 50 V (b) 63.7 V (c) 70.7 V (d) 141.4 V **Answer: (c).** $\text{RMS} = \text{peak} / \sqrt{2} = 0.707 \times \text{peak}$. The root-mean-square value is the DC voltage that would produce the same heating in a resistor, which is why it is the value meters and mains specifications quote. 63.7 V is the *average* over a half cycle — a common distractor.

Q15 medium The Indian mains supply is quoted as 230 V. Its peak value is approximately: (a) 163 V (b) 230 V (c) 325 V (d) 460 V **Answer: (c).** The quoted 230 V is RMS, so $\text{peak} = 230 \times 1.414 = 325$ V. This matters for equipment safety: a capacitor across the mains must be rated well above 325 V, not 230 V, and the reservoir capacitor after a bridge rectifier will charge towards 325 V, not 230 V.

Q16 easy The average value of a sine wave taken over one complete cycle is: (a) 0.707 of peak (b) 0.637 of peak (c) zero (d) equal to the peak **Answer: (c).** The positive and negative halves are equal and opposite, so they cancel exactly. The often-quoted $0.637 \times \text{peak}$ is the average over a *half* cycle, which is what a rectifier delivers and what a moving-coil meter reads after rectification.

Q17 medium The inductive reactance of a 10 μH coil at 1 MHz is approximately: (a) 6.28 ohms (b) 62.8 ohms (c) 628 ohms (d) 1590 ohms **Answer: (b).** $X_L = 2\pi fL = 2\pi \times 10^6 \times 10 \times 10^{-6} = 62.8$ ohms. Inductive reactance rises with frequency, which is the whole basis of a radio-frequency choke: a few microhenries is nearly a short circuit at audio and a serious obstacle at high frequency.

Q18 medium The capacitive reactance of a 100 pF capacitor at 1 MHz is approximately: (a) 1.6 ohms (b) 62.8 ohms (c) 159 ohms (d) 1590 ohms **Answer: (d).** $X_C = 1/(2\pi fC) = 1/(2\pi \times 10^6 \times 100 \times 10^{-12}) = 1591$ ohms. Capacitive reactance *falls* as frequency rises — the opposite behaviour to an inductor, and the reason a small coupling capacitor passes radio frequency while blocking DC.

Q19 medium A series circuit contains $L = 100 \mu\text{H}$ and $C = 100 \text{ pF}$. Its resonant frequency is approximately: (a) 159 kHz (b) 1.59 MHz (c) 15.9 MHz (d) 159 MHz **Answer: (b).** $f = 1/(2\pi\sqrt{LC}) = 1/(2\pi\sqrt{(10^{-4} \times 10^{-10})}) = 1/(2\pi \times 10^{-7}) = 1.59 \text{ MHz}$. Note that resonance depends on the *product* LC , so halving L and doubling C leaves the frequency unchanged — that is how a tuned circuit's impedance is set independently of its frequency.

Q20 medium At series resonance, the impedance of an R-L-C circuit is: (a) maximum, and the current is minimum (b) minimum and purely resistive, and the current is maximum (c) purely inductive (d) zero **Answer: (b).** At resonance X_L and X_C are equal and opposite and cancel, leaving only R . Series resonance therefore gives minimum impedance and maximum current — it is an *acceptor* circuit, used to pass one frequency. It is never truly zero because the coil and wiring always have some resistance.

Q21 medium At parallel resonance (a "tank" circuit), the circuit presents: (a) minimum impedance and maximum line current (b) maximum impedance and minimum line current (c) zero impedance (d) a purely reactive impedance **Answer: (b).** A parallel tuned circuit is a *rejector*: at resonance it looks like a high resistance to the supply while a large circulating current flows round the L-C loop itself. This is why the tank circuit in a transmitter's output stage develops a large radio-frequency voltage from a modest supply current.

Q22 hard A tuned circuit resonant at 7.1 MHz has a bandwidth of 71 kHz between its half-power points. Its Q factor is: (a) 10 (b) 71 (c) 100 (d) 500 **Answer: (c).** $Q = f_0 / \text{bandwidth} = 7,100,000 / 71,000 = 100$. High Q means a narrow, selective response and low losses; low Q means a broad response. A receiver's intermediate-frequency filter needs high Q for selectivity, whereas a transmitter's output network needs enough Q to suppress harmonics but not so much that it is impossible to tune.

Q23 medium The power factor of an AC circuit is: (a) the ratio of reactance to resistance (b) the cosine of the phase angle between voltage and current (c) always equal to 1 (d) the ratio of peak to RMS value **Answer: (b).** Power factor = $\cos \phi$, and true power = $V \times I \times \cos \phi$. In a purely resistive circuit voltage and current are in phase, $\phi = 0$ and $\cos \phi = 1$. In a purely reactive circuit they are 90 degrees apart, $\cos \phi = 0$, and no real power is consumed at all despite current flowing.

Q24 hard A load draws 5 A from a 230 V AC supply at a power factor of 0.8 lagging. The true power consumed is: (a) 1150 W (b) 920 W (c) 184 W (d) 1437 W **Answer: (b).** True power = $V \times I \times \cos \phi = 230 \times 5 \times 0.8 = 920 \text{ W}$. The apparent power is $230 \times 5 = 1150 \text{ VA}$, and the difference is reactive power that shuttles back and forth without doing work. "Lagging" means the current lags the voltage, which indicates an inductive load such as a motor or a transformer.

Q25 medium A transformer has 1000 primary turns and 50 secondary turns. If 240 V AC is applied to the primary, the secondary voltage is: (a) 4.8 V (b) 12 V (c) 24 V (d) 4800 V **Answer: (b).** $V_s/V_p = N_s/N_p$, so $V_s = 240 \times 50/1000 = 12 \text{ V}$. Voltage transforms in direct proportion to the turns ratio; current transforms in inverse proportion, so the secondary can supply twenty times the primary current at one twentieth of the voltage, ignoring losses.

Q26 hard A transformer is used to match a 50 ohm source to an 800 ohm load. The required turns ratio is: (a) 4:1 (b) 1:4 (c) 16:1 (d) 1:16 **Answer: (b).** Impedance transforms as the **square** of the turns ratio: $Z_p/Z_s = (N_p/N_s)^2$. Here $800/50 = 16$, so $N_p/N_s = \sqrt{16} = 4$, meaning 4 secondary turns for every 1 primary turn — a 1:4 step-up. This is the "transformer as a matching device" the General-grade syllabus names, and it is the principle behind a 4:1 balun feeding a folded dipole.

A3. SEMICONDUCTOR DEVICES, POWER SUPPLIES AND RECTIFIERS

Syllabus: Annexure II §2.1 Section A (iii) — diodes and transistors for amplifiers, oscillators, detectors and frequency changers; §2.1 (ii) 4 — rectifiers, voltage regulation and smoothing; and for the General grade §2.2 (iii) semiconductor conduction, symbols and biasing, and §2.2 (iv) power supplies including the bridge rectifier.

Q27 **easy** A semiconductor diode conducts appreciably when: (a) reverse biased (b) forward biased (c) unbiased (d) at any polarity above 0.1 V **Answer: (b).** Forward bias — anode positive with respect to cathode — collapses the depletion region and lets current flow, above about 0.3 V for germanium and 0.7 V for silicon. Reverse bias widens the depletion region and only a tiny leakage current flows. This one-way behaviour is what makes rectification possible.

Q28 **medium** In an NPN transistor operating as a linear amplifier: (a) both junctions are forward biased (b) both junctions are reverse biased (c) the emitter-base junction is forward biased and the collector-base junction is reverse biased (d) the emitter-base junction is reverse biased and the collector-base junction is forward biased **Answer: (c).** This is the definition of the active region. Forward bias on the emitter-base junction injects carriers into the thin base; the reverse-biased collector-base junction then sweeps almost all of them into the collector. Both junctions forward biased is saturation, a closed switch; both reverse biased is cut-off, an open switch.

Q29 **medium** A common-emitter amplifier stage produces an output that is: (a) in phase with the input (b) 90 degrees out of phase (c) 180 degrees out of phase (d) at twice the input frequency **Answer: (c).** Rising base current increases collector current, which increases the drop across the collector load resistor and therefore *lowers* the collector voltage — an inversion. The common emitter is the workhorse configuration because it gives both voltage gain and current gain. The emitter follower (common collector) has no inversion and no voltage gain, but low output impedance.

Q30 **easy** A half-wave rectifier fed from the 50 Hz mains produces a ripple at: (a) 25 Hz (b) 50 Hz (c) 100 Hz (d) 200 Hz **Answer: (b).** A half-wave rectifier passes only one half of each cycle, so there is one output pulse per input cycle — 50 pulses per second. A full-wave or bridge rectifier uses both halves and produces 100 Hz, which is easier to smooth because the capacitor has half as long to discharge between pulses.

Q31 **medium** How many diodes does a bridge rectifier use, and what is its advantage over a centre-tapped full-wave rectifier? (a) Two; it needs no transformer (b) Four; it uses the whole secondary winding and needs no centre tap (c) Four; it produces a higher ripple frequency (d) Two; it has lower forward voltage drop **Answer: (b).** The bridge uses four diodes in a ring, so the full secondary voltage appears at the output on both half cycles and no centre tap is needed. Its penalty is two diode drops in series instead of one. Both circuits give the same 100 Hz ripple frequency from 50 Hz mains, so (c) is wrong.

Q32 **medium** The purpose of the capacitor across the output of a rectifier is to: (a) increase the output voltage only (b) store charge between rectifier pulses and so reduce ripple (c) block DC (d) protect the diodes **Answer: (b).** The reservoir capacitor charges to near the peak on each pulse and discharges into the load between pulses, filling in the gaps. Its side effect is that the output rises towards the *peak* value rather than the average, which is why an unloaded 12 V transformer with a bridge and a capacitor can measure nearly 17 V.

Q33 **medium** A Zener diode used as a voltage regulator is connected: (a) in series with the load, forward biased (b) across the load, reverse biased (c) across the load, forward biased (d) in series with the load, reverse biased **Answer: (b).** A Zener is operated in reverse breakdown,

where its voltage is nearly constant over a wide range of current. Placed in parallel with the load and fed through a series resistor, it absorbs whatever current the load does not take, holding the voltage steady. Forward biased it is just an ordinary diode.

Q34 easy Voltage regulation of a power supply refers to: (a) the ripple content of the output (b) how much the output voltage changes between no load and full load (c) the maximum current available (d) the efficiency of the rectifier **Answer: (b).** Regulation is usually expressed as a percentage: $(V_{\text{no load}} - V_{\text{full load}})/V_{\text{full load}} \times 100$. A supply with poor regulation sags when the transmitter keys up, which pulls the frequency of a free-running oscillator and can distort audio. Ripple is a separate defect, measured separately.

Q35 medium A transistor used as a frequency changer (mixer) in a receiver produces at its output: (a) only the original two frequencies (b) the sum and difference of the two input frequencies, as well as the originals (c) only the difference frequency (d) a frequency equal to the average of the two inputs **Answer: (b).** Mixing is a non-linear process, and non-linearity generates sum and difference products. A superheterodyne receiver selects the difference product with a fixed intermediate-frequency filter and discards the rest. Because both the sum and the difference exist, two different incoming frequencies can produce the same intermediate frequency — which is the origin of image interference.

Q36 hard In an N-type semiconductor, the majority carriers are electrons because the silicon has been doped with: (a) a trivalent impurity such as boron (b) a pentavalent impurity such as phosphorus (c) a metal (d) nothing; N-type silicon is pure **Answer: (b).** A pentavalent atom has five valence electrons; four form covalent bonds with neighbouring silicon atoms and the fifth is loosely bound and free to conduct — hence "donor" doping and negative majority carriers. Trivalent doping with boron leaves a hole and gives P-type material. Pure silicon is "intrinsic" and a poor conductor.

A4. TRANSMITTERS, RECEIVERS AND MODULATION

Syllabus: Annexure II §2.1 Section A (iv) radio receivers and (v) transmitters; plus, for the General grade, §2.2 (i) principles of communications — analogue and digital, need for modulation, amplitude, frequency and pulse modulation, demodulation — and §2.2 (vii) facsimile and television.

Q37 easy The principal advantage of the superheterodyne receiver over the tuned-radio-frequency receiver is that: (a) it needs fewer components (b) all the gain and selectivity are obtained at one fixed intermediate frequency, so performance is uniform across the band (c) it does not need an antenna (d) it can receive only CW **Answer: (b).** A tuned-radio-frequency receiver must retune every tuned stage together as you move across a band, and its bandwidth widens as frequency rises. The superheterodyne converts every incoming signal to a single fixed intermediate frequency, so one carefully designed filter provides the same selectivity everywhere. The cost is the image response.

Q38 medium A receiver tuned to 7.100 MHz uses an intermediate frequency of 455 kHz with the local oscillator above the signal frequency. The local oscillator frequency is: (a) 6.645 MHz (b) 7.100 MHz (c) 7.555 MHz (d) 8.010 MHz **Answer: (c).** High-side injection means local oscillator =

signal + intermediate frequency = $7.100 + 0.455 = 7.555$ MHz. The mixer's difference product is then exactly 455 kHz. Low-side injection at 6.645 MHz would work equally well arithmetically but is less common because it demands a wider oscillator tuning range as a percentage.

Q39 **hard** For the receiver in Q38, the image frequency is: (a) 6.645 MHz (b) 7.555 MHz (c) 8.010 MHz (d) 6.190 MHz **Answer: (c).** The image lies **twice the intermediate frequency away** from the wanted signal, on the far side of the local oscillator: $7.100 + (2 \times 0.455) = 8.010$ MHz. A signal there also mixes with the 7.555 MHz oscillator to give 455 kHz and passes straight through the filter. Only tuned circuits *ahead* of the mixer — the radio-frequency preselector — can reject it, which is why a higher intermediate frequency, or a double-conversion design, gives better image rejection.

Q40 **easy** The selectivity of a receiver is its ability to: (a) receive very weak signals (b) reject signals on adjacent frequencies (c) reproduce audio faithfully (d) operate over a wide temperature range **Answer: (b).** Selectivity is about discrimination between channels and is set by the intermediate-frequency filter bandwidth. Sensitivity (a) is about weak-signal reception and is set by gain and noise figure. Fidelity (c) is faithfulness of reproduction. The examiner routinely offers all three together, so learn the distinction rather than the words.

Q41 **medium** The purpose of automatic volume control, also called automatic gain control, in a receiver is to: (a) mute the receiver when no signal is present (b) keep the audio output roughly constant despite large variations in received signal strength (c) improve selectivity (d) generate the beat frequency for CW reception **Answer: (b).** Automatic volume control rectifies part of the intermediate-frequency signal and feeds it back as bias to reduce the gain of earlier stages when the signal is strong. Without it, a fading HF signal would swing between inaudible and deafening. Option (a) describes squelch, which is a different circuit with a different job.

Q42 **easy** A beat frequency oscillator is required in a receiver in order to: (a) receive AM broadcast stations (b) make CW (A1A) and single-sideband signals audible (c) suppress noise (d) tune the antenna **Answer: (b).** A keyed unmodulated carrier contains no audio; passed through an envelope detector it produces only thumps. The beat frequency oscillator injects a signal offset by a few hundred hertz from the intermediate frequency, and the difference between them is an audible tone. The same principle recovers single sideband, where the carrier was suppressed at the transmitter.

Q43 **medium** The function of squelch, also called muting, in an FM receiver is to: (a) reduce the transmitter power (b) silence the audio output when no signal is being received (c) limit the deviation (d) improve the frequency stability **Answer: (b).** An FM receiver with no signal produces full-volume hiss, because the limiter and discriminator amplify noise to the same level as a signal. The squelch circuit senses the absence of a carrier, or the presence of high-frequency noise, and mutes the audio stage. This is why every VHF handheld has a squelch control, and why setting it too tight loses weak stations.

Q44 **medium** Why is modulation necessary in radio communication? (a) Because audio frequencies would require impractically large antennas and could not be separated into channels (b) Because audio cannot travel through air (c) To increase the transmitter's power (d) To reduce the receiver's noise **Answer: (a).** A 3 kHz audio signal has a wavelength of 100 km, so

a half-wave antenna would be 50 km long. Modulation shifts the information onto a radio-frequency carrier that can be radiated efficiently from a practical antenna, and it allows many users to share the spectrum by occupying different carrier frequencies.

Q45 medium In amplitude modulation, the bandwidth occupied is: (a) equal to the highest modulating frequency (b) twice the highest modulating frequency (c) equal to the carrier frequency (d) independent of the modulating frequency **Answer: (b).** Amplitude modulation produces an upper and a lower sideband, each extending from the carrier by the highest modulating frequency. Speech limited to 3 kHz therefore occupies 6 kHz of spectrum in AM. Single sideband transmits one sideband only and needs about 3 kHz — which is the main reason amateurs use J3E on HF.

Q46 hard An AM transmitter with a carrier power of 100 W is modulated 100 %. The total radiated power is: (a) 100 W (b) 125 W (c) 150 W (d) 200 W **Answer: (c).** Total power = $P_{\text{carrier}} \times (1 + m^2/2) = 100 \times (1 + 0.5) = 150$ W, with 25 W in each sideband. Note that at 100 % modulation two thirds of the transmitted power is still in the carrier, which carries no information at all. That inefficiency is exactly what single sideband eliminates by suppressing the carrier and one sideband.

Q47 hard An FM transmitter has a peak deviation of 5 kHz and the highest modulating frequency is 3 kHz. By Carson's rule the bandwidth is: (a) 5 kHz (b) 8 kHz (c) 16 kHz (d) 30 kHz **Answer: (c).** Carson's rule gives $BW = 2 \times (\Delta f + f_m) = 2 \times (5 + 3) = 16$ kHz. This is why amateur VHF FM sits comfortably in 25 kHz channels, and why "narrow FM" with ± 2.5 kHz deviation is used where 12.5 kHz channels apply. Over-deviating widens the signal and splatters into the adjacent channel.

Q48 medium The chief advantage of frequency modulation over amplitude modulation for VHF voice is: (a) it uses less bandwidth (b) it is much less affected by amplitude noise and impulse interference (c) it needs no antenna tuning (d) it travels further **Answer: (b).** Information in FM is carried by frequency, so the receiver's limiter can clip away every amplitude variation — including ignition noise and static — before demodulation. The price is bandwidth: FM needs several times the spectrum of single sideband. FM also shows the "capture effect", where the stronger of two co-channel signals suppresses the weaker entirely.

Q49 medium A crystal oscillator is preferred over an LC oscillator in a transmitter because: (a) it is cheaper (b) quartz has a very high Q, giving far better frequency stability (c) it can be tuned over a wide range (d) it produces more power **Answer: (b).** A quartz crystal behaves as an extremely high-Q resonator — Q in the tens of thousands against perhaps 200 for a good LC circuit — so it holds frequency tightly against changes in supply voltage, temperature and loading. Its drawback is precisely the opposite of (c): it is essentially fixed in frequency.

Q50 medium A buffer amplifier is placed between the oscillator and the power amplifier of a transmitter in order to: (a) increase the output power (b) isolate the oscillator so that changes in the load do not pull its frequency (c) filter harmonics (d) modulate the carrier **Answer: (b).** Without a buffer, the varying load presented by the driven stage — and by the antenna as

standing wave ratio changes — reacts back on the oscillator and shifts its frequency, an effect called pulling. The buffer's high input impedance and low output impedance break that coupling. It provides some gain as a side effect, not as its purpose.

Q51 **hard** Which class of amplifier is unsuitable for amplifying a single-sideband (J3E) signal, and why? (a) Class A, because it is inefficient (b) Class AB, because it is biased near cut-off (c) Class C, because it is non-linear and would destroy the amplitude information (d) Class B, because it draws no quiescent current **Answer: (c).** Single sideband carries its information in the amplitude envelope, so the amplifier must be **linear** — Class A or Class AB. Class C conducts for less than half a cycle and is highly non-linear; it is efficient and perfectly good for CW and FM, where amplitude carries nothing, but on single sideband it produces severe distortion and splatter into adjacent channels.

A5. PROPAGATION AND THE ELECTROMAGNETIC SPECTRUM

Syllabus: Annexure II §2.1 Section A (vi) — electromagnetic spectrum, wavelength and frequency, frequency bands, ground, sky and space waves, skip distance, skip zone, fading; and for the General grade §2.2 (viii) — characteristics of the ionosphere and troposphere, properties of the ionospheric layers, critical frequency, and day and night frequencies.

Q52 **easy** The wavelength corresponding to a frequency of 145 MHz is approximately: (a) 0.2 m (b) 2.07 m (c) 20.7 m (d) 145 m **Answer: (b).** λ in metres = $300 / f$ in MHz = $300/145 = 2.07$ m. This is why 144–146 MHz is called the "2 metre band". The same formula run backwards tells you the 20 m band is around $300/20 = 15$ MHz, which is indeed inside 14.000–14.350 MHz.

Q53 **easy** According to ITU Article 2, the VHF band covers: (a) 3–30 MHz (b) 30–300 MHz (c) 300–3000 MHz (d) 3–30 GHz **Answer: (b).** VHF is 30–300 MHz, metric waves, band 8; 300–3000 MHz is UHF, decimetric waves, band 9. [VERIFIED — ITU Radio Regulations, Edition of 2024, Article 2, table at RR 2.1] **Trap:** the NIAR Study Manual's own model answer for this question says "300 to 3000 MHz", which is wrong. Go with 30–300 MHz.

Q54 **medium** The ionospheric layer chiefly responsible for absorbing HF signals during daylight hours is the: (a) D layer (b) E layer (c) F1 layer (d) F2 layer **Answer: (a).** The D layer forms at roughly 60–90 km, is ionised only by direct sunlight, and absorbs rather than refracts — most strongly at the lower frequencies. It is why 80 m and 40 m are short-range by day and open up dramatically after sunset when the D layer disappears.

Q55 **medium** The critical frequency of an ionospheric layer is: (a) the lowest frequency that will penetrate it (b) the highest frequency returned to earth when transmitted vertically upward (c) the frequency of maximum absorption (d) the frequency at which the layer disappears **Answer: (b).** Send a signal straight up and increase the frequency; at some point the layer no longer bends it back and it escapes into space. That threshold is the critical frequency f_c . At oblique incidence a higher frequency can still be returned, which gives the maximum usable frequency: $MUF = f_c / \cos \theta$, where θ is the angle from the vertical.

Q56 **medium** Skip distance is: (a) the distance the ground wave travels (b) the shortest distance from the transmitter at which a sky wave of a given frequency is returned to earth (c) the distance between two successive hops (d) the maximum range of a VHF signal **Answer: (b).**

Below a certain angle the wave penetrates the ionosphere and is lost; the first place the sky wave comes back down defines the skip distance. Between the outer limit of the ground wave and that point lies the **skip zone**, where the station is inaudible even though it is perfectly audible both nearer and further away.

Q57 medium For long-distance HF communication, which is generally true? (a) Higher frequency bands work better at night, lower bands by day (b) Higher frequency bands work better by day, lower bands at night (c) Frequency makes no difference (d) Only the 2 m band is usable

Answer: (b). Daytime solar ionisation raises the maximum usable frequency, so 20 m, 15 m and 10 m open; at night ionisation decays, the maximum usable frequency falls, and the higher bands close while 80 m and 40 m come alive as D-layer absorption vanishes. The syllabus names this explicitly as "critical frequency and day / night frequencies".

Q58 easy VHF and UHF signals normally propagate by: (a) ground wave following the earth's curvature (b) sky wave refracted by the F2 layer (c) space wave, essentially line of sight with some tropospheric bending (d) conduction through the ground **Answer: (c).** Above about 30 MHz the ionosphere usually lets signals straight through, so 2 m and 70 cm behave more like light — a direct path plus a little refraction in the troposphere, which extends the radio horizon roughly 15 % beyond the optical one. Height, not power, is what buys range on these bands.

Q59 hard A VHF antenna is mounted 100 m above ground. Its approximate radio horizon is: (a) 10 km (b) 36 km (c) 41 km (d) 100 km **Answer: (c).** The usual approximation is $d \text{ in km} \approx 4.12 \times \sqrt{h \text{ in metres}}$ = $4.12 \times 10 = 41.2 \text{ km}$. The corresponding *optical* horizon would be $3.57 \times \sqrt{h} = 35.7 \text{ km}$; the difference is tropospheric refraction bending the wave slightly around the curve. Both stations' heights add, so two 100 m sites can work over about 82 km.

Q60 medium Fading of an HF signal is most commonly caused by: (a) transmitter instability (b) two or more components of the same signal arriving by different paths and interfering (c) receiver automatic volume control action (d) antenna misalignment **Answer: (b).** Multipath is the usual culprit: a single-hop and a two-hop path, or an ordinary and an extraordinary ray, arrive with continuously changing relative phase and alternately reinforce and cancel. Absorption fading (slow, as the D layer thickens) and selective fading (different parts of the same sideband fading independently, which sounds like distortion) are the other named types.

Q61 medium Tropospheric ducting on VHF is associated with: (a) high sunspot numbers (b) a temperature inversion trapping the signal in a layer near the surface (c) auroral activity (d) the F2 layer **Answer: (b).** When a warm layer sits above cooler air — common over the sea and on still evenings — the refractive index gradient reverses and VHF signals are guided along inside the layer, sometimes for several hundred kilometres. It is a *tropospheric* phenomenon, entirely independent of the ionosphere and of the sunspot cycle.

A6. ANTENNAS AND TRANSMISSION LINES

Syllabus: Annexure II §2.1 Section A (vii) — common types of transmitting and receiving aerials; and for the General grade §2.2 (ix) — principles of radiation, and aerials for different frequency bands including aerials for microwave. Transmission lines are not separately named in the syllabus; they are treated here as part of "aerials".

Q62 medium The approximate physical length of a half-wave dipole for 14.2 MHz is: (a) 5 m (b) 10 m (c) 21 m (d) 42 m **Answer: (b).** $\lambda = 300/14.2 = 21.1$ m, so a half wavelength in free space is 10.6 m. A real wire is electrically slightly longer than its physical length, the end effect, so it is cut about 5 % short: roughly $143/f$ in MHz = 10.1 m. Answer (c) is the *full* wavelength halved incorrectly — a classic distractor.

Q63 easy The feedpoint impedance of a half-wave dipole in free space is approximately: (a) 25 ohms (b) 50 ohms (c) 73 ohms (d) 300 ohms **Answer: (c).** About 73 ohms resistive at resonance, which is why 75 ohm coaxial cable was historically the natural feeder and why 50 ohm cable gives a perfectly usable standing wave ratio of about 1.5:1. A quarter-wave vertical over a good ground plane is roughly half that, around 36 ohms; a folded dipole is four times it, around 300 ohms.

Q64 medium In a Yagi-Uda array, the reflector is: (a) shorter than the driven element and placed in front of it (b) longer than the driven element and placed behind it (c) the same length as the driven element (d) connected directly to the feeder **Answer: (b).** The reflector is roughly 5 % longer than the driven element, making it inductive, and it sits on the opposite side from the direction of maximum radiation. Directors are about 5 % shorter, making them capacitive, and sit in front. Only the driven element is connected to the feeder; the rest are parasitic.

Q65 medium A standing wave ratio of 1:1 on a feeder indicates that: (a) the antenna is radiating no power (b) the load impedance exactly matches the line's characteristic impedance and there is no reflected power (c) the feeder is open circuit (d) the transmitter is faulty **Answer: (b).** Standing wave ratio is the ratio of maximum to minimum voltage along the line, and it is 1:1 only when all forward power is absorbed by the load. A 150 ohm load on 50 ohm cable gives $150/50 = 3:1$. Note that a **dummy load gives a perfect 1:1 while radiating nothing at all**, so a good reading is not by itself proof that the antenna is working.

Q66 medium The velocity factor of a typical solid polyethylene coaxial cable is about: (a) 0.25 (b) 0.66 (c) 1.0 (d) 1.5 **Answer: (b).** Signals travel at roughly 66 % of the speed of light in solid polyethylene dielectric, about 80 % in foam types. This matters whenever you cut a cable to an electrical length — a quarter-wave matching section for 145 MHz is not 0.52 m of cable but $0.52 \times 0.66 = 0.34$ m. Nothing travels faster than light, so (c) and (d) are impossible.

Q67 medium A balun is used between a coaxial feeder and a dipole in order to: (a) increase the gain of the antenna (b) connect an unbalanced line to a balanced antenna and stop radio frequency flowing on the outside of the braid (c) protect against lightning (d) measure standing wave ratio **Answer: (b).** "Balun" is short for balanced-to-unbalanced. Without one, current can flow back along the outside of the coaxial braid, making the feeder part of the radiating system — which distorts the pattern, brings radio frequency into the operating room, and makes standing-wave-ratio readings depend on where you stand. A choke balun of ferrite beads or coiled coaxial cable is the simplest form.

Q68 hard The gain of a half-wave dipole compared with an isotropic radiator is: (a) 0 dB (b) 2.15 dB (c) 3 dB (d) 6 dB **Answer: (b).** An isotropic radiator is a theoretical point source radiating equally in all directions; a real dipole concentrates energy broadside and so has 2.15 dB gain over it. This is the difference between dBi and dBd: $\text{dBd} = \text{dBi} - 2.15$. An antenna advertised as "3 dB gain" is worth checking — 3 dBi is barely better than a dipole, while 3 dBd is 5.15 dBi.

Q69 medium Two stations use antennas of opposite polarisation, one horizontal and one vertical. The likely result is: (a) no difference (b) a substantial loss, often around 20 dB on a direct path (c) improved signal strength (d) damage to the transmitter **Answer: (b).** Cross-polarisation loss on a clean line-of-sight VHF path is severe. This is why Indian VHF practice, like most of the world's, uses **vertical** polarisation for FM mobile and repeater work and horizontal for single-sideband and weak-signal work — mixing them wastes most of your signal. On HF sky-wave paths the ionosphere scrambles polarisation, so it matters much less.

Q70 easy The "rubber duck" antenna supplied with a handheld transceiver is: (a) a full half-wave dipole (b) an electrically shortened, deliberately inefficient antenna traded for convenience (c) more efficient than a quarter-wave whip (d) a directional antenna **Answer: (b).** It is a helically loaded element far shorter than a quarter wave, with correspondingly low radiation resistance and high losses — typically several decibels down on a simple quarter-wave whip. Replacing it with a proper whip or, better, connecting the handheld to an outdoor antenna, buys more range than any legal increase in power could.

Q71 hard An antenna is fed with 10 W and the feeder has 3 dB of loss. The power reaching the antenna is: (a) 7 W (b) 5 W (c) 3 W (d) 1 W **Answer: (b).** 3 dB is a factor of two in power, so half of it is lost as heat in the cable and 5 W arrives. This is why long runs of thin coaxial cable at UHF are so damaging — at 435 MHz, 30 m of RG-58 can easily eat more than half your transmitter output before it reaches the antenna.

A7. DECIBELS AND POWER CALCULATIONS

Not a separately named syllabus heading, but decibel questions appear in the official WPC sample paper and the concept underpins "power and energy — definition, units and simple applications" (Annexure II §2.1 Section A (i) 5).

Q72 easy A power gain of 3 dB corresponds to a power ratio of approximately: (a) 1.5 (b) 2 (c) 3 (d) 10 **Answer: (b).** $\text{dB} = 10 \log_{10}(P_2/P_1)$. For a ratio of 2, $10 \log 2 = 3.01$ dB. The three ratios worth memorising are 3 dB = $\times 2$, 10 dB = $\times 10$, 20 dB = $\times 100$. Everything else can be built from them: 13 dB = $2 \times 10 = \times 20$.

Q73 medium An amplifier increases power from 1 W to 100 W. Its gain in decibels is: (a) 10 dB (b) 20 dB (c) 40 dB (d) 100 dB **Answer: (b).** $10 \log_{10}(100/1) = 10 \times 2 = 20$ dB. Note that the *voltage* formula uses 20 log, not 10 log, so a hundredfold voltage gain would be 40 dB. Using the wrong one is the commonest decibel mistake — power uses 10, voltage and current use 20.

Q74 medium A power level of 10 W expressed in dBm is: (a) 10 dBm (b) 20 dBm (c) 40 dBm (d) 100 dBm **Answer: (c).** dBm is referenced to 1 milliwatt, and 10 W = 10,000 mW, so $10 \log_{10}(10,000) = 40$ dBm. Handy anchors: 0 dBm = 1 mW, 30 dBm = 1 W, 40 dBm = 10 W, 50 dBm = 100 W.

Q75 hard The amateur-satellite service is permitted to Indian General-grade holders at a maximum effective isotropic radiated power of 30 dBW. In watts that is: (a) 30 W (b) 100 W (c) 300 W (d) 1000 W **Answer: (d).** dBW is referenced to 1 watt, so $30 \text{ dBW} = 10^3 = 1000 \text{ W}$. Note that this is *effective isotropic radiated power*, which includes antenna gain — so a 25 W transmitter behind a 16 dBi antenna is already close to the limit. [VERIFIED — WPC OM L-14011/255/2004-AMT dated 2010-08-13, Annexure-I, Note (i)]

Q76 medium A signal report improves by two S-units. By the usual convention this represents a change of about: (a) 2 dB (b) 6 dB (c) 12 dB (d) 20 dB **Answer: (c).** One S-unit is conventionally 6 dB, so two S-units is 12 dB — a factor of about 16 in power. This is worth internalising before you spend money: doubling your transmitter power gains 3 dB, which is **half an S-unit** and usually inaudible. A better antenna almost always beats a bigger amplifier.

Q77 hard A transmitter's final stage draws 10 W of DC input power and the stage is 60 % efficient. The RF output power is approximately: (a) 4 W (b) 6 W (c) 10 W (d) 16.7 W **Answer: (b).** Output = input $\times$ efficiency = $10 \times 0.6 = 6 \text{ W}$; the other 4 W becomes heat in the device. This distinction is not academic in India: the Restricted-grade VHF and UHF limit of **10 W is DC input to the final stage**, so a radio marketed as "10 W RF output" would need roughly 15–25 W of DC input (the range used throughout this report, at 40–65 % amplifier efficiency) and would exceed the limit. [VERIFIED — Annexure-I, Note (iv): "DC input power is the total direct current power input to the final stage of the transmitter." The efficiency range and the resulting DC figure are engineering inference, not law — **[UNCERTAIN]**; see Part 1 §1.8 and Part 3 §3.4.4.]

Q78 hard Two signals are received, one at 5 microvolts and one at 50 microvolts, into the same impedance. The difference between them is: (a) 10 dB (b) 20 dB (c) 40 dB (d) 100 dB **Answer: (b).** These are voltages, so use $20 \log_{10}(50/5) = 20 \times 1 = 20 \text{ dB}$. In power terms the ratio is 100:1, which also comes to 20 dB — the two formulas agree, as they must, provided the impedance is the same at both points. If it is not, you must convert to power first.

A8. MEASUREMENT AND TEST EQUIPMENT

Syllabus: Annexure II §2.1 Section A (viii) — "Frequency Measurement: Measurement of frequency and use of simple frequency meters". Note: the separate "Measuring Instruments" topic (moving-coil and moving-iron meters) that existed in the pre-2010 Advanced-grade syllabus was deleted in the 2010 revision and should not be examined. [VERIFIED — comparison of the 2010 Annexure II with the pre-2010 syllabus]

Q79 easy The most accurate instrument for measuring the frequency of a transmitter is a: (a) absorption wavemeter (b) digital frequency counter (c) grid-dip oscillator (d) field strength meter **Answer: (b).** A frequency counter gates the incoming signal against a crystal time base — often oven-controlled or disciplined by satellite navigation — and reads out directly, to parts per million or better. An absorption wavemeter is a broad tuned circuit good only for confirming roughly which band you are on; a dip meter is similar in accuracy.

Q80 medium A grid-dip meter, or gate-dip meter, is used to: (a) measure transmitter power (b) find the resonant frequency of an unpowered tuned circuit or antenna (c) measure standing wave ratio (d) generate a modulated test signal **Answer: (b).** The instrument is a tunable

oscillator with a coupling coil. When its frequency matches the resonance of a nearby circuit, that circuit absorbs energy and the meter reading dips. It is the classic way to check where a home-made trap, tank circuit or antenna actually resonates before applying any power to it.

Q81 easy A dummy load, or artificial aerial, is used in order to: (a) increase transmitter output (b) allow the transmitter to be tested and adjusted at full power without radiating a signal (c) match the antenna (d) measure frequency **Answer: (b).** It is a non-inductive resistor, usually 50 ohms, that converts radio frequency into heat instead of radiation. Indian rules make this more than good manners: **test signals may not exceed 30 seconds, and for tests longer than that an artificial aerial must be used.** [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A, rule 9(4)(a) and (b)]

Q82 medium A standing-wave-ratio meter inserted in the feeder measures: (a) transmitter frequency (b) the ratio between forward and reflected power on the line (c) antenna gain (d) receiver sensitivity **Answer: (b).** A directional coupler samples the forward and reverse travelling waves separately; the instrument computes the ratio from them. Remember its blind spots: it tells you nothing about whether the power is being *radiated* — a dummy load reads 1:1 — and a lossy feeder masks a bad antenna by attenuating the reflection on its way back.

Q83 medium To measure the DC input current to a transmitter's final stage, an ammeter should be connected: (a) across the supply, in parallel (b) in series with the supply lead to that stage (c) between the antenna and the feeder (d) across the output tank circuit **Answer: (b).** An ammeter is a low-resistance instrument and must carry the current it measures, so it goes in series; a voltmeter is high-resistance and goes in parallel. Connecting an ammeter across a supply short-circuits it. For an Indian licensee this measurement is directly relevant, since the authorised power limit is expressed as **DC input to the final stage** — the product of the voltage at that stage and the current through it. [VERIFIED — Annexure-I, Note (iv)]

Q84 medium The Rules require every Indian amateur station to have: (a) an oscilloscope (b) a reliable frequency measuring device, checked each time the transmitter frequency is changed (c) a spectrum analyser (d) a calibrated dummy load **Answer: (b).** Annexure A rule 4(2) requires "a reliable frequency measuring equipment to verify, each time the frequency of the transmitter is changed". It is an anachronism in the era of synthesised handhelds with a digital display, but it is still in the current Rules. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A, rule 4(2)]

Q85 hard A field strength meter placed a fixed distance from an antenna is most useful for: (a) measuring absolute radiated power in watts (b) making *relative* comparisons — before and after an antenna adjustment, or in different directions (c) measuring frequency (d) measuring the DC input to the final stage **Answer: (b).** A simple field strength meter is uncalibrated, so its absolute reading is meaningless; its value lies entirely in comparison. Move the antenna, retune the matching network or rotate the beam, and watch whether the reading rises. Absolute field strength measurement requires a calibrated antenna and receiver.

B.2 GROUP C — INTERFERENCE AND ELECTROMAGNETIC COMPATIBILITY

This topic straddles both halves of the paper. On the Section A side the syllabus names "adjacent channel and image interference" under radio receivers (Annexure II §2.1 Section A (iv) 3). On the Section B side it names "Interference, measures against interference & tests — Article 15" of the ITU Radio Regulations, and the national non-interference condition sits in Annexure A rule 5 of the 2024 Rules. [VERIFIED — both instruments]

Q86 easy In the Q-code system, interference caused by other stations or by man-made electrical sources is reported as: (a) QRN (b) QRM (c) QRT (d) QSB **Answer: (b).** QRM means "Are you being interfered with?" or "I am being interfered with", and refers to man-made interference. QRN is the natural equivalent — "Are you troubled by atmospherics?", meaning static from lightning and similar sources. Both take a 1–5 severity number: 1 not at all, 5 very strongly. [VERIFIED — both codes are in the official thirty-code examined list, OM 2010-08-13, Annexure II; meanings REPORTED — NIAR Study Manual pp.122–123]

Q87 medium A transmitter operating on 7.050 MHz is found to be causing interference on 14.100 MHz. The most likely cause is: (a) an image response in the victim receiver (b) radiation of the second harmonic (c) cross-modulation (d) key clicks **Answer: (b).** 14.100 is exactly twice 7.050. Every non-linear stage generates harmonics at integer multiples of the fundamental, and the second harmonic is usually the strongest. The cure is a low-pass filter and adequate output-network selectivity at the transmitter. This is also a licence matter in India: the transmitter must be tuned so that **no energy is radiated outside the authorised bands**. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A, rule 4(1)]

Q88 medium "Splatter" — a signal audible several kilohertz either side of a single-sideband transmission — is normally caused by: (a) a faulty antenna (b) overdriving the transmitter, producing intermodulation distortion (c) an inaccurate frequency counter (d) high standing wave ratio **Answer: (b).** Driving a linear amplifier beyond its linear range creates intermodulation products that fall outside the intended 3 kHz channel. The fix is to reduce the microphone gain or drive level until the automatic level control just begins to act. On FM the analogous fault is over-deviation, which widens the signal beyond the channel in the same way.

Q89 medium A neighbour's television suffers interference only when you transmit. The most appropriate first steps are to: (a) increase your transmitter power to override it (b) confirm your own transmitter is clean, then fit a high-pass filter and ferrite chokes at the television and check its antenna downlead (c) change your call sign (d) transmit only at night **Answer: (b).** Television interference has two possible origins: spurious emissions from your transmitter, or inadequate immunity in the victim equipment, usually fundamental overload arriving on the mains lead or the antenna downlead. Establish which by testing into a dummy load. Note that a licence condition requires the amateur station to be built and worked so as **not** to cause interference. [VERIFIED — Annexure A, rule 5]

Q90 hard Cross-modulation in a receiver is: (a) two stations transmitting on the same frequency (b) the modulation of a strong unwanted signal being transferred onto the wanted signal by non-linearity in the receiver front end (c) the same as image interference (d) caused by the beat frequency oscillator **Answer: (b).** A strong nearby signal drives a radio-frequency or mixer

stage into its non-linear region, and the resulting mixing impresses that station's audio onto everything you tune. The tell-tale sign is that the intruding audio disappears when you switch in the receiver's attenuator — the cure is less front-end gain, better selectivity ahead of the mixer, or an attenuator, never more gain.

Q91 **medium** The definition of *harmful interference* in the ITU Radio Regulations turns on whether the interference: (a) is intentional (b) endangers the functioning of a radionavigation or other safety service, or seriously degrades, obstructs or repeatedly interrupts a radiocommunication service (c) exceeds a specified field strength (d) lasts more than 30 seconds

Answer: (b). Harmful interference is defined by its **effect**, not by the intent of the operator and not by any numeric threshold. That is why an entirely accidental spurious emission can still be harmful interference, and why "I did not mean to" is no answer to a monitoring station.

[VERIFIED — that interference is governed by Article 15 of the ITU Radio Regulations, Edition of 2024, and is a named syllabus item] **[UNCERTAIN]** — the definition itself sits at RR 1.169 in Article 1, which was **not** among the provisions fetched during the research. The wording above is the standard formulation, but check it against RR 1.169 before quoting it as verbatim treaty text.

Q92 **medium** Under the Indian licence conditions, if the Central Government or a land station asks an amateur to stop or restrict transmissions because of interference, the ASOC holder must: (a) comply (b) continue but reduce power by half (c) ignore the request unless it is in writing from a court (d) refer the matter to the ITU **Answer: (a).** Annexure A rule 5 requires the station to be designed, constructed, erected, maintained and worked so as not to cause interference, and to **discontinue or restrict transmissions on request**. There is no discretion in the text. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A, rule 5]

Q93 **medium** Before transmitting on a frequency, an Indian amateur is required to: (a) announce their call sign three times regardless of who else is present (b) take precautions to ensure their emissions will not interfere with transmissions already in progress, and if interference is likely, wait for an appropriate break (c) obtain permission from the nearest monitoring station (d) transmit a test carrier for 30 seconds **Answer: (b).** This is the "listen before transmit" rule, and in India it is an express licence condition, not merely etiquette. The Q-code for asking is **QRL?**, "Is this frequency in use?". [VERIFIED — Annexure A, rule 9(1)(a)]

Q94 **medium** Damped-wave emissions are: (a) permitted on HF only (b) permitted for General grade only (c) prohibited (d) required for distress calls **Answer: (c).** Damped waves — the spark-transmitter emissions of the earliest era, designated class B — occupy enormous bandwidth and are prohibited under the Indian licence conditions and internationally. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, Annexure A, rule 5]

Q95 **hard** An ASOC holder must not emit a carrier wave unless: (a) it is on an authorised frequency (b) it is subjected to intelligible modulation (c) the power is below 10 W (d) the station log has been signed **Answer: (b).** Annexure A rule 9(4)(c) states it directly: "The ASOC holder shall not emit carrier wave unless such carrier wave is subjected to intelligible modulation." An

unmodulated carrier left on the air occupies a channel while conveying nothing, and it makes the source hard to identify. The narrow exception is the permitted test signal — VVV plus the call sign, for no more than 30 seconds. [VERIFIED — Annexure A, rules 9(4)(a) and 9(4)(c)]

B.3 GROUP D — SAFETY

Read this heading carefully. Safety is **not an itemised topic in the ASOC syllabus**. Neither radio-frequency exposure nor mains safety nor antenna safety appears anywhere in Annexure II of WPC OM L-14011/255/2004-AMT dated 2010-08-13, and none of the ITU items in the Section B table covers them. [VERIFIED — the full text of Annexure II was read]

These nine questions are included because the reader of this report is about to buy and operate a transmitter, and because Q104 does rest on a real Indian regulatory requirement. Do not treat this group as examination preparation. If you are short of study time, skip to B.4.

Q96 easy Radio-frequency energy is described as *non-ionising* radiation because: (a) it cannot be detected by instruments (b) its photons lack the energy to strip electrons from atoms, so its principal biological effect is heating (c) it is always safe at any power level (d) it does not travel through air **Answer: (b)**. Unlike X-rays or gamma rays, radio-frequency photons cannot ionise matter and so cannot cause the direct genetic damage that ionising radiation does. That does **not** make it harmless: at sufficient power density radio frequency deposits heat in tissue, and the eyes and testes are particularly vulnerable because they are poorly cooled by blood flow. "Non-ionising" and "safe" are different statements.

Q97 medium If you double your distance from a radiating antenna, the power density falls to: (a) one half (b) one quarter (c) one eighth (d) it is unchanged **Answer: (b)**. In the far field, power spreads over the surface of a sphere whose area goes as r^2 , so power density follows the inverse square law. Distance is by far the cheapest and most effective protection: moving from 0.5 m to 2 m from an antenna cuts exposure by a factor of 16.

Q98 medium For a handheld transceiver, the single most effective everyday radio-frequency exposure precaution is to: (a) reduce the squelch setting (b) hold the radio so the antenna is away from your head and eyes, and use low power when it will do (c) use a longer antenna (d) transmit for longer periods less often **Answer: (b)**. A handheld's antenna is operated within centimetres of the user, in the near field where exposure is highest. Keeping the antenna away from the head and eyes, using a speaker-microphone, and dropping to the low-power setting when the contact does not need full power all reduce exposure directly. Duty cycle matters too — you transmit only a fraction of the time, and averaged exposure scales with that fraction.

Q99 medium Transmitting into no antenna at all, or into a badly mismatched load, risks: (a) nothing; the transmitter simply radiates less (b) damage to the final amplifier from reflected power and excessive dissipation (c) an improved signal report (d) reduced harmonic output **Answer: (b)**. Power that is not absorbed by the load is reflected back into the output stage,

where it appears as heat and as voltage stress on the devices. Most modern solid-state radios include protection that folds back power on high standing wave ratio, but it is not to be relied on. Always have an antenna or a dummy load connected before keying up.

Q100 medium Before working inside a mains-powered linear amplifier or power supply that has been switched off, you must: (a) wait ten seconds (b) disconnect it from the mains and confirm that the high-voltage capacitors have discharged (c) wear rubber gloves and proceed immediately (d) short the antenna socket **Answer: (b).** Reservoir and smoothing capacitors can hold a lethal charge for a long time after the supply is disconnected. Bleeder resistors are fitted to drain them, but they fail, and the only safe practice is to measure the voltage before touching anything. Switching off is not the same as disconnecting, and disconnecting is not the same as discharged.

Q101 medium In a modern Indian installation using the harmonised IEC colour coding, the earth conductor is: (a) red (b) black (c) green, or green with a yellow stripe (d) blue **Answer: (c).** Under the current harmonised code, brown is live, blue is neutral and green/yellow is earth. The older Indian domestic convention was red live, black neutral, green earth — and both are still found in service, so **verify with a meter rather than trusting the colour** in any installation you did not wire yourself.

Q102 medium The metal chassis of mains-powered radio equipment should be: (a) left floating (b) connected to the protective earth conductor (c) connected to the neutral conductor (d) connected to the antenna **Answer: (b).** Earthing the chassis means that a fault between a live conductor and the case draws a large fault current, which blows the fuse or trips the residual current device, rather than leaving the case live at 230 V waiting for someone to touch it. Connecting the chassis to *neutral* instead is dangerous, because a broken neutral would leave the chassis live.

Q103 medium The most serious hazard when erecting an outdoor antenna is: (a) radio-frequency exposure (b) contact between the antenna or mast and overhead power lines (c) static build-up (d) standing wave ratio **Answer: (b).** Antenna work kills people through electrocution and falls, not through radio frequency. The standard rule is to site any mast so that it cannot reach a power line even if it falls in the worst direction, and never to raise a mast alone. Add lightning protection and a proper earth, and disconnect the feeder during thunderstorms.

Q104 hard India's licence-exemption notifications for short-range devices deal with safety by: (a) specifying an exact maximum permitted specific absorption rate in the notification itself (b) requiring that safety-related requirements be met as per international or national standards such as ITU, ETSI, ANSI, BIS or ICNIRP (c) not mentioning safety at all (d) requiring an annual medical examination of the user **Answer: (b).** The de-licensing rules delegate rather than legislate the numbers. G.S.R. 1047(E) states: "The safety related requirements shall be as per the International or National standards such as ITU/ETSI/ANSI/BIS/ICNIRP for the respective devices and frequency bands." (ETSI is the European Telecommunications Standards Institute; ANSI the American National Standards Institute; BIS the Bureau of Indian Standards; ICNIRP the International Commission on Non-Ionizing Radiation Protection.) [VERIFIED — G.S.R. 1047(E) dated 2018-10-18; the same clause appears in G.S.R. 45(E), G.S.R. 1048(E) and G.S.R. 853(E)]

[UNCERTAIN] — no fetchable primary source stating India's numeric electromagnetic-field exposure limits for amateur or base-station use could be obtained during the research; the DoT EMF portal and the Telecommunication Engineering Centre pages were not retrievable. Do not quote a number without checking it.

B.4 The paper's SECTION B — RADIO REGULATIONS

50 marks of the paper — the same weight as the whole of radio theory, and the entire written paper for a candidate exempt from the paper's Section A. There is no exemption from Section B for anybody. [VERIFIED — official WPC sample question paper instruction block; OM 2010-08-13, Annexure II §1 Note]

Currency warning built into every question below. The syllabus still names "the Indian Wireless Telegraphs (Amateur Service) Rules, 1978 and amendments". **That instrument no longer exists.** It was superseded on 2024-10-29/30 by the **Telecommunications (Amateur Services) Rules, 2024**, G.S.R. 675(E), made under section 47 of the Telecommunications Act, 2023. Every answer below states the **2024** position. If an examiner marks against the 1978 text, the answers on grades, fees, validity and age will differ — flag it and cite the gazette. [VERIFIED — G.S.R. 675(E) preamble]

The **Indian Wireless Telegraph Rules, 1973** — syllabus item Section B(a)(i) — do survive, saved by section 61 of the Telecommunications Act, 2023. [VERIFIED — Telecommunications Act, 2023, s.61]

[UNCERTAIN] — no authoritative full text of the 1973 Rules could be located online during this research, which is a real gap in preparation for that one syllabus item.

B1. INDIAN LEGAL FRAMEWORK, GRADES, ELIGIBILITY, FEES AND VALIDITY

Q105 **easy** Indian amateur radio is currently governed by: (a) the Indian Wireless Telegraphs (Amateur Service) Rules, 1978 (b) the Telecommunications (Amateur Services) Rules, 2024 (c) the Indian Telegraph Act, 1885 alone (d) the ITU Radio Regulations directly **Answer: (b).** G.S.R. 675(E) dated 2024-10-29, published in the Gazette of India Extraordinary No. 617 on 2024-10-30, made under section 47 of the Telecommunications Act, 2023, expressly supersedes the 1978 Rules. Almost every club website and coaching page still describes the old regime. [VERIFIED — G.S.R. 675(E), preamble and gazette citation]

Q106 **easy** How many categories of Amateur Station Operator's Certificate does India currently issue? (a) One (b) Two — General and Restricted (c) Four — Advanced, Grade I, Grade II and Restricted (d) Five, including a Short Wave Listener licence **Answer: (b).** Rule 3(2) provides for exactly two: ASOC (General) and ASOC (Restricted). The four-grade structure in option (c) was abolished with effect from 2010-04-01, and the Short Wave Listener licence went with it — **receiving requires no licence in India today.** [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, r.3(2); WPC OM 2010-08-13, para 3 table]

Q107 **medium** Under the 2024 Rules, an ASOC holder additionally requires: (a) a separate Amateur Wireless Telegraph Station Licence (b) nothing further — the ASOC is itself the operating authorisation and carries the call sign (c) an annual renewal certificate (d) a police clearance certificate **Answer: (b).** This is the biggest structural change of 2024. Under the old

regime a licensee held both a certificate, won by examination, and a separate station licence. Rule 6(1) now issues the certificate "along with unique call signs for amateur station operators" — **one application, one fee, one document**. Any guidance telling you to apply separately for a station licence is describing the pre-2024 regime. [VERIFIED — r.3(1) and r.6(1)]

Q108 easy The call sign block reserved for the General grade is: (a) VU2 (b) VU3 (c) VU4 (d) AT1 **Answer: (a)**. VU2 is General, VU3 is Restricted. [VERIFIED — WPC OM L-14011/255/2004-AMT dated 2010-08-13, para 11: "The call-sign block VU2 shall be reserved for 'General' grade and block VU3 to 'Restricted' Grade for future licences."] Older call signs issued before the 2010 restructuring do not follow this split, so a VU2 call sign is not by itself proof of current grade.

Q109 medium The minimum age for the grant of an ASOC is: (a) 12 years (b) 14 years (c) 18 years, with an exception from 12 on production of a guardian's certificate (d) 21 years **Answer: (a)**. Rule 4(1)(a): "is not less than twelve years of age". The old 18-year default with a 12-to-18 "bona fide experimenter" exception requiring a certificate from a head of school or guardian was abolished in 2024 — twelve is now a flat floor with no guardian certificate required by the Rules. [VERIFIED — r.4(1)(a)]

Q110 medium Who may be granted an ASOC? (a) Any person resident in India (b) A citizen of India; a non-citizen only with administrative clearance of the Ministry of External Affairs (c) Any person of any nationality, without restriction (d) Indian citizens over 18 only **Answer: (b)**. Rule 4(1) grants the ASOC "to any person, being a citizen of India" meeting the age and examination conditions. The proviso to rule 4(1) and rule 4(2) allow a non-citizen holding an equivalent foreign certificate to apply, subject to Ministry of External Affairs administrative clearance and a fee of ₹1,000; such a certificate runs co-terminus with the visa or for twelve months, whichever is less. [VERIFIED — r.4(1), r.4(2), r.6(3)] **[UNCERTAIN]** — rule 4(3) allows the Government to specify a list of recognised countries; no such list could be found published.

Q111 easy The examination fee for the ASOC is: (a) ₹10 (b) ₹100 (c) ₹1,000 (d) ₹2,000 **Answer: (b)**. Rule 5(2): "An application under sub-rule (1) shall be accompanied by examination fee of one hundred rupees." That is per grade. Note that the application must be made **one month before the date of the examination**. [VERIFIED — r.5(1) and r.5(2)]

Q112 medium The fee for an ASOC valid for twenty years, and for one with lifetime validity, are respectively: (a) ₹100 and ₹200 (b) ₹500 and ₹1,000 (c) ₹1,000 and ₹2,000 (d) ₹2,000 and ₹5,000 **Answer: (c)**. Rule 6(1) sets ₹1,000 for twenty years and ₹2,000 for lifetime, identically for both grades. The Explanation to rule 6(1) defines "lifetime" as **until the ASOC holder attains the age of eighty years** — so it is not literally for life. [VERIFIED — r.6(1) and Explanation]

Q113 hard A candidate passes the ASOC examination but does not pay the certificate fee. After what period does the pass lapse? (a) Six months (b) One year (c) Two years from the date of declaration of results (d) It never lapses **Answer: (c)**. Rule 6(2): payment "shall be made within a period of two years from the date of declaration of results ... failing which no certificate shall be granted". The candidate must then sit the examination again. This is a genuine trap for someone who passes and then gets busy. [VERIFIED — r.6(2)]

Q114 medium An ASOC holder should apply for renewal: (a) at any time after expiry (b) at least twelve months before the date of expiry (c) on the day of expiry (d) within thirty days of expiry **Answer: (b).** Rule 8(1) requires the application "at least twelve months prior to the date of expiry". A late application is accepted up to **two years after expiry** on payment of a late fee of ₹1,000, and **no application is considered after two years from expiry**. A lifetime ASOC may be extended ten years at a time without further fee on request. [VERIFIED — r.8(1) and its two provisos, r.8(2)]

Q115 medium The pass criteria for the ASOC examination are: (a) 40 % in each part and 50 % overall for Restricted; 50 % in each part and 60 % overall for General (b) 40 % in each part and 40 % overall, for both grades (c) 50 % overall for both grades (d) 33 % overall **Answer: (b).** Rule 5(5) requires "a minimum of forty per cent of the total marks in respect of each category of ASOC", and DoT's own FAQ of 2024-12-05 spells it out as 40 % in both Part A and Part B and 40 % overall, for both grades. **Option (a) is the superseded 2010 figure**, and it is still printed by NIAR, SIARS, DARTS, RIRCOM and by DoT's own stale [FAQ_for_HAM.pdf](#), which remains live on the Saral Sanchar portal. The 2024 Rules are subordinate legislation and prevail over a 2010 executive memorandum. [VERIFIED — r.5(5); DoT WPC FAQ No. P-14036/02/2023-COP dated 2024-12-05, Q8]

Q116 medium The Restricted-grade written examination consists of: (a) 25 questions from each section, two marks each, one hour (b) 50 questions from each section, one mark each, two hours (c) 100 questions, three hours (d) 25 questions in total, one hour **Answer: (a).** The paper is a single 100-question paper — 50 in Section A, 50 in Section B. Restricted candidates answer **only the first 25 of each section at two marks each** in one hour; General candidates answer **all 50 of each at one mark each** in two hours. Both come to 100 marks. There is no separate, shorter Restricted paper, and there is no three-hour option — that belonged to the pre-2010 Advanced grade. [VERIFIED — official WPC sample question paper instruction block; corroborated by DoT/ Saral Sanchar [FAQ_for_HAM.pdf](#) Q5]

Q117 medium A Morse test is required: (a) for both grades (b) for the General grade only, at 8 words per minute in both sending and receiving (c) for the Restricted grade only (d) for neither grade — Morse was abolished in India **Answer: (b).** Morse has **not** been abolished in India, contrary to widespread belief. Rule 5(4) of the 2024 Rules preserves a Morse requirement for ASOC (General) "as may be specified", and DoT's December 2024 FAQ specifies a minimum of 8 words per minute in both transmission and reception. There is no Morse test for Restricted. [VERIFIED — r.5(4); DoT WPC FAQ dated 2024-12-05, Q8]

Q118 medium The language of the ASOC examination is: (a) English only (b) Hindi only (c) English or Hindi at the candidate's choice (d) any scheduled language **Answer: (a).** Rule 5(4) states flatly: "The language of such examination shall be English". There is no Hindi paper. [VERIFIED — r.5(4)]

Q119 hard Which candidate is exempt from the paper's Section A (Radio Theory and Practice)? (a) Any graduate (b) A holder of a degree in Engineering or Science, or a Diploma in Engineering, who has studied electronics or telecommunications (c) A holder of any existing amateur licence (d) Nobody — there is no exemption **Answer: (b).** The Note to Annexure II §1

grants the exemption, and DoT's Saral Sanchar FAQ Q12 lists the qualifications accepted: B.Tech or BE in Electronics or Telecommunication; B.Sc or M.Sc with Electronics or Telecommunication; Diploma in Engineering with Electronics or Telecommunication. Proof must be produced to the examiner on the day. For such a candidate the written paper reduces to Section B alone. [VERIFIED — OM 2010-08-13, Annexure II §1 Note; [FAQ_for_HAM.pdf](#) Q12 and Q16] **[UNCERTAIN]** — whether the exemption survives into any syllabus reissued under the 2024 Rules; no post-2024 syllabus has been published.

B2. AUTHORISED BANDS, EMISSIONS AND POWER

The single most important structural fact in this group: the **2024 Rules contain no band table, no emission list and no power figures at all** — the words "watt", "MHz" and "kHz" do not appear anywhere in them. Rule 7(1) and Annexure A rule 3 delegate all of it to be "specified by the Central Government". The operative specification remains **Annexure-I to WPC OM No. L-14011/255/2004-AMT dated 2010-08-13**, which WPC's own Central Public Information Officer confirmed by right-to-information reply dated 2020-12-28 is the authoritative list. [VERIFIED — r.7(1), Annexure A r.3, the 2010 OM, and WPC letter No. P-14033/02/2020-COP dated 2020-12-28] **[UNCERTAIN]** — no post-2024 replacement specification could be located. Ask WPC in writing to confirm the current table, citing rule 7(1).

Q120 **easy** The maximum power authorised to an ASOC (Restricted) holder on the 144–146 MHz band is: (a) 5 W (b) 10 W (c) 25 W (d) 50 W **Answer: (b).** 10 watts, and the same on 50–54 MHz and 434–438 MHz. This is exactly the figure a 10 W walkie-talkie needs, and it is available at the **lowest** grade of Indian amateur licence. [VERIFIED — WPC OM 2010-08-13, Annexure-I, Restricted Grade table]

Q121 **medium** The maximum power authorised to an ASOC (Restricted) holder on the HF bands is: (a) 10 W (b) 25 W (c) 50 W (d) 100 W **Answer: (c).** 50 W on all ten authorised HF bands. Two common errors to avoid: the claim that "Restricted grade is VHF-only" — it is not; it has ten HF bands — and the claim that Restricted grade gets 100 W, which it does not. [VERIFIED — Annexure-I, Restricted Grade table]

Q122 **medium** The maximum power authorised to an ASOC (General) holder on HF, and on 144–146 MHz, are respectively: (a) 400 W and 25 W (b) 400 W and 100 W (c) 1000 W and 50 W (d) 150 W and 25 W **Answer: (a).** General grade gets 400 W on HF and 25 W on 6 m, 2 m, 70 cm and the 5725–5840 MHz band. Note how modest the VHF and UHF figure is: upgrading from Restricted to General multiplies your HF allowance by eight but your VHF allowance only by two and a half. [VERIFIED — Annexure-I, General Grade table]

Q123 **hard** The power limits in Annexure-I are expressed as: (a) RF output power at the antenna socket (b) effective radiated power (c) total DC power input to the final stage of the transmitter (d) peak envelope power **Answer: (c).** Note (iv) to Annexure-I says so in terms: "DC input power is the total direct current power input to the final stage of the transmitter." A stage taking 10 W of DC and running at 60 % efficiency puts out about 6 W of radio frequency — so **a radio advertised as "10 W output" would exceed the Restricted limit.** [VERIFIED — Annexure-I,

Note (iv)] **[UNCERTAIN]** — how WPC measures or enforces this on a modern solid-state handheld, where the manufacturer does not publish DC input to the power-amplifier stage. This is a good question to put to WPC in writing.

Q124 medium Which classes of emission are authorised to Indian amateurs on 144–146 MHz? (a) A3E, H3E, J3E, R3E (b) F1B, F2B, F3E, F3C (c) A1A only (d) All classes **Answer: (b).** Both grades are limited on 6 m, 2 m and 70 cm to F1B, F2B, F3E and F3C — that is, frequency modulation and frequency-shift-keyed digital modes only. **Single sideband and amplitude modulation are not authorised on VHF or UHF in India for either grade**, which surprises visitors from countries where 2 m single sideband is routine. A handheld FM walkie-talkie is F3E and sits squarely inside the permitted set. [VERIFIED — Annexure-I, both grade tables]

Q125 hard A newly qualified ASOC (Restricted) holder wishes to work CW (A1A) on 40 metres. Under the Annexure-I table this is: (a) permitted, at 50 W (b) permitted, at 10 W (c) not among the emissions listed for the Restricted grade on HF (d) permitted only with General grade equipment **Answer: (c).** The Restricted-grade HF row lists only **A3E, H3E, J3E and R3E** — the amplitude and single-sideband telephony classes. A1A appears only in the "Remarks" column, which authorises it for **old Grade II licensees** migrated into the Restricted category, not for someone qualifying afresh. General grade gets A1A and A2A on HF as of right. [VERIFIED — Annexure-I, Restricted Grade table and its Remarks column]

Q126 medium Which of these bands is authorised to Indian amateurs? (a) 27.000–27.500 MHz (b) 14.000–14.350 MHz (c) 446.0–446.2 MHz (d) 88–108 MHz **Answer: (b).** 14000–14350 kHz is the 20 m amateur band and appears in Annexure-I for both grades. 26.957–27.283 MHz is the licence-exempt Citizen Band, not an amateur band; 446.0–446.2 MHz is licence-exempt PMR446 (Personal Mobile Radio 446 MHz, the term G.S.R. 1047(E) Table V itself uses), not an amateur band; 88–108 MHz is FM broadcasting. [VERIFIED — Annexure-I; G.S.R. 533(E) dated 2005-08-12 for Citizen Band; G.S.R. 1047(E) dated 2018-10-18 for PMR446]

Q127 medium The 70 cm band authorised to Indian amateurs is: (a) 420–450 MHz (b) 430–440 MHz (c) 434–438 MHz (d) 440–450 MHz **Answer: (c).** Just 4 MHz, from 434 to 438 MHz, and on a **secondary basis**. This is much narrower than the 420–450 MHz allocation familiar in ITU Region 2, so a radio programmed with an American band plan will happily transmit outside the Indian authorisation. [VERIFIED — Annexure-I, with footnote "@ On secondary basis as per the relevant provisions of Radio Regulations"]

Q128 hard In Annexure-I, the footnote symbol "\$" against a band means that authorisation is: (a) on a primary shared basis (b) on a non-interference and non-protection basis (c) on a secondary basis (d) for the General grade only **Answer: (b).** Annexure-I uses three footnote markers: * = primary shared basis, \$ = non-interference and non-protection basis, @ = secondary basis. Non-interference, non-protection is the weakest status of the three — you must not interfere with anyone, and you have no right to complain if someone interferes with you. [VERIFIED — Annexure-I footnotes, read verbatim]

Q129 medium The 5725–5840 MHz band is authorised: (a) to both grades at 25 W (b) to the General grade only, at 25 W (c) to the Restricted grade only (d) to neither grade **Answer: (b).** It appears only in the General Grade table, at 25 W and on a secondary basis, with the same F1B/F2B/F3E/F3C emission set. It is the only band in the whole table that is closed to Restricted holders. [VERIFIED — Annexure-I, General Grade table]

Q130 hard Amateur-satellite operation in India is: (a) permitted to both grades without limit (b) permitted to the General grade in the appropriate sub-bands, at a maximum effective isotropic radiated power of 30 dBW (c) prohibited (d) permitted only with a special event call sign **Answer: (b).** Note (i) to Annexure-I: "Amateur satellite service is permitted for General Grade in the appropriate sub bands in accordance with Radio Regulations and in those cases the maximum output RF power (eirp) is 30 dBw." 30 dBW is 1000 W effective isotropic radiated power — a limit on radiated power including antenna gain, quite unlike the DC-input limits elsewhere in the table. [VERIFIED — Annexure-I, Note (i)]

Q131 hard A person who wants a 10 W walkie-talkie without taking any examination should understand that: (a) 10 W is available licence-exempt on PMR446 (b) no Indian licence exemption permits 10 W for any two-way voice radio; the highest exempt figure is 5 W effective radiated power on Citizen Band, and PMR446 is limited to 500 mW effective radiated power (c) any radio under 10 W is exempt (d) an exemption applies to imported radios only **Answer: (b).** There is no licence-exempt route to 10 W in India. PMR446 (446.0–446.2 MHz) is capped at **500 mW effective radiated power**, hand-portable with an integral antenna, no base station and no repeater; Citizen Band (26.957–27.283 MHz) is capped at **5 W effective radiated power** with a built-in antenna. A 10 W walkie-talkie always requires an authorisation — amateur, captive or commercial. Separately, licence-exemption does **not** exempt the equipment: every exemption rule still requires WPC Equipment Type Approval. [VERIFIED — G.S.R. 1047(E) dated 2018-10-18; G.S.R. 533(E) dated 2005-08-12]

B3. LICENCE CONDITIONS, STATION RECORDS AND OPERATING PROCEDURE

All of this comes from Annexure A to the Telecommunications (Amateur Services) Rules, 2024, headed "Terms and conditions for the installation and operation of amateur station", invoked by rule 7(2)(b). It is the current replacement for the 1978 Rules that the syllabus still names, and it is the densest examinable text in the paper's Section B. [VERIFIED — G.S.R. 675(E), Annexure A]

Q132 medium The station log must be kept: (a) in any convenient form, including electronically (b) in a bound book, not loose-leaf (c) on the reverse of QSL cards (d) only for contest operation **Answer: (b).** Annexure A rule 6(1) requires "a chronological record of all transmissions emanating from or received at the amateur station ... in bound book (not loose-leaf)". The requirement survived into the 2024 Rules unchanged, and the Rules do not authorise an electronic log. [VERIFIED — Annexure A, r.6(1)] **[UNCERTAIN]** — whether WPC accepts an electronic log in practice.

Q133 medium All times entered in the station log must be stated in: (a) UTC (b) Indian Standard Time (c) the operator's local time (d) either UTC or IST, provided the choice is stated **Answer: (b).** Annexure A rule 6(2): "All times in the log shall be stated in the Indian Standard Time." This is an Indian departure from the international amateur convention of logging in UTC — note that

ITU RR 2.6 requires UTC for international radiocommunication activities generally, but the Indian *log* rule specifies IST. **This is the single sharpest conflict between the current Rules and the mass-market study material, which teaches UTC throughout (see A.4.5 and A.6.3).** Know both, and answer IST for the log. [VERIFIED — Annexure A, r.6(2); ITU RR 2.6]
[UNCERTAIN — which answer a 2026 examiner will mark correct.]

Q134 medium The station log must be preserved for: (a) three months from the date of the last entry (b) one year from the date of the last entry (c) five years (d) the life of the certificate
Answer: (b). Annexure A rule 6(4) requires preservation "for a period of one year from the date of last entry therein before it is destroyed". Rule 6(3) additionally requires that no gaps be left between entries and that entries be made **at the time** of receiving and transmitting, not written up afterwards. [VERIFIED — Annexure A, r.6(3) and r.6(4)]

Q135 medium Which of the following must the station log contain? (a) The date and time of each transmission, a summary of communications, the call signs of stations worked with times and emission types, and the times of opening and closing down (b) Only the call signs of stations worked (c) Only contacts made outside India (d) Only the operator's name and address **Answer: (a).** Annexure A rule 6(1) lists six items: date and time of each transmission; summary of communications exchanged; brief description of experiments and tests undertaken; call sign of the stations worked with times and type of emission; times of opening and closing down; and, for a portable or mobile station, the particulars of the temporary location. [VERIFIED — Annexure A, r.6(1)(a)–(f)]

Q136 medium An ASOC holder must send their call sign: (a) once at the start of each day's operation (b) at the beginning and end of each period of transmission, and repeated whenever the period exceeds ten minutes (c) every two minutes (d) only when requested by another station **Answer: (b).** Annexure A rule 9(1)(b) says exactly this and adds that the holder "shall not make transmission without identification or with false identification". The ITU requirement is more loosely worded — RR 25.9 says amateur stations "shall transmit their call sign at short intervals" — so the Indian ten-minute rule is the specific figure to remember. [VERIFIED — Annexure A, r.9(1)(b); ITU RR 25.9]

Q137 hard Under the Indian call and reply procedure, the call sign of the station being called may be sent: (a) once only (b) not more than three times (c) not more than five times (d) as many times as necessary **Answer: (b).** Annexure A rule 9(2)(a): the call consists of the called station's call sign **not more than three times**, the word DE in Morse or the words "This is" in telephony, and the calling station's call sign not more than three times. Rule 9(2)(c) adds that the call may be sent three times at two-minute intervals, after which it must not be repeated for ten minutes, during which the operator listens. Rule 9(1)(c) requires that prolonged calls be avoided. [VERIFIED — Annexure A, r.9(2)]

Q138 medium A general call to all stations is made using: (a) SOS (b) CQ in radiotelegraphy, and "Hello all stations" or CQ in radiotelephony (c) QRZ (d) VVV **Answer: (b).** Annexure A rule 9(2) (d) replaces the called station's call sign with the signal CQ, or with the words "Hello all

stations", in a general call. **QRZ** means "who is calling me?" and is a response, not a general call. **VVV** is the test signal. **SOS** is the distress signal and must never be used casually. [VERIFIED — Annexure A, r.9(2)(d)]

Q139 medium The end of a message and the end of work between two stations are signalled respectively by: (a) AR / "Over" and VA / "Out" (b) K / "Go ahead" and CL / "Closing" (c) SK / "Finish" and 73 (d) AS / "Wait" and NIL **Answer: (a).** Annexure A rule 9(3): a message is terminated by the signal **AR** in Morse or the word "**Over**" in telephony; the end of work between two stations is indicated by each of them with the signal **VA** in Morse or the word "**OUT**" — VA being spoken as "Victor Alfa". [VERIFIED — Annexure A, r.9(3)(a) and (b)]

Q140 medium Test signals from an amateur station must: (a) not exceed 30 seconds and must consist of a series of VVV followed by the station's call sign (b) not exceed five minutes (c) consist of an unmodulated carrier (d) be made only on 2 metres **Answer: (a).** Annexure A rule 9(4)(a). In radiotelephony the VVV series is replaced by the figures 1, 2, 3, 4 and so on, spoken in the figure code, still followed by the call sign. **For tests exceeding 30 seconds an artificial aerial must be used** — rule 9(4)(b). [VERIFIED — Annexure A, r.9(4)]

Q141 medium SACFA clearance for an amateur station is required: (a) always (b) never (c) if the antenna is installed on a tower, building or mast (d) only for General grade holders **Answer: (c).** Annexure A rule 4(3) requires clearance from SACFA — the Standing Advisory Committee on Frequency Allocation — "as applicable", and DoT's 2024 FAQ scopes it: clearance is required "if the amateur station antenna is installed on a tower, building, or mast". A handheld with a rubber-duck antenna needs none; a rooftop mast does. Application is made online through the portal. [VERIFIED — Annexure A, r.4(3); DoT WPC FAQ dated 2024-12-05, Q6 and Q7]

Q142 hard An ASOC holder wishes to operate from a holiday address for three weeks. They must: (a) apply for a change of location first (b) do nothing further, provided the area is not one restricted by the Government and the equipment is returned to the registered location afterwards (c) surrender the ASOC (d) obtain a special event call sign **Answer: (b).** DoT's 2024 FAQ Q1, Q3 and Q4 confirm that the Rules do not prohibit operating away from the address on the certificate, provided it is for a short period, outside government-restricted areas, and the equipment is restored to the original location afterwards. The FAQ fixes "short period" at **90 days or less**; beyond that a change of location must be applied for under rule 7(6), fee ₹1,000. [VERIFIED — DoT WPC FAQ dated 2024-12-05, Q1, Q3, Q4 and Q5; r.7(6)]

Q143 medium The fee for a duplicate ASOC, or for a change of the station's location, is: (a) ₹100 (b) ₹200 (c) ₹1,000 (d) free **Answer: (c).** Rule 7(6) sets ₹1,000 for either. **Trap:** NIAR's material and DoT's own stale [FAQ_for_HAM.pdf](#) still print the old figures of ₹200 for a change of location and ₹100 for a duplicate. Those were superseded in 2024. [VERIFIED — Telecommunications (Amateur Services) Rules, 2024, r.7(6)]

Q144 hard To form an amateur society (club station), the minimum number of ASOC holders required and the fee are: (a) two holders, ₹1,000 (b) four holders, ₹2,000 (c) ten holders, ₹5,000 (d) any number, free **Answer: (b).** Rule 12(1) requires a minimum of four ASOC holders and a fee of ₹2,000, with one holder named as **custodian**. The permission runs for twenty years or co-terminus with the custodian's own ASOC, whichever is less; changes of custodianship or

membership must be notified within thirty days; and the permission lapses if fewer than four of the original applicants remain. Separately, a **special event call sign** costs ₹200, must be applied for at least 90 days before the event, and is valid for 90 days from commencement (rule 13). [VERIFIED — r.12 and r.13]

B4. PERMITTED AND PROHIBITED TRANSMISSIONS

This is the group that most often surprises a new licensee, and it matters more than any other to anyone whose real aim is a pair of walkie-talkies. All of it is Annexure A rule 2 unless otherwise stated. [VERIFIED — G.S.R. 675(E), Annexure A, r.2]

Q145 medium The Rules define "amateur services" as radio communication for: (a) any lawful private purpose (b) self-training, intercommunication and technical investigations, carried out solely with a personal aim and without any pecuniary interest (c) emergency communication only (d) commercial trials of radio equipment **Answer: (b).** Rule 2(1)(c) of the 2024 Rules. Every restriction that follows — no business, no third-party traffic, no broadcasting — flows from those two phrases, "solely with a personal aim" and "without any pecuniary interest". This is also the ITU definition at RR 1.56. [VERIFIED — r.2(1)(c); ITU RR 1.56]

Q146 easy An amateur station may exchange radio communications with: (a) anyone with a radio (b) other authorised amateur stations (c) any station on the same frequency (d) commercial land mobile stations **Answer: (b).** Annexure A rule 2(1)(a) permits exchanges with "other authorised amateur stations" only, and not with amateur stations of countries whose administrations have notified an objection to such communications. Combined with the ban on third-party messages, **every party to the contact needs their own certificate.** [VERIFIED — Annexure A, r.2(1)(a) and r.2(2)(b)]

Q147 hard A licensed ASOC holder buys a second handheld so that an unlicensed family member can talk to them. This is: (a) permitted, because the licensee is present (b) permitted for emergencies only (c) not permitted — the other station must itself be an authorised amateur station, and third-party messages are prohibited (d) permitted if the family member does not give a call sign **Answer: (c).** This is the single commonest misconception among people buying a pair of walkie-talkies. Annexure A rule 2(1)(a) allows communication with authorised amateur stations only, and rule 2(2)(b) expressly bans third-party messages. The unlicensed person is operating an unauthorised station in their own right. If both people need to talk, both need an ASOC. [VERIFIED — Annexure A, r.2(1)(a) and r.2(2)(b)]

Q148 medium Which of the following may an ASOC holder transmit? (a) Recorded music to test audio quality (b) A relay of a news broadcast (c) An advertisement for a friend's shop (d) None of the above **Answer: (d).** Annexure A rule 2(2)(a) bans reproduction of broadcast programmes, tape recordings and transmissions of entertainment value or music; rule 2(2)(b) bans false or misleading calls, signals and news, advertisements, communications of business, statements on topics of political or industrial controversy, and third-party messages; rule 2(2)(c) bans indecent, obscene or offensive language or signals. [VERIFIED — Annexure A, r.2(2)]

Q149 **hard** May an amateur station be used to coordinate a small business, such as a warehouse or a security detail? (a) Yes, if the operators all hold ASOCs (b) Yes, at Restricted grade only (c) No — "communications of business" are expressly prohibited and the service definition excludes any pecuniary interest (d) Yes, with prior intimation to WPC **Answer: (c).** Two independent bars: the definition of amateur services in rule 2(1)(c) confines it to a personal aim without pecuniary interest, and Annexure A rule 2(2)(b) expressly forbids "communications of business". Holding certificates does not cure it — the *purpose* is what is prohibited. A business needs a captive or commercial authorisation, and under the 2026 captive-service rules those are open only to companies and government bodies, not to individuals. [VERIFIED — r.2(1)(c); Annexure A r.2(2)(b); Telecommunications (Authorisation for Captive Telecommunication Services) Rules, 2026, G.S.R. 511(E) dated 2026-06-23]

Q150 **hard** When may an ASOC holder handle third-party messages? (a) Never, under any circumstances (b) Whenever the message is not commercial (c) On failure of normal telecommunication facilities, if permitted by the Central or State Government, for calamity traffic originating from or addressed to a District Magistrate, Deputy Commissioner, Collector or an officer authorised by them (d) At the operator's discretion during any emergency **Answer: (c).** Annexure A rule 2(3) sets four cumulative conditions: normal telecommunication must have failed; government permission is required; the traffic must relate to a natural or manmade calamity; and it must originate from or be addressed to the named class of civil authority. Every element must be satisfied. This mirrors ITU RR 25.3, which permits third-party international traffic "only in case of emergencies or disaster relief". [VERIFIED — Annexure A, r.2(3); ITU RR 25.3]

Q151 **medium** If an ASOC holder accidentally receives a message they are not entitled to receive, they must: (a) publish it so others are warned (b) not disclose its contents, origin, destination, existence or receipt to anyone other than an officer duly authorised by the Central Government or a court of law (c) reply to the sender (d) log it and forward it to the nearest police station **Answer: (b).** Annexure A rule 8, the secrecy-of-correspondence condition, also forbids reproducing, copying or making any use of such a message. It is one of the oldest conditions in radio regulation and it applies whether the interception was deliberate or accidental. [VERIFIED — Annexure A, r.8]

Q152 **medium** During a natural calamity, at the request of the Central or a State Government, an ASOC holder: (a) may charge a reasonable fee for communication services (b) shall provide radio communication services in the amateur bands without any fee or charge (c) is exempt from all licence conditions (d) must surrender the certificate **Answer: (b).** Rule 7(7) creates a positive duty, new in 2024: the holder "shall provide radio communication services in amateur services frequency band, at the request of the Central Government, or a State Government, or any officer specially authorised in this behalf ... during natural calamities and disasters, **without any fees or charge**". [VERIFIED — r.7(7)]

B5. PENALTIES AND ENFORCEMENT

A structural caution before you answer these. India is mid-transition. The Telecommunications Act, 2023 has been commenced only in parts. Sections 42–44 (offences and search) came into force on 2024-06-26 and section 3(1) and 3(6) on 2026-06-23, but **Chapter VIII (sections 31–41), which builds the civil-penalty machinery including the Third Schedule, appears not to have been notified**, and **section 60 (the repeal of the 1885 and 1933 Acts) appears not to have been commenced** — WPC was still making rules under the old Acts as recently as G.S.R. 466(E) of 2026-06-10. So the old criminal provisions and the new civil ones may both be live. [VERIFIED — the 2026 gazette recital; REPORTED — commencement analysis by Trilegal and others; UNCERTAIN — no definitive gazette commencement index was obtained]

Q153 medium Possessing a wireless transmitter without a licence is an offence under section 6 of the Indian Wireless Telegraphy Act, 1933, punishable with: (a) a fine of ₹100 only (b) imprisonment up to three years, or a fine up to ₹1,000, or both (c) imprisonment up to seven years (d) confiscation only **Answer: (b).** Section 6(1-A) covers apparatus "capable of being used for transmission" and carries up to three years' imprisonment, a fine up to ₹1,000, or both. Section 6(1), which covers non-transmitting wireless apparatus, carries only a fine of up to ₹100 for a first offence and ₹250 thereafter. **The money is trivial; the exposure is a criminal record and a custodial term.** The ₹1,000 ceiling dates from 1933 and has never been raised. [VERIFIED — Indian Wireless Telegraphy Act, 1933, s.6]

Q154 medium Working an unauthorised wireless telegraph is punishable under section 20(1) of the Indian Telegraph Act, 1885 with: (a) a fine of ₹50 only (b) imprisonment up to three years, or a fine with no stated ceiling, or both (c) a warning (d) suspension of the ASOC only **Answer: (b).** Section 20(1) distinguishes wireless from wireline: for a **wireless** telegraph the fine has **no ceiling**, in contrast to the ₹1,000 cap for wireline. In practice courts have imposed small sums — ₹100 to ₹500 in the reported cases — but the statutory exposure is open-ended and includes three years' imprisonment. [VERIFIED — Indian Telegraph Act, 1885, s.20(1)]

Q155 hard Under the Third Schedule to the Telecommunications Act, 2023, possessing radio equipment without an authorisation or an exemption attracts: (a) three years' imprisonment (b) a civil penalty up to ₹50,000 for a first offence and up to ₹2 lakh for each subsequent instance (c) a fine of ₹2 crore (d) no penalty **Answer: (b).** Third Schedule, Sl. No. 1(a). Parliament deliberately routed ordinary possession into the **civil** track rather than the criminal one — the entry's own carve-out ("except for the offence under clauses (d) and (f) of sub-section (3) of section 42") makes that explicit. Note that section 57 allows the Schedules to be amended by notification, subject only to a ceiling of ₹10 crore. [VERIFIED — Telecommunications Act, 2023, Third Schedule Sl. 1(a) and s.57] **[UNCERTAIN]** — whether Chapter VIII, which operates the Third Schedule, has been notified.

Q156 medium Section 42(1) of the Telecommunications Act, 2023 makes unauthorised provision of telecommunication services, or unauthorised network establishment, punishable with: (a) a civil penalty of ₹50,000 (b) imprisonment up to three years, or a fine up to ₹2 crore, or both (c) imprisonment up to seven years (d) a warning for a first offence **Answer: (b).** This is the modern successor to Telegraph Act s.20(1), and the fine ceiling has jumped from nothing-stated

to ₹2 crore. Sections 42–44 were among those commenced on 2024-06-26, so they are live. [VERIFIED — Telecommunications Act, 2023, s.42(1); REPORTED — the 2024-06-26 commencement list]

Q157 **medium** An ASOC may be suspended or cancelled for: (a) any reason, without notice (b) breach of the terms of the ASOC, breach of the ITU Convention or Radio Regulations, or wilfully furnishing false information — and only after a reasonable opportunity to be heard (c) failing to make 40 contacts in a year (d) changing address **Answer: (b)**. Rule 9 lists the grounds and requires a hearing: "no order ... shall be made unless the ASOC holder has been given a reasonable opportunity of making a representation and being heard." Rule 10 provides that no fees are refunded on suspension or cancellation, and rule 11(1) makes the ASOC non-transferable. The 40-contacts-a-year renewal condition of the earlier Rules was **dropped** in 2024. [VERIFIED — r.9, r.10, r.11(1)]

Q158 **medium** An authorised officer arriving at an amateur station may require the holder to: (a) cease operating permanently (b) produce the certificate, the station log and other records, and permit inspection, examination or testing of the apparatus (c) surrender the equipment on the spot (d) nothing, without a court order **Answer: (b)**. Rule 14 of the 2024 Rules obliges every ASOC holder to produce records and equipment for inspection, and Annexure A rule 10 repeats it for the certificate and the station log. This is why the log and the equipment register are not paperwork for its own sake — they are the documents you will be asked for. [VERIFIED — r.14; Annexure A, r.10]

Q159 **hard** The equipment register that an ASOC holder must maintain records, among other things: (a) only the radios currently owned (b) the make, model, type and serial number of each apparatus, from whom it was received or that it is self-made, the date of receipt or assembly, and details of any sale or transfer including the purchaser's certificate particulars (c) only equipment purchased in India (d) nothing — no register is required **Answer: (b)**. Annexure A rule 7(4) prescribes a nine-column register. The transfer columns matter: you must record to whom you sold or transferred equipment, when, and the particulars of the certificate issued in the purchaser's name — which in effect means you should not transfer transmitting equipment to someone who holds no certificate. [VERIFIED — Annexure A, r.7(4)]

Q160 **hard** Which statement about importing amateur radio equipment is correct for an ASOC holder? (a) A separate WPC import licence is required for every transceiver (b) No separate authorisation for the import of amateur radio equipment is required (c) Import is prohibited (d) Import requires SACFA clearance **Answer: (b)**. Rule 7(3) states it directly: "An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment." This codifies what WPC had already confirmed by right-to-information reply in 2019. **But note the limit of that statement:** it removes the WPC *import* requirement only. It does not resolve Equipment Type Approval, customs, or the legality of a radio capable of transmitting outside the amateur bands — which is the trap that catches cheap imported handhelds. [VERIFIED — r.7(3); WPC right-to-information reply dated 2019-12-04]

B6. ITU RADIO REGULATIONS

The syllabus specifies exactly ten ITU topics, each mapped to an Article or Appendix. All ten mappings were re-verified against the ITU Radio Regulations, Edition of 2024 and all are still correct, despite the syllabus citing the 2008 edition. [VERIFIED]

Syllabus topic	ITU provision
Designation of emission	Appendix 1
Phonetic alphabet and figure code	Appendix 14
Nomenclature of frequency and wavelength	Article 2
Frequency allocation for amateur services	Article 5
Interference, measures against interference and tests	Article 15
Identification of stations	Article 19
Distress and urgency signal, call and message transmissions	Articles 30, 31, 32, 33
Amateur station	Article 25
Call sign series allocated to India	Appendix 42

Q161 easy The ITU Radio Regulations Article dealing specifically with the amateur services is: (a) Article 2 (b) Article 15 (c) Article 19 (d) Article 25 **Answer: (d).** Article 25 is headed "Amateur services", with Section I on the amateur service and Section II on the amateur-satellite service. Article 2 is nomenclature, Article 15 is interference, Article 19 is identification of stations. [VERIFIED — ITU RR, Edition of 2024, Vol. 1, Article 25]

Q162 medium Under ITU RR 25.7, the maximum power of amateur stations is: (a) fixed at 1500 W worldwide (b) fixed by the administrations concerned (c) unlimited (d) fixed by the IARU **Answer: (b).** RR 25.7 reads: "The maximum power of amateur stations shall be fixed by the administrations concerned." This is why India's limits — 50 and 400 W on HF, 10 and 25 W on VHF and UHF — can differ so sharply from those in other countries: there is no international figure. It is also why you cannot cite a foreign limit as authority for operating at that power in India. [VERIFIED — ITU RR 2024, 25.7]

Q163 hard Under ITU RR 25.2A, transmissions between amateur stations of different countries: (a) may be encoded provided both operators agree (b) shall not be encoded for the purpose of obscuring their meaning, except for control signals between earth command stations and amateur-satellite space stations (c) must always be in Morse code (d) must be in English **Answer: (b).** The prohibition is on codes intended to **obscure meaning**. Standard published codes — Q-codes, the RST system, digital protocols with public specifications such as radioteletype or PSK31 — are not obscuring anything and are entirely proper. Encryption is. [VERIFIED — ITU RR 2024, 25.2A]

Q164 medium According to ITU Article 2, the frequency range 3–30 MHz is designated: (a) MF, hectometric waves (b) HF, decametric waves (c) VHF, metric waves (d) UHF, decimetric waves **Answer: (b).** Band 7, HF, 3–30 MHz, decametric waves — so called because the wavelengths run from 100 m down to 10 m. Learn the metric names alongside the letter abbreviations: MF is hectometric, VHF metric, UHF decimetric, SHF centimetric, EHF millimetric. [VERIFIED — ITU RR 2024, Article 2, table at RR 2.1]

Q165 medium ITU RR 2.6 requires that time in international radiocommunication activities be expressed as: (a) local time, in twelve-hour format (b) UTC, as a four-digit group from 0000 to 2359 (c) Indian Standard Time (d) sidereal time **Answer: (b).** RR 2.6 also directs that "the abbreviation UTC shall be used in all languages". Note the contrast with the Indian **station log**, which Annexure A rule 6(2) requires to be kept in IST. IST = UTC + 5:30, so 1200 UTC is 1730 IST. [VERIFIED — ITU RR 2024, 2.6; Annexure A r.6(2)]

Q166 hard In the emission designator **3K00A3E**, the "3K00" indicates: (a) a frequency of 3 kHz (b) a necessary bandwidth of 3000 Hz (c) a power of 3 kW (d) three channels **Answer: (b).** Under RR Appendix 1, necessary bandwidth is written as three numerals and one letter, the letter occupying the position of the decimal point and giving the unit: H for hertz, K for kilohertz, M for megahertz, G for gigahertz. So 3K00 = 3.00 kHz = 3000 Hz, and 12K5 = 12.5 kHz. **The bandwidth goes immediately before the class symbols**, so "A3E3K00" is wrong — even though it appears that way in the official WPC sample paper. [VERIFIED — ITU RR 2024, Vol. 2, Appendix 1, §2]

Q167 medium In the emission designator **J3E**, the three symbols mean respectively: (a) single-sideband suppressed carrier; single channel containing analogue information; telephony (b) frequency modulation; digital; telegraphy (c) amplitude modulation; two channels; television (d) pulse modulation; no modulating signal; data **Answer: (a).** The first symbol is the type of modulation of the main carrier (J = single sideband, suppressed carrier), the second the nature of the modulating signal (3 = single channel containing analogue information), the third the type of information (E = telephony). Applying the same rules, **F3E** is FM telephony — what a walkie-talkie uses — and **A1A** is double-sideband, single channel digital without sub-carrier, telegraphy for aural reception, that is, ordinary CW. [VERIFIED — ITU RR 2024, Vol. 2, Appendix 1, §5 and the three symbol tables]

Q168 medium The international call sign series allocated to India by ITU Appendix 42 are: (a) VU2 and VU3 only (b) ATA–AWZ, VTA–VWZ and 8TA–8YZ (c) VU only (d) AT, VT and 8T only **Answer: (b).** Three blocks, and India's amateur call signs (VU2xxx, VU3xxx) sit inside the VTA–VWZ block. **Note a defect in the official WPC sample paper:** its option (c) prints "8TA – 8WZ", which is wrong — the ITU allocation runs to **8YZ**. [VERIFIED — ITU RR 2024, Vol. 2, Appendix 42]

Q169 hard ITU RR 19.69 provides that the prohibition on using the digits 0 and 1 in a call sign: (a) applies strictly to amateur stations (b) does not apply to amateur stations (c) applies only to stations in Region 1 (d) applies only to satellites **Answer: (b).** RR 19.68 sets the general amateur call-sign format — one or two characters plus a single digit other than 0 or 1, followed by a group of not more than four characters ending in a letter — and then RR 19.69 immediately

disapplies the 0/1 restriction for amateur stations. That is what makes call signs such as VU2 and VU3, and elsewhere W1 and K0, possible. RR 19.68A permits longer call signs on special occasions. [VERIFIED — ITU RR 2024, Vol. 1, Article 19, 19.68, 19.68A, 19.69]

Q170 **medium** Which ITU Region does India belong to? (a) Region 1 (b) Region 2 (c) Region 3 (d) India spans two regions **Answer: (c).** Region 1 is Europe, Africa, the Middle East west of the Persian Gulf and northern Asia; Region 2 is the Americas; Region 3 is most of Asia and Oceania, including India. The distinction matters because the amateur allocations in ITU Article 5 differ between regions — the 40 m and 80 m band edges in particular. India is also in **IARU** Region 3, which is a separate, voluntary organisation. [REPORTED — the three-region division is set by ITU RR Article 5 and India's placement in Region 3 is not in dispute, but Article 5's regional boundary text was not itself fetched during this research]

Q171 **medium** ITU RR 25.9B provides that whether a visiting foreign amateur may operate while temporarily in a country is: (a) automatic under the Radio Regulations (b) a matter for the administration of the country being visited, which may impose conditions or restrictions (c) decided by the IARU (d) never permitted **Answer: (b).** Reciprocal operating is permissive, not automatic. In India the corresponding provision is rule 4(2) of the 2024 Rules, which requires Ministry of External Affairs administrative clearance and a ₹1,000 fee, and rule 6(3), which limits such a certificate to the visa duration or twelve months, whichever is less. [VERIFIED — ITU RR 2024, 25.9B; 2024 Rules r.4(2) and r.6(3)]

Q172 **hard** ITU RR 25.5 provides that whether an amateur must demonstrate Morse ability is: (a) mandatory worldwide (b) abolished worldwide since WRC-03 (c) for each administration to determine (d) determined by the licence grade **Answer: (c).** RR 25.5 reads: "Administrations shall determine whether or not a person seeking a licence to operate an amateur station shall demonstrate the ability to send and receive texts in Morse code signals." The 2003 World Radiocommunication Conference removed the *international mandate*, and most administrations then dropped Morse — but **India did not**, and 25.5 is the provision that lets it keep the requirement for the General grade. RR 25.6 separately requires administrations to verify operational and technical qualifications. [VERIFIED — ITU RR 2024, 25.5 and 25.6]

B7. INTERNATIONAL OPERATING PRACTICE

Syllabus: Annexure II §2.1 Section B (c) standard frequency and time signal services, and (d) the closed list of 30 Q-codes and 17 abbreviations; plus the distress and urgency material at Articles 30–33 and the phonetic alphabet at Appendix 14. The RST system, the HX/H24/HJ/HN designators and repeater practice are not named in the syllabus but appear in the official sample paper and in every study manual.

Q173 **easy** QTH means: (a) What is the correct time? (b) What is your position or location? (c) Shall I stop sending? (d) Who is calling me? **Answer: (b).** QTH is position or location. The near neighbours worth learning together are **QTR** (correct time), **QTC** (how many messages have you to send), **QRA** (name of your station) and **QRG** (exact frequency). [Q-code list VERIFIED — OM 2010-08-13, Annexure II; meanings REPORTED — NIAR Study Manual pp.122–123]

Q174 medium Which of these Q-codes appears in the **official** examined list? (a) QRO (b) QRQ (c) QRP (d) QST **Answer: (b). QRQ**, "shall I send faster?", is in the official thirty-code list. QRO, QRP, QRB, QSP, QST, QTA and QSK are **not**, and neither are the numeric codes 73 and 88. This is a genuine trap: NIAR, SIARS and DARTS all reproduce QRO in place of QRQ, a shared transcription error that has been copied across the Indian study material for years, and NIAR's Q-code table then defines QRO and omits QRQ entirely. **Learn both meanings** — QRQ send faster, QRO increase power — since QRO is what the mass-market material teaches. [VERIFIED — OM 2010-08-13, Annexure II, re-read from a 220 dots-per-inch render of the scan]

Q175 medium A station sends "QRZ?". This means: (a) I am closing down (b) Who is calling me? (c) Shall I increase power? (d) Your signal is fading **Answer: (b). QRZ** asks who is calling. Distinguish it from **QRL?** (is the frequency in use, or are you busy), **QSB** (your signal is fading), **QRT** (stop sending) and **QRU** (have you anything for me, or I have nothing for you). A Q-code with a question mark is the question form; without it, the answer or instruction form. [Official list VERIFIED; meanings REPORTED — NIAR Study Manual]

Q176 medium In the ITU phonetic alphabet, the letters Q and V are spoken as: (a) Queen and Victory (b) Quebec and Victor (c) Quick and Violet (d) Quaker and Vienna **Answer: (b).** Appendix 14 fixes the twenty-six code words — Alfa, Bravo, Charlie, Delta, Echo, Foxtrot, Golf, Hotel, India, Juliett, Kilo, Lima, Mike, November, Oscar, Papa, Quebec, Romeo, Sierra, Tango, Uniform, Victor, Whiskey, X-ray, Yankee, Zulu. Note the ITU spellings "Alfa" and "Juliett", which differ from ordinary English. A station called VU3ABC identifies as "Victor Uniform Three Alfa Bravo Charlie". [VERIFIED — ITU RR 2024, Vol. 2, Appendix 14 (REV.WRC-23), §1]

Q177 hard In the ITU figure code of Appendix 14, the digit 9 is spoken as: (a) Niner (b) Novenine (c) Nine (d) Nadanine **Answer: (b).** The ITU figure code is a separate table from the letter table: Nadazero, Unaone, Bissotwo, Terrathree, Kartefour, Pantafive, Soxisix, Setteseven, Oktoeight, Novenine, plus "Decimal" for the decimal point and "Stop" for the full stop. ITU prints them as single words; NIAR prints them as two ("Nova Nine") and omits Decimal and Stop; the WPC sample paper uses two-word forms. Accept either spacing. "Niner" is the aviation convention, not the ITU one. [VERIFIED — ITU RR 2024, Appendix 14, §2]

Q178 easy The distress signal in radiotelephony is: (a) PAN PAN (b) SECURITE (c) MAYDAY, spoken three times (d) SOS, spoken three times **Answer: (c).** MAYDAY, from the French *m'aider*, is the spoken distress signal; SOS is its Morse equivalent. Learn the full hierarchy: **distress** MAYDAY / SOS has absolute priority over everything; **urgency** PAN PAN / XXX comes second; **safety** SÉCURITÉ / TTT comes third. All stations hearing a distress call must immediately stop any transmission liable to interfere, keep listening, and render assistance if they can. [Priorities and signals REPORTED — NIAR Study Manual p.124; underlying provisions VERIFIED — ITU RR 2024, Articles 30–33]

Q179 medium The urgency signal in radiotelegraphy is: (a) SOS (b) XXX sent three times (c) TTT sent three times (d) VVV **Answer: (b).** XXX, spoken PAN PAN, is urgency — a very urgent message concerning the safety of a station or a person, short of imminent danger. TTT, spoken

SÉCURITÉ, is the safety signal, used for navigational and meteorological warnings. VVV is not a priority signal at all; it is the **test** signal. [REPORTED — NIAR Study Manual p.124; ITU RR Articles 30–33 VERIFIED]

Q180 medium In the RST reporting system, the three figures represent: (a) Range, Signal, Time (b) Readability, Strength, Tone (c) Reception, Sensitivity, Transmission (d) Rate, Speed, Timing **Answer: (b).** Readability runs 1–5, Signal strength 1–9 and Tone 1–9. **Voice contacts use only R and S;** tone applies to a keyed carrier, so it is reported on CW only. A report of "59" on single sideband means perfectly readable, extremely strong signals. [REPORTED — NIAR Study Manual p.125]

Q181 hard In the RST system, a signal strength of 9 means: (a) strong signals (b) extremely strong signals (c) moderately strong signals (d) fair signals **Answer: (b).** The scale runs 1 faint or barely perceptible, 2 very weak, 3 weak, 4 fair, 5 fairly good, 6 good, 7 moderately strong, **8 strong, 9 extremely strong.** **Trap:** NIAR's own exercise answer key on page 134 says S9 means "strong signals", which contradicts NIAR's own RST table on page 125. The table is right. [REPORTED — NIAR Study Manual p.125; the internal contradiction is documented at A.4.2 above]

Q182 medium Which of the following is in the official examined list of abbreviations? (a) BK (b) 73 (c) AR (d) OM **Answer: (c).** The official seventeen are **AA, AB, AR, AS, C, CFM, CL, CQ, DE, K, NIL, OK, R, TU, VA, WA, WB.** AR is the ending signal for a message. BK (break), 73 (best regards), 55 (good luck), OM (old man) and YL (young lady) are all taught and widely used but are **not** in the syllabus list — although BK does appear in the official WPC sample paper, which is a genuine inconsistency in the official material. Learn the seventeen, and know the common extras anyway. [VERIFIED — OM 2010-08-13, Annexure II]

Q183 hard In Morse code timing as taught for the Indian examination, the space between two words is: (a) one unit (b) three units (c) five units (d) seven units **Answer: (c).** The timing framework is: dot = 1 unit, dash = 3 units, space between elements of a character = 1 unit, space between letters = 3 units, **space between words = 5 units. This diverges from the international standard of seven units,** and NIAR's own manual states seven on page 137 and five on page 139. The official WPC sample paper offers "five dots" as the answer, so **answer five in the examination** and know that seven is the international figure. [VERIFIED — WPC sample paper, the paper's Section B Q9; NIAR Study Manual p.139]

Q184 hard The Morse receiving test for ASOC (General) consists of a plain-language passage of: (a) 125 characters in three minutes (b) 200 characters, sent for five consecutive minutes at 8 words per minute (c) 300 characters in five minutes at 12 words per minute (d) 500 characters in ten minutes **Answer: (b).** The arithmetic checks: 8 words per minute × 5 minutes × 5 characters per word = 200 characters. Each figure and each punctuation mark counts as **two** characters. More than five errors disqualifies, but a candidate who receives any one continuous minute without error is declared successful. The sending test is at not less than 8 words per minute for five consecutive minutes, and a candidate must pass receiving and sending together. Options (a) and (c) are the pre-2010 Grade II and Grade I figures. [VERIFIED — OM 2010-08-13, Annexure II, Part II]

Q185 **hard** The IARU Region 3 band plan is: (a) legally binding on Indian amateurs (b) a voluntary agreement on mode segments, not law, and not part of the ASOC syllabus (c) issued by WPC (d) the source of India's power limits **Answer: (b)**. The International Amateur Radio Union is a non-governmental federation of national amateur societies. Its band plans are gentlemen's agreements about which parts of a band to use for which mode, and they carry no legal force in India. What binds an Indian amateur is Annexure-I to the 2010 WPC Office Memorandum for bands, emissions and power, and Annexure A to the 2024 Rules for operating conditions. Following the IARU plan is good practice and good manners; confusing it with law is an error. [REPORTED — IARU Region 3 Interim Band Plan; VERIFIED that it does not appear anywhere in Annexure II of the syllabus]

B.5 Practice-question answer key at a glance

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	38	c	75	d	112	c	149	c
2	c	39	c	76	c	113	c	150	c
3	c	40	b	77	b	114	b	151	b
4	b	41	b	78	b	115	b	152	b
5	c	42	b	79	b	116	a	153	b
6	b	43	b	80	b	117	b	154	b
7	b	44	a	81	b	118	a	155	b
8	c	45	b	82	b	119	b	156	b
9	b	46	c	83	b	120	b	157	b
10	a	47	c	84	b	121	c	158	b
11	b	48	b	85	b	122	a	159	b
12	b	49	b	86	b	123	c	160	b
13	b	50	b	87	b	124	b	161	d
14	c	51	c	88	b	125	c	162	b
15	c	52	b	89	b	126	b	163	b
16	c	53	b	90	b	127	c	164	b
17	b	54	a	91	b	128	b	165	b
18	d	55	b	92	a	129	b	166	b
19	b	56	b	93	b	130	b	167	a
20	b	57	b	94	c	131	b	168	b

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
21	b	58	c	95	b	132	b	169	b
22	c	59	c	96	b	133	b	170	c
23	b	60	b	97	b	134	b	171	b
24	b	61	b	98	b	135	a	172	c
25	b	62	b	99	b	136	b	173	b
26	b	63	c	100	b	137	b	174	b
27	b	64	b	101	c	138	b	175	b
28	c	65	b	102	b	139	a	176	b
29	c	66	b	103	b	140	a	177	b
30	b	67	b	104	b	141	c	178	c
31	b	68	b	105	b	142	b	179	b
32	b	69	b	106	b	143	c	180	b
33	b	70	b	107	b	144	b	181	b
34	b	71	b	108	a	145	b	182	c
35	b	72	b	109	a	146	b	183	c
36	b	73	b	110	b	147	c	184	b
37	b	74	c	111	b	148	d	185	b

The heavy preponderance of (b) is an artefact of how these questions were drafted, not a hint about the real paper. Do not read anything into it, and do not let it train you to guess (b).

5.3 Critical uncertainties a reader must carry away from this part

#	Uncertainty	Status	What to do
1	No official past paper or answer key exists for the post-2010 ASOC. Every "answer" attached to a documented question in Part 5A is a derivation, an institute's answer or a club's answer — never the Government's.	[VERIFIED negative]	Treat all keys as informed opinion. Where a key conflicts with the Rules, follow the Rules.

#	Uncertainty	Status	What to do
2	Log timekeeping: IST or UTC. The 2024 Rules say IST verbatim; every study manual and club paper says UTC.	[VERIFIED conflict]	Keep your own log in IST — that is what the Rules require. Expect that an examiner using stale material may key UTC.
3	Pass mark: 40/40 or 50/60. The 2024 Rules and the December 2024 DoT FAQ say 40 % per part and 40 % overall for both grades. NIAR, SIARS, DARTS, RIRCOM and DoT's own stale FAQ still print 50 %/ 60 %.	[VERIFIED conflict; the Rules prevail]	Aim well above 50 % and the conflict never bites you.
4	Q-code list: QRQ or QRO. The official examined list has QRQ; the mass-market material prints QRO.	[VERIFIED conflict]	Learn both meanings.
5	Morse word spacing: five units or seven. The Indian material and the official sample paper teach five; the international standard is seven.	[VERIFIED conflict]	Answer five in the hall.
6	The 30 metre band, 10100–10150 kHz. One research thread records that it is absent from Annexure-I of the 2010 OM for both grades; another reproduces NIAR's table, which shows it.	[UNCERTAIN]	Do not transmit on 30 m until WPC confirms in writing, citing rule 7(1).
7	The bands, emissions and power table rests on a 2010 executive memorandum issued under Rules that were superseded in 2024, and no replacement specification has been published.	[UNCERTAIN]	Confirm the current authorised bands, emissions and power for your grade with WPC in writing before transmitting.
8	The syllabus has not been reissued since 2010 and still names the superseded 1978 Rules, even though rule 5(3) of the 2024 Rules obliges the Government to publish the syllabus on the portal.	[UNCERTAIN]	Study the 2010 Annexure II, but answer regulations questions from the 2024 Rules and be ready to cite the gazette.
9	Indian telecom law is mid-transition. The Telecommunications Act, 2023 has been commenced only in parts; the old criminal provisions and the new civil ones may both be live.	[UNCERTAIN]	Assume the stricter reading applies to you.
10	One candidate recollection could not be retrieved — a Reddit post on a Restricted-grade ASOC experience returned HTTP 403 by every route tried.	[UNCERTAIN]	It is the single highest-value remaining gap; worth one manual look in a browser.

5.4 Disclaimer for this part

This part is a research compilation, not legal advice and not an official question bank. **No question in it is guaranteed to appear in any examination.** Part 5A reproduces published documents; Part 5B is constructed practice, clearly labelled as such, and no question in Part 5B is a past paper or a recollection of one.

Fees, bands, power limits, validity periods and the governing Rules referenced inside these questions change, and several of the documented sources predate the Telecommunications (Amateur Services) Rules, 2024 and are marked accordingly. Where the primary sources conflict with the popular study material, this part follows the primary source and says so — but a real examiner might mark differently, particularly on the pass mark, the QRQ/QRO question, Morse word spacing and log timekeeping.

Confirm anything load-bearing — anything that would change what you buy, what power you run or what you say on the air — with the WPC Wing of DoT or a licensed amateur radio club before acting on it.

Part 6 — Penalties, enforcement, risk, and appendices

This part sets out what actually happens if you hold or use a radio you are not authorised to hold or use: the exact penal provisions and amounts, who may search and seize, what real enforcement has looked like, and an honest assessment of the risk. It then closes the report with a glossary, a consolidated table of forms and fees, contact addresses, a grouped list of every source URL relied on, and the legal disclaimer.

Evidence tags are carried through from the underlying research. **[VERIFIED]** means the claim is stated in a primary or official source that was fetched (an Act text, a gazette PDF, a Press Information Bureau release, a court judgment). **[REPORTED]** means a credible named secondary source. **[UNCERTAIN]** means it could not be confirmed, or sources disagree.

6.1 Which law bites today, 2026-08-23

Everything in this part depends on a single structural fact: **India currently has two live penal regimes for radio, not one.** The Telecommunications Act 2023 has been commenced in stages, and the section that repeals the two old Acts has not been among the stages commenced so far.

Provision	Commencement status as at 2026-08-23	Evidence
Telecommunications Act 2023, ss. 1, 2, 10–30, 42–44 , 46, 47, 50–58, 61, 62	In force from 2024-06-26	[VERIFIED] S.O. 2408(E), dated 2024-06-21, Gazette of India Extraordinary Part II eservices.dot.gov.in/sites/default/files/circular-notifications/Notification%20for%20enforcement%20of%20the%20Telecommunications%20Act%202023.pdf
Telecommunications Act 2023, ss. 6–8, 48, 59(b)	In force from 2024-07-05	[VERIFIED] S.O. 2623(E), dated 2024-07-04 — https://eservices.dot.gov.in/sites/default/files/circular-notifications/Notification%20for%20enforcement%20of%20the%20Telecommunications%20Act%202023.pdf
Telecommunications Act 2023, s. 3(1) (authorisation, including s. 3(1)(c) "possess radio equipment") and s. 3(6)	In force from 2026-06-23	[VERIFIED] S.O. 3368(E), dated 2026-06-23, Gazette of India Extraordinary Part II E-23062026-273762 — https://egazette.gov.in/WriteReadData/2026/273762.pdf
Telecommunications Act 2023, Chapter VIII (ss. 31–41) — Adjudicating Officer, civil penalties, the machinery that alone can impose Second and Third Schedule penalties	Apparently still not commenced	[UNCERTAIN] , resting on a negative. Not in either 2024 commencement notification or 2024 enforcement notification. Trilegal's July 2024 update recorded ss. 31–41 as not commenced. A notification was found — https://trilegal.com/wp-content/uploads/2024/07/Trilegal-Notifies-Sections-of-the-Telecommunications-Act-2023-For-publication.pdf
Telecommunications Act 2023, s. 60 (repeal of the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933)	Apparently still not commenced	[UNCERTAIN] as a negative, but with strong positive corroboration — see below
Indian Telegraph Act 1885 and Indian Wireless Telegraphy Act 1933	Still in force	Follows from the above; corroborated by continued rule-making and charging under the Act

Why the conclusion that section 60 has not been commenced is reasonably solid, despite being a negative. Three independent pieces of positive evidence point the same way:

1. DoT was still making rules **under** both old Acts in June 2026. G.S.R. 468(E) dated 2026-06-11 opens: *"In exercise of the powers conferred by sections 4 and 7 of the Indian Telegraph Act, 1885 (13 of 1885) and sections 4 and 10 of the Indian Wireless Telegraphy Act, 1933 (17 of 1933) ..."* [VERIFIED] — <https://www.dot.gov.in/static/uploads/2026/06/d44ff2aec0505d538d9f3d51eba88936.pdf>. You cannot exercise powers under a repealed Act.
2. Rule 10(a) of the Telecommunications (Radio Equipment Possession Authorisation) Rules 2026, in force from 2026-07-08, expressly preserves the position of a person who *"holds a licence, registration, permission, by whatever name called, under the Indian Telegraph Act, 1885 ... or the Indian Wireless Telegraphy Act, 1933 ... and continues to operate under the terms and conditions of such licence"*. [VERIFIED]
3. Police were still charging under the old Act in 2026. An FIR (First Information Report) in Meerut, Uttar Pradesh was *"registered under relevant sections of the Bharatiya Nyaya Sanhita, Information Technology Act, Telecommunications Act 2023, and the Indian Telegraph Act"*. [VERIFIED as to the report text] — *Hindustan Times*, 2026-05-21.

A conflict in the underlying research, surfaced. One research file (the equipment-legality note) states flatly that the 1933 Act *"is repealed by s.60 of the 2023 Act"* and that bare possession is therefore only a civil penalty. That is **not supported** by the commencement record. Four other research files — the legal-framework, licence-exempt, recent-changes and penalties notes — reach the opposite conclusion, and they do so on primary evidence (gazette notifications, June 2026 rule-making under the old Acts, and a 2026 FIR under the Telegraph Act). **Treat the old Acts as alive.** Anyone who tells you the maximum exposure for unlicensed possession is a ₹50,000 civil penalty is relying on a repeal that has not been brought into force.

Cumulative liability is express. Telecommunications Act 2023, s. 49(2) [VERIFIED]:

"(2) The provisions of this Act are in addition to and without prejudice to any other liability which a person may have incurred under any other law for the time being in force."

and s. 52(1)–(2) [VERIFIED]:

"52. (1) The provisions of this Act shall be in addition to, and not be construed in derogation of the provisions of any other law, and shall be construed as consistent with such law, for the time being in force. (2) If any conflict arises between a provision of this Act and a provision of any other law for the time being in force ... the provision of this Act shall prevail to the extent of such conflict."

Practical consequence. A person found today with an unlicensed 10 W hand-held can, on the face of the statute book, be proceeded against under **any or all** of: Indian Wireless Telegraphy Act 1933 s. 6(1-A); Indian Telegraph Act 1885 s. 20(1); and Telecommunications Act 2023 s. 42(1). The prosecutor chooses.

6.2 The penalty provisions, exactly

6.2.1 INDIAN WIRELESS TELEGRAPHY ACT 1933 (IWTA) — THE POSSESSION OFFENCE

Text below is quoted from the Act as hosted by the Telecom Regulatory Authority of India (TRAI) at https://www.trai.gov.in/sites/default/files/2024-10/Indian_Wireless_Telegraphy_Act_1933.pdf, cross-checked word-for-word against the consolidated text on Indian Kanoon (<https://indiankanoon.org/doc/251935/>). Both fetched 2026-08-23; the two texts agree. [VERIFIED]

Section 3 — the prohibition:

"3. Prohibition of possession of wireless telegraphy apparatus without licence. — Save as provided by section 4, no person shall possess wireless telegraphy apparatus except under and in accordance with a licence issued under this Act."

The definitions that put a walkie-talkie in the severe limb (s. 2):

"(2) 'wireless telegraphy apparatus' means any apparatus, appliance, instrument or material used or capable of use in wireless communication ... (2-A) 'wireless transmitter' means any apparatus, appliance, instrument or material used or capable of use for transmission or emission of wireless communication;"

A hand-held two-way radio is, without argument, a "wireless transmitter". That matters, because it places it in the three-year limb of s. 6, not the ₹100 limb.

Section 6 — the penalty, in full:

"6. Offence and penalty. — (1) Whoever possesses any wireless telegraphy apparatus, other than a wireless transmitter, in contravention of the provisions of section 3 shall be punished, in the case of the first offence, with fine which may extend to one hundred rupees, and, in the case of a second or subsequent offence, with fine which may extend to two hundred and fifty rupees.

(1-A) Whoever possesses any wireless transmitter in contravention of the provisions of section 3 shall be punished with imprisonment which may extend to three years, or with fine which may extend to one thousand rupees, or with both.

(2) For the purposes of this section a Court may presume that a person possesses wireless telegraphy apparatus if such apparatus is under his ostensible charge, or is located in any premises or place over which he has effective control.

(3) If in the trial of an offence under this section the accused is convicted the Court shall decide whether any apparatus in respect of which an offence has been committed should be confiscated, and, if it so decides, may order confiscation accordingly."

Three points a reader must not miss:

- **₹1,000 is the maximum fine, and it has never been raised.** The figure dates from the Indian Wireless Telegraphy (Amendment) Act 1949, which inserted s. 6(1-A). No amending Act increasing it was found. **[VERIFIED — neither the TRAI text nor the Indian Kanoon consolidated text shows a higher figure or an amendment footnote to that effect.]** The money is trivial; the three-year imprisonment limb is not.
- **Section 6(2) reverses the practical burden.** The prosecution need not prove ownership or use. A radio under your "ostensible charge", or in a place over which you have "effective control" — a car boot, a bedroom cupboard — raises the presumption.
- **Confiscation is a decision the court is directed to take.** Read with s. 8, a confiscated set becomes government property.

Section 8 — confiscated apparatus vests in the Central Government [VERIFIED]:

"8. ... All wireless telegraphy apparatus confiscated under the provision of sub-section (3) of section 6, and all wireless telegraphy apparatus having no ostensible owner shall be the property of the Central Government."

Note the second limb: apparatus "**having no ostensible owner**" becomes Central Government property with no conviction at all. An unclaimed radio is simply gone.

Section 10(3) — breach of a subordinate rule [VERIFIED]: a breach of a rule made under the Act (for example the Indian Wireless Telegraphy (Possession) Rules 1965, as distinct from the Act itself) may be made "*punishable with fine which may extend to one hundred rupees*".

Section 11 — the two Acts stack [VERIFIED]:

"11. **Saving of Indian Telegraph Act, 1885.** — Nothing in this Act contained shall authorise the doing of anything prohibited under the Indian Telegraph Act, 1885 (13 of 1885), and no licence issued under this Act shall authorise any person to do anything for the doing of which a licence or permission under the Indian Telegraph Act, 1885 (13 of 1885), is necessary."

This is why **a possession licence is not an operating licence**. Prosecutions routinely charge both Acts together.

Why the offence is framed as possession, not use. The Statement of Objects and Reasons to the 1933 Act says so in terms **[VERIFIED, reproduced in the consolidated text at <https://indiankanoon.org/doc/251935/>]**:

"The detection of unlicensed apparatus and the successful prosecution of offenders is therefore difficult in practice as it is first necessary to locate unlicensed apparatus and then to prove that it has actually been established, maintained and worked. ... It is now proposed to remedy this state of affairs by legislation to prohibit the possession without license of wireless apparatus as distinct from the establishment, maintenance and working of such apparatus."

Parliament deliberately removed the "I never keyed the microphone" defence in 1933.

6.2.2 INDIAN WIRELESS TELEGRAPHY (POSSESSION) RULES 1965 — THE OPERATIVE RULES

[VERIFIED] via the Delhi High Court's verbatim reproduction in *Frederick Joseph Iseman v State NCT of Delhi*, order dated 2024-11-12 (<https://indiankanoon.org/doc/72044028/>) and via Indian Kanoon's rule-by-rule text.

- **Rule 3(1):** *"Save as provided in rule 5, no person shall possess a wireless telegraphy apparatus except under and in accordance with a license issued under these rules."*
- **Rule 3(2):** a dealer may not sell or hire a complete wireless set unless the buyer produces a valid possession licence or a s. 4 Telegraph Act operating licence in the buyer's own name, or the dealer obtains the licence for the buyer within seven days.
- **Rule 4 — imported apparatus:** *"time up to thirty days from the date following the date of clearance of the apparatus by the Customs Authorities shall be allowed to the importer for taking out a valid licence."* This 30-day window is the only genuine grace period in the whole framework, and it is the ground on which the Delhi High Court stayed a s. 6 prosecution in 2024 (§6.4.6).
- **Rule 5 — exemptions**, of which **rule 5(a)** is the important one: a person other than a dealer needs no possession licence for *"such apparatus as is reasonably required for the purpose specified in a current licence issued to him under section 4 of the Indian Telegraph Act, 1885, to establish maintain and work a wireless telegraph"*. Hold the operating licence and the possession side is covered.
- **Rule 16 — inspection:** an Inspector appointed by the Licensing Authority *"may inspect any wireless telegraphy apparatus in the possession of any person"*, and that person must, if required, *"give particulars of the name and addresses of any person to whom he may have transferred or from whom he may have received any wireless telegraphy apparatus"*; a licensee must produce the licence on demand. This is a standing duty that needs no "reason to believe" and is separate from the s. 7 search power.
- **Rule 9** sets the possession licence fee at *"fifteen rupees per year"* — a figure so obsolete that it is itself evidence of how little of this rule-set has been maintained. **[VERIFIED, <https://indiankanoon.org/doc/54047780/>]**

6.2.3 INDIAN TELEGRAPH ACT 1885 (ITA) — THE OPERATING OFFENCE

Source: the Act as hosted by TRAI at https://www.trai.gov.in/sites/default/files/2024-10/Indian_Telegraph_Act_1885.pdf , Part IV ("PENALTIES"), cross-checked against the AdvocateKhoj bare-Act pages. Both fetched 2026-08-23.

Section 4(1) — the exclusive privilege that creates the offence [VERIFIED]:

"4. Exclusive privilege in respect of telegraphs, and power to grant licenses. — (1) Within [India], the Central Government shall have the exclusive privilege of establishing, maintaining and working telegraphs: PROVIDED that the Central Government may grant a license, on such conditions and in consideration of such payments as it thinks fit, to any person to establish, maintain or work a telegraph within any part of [India] ..."

"Telegraph" is defined widely enough to catch every radio, s. 3(1) [VERIFIED]:

"(1) 'telegraph' means any appliance, instrument, material or apparatus used or capable of use for transmission or reception of signs, signals, writing, images, and sounds or intelligence of any nature by wire, visual or other electro-magnetic emissions, radio waves or Hertzian waves, galvanic, electric or magnetic means;"

Section 20 — the main operating offence, in full [VERIFIED]:

"20. Establishing, maintaining or working unauthorized telegraph (1) If any person establishes, maintains or works a telegraph within [India] in contravention of the provisions of section 4 or otherwise than as permitted by rules made under that section, he shall be punished, if the telegraph is a wireless telegraph, with imprisonment which may extend to three years, or with fine, or with both, and, in any other case, with a fine which may extend to one thousand rupees. (2) Notwithstanding anything contained in the Code of Criminal Procedure, 1898 (5 of 1898), offences under this section in respect of a wireless telegraph shall, for the purposes of the said Code, be bailable and non-cognizable. (3) When any person is convicted of an offence punishable under this section, the court before which he is convicted may direct that the telegraph in respect of which the offence has been committed, or any part of such telegraph, be forfeited to government."

Points to note:

- For a **wireless** telegraph the fine has **no stated ceiling** — the section simply says "with fine", in contrast to the ₹1,000 cap for wireline telegraphs. On paper this is the most expensive of the old provisions; in the reported cases courts actually imposed ₹100–₹200.
- s. 20(2) is a deliberate de-escalation: the offence is **bailable and non-cognizable**. Police cannot investigate a standalone s. 20 wireless case without a Magistrate's order. (That protection collapses the moment an IWTA s. 6(1-A) charge is added — see §6.3.)
- s. 20(3) gives **forfeiture to Government** on conviction.

Section 20A — breach of a licence condition [VERIFIED]: *"fine which may extend to one thousand rupees, and with a further fine which may extend to five hundred rupees for every week during which the breach of the condition continues."* This is the provision aimed at a **licensed** operator who exceeds power, uses an off-plan frequency, or lets an unlicensed person transmit.

Section 21 — using someone else's unauthorised telegraph [VERIFIED]: fine *"which may extend to fifty rupees"*. This is the provision that would catch a family member talking on the other end of your unlicensed hand-held. ₹50 is nominal — but it is still a criminal conviction.

Caution: the Andhra Pradesh High Court in *Ramakrishna Dish Antenna System* (1989-03-07, <https://indiankanoon.org/doc/1208726/>) described s. 21's fine as extending "up to Rs. 1,000/-". That appears to be an error in the judgment; two texts of the Act say fifty rupees. **[VERIFIED against the statute; the judicial dictum is [REPORTED] and inconsistent with the statutory text.]**

Section 25 — intentional damage or interference [VERIFIED]: imprisonment up to three years, or fine, or both, for damaging, removing, tampering with or touching any part of a telegraph with intent to obstruct transmission, intercept messages, or commit mischief. Requires intent, so accidental interference does not attract it — but deliberate jamming of a licensed service does, on top of s. 20.

Section 32A — attempts [VERIFIED]: an attempt carries the same punishment as the offence.

6.2.4 TELECOMMUNICATIONS ACT 2023 — THE NEW STRUCTURE

Source for all quotes: the gazette text of Act No. 44 of 2023, dated 2023-12-24, hosted by TRAI at https://www.trai.gov.in/sites/default/files/2024-09/Telecommunications_01012024.pdf, and the e-gazette copy at <https://egazette.gov.in/WriteReadData/2023/250880.pdf>.

Section 3(1) — the new prohibition [VERIFIED]:

"3. (1) Any person intending to — (a) provide telecommunication services; (b) establish, operate, maintain or expand telecommunication network; or **(c) possess radio equipment**, shall obtain an authorisation from the Central Government, subject to such terms and conditions, including fees or charges, as may be prescribed."

"radio equipment" is defined in s. 2(l) as *"telecommunication equipment used or capable of use for telecommunication by means of Hertzian or radio waves"* **[VERIFIED]**. Existing exemptions survive: s. 3(4) preserves *"Any exemption granted prior to the appointed day under the Indian Telegraph Act, 1885 or the Indian Wireless Telegraphy Act, 1933"* **[VERIFIED]** — so the 2018 low-power exemption rules (446.0–446.2 MHz and the rest) carry over.

Section 42 — the criminal offences [VERIFIED], abbreviated to what matters here:

Sub-section	Conduct	Maximum penalty
42(1)	Providing telecommunication services or establishing a telecommunication network without authorisation under s. 3(1); or damaging critical telecommunication infrastructure	3 years imprisonment, or fine up to ₹2 crore , or both
42(2)	Unauthorised access to a network or to an authorised entity's data; unlawful interception	3 years, or ₹2 crore , or both

Sub-section	Conduct	Maximum penalty
42(3) (a)–(f)	Possessing/using jamming equipment; misuse of telecommunication identifiers; possessing radio equipment without authorisation that can accommodate more than the specified number of subscriber identity modules (SIMs) ; obtaining SIMs by fraud; wilfully possessing radio equipment known to use tampered identifiers	3 years, or ₹50 lakh , or both
42(4)	Wilful contravention of national-security measures notified under s. 21	3 years, or ₹2 crore , or both, plus possible suspension or termination of service
42(5)	Damage to a non-critical telecommunication network	Compensation plus fine up to ₹50 lakh
42(6)	Abetment, attempt, conspiracy	Punishment provided for the offence
42(7)	—	All s. 42 offences are cognizable and non-bailable , notwithstanding the Code of Criminal Procedure
42(8)	—	Triable only by a Chief Metropolitan Magistrate or Chief Judicial Magistrate of the first class or above

Read s. 42(3) carefully. Only clauses (d) and (f) mention possessing radio equipment, and both are tied to subscriber identity modules and telecommunication identifiers — that is, SIM boxes and identifier-spoofing gear, not walkie-talkies. **A plain unlicensed walkie-talkie falls within no criminal limb of s. 42(3). [VERIFIED from the text.]**

What could catch a hand-held user is **s. 42(1)**, on the argument that operating a radio "establishes ... a telecommunication network" without authorisation. There is a real argument the other way: s. 2(s) defines "telecommunication network" as a system *"used or intended to be used for providing telecommunication services"*, and a pair of private hand-helds is not obviously providing services to anyone. **[UNCERTAIN — no judgment construing s. 42(1) against a private radio user was found. The point is untested and the downside is 3 years or ₹2 crore.]**

Third Schedule — civil penalties [VERIFIED]:

Sl. No.	Contravention	Civil penalty
1	(a) Possessing radio equipment without an authorisation or an exemption , except for the offences under s. 42(3)(d) and (f); (b) use of SIMs in excess of the notified number	First offence: up to ₹50,000 . Each subsequent offence: up to ₹2 lakh for each such instance
2	Use by any person of a telecommunication service or network knowing, or having reason to believe, that it lacks the required authorisation	Up to ₹10 lakh
3	Contravention of s. 28 (measures for protection of users)	First: up to ₹50,000. Subsequent: up to ₹2 lakh each, or suspension of service, or both
4	Any other contravention of the Act, rules, or terms of an authorisation or assignment for which no penalty is provided elsewhere	First: up to ₹25,000 . Second or subsequent: further penalty up to ₹50,000 for every day after the first during which the contravention continues

This is the most counter-intuitive finding in the whole report. Under the 1933 Act, bare possession of an unlicensed transmitter is a three-year jail offence. Under the 2023 Act, Parliament routed ordinary possession into the **civil** track (Sl. No. 1(a)) and reserved the criminal track for SIM-related gear. Note also Sl. No. 2: merely *using* an unauthorised network knowingly is a civil penalty up to ₹10 lakh — the modern analogue of Telegraph Act s. 21's ₹50, a 20,000-fold increase.

Second Schedule — breach of authorisation terms [VERIFIED]: Severe up to ₹5 crore; Major up to ₹1 crore; Moderate up to ₹10 lakh; Minor up to ₹1 lakh; Non-severe, a written warning. This applies to an **authorised entity or assignee** — a licensed captive or commercial operator — not to an unlicensed individual.

Adversarial fact-check, 2026-08-23 — every penalty figure in this part was re-read from the primary text and no discrepancy was found. The gazette text of Act 44 of 2023 was extracted and read section by section. Confirmed exactly as stated in the tables above: **s. 42(1)** three years / ₹2 crore for providing telecommunication services or establishing a telecommunication network without authorisation under s. 3(1), or damaging critical telecommunication infrastructure; **s. 42(2)** three years / ₹2 crore; **s. 42(3)(a)–(f)** three years / ₹50 lakh, with clause (d) reaching only radio equipment "*that can accommodate more than specified number of subscriber identity modules*" and clause (f) only equipment knowingly using tampered identifiers; **s. 42(4)** three years / ₹2 crore; **s. 42(5)** compensation plus ₹50 lakh; **s. 42(7)** all s. 42 offences cognizable and non-bailable; **s. 42(8)** trial no lower than a Chief Metropolitan Magistrate or a Chief Judicial Magistrate of the first class; **s. 43** search and take possession, expressly conditioned on "*reason to believe*" that equipment "*in respect of which an offence punishable under section 42 has been committed*" is kept or concealed; **Second Schedule** [see ss. 32(1)(a) and 34(8)]; **Third Schedule** [see ss. 33(1),(2), 34(1),(3),(5) and 34(8)] item 1(a) "*Possessing radio equipment without an authorisation or an exemption, except for the*

offence under clauses (d) and (f) of sub-section (3) of section 42", first offence up to ₹50,000 and each subsequent offence up to ₹2 lakh. s. 60(1) repeals the 1885 and 1933 Acts; s. 61 preserves rules made under them.

On the old Acts: **IWTA 1933 s. 6(1)** fine up to ₹100 first and ₹250 subsequent for non-transmitter apparatus; **s. 6(1A)** imprisonment up to three years, or fine up to ₹1,000, or both, for a **wireless transmitter**; **s. 6(2)** the "ostensible charge" / "effective control" presumption; **s. 6(3)** confiscation on conviction; **s. 7** search and take possession where an offence under s. 6 has been committed; **s. 8** confiscated apparatus becomes Central Government property; **s. 10(3)** breach of a rule punishable with fine up to ₹100; **s. 11** a possession licence never authorises anything for which a Telegraph Act licence is needed. All read from DoT's own copy at https://eservices.dot.gov.in/sites/default/files/2024-11/The_Indian_Wireless_Act,_1933.pdf. **ITA 1885 s. 3(1)** (definition of "telegraph"), **s. 4** and **s. 20** were read on a **gov.in** host at <https://wbconsumers.gov.in/writereaddata/ACT%20%20RULES/Relevant%20Act%20%20Rules/THE%20INDIAN%20TELEGRAPH%20ACT,%201885.htm>, confirming imprisonment up to three years for a wireless telegraph with **no stated fine ceiling** on that limb (the ₹1,000 cap attaches only to the non-wireless limb), s. 20(2) bailable and non-cognizable, and s. 20(3) forfeiture on conviction. **This upgrades the 1885 Act section texts in Part 1 §1.4.2 from [REPORTED] to [VERIFIED]**. India Code (indiacode.nic.in) and dot.gov.in/static/uploads/... both returned HTTP 403 to an unauthenticated fetch on 2026-08-23, so those two routes remain closed.

What the fact-check could not resolve, and the reader must treat as open: whether **s. 60** (repeal of the 1885 and 1933 Acts) has been commenced, and whether **Chapter VIII** (ss. 31–41, the adjudication machinery) has been commenced. Both rest on a negative search and no usable public gazette search was reachable. The consequence is stated immediately below and should not be softened in any summary of this report.

But the civil track may not yet be operable. Third Schedule penalties are imposed by an Adjudicating Officer appointed under s. 35, inside Chapter VIII. If Chapter VIII has not been commenced (see §6.1), there is **no officer competent to impose the ₹50,000 penalty**, and the only live sanction for possession is the 1933 Act's three-year offence. **[UNCERTAIN — this is the single most consequential unresolved point in this part. Put it to WPC in writing.]**

No self-disclosure amnesty. The voluntary-undertaking mechanism in s. 34 is open only to an "authorised entity or assignee", and only for s. 32 breaches or Third Schedule Sl. No. 4 — **not** Sl. No. 1(a) possession. **[VERIFIED]** There is no way to come forward, confess and settle.

What the 2023 Act does *not* contain [VERIFIED by exhaustive text search of the gazette PDF — zero hits for each term]:

- **No confiscation or forfeiture power.** The words "confiscat", "forfeit" and "seiz" appear nowhere in the Act. Section 43 says only "take possession thereof", with no provision on what happens to the property afterwards and no return mechanism. This is a regression from IWTA ss. 6(3)/8 and ITA s. 20(3), and it means the fate of a seized hand-held under the new Act is governed by the general property provisions of the Bharatiya Nagarik Suraksha Sanhita 2023 (BNSS) rather than by the telecom statute.
- **No compounding provision.** You cannot buy your way out of a s. 42 prosecution.

- **No requirement of Government sanction to prosecute**, beyond the court-level requirement in s. 42(8).

The schedules can be raised by executive notification. Section 57(1) [VERIFIED] lets the Central Government amend the Second and Third Schedules by notification, *"Provided that penalty or civil penalty specified in such Schedules shall be not exceeding ten crore rupees."* The ₹50,000 possession penalty can therefore be raised to as much as ₹10 crore without returning to Parliament.

Extraterritorial reach. Section 50 [VERIFIED]: the Act applies to an offence or contravention committed outside India where the conduct involves a telecommunication service provided in India, or equipment or a network located in India.

6.2.5 SIDE-BY-SIDE COMPARISON

Conduct	Indian Wireless Telegraphy Act 1933	Indian Telegraph Act 1885	Telecommunications Act 2023
Possessing a transmitter (a walkie-talkie) without licence or authorisation	s. 6(1-A): up to 3 years , or fine up to ₹1,000 , or both	— (possession as such is not an 1885 Act offence)	Third Schedule Sl. 1(a): civil penalty up to ₹50,000 first, up to ₹2 lakh each subsequent. Not a criminal offence
Possessing non-transmitter wireless apparatus without licence	s. 6(1): fine up to ₹100 first, ₹250 subsequent	—	Same Third Schedule Sl. 1(a)
Operating or working an unauthorised wireless set	—	s. 20(1): up to 3 years , or fine with no stated ceiling , or both	s. 42(1): up to 3 years , or fine up to ₹2 crore , or both — if it amounts to establishing a telecommunication network
Talking on someone else's unauthorised set	—	s. 21: fine up to ₹50	Third Schedule Sl. 2: civil penalty up to ₹10 lakh
Breach of a condition of your licence or authorisation	—	s. 20A: ₹1,000 plus ₹500 per week continuing	s. 32 with Second Schedule: ₹1 lakh to ₹5 crore by severity
Breach of a subordinate rule	s. 10(3): fine up to ₹100	—	Third Schedule Sl. 4: ₹25,000 first; ₹50,000 per day continuing
Cognizable?	s. 6(1-A) yes (default classification); s. 6(1) no	s. 20 expressly no (s. 20(2))	Yes (s. 42(7))
Bailable?	s. 6(1-A) non-bailable ; s. 6(1) bailable	s. 20 expressly bailable (s. 20(2))	Non-bailable (s. 42(7))

Conduct	Indian Wireless Telegraphy Act 1933	Indian Telegraph Act 1885	Telecommunications Act 2023
Confiscation or forfeiture	ss. 6(3) and 8: yes , becomes Central Government property	s. 20(3): yes , forfeiture to Government on conviction	No provision
Trial court	Magistrate (s. 6(1)); Magistrate first class (s. 6(1-A))	Any Magistrate	CMM / CJM first class or above (s. 42(8))
Compounding available?	No	No	No

All rows **[VERIFIED]** from the statutory texts cited in §§6.2.1–6.2.4, except the cognizability classification of IWTA s. 6(1-A), which derives from the Code of Criminal Procedure First Schedule Part II as applied by the Gujarat High Court in 1997 (see §6.3.1).

6.2.6 ADJACENT REGIMES THAT HURT MORE THAN THE TELECOM ACTS

- **Customs Act 1962.** Illegal **import** is prosecuted under the Customs Act, not the wireless Acts — a far harsher regime (confiscation of goods, penalty up to the value of the goods, arrest, prosecution). This is what actually produced arrests in 2026 (§6.4.8). **[REPORTED — *Deccan Herald* (Press Trust of India), 2026-03-22.]**
- **Consumer Protection Act 2019.** The route by which the Central Consumer Protection Authority (CCPA) fined e-commerce platforms ₹10 lakh each for selling non-compliant walkie-talkies (§6.4.7). Aimed at sellers and platforms, not buyers. **[VERIFIED — PIB Release ID 2215261, 2026-01-16.]**
- **Bharatiya Nyaya Sanhita 2023 and the Information Technology Act 2000.** Routinely added alongside telecom charges where a radio turns up in a wider investigation. **[VERIFIED as to the Meerut FIR text — *Hindustan Times*, 2026-05-21.]**
- **Indian Penal Code abetment provisions (now BNS equivalents).** In the 1997 Gujarat case the distributor and importer were charged with abetting the end user's illegal possession under IPC ss. 109 and 114 read with IWTA s. 6. **[VERIFIED from the judgment.]**

6.3 Search, seizure, inspection and arrest

6.3.1 THE POWERS, SIDE BY SIDE

Power	Provision	Who may exercise it	Where	Warrant needed?	Threshold
Search and take possession of wireless apparatus	IWTA 1933, s. 7	An officer <i>"specially empowered by the Central Government in this behalf"</i>	<i>"any building, vessel or place"</i>	No	<i>"reason to believe"</i> that apparatus in respect of which a s. 6 offence has been committed is kept or concealed
Search and take possession of radio equipment	Telecommunications Act 2023, s. 43	Any officer <i>"authorised by the Central Government in this behalf"</i>	<i>"any building, vehicle, vessel, aircraft or place"</i>	No	<i>"reason to believe"</i> that unauthorised network/ telecommunication equipment/radio equipment in respect of which a s. 42 offence has been committed is kept or concealed
Inspect apparatus, demand production of licence, demand supplier's name and address	Possession Rules 1965, rule 16	An Inspector appointed in writing by the Licensing Authority	Wherever the apparatus is	No	None stated — no "reason to believe" requirement at all
Seize property found in suspicious circumstances	BNSS 2023, s. 106 (ex-CrPC s. 102)	Any police officer	—	No	Property suspected to be stolen or found in circumstances creating suspicion of an offence

Power	Provision	Who may exercise it	Where	Warrant needed?	Threshold
Physical verification of radio equipment and premises	REPA Rules 2026, rule 13(2)	Central Government	Premises of an authorised entity	No	After notice — but the proviso removes the notice requirement " <i>if the Central Government considers that immediate action is necessary or expedient in public interest</i> "
Forfeiture / confiscation on conviction	IWTA s. 6(3) with s. 8; ITA s. 20(3)	The convicting court	—	—	Conviction

All rows **[VERIFIED]** from the statutory and rule texts.

Section 43 of the 2023 Act is wider than IWTA s. 7 in three ways: it adds **vehicle** and **aircraft**; it extends beyond "wireless telegraphy apparatus" to "telecommunication equipment or radio equipment" generally; and it requires only that the officer be "*authorised*", not "*specially empowered*". It is otherwise a straight lift of the 1933 formula, with the same absence of a warrant requirement, reasons-in-writing, or seizure-memo safeguards. The Internet Freedom Foundation flagged exactly this **[REPORTED]**:

"Section 43 confers quasi-judicial powers to any officer authorised by the Union government to 'search any building, vehicle, vessel, aircraft or place ...'. Such search and seizure powers are accompanied by the power to summon information ... Such powers, non-existent in the Telegraph Act, 1885, may be open to misuse due to their ambiguous phrasing ..." — internetfreedom.in, "The Telecom Act, 2023 is ready for (partial) take off", June 2024

There is also a **standing reporting duty** for anyone holding a Radio Equipment Possession Authorisation: rule 9(1)(h) of the REPA Rules 2026 requires the holder to "*forthwith report any instance of loss, theft, loss of possession or suspected misuse of such radio equipment on the portal*". **[VERIFIED]**

6.3.2 CAN AN ORDINARY POLICE OFFICER SEIZE YOUR RADIO? YES.

Divyang R. Parikh v State of Gujarat, Gujarat High Court, judgment dated **1997-04-08** (<https://indiankanoon.org/doc/601828/>), is the most useful authority on procedure. The Court held **[VERIFIED]**:

"On facts seizure has been made under Section 102 of the Cr. P.C. The word 'any offence' employed in Section 102 of the Cr. P.C. shows that even in case of non-cognizable offence, police may seize property found under suspicious circumstances.

Apart from the factual position, Section 7 is only enabling provision. It empowers the Central Government to authorise any officer for the purpose of the Act of 1933 so far as the search of the unauthorised wireless sets are concerned. It does not prohibit the statutory power of the police under Section 156 of the Cr. P.C. of investigation of a cognizable offence and the power under the other provisions of the Cr. P.C. in the matter of seizure."

CrPC s. 102 is now **BNSS s. 106** ("Power of police officer to seize certain property") [VERIFIED, <https://indiankanoon.org/doc/44870458/>], and CrPC s. 155 is now **BNSS s. 174**, whose sub-section (2) still bars police investigation of a non-cognizable case without a Magistrate's order [VERIFIED, via *Prasanth K.S v State of Kerala*, Kerala High Court, 2024-09-05, <https://indiankanoon.org/doc/100888972/>].

6.3.3 COGNIZABLE, NON-BAILABLE, AND HOW THE PROTECTION IN S. 20(2) COLLAPSES

The same Gujarat judgment explains the interaction [VERIFIED]:

"For deciding the same reference will have to be made to Part II of the First Schedule of the Cr. P.C., wherein it is explicitly mentioned that an offence punishable for 3 years and upwards is cognizable and non-bailable."

"In the instant case, by virtue of Sub-section (2) of Section 20, the offence under Section 20(1) is non-cognizable, but offence under Section 6(1-A) of the Act of 1933 is cognizable. Hence, the case shall be deemed to be cognizable. In view of this, there is no illegality in the investigation."

The classification the Court relied on is set out verbatim by the Supreme Court in *Union of India v Ashok Kumar Sharma*, 2020-08-28 (<https://indiankanoon.org/doc/181202406/>) [VERIFIED]:

"If punishable with imprisonment for 3 years, and upwards but not more than 7 years — Cognizable — Non-bailable — Magistrate of the first class."

IWTA s. 6(1-A) carries a maximum of exactly three years, so it falls in that row. **The practical upshot: adding a s. 6(1-A) charge to a s. 20 case makes the whole case cognizable and non-bailable.** Anyone repeating the comforting line "it is only a bailable offence" is quoting s. 20(2) in isolation.

Under the 2023 Act the position is inverted from the outset: s. 42(7) makes **every** s. 42 offence cognizable and non-bailable notwithstanding the CrPC, removing the 1885 Act's deliberate softening.

6.3.4 WHO INVESTIGATES AND PROSECUTES

Divyang R. Parikh reproduces paragraph 35 of Chapter VIII of the **Wireless Licensing Manual** issued by the Indian Posts and Telegraphs Department **[VERIFIED as reproduced in the judgment; the Manual itself could not be obtained, and whether it is still current is [UNCERTAIN]]**:

"35. The penalties under Section 20 of the Indian Telegraph Act and Section 6 of the Indian Wireless Telegraphy Act in respect of wireless transmitters include imprisonment and the complaints under these sections will be treated as warrant cases ... **The investigation into the cases as well as the prosecution is to be conducted by the Posts and Telegraphs Department.**"

The Court held that the Manual's assertion that *all* such offences are non-cognizable is legally wrong and cannot override the CrPC. But the paragraph remains good evidence of departmental practice: DoT's own wireless wing runs the investigation and prosecution in the ordinary case, and a transmitter case is a **warrant case** (the more serious of the two magistrate-trial procedures). Consistent with that, the Supreme Court's 1962 *Mangal Sao* case records the detection being made by "*an Inspector of Wireless Telegraph*" who "*visited his shop*" **[VERIFIED]**.

The technical detection capability is the **Wireless Monitoring Organisation (WMO)**, described by the Comptroller and Auditor General of India as "*the field organization of the WPC wing*", and by PIB as comprising one International Satellite Monitoring Earth Station, five International Monitoring Stations and 22 Wireless Monitoring Stations. **[VERIFIED — PIB Release ID 1988428, 2023-12-19.]**

6.3.5 LIMITATION

[UNCERTAIN — this is an inference from the limitation provisions, not a proposition found stated in any source.] Under CrPC s. 468 (now BNSS s. 514) the limitation period tracks the maximum punishment: six months for fine-only offences, one year where imprisonment does not exceed one year, three years where imprisonment exceeds one year and does not exceed three years. On that scale IWTA s. 6(1) (fine only) would carry six months, and IWTA s. 6(1-A), ITA s. 20(1) and Telecommunications Act s. 42 would each carry three years. **Do not rely on this.** Two things must be checked with a lawyer: the BNSS numbering, and whether unlicensed *possession* is a **continuing offence** — if it is, limitation never starts to run at all.

6.3.6 CHALLENGE AND APPEAL ROUTES

Matter	First instance	Appeal 1	Appeal 2
Third Schedule civil penalty (including possession)	Adjudicating Officer, s. 33	Designated Appeals Committee within 30 days, s. 36	Any civil court having jurisdiction , s. 40 — <i>not</i> TDSAT
Second Schedule / breach of authorisation terms	Adjudicating Officer, s. 32	Designated Appeals Committee, s. 36	TDSAT within 30 days, s. 39

Matter	First instance	Appeal 1	Appeal 2
Criminal charge under s. 42	CMM / CJM first class or above, s. 42(8)	Ordinary criminal appellate hierarchy	—
Criminal charge under IwTA s. 6 / ITA s. 20	Magistrate	Sessions	High Court revision; quashing petition under BNSS (ex-CrPC s. 482)

[VERIFIED] for the statutory routes. Section 41 bars civil court jurisdiction over anything the Adjudicating Officer, the Designated Appeals Committee, the Central Government or TDSAT is empowered to determine under Chapter VIII. **[VERIFIED]**

6.4 Real enforcement — what has actually happened, with dates

6.4.1 JETHANAND BETAB V THE STATE OF DELHI — SUPREME COURT, 1959-09-15

1960 AIR 89; 1960 SCR (1) 755. <https://indiankanoon.org/doc/125699/> **[VERIFIED]**

The leading authority that IwTA s. 6(1-A) is valid law. Verbatim from the judgment:

"Jethanand, the appellant herein, was prosecuted, along with another, in the Court of the Magistrate, First Class, Delhi, under s. 6(1-A) of the Act for possessing a wireless transmitter in contravention of the provisions of s. 3 of the Act, and was sentenced to six months rigorous imprisonment. On appeal, the learned First Additional Sessions Judge, Delhi, upheld the conviction but reduced the sentence to the period of imprisonment already undergone plus a fine of Rs. 500. On revision, the High Court confirmed both the conviction and the sentence."

Offence date 1953-07-31. **Outcome: conviction and sentence upheld.** The Court noted in passing that a transmitter can be charged under either limb of s. 6, and that the prosecution's choice of limb determines whether imprisonment is on the table at all.

6.4.2 STATE OF BIHAR V MANGAL SAO — SUPREME COURT, 1962-03-29

1963 AIR 445; 1963 SCR (1) 148. <https://indiankanoon.org/doc/766168/> **[VERIFIED]**

A Patna shopkeeper was found playing an unlicensed radio **receiver** in his shop in November 1955 by an Inspector of Wireless Telegraph. Convicted; fined ₹200 under ITA s. 20, reduced by the High Court to ₹100 under IwTA ss. 3 and 6. Headnote holdings **[VERIFIED]**:

"Held, that a Radio Receiving Set is a 'telegraph' within the meaning of s. 3(1) of the Indian Telegraph Act. Held, further, that using and keeping a Radio Set amounted to 'maintaining' and 'working' a 'telegraph' under s. 3(1) of the Indian Telegraph Act, 1885."

This is the clearest authority that merely **keeping and using** a radio is "working a telegraph". If that is true of a domestic receiver, it is a fortiori true of a transmitter.

6.4.3 DIVYANG R. PARIKH V STATE OF GUJARAT — GUJARAT HIGH COURT, 1997-04-08

<https://indiankanoon.org/doc/601828/> [VERIFIED]

The Ahmedabad distributor of Motorola wireless sets challenged an investigation. The FIR arose when a police sub-inspector found a control-room employee at Saurashtra Cement Ltd holding an unlicensed Motorola walkie-talkie; six further GP-300 sets (₹1.5 lakh) and 16 ECIL C-5020 sets (₹80,000) were seized. The charges, verbatim [VERIFIED], show the full enforcement model:

"(1) Saurashtra Cement Ltd. - committed offence under Sections 6(1) & 6(1-A) of Indian Wireless Telegraphy Act, 1933 by possessing transreceiver wireless sets ... without licence ... and they also committed offence under Section 20(1) of the Indian Telegraph Act, 1885 by using the said wireless sets without licence required under Section 4 ... (2) M/s. Veearsons - committed offence under Sections 6(1) and 6(1-A) ... read with Sections 109 and 114 of I.P.C. by abetting the illegal possession ... (3) Adino Telecom of Bombay - committed an offence under Sections 109 & 114 of the Indian Penal Code read with Sections 6(1) & 6(1-A) ... by supplying the said wireless sets ..."

Outcome: the petition to quash was rejected — "Rule discharged. The interim relief stands vacated." The end-user company, the distributor and the importer were all charged; the dealers via abetment. The investigating officer's affidavit recorded the security rationale: possession of such sets by unauthorised persons *"in the sensitive border State ... has grave consequences for National Security."* [VERIFIED]

6.4.4 STATE OF GUJARAT V ANWAR OSMAN SUMBHANIYA — SUPREME COURT, 2019-02-27

<https://indiankanoon.org/doc/120823041/> [VERIFIED as to charges and the acquittal below]

A prosecution under the Terrorist and Disruptive Activities (Prevention) Act (TADA) arising from a 1993 raid in Jamnagar. Charge-sheets included ITA s. 20 and IWTA s. 6(1-A) alongside IPC ss. 121/121A/122, TADA and Arms Act charges. Seized items included *"One – Transmitter walky-talky set worth Rs. 1 lakh."* The accused had been acquitted by the Designated Court and the State's appeal was before the Supreme Court in 2019 — a 1993 seizure still in litigation 26 years later. [UNCERTAIN as to the Supreme Court's final disposal — the operative paragraph was not read.]

The lesson: walkie-talkie charges ride along with arms and terror charges and are never the main event, but they are routinely added.

6.4.5 RAMAKRISHNA DISH ANTENNA SYSTEM V DEPUTY SUPERINTENDENT OF POLICE — ANDHRA PRADESH HIGH COURT, 1989-03-07

<https://indiankanoon.org/doc/1208726/> [VERIFIED that the judgment discusses ITA ss. 20, 20-A, 21 and 32-A and the confiscation power.] Its statement that s. 21's fine extends to ₹1,000 conflicts with the statutory text (₹50) and should not be relied on.

6.4.6 FREDERICK JOSEPH ISEMAN V STATE NCT OF DELHI — DELHI HIGH COURT, 2024-11-12

CRL.M.C. 8739/2024, Bhambhani J. <https://indiankanoon.org/doc/72044028/> [VERIFIED]

This is the most instructive modern case for a private individual, and the single best answer to "what actually happens to me?" Verbatim from the order:

"the matter relates to seizure of a satellite phone from the petitioner, who is a citizen of the United States of America, on 30.12.2018 ... the petitioner is about 72 years old and was travelling from Sri Lanka to Bhutan via Delhi on a private chartered flight, in the course of which travel, the petitioner had a 01-day layover in Delhi ..."

"at the time the satellite phone was recovered and seized from the petitioner's baggage during screening at the IGI Airport, New Delhi the petitioner had not used the satellite phone, which was in a discharged state and was therefore inoperable. The recovery of the phone however also led to the seizure of the petitioner's passport for the purported violation of section 6 of the Indian Wireless Telegraphy Act, 1933."

The defences run were ignorance of the prohibition, the Rule 3 / Rule 4 thirty-day window, and the bar on police investigation of a non-cognizable offence without a Magistrate's order. The Court's interim relief rested on **Rules 3 and 4**, not on ignorance:

"On a prima-facie view of the matter and in light of Rules 3 and 4 of the Indian Wireless Telegraphy (Possession) Rules 1965, further proceedings in kalandra dated 02.01.2019 registered under section 6 of the Indian Wireless Telegraphy Act, 1933 ... pending before the learned ACMM are stayed, till the next date of hearing before this court."

Re-notified for 2025-02-12. **[UNCERTAIN — no later order was found, so the final outcome is unknown.]** The Court also recorded that a co-ordinate bench had deferred trial-court proceedings in two matters "almost on all fours" by order dated 2023-07-19 in CRL.M.C. Nos. 4898/2023 and 4909/2023 — there is a small line of similar satellite-phone quashing petitions in Delhi.

Read this case for the timeline, not the law. Seizure 2018-12-30; kalandra 2019-01-02; summons 2019-01-05; passport impounded; still live in the High Court in November 2024. Nearly six years of criminal exposure over an inoperable device with a flat battery that the traveller never switched on. **That is the real penalty.**

6.4.7 THE CCPA CAMPAIGN AGAINST E-COMMERCE PLATFORMS, 2025-05 TO 2026-03

This is the largest, best-documented and most recent enforcement campaign specifically about walkie-talkies in India. It runs under the **Consumer Protection Act 2019** and is aimed at **platforms and sellers**, not users — but it is the Government's clearest recent statement of the legal position.

Stage 1 — thirteen notices, 2025-05-09. [VERIFIED — PIB Release ID 2127931, Ministry of Consumer Affairs, Food & Public Distribution, dated 09 MAY 2025]

"The Central Consumer Protection Authority (CCPA) has issued thirteen notices to leading digital marketplaces against the listing and sale of walkie-talkies on e-commerce platforms without proper frequency disclosure, licensing information, or Equipment Type Approval (ETA) which constitutes a violation of the Consumer Protection Act, 2019."

"These platforms include Amazon, Flipkart, Meesho, OLX, TradeIndia, Facebook, Indiamart, VardaanMart, Jiomart, Krishnamart, Chimiya, Talk Pro Walie Talkie and MaskMan toys."

Scale at that date: approximately 467 listings on Amazon, 314 on Flipkart, 489 on Meesho and 423 on TradeIndia.

Stage 2 — Guidelines notified, 2025-05-27; announced by PIB on 2025-05-30. [VERIFIED as to the notification date and the announcement — PIB Release ID 2132575, 2025-05-30, quoted at length in Part 2 §2.10.1; the notified PDF is itself named [Guidelines_27052025.pdf](#). The commentary below is [REPORTED] — MediaNama, 2025-06-03.] The CCPA notified the "Guidelines for the Prevention and Regulation of Illegal Listing and Sale of Radio Equipment including Walkie Talkies on E-Commerce Platforms, 2025". Platforms must list only authorised, compliant equipment on permitted frequencies; devices needing a frequency assignment from DoT may be bought only through an authorised dealer listed on the Saral Sanchar portal. MediaNama records the trigger: *"This regulatory crackdown coincided with broader government responses following the terrorist attack in Pahalgam in April 2025."* The text of the Guidelines themselves could not be fetched (doca.gov.in returned HTTP 403), so their exact wording is [UNCERTAIN].

Stage 3 — final orders and penalties, 2025-11 to 2026-01. [VERIFIED — PIB Release ID 2215261, dated 16 JAN 2026, "CCPA takes action against Unauthorized Sale of Walkie-Talkies on E-Commerce Platforms"]

"The action follows the identification of over 16,970 non-compliant walkie-talkie listings across platforms."

"The Authority imposed penalties of ₹10 lakh each on Meesho, Flipkart, Amazon and Meta Platforms Inc, and ₹1 lakh each on Chimiya, JioMart, Talk Pro and MaskMan Toys. Several platforms have already paid the penalty amounts, while payment from the remaining entities is awaited."

"Some e-commerce platforms claimed that they were only intermediaries and were not responsible for products listed by third-party sellers. The CCPA rejected this claim and clarified that platforms which allow the listing, promotion and sale of regulated products cannot avoid responsibility."

The same release states the licensing position that matters most to this reader:

"Under these rules, only walkie-talkies operating strictly within the 446.0–446.2 MHz frequency band are exempted from licensing. Even such licence-exempted devices must obtain ETA certification before being imported or sold in India."

And, on specific sellers:

"CCPA found that Talk Pro (Iconet Services Pvt. Ltd.) listed and sold walkie-talkie devices operating across wide frequency ranges, including UHF 400–1200 MHz, while falsely representing them as '100% legal' and 'license-free.'"

"In the matter concerning Flipkart, ... Data furnished revealed that 42,275 units were sold with the exact license-exempt frequency range disclosed, whereas 65,931 units were sold where the frequency range was either left blank or was outside the exempted range."

"The Authority noted that unauthorised radio devices can cause harmful interference with communication systems used by law-enforcement agencies, disaster management authorities and emergency services, posing serious risks to public safety, public order and national security."

Read the Flipkart figure again: **65,931 units** sold on one platform with the frequency range blank or outside the exempted range. Those radios are in Indian homes now, and the great majority of their owners have no authorisation of any kind.

Stage 4 — Meta's challenge, Delhi High Court, 2026-03-25. [VERIFIED — <https://indiankanoon.org/doc/37518660/> , *Meta Platforms, Inc v Union of India & Anr, Kaurav J.*] The writ petition was withdrawn as to the penalty and self-audit directions with liberty to appeal to the National Consumer Disputes Redressal Commission (NCDRC); the Court recorded that Meta had already complied with the listing prohibition and that *"Facebook Marketplace is not permitting the listing, advertising, and hosting of walkie-talkies on its platform."*

[REPORTED] The campaign continued: *Livemint*, 2026-02-12, reported that CCPA fined TradeIndia ₹10 lakh over uncertified walkie-talkie sales.

6.4.8 DRI SEIZURE AT NHAVA SHEVA — 2026-03-21

[REPORTED — *Free Press Journal*, 2026-03-22; *Deccan Herald* (PTI), 2026-03-22]

"The Directorate of Revenue Intelligence (DRI) has seized 11,060 prohibited walkie-talkie sets and over 52,000 second-hand Hard Disk Drives (HDDs), collectively valued at Rs 9.25 crore, in a major operation in Mumbai. Two persons, identified as owners of importing firms based in Lamington Road, have been arrested in connection with the case"

"the seized walkie-talkies are of the **Baofeng BF-888S** model, which has been **blacklisted by the government**. These devices, valued at around Rs 2.5 crore, are prohibited as they operate on frequency ranges beyond permissible limits and require specific licences for use."

"The devices reportedly operate in the 400–480 MHz frequency range, in violation of norms laid down under the Wireless Planning and Coordination (WPC) Wing ... As per existing rules, licence-free operation is allowed only within the 446.0–446.2 MHz band."

The PTI copy adds the charging provision: the father-and-son importers were **arrested under the Customs Act**. This is the crucial detail for the risk picture — **import offences go to Customs, not to the wireless Acts**, and the Customs Act is by far the harsher regime.

6.4.9 OTHER REPORTED INCIDENTS

[REPORTED], at varying confidence. Where the article body could not be fetched, only the outlet and date are confirmed from feed metadata; those items should be re-verified before being quoted elsewhere.

Date	Incident	Source	Confidence
2026-05-21	Illegal telephone exchange busted in Meerut; FIR under BNS, IT Act, Telecommunications Act 2023 and Indian Telegraph Act ; action taken by police, cyber cell and SWAT team on DoT information	<i>Hindustan Times</i>	Article text fetched [VERIFIED as to the report]
2026-04-19	US citizen arrested at Srinagar airport carrying a satellite phone	organiser.org	Headline and date only
2025-06-29	Seven arrested for carrying banned walkie-talkie sets on the India–Nepal border	NDTV	Headline and date only
2025-02-21	Israeli national arrested in Kerala for unauthorised use of a satellite phone	<i>Deccan Herald</i>	Headline and date only
2025-01-02	UK-based hiker detained at Delhi airport over an "illegal" GPS device	News18	Headline and date only
2024-12-10	American citizen caught with a satellite phone at Dehradun airport, arrested	<i>Hindustan Times</i>	Headline and date only
2023-06-03	Seven bootleggers in Thanjavur used walkie-talkies to track police; arrested	DT Next	Headline and date only
2022-10-26	A British executive was jailed in India for using a satellite phone	qz.com	Headline and date only

6.4.10 THE GAP IN THE RECORD — AND WHAT IT DOES AND DOES NOT MEAN

No reported Indian judgment or news report was found of an individual hobbyist being prosecuted purely for a hand-held VHF/UHF walkie-talkie, with no other alleged offence. Every individual prosecution located involved either a satellite phone, or a walkie-talkie found alongside arms/smuggling/terror charges, or a commercial user or dealer. **[UNCERTAIN]**

Do not over-read that. Two explanations fit the evidence equally well, and the second is at least as likely as the first:

1. It genuinely does not happen to lone hobbyists.
2. It happens and is invisible. A s. 6(1) or s. 20 case ending in a ₹100–₹200 fine before a magistrate generates no reportable judgment and no news coverage. **Absence of reported cases is not evidence of absence of prosecutions.**

There are also **no official statistics** on prosecutions, convictions or seizures under the 1933 Act. No parliamentary answer or DoT annual report giving numbers could be reached.

[UNCERTAIN]

6.5 Defences that do not work

Argument	Status	Why
"I did not know it was illegal"	No defence	Ignorance of the law is no excuse, and nothing in the 1933, 1885 or 2023 Acts displaces that rule. In <i>Iseman</i> the argument was advanced; the Court's interim relief rested on the Rule 4 import window instead [VERIFIED from the order]
"I never transmitted"	No defence to possession	The offence is bare possession, and the 1933 Statement of Objects and Reasons says Parliament chose possession precisely because proving use was too hard [VERIFIED]
"The battery was flat / it does not work"	Probably no defence	The definition is " <i>used or capable of use in wireless communication</i> " — capability, not operation, is the test. In <i>Iseman</i> the phone was discharged and inoperable and the prosecution still ran for six years [VERIFIED as to the definition and the facts; [UNCERTAIN] whether any court has finally decided the point]
"It is not mine / it was in the cupboard"	Weak	IWTA s. 6(2) lets the court presume possession from " <i>ostensible charge</i> " or " <i>effective control</i> " over the premises [VERIFIED]
"It is only a hobby"	Not a defence — it is a licence category	The 2023 Act, s. 47, defines amateur services as carried out by " <i>duly authorised person</i> ". Hobby status exists only inside the licence [VERIFIED]
"It is personal, not commercial"	No defence	Nothing in IWTA ss. 3/6, ITA ss. 4/20, or Telecommunications Act s. 3(1)(c) turns on commerciality [VERIFIED]
"I have a possession licence"	Only covers possession	IWTA s. 11: a possession licence does not authorise anything requiring a licence under the 1885 Act [VERIFIED]
"I bought it on Amazon / Flipkart, it said '100% legal'"	No defence against the State	The CCPA's own findings record platforms marketing UHF sets as "100% legal" and "license-free" when they were not [VERIFIED — PIB, 2026-01-16] . That gives you a consumer complaint against the platform. It gives you nothing against a prosecutor
"I turned my 10 W radio down to 0.5 W"	No defence	The PMR446 exemption also requires an integral antenna, hand-portable form, no repeater capability, operation confined to 446.0–446.2 MHz, and ETA for that exact model. A frequency-agile radio with a detachable antenna fails on every limb [VERIFIED]

The defences that do exist, and they are narrow:

1. **Genuinely falling inside a licence exemption** — the device really is within 446.0–446.2 MHz at no more than 500 mW effective radiated power, with a valid ETA. A 10 W set can never qualify.

2. **Rule 5(a) of the Possession Rules 1965** — you hold a s. 4 Telegraph Act operating licence (or, under the new regime, an Amateur Station Operator Certificate, which rule 10(f) of the REPA Rules 2026 exempts from any possession authorisation). **[VERIFIED]**
3. **Rule 4 of the Possession Rules 1965** — the 30-day window after customs clearance for imported apparatus. This is what provisionally saved the petitioner in *Iseman*.
4. **Procedural challenges** — for example investigation of a non-cognizable offence without a Magistrate's order. This failed in *Divyang R. Parikh* because a cognizable s. 6(1-A) charge was also laid.

6.6 Honest risk assessment of operating unlicensed

Everything in this sub-section beyond the tagged facts is judgement, not law. The underlying facts are tagged in §§6.1–6.5.

6.6.1 WHAT IS MOST LIKELY TO HAPPEN

Nothing, for a long time — and then something abrupt and disproportionate. There is no routine door-to-door enforcement against private radio owners in India. The State's attention is on the **supply side** (imports, e-commerce platforms, dealers) and on radios discovered **incidentally** during some other interaction with authority.

That shape of enforcement produces a specific, dangerous illusion: years of uneventful use feel like evidence of legality. They are not. They are evidence of low detection probability, which can change overnight after a security incident — as it did after April 2025, when the CCPA campaign began.

6.6.2 THE REALISTIC TRIGGER EVENTS, IN ROUGH ORDER OF LIKELIHOOD

#	Trigger	Why it is likely	Evidence base
1	Airport or customs screening	Best-documented path by a wide margin. Screening picks up the radio, it is seized, a kalandra or FIR follows, you are summoned	<i>Iseman</i> plus a continuing string of satellite-phone arrests, 2022–2026
2	Importing or ordering from abroad	Customs interception exposes you to the Customs Act 1962 on top of the wireless Acts — confiscation, penalty up to the value of the goods, arrest	DRI Nhava Sheva, 2026-03-21: arrests made under the Customs Act
3	Police contact for an unrelated reason — traffic stop, venue raid, crowd control, election-time vehicle check	Any police officer can seize under BNSS s. 106 with no special empowerment; adding a s. 6(1-A) charge makes the case cognizable	<i>Divyang R. Parikh</i> , 1997

#	Trigger	Why it is likely	Evidence base
4	A complaint or interference report	DoT's monitoring machinery can direction-find a persistent transmitter; the "Inspector of Wireless Telegraph" model of enforcement is old and still real	<i>Mangal Sao</i> (1962); WMO monitoring network (PIB, 2023-12-19)
5	Being visibly organised	Radios used at events, by private security, by a shop, or across a group draw attention and generate the paper trail (invoices, staff, group chats) that turns a seizure into a chargeable case	<i>Divyang R. Parikh</i> , where a sub-inspector walked into a control room and asked to see the licence

6.6.3 WHAT THE EXPOSURE ACTUALLY CONSISTS OF

Consequence	Severity	Likelihood if caught
The radio is seized and never comes back (IWTA s. 8 makes confiscated or unclaimed apparatus Central Government property)	Low financial	Very high
A criminal case that runs for years, requiring repeated court appearances	High disruption	High
Passport impounded or travel restricted while the case is live	High	Moderate — it happened in <i>Iseman</i>
Arrest, with a non-bailable classification (IWTA s. 6(1-A); all Telecommunications Act s. 42 offences)	High	Low to moderate
The actual fine on conviction (₹100–₹1,000 under the old Acts; reported sentences were ₹100–₹500)	Trivial	Certain on conviction
A criminal conviction on record — visas, government tenders, security clearances, professional licensing	High and long-lived	Certain on conviction
Civil penalty up to ₹50,000, or ₹2 lakh per subsequent instance, under the 2023 Act's Third Schedule	Moderate	Currently uncertain — Chapter VIII appears un-commenced, so there may be no officer competent to impose it
Fine up to ₹2 crore under s. 42(1)	Catastrophic on paper	Very low for a private user; that limb targets network operators
Customs Act consequences if the set was imported	Severe	High if intercepted at import
Additional charges if the radio is found alongside anything else (Arms Act, smuggling, historically TADA)	Extreme	Only where other offending exists

6.6.4 THE BLUNT SUMMARY

The statutory fine for owning an unlicensed walkie-talkie in India is smaller than the price of the radio. The real cost is the process — seizure, an FIR or kalandra, a warrant case before a magistrate, years of hearing dates, possibly a passport hold, and a permanent record. Nobody should reason "the maximum fine is ₹1,000, so my risk is ₹1,000." In *Iseman* the maximum realistic financial penalty was ₹1,000 and the actual cost was six years of criminal exposure, a seized passport, and High Court litigation, over a device that had never been switched on in India.

And one conclusion the reader must not draw. The fact that 2025–2026 enforcement has been aimed at Amazon, Flipkart, Meesho, Meta and importers rather than at end users is a **prosecutorial priority, not a legal safe harbour**. The CCPA's own releases repeatedly warn consumers about "*the potential legal consequences of unauthorized use*" and describe buyers as exposed "*to legal and regulatory risks*" [VERIFIED — PIB, 2025-05-09 and 2026-01-16]. The State's stated position is that the end user is also acting unlawfully.

6.6.5 WHAT WOULD CHANGE THIS ASSESSMENT

Four developments would materially move the risk picture, and a reader relying on this report in future should check each:

1. **Commencement of Chapter VIII (ss. 31–41)** and appointment of an Adjudicating Officer. This would switch on the ₹50,000 / ₹2 lakh civil penalty and create an administrative enforcement route that does not require a criminal court — cheaper for the State to use, therefore likely to be used more often.
2. **Commencement of s. 60**, repealing the 1885 and 1933 Acts. This would remove the three-year imprisonment exposure for bare possession, leaving the civil penalty (and s. 42(1) for operation). Counter-intuitively, this would *reduce* individual criminal risk.
3. **A notification under s. 57 raising the Third Schedule figures**, permissible up to ₹10 crore without returning to Parliament.
4. **A reported judgment construing s. 42(1) against a private radio user**, which would settle whether operating a pair of hand-helds "establishes a telecommunication network".

Appendix A — Glossary of abbreviations

Every abbreviation that carries weight in this report, expanded and explained. Abbreviations that appear only inside a verbatim quotation, a proper name, a URL, or an examination question's answer options are not repeated here.

A.1 Institutions and bodies

Abbreviation	Expansion	What it is
AAI	Airports Authority of India	Statutory body; a SACFA member consulted on siting near airports
AARS	ActiveHams Amateur Radio Society, Kerala	Publishes free study material at aars.in ; also styled AARS-KL
ARCAD	Amateur Radio Club, Aniruddha's Academy of Disaster Management, Mumbai	Club; runs workshops, publishes a free self-assessment quiz
ARSI	Amateur Radio Society of India	India's IARU member society, based in Bangalore. Not a government body
CBIC	Central Board of Indirect Taxes and Customs	Circulated the 2024 ETA self-declaration memorandum to Customs formations
CPIO	Central Public Information Officer	The officer who answers Right to Information requests; WPC's CPIO confirmed the 2010 Annexure-I in 2020
DARTS	<i>Expansion unconfirmed</i> [UNCERTAIN]	The amateur radio club at darts.org.in , Delhi National Capital Region. Publishes three sample papers and five online quizzes. See the note at the end of A.5
DGFT	Directorate General of Foreign Trade	Sets the import policy category (FREE / Restricted) that decides which ETA and import route applies
IACDM	Indian Academy of Communication and Disaster Management, Kolkata	Runs the most structured ASOC course found, with the West Bengal Radio Club
NBARS	North Bengal Amateur Radio Society, Siliguri	Free informal coaching and exam facilitation
PHARC	Pune HAMS Amateur Radio Club (VU2PHQ)	Exam facilitation and monthly meets
RIRCOM	International Radio Communicators Club (VU2IRD), Rajapalayam	Free video tutorials on the paper's Section B
SIARS	South India Amateur Radio Society, Chennai	Publishes an answered ASOC query page
WBRC	West Bengal Radio Club, Kolkata	Runs the IACDM course
BCAS	Bureau of Civil Aviation Security	Ministry of Civil Aviation body; issued the AVSEC circular under which satellite phones are seized at airports
CAG	Comptroller and Auditor General of India	Constitutional auditor; audited WPC spectrum management in Report No. 21 of 2018

Abbreviation	Expansion	What it is
CCA	Controller of Communication Accounts	DoT's regional accounting authority; receives ASOC fees through Bharatkosh
CCPA	Central Consumer Protection Authority	Regulator under the Consumer Protection Act 2019; ran the 2025–2026 walkie-talkie e-commerce campaign
DoT	Department of Telecommunications	Part of the Ministry of Communications; WPC sits inside it
DRI	Directorate of Revenue Intelligence	Customs intelligence agency; seized 11,060 Baofeng sets at Nhava Sheva in March 2026
IARU	International Amateur Radio Union	Global federation of national amateur radio societies
IFF	Internet Freedom Foundation	Indian digital-rights advocacy organisation
ITU	International Telecommunication Union	UN specialised agency for radio; its Radio Regulations are examinable in the ASOC syllabus
MeitY	Ministry of Electronics and Information Technology	Separate from DoT; relevant to IT Act matters
MHA / MoHA	Ministry of Home Affairs	Formerly gave security clearance for amateur licences; that requirement was removed in 2016
NCDRC	National Consumer Disputes Redressal Commission	Appellate forum from CCPA orders
NIAR	National Institute of Amateur Radio	Hyderabad-based non-government institute; the most reliable unofficial help desk for ASOC candidates
PIB	Press Information Bureau	Government of India's official press release service
PTI	Press Trust of India	News agency
RLO	Regional Licensing Office	WPC's regional licensing arm (Northern, Eastern, Western, Southern)
SACFA	Standing Advisory Committee on Frequency Allocation	Inter-ministerial committee giving <i>siting</i> clearance for wireless installations; WPC provides its secretariat
TDSAT	Telecom Disputes Settlement and Appellate Tribunal	Hears appeals on authorisation-breach penalties (s. 39), not Third Schedule possession penalties
TRAI	Telecom Regulatory Authority of India	Statutory regulator; hosts authoritative copies of the Acts
WMO	Wireless Monitoring Organisation	WPC's field and enforcement unit, established 1952
WPC	Wireless Planning and Coordination Wing	The spectrum licensing and planning wing inside DoT. Not a separate statutory body; it acts for the Central Government

A.2 Statutes, rules and instruments

Abbreviation	Expansion
BNS	Bharatiya Nyaya Sanhita 2023 — replaced the Indian Penal Code
BNSS	Bharatiya Nagarik Suraksha Sanhita 2023 — replaced the Code of Criminal Procedure
CrPC	Code of Criminal Procedure 1973 (superseded by BNSS, but still cited in older judgments and in s. 42(7) of the 2023 Act)
CPA 2019	Consumer Protection Act 2019
G.S.R.	General Statutory Rules — the gazette series for subordinate legislation (rules)
IPC	Indian Penal Code 1860 (superseded by BNS)
ITA 1885	Indian Telegraph Act 1885
IWTA 1933	Indian Wireless Telegraphy Act 1933
NFAP	National Frequency Allocation Plan — current edition NFAP 2022
OM	Office Memorandum — the instrument by which WPC issues most administrative instructions
REPA	Radio Equipment Possession Authorisation, under the Telecommunications (Radio Equipment Possession Authorisation) Rules 2026, G.S.R. 592(E), 2026-07-08
S.O.	Statutory Order — the gazette series for orders, including commencement notifications
TADA	Terrorist and Disruptive Activities (Prevention) Act — repealed, but cases from the 1990s were still live in 2019

A.3 Licences, authorisations and certificates

Abbreviation	Expansion	Who it is for
AIP	Advance Import Permission (issued by WPC in the legacy network-licence workflow)	Applicant procuring equipment before the operating licence issues
ASOC	Amateur Station Operator Certificate	The individual amateur radio certificate. Two grades: Restricted and General
CMRTS	Captive Mobile Radio Trunking Service	Companies and government bodies only
CNPN	Captive Non-Public Network	Companies and government bodies only
CUG	Closed User Group — the colloquial term for a captive private network	—
DL	Decision Letter — the WPC document issued after payment of spectrum charges, before the operating licence	Legacy network-licence applicants

Abbreviation	Expansion	Who it is for
DPL	Dealer Possession Licence, under the Possession Rules 1965	Dealers. Applications stopped 2026-07-03
ETA	Equipment Type Approval	Importer or Indian representative of any RF device in a de-licensed band. Two flavours: SD-ETA (self-declaration) and scrutiny-based ETA
GMDSS	Global Maritime Distress and Safety System (operator certificate)	Maritime operators
LoI	Letter of Intent — the first WPC output in the legacy frequency-assignment chain	Network-licence applicants
MMSL	Maritime Mobile Station Licence	Ships, boats, fishing trawlers
MTCTE	Mandatory Testing and Certification of Telecom Equipment	Manufacturers and importers of notified equipment
NDPL	Non-Dealer Possession Licence	Non-dealer holders of wireless equipment. Applications stopped 2026-07-03
PMRTS	Public Mobile Radio Trunking Service	Commercial trunking operators
RTR / RTR(A)	Radio Telephone Operator's Restricted Certificate of Proficiency (Aeronautical)	Aeronautical operators. The maritime equivalent is the GMDSS certificate, not RTR
USR	UHF Short Range Radio	A WPC service category in 350–351 MHz
VNO	Virtual Network Operator	A licence/authorisation flavour
VSAT	Very Small Aperture Terminal	Satellite ground terminal
WOL	Wireless Operating Licence — the final legacy instrument permitting operation	Network-licence holders

A.4 Technical terms

Abbreviation	Expansion	Note
CW	Continuous Wave — Morse code telegraphy	Tested at 8 words per minute for the General grade ASOC
DC input power	Direct current power into the transmitter's final stage	The metric the Indian amateur power table uses — not RF output. A radio sold as "10 W output" typically exceeds a 10 W DC-input ceiling
e.r.p. / ERP	Effective Radiated Power	The metric used in India's licence-exemption notifications
EIRP	Equivalent Isotropically Radiated Power	Used in SACFA exemption thresholds

Abbreviation	Expansion	Note
F3E	ITU emission designator for frequency-modulated (FM) telephony	The mode a voice walkie-talkie uses
HF / VHF / UHF	High Frequency (3–30 MHz) / Very High Frequency (30–300 MHz) / Ultra High Frequency (300 MHz–3 GHz)	—
PMR446	Personal Mobile Radio, 446 MHz	The European licence-free hand-held standard adopted in India at 446.0–446.2 MHz, 500 mW e.r.p. "Personal", not "Private", is the wording of G.S.R. 1047(E) Table V
RF	Radio Frequency	—
HSN	Harmonised System of Nomenclature	The customs tariff coding system; amateur equipment is reported at 8525 60 00 [REPORTED]
OEM	Original Equipment Manufacturer	Relevant to the import exemptions in the 2022 WPC compendium
SIM	Subscriber Identity Module	Relevant because the only criminal <i>possession</i> limbs in the 2023 Act are SIM-related
SRD	Short Range Device	The class of equipment covered by the licence-exemption rules
wpm	Words per minute — the Morse code speed unit	—

A.5 Process and payment terms

Term	Meaning
Bharatkosh	The Non-Tax Receipt Portal (NTRP) at bharatkosh.gov.in , through which all WPC fees are paid
Cognizable offence	An offence for which police may arrest without a warrant and investigate without a Magistrate's order
Compounding	Settling a criminal charge by payment. Not available under any of the three telecom statutes
DDO	Drawing and Disbursing Officer — a code required when paying on Bharatkosh
FIR	First Information Report — the police record that starts a criminal investigation
Kalandra	A police report or complaint placed before a magistrate (the term used in the Delhi <i>Iseman</i> proceedings)
Non-bailable offence	An offence where bail is at the court's discretion rather than a matter of right

Term	Meaning
NSWS	National Single Window System — the Government's consolidated clearance portal, which also stopped accepting DPL/NDPL applications from 2026-07-03
NTRP	Non-Tax Receipt Portal — the formal name for Bharatkosh
OCR	Optical character recognition — relevant because several of the source PDFs are image-only scans
RTI	Right to Information (Act, 2005) — the route by which WPC's 2020 confirmation of the Annexure-I band table was obtained
PAO	Pay and Accounts Officer — a code required when paying on Bharatkosh
RHQ	Regional Headquarters of the Wireless Monitoring Organisation (Western, Northern, Southern, Eastern)
Saral Sanchar	saralsanchar.gov.in — the legacy DoT licensing portal, still used for ASOC
Authorisation Portal	eservices.dot.gov.in/authorisation-portal — the new portal under the 2023 Act, live 2026-06-25
Summons case / warrant case	Two procedures for magistrate trials. A wireless-transmitter case is a warrant case , the more serious procedure
WMS	Wireless Monitoring Station — the 27 centres at which the ASOC examination is held

Note on one name: the abbreviation **DARTS** is expanded three different ways across the sources consulted — "Delhi Amateur Radio Society", "Delhi Amateur Radio Technical Society" and "Deccan Amateur Radio & Technical Society". None could be confirmed from the club's own pages. **[UNCERTAIN]** The club operates at darts.org.in and is based in the Delhi National Capital Region. **This report therefore uses "DARTS" unexpanded throughout**, and earlier drafts of Parts 4 and 5 that printed one of the expansions as fact have been corrected.

Note on a second name: **SACFA** is written "Standing Advisory Committee on Frequency Allocation" on DoT's own service page and is used in that form throughout this report; the 2010 WPC Office Memorandum's Annexure-I writes it as "Standing Advisory committee on Radio frequency allocation". Same body. **[VERIFIED — both documents read]**

Appendix B — Forms, fees and payment heads

B.1 The amateur radio route (the realistic individual route to 10 W)

All figures from the Telecommunications (Amateur Services) Rules 2024, G.S.R. 675(E), notified 2024-10-29, unless noted. **[VERIFIED]**

#	Item	Fee	Authority
1	ASOC examination, one grade (Restricted or General)	₹100	Rule 5(2)
2	ASOC examination, both grades in one sitting	₹200	Saral Sanchar FAQ Q3 [VERIFIED] ; NIAR advises paying ₹200 so that a Morse failure still leaves a Restricted pass [REPORTED]
3	ASOC (General or Restricted), 20-year validity	₹1,000	Rule 6(1)
4	ASOC (General or Restricted), lifetime validity	₹2,000	Rule 6(1). "Lifetime" means until the holder turns 80
5	Renewal, 20 years	₹1,000	Rule 8(1) with 6(1)
6	Renewal, lifetime	₹2,000	Rule 8(1) with 6(1)
7	Extension of a lifetime ASOC beyond age 80	Nil , 10 years at a time	Rule 8(2)
8	Late-renewal surcharge, up to 2 years after expiry	₹1,000	Rule 8(1), first proviso
9	Renewal more than 2 years after expiry	Not possible	Rule 8(1), second proviso
10	Permanent change of station location	₹1,000	Rule 7(6)
11	Duplicate ASOC	₹1,000	Rule 7(6)
12	Special (event) call sign, up to 90 days	₹200	Rule 13(1). Apply at least 90 days before the event [REPORTED — NIAR]
13	Amateur society (club) formation, minimum 4 ASOC holders	₹2,000	Rule 12(1)
14	Reciprocal ASOC for a foreign national	₹1,000	Rule 4(2)
15	SACFA processing fee, if an antenna goes on a tower, building or mast	₹1,000	[REPORTED — NIAR fee sheet and SACFA guide; not in the Amateur Rules]
16	Refunds on suspension, cancellation or surrender	None	Rules 10 and 11(2)

Two deadlines that catch people **[VERIFIED]**:

- Rule 6(2): after passing, **payment for the certificate must be made within two years of the declaration of results**, failing which no certificate is granted and the exam must be retaken.

- Renewal must be applied for at least **12 months before expiry [REPORTED — NIAR fee sheet]**, and after expiry there is a hard **two-year** cut-off.

Obsolete figures still circulating [VERIFIED as superseded]: change of location ₹200, duplicate licence ₹100, late renewal ₹100 per half-year, and a ₹200 "mobile endorsement". These come from WPC OM L-14011/255/2004-AMT dated 2010-08-13 and still appear in a WPC-authored FAQ that clubs redistribute, and in the Saral Sanchar FAQ PDF created 2020-08-24.

Where the FAQ and the 2024 Rules conflict, the Rules win.

B.2 Equipment and possession permissions

Item	Fee	Status as at 2026-08-23	Source
Equipment Type Approval (ETA)	₹10,000 per application; validity "lifetime unless revoked"	Open; licence-exempt ETA is granted on self-declaration since 2024-09-09	[VERIFIED] DoT "Fees and Charges for Services"
WPC Import Licence	₹500 per licence; 1 year	Closed to new Telegraph Act applications; migrating to REPA	[VERIFIED]
Dealer Possession Licence (DPL)	<i>"nominal fees of Rs. 60";</i> 1 year	Applications and renewals stopped from 2026-07-03	[VERIFIED]
Non-Dealer Possession Licence (NDPL)	Same page as DPL	Stopped from 2026-07-03	[VERIFIED]
Radio Equipment Possession Authorisation (REPA) — application	₹1,000 non-refundable	Open from 2026-08-06 on the Authorisation Portal	[VERIFIED] REPA Rules 2026, rule 4(3)(b)
REPA — dealers (manufacture, purchase or import for sale, hire, repair, test or demonstration)	₹10,000 per annum; 1–5 years	Open	[VERIFIED] rules 5(2)(a) and 6
REPA — testing or demonstration only	₹2,000 per annum or pro rata, minimum ₹500; up to 12 months	Open	[VERIFIED] rules 5(2)(a) and 6
REPA — modification	₹1,000	Open	[VERIFIED]
Possession licence under the Possession Rules 1965	<i>"fifteen rupees per year"</i> (rule 9)	Legacy; the figure is a museum piece	[VERIFIED]

Critical point for an end user. REPA covers **dealers and testers only**. Rule 10 lists who does **not** need a possession authorisation, and it includes — at rule 10(f) — anyone who *"holds an Amateur Station Operator Certificate in accordance with the Telecommunications (Amateur*

Services) Rules, 2024", and at rule 10(e) anyone who has spectrum assigned. **[VERIFIED]** Get the ASOC (or the assignment) and the possession side is covered; there is no separate end-user possession licence under the new regime.

Rule 9(2)(b) sets the threshold at which DoT stops treating radiated power as trivial: equipment radiating more than **100 milliwatt** must adhere to the terms of a spectrum assignment.

[VERIFIED]

B.3 Captive and commercial routes (for completeness — closed to natural persons)

From the Schedule to the Telecommunications (Authorisation for Captive Telecommunication Services) Rules 2026, G.S.R. 511(E), 2026-06-23. **[VERIFIED]**

Authorisation	Processing fee	Entry fee	Initial guarantee	Annual authorisation fee
Captive mobile radio trunking service (CMRTS)	₹10,000	Nil	₹20,000	₹300 per user terminal, minimum ₹5,000; from the fourth year, the higher of ₹300 per terminal or ₹25,000. Police, fire and government security services are exempt
Captive non-public network (CNP)	₹10,000	Nil	Nil	Nil (rule 46)
Captive VSAT	₹10,000	₹7.5 lakh	₹3 lakh	₹10,000 per terminal (rule 51(1))
Captive general service	Nil	Nil	Nil	Nil (rule 59)

Eligibility closes the door on individuals. Rule 5 confines eligibility to companies and government bodies, and further bars any non-government company from captive *general* service. **A natural person is not eligible for any captive authorisation. [VERIFIED]**

Legacy WPC network licence charges (Indian Telegraph Act era; whether the counter is still open is **[UNCERTAIN]**), from WPC Order No. P-11014/34/2009-PP dated 2023-12-11, Schedule II Part II **[VERIFIED]**:

Item	Annual fee
Base or fixed station licence	₹500 per station
Vehicle or hand-held mobile station licence	₹250 per station
Royalty	Formula-based on band popularity, coverage category, channel bandwidth and city population. Minimum charging unit 12.5 kHz
Modification of an existing assignment	₹1,000

A worked estimate for a small hand-held network on the UHF Short Range Radio route came to roughly **₹3,500–₹4,500 per year** all in — consistent with the ₹2,000–₹3,000 royalty plus per-station fees quoted in Part 1 §1.6. **[UNCERTAIN — the royalty tables are in a scanned PDF with unreliable optical character recognition (OCR), and the coverage-category rules do not say how to classify a network with no base station.]**

Other published WPC fees **[VERIFIED — DoT "Fees and Charges for Services"]**: SACFA general application ₹1,000 (₹100 per small cell or additional antenna); Experimental / Technology Trial Licence ₹5,000 per band; Demonstration Licence ₹5,000 per band; Non-Network Aero Mobile Station Licence ₹5,000 per station; MMSL ₹5,000; MMSL for a fishing trawler or boat ₹500.

B.4 Forms — there is no numbered form

[VERIFIED] The Telecommunications (Amateur Services) Rules 2024 never name a form. Rules 4(2), 5(1), 8(1), 8(2), 12(1), 12(2), 12(4) and 13(1) all say the application is made *"in the form specified for this purpose in the portal"*, and rule 15 requires the Central Government to notify a portal for digital implementation. **The answer to "what is the form number?" is: none.**

What you will actually encounter:

Artefact	Exact label	Evidence
Exam application (web form)	"Application for Amateur Exam"	[REPORTED — NIAR screenshots, 2025-03-21]
Exam application ID	EXAM + year + serial, e.g. EXAM20250200019	[REPORTED — NIAR]
Admit card	"e-ADMIT CARD / FOR AMATEUR STATION OPERATING CERTIFICATE EXAMINATION- 2025"	[REPORTED — NIAR model admit card]
Certificate application (web form), PDF rendering	"APPLICATION FOR WPC (Amateur Station Operator's Certificate)", application ID pattern HAM<YYYYMMDD>0000NN	[VERIFIED — screenshots in NIAR's License.pdf, 2025-03-01]
Nationality proforma A	"SELF DECLARATION OF CHARACTER AND NATIONALITY (in case valid Indian passport is available)"	[REPORTED — DoT OM 2025-02-28 enclosure, ARSI-hosted scan]
Nationality proforma B	"NATIONALITY CERTIFICATE (in case valid Indian passport is not available)" — to be signed by a Gazetted officer, MP/MLA, or SDM/Tahsildar	[REPORTED — same]
Minor's undertaking (ages 12–18)	"ASOC EXAMINATION UNDERTAKING (To be issued by Parent/Guardian/Head of Institution ...)"	[REPORTED — DARTS archive of the pre-2020 form set]

Obsolete forms still circulating online — do not use them [REPORTED — DARTS archive, 2018, <https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Form.pdf>]: "ANNEXURE II — Application from an individual for a licence to Establish, Maintain and Work an Amateur Wireless Telegraph Station in India"; "APPENDIX III — NATIONALITY CERTIFICATE"; the four-

copy personal-particulars sheet asking for height, eye colour and identification marks; and the Section A exemption letter. The old demand-draft route (State Bank of India draft in favour of "PAO (HQ), DOT") has been replaced entirely by Bharatkosh.

Attestation by a Gazetted officer is no longer required for new exam/certificate applications (OM dated 2025-02-28, ¶2.2) — self-certification plus originals produced at the exam hall. Most club guides still say otherwise because they predate that OM. **[VERIFIED as to the OM; [UNCERTAIN] as to residual portal validation behaviour.]**

B.5 Bharatkosh payment heads

All WPC fees are paid on the Non-Tax Receipt Portal, <https://bharatkosh.gov.in>, now embedded in Saral Sanchar. If you ever need the heads manually:

Examination fee — [VERIFIED, Saral Sanchar FAQ Q24]

- Purpose head: *"Amateur Exam conducted by Wireless Monitoring Stations and RHQs"*
- Payment type: *"Exam fee"*
- Function head: *"127500103050100 - Proficiency Certificate"*
- PAO and DDO vary by **region of your exam centre**:

Regional HQ	PAO	DDO
Western RHQ, Mumbai	077148 — Controller of Communication Accounts, Maharashtra	201562 DDO CCA
Northern RHQ, New Delhi	077177 — Controller of Communication Accounts, New Delhi	201540 DDO CCA
Southern RHQ, Chennai	077153 — Controller of Communication Accounts, Tamil Nadu	201579 DDO CCA
Eastern RHQ, Kolkata	077155 — Controller of Communication Accounts, West Bengal	201589 DDO CCA

Certificate, renewal, late fee and duplicate — [VERIFIED, Saral Sanchar FAQ Q25; DDO code [REPORTED] from the WPC instruction sheet]

- Purpose head: *"Amateur License"*
- Payment type: *"License fee" / "License renewal fee" / "Late fee" / "Fee for issue of duplicate license"*
- Function head: 127500103050100
- PAO: **077188 — Controller of Communication Accounts, PAO HQ**
- DDO: **205155 — DDO, SO(Cash), New Delhi**

Appendix C — Key contacts

C.1 WPC headquarters and the amateur section

Role	Details	Evidence
Licensing authority (amateur)	The Assistant Wireless Adviser to the Government of India, Ministry of Communications, Department of Telecommunications, Wireless Planning & Coordination Wing, Amateur Section, 6th Floor, Sanchar Bhavan, 20 Ashoka Road, New Delhi – 110001	[REPORTED — NIAR guides 2024-11 to 2025-09; the email also appears on the [VERIFIED] WPC FAQ and the 2025 OMs]
Telephone (amateur / COP section)	011-2303 6181 , 011-2303 6539, 011-2335 5441	[REPORTED]
Email (amateur / COP section)	awacop-dot@gov.in	[VERIFIED] — appears on the WPC FAQ dated 2024-12-05 and on the 2025 OMs
Engineer (COP), signatory of the 2024 FAQ	Rahul Sondhara, 011-2303 6181, awacop-dot@gov.in	[VERIFIED]
DoT eServices nodal cell for ASOC	Sr. Deputy Wireless Advisor, 011-23036540 , umang.srivastava@nic.in	[VERIFIED — https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc , fetched 2026-08-23]
Saral Sanchar helpdesk	011-23350020 / 011-23350025 , Monday–Friday 09:30–17:30	[VERIFIED — same page]
General WPC queries	011-2303 6534 or 011-2303 6539	[REPORTED — WPC-authored FAQ]
Postal identity used on WPC orders	Government of India / Ministry of Communications / Department of Telecommunications / Wireless Planning and Coordination Wing / 20, Ashoka Road, Sanchar Bhawan, New Delhi	[VERIFIED — header of WPC Order P-11014/34/2009-PP, 2023-12-11]

A practical warning about WPC's web presence [VERIFIED by direct testing, 2026-08-23]: the historic standalone site wpc.dot.gov.in **no longer resolves**. Every link to wpc.dot.gov.in/content/... in older documents and search results is dead. Current content lives at dot.gov.in (a JavaScript application that is difficult to fetch programmatically) and, for services, at eservices.dot.gov.in.

C.2 WPC field units — the Wireless Monitoring Organisation

The WMO is "*the field unit of WPC Wing*" [VERIFIED — PIB Release ID 1988428, 2023-12-19], and comprises one International Satellite Monitoring Earth Station, five International Monitoring Stations and 22 Wireless Monitoring Stations.

A count that does not reconcile, stated openly. That PIB enumeration totals **28** facilities, whereas the Saral Sanchar FAQ's list of **ASOC examination centres names 27**. The two documents are both official and neither explains the difference; the obvious guess is that the satellite earth station is not an examination venue, but no source says so. [UNCERTAIN] Use the FAQ's list of 27 for examination purposes, and confirm the venue printed on your own admit card.

The 27 centres [VERIFIED — Saral Sanchar FAQ Q4, https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf]:

#	Centre	#	Centre	#	Centre
1	Ahmedabad	10	Gorakhpur	19	New Delhi
2	Ajmer	11	Hyderabad	20	Patna
3	Bengaluru	12	Jalandhar	21	Raipur
4	Bhopal	13	Jammu	22	Ranchi
5	Bhubaneswar	14	Kolkata	23	Shillong
6	Chennai	15	Lucknow	24	Siliguri
7	Dehradun	16	Mangalore	25	Thiruvananthapuram
8	Dibrugarh	17	Mumbai	26	Vijayawada
9	Goa	18	Nagpur	27	Vishakapatnam

Regional Headquarters mapping (this determines your Bharatkosh PAO/DDO) [VERIFIED — same FAQ, Q24]:

Regional HQ	Centres served
Western RHQ, Mumbai	Ahmedabad, Bhopal, Goa, Mumbai, Nagpur, Raipur
Northern RHQ, New Delhi	Ajmer, Dehradun, Gorakhpur, Jalandhar, Jammu, Lucknow, New Delhi

Regional HQ	Centres served
Southern RHQ, Chennai	Bengaluru, Chennai, Hyderabad, Mangalore, Thiruvananthapuram, Vijayawada, Vishakhapatnam
Eastern RHQ, Kolkata	Bhubaneswar, Dibrugarh, Kolkata, Patna, Ranchi, Shillong, Siliguri

Addresses of the four Regional Headquarters and two other major stations [REPORTED — NIAR's compiled list, <http://www.niar.org/downloads/ham-downloads/List-of-Wireless-Monitoring-Stations.pdf>, undated]:

Station	Address	Telephone
Northern RHQ, New Delhi	NRHQ, IMS Campus, Ghitorni, PO Mehrauli, New Delhi – 110030	011-26502658
Western RHQ, Mumbai	WRHQ, Gorai Road, Borivli West, Mumbai – 400092, Maharashtra	022-28671626
Southern RHQ, Chennai	SRHQ, Near FACIT Asia, Kandanchavadi, Perungudi, Chennai – 600096, Tamil Nadu	044-24960275
Eastern RHQ, Kolkata	ERHQ, Gopalpur, Sarkarpool PO, 24 Parganas, Kolkata – 700143, West Bengal	033-24010151
Ahmedabad	Opp. Jagat Park Society, Ghatlodiya, Ahmedabad – 380061, Gujarat	079-27603444
Hyderabad	Room No. 302, 3rd Floor, Telecom Engineering Centre Building, Charlapalli, Hyderabad – 500051, Telangana	040-27261516 (telefax), 27261517, 27261518

[UNCERTAIN] The NIAR list is undated, and its Dehradun, Lucknow, Patna and Raipur entries carry no address or telephone number. **Confirm the address printed on your own admit card** rather than relying on this list.

Regional Licensing Offices (RLOs) — WPC's regional *licensing* arms, distinct from the monitoring stations. The DoT service page for the hand-held radio licence lists four: **Northern** (Ghitorni, New Delhi), **Eastern** (Sarkarpool, Gopalpur, Kolkata), **Western** (Charkop, Kandivali West, Mumbai) and a Southern office. **[VERIFIED — <https://eservices.dot.gov.in/network-short-range-uhf-hand-held-radio>]**

C.3 NIAR — National Institute of Amateur Radio

[VERIFIED — header of every page at <http://www.niar.org>]

"National Institute of Amateur Radio 6-3-1092/93, Raj Bhavan Road, Somajiguda, Hyderabad-500082, Telangana, India. +91 40 2331 0287 +91 40 2331 0787 info@niar.org"

Founded 1983 (formerly the A.P. Amateur Radio Society, 1975). Its campus includes a library, laboratory, museum, club station, classrooms, a QSL bureau and guest-house facilities, so residential study is physically possible. [REPORTED — <http://www.niar.org/facilities.html>]

Two practical warnings [VERIFIED by direct testing]:

1. The site is **broken over HTTPS** — its TLS certificate covers only [cpanel.](#) and [mail.](#) subdomains, not [niar.org](#) or [www.niar.org](#). Browsers will warn. Fetch over plain HTTP or accept the warning.
2. Pages carry "© Copyrights 2014", the HTTP [Last-Modified](#) header is 2024-07-23 on every page checked, and the news page stops at 2018. **Treat all NIAR prices and course details as needing a telephone confirmation.** Neither course fee, course length nor a class calendar is published. [UNCERTAIN]

NIAR study material prices, as published [REPORTED — <http://www.niar.org/training-material.html> , prices undated on the page]:

Material	NIAR members (₹)	Non-members (₹)	Postage (₹)
Study Manual	140	150	50
Morse Code Practice Book	40	50	50
Morse Key	190	200	—
Morse Code Practice Oscillator	290	300	—

Payment is by demand draft in favour of "National Institute of Amateur Radio", payable at Hyderabad. No online payment is offered. The three NIAR study manuals are also available as **free PDF downloads** — see Appendix D.9.

C.4 ARSI — Amateur Radio Society of India

India's IARU member society. [REPORTED — <https://arsi.info/contact-us/>]

"Amateur Radio Society Of India Ramesh Kumar VU2LU, President 635, 6th Main Road, Next to Bank of India Hanumanthnagar, Bangalore-560019."

ARSI does **not** run ASOC classes and does not publish an ASOC textbook. Its value is its document library (the best-curated collection of WPC source documents in India, at <https://arsi.info/important-wpc-documents/>), its Morse training videos, and its affiliated-club network.

Ordinary ARSI membership requires a valid amateur licence, so a pre-exam candidate cannot join as an Annual Member — ARSI is a post-exam organisation for this reader.

[REPORTED — <https://arsi.info/arsi-membership/>]

Contacting ARSI is nonetheless the most reliable single way to be pointed at a club near a given city; its published affiliated-club list (12 clubs) is not a complete map of Indian ham clubs.

Appendix D — Sources, grouped by topic

Every URL below was fetched during the research for this report on **2026-08-23** unless a different date is given. Where a source could not be fetched, it is listed in D.11 so that nobody repeats the attempt.

D.1 Statutes

Document	Date	URL
Indian Wireless Telegraphy Act 1933 (Act 17 of 1933) — TRAI copy	Act of 1933	https://www.trai.gov.in/sites/default/files/2024-10/Indian_Wireless_Telegraphy_Act_1933.pdf
Indian Wireless Telegraphy Act 1933 — consolidated text with Statement of Objects and Reasons	published 1933-09-11	https://indiankanoon.org/doc/251935/
Indian Wireless Telegraphy Act 1933 — DoT's own copy	—	https://eservices.dot.gov.in/sites/default/files/2024-11/The_Indian_Wireless_Act,_1933.pdf
Indian Telegraph Act 1885 — TRAI copy	Act of 1885	https://www.trai.gov.in/sites/default/files/2024-10/Indian_Telegraph_Act_1885.pdf
Indian Telegraph Act 1885 — bare-Act sections 3, 4, 20, 21, 25 and index	undated	https://www.advocatekhoj.com/library/bareacts/indiantelegraph/index.php
Telecommunications Act 2023 (Act 44 of 2023) — gazette text, TRAI copy	assented 2023-12-24	https://www.trai.gov.in/sites/default/files/2024-09/Telecommunications_01012024.pdf
Telecommunications Act 2023 — egazette copy, CG-DL-E-24122023-250880	2023-12-24	https://egazette.gov.in/WriteReadData/2023/250880.pdf
Bharatiya Nagarik Suraksha Sanhita 2023, s. 106 (police power to seize)	—	https://indiankanoon.org/doc/44870458/
TRAI "Acts & Policies" index	—	https://www.trai.gov.in/about-us/acts-policies

D.2 Commencement notifications

Document	Date	URL
S.O. 2408(E) — commences ss. 1, 2, 10–30, 42–44, 46, 47, 50–58, 61, 62 from 2024-06-26	2024-06-21	https://eservices.dot.gov.in/sites/default/files/circular-notifications/Notification%20for%20enforcement%20of%20sections%20of%20the%20Telecommunications%20Act%202023.pdf
S.O. 2623(E) — commences ss. 6–8, 48, 59(b) from 2024-07-05	2024-07-04	https://eservices.dot.gov.in/sites/default/files/circular-notifications/Notification%20for%20enforcement%20of%20sections%20of%20the%20Telecommunications%20Act%202023.pdf
S.O. 3368(E) — commences s. 3(1) and s. 3(6) from 2026-06-23	2026-06-23	https://egazette.gov.in/WriteReadData/2026/273762.pdf and https://www.dot.gov.in/static/uploads/2026/06/35e1d307f3cbb6396b4c48802ed4b0.pdf
S.O. 3370(E) — notification of the Telecom eServices Portal under s. 53	2026-06-23	https://eservices.dot.gov.in/sites/default/files/circular-notifications/notification%20of%20the%20Telecom%20eServices%20Portal%20under%20s.%2053.pdf
Telecommunications Act 2023, s. 1 with commencement footnotes	—	https://www.advocatekhoj.com/library/bareacts/telecommunications/1.php

D.3 Subordinate rules

Document	Date	URL
Telecommunications (Amateur Services) Rules 2024, G.S.R. 675(E)	notified 2024-10-29, published 2024-10-30	https://eservices.dot.gov.in/sites/default/files/circular-notifications/amateur-services-rules-2024.pdf
Telecommunications (Radio Equipment Possession Authorisation) Rules 2026, G.S.R. 592(E)	2026-07-08	https://eservices.dot.gov.in/sites/default/files/circular-notifications/telecommunications-radio-equipment-possession-authorisation-rules-2026.pdf
Telecommunications (Authorisation for Captive Telecommunication Services) Rules 2026, G.S.R. 511(E)	2026-06-23	https://eservices.dot.gov.in/sites/default/files/reference-doc/captive-telecommunications-services-rules-2026.pdf

Document	Date	URL
Telecommunications (Authorisation for Provision of Miscellaneous Telecommunication Services) Rules 2026, G.S.R. 512(E)	2026-06-23	https://eservices.dot.gov.in/sites/default/files/circular-notifications/miscellaneous-telecom-services-rules-2026.pdf
Telecommunications (Authorisation for Provision of Principal Telecommunication Services) Rules 2026, G.S.R. 513(E)	2026-06-23	https://eservices.dot.gov.in/sites/default/files/circular-notifications/principal-telecom-services-rule-2026.pdf
Telecommunications (Terms and Conditions for Migration) Rules 2026, G.S.R. 510(E)	2026-06-23	https://eservices.dot.gov.in/sites/default/files/circular-notifications/migration-rules-2026.pdf
Draft Captive Telecommunication Services Rules 2025, G.S.R. 614(E) (superseded)	2025-09-10	https://thc.nic.in/Central%20Governmental%20Rules/Telecommunications%20(Authorisation%20for%20Captive%20Telecommunication%20Services)%20Rules,%202025.pdf
Indian Wireless Telegraphy (Possession) Rules 1965 — rule 3	1965	https://indiankanoon.org/doc/90497701/
Possession Rules 1965 — rule 5 (exemptions)	1965	https://indiankanoon.org/doc/77506268/
Possession Rules 1965 — rule 9 (fee)	1965	https://indiankanoon.org/doc/54047780/
Possession Rules 1965 — rule 16 (inspection)	1965	https://indiankanoon.org/doc/172541687/
Possession Rules 1965 — rule 19 (surcharge)	1965	https://indiankanoon.org/doc/93223764/

D.4 Licence-exemption notifications

Document	Date	URL
G.S.R. 1047(E) — Use of Low Power and Very Low Power Short Range Radio Frequency Devices (Exemption from Licensing Requirement) Rules 2018; Table V is PMR446, 446.0–446.2 MHz, 500 mW e.r.p.	2018-10-18	https://www.legitquest.com/act/use-of-low-power-and-very-low-power-short-range-radio-frequency-devices-exemption-from-licensing-requirement-rules-2018/9F4C
G.S.R. 271(E) — 2024 amendment to the 2018 exemption rules	2024-05-09	https://eservices.dot.gov.in/sites/default/files/circular-notifications/use-of-low-power-and-very-low-power-short-range-radio-frequency-devices-exemption-from-licensing-requirement-amendment-rules-2024.pdf
G.S.R. 468(E) — Short-Range Automotive Radar 77–81 GHz exemption rules, made under the 1885 and 1933 Acts (key evidence that s. 60 is not commenced)	2026-06-11	https://www.dot.gov.in/static/uploads/2026/06/d44ff2aec0505d538d9f3d51eba88936.pdf
National Frequency Allocation Plan 2022 (mirror hosted by the Cellular Operators Association of India, COAI; the dot.gov.in copy 404s)	2022	https://coai.com/storage/files/2/spectrum/NFAP%202022%20Document%20for%20e-release.pdf

D.5 WPC orders, office memoranda and FAQs

Document	Date	URL
WPC Order P-11014/34/2009-PP — Spectrum Charges for Assignment of Frequencies to Captive Users	2023-12-11	https://eservices.dot.gov.in/sites/default/files/circular-notifications/Spectrum%20Charges%20for%20Assignment%20of%20Frequencies%20to%20Captive%20Use
WPC Compendium on import licensing, R-11018/02/2017-PP	2022-07-06	https://eservices.dot.gov.in/sites/default/files/circular-notifications/Compendium%20of%20Orders%20related%20import%20licence%20-signed%20copy%2006072

Document	Date	URL
WPC OM R-11017/01/2018-PP(part-1) — ETA for licence-exempt devices on self-declaration	2024-09-09	https://eservices.dot.gov.in/sites/default/files/2024-11/Issuance%20of%20Equipment%20Type%20Approval%20%28ETA%29%20forLicense-Exempt.pdf
WPC OM K-19013/13/2005-CFA — simplification of SACFA siting clearance (₹100 rates)	2022-05-09	https://eservices.dot.gov.in/sites/default/files/2024-11/Simplification%20of%20SACFA%20siting%20clearance%20guidelines.pdf
WPC OM K-19012/16/2019-CFA — simplified SACFA clearance for towers (deemed clearance after 30 days)	2022-01-12	https://eservices.dot.gov.in/sites/default/files/2024-11/Guidelines%20for%20simplified%20SACFA%20clearance%20process%20for%20installing.pdf
WPC OM K-19013/13/2005-CFA — SACFA self-declaration / automated approvals	2021-10-06	https://eservices.dot.gov.in/sites/default/files/2024-11/Simplification%20of%20SACFA%20clearance%20process%20for%20installing%20towers.pdf
WPC OM P-14036/03/2023-COP — launch of the modified Amateur/HAM module, with the DoT user manual and the two nationality proformas	2025-02-28	https://arsi.info/wp-content/uploads/2025/03/Launch-of-Modified-Amateur_HAM-MoD-with-User-manual.pdf
WPC OM P-14036/03/2023-COP — launch of the Amateur/HAM migration module	2025-03-11	https://arsi.info/wp-content/uploads/2025/03/WPC-Migration-to-Saral-Sanchar.pdf

Document	Date	URL
WPC OM P-14036/03/2023- COP — migration deadline extended to 2025-12-15	2025-10-09	https://saralsanchar.gov.in/circular/HAM/Extension_HAM_Mig.pdf
WPC OM L-14011/01/2023- COP — migration deadline extended to 2026-06-15	2026-01-05	https://qsl.net/vu2jos/Extension.pdf
WPC OM R-11014/03/2015-PP — removal of security clearance for amateur licences	2016-03-10	https://arsi.info/wp-content/uploads/2020/11/Security_Clearance-OM-R-11014032
WPC OM P-11015/01/2019- COP — amateur stations may operate anywhere in India except restricted locations	2019-06-07	https://arsi.info/wp-content/uploads/2020/11/OM-N0.P-11015.01.2019-COP-dated0
WPC OM L-14011/255/2004- AMT — guidelines with Annexure I (bands, emissions, power) and Annexure II (syllabus and exam scheme)	2010-08-13	https://arsi.info/wp-content/uploads/2020/11/Amateur_New_Guidelines-L-14011-2
WPC FAQ P-14036/02/2023- COP — FAQ on the Telecommunication (Amateur Service) Rules 2024 (image- only scan)	2024-12-05	https://arsi.info/wp-content/uploads/2024/12/Frequently-Asked-Questions-on-the-Amateur-Service-Rules-2024.pdf

Document	Date	URL
Saral Sanchar official FAQ for amateur examination and licence (exam centres, fees, Bharatkosh heads)	PDF created 2020-08-24	https://saralsanchar.gov.in/circular/licenses_issued/FAQ_for_HAM.pdf
WPC instruction sheet for online ASOC filing (Bharatkosh PAO/DDO tables)	undated	https://qsl.net/vu2jos/WPC2.pdf

D.6 DoT and portal service pages (all fetched 2026-08-23)

Page	URL
Amateur Station Operator Certificate (ASOC)	https://eservices.dot.gov.in/amateur-station-operator-certificate-asoc
Certificate of proficiency (ASOC, RTR, GMDSS)	https://eservices.dot.gov.in/certificate-proficiency-asoc-rtr-gmdss
Fees and Charges for Services	https://eservices.dot.gov.in/fees-charges-for-services
Document Required for Services	https://eservices.dot.gov.in/document-required-for-services
Equipment Type Approval (ETA)	https://eservices.dot.gov.in/equipment-type-approval-eta
Import Licence	https://eservices.dot.gov.in/import-license
Dealer / Non-Dealer Possession Licence	https://eservices.dot.gov.in/dealer-possession-license
Radio Equipment Possession Authorisation	https://eservices.dot.gov.in/radio-equipment-possession-authorisation-services
Network (index of the nine WPC network sub-categories)	https://eservices.dot.gov.in/network
Network — Short Range UHF Hand Held Radio	https://eservices.dot.gov.in/network-short-range-uhf-hand-held-radio
Network — Fixed/Land Mobile HF/VHF/UHF below 806 MHz	https://eservices.dot.gov.in/network-license-fixedland-mobile-hfvhfuhf-below-806-mhz
Non-Network (MMSL, Aero Mobile)	https://eservices.dot.gov.in/non-network

Page	URL
SACFA clearance	https://eservices.dot.gov.in/sacfa-standing-advisory-committee-frequency-allocation-clearance
CMRTS authorisation (new regime)	https://eservices.dot.gov.in/captive-mobile-radio-trunking-service-cmrts-authorisation
Captive General Service authorisation	https://eservices.dot.gov.in/captive-general-service-authorisation
Captive Non-Public Network authorisation	https://eservices.dot.gov.in/captive-non-public-network-cnpn-service-authorisation
Document checklist (corporate)	https://eservices.dot.gov.in/saral/vps/document-checklist
Authorisation Portal (live 2026-06-25; REPA from 2026-08-06)	https://eservices.dot.gov.in/authorisation-portal
Acts, Rules & Notifications index	https://eservices.dot.gov.in/act-and-rules
Saral Sanchar portal home (carries the transition notice)	https://saralsanchar.gov.in/
Saral Sanchar WPC user-manual index (54 modules)	https://www.saralsanchar.gov.in/wpc_usermanual_list_for_home.php
DoT user manual, "Online Registration for Amateur Examinations and Certificates"	https://eservices.dot.gov.in/sites/default/files/user-mannual/ASOC%20Manual%20(2).pdf
Bharatkosh (Non-Tax Receipt Portal)	https://bharatkosh.gov.in

D.7 Official press releases and audit reports

Document	Date	URL
PIB Release ID 2127931 — CCPA issues 13 notices over illegal walkie-talkie listings	2025-05-09	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2127931&reg=3&lang=2
PIB Release ID 2215261 — CCPA final orders and penalties on walkie-talkie sales	2026-01-16	https://www.pib.gov.in/PressReleasePage.aspx?PRID=2215261&reg=3&lang=1 (PDF copy: https://consumeraffairs.gov.in/public/upload/admin/cmsfiles/pressRelease/CCPA_takes_action_against_Unauthorized_Sale_of_Walkie-Talkies_on_E-Commerce_Platformspress_release.pdf)
PIB Release ID 1988428 — WPC and WMO, International Monitoring Station Day (WMO structure)	2023-12-19	https://www.pib.gov.in/PressReleaseIframePage.aspx?PRID=1988428&reg=48&lang=2

Document	Date	URL
PIB Release ID 2277786 — DoT opens the Authorisation Portal	2026-06-25	https://www.dot.gov.in/static/uploads/2026/06/d3e5ac6a965d27b215b5aad8a8d0e34c.pdf
CAG Report No. 21 of 2018, Chapter II — DoT spectrum management, WPC process, SACFA, RLOs	2018	https://cag.gov.in/uploads/download_audit_report/2018/Chapter_2_Ministry_of_Communications_Department_of_Telecommunications_of_Report_No_21_of_2018_Complia.pdf
TRAI, Recommendations on Ease of Doing Business in Telecom and Broadcasting	2023-05-02	https://www.trai.gov.in/sites/default/files/2024-09/Recommendations_02052023.pdf

D.8 Judgments

Case	Court and date	URL
<i>Jethanand Betab v The State of Delhi</i> — 1960 AIR 89	Supreme Court, 1959-09-15	https://indiankanoon.org/doc/125699/
<i>State of Bihar v Mangal Sao</i> — 1963 AIR 445	Supreme Court, 1962-03-29	https://indiankanoon.org/doc/766168/
<i>Ramakrishna Dish Antenna System v Deputy Superintendent of Police</i>	Andhra Pradesh High Court, 1989-03-07	https://indiankanoon.org/doc/1208726/
<i>Divyang R. Parikh v State of Gujarat</i>	Gujarat High Court, 1997-04-08	https://indiankanoon.org/doc/601828/ (duplicate at https://indiankanoon.org/doc/648474/)
<i>State of Gujarat v Anwar Osman Sumbhaniya</i>	Supreme Court, 2019-02-27	https://indiankanoon.org/doc/120823041/
<i>Union of India v Ashok Kumar Sharma</i> — CrPC First Schedule Part II classification	Supreme Court, 2020-08-28	https://indiankanoon.org/doc/181202406/
<i>Prasanth K.S v State of Kerala</i> — BNSS s. 174	Kerala High Court, 2024-09-05	https://indiankanoon.org/doc/100888972/
<i>Frederick Joseph Iseman v State NCT of Delhi</i> , CRL.M.C. 8739/2024	Delhi High Court, 2024-11-12	https://indiankanoon.org/doc/72044028/
<i>Meta Platforms, Inc v Union of India & Anr</i>	Delhi High Court, 2026-03-25	https://indiankanoon.org/doc/37518660/

D.9 Study resources, clubs and exam material

Resource	URL
NIAR Study Manual (Restricted & General Grade), 147 pp	http://www.niar.org/downloads/Study-Manual-General-Grade-22.pdf
NIAR Study Manual (Restricted Grade), 117 pp	http://www.niar.org/downloads/Study-Manual-Restricted-Grade-22.pdf
NIAR Study Manual (older edition), 158 pp	http://www.niar.org/downloads/ham-downloads/Study-Manual.pdf
NIAR training material and prices	http://www.niar.org/training-material.html
NIAR list of Wireless Monitoring Stations (addresses)	http://www.niar.org/downloads/ham-downloads/List-of-Wireless-Monitoring-Stations.pdf
NIAR "How to be a HAM"	http://www.niar.org/how-to-be-a-ham.html
Official WPC sample question paper (Wayback copy of the retired WPC site)	https://web.archive.org/web/20160804195151if_/http://wpc.dot.gov.in/DocFiles/Saple%20Q-Paper%20for%20ASOC.pdf
VU2JOS syllabus and examination details (PDF created 2024-11-13)	https://qsl.net/vu2jos/Syllabus.pdf
VU2JOS guide to the online exam application, 2025-03-21	https://www.qsl.net/vu2jos/Exam.pdf
VU2JOS guide to the online licence application, 2025-03-01	https://www.qsl.net/vu2jos/License.pdf
VU2JOS revised fee structure with effect from 2024-10-30	https://qsl.net/vu2jos/VU_Fees_24.pdf
VU2JOS SACFA clearance for radio amateurs	https://qsl.net/vu2jos/SACFA.pdf
VU2JOS new call-sign prefixes and issued call-sign list	https://qsl.net/vu2jos/New_Bio.pdf and https://qsl.net/vu2jos/New_Calls.htm
ARSI important WPC documents (the best-curated legal reading list)	https://arsi.info/important-wpc-documents/
ARSI procedure for ASOC exam and certificate	https://arsi.info/asoc/
ARSI affiliated clubs	https://arsi.info/affiliated-clubs/
ARSI membership	https://arsi.info/arsi-membership/
DARTS question bank (unsolved) and sample papers	https://darts.org.in/question-papers-unsolved/
DARTS pre-2020 paper form set (obsolete — reference only)	https://darts.org.in/wp-content/uploads/2018/02/ASOC-Exam-Form.pdf

Resource	URL
Indian Institute of Hams FAQ (WPC-authored, pre-2024 fee figures)	https://www.indianhams.com/faq.pdf

D.10 Secondary analysis and news

Source	Date	URL
Internet Freedom Foundation — "The Telecom Act, 2023 is ready for (partial) take off"	June 2024	https://internetfreedom.in/the-telecom-act-2023-partial-notification/
Trilegal — "Government notifies sections of the Telecommunications Act, 2023"	2024-07-26	https://trilegal.com/wp-content/uploads/2024/07/Trilegal-Update_Government-notifies-sections-of-the-Telecommunications-Act-2023-For-publication-final.pdf
AZB & Partners — staggered enforcement of the Telecommunications Act 2023	June 2024	https://www.azbpartners.com/wp-content/uploads/2024/06/Update-Staggered-Enforcement-of-Telecommunications-Act-2023.pdf
Khaitan & Co ERGO — Captive Service Authorisation Rules	2026-06-25	https://www.khaitanco.com/sites/default/files/2026-06/ERGO_Captive%20Service%20Authorisation%20Rules_25062026.pdf
Mondaq — notification of s. 3(1) and s. 3(6)	2026	https://www.mondaq.com/india/telecoms-mobile-cable-communications/1823106/notification-of-section-31-and-section-36-of-telecommunications-act-2023-and-operative-rule
MediaNama — CCPA 2025 guidelines on illegal radio-equipment sales	2025-06-03	https://www.medianama.com/2025/06/223-ccpa-2025-guidelines-illegal-radio-sales/
MediaNama — DoT notifies network authorisation rules	2026-07-24	https://www.medianama.com/2026/07/223-dot-telecommunications-authorisation-rules-2026/
The Hindu BusinessLine — cluster of notifications under the Telecommunications Act	2026-06-23	https://www.thehindubusinessline.com/info-tech/centre-releases-cluster-of-notifications-under-telecommunications-act-2023/article71141158.ece
The Hindu — Explained: the Telecommunications Act 2023 comes into force	2026-06-27	https://www.thehindu.com/news/national/explained-the-telecommunications-act-2023-comes-into-force/article71153558.ece
Free Press Journal — DRI seizes banned walkie-talkies and e-waste HDDs, 2 arrested	2026-03-21/22	https://www.freepressjournal.in/mumbai/mumbai-dri-flags-illicit-imports-from-china-seizes-banned-walkie-talkies-e-waste-hdds-worth-925-crore-2-arrested

Source	Date	URL
Deccan Herald (PTI) — same DRI seizure, arrests under the Customs Act	2026-03-22	https://www.deccanherald.com/india/maharashtra/prohibited-walkie-talkies-e-waste-hard-disks-imported-from-china-seized-by-dri-in-mumbai-2-held-3940157
Hindustan Times — Meerut illegal exchange; FIR under the Telecommunications Act 2023 and the Indian Telegraph Act	2026-05-21	https://www.hindustantimes.com/cities/illegal-telephone-exchange-fake-online-loan-racket-busted-in-up-s-meerut-3-arrested-101779377479109.html
Livemint — CCPA fines TradeIndia ₹10 lakh	2026-02-12	https://www.livemint.com/news/india/ccpa-widens-crackdown-fines-tradeindia-10-lakh-over-uncertified-walkie-talkie-sale-11770878341945.html
Livemint — why India is cracking down on walkie-talkie sales online	2026-01-20	https://www.livemint.com/industry/india-ccpa-crackdown-illegal-walkie-talkie-sales-11768904205794.html
MPR India Certification — DoT OM R 11018/13/2026-PP suspending DPL/NDPL processing (OM itself not obtained)	on or after 2026-07-02	https://www.certification-india.com/en/india-suspends-key-wpc-licenses-under-regulatory-update/
Centre for Internet and Society — section-by-section summary of the IWTA 1933 (undated)	—	https://cis-india.org/telecom/resources/indian-wireless-telegraphy-act

D.11 Sources that could not be fetched

Recorded so that no one repeats the attempt. Several of these are the *most* authoritative copies of documents quoted above, and a reader with an ordinary browser may succeed where automated fetching failed.

Source	Failure mode
https://www.indiacode.nic.in/ ... (the official India Code copies of all three Acts)	HTTP 403 / 504, Akamai block
https://egazette.gov.in/WriteReadData/2023/250880.pdf (on some attempts)	TLS certificate verification failure
https://dot.gov.in/telecommunications-act-2023 , https://dot.gov.in/actrules/india-wireless-act-1933 , https://dot.gov.in/spectrum-management/2457	JavaScript single-page application; empty response to automated fetching, HTTP 403 to WebFetch
http://wpc.dot.gov.in/	Does not respond; the standalone WPC site appears retired

Source	Failure mode
https://dca.gov.in/ccpa/orders-advisories.php — the index of CCPA final orders against Amazon, Flipkart, Meesho, Meta, JioMart, Talk Pro, Chimiya and MaskMan Toys	HTTP 403. Worth retrieving from a normal browser — these are the primary orders behind the penalties described in §6.4.7
https://www.pib.gov.in/PressReleasePage.aspx?PRID=2132575 (CCPA guidelines release)	HTTP 403 on the routes tried for this part . Note the conflict: the research behind Part 2 §2.10.1 did retrieve this release and quotes it verbatim, so the failure is route-specific, not absolute. The release is treated as [VERIFIED] on the strength of that retrieval
https://consumeraffairs.nic.in/.../Guidelines_27052025.pdf (text of the CCPA 2025 Guidelines)	HTTP 403 / no response
https://www.niar.org/downloads/ over HTTPS	TLS certificate mismatch; use plain HTTP
https://dot.gov.in/rtis/dotelr201950891 (a WPC RTI reply on walkie-talkie licensing)	HTTP 403

Appendix E — Legal disclaimer

This report is a research summary compiled on 2026-08-23. It is not legal advice, and it must not be relied on as a substitute for advice from a qualified Indian telecom lawyer or from the Wireless Planning and Coordination Wing itself.

Specific limitations the reader must keep in mind:

- 1. Indian radio law is mid-transition.** The Telecommunications Act 2023 is being commenced in stages. Sections 3(1) and 3(6) came into force only on 2026-06-23; the Radio Equipment Possession Authorisation Rules only on 2026-07-08, with the application portal opening on 2026-08-06. Commencement notifications, rules, portals and fees are changing month by month.
- 2. The whole penalty analysis turns on three commencement questions that could not be settled from the available sources,** and each is flagged **[UNCERTAIN]** in §6.1: whether **section 60** (repealing the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933) has been commenced; whether **Chapter VIII (sections 31–41)**, the Adjudicating Officer machinery that alone can impose Third Schedule civil penalties, has been commenced; and whether an Adjudicating Officer has actually been appointed. **Put all three to WPC in writing before relying on anything in this part.** Official sources for these questions (India Code, the eGazette search and dot.gov.in) were unreachable during the research; the conclusions rest on gazette notifications, corroborating rule-making under the old Acts, and named law-firm commentary.

3. **Evidence tags are not decoration.** A claim marked **[REPORTED]** rests on a secondary source and may be wrong. A claim marked **[UNCERTAIN]** could not be confirmed at all. Only claims marked **[VERIFIED]** are traceable to a primary or official source that was actually fetched, and even those carry the date of the document they came from — fees, forms, portals, band plans and penalties all change.
4. **Where the research sources disagreed, both positions have been shown** rather than silently resolved. The most important such conflict is whether the 1933 Act is still in force: this report proceeds on the basis that it **is**, because that is where the primary evidence points, but a contrary view exists.
5. **No absence of evidence has been treated as evidence of absence.** In particular, no reported prosecution of an individual hobbyist for a plain walkie-talkie alone was found — that is a gap in the record, not a licence to assume immunity.
6. **Nothing here authorises anything.** Possessing or operating radio equipment outside a licence, authorisation or exemption exposes you to seizure of the equipment and to criminal proceedings, irrespective of intent, of personal or non-commercial use, or of what the seller told you.

Before acting, confirm the current position with the WPC Amateur Section (awacop-dot@gov.in, 011-2303 6181), with the DoT eServices nodal cell (011-2303 6540), or with a licensed amateur radio club such as NIAR (info@niar.org, +91 40 2331 0287) or ARSI.

Part 7 — Choosing a radio: a worked example

Parts 1 to 6 set out the law. This part applies it to one concrete decision, because that is the form in which the question is actually faced: **should I spend ₹14,449 on a Baofeng UV-32 from the Baofeng India storefront, mount it in my vehicle, and use it for overland expeditions and for learning radio?**

The answer is no — but the reasons are more useful than the verdict, because three of them generalise to every radio purchase an Indian buyer will ever make.

Position as at 2026-08-23. All prices, listings and web pages described here were fetched on that date and are volatile. Evidence tags carry the same meaning as in Part 1 §1.1: **[VERIFIED]** = a primary or official source actually fetched, with URL and document date; **[REPORTED]** = a named credible secondary source, or an official document obtained only from a mirror; **[UNCERTAIN]** = could not be confirmed, sources disagree, or the source is undated. One further tag is used in this part where the underlying research used it: **[ENGINEERING ESTIMATE]** — a range or performance figure derived from propagation and circuit fundamentals, which is a technical judgement and never a legal claim.

Abbreviations expanded on first use in this part: **ASOC** = Amateur Station Operator Certificate, the Indian amateur radio licence (Part 3). **WPC** = Wireless Planning and Coordination Wing of the **DoT** = Department of Telecommunications. **ETA** = Equipment Type Approval. **DPL** = Dealer

Possession Licence. **RF** = radio frequency; **DC** = direct current. **VHF / UHF** = very high frequency (30–300 MHz) / ultra high frequency (300–3000 MHz). **PMR446** = Personal Mobile Radio in the 446 MHz band. **CB** = Citizen Band. **SMA** = the miniature screw antenna connector used on hand-held radios. **e.r.p.** / **ERP** = effective radiated power, referenced to a half-wave dipole and including antenna gain. **FCC** = the United States Federal Communications Commission. **CIN** = Corporate Identity Number, the Indian company registry identifier. **MCA** = Ministry of Corporate Affairs. **CCPA** = Central Consumer Protection Authority. **VFO** = variable frequency oscillator, the mode in which a radio is tuned to an arbitrary frequency from the keypad rather than to a stored channel.

7.1 If you only read one page

7.1.1 THE VERDICT TABLE

The question actually being asked	Answer	Evidence
Is the UV-32 legal to buy in India?	For a seller with a valid dealer authorisation, selling to an ASOC holder — yes. For you, today, with no ASOC — no lawful possession route exists. The storefront's own page data does not enforce this; the law does.	[VERIFIED] on the seller's compliance documents and the page data; the ASOC possession gateway is [VERIFIED] on the original retrieval of rule 10(f) and [UNCERTAIN] after the 2026-08-23 fact-check — Part 1 §1.5 Group B, Part 2 §2.9.3
Is it legal to use?	Only with an ASOC, only inside 144–146 MHz and 434–438 MHz , and — for the entry-level Restricted grade — realistically only on its lowest power step .	[VERIFIED] on the bands and the ceiling; the mapping of the ceiling onto this radio's power steps is inference, [UNCERTAIN]
How much of the radio may you lawfully key up?	About 3 % . Advertised transmit coverage is roughly 198 MHz wide; the Indian amateur portion of it is 6 MHz .	[VERIFIED] — §7.3.1
Will it solve the convoy problem?	No, and no radio on the amateur bands will. Amateur transmissions may be exchanged only with other authorised amateur stations, and third-party messages are expressly banned.	[VERIFIED] — §7.2.2
Will it solve the vehicle-mount problem?	Yes in law — a vehicle-mounted amateur station needs nothing beyond the ASOC — but a hand-held is the wrong tool, and this hand-held is the wrong hand-held.	[VERIFIED] — §7.2.1

The question actually being asked	Answer	Evidence
Will it solve the learning problem?	Partly, and less well than cheaper alternatives. Self-training is the statutory purpose of the amateur service; but a Restricted ASOC's genuinely useful expedition capability is 50 W DC input on ten high-frequency bands , which this radio cannot reach at all.	[VERIFIED] — §7.2.3
Does the FCC ID on the label mean anything?	No. It covers the Bluetooth module and the receiver. The VHF/UHF transmitter has no transmitter grant of any kind, and no United States or European mark has any legal effect in India.	[REPORTED] on the grant contents (mirror); [VERIFIED] that neither mark operates in India
Is there an Indian ETA for it?	None, and none can exist. ETA certifies conformity with a <i>licence-exempt band</i> notification. This is licensed-band equipment.	[VERIFIED] as a matter of law — Part 2 §2.7.1; [UNCERTAIN] as a matter of register record, because the ETA register is CAPTCHA-gated
Is the India storefront a scam?	No. It is a real Indian company with a real DPL number, real ETA certificates on its 0.5 W models, and honest labelling of the UV-32 as a licence-band radio. That protects the seller, not you.	[VERIFIED] / [REPORTED] — §7.4

7.1.2 THE RECOMMENDED PLAN, IN ONE TABLE

Order	Action	Cost	Why this order
1	Sit the ASOC (Restricted) examination. No Morse code, 40 % pass mark, minimum age 12.	₹100 examination fee + ₹1,000 (20 years) or ₹2,000 (lifetime)	Nothing downstream — possession, import, transmission — becomes lawful before this, and owning hardware first makes the exam no easier. It does make you unlawful in the interval. [VERIFIED] — Part 3 §3.7
2	Buy licence-free radios for the convoy now , in parallel with step 1. Citizen Band at 5 W ERP if the antenna question at §7.6.1 resolves acceptably; ETA-approved PMR446 at 500 mW otherwise or as well.	₹6,000–₹12,000 per vehicle (CB) or ₹3,000–₹9,000 per handset (PMR446)	This is the <i>only</i> lawful way to talk to unlicensed travelling companions, and it is available today with no exam and no licence. [VERIFIED]

Order	Action	Cost	Why this order
3	After the ASOC issues, buy a band-limited amateur radio — a European-version Yaesu FT-65E or FT-4XE class hand-held, or a proper mobile transceiver for the vehicle.	Imported; higher than ₹14,449	12 MHz of transmit coverage instead of 198 MHz. That single choice removes almost all of the equipment risk in this part. [VERIFIED] on the coverage figures — \$7.5
4	Get the rest of the convoy licensed if you want lawful high-power in-convoy communication.	₹1,100 or ₹2,100 per person	Once every vehicle has its own ASOC holder at its own radio, 2 metre and 70 centimetre working at 10 W becomes lawful for the group. This is the only route to it. [VERIFIED] — \$7.2.2
5	Buy the UV-32 only if you still want it , after the ASOC, from a dealer that checks the certificate, run on low power, memories programmed to Indian amateur frequencies.	₹14,449	Defensible, but you are paying three times the manufacturer's direct price for a radio whose wideband transmit capability is a permanent liability no configuration removes. [VERIFIED] on the price comparison

The blunt version. The UV-32 does not do the thing the reader most wants a radio to do — talk to the other vehicles — and it cannot be made to do it by any licence, any configuration or any purchase channel. It does the thing the reader can already do more cheaply (learn radio) while carrying more legal risk than any alternative. Spending ₹14,449 on it now, before the examination, buys an unlawful possession, an unreturnable serial number on a dealer's traceability register, and no lawful transmission at all.

7.2 The three intended uses, judged separately

These three uses are usually described in one breath — "a radio for the truck, for the convoy, and to learn with" — and they have three completely different answers. Separating them is the single most useful thing this part does.

7.2.1 USE 1 — MOUNTING A RADIO IN A VEHICLE AND OPERATING MOBILE

Verdict: lawful, and easier than most people expect. With an ASOC, on amateur frequencies, within the power ceiling, there is nothing else to obtain — no separate mobile authorisation, no notification, no fee, no siting clearance.

This is a recent liberalisation and is worth stating precisely, because superseded advice is still in wide circulation.

Rule 7(2) of the Telecommunications (Amateur Services) Rules, 2024 reads, on its face, like a bar:

"(2) Every amateur station shall be installed and operated at the location specified in the ASOC in accordance with,- (a) the provisions of the Act and any rules made thereunder, as may be applicable; (b) the terms and conditions specified in Annexure A to these rules; and (c) the provisions of the International Telecommunication Convention or Radio Regulations."

[VERIFIED] — G.S.R. 675(E), 2024-10-29, rule 7(2), gazette PDF read 2026-08-23.

WPC has said in writing that it is not a bar. From WPC's FAQ No. P-14036/02/2023-COP dated 2024-12-05, signed by the Engineer (COP):

"**Question 2:** Sub-rule 2 of Rule 7 mentions that every amateur station shall be installed and operated at the location specified in the ASOC. **Reply:** Sub-rule 2 of Rule 7 specifies the usual operating location of the ASOC holder, which is the place from where the holder typically operates the amateur station."

"**Question 5:** Can amateur station equipment be carried while traveling outside the ASOC location? **Reply:** The Amateur Service Rules, 2024 do not prohibit an ASOC holder from carrying amateur station equipment while traveling outside the specified ASO Certificate location, **except in areas declared restricted by the Government.**"

[VERIFIED] — FAQ pages 1–2, read 2026-08-23 from a byte-identical public mirror because eservices.dot.gov.in returned HTTP 403 to the automated fetcher; cross-checked against the National Institute of Amateur Radio (NIAR) compilation's index, which lists the same FAQ at the same date.

What a permanently vehicle-mounted amateur station needs beyond the ASOC:

Requirement	Applies to a vehicle-mounted amateur station?	Authority
Operating authorisation	Yes — your ASOC. That is the whole permission.	Rules 3(1) and 7(1) [VERIFIED]
Separate mobile or portable authorisation	No — abolished. The 1978 Rules required a special authorisation, a ₹20 fee, an "/MO" callsign suffix and a test report within 15 days. All of it went when the 1978 Rules were superseded on 2024-10-30.	G.S.R. 675(E) preamble; nothing in rules 1–15 [VERIFIED]. The 1978 rule 18 text is [REPORTED] — a transcript, not a gazette scan
Change-of-location filing (₹1,000)	Only if you park and operate at one non-ASOC site for more than 90 days	Rule 7(6) plus FAQ Q4 [VERIFIED]
SACFA (Standing Advisory Committee on Frequency Allocation) siting clearance	No. WPC's FAQ Q6: clearance is required "if the amateur station antenna is installed on a tower, building, or mast ". A magnetic-mount or boot-lip whip on a car is none of the three.	Annexure A para 4(3) plus FAQ Q6 [VERIFIED]; see Part 1 §1.5 Group D

Requirement	Applies to a vehicle-mounted amateur station?	Authority
Radio Equipment Possession Authorisation	No , on rule 10(f) of the 2026 Rules	[VERIFIED] on the original retrieval; [UNCERTAIN] after the fact-check — Part 1 §1.5 Group B
Import authorisation	No . "An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment."	Rule 7(3) [VERIFIED]
ETA	Not applicable and not obtainable for amateur-band equipment	Part 2 §2.7.1 [VERIFIED]
Road transport authority endorsement for a fitted transmitter	Nothing found in the Central Motor Vehicles Rules	[UNCERTAIN] — absence of evidence, not evidence of absence

Four obligations *do* follow you into the vehicle, and they are easy to forget because they feel like home-station paperwork:

1. **The station log, in a bound book.** Annexure A para 6(1) requires a chronological record of all transmissions "in **bound book (not loose-leaf)**", in Indian Standard Time, with no gaps, made at the time of transmitting, preserved one year from the last entry — and, in the clause written for exactly this use case, "**(f) in case of portable or mobile amateur station, the particulars of temporary location.**" **[VERIFIED]** The Rules do not authorise an electronic log; whether WPC accepts one in practice is **[UNCERTAIN]**. Carry a physical notebook.
2. **The equipment register.** Annexure A para 7(4): make, model, serial number, from whom received, date, seller's receipt and certificate numbers. **[VERIFIED]** Enter the radio on the day you acquire it.
3. **Inspection on demand.** Rule 14 makes it "the duty of every ASOC holder to produce for inspection, records, as well as, amateur radio equipment". **[VERIFIED]** Keep the ASOC, the log and the register **in the vehicle**, not at home. On an overland trip you are the most likely licensee in India to be asked.
4. **Restricted areas.** FAQ Q3 and Q5 both carve out "areas declared restricted by the Government". **[VERIFIED]** **No published list of those areas could be found.** **[UNCERTAIN]** This is the single biggest practical risk for an overland expedition, because Indian overlanding routes run through precisely the sensitive belts — Ladakh, the Line of Control districts, Arunachal Pradesh and the Inner Line Permit states, the Andaman and Nicobar Islands. Note also that Protected Area Permit and Inner Line Permit regimes, and section 163 of the Bharatiya Nagarik Suraksha Sanhita prohibitory-order power, operate entirely independently of telecom law: a District Magistrate can lawfully ban radio equipment in a district for reasons that have nothing to do with WPC.

And one that is not telecom law at all. Section 184 of the Motor Vehicles Act, 1988, as amended in 2019, contains a **deeming provision**:

"[Explanation. — For the purpose of this section,- ... **(c) use of handheld communications devices while driving;** ... shall amount to driving in such manner which is dangerous to the public.]"

[VERIFIED] — Motor Vehicles Act 1988, s.184, fetched 2026-08-23. The first-offence penalty includes imprisonment "which may extend to one year but shall not be less than six months", or a fine of ₹1,000 to ₹5,000, or both.

A hand-held radio keyed by the driver of a moving vehicle is a "handheld communications device" on any ordinary reading of those words, and the prosecution does not have to prove the use was in fact dangerous — the Explanation deems it. **[VERIFIED]** as to the text; **[UNCERTAIN]** as to whether any Indian court has applied clause (c) to a two-way radio rather than a mobile telephone, because no reported decision was found. Whether a hook-mounted fist microphone on a fixed rig falls inside clause (c) is **[UNCERTAIN]** and there is no authority either way.

The practical rule to adopt: do not transmit while the vehicle is moving unless the front passenger is doing it, or you have a genuine hands-free installation. On an expedition this costs nothing — designate the passenger as the radio operator — and it is also the correct answer under Annexure A, since whoever keys the transmitter must be the ASOC holder.

Where this leaves the UV-32 specifically. It is a legal vehicle radio for a licensee, and its detachable SMA antenna makes a roof mag-mount trivial to fit. It is also, for the same reason, a *visible* installation: a whip on the roof converts a pocket radio into an identifiable station that invites the question "may I see your licence?". A purpose-built mobile transceiver would be better on audio, heat dissipation and transmit duty cycle (\$7.5.2), and a hand-held is a compromise in a vehicle whatever its badge.

7.2.2 USE 2 — CONVOY COMMUNICATION WITH UNLICENSED COMPANIONS

Verdict: not lawful. This is not a close question, and it should not be softened.

This is the use most people actually buy the radio for, and it is the one the amateur licence categorically does not deliver. Four independent provisions each prohibit it, and all four would have to be defeated.

Bar 1 — the purpose limitation. The service is defined by reference to *authorised persons at both ends*:

"(c) '**amateur services**' means radio communication services for the purpose of **self-training, intercommunication and technical investigations** carried out by an amateur, that is, by a duly **authorised person** interested in radio technique, **solely with a personal aim and without any pecuniary interest**;"

[VERIFIED] — G.S.R. 675(E), rule 2(1)(c). It mirrors International Telecommunication Union Radio Regulation No. 1.56 word for word. **[VERIFIED]** — ITU Radio Regulations, Edition of 2024, Volume 1, fetched 2026-08-23.

Bar 2 — who you may talk to. This is the operative sentence, and it is a closed list of one item:

"2. Messages. - (1) (a) Radio communications may be exchanged with **other authorised amateur stations**. The amateur stations shall not communicate with amateur stations of countries whose administrations have notified the International Telecommunication Convention of their objection to such radio communications;"

[VERIFIED] — G.S.R. 675(E), Annexure A, para 2(1)(a).

An unlicensed person holding a handset in the second vehicle is not an authorised amateur station. Their station is unauthorised, and your transmission to them falls outside the permission.

Bar 3 — third-party messages, expressly banned.

"(2) The authorised entity shall **not transmit messages containing**, - (a) reproduction of broadcast programmes, tape recordings or transmissions of entertainment value or music; (b) false or misleading calls, signals, news, advertisements, **communications of business**, statements on topics of political or industrial controversy, or **third party messages**; or (c) use of indecent, obscene, offensive language, or signals."

[VERIFIED] — G.S.R. 675(E), Annexure A, para 2(2).

"Third party messages" is not defined in the Rules — the word "third" appears only in paragraphs 2(2) and 2(3). **[VERIFIED]** In international amateur usage it means traffic passed on behalf of, or to, someone who is not a licensed operator at either station. **Convoy coordination with unlicensed crews is third-party traffic in its purest form:** you transmit messages *to* non-amateurs and relay messages *from* them.

Bar 4 — content limited to technical matters.

"(b) Transmissions shall be made in **plain language and limited to messages of a technical nature**."

[VERIFIED] — G.S.R. 675(E), Annexure A, para 2(1)(b).

"Landslide at kilometre 42, close up to 200 metre spacing, tail vehicle confirm" is operational traffic, not a message of a technical nature about radio. **An honesty point about Bar 4:** the 2024 Rules are *stricter in drafting* than the 1978 conditions they replaced, which permitted "messages of a technical nature relating to tests **and to remarks of personal character**". The 2024 text drops the personal-remarks clause entirely. **[VERIFIED]** on the comparison. Read literally, ordinary amateur conversation is now outside the permitted content — which is why no WPC clarification was found and why ragchewing is not enforced against. **[UNCERTAIN]** on enforcement. **Bar 4 on its own is therefore a weak bar. Bars 1, 2 and 3 are not weak, and any one of them is sufficient.** See also Part 1 §1.7 and Part 3 §3.6, which reach the same conclusion about talking to unlicensed family members.

Every vehicle needs its own licensed operator. Two consequences people consistently get wrong:

- **A licensed amateur cannot "lend" coverage to an unlicensed crew.** There is no Indian equivalent of the United States "control operator" doctrine that permits an unlicensed person to speak into a licensed operator's microphone under supervision. Rule 3(1) prohibits any person from *operating* an amateur station without an ASOC, and the nearest thing to a supervision concept in the Rules — Annexure A para 7(3), requiring equipment to be "housed in such manner as to **preclude access to unauthorised persons**" — cuts the other way. **[VERIFIED]** Whether WPC would treat an unlicensed passenger speaking on a station controlled by an ASOC holder as a breach is **[UNCERTAIN]**; the Rules do not address it and no plan should be built on it.
- **Family members are not an exception.** A spouse or child in the second vehicle needs their own ASOC. Minimum age is 12. **[VERIFIED]** — rule 4(1)(a).

The exposure runs in two directions, and the second is the serious one. Operating outside your certificate conditions is a ground for suspension or cancellation of the ASOC under rule 9(a), after a hearing. **[VERIFIED]** But the unlicensed crews in the other vehicles hold no authorisation at all — they are in unauthorised possession and unauthorised operation, with the consequences set out in Part 2 §2.10 and Part 6 §6.2. **Equipping other people to commit contraventions is the sharpest way to see why this plan does not work.**

The calamity carve-out, covered honestly

The word "emergency" does not appear anywhere in the Rules. A search of the complete gazette text of G.S.R. 675(E) — rules 1 to 15 and Annexure A — for "emergenc", "distress" and "safety of life" returns **zero occurrences of any of them**. **[VERIFIED]** India's amateur rules have no general emergency clause. This is a real difference from, for example, FCC Part 97, which has express provisions for distress calls.

What India has instead is one narrow, government-mediated permission:

"(3) In case of failure of normal telecommunication facilities, an ASOC holder maybe permitted by the Central Government or State Government to handle third party messages pertaining to natural or manmade calamities, originating from and addressed to a competent civil authority namely, District Magistrates or Deputy Commissioners or Collectors of the district or any other officer authorised by them."

[VERIFIED] — G.S.R. 675(E), Annexure A, para 2(3).

Every limb is a condition and all of them must be satisfied simultaneously:

Condition	What it actually requires
"In case of failure of normal telecommunication facilities"	Normal telecommunications must have failed . Poor mobile coverage on a remote track is not a failure of facilities — it is the absence of facilities. [UNCERTAIN] whether WPC would distinguish the two; the natural reading distinguishes them
"maybe permitted by the Central Government or State Government"	Permission is required. This is not a self-executing right exercised at the roadside
"third party messages pertaining to natural or manmade calamities"	Calamity traffic only. A vehicle breakdown, a lost driver, or an injured member of your own party is not a calamity
"originating from and addressed to a competent civil authority namely, District Magistrates or Deputy Commissioners or Collectors ... or any other officer authorised by them"	The counterparty must be the district administration. Not your convoy. Not your family. Not a hospital. Not even the police, unless authorised by the District Magistrate

This provision does not help a routine expedition, and it is not designed to. Its counterparty is a Collector, not a companion; its trigger is a declared calamity, not a puncture; and it requires a government permission you will not have on a Tuesday afternoon in a side valley.

Two related provisions complete the picture and neither changes it:

- **Rule 7(7) is an obligation on you, not a permission for you.** "The ASOC holder **shall provide radio communication services ...** at the request of the Central Government, or a State Government ... **during natural calamities and disasters, without any fees or charge.**" **[VERIFIED]** It is activated by a government request, not by your own assessment of an emergency.
- **ITU Radio Regulation 4.9 is the genuine safety valve, and it is a narrow one.** Rule 7(2)(c) incorporates the Radio Regulations into your certificate conditions **[VERIFIED]**, and RR 4.9 provides that "No provision of these Regulations prevents the use by a **station in distress**, or by a station providing assistance to it, of **any means of radiocommunication at its disposal**". **[VERIFIED]** — ITU RR 2024, Article 4. How much comfort to take: some, but not as a plan. RR 4.9 addresses genuine, immediate danger to life. Whether an Indian authority would accept it as a defence to a breach of Annexure A para 2 is **[UNCERTAIN]** — no Indian case, WPC clarification or departmental guidance applying it was found, and India has exercised its RR 25.3 option to determine the applicability of third-party emergency traffic by enacting para 2(3), which is much narrower than RR 25.3 itself. **Practically: if someone in your party is dying and a ham frequency is the only means at your disposal, use it and log it in detail. No sensible regulator prosecutes that. But a communications plan is by definition the routine case, and RR 4.9 addresses only the distress case.**

The lawful version of this use exists and is cheap. If *every* vehicle carries an ASOC holder who is the person operating that station, Bars 1, 2 and 3 are all satisfied, and 2 metre and 70 centimetre working at 10 W becomes lawful for the group. **[VERIFIED]** The one absolute

condition is that the trip must be personal: "solely with a personal aim and without any pecuniary interest" in rule 2(1)(c), plus the express ban on "communications of business" in Annexure A para 2(2)(b), rule out a paid tour, a commercially organised expedition, a sponsored trip, a film shoot, or a support vehicle for a paying client. **[VERIFIED]** There is no workaround for that. A four-vehicle convoy of licensed amateurs costs under ₹9,000 in government fees and is entirely lawful.

7.2.3 USE 3 — LEARNING ABOUT RADIO

Verdict: lawful, and this is literally what the amateur service is for. You do not need to justify it. The statutory definition — rule 2(1)(c), quoted above — is "self-training, intercommunication and technical investigations".

Everything the Rules ask of you points the same way. The station log must record "a **brief description of the experiments and tests undertaken**" (Annexure A para 6(1)(c)). **[VERIFIED]** The equipment register has a column for gear that is "**self-made**" (Annexure A para 7(4)) — homebrew construction is expressly contemplated. **[VERIFIED]** Annexure A para 9 sets out test-signal procedure, including the use of an artificial aerial for longer tests. **[VERIFIED]** These are the rules of a service designed around building things and finding out how they work.

But the UV-32 is a mediocre learning platform for an Indian licensee, for a reason that has nothing to do with legality. A Restricted ASOC authorises **50 W DC input on ten high-frequency bands** — 1820–1860, 3500–3700, 3890–3900, 7000–7100, 7100–7200, 14000–14350, 18068–18168, 21000–21450, 24890–24990 and 28000–29700 kHz, in voice emissions A3E/H3E/J3E/R3E. **[VERIFIED]** — WPC Office Memorandum No. L-14011/255/2004-AMT, 2010-08-13, Annexure-I; see Part 3 §3.4, and note the **[UNCERTAIN]** at Part 1 §1.8 as to whether that 2010 table is still the operative specification under rule 7(1) of the 2024 Rules.

From a valley in Ladakh, 40 metre or 20 metre single-sideband will reach hundreds or thousands of kilometres when nothing else will. **That, not UHF, is the amateur service's real expedition capability, and a VHF/UHF hand-held cannot touch it.** India's VHF and UHF repeater network — of the order of 50-plus machines on 144–145 and 434–437 MHz — is concentrated in Mumbai, Bengaluru, Chennai, Kolkata, Pune, Delhi, Hyderabad, Kochi, Coimbatore, Mysore and Trivandrum. **[REPORTED]** — Amateur Radio Society of India (ARSI) repeater listing, fetched 2026-08-23. It is an **urban** network. It will get you a contact as you approach a city; it will not cover a Himalayan or Rann route, and no repeater coverage information for Ladakh or the Northeast could be found **[UNCERTAIN]**.

If the goal is genuinely to *learn*, the cheapest high-value purchases are, in order: the exam itself; a ₹2,000-class licence-free handset to understand propagation with no legal exposure at all; and — once licensed — an HF radio or a kit. **HF Signals** of Hyderabad, founded by Ashhar Farhan VU2ESE, manufactures the **uBITX** family of HF transceivers in India, which sidesteps the import question entirely. **[REPORTED]** — product line not independently verified in this research.

Two features of the UV-32 are actively awkward for a learner operating under an Indian ASOC:

- **Encryption.** Menu 6.10 offers "ON: Activating encryption mode ensures the privacy of your conversations." **[VERIFIED]** — UV-32 user manual, Appendix C. Annexure A para 2(1)(b) requires plain language. Merely possessing the feature is not an offence, but using it on the amateur bands would breach the ASOC, and its presence makes the radio look like a security product rather than an amateur one.
- **Bluetooth over-the-air programming.** Menu 3 permits writing frequencies into the radio from a phone application. **[VERIFIED]** Convenient — and it means the channel plan can be changed in seconds by anyone with the app, which undercuts any argument that the set is "configured for amateur use only".

7.3 The UV-32 examined

The Baofeng UV-32 (also sold as the **AR-32**; the manual covers both) is a 182 × 67 × 41 mm, 350 gram hand-held with a 2.0 inch colour screen, GPS, Bluetooth, a full keypad, a 7.4 V 2500 mAh lithium-ion battery, USB Type-C charging and 1000 memory channels in 10 banks. **[VERIFIED]** — UV-32 user manual, document [Baofeng_UV-32_User_Manual_20250210.pdf](#), header "UV-32_USER'S MANUAL EN 241124", Appendix B and §1.3. The manufacturer of record for regulatory purposes is *PO FUNG ELECTRONIC (HK) INTERNATIONAL GROUP COMPANY LIMITED* — the misspelling is in the FCC filing. **[REPORTED]** — FCC grantee record as mirrored on [fccid.io](#).

It is not a bad radio. It is the wrong radio.

7.3.1 TRANSMIT COVERAGE, BAND BY BAND, AGAINST INDIA'S ALLOCATIONS

India's amateur VHF/UHF allocations, from the 2010 WPC Office Memorandum Annexure-I:

Grade	VHF/UHF bands	Maximum DC input power	Emissions
ASOC (Restricted) , callsign block VU3	50–54, 144–146 , 434–438 MHz	10 W	F1B, F2B, F3E , F3C
ASOC (General) , callsign block VU2	50–54, 144–146 , 434–438 , 5725–5840 MHz	25 W	same

[VERIFIED] — WPC OM No. L-14011/255/2004-AMT, 2010-08-13, Annexure-I, read page by page from the scan; see Part 3 §3.4. Note that 434–438 MHz is **secondary** in India, carrying the footnote "On secondary basis as per the relevant provisions of Radio Regulations". **[VERIFIED]**

Baofeng's own product page for the UV-32 gives transmit coverage of "**136-174 MHz, 220-260 MHz, 400-520 MHz**". **[VERIFIED]** — [baofengradio.com/products/uv-32](#), fetched 2026-08-23; the same figures are published by Radioddity, a principal authorised retailer **[REPORTED]**, and by Baofeng's own United Kingdom arm **[VERIFIED]**.

Mapping that onto Indian law:

UV-32 transmit segment (as advertised)	Width	Lawful for an Indian amateur?	What else is in there
136–144 MHz	8 MHz	No	
144–146 MHz	2 MHz	Yes — 2 metre, primary allocation	
146–174 MHz	28 MHz	No	Marine VHF 156–162 MHz; state and central government VHF
220–260 MHz	40 MHz	No	India has no amateur allocation anywhere near 220 MHz
400–434 MHz	34 MHz	No	Includes the 433.05–434.79 MHz 10 mW short-range-device band
434–438 MHz	4 MHz	Yes — 70 centimetre, secondary allocation	
438–520 MHz	82 MHz	No	Includes PMR446 (446.0–446.2 MHz, 500 mW, integral antenna only) and government and paramilitary UHF

Total advertised transmit coverage $\approx$ 198 MHz. Lawful for an Indian amateur: 6 MHz. About 3 %. Roughly 97 % of what this radio can transmit on is spectrum an Indian amateur may never touch. [VERIFIED] — against Annexure-I and the manufacturer's published coverage.

It also **cannot reach 50–54 MHz**, which is the one Indian VHF amateur band offering 10 W on a genuinely quiet band with propagation useful for overland work.

Two source conflicts the reader must know about, neither of which rescues the radio.

First, Baofeng contradicts Baofeng. Appendix B of the user manual gives, verbatim:

"**Frequency Range (TX)** 144 to 148 MHz, 420 to 450 MHz (America version) / 144 to 148 MHz, 430 to 450MHz (Canadian version) / 144 to 146 MHz, 430 to 440MHz (European Union version)*"

[VERIFIED] — manual Appendix B, p.21. Note what those lines *are*: the amateur allocations of the United States, Canada and the European Union. They describe what the buyer is assumed to be licensed for, not what the hardware does, and the manual never claims the firmware enforces them. Baofeng's own UK site sells the "European Union version" while advertising 136–174 MHz transmit — flatly contradicting the manual. **[VERIFIED]**

The honest resolution: the manufacturer publicly advertises 136–174 / 220–260 / 400–520 MHz on its own website, through its own UK arm and through its principal retailer. Even if an individual unit proved narrower, **that is the specification Baofeng puts its name to, and it is the specification a customs officer, a police officer or a WPC inspector reads off the box.**

[VERIFIED] And **there is no version of this radio, on any source, that fits inside Indian amateur spectrum**: the narrowest claim anywhere — the manual's EU line of 144–146 and 430–440 MHz — still puts 430–434 and 438–440 MHz outside India's 434–438 MHz allocation.

[VERIFIED]

Second, the India listing says something different again. The storefront states "400–520 MHz" and "128 channels", against the manufacturer's tri-band, 1000-channel specification. **[VERIFIED]** that both pages say what they say. Either India receives a different, UHF-only variant, or the listing is materially incomplete. **[UNCERTAIN]** — this cannot be resolved from outside, and it matters both ways: a genuinely UHF-only 400–520 MHz unit would not cover **144–146 MHz** at all, halving its usefulness to a VU3 operator, since 2 metre is where most Indian repeater and simplex activity is. **Ask the seller in writing which variant ships and whether it transmits on 144–146 MHz, before paying, and keep the reply.**

Two further notes on coverage. The **220–260 MHz** transmit claim is doubtful in practice — owners report the radio displaying 220 MHz frequencies while transmitting "at exactly twice the displayed frequency", and others could not raise 220 MHz repeaters at all. **[REPORTED / UNCERTAIN]** — RadioReference thread 484874. It is legally irrelevant in India either way. And the **aviation band** at 108–136 MHz is **receive only**: the radio's only emission designators are 16K0F3E and 11K0F3E, wide and narrow FM telephony, and an FM-only transmitter cannot produce the amplitude modulation the air band uses. **[VERIFIED]** — manual Appendix B. Marketing that calls this an "air band radio" is describing the receiver.

One thing does work. F3E — analogue FM telephony — is expressly authorised to both ASOC grades. **[VERIFIED]** Unlike its digital sibling the DM-32UV, the UV-32 raises no question about whether digital voice modes fall inside the permitted F1B/F2B/F3E/F3C emission set.

7.3.2 THERE IS NO BAND-LOCK, AND BAOFENG SAYS SO IN ITS OWN MANUAL

This is the cleanest single piece of evidence in the whole equipment analysis.

Manual §5.5, on VFO tuning:

"In Frequency (VFO) mode you can navigate up and down the band by using the ▲ or ▼ keys. ... **You can also input frequencies directly on your numeric keypad with kilohertz accuracy.**"

And immediately after the worked example, the manual prints this warning verbatim:

"WARNING! Just because you can program in a channel does not mean you're automatically authorized to use that frequency. Transmitting on frequencies you're not authorized to operate on is illegal, and in most jurisdictions a serious offence. However, it is legal in most jurisdictions to listen. Contact your local regulatory body for further information on what laws, rules and regulations apply to your area."

[VERIFIED] — manual §5.5, p.10.

That warning is Baofeng conceding, in its own documentation, that the radio does not enforce band limits — the operator does.

The complete menu tree from Appendix C — 7 top-level menus with 27 items under Radio Setting and 17 under Program Channel — was transcribed in the underlying research. **There is no band-limit, transmit-inhibit, region-lock or amateur-band-only item anywhere in it. [VERIFIED]** — manual Appendix C, pp.21–26. The only frequency-range setting in the radio is SCAN → Freq Ranger, which bounds a *scan sweep* and has no effect on transmit. **[VERIFIED]** The transmit frequency of a stored channel is itself directly editable from the front panel (menu 6.3). **[VERIFIED]**

Can it be restricted in software? **Officially, no** — no documented Baofeng programming-software option, menu item or service procedure restricts transmit to amateur bands. **[VERIFIED]** by the absence of any such item in the manual; **[UNCERTAIN]** as to undocumented programming-software or service-mode behaviour, which could not be tested. **Unofficially, the community traffic goes the other way:** the documented activity around this platform is *widening* coverage by hex-editing the firmware image, not narrowing it. **[REPORTED]** — RadioReference thread 484874, pages 3 and 6.

The practical consequence for a licensee. You cannot make this radio safe by configuration. If a WPC or police inspection asks "can this transmit on 156 MHz?", the honest answer is yes, and no setting you can point to says otherwise. Meanwhile Annexure A para 4(1) requires the apparatus to be "tuned as accurately as possible to ensure that **no energy is radiant on any frequency outside the limits of the authorised frequency bands**". **[VERIFIED]** You are one misplaced VFO entry away from transmitting on police, railway or commercial spectrum.

The antenna compounds this. It is **detachable** — "Screw the antenna into the connector on the top of the transceiver ... Turn the antenna counter-clockwise to remove it" **[VERIFIED]**, manual §3.2 — on an SMA fitting of the type that takes the "SMA-Female" antennas sold across the UV-5R family **[REPORTED]**. That has two legal consequences. It **disqualifies the radio from the PMR446 exemption by construction**, independently of power, because Table V of G.S.R. 1047(E) defines the exempt device as one that "uses integral antennas only" **[VERIFIED]** — so turning a UV-32 down to 0.5 W does *not* make it a licence-free radio (Part 2 §2.5.1). And it makes the reader's stated plan — a roof whip — trivially easy to execute and just as easy to see.

7.3.3 RATED POWER, MEASURED POWER, AND THE DC-INPUT TRAP

This is where the reader's "10 W" and India's "10 W" turn out to be two different numbers.

What is claimed:

Source	Claimed power
Baofeng product page	"Output Power: 10W" [VERIFIED]
Baofeng UK specification table	"High: 10W / Mid: 5W / Low: 1W" [VERIFIED]
twowayradio.us	"UHF 10W / VHF 8W" [REPORTED]
defensedistributors.com	"10W / 5W / 2W selectable" [REPORTED]
Baofeng India storefront	"Transmit Power: 5W-10W" [VERIFIED] that the page says this

Source	Claimed power
UV-32 manual, Appendix B	No transmitter power figure at all. The TRANSMITTER block gives modulation, spurious emission, frequency stability, audio distortion and hum — and no output power [VERIFIED]

The manual's silence is worth dwelling on. The only power figure anywhere in the manual is in the GENERAL block: "Battery Life: 5% TX, 5% RX, 90% Standby — Li-on: 18 hours @ **5 watts**". **[VERIFIED]** Baofeng's own engineering documentation characterises the radio at 5 watts while the box says 10.

What is measured: independent testing is reported at about **5 W on 2 metre and 3 W on 70 centimetre** — roughly half the rated figure. **[REPORTED]** — mobitex.org, echoed by besthamradio.com. One outlier report of 13 W on 2 metre circulates and is inconsistent with the rest; it should not be relied on. **[REPORTED / UNCERTAIN]** No laboratory measurement from a named accredited source could be found. **[UNCERTAIN]**

Now the trap. India's amateur power ceilings are expressed as **DC input to the final stage**, not RF output. The column heading in Annexure-I is "*Max DC Input Power unless Otherwise specified*" and note (iv) reads "*DC input power is the total direct current power input to the final stage of the transmitter.*" **[VERIFIED]** Consumer hand-helds are universally advertised by **RF output at the antenna socket**, and nothing in Baofeng's material suggests otherwise; the manual publishes no DC input figure and no transmit current, so the DC input must be inferred. **[VERIFIED]** that the figures are presented as output and that no DC figure exists; the inference below is **[UNCERTAIN]**.

One manufacturer-published data point makes the conversion concrete without guesswork. Yaesu publishes the FT-65's transmit current draw:

"Current Consumption (Approx. @7.4V): ... 1.5 A (5 W Tx , 144 MHz) 7.4 V DC / 1.7 A (5 W Tx , 430 MHz) 7.4 V DC"

[VERIFIED] — Yaesu FT-65R/FT-65E Operating Manual, Specifications p.46, PDF fetched from yaesu.com and text-extracted. That is **11.1 W** of DC into the whole radio for 5 W out on 2 metre, and **12.6 W** for 5 W out on 70 centimetre. Subtract a little for receiver, display and logic and the final stage is drawing somewhere around 10–12 W DC to deliver 5 W of RF. **The current figures are [VERIFIED]; the apportionment between the final stage and the rest of the radio is engineering inference and is [UNCERTAIN].**

Applying that arithmetic to the UV-32:

UV-32 power step	RF out (claimed)	RF out (measured, 2 m)	Implied DC input to the final stage	Restricted (10 W DC)	General (25 W DC)
High	10 W	~5 W	roughly 18–25 W if it really makes 10 W; roughly 10–12 W if it makes the measured 5 W	Over	Inside
Mid	5 W	—	roughly 10–12 W	At or just over	Inside
Low	1 W (some sources say 2 W)	—	roughly 2–5 W	Inside	Inside

The two legal ceilings are [VERIFIED]. Every figure in the "implied DC input" column is inference from power-amplifier efficiency and the Yaesu data point above, and is [UNCERTAIN].

Three readings follow:

1. A **Restricted (VU3) licensee should treat the UV-32 as a low-power-only radio**. Even the 5 W mid setting sits at or over the 10 W DC ceiling on this arithmetic.
2. A **General (VU2) licensee has headroom** at mid power and is probably still inside 25 W DC at high power, though at a claimed 10 W output that becomes marginal.
3. **Nobody can demonstrate compliance without measuring**, because the radio publishes neither its DC input nor a reliable output figure. That is an uncomfortable position for a licensee whose rule 14 obligations include producing the equipment for inspection. [VERIFIED]

This is the same point Part 1 §1.8 flags as the first of three uncertainties a reader must not gloss over, and it is the reason §7.7.2 puts the measurement question to WPC in writing.

7.3.4 THE FCC ID COVERS THE BLUETOOTH RADIO AND THE RECEIVER — NOTHING ELSE

FCC ID: 2AJGM-NA32UV, printed on the product page and, per the filing, on the device label. [VERIFIED] that the product page displays this identifier.

The grant record, as mirrored on fccid.io and fetched 2026-08-23:

Frequency range	Rule part	Output power	What it actually is
2402.0–2480.0 MHz	15C	0.001 W conducted	The Bluetooth radio
108.0–136.0 MHz	15B	—	Scanning receiver
136.0–174.0 MHz	15B	—	Scanning receiver
220.0–260.0 MHz	15B	—	Scanning receiver

Frequency range	Rule part	Output power	What it actually is
350.0–390.0 MHz	15B	—	Scanning receiver
400.0–520.0 MHz	15B	—	Scanning receiver

Applicant PO FUNG ELECTRONIC (HK); grant date **2025-04-18**; equipment classes *Digital Transmission System* and *Scanning Receiver*; test laboratory Bay Area Compliance Laboratories Corp. (Xiamen). **[REPORTED]** — fccid.io mirrors the FCC Office of Engineering and Technology equipment authorisation database; apps.fcc.gov returned HTTP 403 and could not be fetched directly, so this is a mirror rather than the primary record.

Read that table carefully. There is no transmitter grant of any kind for the VHF/UHF transmitter. The only transmitter the FCC certified is the 1 milliwatt Bluetooth module; everything else is a receiver certification. That is not an oversight — in the United States, transceivers built for the amateur service are exempt from equipment certification, because 47 CFR Part 97 puts the compliance burden on the licensed operator rather than on the equipment. So an FCC identifier on an amateur hand-held tells you the Bluetooth and the receiver were tested and **tells you nothing at all** about the legality or the emissions quality of the transmitter. **[This characterisation of the US regulatory structure is [UNCERTAIN] in the sense that Part 97 itself was not fetched; the grant contents are [REPORTED].]**

The CE mark is no better. It rests on a **manufacturer self-declaration** under Directive 2014/53/EU, reproduced in the manual's Declaration of Conformity **[VERIFIED]**, and it is not a third-party approval. The same manual states, in bold: **"WARNING: European Users should note that operation of this unit in Transmit mode requires the operator to have a valid Amateur Radio License ..."** **[VERIFIED]**, and Baofeng's UK arm says plainly that **"Transmission on PMR446 (Licence-Free) frequencies is not permitted with this device."** **[VERIFIED]**

Neither mark has any legal effect in India. They do not substitute for an ETA, do not authorise import, do not authorise possession and do not authorise transmission. This matters because an FCC identifier on a label is exactly the kind of thing a buyer reads as "approved" — and it is the same mistake as the ETA-sticker mistake in Part 1 §1.7.

7.3.5 WHY NO INDIAN ETA EXISTS, AND WHY NONE COULD

No Indian ETA was found for the UV-32, and none is legally available.

ETA exists to certify compliance with the technical limits of a **licence-exempt band** notification. WPC's own compendium: *"equipment complying with such notifications need to be type approved and accordingly Equipment Type approval (ETA) certificate is issued to such compliant RF devices"* — and for licensed-band devices the control is an operating licence instead. **[VERIFIED]** — WPC Compendium F. No. R-11018/02/2017-PP, 2022-07-06, §§2.1–2.2; see Part 2 §2.7.1.

A 136–174 / 400–520 MHz, 10 W, detachable-antenna transmitter is **licensed-band equipment**. There is no de-licensed Indian band it fits — Part 2 §2.6.1 establishes the complete licence-exempt ladder by exhaustive enumeration, and its ceiling for a UHF hand-held is 500 mW e.r.p.

So there is no ETA to obtain, and **any seller claiming an ETA for a UV-32 is either confused or misrepresenting. [VERIFIED]** as a matter of law. A negative check against WPC's published register could not be run, because sara.sanchar.gov.in/wpc_eta_report.php is CAPTCHA-protected and cannot be queried programmatically — so the absence of a UV-32 ETA is **[UNCERTAIN]** as a matter of record, though certain as a matter of law.

Do not read the absence of an ETA on the India listing as sloppiness. Read it as confirmation that the seller has classified the radio correctly. The storefront makes no ETA claim for the UV-32 and has no compliance tab on its page **[VERIFIED]** — which is exactly right.

7.3.6 POSSESSION, AND RULE 10(F)

Situation	Position
You hold an ASOC	Lawful , on rule 10(f) of the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026, G.S.R. 592(E), 2026-07-08: an authorisation is not required if a person " <i>holds an Amateur Station Operator Certificate in accordance with the Telecommunications (Amateur Services) Rules, 2024</i> ". Note that rule 10 is unqualified by frequency — it exempts the person , not the equipment. [VERIFIED] on the original retrieval of the final rules. But see the hedge below. Keep the ASOC with the radio and enter the set in the Annexure A para 7(4) equipment register on the day you acquire it
You do not hold an ASOC	Unlawful, with no available fix. Rule 4(1) offers possession authorisations only for dealing or for testing and demonstration, and rule 4(2) restricts eligibility to companies, limited liability partnerships, trade-licensed persons and government. A private hobbyist fits none of them. [VERIFIED] Exposure: civil penalty up to ₹50,000 first instance and up to ₹2,00,000 for each subsequent instance under Third Schedule item 1(a) of the Telecommunications Act 2023 [VERIFIED] , subject to the commencement question at Part 6 §6.1; plus, if the Indian Wireless Telegraphy Act 1933 still stands, up to three years' imprisonment under its s.6(1A) [VERIFIED] on the text, [UNCERTAIN] on the repeal — Part 1 §1.8
You import it yourself without an ASOC	Worst case. This moves you out of the civil-penalty regime and into the Customs Act, which is what actually produced arrests in the March 2026 Nhava Sheva seizure [REPORTED] . Buying domestically avoids this entirely — see §7.4.4

The hedge, and it must not be glossed. Rule 10(f) is recorded as **[VERIFIED]** in the underlying equipment research from a direct retrieval of G.S.R. 592(E). The adversarial fact-check of 2026-08-23 could **not** re-fetch the final 2026 Rules from any gov.in host, and the February 2025 *draft* of the same rules numbers the exemption differently (rule 4, not rule 10) and does not contain an express ASOC gateway. **The report therefore carries this as [UNCERTAIN] — see Part 1 §1.5 Group B and Part 2 §2.9.3 — and it is one of the two questions to put to WPC in writing before buying hardware.** Do not spend ₹14,449 on the strength of a rule you cannot currently re-read.

Two further risks specific to this radio, stated for completeness.

Seizure is independent of penalty. Section 43 of the 2023 Act allows an authorised officer to search a building or **vehicle** and take possession of equipment. **[VERIFIED]** on the text. Part 1 §1.7 records the correction that s.43 is *conditional* on reason to believe an offence punishable

under s.42 has been committed, and that bare unauthorised possession of an ordinary hand-held is a Third Schedule civil contravention rather than an s.42 offence — so s.43 does not plainly reach it, while s.7 of the 1933 Act does. **[UNCERTAIN]** which power an officer would in practice invoke. A radio mounted in a vehicle is nonetheless squarely within the wording of both.

The BF-888S precedent. Press Trust of India reported in March 2026 that "*The Baofeng BF-888S walkie-talkies have been blacklisted by the Department of Telecommunications (DoT) for operating beyond permissible frequencies ...*" **[REPORTED]**. **No gazette notification or published DoT list naming specific models could be located**, so the legal status of that "blacklist" is **[UNCERTAIN]** — Part 2 §2.10. But the reasoning DoT is reported to have applied — *operating beyond permissible frequencies* — describes the UV-32 precisely, and the brand name on the case is the same one that was in the newspapers.

Also note the reception side, which is not an offence but is not neutral either. The UV-32 receives 350–390 MHz, taking in the 350.1625–350.5375 MHz short-range-radio channels of the National Frequency Allocation Plan 2022 and the 351–358 / 361–368 MHz captive mobile radio trunking pairs. **[VERIFIED]** — NFAP-2022 Annexure-2. Receiving is not transmitting, and questions about reception of transmissions not intended for you are outside this report's scope — but a radio that scans government trunking allocations is not a neutral object to be found holding at a checkpoint in a border district.

7.4 The seller: who actually runs the India storefront

The general lesson of this section can be stated before the evidence, because it is the part that generalises: **seller legality and operator legality are two different findings, and no amount of the first produces any of the second.**

7.4.1 THERE IS ONE STOREFRONT, UNDER THREE NAMES

baofengradios.in and baofengradios.co.in are HTTP 301 redirects to baofengradios.com. baofeng.in redirects to an unrelated Mexican general-merchandise shop and is not a Baofeng India store at all. baofengindia.com, baofengindia.in and baofengradio.in do not resolve. **[VERIFIED]** — redirect chains traced with `curl -sL -D -` on 2026-08-23. Anything presenting itself as a different "Baofeng India" site is either a marketplace listing or unrelated.

7.4.2 IT IS RUN BY AN INDIAN COMPANY, NOT BY BAOFENG

The site's footer, verbatim on every page:

"G S Tele Communication Limited, Unit No. 614, Tower B4, Spaze i-Tech Park, Sector 49, Gurgaon, Haryana – 122018" / "DPL: EDPL-14-F-202500053" / "support@baofengradios.com"

[VERIFIED] — fetched 2026-08-23.

Field	Value	Tag
Operator	G S Tele Communication Limited	[VERIFIED] — site's own disclosure
CIN	U64203DL2007PLC161534	[REPORTED] — Tofler, an MCA mirror
Incorporated	2007-04-02; unlisted public limited company	[REPORTED]
Registered office	Rithala, New Delhi – 110085 (the website gives a Gurugram corporate office; both localities recur on the company's own ETA certificates)	[REPORTED] / [VERIFIED] respectively
Paid-up capital	₹14.3 lakh	[REPORTED]
Brand rights	Granted by Shopperclue Webtrade India Private Limited , CIN U74999HR2015PTC056598, a Gurgaon company holding Indian BAOFENG trade mark filings — not by Fujian Baofeng Electronics Co., Ltd., the Chinese manufacturer	[VERIFIED] for the site's own statement of the chain; [REPORTED] for the registry data

The MCA rows are [REPORTED] rather than [VERIFIED] because they come from a commercial registry mirror; the MCA portal itself could not be queried, ZaubaCorp returned HTTP 403 and cleartax returned HTTP 410.

The About Us page grants the operator "perpetual, exclusive, irrevocable, and unrestricted rights" over the brand in India — [VERIFIED] — and it is worth noting what that paragraph never claims: any relationship whatsoever with Fujian Baofeng. The entire chain of authority runs G S Tele ← Shopperclue ← an Indian trade mark registration. The word "official" in the site's page titles is self-conferred on that basis.

The trade mark position is narrower than the site's flat assertion that "BAOFENG® is a registered trademark in India" suggests: the one mark shown as **Registered** (application 5014095, Class 9, 2021) covers **accessories** — chargers, earphones, cables — not walkie-talkies; the application that would cover radios (6501267) is shown as **Opposed** and the walkie-talkie retail application (6604211) as pending. [REPORTED] — trade mark mirrors fetched 2026-08-23; tmrsearch.ipindia.gov.in is CAPTCHA-gated and could not be queried, so exact current status is [UNCERTAIN].

This is not a consumer-safety point. It is a "who am I actually dealing with" point.

7.4.3 THE MANUFACTURER'S AUTHORISED-DEALER LIST NAMES NO INDIAN DEALER

Fujian Baofeng's official page baofengradio.com/blogs/news/baofeng-official-authorized-dealer, marked "Latest Updated: July.2025", lists dealers by marketplace — Amazon stores, AliExpress, Mercado Libre, TikTok Shop, Walmart, Shopee, eBay, Wish, Temu, Alibaba, Allegro, Joom, Emag, Coupang. **No Indian dealer is listed.** baofengradios.com, **G S Tele Communication and Shopperclue do not appear.** [VERIFIED] — fetched 2026-08-23.

The same page states that "Baofeng's limited warranty applies only to brand-new items purchased directly from baofengradio.com. For purchases made through authorized dealers, warranty services are provided directly by the respective dealer." **[VERIFIED]** So the manufacturer's warranty does not reach this purchase; the reader's only warranty is G S Tele's, whose term for the UV-32 is not stated on the product page at all **[UNCERTAIN]** — get it in writing before paying.

How much weight to give the dealer list: it is a marketing page, it is 13 months stale, and it lists platforms rather than country distributors. Absence from it is **not** proof that G S Tele is unauthorised. Combined with the About Us page tracing authority only to an Indian trade mark, and with the manufacturer never being mentioned anywhere on the India site, the honest reading is **an independent Indian brand-licence operation, not a Baofeng subsidiary or appointed distributor.** **[UNCERTAIN]** as to whether some private supply arrangement nonetheless exists.

7.4.4 THE CHECKOUT DOES NOT ENFORCE THE LICENCE IT DISPLAYS

This is the finding with the most practical weight, so the evidence is given in full.

On the UV-32 page the site displays a compliance box, verbatim:

"NOTE : User needs the following documents: **WPC User License** [text field: 'Enter licence number', max 20 chars] [Upload button] If you do not have the required licence or documents, click **HERE** to apply. If you want to know more about WPC user licence, **CLICK HERE.**"

[VERIFIED] — page source, 2026-08-23.

But the page's own embedded product data marks that document optional:

```
"complianceDocuments":[{"code":"$undefined","complianceId":"69ff4790d6920afe40db6e32",
"documentName":"WPC User License","isMandatory":false,"userMustProvide":"$undefined"}]
```

[VERIFIED] — page source, 2026-08-23.

And the site's own JavaScript gates the purchase on `isMandatory === true` only. From the product-page bundle `/_next/static/chunks/5279214e4aa3feaa.js`:

```
nz = (tQ?.complianceDocuments ?? []).filter(e => e.isMandatory === true),
nH = nz.length > 0
```

with, in both the Add-to-Cart and Buy-Now handlers:

```
if (nH && !nc) return nC(eL,"error"), document.getElementById(tE)?.scrollIntoView(...),
[2];
```

[VERIFIED] — bundle fetched and read 2026-08-23. Since `isMandatory` is `false` for the UV-32's "WPC User License", the flag `nH` is false, and **neither Add to Cart nor Buy Now is blocked.**

Stated honestly: the client-side purchase flow does not require a licence, and the site's own data marks the licence non-mandatory. **[VERIFIED]** for that. No order was placed, no account created and no checkout attempted, so a server-side check at order confirmation or dispatch cannot be excluded — and the Shipping Policy does say orders "may be delayed where identity verification, address confirmation or **regulatory review** is reasonably required" **[VERIFIED]**. Whether anything downstream catches it is **[UNCERTAIN]**.

The site publishes the rule and does not implement it on this product. Its own blog article "Licence-Free vs Licensed Walkie Talkie", published 2026-05-02, lists as best practice for licensed models: "Clearly label: 'Licensed Radio – For WPC Licence Holders Only' · **Add a mandatory checkbox at checkout** · Request licence details for B2B sales", and lists among "the most common violations": "**Selling licensed radios to buyers without licence → Illegal**". **[VERIFIED]** That is the single cleanest demonstration in this whole report of why a seller's stated compliance posture is not a substitute for the buyer's own.

Two further problems with the box, from the reader's point of view:

- **Neither link points at the route he needs.** "HERE (to apply)" resolves to the National Single Window System page for the WPC Network licence, Fixed / Land Mobile HF/VHF/UHF below 806 MHz — the **captive/commercial route**, which Part 1 §1.7 establishes is not realistically available to an individual. "CLICK HERE (to learn more)" resolves to a 33-page scan of DoT Order No. P-11014/34/2009-PP dated 2023-12-11 on *spectrum charges for captive users* — not a guide at all. **[VERIFIED]** — both targets traced and the PDF text-extracted 2026-08-23. **The site never mentions amateur licensing in its compliance box**, even though it runs a "HAM Corner" category.
- **The captive route those links point at is a pre-2023-Act instrument administered through portals DoT has been closing.** New applications under the Indian Telegraph Act 1885 through Saral Sanchar and the National Single Window System stopped from 2026-06-25 **[VERIFIED]** — Part 1 §1.9. A "click HERE to apply" link into that portal in August 2026 may be pointing at a closed door. **[UNCERTAIN]** — the page is client-rendered and could not be read to confirm.

The same box on a genuinely licence-free product renders as "No compliance documents listed for this product" — while still saying "If you do not have the required licence or documents, click HERE to apply". **[VERIFIED]** The template is applied indiscriminately. Confusing rather than harmful, but it indicates how much curatorial attention the compliance furniture has received.

7.4.5 THE SELLER'S OWN DISCLAIMER ANSWERS THE QUESTION

Bundled into the site's privacy-policy page, the Regulatory Compliance Policy states, verbatim:

"3. Dealer Possession Licence (DPL) The Company operates under Dealer Possession Licence (DPL) No. EDPL-14-F-202500053. The existence of this licence authorises the Company to undertake activities permitted under the licence. **It does not automatically extend regulatory privileges to purchasers beyond those provided by applicable law."**

[VERIFIED] — fetched 2026-08-23.

The Website Legal Disclaimer goes further and covers all three gates:

"3. Regulatory Disclaimer ... Customers remain responsible for ensuring that the purchase, possession and operation of radio equipment complies with applicable legal requirements. Nothing on this Website should be interpreted as authorising unlawful operation or modification of radio equipment."

[VERIFIED]

That is the whole answer to "does buying from the official India store make it legal?" — the seller says no. And it agrees with the more authoritative statement of the same principle, which appears on the face of a WPC import licence itself and is quoted in Part 2 §2.8.2: *"This is only a clearance from technical angle ... the grant of import license does not imply permission to establish, maintain and operate a Wireless Telegraph equipment. A separate license will be required for the usages of the equipment."* [VERIFIED] When the seller and the regulator agree that the seller's licence does nothing for you, the point is settled.

One caveat the site does not give about its own DPL: DoT stopped accepting DPL applications **and renewals** from 2026-07-03 and replaced the regime with the Radio Equipment Possession Authorisation (Part 1 §1.9), and DoT's own page gives DPL validity as **one year**. The number **EDPL-14-F-202500053** embeds **2025**. **The site publishes no issue date and no expiry date.** [UNCERTAIN] — a fair question to put to the seller in writing.

7.4.6 THE GENUINE ETA CERTIFICATES — AND WHAT THEY PROVE

The licence-free models carry real, checkable certificates. Six models, six distinct registration numbers, all in the name of G S Tele Communication Ltd:

Model	ETA registration number	Certificate date	Frequency	Power
BF-888s	ETA-SD-20260100411	2026-01-16	446–446.2 MHz	0.5 W ERP
MP-31	ETA-SD-20260201589	2026-02-18	446–446.2 MHz	0.5 W ERP
BF-358	ETA-SD-20260504660	2026-05-19	446–446.2 MHz	0.5 W ERP
BF-1901	ETA-SD-20260504748	2026-05-25	446–446.2 MHz	0.5 W ERP
T-20	ETA-SD-20260504812	2026-05-25	446–446.2 MHz	0.5 W ERP
GT-1	ETA-SD-20260504849	2026-05-25	446–446.2 MHz	0.5 W ERP

[VERIFIED] that these PDFs are published and contain these values — each downloaded from images.baofengradios.com and text-extracted 2026-08-23. Each is on WPC letterhead, headed "[Generation of Equipment Type Approval (ETA) through self-declaration issued under O.M. No. ETA-WPC/Policy/2018-19 dated 26 February, 2019]", with equipment category "**Walkie Talkie Radio**". **[VERIFIED]** **[UNCERTAIN]** whether the certificates are genuine, because the WPC public register is CAPTCHA-gated and cannot be queried programmatically.

An incidental correction to Part 2 §2.11.1. That section records that the Saral Sanchar ETA search's equipment-category dropdown has no walkie-talkie entry, so PMR sets would be filed under "Other". These certificates show an explicit "Walkie Talkie Radio" category in use in 2026, while a re-fetch of saralsanchar.gov.in/wpc_eta_report.php on 2026-08-23 shows the dropdown still lacking it. **[VERIFIED]** on both observations; **[UNCERTAIN]** on the reason — either the public search dropdown lags the back-office category list, or the certificates use a different taxonomy.

Three things on those certificates the reader should internalise, all **[VERIFIED]** from the GT-1 certificate page 2:

- **Clause (i) and (iii): the ETA covers a *configuration*, not a *box*.** "This certificate will not be valid in case any change in the above parameters ..." and "**Use of such equipment in case not conforming to above notification will require a specific wireless operating license**". The moment a licence-free Baofeng is reprogrammed off 446 MHz or fitted with a different antenna, the ETA stops protecting it. That is the mechanism by which a perfectly legal ₹3,399 GT-1 becomes an illegal transmitter after ten minutes with programming software.
- **Clause (vi) puts liability on the applicant, not the regulator.** "The applicant is liable for prosecution under Indian Law in case of any wrong declaration ..." A self-declaration ETA is not WPC saying "we tested this"; it is the applicant saying "we declare this", with a private laboratory's report behind it. All ETAs for licence-exempt equipment have been granted on self-declaration since 2024-09-09 (Part 2 §2.7.2). **An ETA number on a listing is a declaration, not a test result.**
- **One of the six published files is incomplete.** The BF-358 PDF is named [BF-358_removed.pdf](#) and contains only page 1 of a document paginated "1/2" — the terms-and-conditions page has been stripped. **[VERIFIED]** Most likely a careless PDF split rather than anything sinister, but worth knowing that at least one published "compliance document" is a partial copy.

Do the five-minute check yourself. Open https://saralsanchar.gov.in/wpc_eta_report.php, choose "List of Issued ETA (Self-Declaration)", enter the number for whichever model you are considering, solve the CAPTCHA, and confirm that holder, model, frequency and power match the PDF. Until that is done these certificates are **[UNCERTAIN]** — seller-hosted PDFs on the seller's own content delivery network, self-consistent and plausible, but unverified against the register. Part 2 §2.11.1 sets out the same procedure generally.

7.4.7 RETURNS, TRACEABILITY, AND THE SEQUENCE YOU ARE COMMITTING TO

Three clauses interact, and together they make "buy now, sort the licence out later" a one-way door.

- **Change-of-mind returns are refused, expressly on regulatory-traceability grounds.** The Return Policy declines return, refund or replacement merely because "The buyer no longer requires the product. The buyer has changed their mind after purchase. ... The buyer ordered an incorrect product without any fault on the part of the Company." **[VERIFIED]** "I bought it and then found out I need a licence" is squarely a change-of-mind request.
- **Serial numbers are recorded and may be produced to WPC.** Return Policy §3: serial numbers "are recorded and maintained as part of the Company's internal compliance and traceability processes, and certain records **may be required to be maintained or produced before the competent regulatory authorities, including the Wireless Planning & Coordination (WPC) Wing**". **[VERIFIED]**
- **Cancellation closes when the serial number is allocated.** Cancellation "may not be accepted once an order has been packed, assigned to a courier, dispatched, **or where serial numbers have been allocated and recorded**". **[VERIFIED]**

Read together: from the moment a serial number is allocated to your order, you cannot cancel, you cannot return, and your name is attached to that serial number in a register the company says it may have to produce to WPC. A lawful, compliant dealer is, from the buyer's point of view, a dealer who can be asked who bought serial number X. That is a *good* thing about this seller and a *bad* thing about buying before you are licensed.

Note also that buying is itself deemed a declaration: the Product Compliance Customer Declaration Policy provides that "By purchasing any product from the Website, the customer declares ... that the product will be used only for lawful purposes and in accordance with all applicable Indian laws". **[VERIFIED]** So an unlicensed buyer who then uses the radio has (a) possessed radio equipment without authorisation or exemption, (b) transmitted without an assignment, and (c) made a contractual declaration to the seller that he was doing neither.

Two further weaknesses worth knowing. The Terms and Conditions are self-labelled an "**Expanded Foundation Draft**" and end with a note that "the final enterprise edition will be significantly more detailed (approximately 40–50 pages)"; the Shipping, Return, Warranty, Cancellation and Privacy documents are all self-labelled "Foundation Draft". **[VERIFIED]** And **nothing anywhere in the site's terms addresses what happens if WPC or the police take the radio off you** — there is no seizure, confiscation or forfeiture clause; risk passes on delivery, consequential loss is disclaimed, and compliance is the customer's. **[VERIFIED]** by exhaustive reading of the fetched policy pages. **If the radio is confiscated, the seller owes you nothing under these terms.** The warranty exclusion for "modifications including unauthorized firmware or hardware alterations" **[VERIFIED]** is also the one that bites an experimenter, since reprogramming a Baofeng is most of the point of buying one to learn with.

7.4.8 WHAT WEIGHT TO PUT ON EACH OF THE SELLER'S CLAIMS

Claim	Weight a buyer can put on it
"We hold DPL EDPL-14-F-202500053"	Moderate. Plausible, specific, consistent with a real 2007-incorporated telecom company. But undated, not publicly verifiable, and DPLs are a one-year instrument in a regime DoT closed on 2026-07-03
"Our licence-free models have ETA"	Good, and checkable. Six certificates, distinct numbers, dates, a named accredited test laboratory and a gazette reference. Verify the number before relying on it
"UV-32 is a Licence Band Walkie Talkie"	High — and it is against the seller's interest. Believe it. It is the most useful sentence on the page
"License Radios ... must be sold and operated only by valid WPC license holders"	High as a statement of law. Low as a description of the site's own conduct — the site does not enforce it at checkout
The "WPC User License" box on the UV-32 page	Near zero as a gate. Flagged <code>isMandatory: false</code> . Treat it as a notice, not a control
"Click HERE to apply for a licence"	Low. Points at the captive route, which is not available to an individual, on a portal that may be closed
"Country of Origin: India / Manufacturer: Shopperclue"	Low. A ₹1 lakh paid-up-capital Gurgaon company is not the manufacturer of a tri-band GPS transceiver carrying a United States FCC identifier issued to a Chinese firm; the listing's importer fields are present in the template and blank . [VERIFIED] on the blank fields; [UNCERTAIN] on how to read the origin declaration
"Frequency Range 400–520 MHz"	Unreliable. Contradicted by the manufacturer's own tri-band specification. Get it in writing
"Official Online Shop ... INDIA"	Self-conferred, from an Indian trade mark, not from Fujian Baofeng
Terms, Shipping, Return, Warranty, Privacy	Weak instruments. All self-labelled drafts; the Terms say the real version is still to be written

Two points in the seller's genuine favour, stated fairly. First, neither G S Tele Communication nor Shopperclue nor `baofengradios.com` appears anywhere in the CCPA walkie-talkie action of 2026-01-16, in which eight entities were penalised ₹44 lakh in total after 16,970 listings were examined. **[VERIFIED]** as a negative against four independent reports of the list contents; **[REPORTED]** for the list itself. No consumer complaint, legal notice or enforcement action naming these sellers could be found — **[UNCERTAIN]**, absence of evidence rather than evidence of absence, and a storefront whose content dates from 2026 would not yet have a track record either way. Second, **this is a domestic Indian dispatch, not a cross-border one:** stock sits at a named pickup point, products declare Indian origin, and there is no customs clause anywhere in the shipping policy. **[VERIFIED]** The consequence is a real advantage over buying abroad — **no consignment of yours crosses a border, and there is no scenario in which your parcel is seized at Nhava Sheva.** Whatever import happened, happened upstream and is the

seller's exposure. The residual risk, stated carefully, is that Indian customs law generally allows confiscation of improperly imported goods in the hands of a subsequent possessor; Part 2 §2.8 declined to paraphrase the relevant Customs Act sections without having read them, and that position stands — **[UNCERTAIN]**, and the reader should not be told either that he is safe or that he is exposed. What can be said is that a domestically purchased, invoiced, serial-numbered unit from a company with a DPL is a materially better evidential position than a grey-market parcel.

7.4.9 THE GENERAL LESSON

India's regime separates the gates deliberately (Part 2 §2.2). A seller's DPL, its ETAs and its Indian incorporation speak to **gate 1 (type approval)**, **gate 2 (import)** and **the seller's own gate 3 (possession as a dealer)**. **None of them speaks to your gate 3 (your possession) or your gate 4 (your transmission).**

An ETA certificate is a statement that a *model* conforms to a de-licensing notification. A DPL is a statement that a *dealer* may stock and sell. **Neither is a permission addressed to the purchaser. There is no document a shop in India can give you that authorises you to possess or operate a 10 W transceiver** — that authorisation is issued to a *person*, by WPC, after an examination.

So the correct reading of this storefront is: a real Indian company, better than a random marketplace listing, with honest-enough labelling and verifiable certificates on the models that need them — and **not one gram of authorisation for you.**

7.5 Better hardware for a licensed Indian amateur

7.5.1 THE UNCOMFORTABLE TRUTH FIRST

No mass-market radio is band-limited to India's amateur allocations. India's 70 centimetre allocation is **434–438 MHz**, which matches no other country's band plan: the European Union uses 430–440, the United States 430–450 or 420–450. So even the most tightly restricted imported amateur radio will transmit somewhat outside Indian amateur spectrum. **[VERIFIED]** — Annexure-I compared against the manufacturer specifications below.

The realistic goal is therefore minimising out-of-band transmit capability, not eliminating it. On that measure the difference between a European-version Yaesu FT-65E (6 MHz of transmit range outside Indian amateur spectrum) and a UV-32 (roughly 192 MHz outside) is enormous — and it is exactly the difference a customs officer or an inspecting official would notice.

There is a second, subtler reason band-limiting matters: **the import scope question.** Rule 7(3) exempts an ASOC holder from separate import authorisation, but rule 2(1)(b) scopes that to "a radio equipment required for operating an amateur station". **[VERIFIED]** Practitioner guidance circulated by NIAR puts the point bluntly:

"Every declaration for import of Ham equipment must be as AMATEUR RADIO EQUIPMENT and must be within the allowed frequency. It is always better to declare frequency like 144-146 MHz for VHF 434-438 MHz for UHF ... **and anything out of this frequency is technically disqualify as HAM equipment.**"

[REPORTED] — Devadas K.M. VU2DH, *Import of Amateur Radio Equipment: 2023-24*, circulated by NIAR. A radio whose manufacturer advertises 136–174 / 220–260 / 400–520 MHz transmit is, on that test, not "within the allowed frequency". Whether rule 7(3)'s privilege covers a wideband consumer hand-held is **[UNCERTAIN]** and is an open question across this report (Part 2 §2.8.3). Buying domestically sidesteps it for this purchase; buying a band-limited radio removes it permanently.

7.5.2 THE COMPARISON

	Baofeng UV-32	Yaesu FT-65E (European version)	TalkPRO LF446
Transmit coverage	~136–174, 220–260, 400–520 MHz (≈198 MHz)	144–146, 430–440 MHz (12 MHz)	446.0–446.2 MHz (0.2 MHz)
Of which lawful for an Indian amateur	6 MHz (≈3 %)	6 MHz (50 %)	not applicable — lawful for anyone
Power	10 / 5 / 1 W claimed; ~5 W and 3 W measured	5 / 2.5 / 0.5 W	0.5 W
Lawful power step for VU3 (10 W DC input)	low only	low, arguably mid	not applicable
Licence needed	ASOC	ASOC	none
Indian ETA	none, and none possible	none, and none possible	ETA SD 20250100112
Antenna	detachable SMA	detachable SMA	detachable — see caveat below
Front panel band-locked	no	to the amateur bands	to 16 channels
Approximate price in India	₹14,449	imported; higher	₹9,000 + taxes
Risk profile at a checkpoint	worst	moderate	negligible

Evidence: the FT-65E specification, including the "guaranteed within the 144 and 430 MHz amateur bands only" language and the transmit current figures used in §7.3.3, is **[VERIFIED]** from the Yaesu FT-65R/FT-65E Operating Manual, Specifications p.46, PDF fetched from yaesu.com. The UV-32 figures are as sourced in §7.3. The TalkPRO figures and ETA number are **[VERIFIED]** as published on talkpro.in, fetched 2026-08-23 — but see the antenna caveat at §7.6.2.

A trap worth naming. The Yaesu FT-65 "Asian version" transmits 136–174 / 400–480 MHz — wideband. [VERIFIED] — same manual. **Buying a Japanese brand is not by itself a guarantee of band-limiting; you must check the market version.** The cheapest band-limited Yaesu is the FT-4XE (European version), 144–146 / 430–440 MHz at 5 W [REPORTED] — manual summarised from search results, the PDF was not extracted. Icom, Kenwood and Alinco publish equivalent European variants in their IC-, TH- and DJ- series; their specifications were not verified and figures that have not been read will not be listed here [UNCERTAIN]. The version caveat applies to all of them.

7.5.3 VEHICLE-MOUNTED OPTIONS, GIVEN THE OVERLAND USE CASE

A hand-held on a magnetic mount works. A purpose-built mobile transceiver is better on audio, heat dissipation and transmit duty cycle — and it removes the accidental-VFO risk of §7.3.2 if it is a band-limited version.

Radio	Transmit	RF out	The Indian power problem
Yaesu FTM-6000E (European version)	144–146 / 430–440 MHz	50 / 25 / 5 W, SO-239 antenna socket, 13.8 V, about 10 A maximum on transmit	At 5 W the DC draw is roughly 12–15 W — around the Restricted ceiling. At 25 W or 50 W it is far over both grades. A 50 W mobile is only lawful in India if you keep it on its lowest setting and can show that. [REPORTED] for the specifications — dealer and rig-database pages; [VERIFIED] for the ceilings; the RF-to-DC mapping is inference, [UNCERTAIN]

The general point for any mobile transceiver. Every 2 metre / 70 centimetre mobile on the market starts at 20–50 W and steps down to 5 W. India's Restricted grade allows 10 W **DC input**, which is roughly 4–6 W RF out. So a mobile is lawful for a VU3 licensee **only on its lowest power step**, and lawful for a VU2 licensee up to roughly 10–16 W RF out. Buying a 50 W radio and using 5 W of it is legal but wasteful, and it means the front panel can be nudged into illegality with one button press. **[The ceilings are [VERIFIED]; the RF-output equivalents are inference and [UNCERTAIN].]**

7.5.4 ARE THERE INDIAN TYPE-APPROVED AMATEUR OPTIONS?

- **For licence-exempt 446 MHz: yes.** ETA-holding models exist and can be bought by anybody. [VERIFIED] for the existence of ETA'd 446 MHz product.
- **For amateur bands: no, and the question is malformed.** ETA is issued only for licence-exempt bands. Amateur equipment is not type-approved in India and cannot be; its legality flows entirely from the operator's ASOC. [VERIFIED] — Part 2 §2.7.1. **Any dealer offering an "ETA-approved ham radio" is selling something that does not exist.**

7.6 What to actually do for convoy communication

Every lawful way to get voice between vehicles in India where **not everyone is licensed**:

#	Option	Licence needed	Power ceiling	Equipment cost	Verdict for a convoy
A	Citizen Band, 26.957–27.283 MHz	None	5 W ERP	₹5,800–₹12,000 per set plus antenna	Best licence-free option. Highest legal licence-free power in India, and the rule was written for vehicles
B	PMR446, 446.0–446.2 MHz	None	500 mW e.r.p. , integral antenna only	₹3,000–₹9,000 per handset	Workable for a tight convoy on open ground; marginal in hills
C	LPD433, 433.05–434.79 MHz	None	10 mW e.r.p.	₹500–₹2,000 per pair	Not usable. A few hundred metres
D	Mobile phone / push-to-talk over cellular	None	not applicable	you already own it	Excellent where there is coverage; zero where there is not
E	PMRTS subscription (Public Mobile Radio Trunking Service)	None of your own — you subscribe	the operator's	commercial tariff	Urban and industrial coverage only; not a wilderness answer
F	Captive / closed-user-group frequency assignment	Yes, and not obtainable by an individual	assigned	high	Part 1 §1.7. Not a route for a private person
G	Satellite messenger or satellite phone	Yes — DoT permission	not applicable	not applicable	Do not. See §7.6.4

Options A, B and C rest on the de-licensing notifications catalogued in Part 2 §2.4, all **[VERIFIED]** from gazette text.

7.6.1 OPTION A — CITIZEN BAND AT 5 W ERP, AND ITS ONE REAL PROBLEM

The rule, verbatim:

"3. Use of wireless equipment in the band 26.957 – 27.283 MHz. - Notwithstanding anything contained in any law for the time being in force, no licence shall be required by any person to establish, maintain, work, possess or deal in any wireless equipment intended to be used

while in motion or during halts, on non-interference, non-protection and shared (non-exclusive) basis, in the frequency band 26.957 – 27.283 MHz with **5 Watt Effective Radiated Power and built-in antenna**"

[VERIFIED] — G.S.R. 533(E), 2005-08-12, rule 3, *Use of low power equipment in the Citizen band 26.957 – 27.283 MHz (Exemption from Licensing Requirement) Rules, 2005*. Amended only as to the interference remedy by G.S.R. 35(E), 2007-01-10; **the 5 W figure is unchanged**. **[VERIFIED]** Rule 2(b) confirms that "'Effective Radiated Power' includes the gain of the antenna, if any". **[VERIFIED]**

Why this is the right instrument for a convoy. Read the operative words again: wireless equipment "*intended to be used while in motion or during halts*". **This exemption was drafted for vehicles**. It is also **five watts** — ten times PMR446 — and it is the highest-power licence-free two-way voice allocation in the entire Indian de-licensed corpus (Part 2 §2.6.1 establishes that by exhaustive enumeration). **[VERIFIED]**

The problem that must not be glossed: "built-in antenna". The exemption says 5 W ERP **and built-in antenna**. A vehicle CB installation with a boot-lip or roof whip and a coaxial feed does **not** have a built-in antenna, and on the face of the gazette such an installation falls outside the exemption. **[VERIFIED]** from the wording.

Against that: the same rule contemplates use "while in motion", which is very hard to reconcile with a genuinely built-in antenna at 27 MHz, where a quarter wave is 2.75 metres; President Electronics, a manufacturer selling into India, states it holds Indian ETAs for its CB portfolio and publishes detailed vehicle-antenna mounting guidance **[REPORTED]** — mera-cb.com/in/FAQ.php, fetched 2026-08-23; and Indian retailers sell mobile CB sets openly and describe them as free to use vehicle to vehicle **[REPORTED]**.

[UNCERTAIN] — **market practice and the gazette wording do not line up, and no WPC clarification, test method or enforcement precedent on CB mobile antennas could be found**. Part 2 §2.12 lists the same question as one it could not settle. **This is exactly the point on which to get a written answer from WPC before spending money** — see §7.7.2, question 4.

Real-world range [ENGINEERING ESTIMATE] — derived from propagation fundamentals, not law and not vendor claims:

Scenario	Realistic range
Vehicle to vehicle, proper roof-mount antennas, flat open terrain (desert, plains, salt flat)	5–12 km
Vehicle to vehicle, open highway, rolling country	3–8 km
Vehicle to vehicle, forest or dense scrub	2–5 km
Vehicle to vehicle, mountain switchbacks with a ridge between you	0.5–3 km , sometimes nothing
Across a valley, both vehicles in line of sight	15–40 km

Why 27 MHz beats 446 MHz for this job: ten times the power; an antenna that can actually be a metre or two long instead of 50 millimetres of helical; and better diffraction around terrain obstacles at 11 metres wavelength than at 67 centimetres. Against it: 27 MHz picks up ignition noise, electrical noise and occasional long-distance skip interference — which is why you will sometimes hear a station 3,000 km away and nothing from the vehicle 2 km ahead — and it needs real antenna installation work. **Ignore any "3 km to 160 km" range claim on a retail page. [REPORTED] — that is marketing.**

Indian retail prices, **[REPORTED]** from NPC Wireless, fetched 2026-08-23 and all listed as sold out at the time, so treat them as indicative: President Tommy ASC (40-channel AM) ₹5,799; President Bill II ₹9,500; Luiton base station ₹6,999. Add roughly ₹1,500–₹4,000 per vehicle for a decent mobile antenna, mount and coaxial cable. **[ENGINEERING ESTIMATE]**

Note one loose end: G.S.R. 533(E) unusually contains **no express type-approval rule of its own** (Part 2 §2.12, question 2). **[UNCERTAIN]** That does not remove the general ETA obligation but it makes the chain less tidy than for PMR446. Ask the seller in writing for the ETA number of the exact model and verify it by hand.

7.6.2 OPTION B — PMR446 AT 500 MW, AND THE FRANK ENGINEERING JUDGEMENT

Table V of G.S.R. 1047(E), 2018-10-18, with its definitional note, verbatim:

Frequency range	Transmit power limit	Additional parameters
446.0–446.2 MHz	500 mW e.r.p.	Channel spacing 6.25 kHz and 12.5 kHz

"Note: For the purpose of this Table, personal mobile radio 446 MHz device means **hand portable radio with no base station or repeater use and uses integral antennas only** in order to maximise sharing and minimise interference, and which operates in **short range peer-to-peer mode** and shall be used **neither as a part of infrastructure network nor as a repeater;**"

[VERIFIED] — G.S.R. 1047(E), Table V and its note; full text quoted in Part 2 §2.5.1.

Four conditions bind simultaneously: the band; 500 mW e.r.p.; hand-portable with an **integral** antenna; no base station, no repeater, no infrastructure. Break any one and the exemption is gone.

Is 500 mW enough for a convoy? The honest answer is: it depends entirely on how tightly you drive, and the integral-antenna rule is the reason it often is not.

The killer is not the power — it is that you may not fit an external antenna. A PMR446 handset used *inside a steel vehicle* is transmitting through a Faraday cage. Vehicle body shielding costs roughly **10–20 dB** for a handset held at seat level, and you cannot buy that back with a roof antenna, because a roof antenna would take you outside the exemption. **[ENGINEERING ESTIMATE]**

Scenario	Realistic range [ENGINEERING ESTIMATE]
Two people standing outside, flat open ground, line of sight	2–5 km
Handset inside vehicle to handset inside vehicle, open highway	0.5–2 km
Inside vehicles, rolling or wooded terrain	0.3–1 km
Inside vehicles, mountain switchbacks, ridge in between	100–500 m, frequently nothing
Hilltop to hilltop, handsets outside	10–30 km

Cross-checked against independent secondary sources, which typically quote 1.6–4.8 km in open terrain and under 1.6 km in urban or wooded areas, with the explicit warning that manufacturer claims of "10 km" or "16 km" assume high ground and clear line of sight. **[REPORTED]** — walkie-talkies.net range guide and the PMR446 Wikipedia article, 2026-08-23. The in-vehicle figures above are lower than those open-terrain figures because of body shielding, which those sources do not model.

Verdict, stated bluntly. PMR446 is genuinely useful for a convoy that stays within about a kilometre — a tight group on a highway, a group parked at a campsite, or spotters walking a vehicle through an obstacle, where everyone is outside the vehicle and it works well. **It is not adequate for a convoy strung out over 5 km of Himalayan switchbacks, which is the situation in which you most need communications. Anyone telling you 500 mW will hold a mountain convoy together is selling you radios.**

Indian retail **[REPORTED]**, fetched 2026-08-23: **TalkPRO LF446**, 446.0–446.2 MHz at ≤0.5 W, claimed 4–6 km, quoting **ETA SD 20250100112, ₹9,000** plus tax **[VERIFIED]** that the page publishes this; **BRUTFORCE BFR-001 / 002**, 446–446.2 MHz, "WPC approved, ETA included", ETA number not published on the pages reached **[UNCERTAIN]**. **A vendor quoting an actual ETA number is a good sign — you can verify it. A vendor who just says "WPC approved" is asking you to take their word.**

Two cautions specific to this option:

1. **The TalkPRO page describes its antenna as detachable**, while Table V defines the exempt device as one that "uses integral antennas only". **[VERIFIED]** on both texts. That is a real discrepancy to raise with the seller before buying. **[UNCERTAIN]** as to how WPC reads it in practice.
2. **Two mis-selling traps are live in this market. The BF-888S is widely sold in India as a "licence-free 446 MHz walkie-talkie" and is not one** — it is the exact model of which the Directorate of Revenue Intelligence seized 11,060 units at Nhava Sheva in March 2026 **[REPORTED]**. And **"446 MHz base station" products are being sold as licence-free**, when Table V's note expressly excludes base station use, repeater use, infrastructure use and non-integral antennas. **[VERIFIED]** Part 2 §2.11.2 lists the full set of listing red flags.

7.6.3 THE LAYERED ARCHITECTURE THAT ACTUALLY WORKS

Cellular is better than any radio wherever there is coverage — unlimited range, text fallback, a written record, and no equipment cost. Its weakness is precisely the expedition case: it fails at exactly the moment the terrain gets interesting. Indian national highway coverage is good and improving; Himalayan side valleys, the Rann, deep forest and high-altitude cold deserts are not covered.

What experienced overlanders actually run is a layered system: cellular as the primary link, radio as the short-range in-convoy link, and a **written convoy standard operating procedure** as the fallback that works with no radio at all — fixed rally points at every junction and fuel stop, a maximum interval, a named tail vehicle, and a stop-and-wait rule if the vehicle behind is not seen for N minutes.

The procedure matters more than the radio. A disciplined convoy with no radios loses nobody; an undisciplined convoy with radios loses people the moment the radios fail.

7.6.4 SATELLITE: DO NOT

Iridium, Thuraya and similar handsets, and **Garmin inReach and SPOT-type satellite messengers**, require prior DoT authorisation; without it they are dealt with under the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933. Enforcement is active at airports: a Canadian visitor was detained at Dabolim over a Garmin inReach Mini, and in March 2025 three American travellers and one Australian were booked at Delhi over Garmin units and an Iridium handset. **[REPORTED]** — multiple outlets including Explorersweb and MediaNama. Apple's Emergency SOS via satellite is **not available in India** as at 2026-08. **[REPORTED]**

[UNCERTAIN] — no DoT primary source for the satellite-device position was fetched, because it falls outside this report's scope. The position is well attested in secondary reporting and is consistent with the statutory scheme. **If satellite communication matters to your expedition, it needs its own research before you buy or import anything.**

7.7 The plan, the sequence, and what to confirm before spending money

7.7.1 STEP BY STEP, WITH ROUGH TIMELINES

Timelines are judgement drawn from the research, not published service standards, except where marked otherwise.

#	Step	When	Cost	Notes
0	Decide what each use actually requires. Vehicle mobile → ASOC. Convoy with unlicensed people → licence-free only. Learning → ASOC, plus HF for the expedition case.	Today	₹0	\$7.2

#	Step	When	Cost	Notes
1	Buy licence-free convoy radios now. CB at 5 W ERP if §7.7.2 question 4 resolves acceptably; ETA-approved PMR446 handsets otherwise or in addition. Verify every ETA number by hand on Saral Sanchar before paying.	Week 1	₹6,000–₹12,000 per vehicle (CB) or ₹3,000–₹9,000 per handset	This is the only lawful answer to the convoy problem and it needs no exam. [VERIFIED]
2	Write the convoy standard operating procedure. Rally points, maximum interval, tail vehicle, stop-and-wait rule.	Week 1	₹0	Worth more than either radio
3	Send the written questions in §7.7.2 to WPC.	Week 1	₹0	Send before the money, not after
4	Register on Saral Sanchar and apply for the ASOC (Restricted) examination. Since March 2025 the examination and certificate applications are a single step with self-certified documents.	Week 1–2	₹100	[VERIFIED] — Part 3 §3.9
5	Study and sit the examination at a Wireless Monitoring Station. One paper, two parts: radio theory and practice, and radio regulations. No Morse for Restricted.	Weeks 2–16	included	Examinations run "as and when there are sufficient number of candidates" [VERIFIED] — Part 3 §3.7. Note the [UNCERTAIN] pass-mark conflict recorded at Part 1 §1.2 and Part 3 §3.7.12: assume the harder figures
6	Pay the certificate fee; the ASOC issues with your callsign (VU3 block for Restricted). Payment must be made within two years of the result.	+ weeks	₹1,000 (20 years) or ₹2,000 (lifetime, to age 80)	[VERIFIED]
7	Only now buy licensed-band hardware. Preferably a band-limited European-version hand-held or a band-limited mobile transceiver, not a UV-32. Insist on a tax invoice naming model and serial number, and keep it with the ASOC.	After the certificate	varies	§7.5

#	Step	When	Cost	Notes
8	Set up the station properly. Bound-book log in the vehicle; equipment register entry on the day of acquisition; ASOC (or a print of the digital certificate) in the vehicle; memories programmed only to Indian amateur frequencies; hands-free or passenger-operated push-to-talk.	Same week	₹0	\$7.2.1
9	Get the rest of the convoy licensed if you want lawful amateur-band convoy working.	Ongoing	₹1,100 or ₹2,100 each	\$7.2.2
10	Add HF when the expedition case justifies it. Restricted grade already authorises 50 W DC input on ten HF bands — the capability that actually matters in remote terrain.	Later	varies	\$7.2.3

Realistic total elapsed time to a lawful 10 W station: roughly three to nine months, dominated by examination scheduling and result processing (Part 1 §1.2). **Total government outlay: ₹1,100 to ₹2,100.** The convoy problem, by contrast, is solved in week one for the price of the radios — which is the strongest practical argument for doing the two things in parallel rather than in sequence.

What the sequence deliberately avoids. Buying the UV-32 first and "sorting the licence out later" is the one path that is strictly worse than both alternatives: the site will let you complete the purchase (§7.4.4); from the moment the serial number is allocated you cannot cancel or return it (§7.4.7); your name is attached to that serial number in a register the seller says it may produce to WPC; and in the interval you are in possession of radio equipment with neither authorisation nor exemption.

7.7.2 THE SPECIFIC UNCERTAINTIES TO CONFIRM WITH WPC BEFORE SPENDING MONEY

Send these to the WPC COP Section — the FAQ gives **Engineer (COP), 011-23036181, awacop-dot@gov.in [VERIFIED]** — and to the nodal officer named on DoT's ASOC service page, **Senior Deputy Wireless Adviser, 011-23036540, umang.srivastava@nic.in [VERIFIED]**. Keep every reply with the licence. Part 2 §2.12 and Part 3 §3.12 list further questions; these seven are the ones that bear directly on this purchase.

#	Question to ask	Why it decides whether to spend the money	Tag today
1	Possession. "Does rule 10(f) of the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026 (G.S.R. 592(E), 2026-07-08) exempt an ASOC holder from any requirement for a radio equipment possession authorisation, and does that exemption extend to amateur radio equipment whose transmit coverage extends beyond the amateur allocations?"	If the answer is no, there is no lawful possession route for this radio at all, licence or not	[UNCERTAIN] — Part 1 §1.5 Group B, §7.3.6
2	The power table. "Please confirm whether Annexure-I to OM No. L-14011/255/2004-AMT dated 13.08.2010 continues to be the operative specification of frequency bands, emissions and transmit power under rule 7(1) of the 2024 Rules, or whether a revised specification has been issued."	Every band and power figure in this part rests on a 2010 executive memorandum	[UNCERTAIN] — Part 1 §1.8
3	DC input measurement. "For ASOC (Restricted), Annexure-I specifies 10 watts 'Max DC Input Power' on 144–146 MHz and 434–438 MHz. For a modern solid-state transceiver, how is DC input power to the final stage determined for compliance, and what RF output does WPC treat as corresponding to 10 W DC input?"	Determines which power step of any radio you may lawfully use, and whether a "10 W" hand-held is lawful at all for VU3	[UNCERTAIN] — §7.3.3
4	CB and external antennas. "Rule 3 of G.S.R. 533(E) dated 12.08.2005 exempts equipment at 5 W Effective Radiated Power with built-in antenna . Does a Citizen Band transceiver installed in a road vehicle with an externally mounted antenna remain within the exemption, provided ERP does not exceed 5 W? If not, on what basis has ETA been granted to vehicle-mounted CB equipment?"	Decides whether the recommended convoy solution is solid or merely conventional	[UNCERTAIN] — §7.6.1
5	Restricted areas. "Please provide the current list of areas in which the Government has restricted the installation or operation of amateur stations, or the carriage of amateur station equipment, as referred to in Questions 3 and 5 of the FAQ dated 05.12.2024."	Overland routes run through exactly these belts; no published list could be found	[UNCERTAIN] — §7.2.1
6	Mobile operation and the 90-day gloss. "Where an amateur station is <i>mobile</i> — installed in a road vehicle continuously travelling and not stationed at any single location — does the 90-day period in FAQ Q4 apply at all, and if so how is it computed?"	Matters only for trips beyond three months, but the 90-day figure is an administrative gloss borrowed from rule 13(3) and appears nowhere in the Rules for mobile operation	[UNCERTAIN]

#	Question to ask	Why it decides whether to spend the money	Tag today
7	Blacklisted models. "Is there a published list of radio models blacklisted or prohibited by the Department, and where is it published?"	Press reports the BF-888S as blacklisted; no gazette notification or published list could be located	[UNCERTAIN] — Part 2 §2.10

And four questions to put **to the seller in writing**, before paying, keeping the replies:

1. **The exact transmit frequency ranges of the UV-32 they ship** — specifically, does it transmit on **144–146 MHz**? (§7.3.1: the listing, the manual and the manufacturer's page disagree.)
2. **The warranty term for this model, and whether reprogramming voids it.** No duration is stated on the product page, and the warranty excludes "unauthorized firmware alterations". (§7.4.3, §7.4.7.)
3. **The DPL issue and expiry dates, or the corresponding Possession Authorisation number under the 2026 Rules.** (§7.4.5.)
4. **Whether they will sell a licence-band radio to a buyer with no licence, and on what basis** — given rule 8(1) of the 2026 Rules, under which an authorised dealer may give possession only to a rule 10 person or an authorisation holder. **[VERIFIED]** — Part 2 §2.9.2.

7.7.3 THE SHORT VERSION OF PART 7

1. **The UV-32 is the wrong radio for this reader.** Not because it is bad, and not because the seller is dishonest, but because roughly 97 % of its transmit coverage is spectrum an Indian amateur may never touch, it has no band-lock and cannot be given one, its lawful power step for a Restricted licensee is the lowest one, and its "10 W" is an RF-output figure measured against a DC-input ceiling. **[VERIFIED]** on the coverage; **[UNCERTAIN]** on the power arithmetic.
2. **The seller is real and the seller's compliance is the seller's.** A DPL, six ETA certificates and an Indian incorporation are findings about G S Tele Communication Limited. They authorise nothing for you, and the company's own Regulatory Compliance Policy says so in terms. **[VERIFIED]**
3. **The convoy use is unlawful on amateur frequencies and no purchase changes that.** Communication only with other authorised amateur stations; third-party messages expressly banned; the calamity carve-out requires a government permission and a District Magistrate at the other end. **[VERIFIED]**
4. **The lawful convoy answer exists today and costs less than the UV-32.** CB at 5 W ERP, subject to the built-in-antenna question; PMR446 at 500 mW as a fallback and for out-of-vehicle work; a written convoy procedure that works with no radio at all. **[VERIFIED]** on the legal position; **[ENGINEERING ESTIMATE]** on the ranges.
5. **Do the examination first.** ₹1,100 or ₹2,100 buys the only thing on this page that actually authorises anything. Everything else is cheaper, simpler and safer on the other side of it.

7.8 Disclaimer for this part

This part is a research summary compiled on 2026-08-23. **It is not legal advice**, and nothing in it authorises the purchase, import, possession or operation of any radio. It rests in places on manufacturer marketing material that contradicts the same manufacturer's user manual; on a mirrored rather than primary FCC record; on a commercial mirror of Ministry of Corporate Affairs registry data; on seller-hosted ETA certificates that could not be checked against the WPC register, which is CAPTCHA-gated; and on an amateur band and power table (a WPC Office Memorandum of 2010-08-13) whose continued validity under the 2024 Amateur Services Rules could not be independently confirmed. The power arithmetic in §7.3.3 is engineering inference, not measurement and not law, and the range figures in §7.6 are engineering estimates, not legal claims. Prices, listings, page contents and web pages described here were fetched on one day and change constantly. Several findings are expressly tagged **[UNCERTAIN]** because they could not be verified against a primary source — in particular the ASOC possession gateway at rule 10(f), which the report's own fact-check could not re-read. Confirm the questions in §7.7.2 with the WPC Wing of DoT in writing, verify any ETA number yourself on saralsanchar.gov.in, and speak to an established amateur radio club such as ARSI or NIAR, before you spend money, before you plan a trip around any of this, and before you transmit.

Part 8 — What Indian expedition groups actually use

Parts 1 to 6 set out the law. Part 7 applied it to one purchase decision. This part answers a different question, and it is the question most people ask first: **what do Indian motorcycle rider groups, four-wheel-drive clubs and commercial expedition operators actually put in their vehicles, and how well does it work?**

That is not the same question as "what does the law permit", and it is not the same question as "what does this report recommend". Occasionally the three answers coincide. More often they do not, and the gaps between them are the most useful thing in this part. Where a widespread practice is unlawful, this part says so once, with the specific provision, and then carries on describing the practice. **Describing what people do is not endorsing it.**

Position as at 2026-08-23. Every retailer page, forum thread, operator inclusion list and product listing described here was fetched on that date, and prices, stock and product ranges in this market change weekly.

Evidence tags in this part, and a warning about their distribution. Tags carry the same meaning as in Part 1 §1.1 and Part 7:

Tag	Meaning
[VERIFIED]	A primary or official source actually fetched, with URL and document date. In this part it is used almost exclusively for the legal overlay , carried from Parts 2, 3, 6 and 7 where the gazette texts were fetched and quoted.

Tag	Meaning
[REPORTED]	A named secondary source — a forum post with a username and date, a retailer page, a magazine review, an operator's inclusion list.
[UNCERTAIN]	Could not be confirmed, sources disagree, or the source is undated.
[ENGINEERING ESTIMATE]	A range or performance figure derived from propagation fundamentals. Not law, not measurement, and not a vendor claim.

Most of this part is [REPORTED], and that is the correct tag rather than a weakness.

Community practice is not recorded in gazettes. It is recorded in forum posts, retailer listings, tour operator inclusion lists and product reviews. Where a rider says a mesh intercom holds 500 to 600 metres on a highway, that is **one rider's impression on one set of rides**. It is not a measurement and this part will not dress it up as one. Where several independent people converge on the same number, the convergence is said explicitly, and that convergence is the strongest evidence available on this subject.

Never read a [REPORTED] practice statement in this part as a statement of law. The law is in Parts 1 to 3 and Part 7. This part describes behaviour.

What could not be reached, and what that costs. Stated plainly, because it shapes everything below.

Source	Status
Team-BHP (team-bhp.com)	Returns HTTP 403 to automated clients. Every Team-BHP quotation in this part was read from an Internet Archive Wayback capture , which preserves usernames, post numbers and post dates. The most recent Team-BHP material available through that route is from 2018 to 2021 , not 2026. Where a Team-BHP post is quoted below, assume the archive caveat applies.
Reddit	Entirely inaccessible. No r/indianbikes, r/IndiaTravel or r/india evidence exists in this part at all. That is where the youngest and largest slice of the Indian riding community talks, and it is a genuine gap.
Quora	HTTP 403. Not used.
YouTube	Video pages render as a JavaScript shell; descriptions, dates and view counts were not retrievable. This part can therefore report only that a genre of Indian "licence-free walkie talkie" review videos exists , from search-result titles — not what any individual video says.
Amazon.in	HTTP 503 to automated clients. Indian pricing below comes from specialist retailers and brand storefronts, not from the marketplaces where most Indians actually buy.
xBhp (xbhp.com)	Fetches directly. Post dates were captured for one thread and not the other , flagged inline at §8.3.6.
Saral Sanchar public ETA register	CAPTCHA-gated (Part 2 §2.7). No ETA number anywhere in this part has been checked against WPC's own database. Where a vendor quotes an ETA number, this part records the number and the fact that the vendor quoted it — nothing more.

The practical consequence: community evidence here skews towards **forum material from 2012 to 2018** and **retailer and vendor material from 2026**. It is thinner and older on the motorcycle side than it should be.

Abbreviations used in this part, expanded here on first use; the master list is at Part 1 §1.10 and Part 6 Appendix A. **WPC** = Wireless Planning and Coordination Wing of the **DoT** = Department of Telecommunications. **ASOC** = Amateur Station Operator Certificate, the Indian amateur radio licence (Part 3). **ETA** = Equipment Type Approval (Part 2 §2.7). **CB** = Citizen Band. **PMR446** = Personal Mobile Radio in the 446 MHz band. **FRS / GMRS** = Family Radio Service and General Mobile Radio Service, two United States licence-free allocations that do **not** exist in India. **ERP / e.r.p.** = effective radiated power, referenced to a half-wave dipole and including antenna gain. **VHF / UHF** = very high frequency (30–300 MHz) / ultra high frequency (300–3000 MHz). **LOS** = line of sight. **PTT** = push-to-talk, the transmit button on a radio. **SOP** = standard operating procedure. **ISM** = industrial, scientific and medical, the class of shared radio bands that includes 2.4 GHz. **DMC** = Dynamic Mesh Communication, Cardo's proprietary mesh protocol. **RIDEGRID** = BluArmor's equivalent. **IP67 / IP65** = ingress protection ratings for dust and water. **BODA** = Bangalore Offroaders, a Bengaluru four-wheel-drive club. **POC** = push-to-talk over cellular, a handset that looks like a radio but is a mobile subscriber device. **GNSS** = global navigation satellite system (GPS, GLONASS, Galileo). **CCPA** = Central Consumer Protection Authority. **DRI** = Directorate of Revenue Intelligence. **MHA** = Ministry of Home Affairs. **SKU** = stock keeping unit, one line on a retailer's catalogue. **MV Act** = Motor Vehicles Act, 1988.

8.1 If you only read one page

8.1.1 THE HEADLINE, IN TWO SENTENCES

The dominant purpose-bought communication device in Indian group riding is a 2.4 GHz helmet intercom, and it is not close. [REPORTED] — evidence at §8.2.2 and §8.3.

But the dominant communication system is not a device at all. It is a phone, a WhatsApp group, and an agreed stop-and-regroup discipline — with an intercom layered on top for the riders who have bought one. Radios of any kind are a **minority practice**, confined to commercial operators, off-road event crews and a small pocket of licensed amateurs. Satellite is effectively absent and legally dangerous. [REPORTED]

8.1.2 THE RANKED PICTURE, COMPRESSED

Rank	What is used	Who uses it	Licence-free?
1	Phone, WhatsApp, and rally-point discipline	Everyone. The default, and the fallback for every other layer	Yes — no radio involved
2	Bluetooth or mesh helmet intercom	Motorcyclists, all segments. The dominant <i>device</i>	Yes — G.S.R. 45(E)

Rank	What is used	Who uses it	Licence-free?
3	Hand signals and a convoy SOP	Four-wheel-drive and off-road clubs. The primary medium for off-road spotting	Yes
4	Licence-free PMR446 handhelds	Commercial tour operators, off-road event crews, a few self-drive convoys	Yes if compliant — G.S.R. 1047(E) Table V
5	Non-compliant handhelds sold as "licence-free" (Baofeng BF-888S class)	Buyers who trusted a marketplace listing	No — §8.4.3
6	Amateur radio (VU2 / VU3)	Bangalore off-road marshals, rally communication crews, a handful of touring hams	Requires an ASOC
7	Citizen Band, 27 MHz	Almost nobody. Effectively dead in India	Yes — G.S.R. 533(E)
8	Satellite messenger or satellite phone	Effectively nobody resident; foreign visitors get arrested	No — prior DoT permission required

[REPORTED] as to the ranking and prevalence; **[VERIFIED]** as to every legal basis, all carried from Part 2 §2.4 and Part 7 §7.6. The detailed version with prevalence evidence is at §8.2.

8.1.3 THE THREE THINGS MOST WORTH KNOWING

1. **The best licence-free option on paper is the one nobody uses.** Part 7 §7.6.1 concluded that Citizen Band at 5 W ERP is the only licence-free route to a genuine multi-kilometre vehicle-to-vehicle link in India. **That conclusion is still correct as a matter of law and physics, and the Indian market has comprehensively ignored it.** Every Indian CB listing found in this research was out of stock, and the one Indian off-road club that tested CB rejected it on performance grounds in hilly wooded terrain. Both findings are true at once, and §8.5 sets out exactly how to hold them together.
2. **The Indian off-road community's radio answer was amateur radio, not CB and not PMR446.** A Bangalore four-wheel-drive club put its marshals through the ASOC examination. That is the most impressive community-practice finding in this part, and it is also the lawful route — provided the traffic stays between licensed operators, which at a club event with unlicensed participants it very likely does not. See §8.4.5.
3. **Community legal advice on this subject is systematically out of date, and the errors run in a consistent direction.** The community's mental model is frozen somewhere around 2014: "only ham and CB are legal", "PMR and FRS are illegal here", "mobile amateur operation needs a special permit", "a licence takes a year and the police interview your neighbours". All four were true once. None is true now. See §8.6.

8.1.4 THE RECOMMENDED ARCHITECTURE, COMPRESSED

Detailed at §8.7, with what to buy for which role.

Layer	What	Who carries it	Cost	Legal basis
0	Written convoy SOP — riding order, named lead and sweep, maximum interval, regroup at every junction and fuel stop, stop-and-wait rule	Everyone	₹0	Not a radio question
1	2.4 GHz mesh intercom , one per motorcycle plus one off-helmet unit per four-wheeler, all the same brand	Every rider and the lead car	₹6,500– ₹11,000 each	G.S.R. 45(E) [VERIFIED]
2	Two to four ETA-quoted PMR446 handhelds — lead, sweep, backup vehicle, spare	Crew roles, not every person	₹7,400 a pair	G.S.R. 1047(E) Table V [VERIFIED]
3	Citizen Band — only if you pass the test at §8.5.6	Four-wheelers only	₹7,500– ₹16,000 per vehicle	G.S.R. 533(E) [VERIFIED], with the antenna question open
4	Your own Restricted ASOC	You	₹1,100 or ₹2,100	G.S.R. 675(E) [VERIFIED]
5	A scheduled check-in plan. No satellite messenger.	Everyone	₹0	§8.7.6

Baseline total for a five-person mixed group: roughly ₹43,500, of which ₹0 is the layer that matters most. That is less than the price of one flagship intercom pair, and every rupee of it is lawful.

8.2 The ranked practice picture, in detail

8.2.1 THE TABLE, WITH PREVALENCE EVIDENCE AND LEGAL BASIS

Rank	What is used	Who uses it	Evidence of prevalence	Licence-free? Legal basis
1	Phone, WhatsApp group, voice call, rally-point discipline	Everyone, in every segment	Present in every Indian thread read for this research, from 2005 to 2023. The commercial operators sell the human version of it — road captain, mechanic, backup vehicle — rather than radios (§8.4.2) [REPORTED]	Yes. Not a radio-licensing question at all: you are a subscriber on a licensed network. Part 7 §7.6 option D

Rank	What is used	Who uses it	Evidence of prevalence	Licence-free? Legal basis
2	Bluetooth or mesh helmet intercom	Motorcyclists, all segments; standard kit in the touring and organised-tour segment	Every Indian motorcycle-gear retailer runs a dedicated intercom category; Sena built an India-specific product line; India has a domestic manufacturer with a five-model range; mainstream Indian motoring press reviews them as ordinary kit [REPORTED] — §8.3.2	Yes. G.S.R. 45(E), 2005-01-28, rule 3 — no licence to "establish, maintain, work, possess or deal in any wireless equipment" in 2.4–2.4835 GHz within the power table. ETA still required — rule 6(1). [VERIFIED] — Part 2 §2.4.1, §2.7
3	Hand signals and a written convoy SOP	Four-wheel-drive and off-road clubs; commercial operators in the form of road captain plus sweeper plus backup vehicle	The most substantive Indian off-road communications thread found opens with hand signals, not radios, and gives cost as the first reason [REPORTED] — §8.4.1	Yes. Nothing is transmitted
4	Licence-free PMR446 handhelds	Commercial tour operators, off-road event crews, a small number of self-drive convoys	Present in roughly one in five operator inclusion lists checked (§8.4.2). Indian brands — TalkPRO, BRUTFORCE, Vertel, Sanchar, Wavex, Convey — sell into event and business-to-business channels [REPORTED]	Yes if compliant. G.S.R. 1047(E), 2018-10-18, Table V: 446.0–446.2 MHz, 500 mW e.r.p. , hand-portable, integral antenna only , no base station, no repeater, no infrastructure; ETA required under rule 5(2). [VERIFIED] — Part 2 §2.5.1, Part 7 §7.6.2
5	Non-compliant handhelds sold as "licence-free" — the BF-888S class	Buyers who trusted a marketplace listing	Unknown but large. On one platform alone, CCPA found 65,931 units sold with the frequency range blank or outside the exempted band [VERIFIED] — PIB release, 2026-01-16, Part 6 §6.4.7	No. 400–470 MHz is not an Indian exemption; no ETA exists or can exist for it; unauthorised possession engages the Telecommunications Act 2023. Part 2 §2.10, Part 6 §6.2

Rank	What is used	Who uses it	Evidence of prevalence	Licence-free? Legal basis
6	Amateur radio (VU2 / VU3 callsigns)	Bangalore off-road marshals; rally communication crews; a handful of touring hams	Small, concentrated and genuinely competent. One club is reported at "70% licensed" in 2015 [REPORTED] , and all its marshals licensed [REPORTED] — §8.4.5	Requires an ASOC , and lawful only between licensed operators. G.S.R. 675(E), Annexure A para 2(1) (a) and 2(2)(b). [VERIFIED] — Part 3 §3.6, Part 7 §7.2.2
7	Citizen Band, 26.957–27.283 MHz	Almost nobody. Long-haul truck drivers are the only user group any source identifies, and no usage statistics exist for them	Effectively dead. Every Indian CB SKU found was out of stock; no organised Indian club or overland group running CB in 2026 could be found [REPORTED] / [UNCERTAIN] — §8.5	Yes. G.S.R. 533(E), 2005-08-12, rule 3: 5 W ERP and "built-in antenna", equipment "intended to be used while in motion or during halts". [VERIFIED] — Part 2 §2.5.2, Part 7 §7.6.1. The built-in-antenna condition is [UNCERTAIN] in application
8	Satellite messenger or satellite phone	Effectively nobody resident. Foreign visitors get arrested at airports	Multiple documented arrests and a 2025 United States Embassy travel alert [REPORTED] — Part 7 §7.6.4	No. Prior DoT permission required; without it, criminal exposure under the Indian Telegraph Act 1885 and the Indian Wireless Telegraphy Act 1933 [REPORTED]

8.2.2 ONE CATEGORY THAT DOES NOT FIT THE RANKING: CELLULAR PUSH-TO-TALK HANDSETS

A whole genre of Indian retail sells **POC** — push-to-talk over cellular — handsets: devices shaped like a walkie-talkie, with a PTT button and a speaker grille, containing a 4G modem and a SIM card and running an application such as Zello or RealPTT. Indian search results advertise them with headline ranges of "**5000 Kms**" and "**10k Kms**". **[REPORTED]** — from search-result titles; the video pages themselves could not be fetched, so this part reports the **existence of the genre**, not the content of any video.

Three things about them, all of which matter:

- **They are not radios in any meaningful sense.** They are Android handsets with a PTT button.
- **They raise no radio-licensing question whatsoever.** You are a subscriber on a licensed network, exactly as with a phone. **[VERIFIED]** as a matter of the scheme in Part 1 §1.4.

- **They fail in precisely the places an expedition needs them.** A POC handset has exactly the coverage of the SIM inside it. On a highway convoy it works well and costs almost nothing; at Khardung La it is a paperweight.

The reason they matter to this report is that their marketing pollutes the search results a buyer sees. A rider searching "licence-free walkie talkie India" gets a mixture of lawful PMR446 sets, unlawful BF-888S sets, and cellular handsets advertising thousand-kilometre range — three completely different products, indistinguishable from the thumbnail.

8.2.3 HOW MUCH CONFIDENCE TO PLACE IN EACH RANK

Being explicit about this is more useful than defending the ranking.

Ranking claim	Confidence	Why
Intercoms are the dominant purpose-bought device for motorcyclists	High	Retail structure, an India-localised Sena product line, a domestic manufacturer with five models, mainstream press coverage, and forum threads that treat the purchase as assumed rather than argued
Phone plus WhatsApp plus stop discipline is the underlying universal layer	High	Present in every thread; the commercial operators sell the human version of it rather than radios
Mesh has overtaken classic Bluetooth pairing in the ₹6,000–₹15,000 band	High	The 2026 BluArmor range, EJEAS mesh at under ₹6,500, and Sena's mesh tiers
Real-world intercom range is about 500–600 m per hop, not the claimed 1.1–1.2 km	Medium-high	Three independent Indian reports converge; none is a measurement
Intercom performance in Himalayan switchbacks	Low — a genuine gap	No Indian source tested it. The degradation prediction at §8.3.5 is physics, not evidence
Radios are a minority practice confined to operators, event crews and hams	Medium-high	Consistent across a decade of threads and confirmed by operator inclusion lists
CB is effectively dead in India	Medium-high	Every Indian CB SKU out of stock; one club tested and rejected it; no active community found. A negative cannot be proved
Amateur radio is small but real, concentrated in Bangalore off-roading and rally communications	Medium	The club figure is one member's 2015 estimate; the rally evidence is solid but dates from 2004
Operators mostly do not disclose licensing	High as to non-disclosure; Low as to what they actually hold	The pages were read. They say nothing

Ranking claim	Confidence	Why
Satellite is absent and legally dangerous	High	Multiple independent reports of arrests, including a 2025 US Embassy alert
Anything at all drawn from Reddit	None	Domain inaccessible. No evidence gathered

8.2.4 THE COUNTERWEIGHT, STATED HONESTLY

The absolute numbers are small against India's two-wheeler base. One paid research firm sizes the **India connected-helmet market** at **USD 6.91 million in 2025**, projected at USD 9.18 million in 2026, with Bluetooth-only devices at 39.28% of 2025 revenue, individual riders 71.88% of it, offline retail 57.13% of volumes, and retrofit kits priced ₹2,500–₹5,999. **[REPORTED]** — Mordor Intelligence, *India Connected Helmet Market*, report dated 2026-08-05, fetched 2026-08-23.

Two caveats before that figure is used for anything. It is a **commercial research estimate, not audited data**, and "connected helmet" is a broader and differently drawn category than "aftermarket intercom". **[UNCERTAIN]** as to how it maps onto intercom unit volumes. Use it directionally only.

So the correct sentence is narrower than the headline. The intercom is the dominant *device* among riders who buy a communication device, and it is standard kit in the touring and organised-tour segment. A large majority of Indian motorcyclists still ride with no intercom at all, and their group communication is phone-and-stops.

8.3 Bluetooth and mesh helmet intercoms

8.3.1 WHY THIS IS LICENCE-EXEMPT IN INDIA, AND UNDER WHICH NOTIFICATION

The instrument is **G.S.R. 45(E) dated 2005-01-28**, the *Use of low power Equipment in the frequency band 2.4 GHz to 2.4835 GHz (Exemption from Licensing Requirement) Rules, 2005*, Gazette No. 36, New Delhi, Friday, January 28, 2005. Rule 3, verbatim:

"3. Use of wireless equipment in the band 2.4 GHz to 2.4835 GHz. - Notwithstanding anything contained in any law for the time being in force, **no licence shall be required by any person to establish, maintain, work, possess or deal in any wireless equipment**, on non-interference, non-protection and shared (non-exclusive) basis, in the frequency band 2.4 GHz to 2.4835 GHz with the transmitter power, Effective Radiated Power and height of antenna as specified in the Table below, namely :-"

Maximum output power of transmitter	Maximum Effective Radiated Power	Height of Antenna
1 W (30 dBm) in Spread of 10 MHz or higher	4 W (36 dBm)	Within 5 metres above the roof top of existing authorized building

"6. Equipment. - (1) The wireless equipment shall be **type approved** and designed and constructed in such a manner that the bandwidth of emission and other parameters shall conform to the limits specified in the Table referred to in rule 3."

Signed "[No. R-11014/17/2004-LR (Pt.)] ASHOK KUMAR, Jt. Wireless Advisor". **[VERIFIED]** — read from the scanned gazette PDF, 2026-08-23; catalogued at Part 2 §2.4.1.

Applying it to a helmet intercom, condition by condition:

Condition in rule 3	What an intercom does	Satisfied?
Band 2400–2483.5 MHz	Cardo Packtalk Edge/Pro operates 2405.0–2475.0 MHz; Sena 50S operates 2410–2475 MHz [REPORTED] — United States equipment authorisation records, fetched 2026-08-23	Yes
Transmitter output ≤ 1 W	0.095 W measured for the Cardo mesh radio; 0.055 W for the Sena. Roughly 10 dB of margin [REPORTED]	Yes, by a wide margin
ERP ≤ 4 W	Same figures, plus a helmet-mounted antenna with no meaningful gain — roughly 16 dB of margin [ENGINEERING ESTIMATE]	Yes
"in Spread of 10 MHz or higher"	Both are spread-spectrum, frequency-hopping devices, granted under a digital transmission system equipment class hopping across roughly 70 MHz of the band [REPORTED]	Yes
Antenna height "within 5 metres above the roof top of existing authorized building"	The column is drafted for a fixed installation. A helmet is not a building, and Part 7 §7.2.1 establishes there is no siting-clearance question for a vehicle or personal installation	Not applicable
Rule 6(1) type approval	Required. The duty sits on the manufacturer, importer or seller, not on you as user — Part 2 §2.7	See §8.3.8

Result: a 2.4 GHz mesh intercom needs no licence, no permission, no notification and no fee, for you or for any member of your group. That is the strongest legal position of any option in this report.

The consistency point with Part 2 §2.4.3

Part 2 §2.4.3 item 2 warns that whether WPC would accept a **2.4 GHz voice two-way radio** under G.S.R. 45(E) is **[UNCERTAIN]**, because the 10 MHz spreading condition excludes conventional narrowband frequency modulation and no such radio is known to hold an Indian ETA. **That warning is about a different product and does not apply to an intercom.** A helmet intercom is not a narrowband FM transceiver: it is a frequency-hopping spread-spectrum device of exactly the class the rule contemplates, hopping across far more than 10 MHz, and it is the same class of device as every Bluetooth headphone and Wi-Fi router sold in India. Part 2's caution stands for a hypothetical narrowband 2.4 GHz walkie-talkie. It does not stand against a Sena, a Cardo, a BluArmor or an EJEAS.

One untidy piece of drafting, disclosed

G.S.R. 1047(E) of 2018-10-18, Table IX row 12, permits only **10 mW e.i.r.p.** at 2400–2483.5 MHz for a *non-specific short range device*. **[VERIFIED]** — Part 2 §2.4.1. A 95 mW intercom exceeds that figure. The resolution is that a Bluetooth-class device relies on G.S.R. 45(E), not on the non-specific short-range-device table: DoT's own consolidated index characterises G.S.R. 45(E) as the Wi-Fi and Bluetooth instrument, G.S.R. 1047(E) contains no supersession clause, and the entire Indian consumer electronics market rests on that reading. **[VERIFIED]** as to the two texts; **[UNCERTAIN]** as to how WPC formally reconciles them, on which no clarification could be found. **Practical risk: nil.** It is listed at §8.7.8 as a question worth asking anyway.

8.3.2 WHY THIS IS CONFIDENTLY THE TOP DEVICE

No survey exists. Nobody publishes one. The confidence comes from convergent structural evidence, all **[REPORTED]** and all fetched 2026-08-23:

1. **Every Indian motorcycle-gear retailer runs a dedicated intercom category.** Moto Central, Riders Junction, Helmetwala, Sprocketz, Motuneo, Always Riding, Sparify, Ride and Rider and Motogear Performance, plus a single-brand EJEAS storefront — all with double-digit SKU counts and rupee pricing.
2. **Sena built a product line specifically for India.** Sena India's site carries a "**Bharat Series**" — the B10 and B20 — described verbatim as *"The first product made for the Indian market (#Bharat Version)"*, with *"English and Hindi Voice Prompts"* to *"cater to the diverse linguistic landscape of India"*. A global brand does not localise voice prompts for a niche.
3. **India has a domestic manufacturer with a full range.** BluArmor, Bengaluru. In March 2026 it replaced its entire line-up with **five models** on a new mesh platform. **[REPORTED]** — Autocar India, 2026-03-30.
4. **Indian mainstream motoring media treat them as ordinary kit.** Autocar India, ZigWheels, BikeWale and Team-BHP all run intercom reviews and launch coverage.
5. **Forum threads about group riding default to intercoms.** In a 2023 Team-BHP thread on motorcycle-to-car communication, the opening poster's premise is *"I am planning to buy my first communication system for my rides, and I am considering products from brands like Sena and Cardo"* — the intercom is assumed, and the *car* is treated as the problem. **[REPORTED]** — BHPian MiSiMa, 2023-04-26, read from a Wayback capture.

8.3.3 WHAT THE TECHNOLOGY ACTUALLY IS

It is not Bluetooth mesh. The market calls these "Bluetooth mesh intercoms" and the phrase is misleading. The mesh is a **proprietary protocol in the 2.4 GHz ISM band** — not Bluetooth Mesh Profile and not classic Bluetooth. Cardo calls its version **DMC**; Sena calls its version Mesh Intercom; BluArmor calls its version **RIDEGRID**. Bluetooth is present in the same unit and does a different job: pairing to your phone, to a pillion, and to another brand's headset in a two-way "universal intercom" mode. **[REPORTED]** — Cardo and Sena technical material, fetched 2026-08-23.

Two consequences follow, and the second one decides how you buy.

8.3.4 MESH VERSUS CLASSIC PAIRING — AND THE RULE THAT DECIDES YOUR PURCHASE

	Classic Bluetooth intercom	Mesh intercom
Topology	A chain or star of paired devices, typically 2 to 4 riders	A self-healing network in which every unit is also a relay
When a rider drops out	The chain breaks , and rejoining needs manual re-pairing	The network reroutes ; the rider rejoins automatically on return
Rider action required	Button presses, sometimes an app	None
Group span	Limited to the weakest single hop	Can exceed the per-hop range , because riders relay for each other
Cross-brand	Two-way "universal intercom" fallback exists	None. Mesh does not cross brands

Cardo's own description is the clearest statement of the behaviour: *"Every unit in a DMC group acts as both a receiver and a relay... When a rider drops out of range, the mesh recognises the change and reroutes the signal through other riders... [and on returning] rejoin[s] the mesh automatically, with no button presses, no app interaction, and no pairing process."* **[REPORTED]** — vendor material, fetched 2026-08-23, and consistent with the independent long-distance field report at §8.3.5.

An Indian buyer's guide states the failure mode in exactly the terrain this reader cares about: *"On a Leh trip, the lead rider crests a switchback 800 metres ahead of the sweep rider. With Bluetooth, the chain probably broke 200 metres back. With mesh, the network re-routes."* **[REPORTED]** — EJEAS Store India, fetched 2026-08-23. That is a vendor selling mesh, so treat the framing as marketing; the underlying engineering is nonetheless correct and matches every field account read.

The Indian market has moved decisively to mesh in the ₹6,000–₹15,000 band, and that is the single biggest change since the older forum material. In the 2017–2020 period, Indian forum posters were choosing between two-way and four-way *Bluetooth* conference. By 2026 the mid-market default is mesh: BluArmor's entire 2026 range except the entry model, EJEAS mesh at under ₹6,500, and Sena's mesh tiers. **[REPORTED]**

The practical rule, on which Indian forum consensus is unanimous: everyone in the group must run the same mesh protocol. Cross-brand "universal intercom" fallback exists — Sena's B10 advertises *"Universal Intercom support ... connect with other Bluetooth communication devices from different brands"* **[REPORTED]** — but it drops the group back to classic paired Bluetooth with all its fragility. An Indian buyer's guide puts it bluntly: *"EJEAS mesh won't interoperate with Sena or Cardo mesh systems, though Bluetooth pairing between brands is sometimes possible for two-rider mode only."* **[REPORTED]** Cardo has shipped a "live connection" bridge to Sena Mesh; **[UNCERTAIN]** how well it works in a real mixed group, and no Indian field report of it could be found.

This has a visible group-dynamics effect. One Indian reviewer chose his unit partly because *"all my friends with whom I was riding preferred [it] compared to others."* **[REPORTED]** — Thejesh GN, 2024-10-30. **Indian riding groups converge on one brand, and the brand that wins a group wins every subsequent purchase in it.** Decide the brand before the first person buys.

8.3.5 RANGE — CLAIMED, REPORTED, AND ESTIMATED

What the vendors claim:

Claim	Source
Cardo second-generation DMC: up to 1.6 km unit-to-unit, up to 8 km across the group as the signal relays, up to 31 riders	Cardo, fetched 2026-08-23 [REPORTED]
Sena Mesh 2.0: up to 2 km in open terrain; Group Mesh up to 24 riders; group span "up to 8 km, minimum 6 riders at 1.6 km intervals"	Sena, fetched 2026-08-23 [REPORTED]
BluArmor C30: 1.1–1.2 km line of sight	BluArmor and Indian retailers [REPORTED]
Sena B10 Bharat: <i>"up to 1 km"</i>	Sena India [REPORTED]
EJEAS Q8: <i>"up to 2 km"</i> on the storefront and <i>"1-Km"</i> in the dual-pack listing title — the vendor's own site contradicts itself	riderintercom.com [REPORTED]
EJEAS (India guide): Bluetooth 800 m – 1.5 km; mesh 1.5 km rider-to-rider with hopping	EJEAS Store India [REPORTED]

What Indian users actually report — three independent sources, one number:

Source	Device	Claimed	Reported real-world	Conditions
BikeWale long-term review, 2024-01-03	BluArmor C30	1.2 km	"uninterrupted" within 500 m with obstacles	4 months, Mumbai urban, 2 riders
Team-BHP, BHPian Biker_Ram, 2023-07-28 (Wayback capture)	BluArmor C30	1.2 km	"On highways the range is usually between 500 to 600 meters"	Highway and city
Thejesh GN, 2024-10-30	BluArmor C30	1.1 km	"we could talk up to at least a KM between the riders"	3–4 riders, about 2,000 km

[REPORTED] throughout.

The honest read: a 1–1.2 km line-of-sight claim delivers roughly **500–600 m rider-to-rider on a real road**, and up to about a kilometre when conditions are genuinely open. Three independent Indian riders, three different rides, the same order of magnitude. That is as good as community evidence on this subject gets, and it is not a measurement.

A useful non-Indian datapoint. The most rigorous field test found anywhere is a six-month, 10,000-mile Africa expedition on Honda CRF300Ls covering off-road, highway, rain, sand, freezing conditions and 40 °C, using Cardo Packtalk Edge units. The reviewer's summary is worth more than any specification sheet:

"We've not tested it as far as five miles, and I'd be surprised if it reached that far."

"If you can see each other it's going to work."

[REPORTED] — Mad or Nomad, published 2025-11-24, fetched 2026-08-23. The same review records no dropouts, no water or dust failures across torrential rain and "endless sand rides", a full riding day on one charge, and 25% remaining after 10 hours.

"If you can see each other it's going to work" is the correct mental model, and the physics explains why. At 2.45 GHz, free-space path loss over 1.6 km is about 104 dB. With roughly 20 dBm transmitted and 10–20 dB lost to the helmet shell, the rider's body and off-axis antenna orientation, the receiver sits near its sensitivity floor. Halve the distance and you gain 6 dB and a comfortable margin. **2.4 GHz does not diffract around a rock spur.** It is line of sight plus a little scattering, and nothing more. **[ENGINEERING ESTIMATE]**

Range by scenario:

Scenario	Realistic rider-to-rider range [ENGINEERING ESTIMATE]
Straight open highway, clear line of sight, riders in the same lane	800 m – 1.6 km
Rolling national highway, gentle bends, some traffic between	400 m – 1 km
Ghat section, switchbacks, rock face between the two riders	100–400 m, intermittent
Around a spur with no line of sight at all	Effectively zero — no useful diffraction at 2.4 GHz
Across an open valley, both riders on the road, line of sight	1.5–3 km, occasionally more
Rider to a four-wheeler with the unit inside the cabin	Knock off 5–10 dB; roughly halve the figures above

The mesh relay is real, and it changes the group arithmetic. Because each rider relays, the **first-to-last spread of the group can exceed the per-hop range**. One Indian reviewer puts it directly: *"since it's a mesh, the distance between the first and last rider could be more than 1 KM."*

[REPORTED] — Thejesh GN, 2024-10-30. A six-rider mesh strung out at 400 m intervals is a 2 km chain that still works. **This is the strongest practical argument for mesh over classic Bluetooth in Indian touring**, where convoys stretch on climbs.

Vendor "8 km group range" figures are honest arithmetic — six riders at 1.6 km intervals — and also a description of a group you would never actually ride. In a real Himalayan convoy the riders bunch at the top of a climb and string out on the descent, so the mesh is over-provisioned half the time and broken the other half. **Where mesh genuinely earns its money is not raw distance. It is that the network reroutes instead of collapsing.**

The degradation mode, described from Indian experience: *"Mesh intercom works well when the distance between riders is less. As the distance increases the voice becomes choppy and finally gets disconnected. When the riders are back within the range it automatically connects to mesh."*
[REPORTED] — BHPian Biker_Ram, 2023-07-28, Wayback capture. Graceful, self-healing, requiring no rider action. That is exactly the behaviour a convoy wants.

Two honest gaps in the range evidence:

- 1. **No Indian source measured mesh intercom range in Himalayan switchback terrain.**
BikeWale tested in Mumbai. Biker_Ram tested on highways. Thejesh GN covered 2,000 km without specifying terrain. **[UNCERTAIN]** The switchback row in the table above is physics, not evidence — and it is the row the reader most needs. Expect the link to drop entirely around a rock spur, and expect it to recover the moment you round the corner. The auto-reconnect makes this survivable rather than infuriating, but **an intercom is a line-of-sight device and Himalayan riding is the case where line of sight is least available.** That is the same conclusion Part 7 §7.6.2 reached about PMR446 in hills, for the same reason, one frequency decade higher.
- 2. **Carbon-fibre helmet shells attenuate 2.4 GHz badly**, being conductive. One reviewer reports mesh range dropping from about 2 km to about **400 m** in a carbon helmet.
[REPORTED] — no controlled measurement could be found, so treat the specific numbers as indicative. A polycarbonate or fibreglass Indian-market helmet does not have this problem.

8.3.6 GROUP SIZES — WHAT IS CLAIMED AND WHAT ANYONE HAS ACTUALLY RUN

Device	Stated group capacity
Sena B10 Bharat	4-way
BluArmor C30 / C20	Up to 20 in mesh
BluArmor C40 (2026)	20
BluArmor C60 / C60+ / C60Pro (2026)	"Unlimited" (vendor claim)
Sena Group Mesh	24
Cardo Packtalk Edge	15-rider mesh (international specification); 31 quoted for Mesh Boost
EJEAS Q8	6-rider mesh
EJEAS X10 Mesh 4.0	10
Fodsports M1-S Pro	8-rider Bluetooth
AXOR Mercury (Bluetooth, not mesh)	6, in one-to-five pairing

Device	Stated group capacity
Vnetphone V6 / V8	Conference; V8 to 4 others
ASG SG5	8-rider

[REPORTED] — vendor and retailer pages, fetched 2026-08-23.

Treat the large numbers with scepticism. The only Indian real-world group-size datapoint found is one reviewer's honest disclaimer: *"I have tested this with only three or four folks in the group. I am yet to travel with a larger group."* [REPORTED] — Thejesh GN, 2024-10-30. **Nobody in any Indian source reports running a 20-rider mesh**, and no field report of anyone testing the 24- or 31-rider ceilings could be found anywhere.

Synthesising the sources: **mesh is worth the money at three riders and above**. Below that, a chained Bluetooth pair is fine. [REPORTED]

[UNCERTAIN] — **dates on one Indian forum thread**. The automated fetcher did not return post dates for the xBhp "Universal: Bluetooth Intercom" thread. The model line-up quoted in it — Sena SMH5, Sena 10S, Sena 30K, Cardo Freecom 4+, Vnetphone V6/V8, purchases via Gearbest — places it roughly **2017–2020**. Every price from that thread in this part is flagged as historical.

8.3.7 THE INDIAN BRANDS AND THE PRICE BANDS

(a) BluArmor — the Indian brand, Bengaluru, mesh-first, priced under Sena and Cardo

Model	Price (₹)	Riders	Claimed range	Source
C30	9,999 / 10,999 depending on retailer	Up to 20 in mesh	1.1–1.2 km LOS	Moto Central; BikeWale 2024-01-03; Thejesh GN 2024-10-30
C20	9,999; launched at 6,999	Mesh	Not captured	Moto Central; ZigWheels
C50 Pro	22,499; launched at 24,999	Mesh	Not captured	Moto Central
HS1 (Bluetooth headset, not an intercom)	2,999	1	Not applicable	Moto Central

The 2026 replacement range, announced 2026-03-30, orders from 2026-04-15 [REPORTED] — Autocar India:

Model	Introductory ₹	Regular ₹	Max riders	Notes
C10	3,499	4,999	2 (one-to-one)	1.2 km claimed; 13 h battery
C40	8,999	12,999	20	RIDEGRID 2.0 mesh

Model	Introductory ₹	Regular ₹	Max riders	Notes
C60	10,999	15,999	"Unlimited"	Private conversation within a mesh group
C60+	12,999	19,999	"Unlimited"	MagDock 2.0
C60Pro	14,999	24,999	"Unlimited"	Flagship

"Unlimited" is a vendor claim with no independent test behind it. **[REPORTED]**

(b) Sena — the mainstream aspirational brand, with an India-specific line

Model	Price (₹)	Riders	Claimed range
B10 #Bharat Version	10,999	4-way intercom	"Working intercom distance: up to 1 km"
B20 #Bharat Version	16,999	Multi	Not captured
R35	23,999	—	—
A10 / M10	8,299 / 7,999	—	—
50S clamp kit with Harman Kardon speakers	15,499 (kit only)	—	—
50S dual pack	"45–50k" [REPORTED] — forum, 2023-04-26	Mesh	—

(c) Cardo — present, premium, and priced out of the Indian mass market

Model	Price (₹)
Spirit (Bluetooth, not mesh)	11,099
Spirit HD (Bluetooth)	16,999
Freecom 2X / 4X (Bluetooth)	23,699 / 28,899
Packtalk Special Edition	26,999
Packtalk Neo (DMC)	33,899
Packtalk Edge (DMC)	39,599
Packtalk Pro (DMC)	46,099
Packtalk Edge Duo (two units)	73,699
Second-helmet kits	8,299 – 13,599

At ₹40,000–46,000 for a flagship — roughly the price of an organised Spiti tour package — Cardo's mesh range is a small enthusiast niche in India, not a volume product. That is an inference from price positioning, not a sales figure. **[UNCERTAIN]**

(d) The value tier — where a lot of Indian riders actually shop

Model	Price (₹)	Riders	Claimed range	Source
EJEAS Q8 5G Mesh	5,999 (riderintercom.com) / 6,499 (ejeasstore.com)	6-rider mesh, 17 h	"up to 2 km" and "1-Km" on the same vendor's pages	Two Indian EJEAS storefronts, both fetched 2026-08-23
EJEAS Q8, two-unit pack	11,499 / 12,499	6	—	As above
EJEAS MS20 Mesh 3.0	11,999	Large groups	—	ejeasstore.com
EJEAS X10 Mesh 4.0	15,499	10	1.5 km claimed	ejeasstore.com
EJEAS V6 Pro Plus	2,999 single / 5,999 dual	Multi-rider Bluetooth	800 m	riderintercom.com
Vnetphone V6 / V8	~2,500 / ~5,000	Conference	—	xBhp, historical (see §8.3.6)
ASG SG5	8,500 locally / ~4,500 imported	8-rider	—	xBhp, historical

Note the discrepancy in row 1. Two Indian storefronts for the same brand quote different prices for the same model on the same day, and one of them contradicts itself on range. That is the normal condition of this market, and it is why every price in this part carries a date.

(e) Indian helmet brands with integrated or own-brand Bluetooth

AXOR sells a **Mercury Intercom** at ₹6,992 — Bluetooth 5.3, up to 6 riders in one-to-five pairing, about 30 hours of battery. It is **Bluetooth, not mesh**, so it has the chain-break failure mode. **[REPORTED]** Other Indian-market headsets: Axor IRIS ₹2,240, Moto Torque ₹1,999, Motocom ₹1,999. Parani M10 ₹7,999 and A10 ₹8,299 have real Indian retail presence. **[REPORTED]** — Moto Central, fetched 2026-08-23.

A distinction that catches convoy planners. Vega, Steelbird, Studds, Axor and Royal Enfield sell Bluetooth *audio* built into the helmet shell. Most of those are **single-user headsets for phone and music, not rider-to-rider intercoms**. They are the largest category by unit count and the least useful for group communication. **Do not confuse a ₹2,000 "Bluetooth helmet" with a mesh intercom.**

(f) The price bands, consolidated

Band (₹)	What you get	Representative models
1,000–3,000	Single-user Bluetooth helmet headset. Not a group intercom	Motocom 1,999; Moto Torque 1,999; Axor IRIS 2,240; BluArmor HS1 2,999; EJEAS V6 Pro Plus 2,999

Band (₹)	What you get	Representative models
3,000–7,000	Entry intercom and entry mesh	BluArmor C10 3,499 introductory; EJEAS Q8 5,999–6,499; AXOR Mercury 6,992
7,000–12,000	The volume band for serious group riders. Real mesh, 6–20 riders, IP67, usable battery	BluArmor C40 8,999 introductory; BluArmor C20/C30 9,999–10,999; Sena B10 Bharat 10,999; Parani M10 7,999; Cardo Spirit 11,099; EJEAS MS20 11,999
12,000–25,000	Premium mesh	BluArmor C60/C60+/C60Pro 10,999–14,999 introductory; EJEAS X10 15,499; Sena B20 16,999; Cardo Spirit HD 16,999; Sena R35 23,999
25,000+	Flagship, enthusiast only in India	Cardo Packtalk Neo 33,899; Packtalk Edge 39,599; Packtalk Pro 46,099; Sena 50S dual pack ~45,000–50,000

[REPORTED] — Moto Central, Riders Junction, RiderIntercom, EJEAS Store India, Sena India, AXOR and Autocar India, all fetched 2026-08-23 unless dated otherwise. Prices move; re-check before quoting.

The band that matters for a convoy planner is ₹7,000–₹12,000. That is where mesh, six or more riders, IP67 and a usable battery appear together, and it is where the Indian market's centre of gravity now sits.

Cost arithmetic for a group: five riders can be fully mesh-equipped for about **₹32,500** on value-tier units, or about **₹198,000** on Cardo flagships. The performance gap between those two numbers is real and nowhere near sixfold. For a first expedition, buy the value tier, find out what actually fails, and upgrade those units.

8.3.8 WHERE RIDERS BUY, AND THE ETA QUESTION NOBODY ASKS

Channels, in rough order of how often they appear in the sources **[REPORTED]**:

1. **Indian motorcycle-gear e-commerce** — Moto Central, Riders Junction, Helmetwala, Sprocketz, Always Riding, Motuneo, Sparify, Motogear Performance, Ride and Rider. Authorised stock, Indian warranty, GST invoice.
2. **Brand-direct Indian storefronts** — senaindia.com, bluarmor.com, the EJEAS storefronts.
3. **Amazon.in and Flipkart** — the budget tier lives here. Not fetchable for this research.
4. **Physical motorcycle-accessory shops** — one market estimate puts offline retail at 57.13% of 2025 connected-helmet volumes. **[REPORTED]**
5. **Grey import** — AliExpress, Gearbest, fc-moto with customs duty, personal imports. Explicit in the Indian forum material: an ASG SG5 at ₹8,500 locally against about ₹4,500 imported; a Sena 10S at ₹12,000 via a European retailer. **[REPORTED]**, historical.

The question nobody in the sources asks. Licence-exemption at 2.4 GHz does **not** remove type approval: rule 6(1) of G.S.R. 45(E) requires it. **[VERIFIED]** — Part 2 §2.7.

No Indian rider anywhere in these sources asks whether their intercom holds a WPC ETA, and no Indian retailer volunteers an ETA number for one. [UNCERTAIN] — the ETA status of not one intercom model was verified, because the Saral Sanchar public register is CAPTCHA-gated.

Two observations that cut in opposite directions:

- A locally manufactured product sold openly through the mainstream Indian retail chain is very likely to have gone through the ETA self-declaration route, which since February 2022 covers equipment operating solely in de-licensed bands. **[REPORTED]** — Part 2 §2.7.
- A ₹4,500 intercom shipped direct to a rider from an overseas marketplace certainly has not. **[ENGINEERING ESTIMATE]** on the certainty; **[VERIFIED]** on the legal requirement.

Practical weight: low, but not zero. Bluetooth at a few tens of milliwatts inside a helmet is not what the Wireless Monitoring Organisation goes looking for, and **no report of enforcement against a helmet intercom in India could be found anywhere.** Contrast that with PMR446, where DoT ran an actual monitoring drive and CCPA named non-compliant products (Part 2 §2.11, Part 6 §6.4.7). Buy from an Indian retailer with an invoice and this is a non-issue in practice.

Do not extrapolate that comfort to a walkie-talkie. A device that looks like a police radio attracts attention a helmet accessory never will.

One further buying caution: counterfeits are a real problem in the Indian market. Multiple Indian buying guides warn that fake Cardo and Sena units circulate. **[REPORTED]** — fetched 2026-08-23. Buy from the brand's Indian storefront or a named retailer, not from a marketplace listing with no seller identity.

8.3.9 WHAT INDIAN RIDERS ACTUALLY COMPLAIN ABOUT

From the Indian forum threads and reviews, unfiltered **[REPORTED]**:

- **Water ingress.** A budget unit failed in heavy rain — *"all the functions and lights went haywire."* Against that, a mid-band unit *"survived heavy rain for over 40 minutes"*, and IP67 is now standard in the ₹7,000–₹12,000 band. **The budget tier is where rain kills units**, and Indian touring means monsoon. Look for IP65 as a floor and IP67 by preference.
- **Random audio dropout** on the cheapest models.
- **Mechanical failure** of microphone and speaker pins after extended use.
- **Glove operability.** *"Depending on the type of gloves you wear, operating the buttons will be either manageable or very difficult... it's difficult to feel the buttons."* — BHPian Biker_Ram, 2023-07-28. A real ergonomic point for cold-weather and high-altitude riding, and one that no specification sheet discloses.
- **Phone pairing conflicts** — a smartwatch stealing the audio route from the intercom.

- **Speaker fitment.** Speakers must sit directly over the helmet's ear pockets. Indian-market helmets from Steelbird, Vega, Studds, SMK and AXOR do have recessed pockets, but liner thickness varies and a 40 mm speaker can press on the ear. **[UNCERTAIN]** — no systematic model-by-model Indian fitment data could be found. Test-fit before you buy, or buy a helmet with a factory intercom cutout. Prefer a **clamp** mount over adhesive: it moves between helmets.
- **Battery is largely a solved problem.** Roughly 10 hours real against 16 claimed on a mid-band unit with mixed mesh and music, and *"on a 5 hour ride ... it went down from 100% to just 80%"*. Vendor figures: 13 h mesh talk time on the Sena 50S, about 12–13 h on the Cardo Packtalk Edge, 17 h claimed on the EJEAS Q8, about 30 h on the AXOR Mercury.

The battery number that actually matters is charge-while-riding. On a Ladakh run with ten-hour saddle days and unreliable hotel power, a unit that accepts a USB feed from the bike while in use converts a 13-hour battery into an unlimited one. Older units shut down when charging. **Check this specifically before buying. [REPORTED]**

8.3.10 HELMET VERSUS VEHICLE — WHERE THE INTERCOM DOES AND DOES NOT BELONG

On a motorcycle, this is the correct primary system and there is no close second. It is full-duplex, hands-free, licence-free, purpose-designed for a helmet and it survives Himalayan weather. Every piece of Indian rider evidence points the same way.

Three properties explain why Indian motorcyclists converged on intercoms while Indian off-road clubs converged on radios, and it is a rational split rather than a fashion:

1. **Full duplex versus half duplex.** An intercom is a telephone call: everyone hears everyone, hands never leave the bars. A radio is push-to-talk: one person at a time, and somebody has to press something. At 80 km/h in a headwind, that difference matters more than a kilometre of range. It is why the continuous low-grade chatter that actually holds a group together works on an intercom and does not work on a radio.
2. **It is the safest option under the Motor Vehicles Act.** Section 184 Explanation (c) deems *"use of handheld communications devices while driving"* to be dangerous driving, with a minimum six-month custodial exposure on a first offence if the court takes that route. **[VERIFIED]** — Part 7 §7.2.1. **Nothing is picked up with an intercom;** the control is a button on the helmet shell. A walkie-talkie keyed by a rider or driver in motion is the risky one. **[UNCERTAIN]** as to whether any Indian court has applied Explanation (c) to a two-way radio — no reported decision was found.
3. **It needs no antenna work, no ground plane and no installation.**

In a four-wheeler it works, but it is not what the vehicle wants. An off-helmet mesh unit worn on a cap or a headset keeps a mixed bike-and-car convoy on **one voice net**, which is a genuine and underrated advantage — it is the single thing no other option in this report does. What it will not do is reach the vehicle 4 km back on a Spiti switchback. For that you need a lower frequency and an outside antenna, which means Citizen Band, and §8.5 explains why that is harder than it sounds.

8.3.11 THE GAP NOBODY HAS SOLVED: BIKE-TO-CAR

This deserves its own note, because it is exactly the problem a mixed overland convoy has and the Indian community has not solved it.

The clearest statement of the problem is a 2023 Team-BHP thread: a rider on a motorcycle and family in a car, on the same trip. Sena and Cardo solve bike-to-bike and do nothing for the car. The workarounds suggested, all **[REPORTED]** from named posters, read from Wayback captures:

- **Feed a Sena base unit into the car's auxiliary input.** The original poster tested it and found it half-works: *"It would only work for output. The car's microphone would not pick up voice as the aux is supposed to be input only."*
- **Just stay on a phone call** — a ₹4,000 basic phone with an unlimited-talk prepaid plan at each end, or a WhatsApp group call, or Discord for up to 25 people. *"In places where there's no data, a more elaborate option could be walkie talkies (which is what car reviewers seem to use when driving separate cars)."*
- **Cellular PTT applications** — judged by one poster *"not suitable for motorcyclist"*.
- **Radios in both**, which is what a professional operator does — §8.4.2.

There is no good consumer product for this in India as at 2026-08. The two things that actually span both are an **off-helmet mesh unit in the car on the same brand as the bikes** (§8.3.10), which works within line of sight, and a **licence-free PMR446 set in each vehicle** with an in-helmet headset for the rider — and Part 7 §7.6.2 established that 500 mW with a mandatory integral antenna gives only a few hundred metres to a couple of kilometres, less inside a steel car body.

8.4 Radios in practice — who really uses them, and what they get wrong

8.4.1 THE SHAPE OF IT: RADIOS ARE A MINORITY PRACTICE, CLUSTERED IN THREE PLACES

Radios cluster in exactly three communities in India:

1. **Commercial expedition operators** who run convoys professionally and repeatedly (§8.4.2).
2. **Off-road event and club marshals**, who need to talk to a spotter 50 m away in a quarry.
3. **A small number of licensed amateurs** who came to the hobby through motorsport or off-roading (§8.4.5).

They are conspicuously absent from private motorcycle group riding. Across every Indian rider thread read for this research, radios come up as a *theoretical* answer to the bike-to-car problem and almost never as something the poster owns.

The most grounding fact in this entire part comes from the most substantive Indian off-road communications thread found — Team-BHP thread 149407, March to June 2014, read from a Wayback capture. Part 1 of that thread is about **hand signals, not radios**, and the author, an event organiser, gives the reason plainly:

"Why are Hand Signals used? 1) Its economic, than equipping each vehicle and spotter/marshal with RT set. 2) It is easier than shouting aloud over the noise of vehicles, over a distance. 3) It is more effective, when there are a large number of participants or a course over few days. 4) It is not as distracting, a RT communication in cabin, in the ear."

[REPORTED] — BHPian ex670c, Chennai, 2014-03-28.

An Indian off-road event organiser's first answer to "how do we communicate" is hand signals, and his first reason is cost.

His equipment history for the events he ran is equally instructive, and it ends somewhere useful:

"From TPC2006-2009 we used the FM Walkie Talkies, they were good as a start, but range was bad, and poor battery life, voice clarity was poor. In 2010 we switched to UHF/VHF Handheld sets and 3 Base Stations, this gave us excellent coverage good battery life and good voice clarity, but the sets required a lot of care and temporary mounting inside the vehicle was not safe. In 2011-12, we went in for the UHF/VHF (Motorola/Siemens) sets, these gave us good coverage, good battery life and good voice clarity, but the handheld sets were quite heavy. **Currently we are using Talkpro**, their biggest advantage is they come with a Hands free set which is clipped to the ear, and a remote PTT Switch this makes it more useful while driving."

[REPORTED] — same poster, same date.

Two observations. **TalkPRO is an Indian brand** whose licence-free model quotes a specific ETA number (§8.4.4). And **the reason he switched was the remote PTT and ear-clip headset** — precisely the hands-free arrangement Part 7 §7.2.1 identified as the legally safer configuration under MV Act s.184. He adopted it for convenience. It happens also to be the right answer.

[UNCERTAIN] — whether the UHF/VHF sets and base stations used at those 2010–2012 events were covered by an event frequency assignment. The thread does not say. **Base stations are in any event outside the PMR446 exemption**, which expressly excludes base station use

[VERIFIED] — so if those were 446 MHz, they were outside it.

8.4.2 WHO REALLY USES PMR446: THE COMMERCIAL OPERATORS, AND HOW THEY USE IT

Operators are the strongest practice signal because they run convoys professionally and repeatedly. The public inclusion lists of a range of Indian Ladakh and Spiti motorcycle and self-drive operators were checked. All **[REPORTED]**, all fetched 2026-08-23.

Operator	Trip	Communications in the inclusions	Verbatim
Care My Tours	Spiti motorcycle expedition, 10 days / 9 nights, ₹50,000 per person	Yes	"Reliable Communication: Two pairs of walkie-talkies provided for seamless connectivity in remote areas, ensuring effective coordination throughout the journey."
Planet Way Round	Spiti Snow Drive Expedition (self-drive four-wheel-drive)	Yes, inclusion item 7	"Walkie-Talkie sets for communication between cars"
Planet Way Round	Ladakh Way Round (self-drive, 11 nights)	Yes, inclusion item 8	"Walkie-Talkie sets for communication between vehicles."
xploearth	Curated Himalayan motorcycle rides	Yes, per a rider who went	"their primary means of comm is via high end radio units... you'll see the units mounted on the bikes... In addition they also have Sena / Cardos"
Universal Adventures	Spiti bike trip, ₹17,999	No	Road captain plus mechanic; backup vehicle for groups over 10 bikes
Adventures Overland	Lahaul Spiti self-drive convoy	No mention	"experienced crew, backup vehicles, and medical support"
MotoTour Ladakh, Ride the Himalayas, Brahmand Tour, JustWrael, Rock and Roll Riders, Spiti Valley Tours, The Leopard Expedition, GoBoundless, La Himalaya	Various	No mention on the pages fetched	—

Roughly one in five operator pages lists communications equipment at all. The overwhelming majority sell the *human* solution instead: **a road captain leading, a mechanic, and a backup vehicle sweeping.** That is a convoy SOP in commercial form, and it is exactly the architecture Part 7 §7.6.3 recommended.

Two findings from that table that should change how a private group buys:

1. **"Two pairs of walkie-talkies" is four radios, not one per rider.** For a group of a dozen bikes, four radios means road captain, sweeper, backup vehicle and a spare. **Radios in Indian commercial practice are a link between crew positions, not a group chat channel.** That is a fundamentally different and far more achievable design than "everyone gets a radio", and it works with a 500 mW handset in a way an all-riders net never will.

2. The one operator described as running "high end radio units" mounted on the bikes runs them alongside Sena and Cardo intercoms, not instead of them. Radio for crew coordination and distance; intercom for rider-to-rider chatter. **This layered architecture is the professional answer**, and it is what §8.7 tells a private expedition to copy.

Do the operators hold anything?

[UNCERTAIN] — and this is a finding, not a shrug.

Not one operator page fetched mentions a WPC licence, an ETA, a frequency assignment or any regulatory approval for its radios. Not one names the make, model or band of the equipment either. Contrast that with the same pages' meticulous itemisation of **Inner Line Permits** and **border entry fees**: these businesses clearly do track regulatory paperwork when they know about it.

Three possibilities, indistinguishable from public sources:

- The radios are ETA-approved licence-free 446 MHz sets, in which case **no operator licence is needed and there is nothing to disclose**. This is the most likely explanation and the cheapest.
- The operator holds a captive or closed-user-group frequency assignment. Plausible for a large operator; nothing on the public pages supports it. Part 1 §1.7 sets out why this route is closed to a private individual in any event.
- The radios are non-compliant hardware bought on a marketplace. **[UNCERTAIN]**

One legal point about operators specifically, because it is absolute. A commercial tour is a business, and that closes the amateur route to it entirely: rule 2(1)(c) of G.S.R. 675(E) requires the activity to be "*solely with a personal aim and without any pecuniary interest*", and Annexure A para 2(2)(b) bans "*communications of business*". **[VERIFIED]** — Part 7 §7.2.2. **An operator running a paid Ladakh trip on amateur frequencies is outside the licence, and so is any licensed amateur who provides communications support to it.** Licence-free 446 MHz, by contrast, is open to "any person" with no restriction by purpose and no distinction between commercial and personal use. **[VERIFIED]** — Part 2 §2.5.1.

If you book a guided trip and care about this, ask the operator for the make, model and WPC ETA number of the radios they will hand you. A compliant operator can answer in one line. Nothing in the public material suggests any of them expect to be asked.

8.4.3 THE MISLABELLED NON-COMPLIANT HARDWARE PROBLEM

This is the largest single practice problem in the Indian radio market, and it is not a fringe issue.

The regulator has quantified it. On **2026-01-16** the CCPA issued final orders in its walkie-talkie campaign, following identification of **over 16,970 non-compliant listings**:

"The Authority imposed penalties of ₹10 lakh each on Meesho, Flipkart, Amazon and Meta Platforms Inc, and ₹1 lakh each on Chimiya, JioMart, Talk Pro and MaskMan Toys."

"In the matter concerning Flipkart, ... Data furnished revealed that 42,275 units were sold with the exact license-exempt frequency range disclosed, whereas **65,931 units were sold where the frequency range was either left blank or was outside the exempted range.**"

[VERIFIED] — PIB Release ID 2215261, 2026-01-16; full treatment at Part 6 §6.4.7. And in **March 2026** the DRI seized **11,060 Baofeng BF-888S sets** at Nhava Sheva, with two importers arrested under the Customs Act; press reporting describes the model as blacklisted by DoT for operating beyond permissible frequencies. **[REPORTED]** — Part 6 §6.4.8, Part 2 §2.10.

Read the Flipkart figure again. Sixty-five thousand nine hundred and thirty-one units, on one platform, with the frequency range blank or outside the exemption. Those radios are in Indian homes now, and the great majority of their owners hold no authorisation of any kind and believe they do not need one, because the listing told them so.

The mislabelling seen in this research, itemised:

What a listing says	Why it is wrong	Tag
"License-free Walkie-Talkie — BaoFeng BF-888S UHF 400-470MHz", sold under a collection literally titled "baofeng license free walkie talkie"	Flatly false, and the most dangerous item here. 400–470 MHz is not the 446.0–446.2 MHz exemption; the BF-888S has a detachable antenna; it can hold no ETA. It is the exact model of which 11,060 units were seized at Nhava Sheva	[VERIFIED] on the legal position — Part 2 §2.10; [REPORTED] on the listing, fetched 2026-08-23
"446.0 – 446.1 MHz"	The gazette says 446.0–446.2 MHz	[VERIFIED] — Part 2 §2.5.1
"446.00–446.02 MHz"	Wrong by a factor of ten on bandwidth	[VERIFIED]
"Maximum Permissible RF Output Power: 0.5 Watts"	The limit is 500 mW e.r.p. — effective radiated power referenced to a half-wave dipole, not conducted output. With a gain antenna these are different numbers	[VERIFIED] — G.S.R. 1047(E) definitions
"connectivity up to 6–8 km" and "5 to 10 kilometres" for a 500 mW handset	Marketing. A licensed Indian amateur describing his own relatives' convoy use gives 1–1.5 km urban, 1–3 km highway	[REPORTED] / [ENGINEERING ESTIMATE] — Part 7 §7.6.2

What a listing says	Why it is wrong	Tag
"446 MHz base station" and "446 Base/Car Radio", sold as licence-free at ₹14,999; a vehicle-mounted "base station" marketed as extending team range "by as much as 5 times" with a high-gain antenna	Table V's note expressly excludes base station use, repeater use, infrastructure use and non-integral antennas. A vehicle-mounted 446 set with a roof antenna is outside the exemption whatever the listing says — and it is precisely the product a convoy buyer is most tempted by	[VERIFIED] — Part 2 §2.5.1, Part 7 §7.6.2
Imported Motorola "TalkAbout" FRS/GMRS consumer sets , used on Indian trips	FRS and GMRS frequencies at 462 MHz are not an Indian exemption . A radio that is licence-free in the United States is not licence-free here	[VERIFIED] — Part 2 §2.4.1

The base rate a buyer should assume: on the CCPA's own findings, **if you buy a "licence-free walkie talkie" from a marketplace listing without checking the band and the ETA, you are more likely than not buying an unlawful radio.**

8.4.4 THE COMPLIANT END OF THE INDIAN MARKET, AND ONE AWKWARD COINCIDENCE

Indian products with a claimed compliance story, **[REPORTED]**, fetched 2026-08-23:

Product	Band / power	Claimed range	Compliance claim	Price
TalkPRO LF446	446.0–446.20 MHz, ≤0.5 W	4–6 km	Quotes ETA SD 20250100112	₹9,000 plus tax
BRUTFORCE BFR-002 (pair)	446–446.2 MHz	Up to 2 km	"WPC approved... ETA Certificate & Manual Book" included; marketed as the first "Make in India" PMR446	₹7,400 for two (MRP ₹15,990)
BRUTFORCE BFR-001 / 001i / 003	446–446.2 MHz	5 / 6 / 8 km claimed	"WPC approved"	Not published on the manufacturer page

Accessories: TalkPRO lists a standard earphone at **₹750** and a remote speaker microphone at **₹850**. **[REPORTED]**

A vendor that quotes an actual ETA number is doing something a vendor that says "WPC approved" is not. You can take a number to WPC and ask. You cannot take a slogan anywhere. Get the number in writing, for the exact model and variant, before you pay.

And one coincidence this report should not step around. The CCPA's 2026-01-16 order imposed a ₹1 lakh penalty on "**Talk Pro**", and the release records that "*CCPA found that Talk Pro (Iconet Services Pvt. Ltd.) listed and sold walkie-talkie devices operating across wide frequency ranges, including UHF 400–1200 MHz, while falsely representing them as '100% legal' and 'license-free.'*" **[VERIFIED]** — PIB release, 2026-01-16, quoted at Part 6 §6.4.7. Part 7 §7.6.2 and §7.5 cite the TalkPRO LF446 as the one Indian PMR446 product that publishes an ETA number.

[UNCERTAIN] — **whether the penalised entity is the same business as the vendor of the LF446 could not be established from public sources.** A brand may sell a genuinely compliant 446 MHz model and also list wide-band models. Both facts can be true of one company. **The practical consequence is the same either way, and it sharpens the advice rather than changing it: judge the model, not the brand.** Ask for the ETA number of the exact model and variant you are buying, in writing, and note that Part 7 §7.6.2 already flags a further discrepancy on this specific product — its own page appears to describe a **detachable** antenna, while Table V permits **integral antennas only**. **[UNCERTAIN]** as to how WPC reads a screw-on stub supplied with the radio.

8.4.5 THE AMATEUR RADIO POCKET, AND THE BANGALORE CLUB THAT SAT THE EXAM

The strongest community-practice finding in this part. Team-BHP thread 167728, 2015-09-03, a distinguished member posting from Noida and Bengaluru:

"70% of B.O.D.A (Bangalore offroaders) are into HAM, have passed the exams, have licenses and each one also has their own radio sets. Some are very serious hobbyist now in this field."

Corroborated independently in a different thread a year earlier by another member writing for the same club: "*All the BODA Marshals have written this exam, and possess this license.*"

[REPORTED] — two members, two threads, a year apart, consistent, both read from Wayback captures.

This is the highest concentration of licensed amateurs in any Indian motoring community found in this research, and it is the correct answer to the problem they had. They needed reliable voice in wooded quarries, CB failed them (§8.5.3), and instead of using something unlawful they sat the examination.

[UNCERTAIN] — the "70%" is one member's estimate in a 2015 forum post, not a club statement, and no membership or licence roster could be found.

Other licensed amateurs visible in the Indian motoring community [REPORTED]: VU3BOJ, a Bengaluru third-generation ham licensed in 2014 who started the Team-BHP amateur radio thread in 2015; VU2ORO of Noida, "*in this hobby for the last 25 years or so, run both home made and commercial radios*"; two further Bengaluru hams active in the off-road threads; and VU2MB, VU2BX and VU2MSY as rally communications coordinators.

Why hams believe in this — the rescue story. A Bengaluru member, posting 2012-04-11:

"2 years back on my way to Calicut in Kerala, going through the rain forest hilly area, I had a flat with no spare. **I transmitted a SOS on my Ham radio via the ham repeater in Calicut, and a very kind gentleman replied to my call.** In an hours time, help came apparently a friend of the Ham who received my distress call and stayed in a nearby town... There was no way a mobile phone could have been of any assistance since I didn't knew anyone to call for help in that area."

[REPORTED] This is exactly the capability Part 7 §7.2.3 described: a licensed traveller reaching a repeater near a town when the phone was useless. Note carefully what made it work — **a repeater, near a city.** §8.5.4 explains why that does not transplant to Ladakh.

The practice that exceeds the rules, stated once

Indian licensed amateurs routinely provide **communications support to motorsport events and rallies**, relaying traffic for drivers and officials who hold no licence. This is documented on a ham's own website. An account of the **Himachal Car Rally / Power M1 Challenge 2004** — a sponsored event, Shimla — records approximately **16 hams** from Agra, Chandigarh, Mumbai and Delhi, VHF equipment deployed after prior *"Radio Mapping"* of the area, stations at Rally HQ and Rally Base and mobile positions in the mountains across a 500 km route with 120 km of special stages, **providing, in the site's own words, "Third Party communication" on behalf of rally drivers and rally officials who were not ham operators.** **[REPORTED]** — qsl.net/vu2msy, fetched 2026-08-23. No special licence or permission is mentioned anywhere in the account.

The legal position, once, clearly. Passing traffic on behalf of unlicensed people on amateur frequencies is **third-party traffic, and it is expressly prohibited:** Annexure A para 2(2)(b) of G.S.R. 675(E) bars transmitting messages containing *"...communications of business, statements on topics of political or industrial controversy, or third party messages"*. **[VERIFIED]** Doing it for a **sponsored commercial motorsport event** additionally engages rule 2(1)(c)'s *"solely with a personal aim and without any pecuniary interest"* and the same paragraph's ban on *"communications of business"*. **[VERIFIED]** The only third-party carve-out — Annexure A para 2(3) — requires a **failure of normal telecommunication facilities, a government permission, and a counterparty who is a District Magistrate, Deputy Commissioner or Collector.** A rally is none of these. **[VERIFIED]** — the full analysis is at Part 7 §7.2.2 and Part 3 §3.6.

The 2004 account predates the 2024 Rules, but the 1978 conditions it was written under prohibited third-party messages too. **[VERIFIED]**

That is the whole of the moralising, and it is not aimed at the hams. The practical point for this reader is the one Part 7 §7.2.2 already made and this part now confirms with a real Indian example: **the amateur licence does not solve the convoy problem, and the Indian amateur community's own habit of behaving as though it does is not a precedent you can rely on.** If a convoy is what you want, every vehicle needs its own ASOC holder, or you use a licence-free band.

[UNCERTAIN] — whether Indian rally and motorsport communications have since moved to assigned frequencies rather than amateur bands. If they have, that is a good-news story nobody has published. If they have not, a lot of well-meaning hams are operating outside their certificate conditions.

Why more Indian overlanders are not licensed — and why their reasons are stale

Consistently, across a decade of threads, the deterrent given is **the wait, not the exam [REPORTED]**:

- One member's account of the full thirteen-step process: *"In my case, the entire process was done in 11 months, and I consider myself very lucky."* (2015)
- Another: *"you wont be handed one right after you pass the test, there will be separate address verification by Police department, who will actually come and 'interview' your neighbor... Infact they came in my absence and interviewed a moron neighbor, who called me to inform that 'Police came in search of me regarding a theft of radio'."* (2014)
- And the resigned version: *"in India, anything which requires a license is best left alone."* (2015)

Both of the big obstacles in those posts are gone. The police and Intelligence Bureau verification was **abolished with effect from 2016-03-10 [REPORTED]** — recorded in the original poster's own later edit to his thread — and since March 2025 the examination and certificate applications are a **single step with self-certified documents [VERIFIED]** — Part 3 §3.9. The current realistic elapsed time is **three to nine months**, dominated by examination scheduling (Part 7 §7.7.1), and the total government outlay is **₹1,100 or ₹2,100** (Part 3 §3.7).

The Indian overlanding community's reasons for avoiding the amateur licence are eight to twenty years out of date. That is worth knowing if you are the reader of this report.

8.4.6 TWO SMALLER PRACTICE FINDINGS WORTH CARRYING

Radios get rented for events. ₹250 per day for VHF sets and ₹300 per day for CB sets was the going Indian rate in the forum-era market. **[REPORTED]** Renting from an operator who holds the underlying authorisation is a materially different legal proposition from buying an unlicensed radio — but **[UNCERTAIN]** whether any given Indian rental operator actually holds one, and the question is worth asking in writing before an event.

Sensitive areas are policed on the ground, not by spectrum law, and that is the risk that actually materialises. Indian riders and drivers report being asked to declare or deposit communication devices at checkpoints. One member recorded that *"a friend of mine who was visiting Leh, with a licensed Sat phone, was asked to surrender it at the army check post, to be picked up on his return."* **[REPORTED]** Another warned as early as 2012-11-21 that in the Northeast, carrying wireless equipment without an amateur licence risks action by security forces operating under the Armed Forces (Special Powers) Act. **[REPORTED]** Part 7 §7.2.1 flags the same thing from the legal side: **restricted areas are the single biggest practical risk for an overland expedition, and no published list of them exists. [UNCERTAIN]** — it is question 5 in Part 7 §7.7.2 for a reason.

Even lawful possession does not guarantee lawful use in a sensitive area. Plan for a checkpoint to take an interest in anything with an antenna, carry paperwork, and declare rather than conceal.

8.5 The Citizen Band correction

This section exists because the report contains two findings about Citizen Band that look like a contradiction and are not. Both are correct. They are answers to different questions. A reader who takes away only one of them will make a bad decision in one direction or the other.

8.5.1 THE TWO FINDINGS, SIDE BY SIDE

	The Part 7 finding	The Part 8 finding
The question it answers	<i>Which licence-free allocation gives a private convoy the best chance of a multi-kilometre vehicle-to-vehicle link?</i>	<i>What are Indian riders and overland groups actually running, and can you buy it?</i>
The answer	Citizen Band, 26.957–27.283 MHz, 5 W ERP. The highest-power licence-free two-way voice allocation in the entire Indian de-licensed corpus, and the only one drafted for vehicles	Effectively dead in India. Every Indian CB listing found was out of stock; no active user community could be located; one Indian off-road club tested it and rejected it
What kind of claim it is	A claim about legal headroom and physics	A claim about market availability and terrain performance
Evidence	[VERIFIED] on the gazette; [ENGINEERING ESTIMATE] on the ranges — Part 7 §7.6.1	[REPORTED] on the retail survey and the club's rejection; [UNCERTAIN] on the negative
Is it still true?	Yes, entirely. Nothing in this part disturbs it	Yes. And it does not disturb Part 7

They are not in conflict, and the reason is simple: "the best licence-free option" and "an option with a live market and a proven record in your terrain" are different properties, and CB has the first without the second.

8.5.2 WHAT REMAINS TRUE FROM PART 7, RESTATED

Nothing in the field research touches the legal analysis, and it should not be diluted.

- **The rule is drafted for vehicles.** G.S.R. 533(E), rule 3, exempts wireless equipment *"intended to be used while in motion or during halts"* at **5 W ERP** with a built-in antenna, in 26.957–27.283 MHz. **[VERIFIED]** — Part 2 §2.5.2, Part 7 §7.6.1.
- **It is the highest-power licence-free two-way voice allocation in India**, ten times PMR446, and Part 2 §2.6 establishes that by exhaustive enumeration of every de-licensing notification DoT publishes. **[VERIFIED]**

- **The physics still favours it for this job.** Ten times the power; an antenna that can be a metre or two long instead of 50 mm of helical; and better diffraction around terrain at an 11-metre wavelength than at 67 cm, let alone at the 12.5 cm of an intercom. **[ENGINEERING ESTIMATE]**
- **Indian field reports from the era when people did use it are consistent with the estimates, and slightly conservative [REPORTED]:**

Reported	Source
<i>"In the city I get between 2-7 km in the open areas. Highways I have seen ranges upto 12-14 kms. Once we could speak from the Timber Trail in Parwanoo right upto Panchkula."</i> (unit ₹12,000, antenna ₹3,000)	Team-BHP, BHPian zipperone, 2012-04-20, Wayback capture
<i>"5-7 km on a good day. In the hills, it is about a couple of km."</i> (President Tommy, 3 ft antenna)	Zigwheels forum, about 2014
CB typically 1–2 km in cities, up to 5 km on highways ; needs a 3–6 ft antenna and is "clunky"	The Quint, 2017-06-06
<i>"3 KMS TO 160 KMS"</i>	Indian retailer collection page — this is marketing; ignore it

Set against Part 7 §7.6.1's estimates — 5–12 km open flat, 3–8 km rolling highway, 2–5 km forest, 0.5–3 km mountain switchbacks, 15–40 km valley line-of-sight — the Indian field reports land inside the range and towards its lower half. **That is the right shape, and it means Part 7's numbers were not optimistic.**

- **The field craft from that era is genuinely good advice and it generalises.** The same 2012 poster: *"I would suggest to carry a copy of the WPC notification in the car and produce it if questioned. Its the document that makes the difference really."* He carried CB to Jammu and Kashmir and Rajasthan, declared it at a Border Security Force post beyond Jaisalmer along with cameras and phones, and was told only not to use any of them beyond the post. **[REPORTED] Carry a printout of the exemption notification. Declare, do not conceal.** That advice applies equally to PMR446 and G.S.R. 1047(E).

8.5.3 WHAT THE FIELD RESEARCH ADDS: THE CLUB THAT TESTED CB AND WALKED AWAY

Team-BHP thread 149407, 2014-04-07, a member writing for BODA, the Bangalore off-road club, read from a Wayback capture:

"BODA does not use CB Radios, but HAM Radio. ... Though CB radios were sampled by BODA, their effectiveness falls short of their expectation, the CB radios weren't performing in wooded and hilly areas. Usually the OTRs are around old quarries where the elevations are pretty different for even for every 200 meters. The HAM radios offer crystal clear communication comparatively."

and, the following day, the blunt version:

"the CBs have limited range, something like around 4-5 kms and that too in clear line of site, you are done in by any change in elevation, no matter how tall your antenna is. **This was tried out in BODA recce, in fact hollering on top of our voice was more effective.**"

[REPORTED] — BHPian Grace, 2014-04-07 and 2014-04-08.

This is a real test by a real club with real equipment, and it is the most consequential practice finding about CB in this report.

8.5.4 WHY THE TWO FINDINGS DO NOT CONTRADICT EACH OTHER

Read BODA's numbers against Part 7's estimate table and the apparent conflict dissolves.

BODA's finding	Part 7 §7.6.1's corresponding estimate	Verdict
"around 4-5 kms and that too in clear line of site"	5–12 km flat open; 3–8 km rolling highway	Consistent , at or slightly below the bottom of Part 7's open-country band
"you are done in by any change in elevation, no matter how tall your antenna is"	0.5–3 km, sometimes nothing on mountain switchbacks with a ridge between	Consistent. Part 7 already predicted this
"quarries where the elevations are pretty different for even for every 200 meters"	The worst case in Part 7's table, encountered continuously rather than occasionally	BODA operates entirely in the terrain where Part 7 says CB is weakest

So BODA did not disprove Part 7. BODA confirmed Part 7's own worst-case row, and then correctly concluded that a technology whose worst case is their normal case was not worth buying.

That is the whole reconciliation, and it generalises into a rule the reader can apply:

Part 7's CB recommendation is a statement about open terrain. BODA's rejection is a statement about broken terrain. Both are true. Which one applies to you is decided by your route, not by the radio.

For a highway, desert, Rann or open-plateau convoy, Part 7 §7.6.1 stands unchanged. For a wooded quarry with 200 m elevation changes every few hundred metres, it does not, and BODA proved it by trying rather than by arguing.

Two footnotes to BODA's story that matter to an expedition planner:

- 1. **Their replacement worked because of repeaters, and India's repeater network is urban.** A volunteer-maintained directory lists roughly **70 amateur repeaters in India**, concentrated in Bangalore, Chennai, Coimbatore, New Delhi, Hyderabad, Kerala and other metropolitan and southern locations, with **none listed in Himachal Pradesh, Ladakh, Jammu and Kashmir, Uttarakhand or any Northeastern state.** [REPORTED] — fetched 2026-08-23; absence from a volunteer directory is weaker evidence than presence in it. This confirms Part 7 §7.2.3's assessment from an independent direction. **A Bangalore quarry is within reach of a Bangalore repeater. Baralacha La is not. The BODA solution does not transplant to a Himalayan expedition.**
- 2. **Their reasoning was about performance, not law.** BODA did not reject CB because it was unlawful — it is not — and they did not adopt amateur radio because it was cheaper. They adopted the harder, licensed route because it worked. That is worth noting in a community where the usual direction of travel is the opposite.

8.5.5 THE MARKET EVIDENCE: WHAT A DEAD CATEGORY LOOKS LIKE

The Indian CB retail channel was surveyed on 2026-08-23. It exists on paper and is close to dormant in practice. All [REPORTED].

Seller	Finding
NPC Wireless (Delhi) — the one Indian retailer with a dedicated 27 MHz CB collection	President Tommy ASC ₹5,799 (from ₹8,000) — sold out . President Bill II ₹9,500 (from ₹12,000) — sold out . An unnamed set at ₹14,999 (from ₹29,999) — sold out . Luiton 27 MHz CB base station ₹6,999 (from ₹12,999) — sold out . Every CB product on the page is out of stock , at heavy discounts off list, on the site of the retailer that promotes the category hardest. The same retailer's UHF handheld pages carry live stock in pack quantities up to 25 units
The Engineer Store	President RANDY III at ₹10,224.76, 3–5 working days dispatch — but a United States band plan (26.965–27.405 MHz plus 162.400–162.550 MHz weather channels, described as an "FCC AM Transceiver") and no statement about Indian licensing at all
President Electronics India (mera-cb.com)	Portal is stale: multimedia content dated to a 2013 car rally, no India price list, no India-specific product filtering. Its FAQ's claim to hold Indian ETAs, quoted at Part 7 §7.6.1, remains the strongest compliance statement in the market
Manas Rekha (Mumbai importer)	Page copyright 2012; lists "CB Radio" and "Randy Walky Talky" with no prices, no power figures and no ETA statement
IndiaMART	President CB listings present and marked "In Stock", no price published

Who actually uses CB in India? The only user group any source identifies is **long-haul truck drivers**, and that source gives **no usage statistics** and has no adoption or history section. [REPORTED], and thin.

Conclusion: CB in India is a legal allocation with a channel plan, an encyclopedia article, a handful of out-of-stock SKUs, and no community. [UNCERTAIN] whether any organised Indian four-wheel-drive club or overland group runs CB in 2026 — none could be found, and absence of evidence in searchable English-language sources is not proof.

Historical pricing, for calibration [REPORTED]: in 2012 a vehicle unit ran about ₹6,000 to ₹13,000 with an antenna at about ₹1,500; one member paid ₹12,000 plus ₹3,000; an online Delhi seller started at about ₹11,000. By around 2014 a President Tommy was quoted at ₹4,500 with a 3 ft antenna at ₹4,000. A convertible base/handheld was quoted at ₹9,800 in 2017. **Budget realistically: ₹6,000–₹12,000 per vehicle for the set and ₹1,500–₹4,000 for the antenna, mount and coaxial cable — and expect to telephone a dealer rather than click "add to cart".**

8.5.6 TWO TRAPS THAT ONLY BITE CB BUYERS

Trap 1 — thirteen of the forty channels on an imported set are outside the Indian exemption.

The Indian de-licensed band is **26.957–27.283 MHz. [VERIFIED]** The international 40-channel CB plan runs from **26.965 MHz (channel 1) to 27.405 MHz (channel 40). [REPORTED]** Do the arithmetic: channel 27 is 27.275 MHz and channel 28 is 27.285 MHz. **Channels 1 to 27 fall inside the Indian exemption. Channels 28 to 40 fall above 27.283 MHz and fall outside it. [VERIFIED]** as to the band edge, **[REPORTED]** as to the channel plan; the arithmetic is this report's.

This is not academic — one currently listed Indian retail product is a United States specification radio whose stated coverage runs to 27.405 MHz plus weather receive channels, sold with no Indian compliance statement (§8.5.5). A seller-supplied "multi-norm" set switched to an India or European band plan is a different proposition, **but you have to ask, and get the answer in writing.**

Trap 2 — the "built-in antenna" condition is still unresolved, and it is the same question Part 7 told you to put to WPC.

The exemption says 5 W ERP **and built-in antenna.** A boot-lip or roof whip on a coaxial feed is not a built-in antenna, and on the face of the gazette the standard vehicle installation everybody actually uses falls **outside** the exemption. **[VERIFIED]** from the wording.

Against that: the same rule contemplates use "while in motion", which is very hard to reconcile with a genuinely built-in antenna at 27 MHz where a quarter wave is 2.75 m; a manufacturer selling into India states it holds Indian ETAs for its CB portfolio and publishes detailed vehicle roof-mounting guidance **[REPORTED]**; and Indian retailers sell mobile CB sets openly as vehicle-to-vehicle products **[REPORTED]**.

[UNCERTAIN] — market practice and gazette wording do not line up, and no WPC clarification, test method or enforcement precedent could be found. This is **question 4 in Part 7 §7.7.2**, which contains the draft wording. **Send it before you spend money on a CB installation, not after.**

One further loose end: G.S.R. 533(E) unusually contains **no express type-approval rule of its own** (Part 2 §2.12). **[UNCERTAIN]** That does not remove the general ETA obligation, but it makes the chain less tidy than for PMR446.

And a practical note about motorcycles: do not. There is no ground plane worth the name, no room for a metre of whip and no good mounting point. A short base-loaded 27 MHz stub on a motorcycle is an extremely inefficient radiator. **[ENGINEERING ESTIMATE]** Note the Indian datapoint: the one motorcycle designed around a factory CB installation arrived in the Indian market without it — the Indian specification did not mention CB and photographs showed no whip. **[REPORTED]** — Team-BHP, 2017-12-07. **CB is a four-wheeler technology in India.**

8.5.7 THE PRACTICAL BOTTOM LINE ON CITIZEN BAND

Pursue CB if all of the following are true:

#	Condition
1	You are running two or more four-wheelers . Not motorcycles — §8.5.6
2	Your route crosses genuinely open terrain where vehicles separate by several kilometres: highway, desert, Rann, salt flat, open plateau
3	You are willing to do antenna work — a roof or boot-lip mount on steel, proper coaxial cable, and a whip long enough to be worth fitting
4	You are willing to telephone dealers and wait, because the online channel is dormant
5	The seller gives you the WPC ETA number in writing for the exact model and variant
6	The seller confirms in writing that the set can be band-limited to 26.957–27.283 MHz — channels 1 to 27, not 1 to 40
7	WPC's answer to Part 7 §7.7.2 question 4 on the built-in-antenna condition comes back acceptably

Do not pursue CB if any of the following is true:

#	Condition
1	Your group is motorcycle-only. There is no workable 27 MHz antenna on a motorcycle
2	Your terrain is wooded, quarried, or switchback with frequent elevation change. That is BODA's terrain, and CB failed in it
3	You need equipment this month. Every Indian listing found was out of stock
4	You are unwilling to leave the built-in-antenna question open, and you cannot wait for a WPC reply
5	You would be buying it instead of the intercoms and the SOP rather than in addition to them

If CB is out, the fallback is not a bigger radio. It is PMR446 for the crew roles plus tighter convoy discipline — §8.7. There is no licence-free 10 W anywhere in India, in any band, under any notification in force; Part 2 §2.6 establishes that by exhaustive enumeration. **[VERIFIED]**

The highest licence-free two-way voice figure in the entire corpus is the 5 W ERP on CB that this section has just told you is hard to buy. That is the honest state of the Indian licence-free market, and no amount of shopping changes it.

8.6 Why community advice on this subject is unreliable

Indian community advice about radio law is not malicious and it is rarely stupid. It is **stale**, and it is stale in a systematic pattern that a reader can learn to recognise and correct for.

8.6.1 THE CORRECTION TABLE

Every left-hand entry was said by a real, named Indian source that was fetched for this research. Every right-hand entry is sourced to a part of this report.

Circulating claim	Who said it, and when	Correct position today, and where this report establishes it
<i>"Only HAM and CB are legal stuff in India. Walkie-Talkie (FM based) is not legal"</i>	BHPian svasantosh, Team-BHP, 2014-04-08	Obsolete. PMR446 at 446.0–446.2 MHz, 500 mW e.r.p., was de-licensed on 2018-10-18 . [VERIFIED] — Part 2 §2.5.1, Part 7 §7.6.2
<i>"These [PMR/FRS] sets are illegal to be used in India"</i>	BHPian sachinpk, a licensed amateur, Team-BHP, 2014-06-19	Correct for FRS (462 MHz is not an Indian exemption) and obsolete for PMR since 2018-10-18. The two must not be lumped together. [VERIFIED] — Part 2 §2.4.1
<i>"'Mobile' use is not really sanctioned by WPC. When stations are to go 'mobile' they are to get a special permit for the same"</i>	BHPian sachinpk, 2014-06-19	Abolished. That was rule 18 of the 1978 Rules, liberalised by WPC Office Memorandum of 2019-06-07 and superseded outright on 2024-10-30 . There is no separate mobile authorisation, no ₹20 fee, no "/MO" suffix and no 15-day test report. [VERIFIED] — Part 7 §7.2.1
<i>"the rule does not permit one to operate the radio at other locations than the home station... one cannot operate 'mobile'"</i>	BHPian bejoy (VU3BOJ), Team-BHP, 2015-09-01	Corrected by the author's own later edit, and now settled by the 2024 Rules read with WPC's FAQ of 2024-12-05 : mobile operation needs no authorisation, no notification, no fee and no siting clearance. [VERIFIED] — Part 7 §7.2.1
<i>"there will be separate address verification by Police department, who will actually come and 'interview' your neighbor"</i>	BHPian Grace, Team-BHP, 2014	Abolished with effect from 2016-03-10 [REPORTED] , and the application is a single self-certified step since March 2025. [VERIFIED] — Part 3 §3.9
<i>"in India, anything which requires a license is best left alone"</i>	BHPian raghu.t.k, Team-BHP, 2015	Attitude, recorded not endorsed. The Restricted ASOC costs ₹100 to sit and ₹1,000 or ₹2,000 to hold , with a 40% pass mark and no Morse. [VERIFIED] — Part 3 §3.7

Circulating claim	Who said it, and when	Correct position today, and where this report establishes it
<i>"CB... Comes with licence paper from WPC is what i understand"</i>	Jaggu, Team-BHP Support, 2012-04-11	The single most common Indian confusion in this area. CB is licence-exempt — there is no user licence to issue. What a compliant set needs is an ETA , which certifies the equipment , not the user. [VERIFIED] — Part 2 §2.7, Part 1 §1.5
<i>"Besides CBs are not illegal to use while driving unlike mobiles"</i>	BHPian zipperone, Team-BHP, 2012-04-12	Wrong. MV Act s.184 Explanation (c) deems <i>"use of handheld communications devices while driving"</i> to be dangerous driving. A fist microphone is a handheld communications device on any ordinary reading. [VERIFIED] — Part 7 §7.2.1; [UNCERTAIN] whether a court has so applied it
<i>"i use some ones elses license, when i take them for a rally"</i>	Badcheetah, Team-BHP, 2005-09-24	Not a thing. Authorisations attach to a person and a station; there is no lending mechanism and no Indian equivalent of a "control operator" doctrine. [VERIFIED] — Part 7 §7.2.2
<i>"Legally, yes, you need a licence... But realistically, just go ahead and use them...but make sure its not in front of a cop"</i>	Rtech, Team-BHP Support, 2005-09-23	Recorded as the community's baseline attitude, not as advice. It was at least <i>coherent</i> in 2005, when no licence-free voice band existed. Since 2018 there is a lawful option , which makes the reasoning obsolete as well as unwise — and enforcement has since become visible: ₹44 lakh in CCPA penalties and 11,060 sets seized. [VERIFIED] — Part 6 §6.4
Hams providing <i>"Third Party communication on behalf of rally drivers and rally officials who were not ham operators"</i>	VU2MSY, Himachal Car Rally, 2004	Prohibited. Annexure A para 2(2)(b) bans third-party messages; a sponsored event also engages the pecuniary-interest bar. [VERIFIED] — Part 7 §7.2.2, §8.4.5
<i>"I'm also not sure if this kind of a radio setup requires any special permissions"</i>	BHPian Axe77, Team-BHP, 2023-05-04, describing a professional operator's installation	Honest, and the single most representative statement of Indian rider awareness on this subject. If the kit is ETA-approved 446 MHz, no permission is needed by anyone. [VERIFIED] — Part 2 §2.5.1
<i>"446.0 – 446.1 MHz"; "446.00–446.02 MHz"; "Maximum Permissible RF Output Power: 0.5 Watts"</i>	Vendor guide, 2025-09-13; Indian product page, 2026-08-23; vendor guide, 2025-09-13	Band is 446.0–446.2 MHz ; limit is 500 mW e.r.p. , not conducted output. [VERIFIED] — Part 2 §2.5.1
<i>"License-free Walkie-Talkie — BaoFeng BF-888S UHF 400-470MHz"</i>	Indian retailer collection, fetched 2026-08-23	Flatly false and the most dangerous claim in this table. See §8.4.3. [VERIFIED] — Part 2 §2.10

Circulating claim	Who said it, and when	Correct position today, and where this report establishes it
"CB radios... (3 KMS TO 160 KMS)"; "connectivity up to 6–8 km" for 500 mW	Indian retailer pages, fetched 2026-08-23	Marketing. Part 7 §7.6.1 and §7.6.2 give the honest figures. [ENGINEERING ESTIMATE]
A phone app promising "Mesh Network Intercom" working "even in the most remote mountain passes with zero network coverage", naming Baralacha La, Tanglang La and Khardung La	Indian convoy app, fetched 2026-08-23	The product is a waitlist, not a shipping product — the site says "Coming Soon" [REPORTED]. And a phone with no network can only reach other phones over Wi-Fi Direct or Bluetooth Low Energy, which at handset power inside a jacket pocket is tens to a low hundreds of metres, not a mountain pass. [ENGINEERING ESTIMATE] / [UNCERTAIN]

8.6.2 THE FOUR REASONS THE ERRORS ARE SYSTEMATIC

Understanding *why* the advice is stale is more useful than memorising the corrections, because it lets a reader triage a source they encounter after this report was written.

- 1. The law changed repeatedly and the threads did not.** Between 2016 and 2026 the position moved at least five times: police verification abolished **2016-03-10**; PMR446 de-licensed **2018-10-18**; mobile amateur operation liberalised by WPC Office Memorandum **2019-06-07**; the 1978 Amateur Rules superseded **2024-10-30**; and the possession regime restated in 2026. A forum answer written in 2014 was correct when written and has been wrong for most of the time since. Nobody goes back and edits it. [VERIFIED] on each date — Parts 2, 3, 7 and Part 1 §1.9.
- 2. Search engines rank the stale answer above the current one.** A 2014 thread with a decade of inbound links, quoted and requoted, outranks a gazette notification that nobody links to and that is published as a scanned PDF. The reader searching in 2026 sees 2014's answer first, and it is written in confident, friendly prose by someone who genuinely knew the law at the time.
- 3. The errors are asymmetric, and the asymmetry is the tell.** Indian community advice is **too pessimistic about the law** — "nothing is legal", "only ham and CB", "a licence takes a year" — and **too credulous about the hardware** — "6 to 8 km on 500 mW", "3 to 160 km on CB", "unlimited riders in mesh". A source that tells you the law is more restrictive than it is, and the equipment better than it is, is displaying the exact signature of this community's failure mode. The report's corrections therefore run in two directions at once: **more is legal than people think, and less works than vendors claim.**
- 4. Nobody has an incentive to fix it.** The vendors selling non-compliant hardware benefit from the confusion — CCPA fined eight of them. The licensed amateurs who do know the law are describing a route most people will not take, so their correct answer reads as unhelpful. And the people who finally learn the current position learn it after they have bought, at which point they stop posting.

8.6.3 WHERE COMMUNITY ADVICE IS ACTUALLY EXCELLENT

This is not a section about a community that gets everything wrong, and treating it that way would cost the reader real knowledge. **Indian rider and off-road forums are unreliable on law and outstanding on field craft.** Everything in the following list came from the same sources as the errors above, and all of it is worth copying:

- **Carry a printed copy of the exemption notification in the vehicle.** *"Its the document that makes the difference really."* [REPORTED] — 2012. Sound then, sound now, and it generalises from CB to PMR446.
- **Declare devices at checkpoints rather than concealing them.** The account of declaring a CB along with cameras and phones at a Border Security Force post, and being allowed to carry it, is a model of how to behave. [REPORTED]
- **Hand signals and a convoy SOP beat radios on cost, on noise, on large groups, and on driver distraction** — the four reasons an Indian event organiser gave in 2014, all still correct. [REPORTED]
- **The ear-clip headset and remote PTT switch** an event organiser adopted for convenience is also the configuration that keeps you clear of MV Act s.184. [REPORTED] / [VERIFIED]
- **Standardise the group on one intercom brand before anyone buys.** [REPORTED]
- **Buy the dual-SIM configuration for Ladakh, with a BSNL postpaid line as the emergency voice line** — and note the fact that catches people every season: **prepaid SIMs issued outside the region are deactivated at the regional border**, so an outside prepaid SIM is dead in Leh. [REPORTED] — Indian coverage guide, fetched 2026-08-23.
- **Download offline vector maps before leaving town, and carry a paper map and a power bank**, because these use the phone's own GNSS receiver and work with no network at all. **Receiving GNSS is not transmitting and raises no licensing question whatsoever.** [REPORTED] / [VERIFIED] on the legal point.

The rule to apply: trust an Indian forum on what to do with your hands, your route and your kit. Verify anything it tells you about the law against Parts 1 to 3 and Part 7 of this report, and against the gazette.

8.7 The recommended architecture for a mixed group

For a group in which most members are unlicensed and the reader may become a licensed amateur. Buy from the bottom up: **each layer works without the ones above it.** This is the private-expedition version of what the one professional operator in §8.4.2 actually runs — radios between crew positions, intercoms among the riders — and it is deliberately not the enthusiast version.

8.7.1 LAYER 0 — THE WRITTEN CONVOY SOP (₹0, AND THE HIGHEST RETURN OF ANYTHING HERE)

A one-page standing operating procedure, circulated before departure and agreed by everyone:

- Riding or driving order, with a **named lead** and a **named sweep**.
- A **maximum interval** between vehicles, stated in seconds or metres, not in feel.
- **Regroup at every junction and every fuel stop**, without exception and without discussion.
- A **stop-and-wait rule**: if the vehicle behind is not seen for N minutes, stop.
- The day's **rally points marked in everyone's offline map** before leaving.
- An agreed **hand-signal set** for the four or five things that matter — slow, hazard, stop, fuel, regroup.

Assume every radio will fail, and plan so that it does not matter. Part 7 §7.6.3 put it correctly and this part's evidence only reinforces it: **a disciplined convoy with no radios loses nobody; an undisciplined convoy with radios loses people the moment the radios fail.** Four out of five Indian commercial operators sell exactly this — road captain, mechanic, sweeper, backup vehicle — and no radio at all, at ₹50,000 a head.

8.7.2 LAYER 1 — THE PRIMARY VOICE NET: 2.4 GHZ MESH INTERCOMS

Buy one mesh unit per motorcycle, plus one off-helmet mesh unit for the lead four-wheeler. Standardise on a single brand across the whole group, because mesh does not cross brands (§8.3.4).

Tier	Model and price	Riders	When it is the right choice
Value	EJEAS Q8 5G Mesh, ₹5,999–₹6,499	6, 17 h	A first expedition. Buy this, find out what fails, upgrade those units
Value	BluArmor C40, ₹8,999 introductory / ₹12,999	20	Indian brand, Indian warranty, Indian service. The safest value choice
Volume	BluArmor C30 / C20, ₹9,999–₹10,999	20	The model with the most Indian field reports behind it (§8.3.5)
Volume	Sena B10 #Bharat, ₹10,999	4-way	If the group is four or fewer and wants the mainstream brand
Larger group	EJEAS X10 Mesh 4.0, ₹15,499	10	Groups of 7 to 10 on one net
Premium	Cardo Packtalk Neo ₹33,899 / Edge ₹39,599	15+	Justified if you ride 10,000 km a year in bad weather. Not justified for a first expedition

Check five things before paying, all of which come from §8.3.9: **IP67**, or IP65 as an absolute floor; **charge-while-riding**; **helmet fitment for the actual helmets in the group**; **glove operability of the buttons**; and that you are buying from the brand's Indian storefront or a named retailer, not an anonymous marketplace listing.

What this layer is for: continuous, hands-free, full-duplex conversation between riders within line of sight of each other — which, on a well-run convoy, is where everybody is supposed to be. It also puts the lead four-wheeler on the same net as the bikes, which is the one thing no other option in this report does.

What it is not for: reaching the vehicle 4 km back around a ridge. Expect 500–600 m per hop on a real road, expect the mesh relay to hold a strung-out group well beyond that, and expect the link to disappear behind a rock spur and restore itself when you round the corner (§8.3.5).

Legal: licence-free under G.S.R. 45(E). **[VERIFIED]** — §8.3.1.

8.7.3 LAYER 2 — THE CREW-ROLE AND OUT-OF-VEHICLE LAYER: PMR446

Buy two to four ETA-quoted handhelds, not one per person. One for the lead, one for the sweep, one for the backup vehicle, one spare. This is the design Indian commercial operators actually use (§8.4.2), and it is the design that works with 500 mW when an all-riders net does not.

- **BRUTFORCE BFR-002, pack of two, ₹7,400** from an Indian outdoor retailer, or **TalkPRO LF446 at ₹9,000 plus tax**, which quotes **ETA SD 20250100112. [REPORTED]** — read §8.4.4 first, including the coincidence flagged there, and get the ETA number in writing for the exact model.
- Add a **₹750 earphone** or **₹850 remote speaker microphone** per unit. **[REPORTED]**
- If a rider will use one on the move, add a two-pin helmet kit with a **handlebar-mounted PTT**. These are a standard international accessory at roughly US\$13–25; **no Indian-domiciled retailer for the specific helmet kit could be found**, so budget for an import or a substitute. **[REPORTED] / [UNCERTAIN]**

What this layer is for: spotting a vehicle through an obstacle, recovery, river crossings, walking a line, camp coordination, and talking to the vehicle parked 800 m away. This is the work PMR446 is genuinely good at — everyone outside the vehicle, within a kilometre — and it is work the intercoms cannot do, because by then everyone has taken their helmet off.

Four rules that keep this layer lawful and safe:

1. **A headset does not touch the integral-antenna condition.** Table V restricts the antenna and says nothing about audio accessories. Fitting an earpiece keeps you inside the exemption. **[VERIFIED]**
2. **Never key a handheld while driving or riding.** A radio clipped in a jacket with a bar-mounted button is not handheld. A radio picked up and keyed at 60 km/h is. **[VERIFIED]** — MV Act s.184 Explanation (c), Part 7 §7.2.1.
3. **Do not buy a "446 base station" and do not fit an external antenna.** Both leave the exemption instantly, and both are sold in India as though they did not. **[VERIFIED]** — §8.4.3.
4. **Assume VOX will not work in a helmet at speed.** Wind noise triggers it and a full-face helmet muffles speech. Plan on the button. **[ENGINEERING ESTIMATE]**

Legal: licence-free only if band, power, integral antenna and ETA all hold simultaneously. **[VERIFIED]** — Part 2 §2.5.1.

8.7.4 LAYER 3 — THE MULTI-KILOMETRE VEHICLE LINK: CITIZEN BAND, CONDITIONALLY

Fit CB only if you pass the seven-point test at §8.5.6. Budget ₹6,000–₹12,000 for the set and ₹1,500–₹4,000 for the antenna, mount and coaxial cable, per vehicle. Expect to telephone dealers. **Skip this layer entirely for a motorcycle-only group**, and skip it for wooded or quarry terrain, where an Indian club has already done the experiment for you.

Legal: licence-free at 5 W ERP; the external vehicle antenna point is genuinely unresolved and is question 4 in Part 7 §7.7.2. **[VERIFIED]** / **[UNCERTAIN]**

8.7.5 LAYER 4 — YOUR OWN RESTRICTED ASOC (₹1,100 OR ₹2,100)

Sit the examination. It is the only item in this entire architecture that authorises anything.

What it is genuinely good for:

Capability	Why it matters on an expedition
50 W DC input on ten HF bands	The only capability in this report that lawfully reaches out of a Himalayan valley. From a camp in Ladakh, 40 m or 20 m single sideband will reach hundreds or thousands of kilometres when nothing else will. [VERIFIED] — Part 3 §3.4, Part 7 §7.2.3
A lawful 10 W mobile station on 2 m and 70 cm	No separate mobile authorisation, no notification, no fee, no siting clearance, no import authorisation. [VERIFIED] — Part 7 §7.2.1
The engineering education	Antenna height, efficiency, polarisation and terrain decide whether a convoy link holds. Watts do not. Every other layer in this architecture works better once you understand this
The option value	If the whole group later licenses at ₹1,100 each, 2 m and 70 cm at 10 W with proper roof antennas becomes lawful for the group and outperforms everything else here in open terrain. [VERIFIED] — Part 7 §7.2.2

What it is not good for, stated as plainly as Part 7 states it:

Not this	Why
Talking to your unlicensed group	Four independent bars, any one of which is sufficient: the purpose limitation, the "other authorised amateur stations" limit, the express ban on third-party messages, and the technical-content limitation. [VERIFIED] — Part 7 §7.2.2, Part 3 §3.6
Anything with a commercial element	"Solely with a personal aim and without any pecuniary interest", plus the ban on "communications of business". A paid tour, a sponsored trip or a film shoot is out, with no workaround. [VERIFIED]

Not this	Why
A general emergency channel	The word "emergency" does not appear in the Rules. The only third-party carve-out needs a facilities failure, a government permission, and a District Magistrate at the other end. [VERIFIED] — Part 7 §7.2.2
A VHF handheld as your expedition radio	India's repeater network is urban : roughly 70 machines, none listed in Himachal Pradesh, Ladakh, Jammu and Kashmir, Uttarakhand or the Northeast. [REPORTED] — §8.5.4. Any range figure quoted "via repeater" is irrelevant to the routes you will drive. HF, not UHF, is the reason to get licensed

Keep the group's traffic on Layers 1 and 2 and keep the amateur station for amateur work. And carry the certificate, the bound-book log and the equipment register **in the vehicle** — on an overland trip you are the most likely licensee in India to be asked. **[VERIFIED]** — Part 7 §7.2.1.

8.7.6 LAYER 5 — EMERGENCY COMMUNICATIONS: A CHECK-IN PLAN, NOT A SATELLITE MESSENGER

Do not carry a Garmin inReach, a SPOT or a satellite phone. Part 7 §7.6.4 gives the position and the arrest record; the enforcement history has continued to grow, and it includes a Canadian ultra-runner arrested at Goa, a Scottish hiker arrested at Delhi after putting an inReach in a security tray for a **domestic** flight, and a 2025 United States Embassy travel alert confirming that enforcement extends to GPS-branded satellite devices. **[REPORTED]**

The cheap, lawful substitute that actually gets a search started:

- A **named contact off the mountain** who is not on the trip.
- **Fixed check-in times**, agreed in advance and kept even when there is nothing to report.
- A **written route card** with bail-out points, circulated to that contact before departure.
- An **agreed overdue-response procedure**: what the contact does, and when, if a check-in is missed.

It costs nothing, it needs no equipment, and it works. Note also that **poor mobile coverage on a remote track is not a "failure of normal telecommunication facilities"** — it is the absence of facilities — so the narrow amateur calamity permission does not open up merely because you have no bars. **[VERIFIED]** — Part 7 §7.2.2.

One open question worth resolving before you buy anything in this category: whether a 406 MHz personal locator beacon carried by a private individual requires DoT authorisation. A beacon transmits to a satellite, and DoT's ETA exclusion list defines satellite equipment as any device that transmits to a satellite — but **no Indian authorisation route, exemption or enforcement precedent for a private PLB could be found either way.** **[UNCERTAIN]** **Do not assume a PLB is treated more kindly than an inReach.** Put it to WPC in writing.

8.7.7 WHAT TO BUY FOR WHICH ROLE

The role table is the part to hand to the group.

Role	Carries	Why this and not something else
Lead rider / road captain	Mesh intercom (same brand as the group) plus a PMR446 handheld with earpiece and a handlebar PTT	Talks continuously to the riders on the intercom; talks to the sweep and the backup vehicle on the radio when the group is strung out beyond intercom range
Sweep rider	Mesh intercom plus a PMR446 handheld with earpiece	The sweep is by definition the person furthest from the lead, and the only person who knows a rider has stopped
Backup / support vehicle driver	Off-helmet mesh unit on a headset plus a PMR446 handheld plus the group's spare radio	Off-helmet mesh keeps the vehicle on the same voice net as the bikes — the single most underrated part of this architecture. The radio reaches the crew when the mesh does not
Other four-wheelers, if two or more and the terrain is open	Off-helmet mesh unit plus CB, if the §8.5.6 test passes	CB is the only licence-free way to hold a multi-kilometre vehicle-to-vehicle link, and it only makes sense between four-wheelers
Ordinary riders	Mesh intercom only	They do not need a radio. Adding radios to the whole group multiplies cost and clutter and adds nothing the mesh does not already do
Front-seat passenger in any vehicle	Nothing extra — but designate them the radio operator	Keeps every transmission out of the driver's hands. Free, and it disposes of the MV Act s.184 problem entirely
The reader	All of the above for their own vehicle, plus an ASOC and, when the budget allows, an HF station for camp	The licence is for you, for HF and for learning. It is not the group's communication system

8.7.8 THE COST, AND WHAT TO CONFIRM BEFORE SPENDING

Total for a five-person mixed group:

Layer	Configuration	Cost
0 — SOP	One page, circulated and agreed	₹0
1 — Mesh intercoms	5 units at ₹6,499	₹32,495
2 — PMR446	1 pair at ₹7,400 plus 2 earpieces at ₹750	₹8,900
3 — CB (<i>only if two or more four-wheelers and open terrain</i>)	2 sets at ₹8,000 plus 2 antennas at ₹2,500	₹21,000 (<i>optional</i>)
4 — Your ASOC	Examination ₹100 plus lifetime certificate ₹2,000	₹2,100
5 — Check-in plan	A phone call and a shared document	₹0

Layer	Configuration	Cost
Baseline total (Layers 0, 1, 2, 4, 5)		≈ ₹43,500

That is less than the price of one flagship intercom pair, and every rupee of it is lawful.

Ask every seller, in writing, before paying:

1. The **WPC ETA number** for the exact model and variant, and the name of the ETA holder.
2. For PMR446: whether the antenna is **integral or detachable**, and confirmation the unit is sold as a hand-portable and **not** as a base station.
3. For CB: the **exact frequency coverage as shipped**, and whether the set can be band-limited to **26.957–27.283 MHz**.
4. For any imported intercom: whether the unit was **imported against an Indian ETA** or is a personal import.

Ask WPC, in writing. Part 7 §7.7.2 lists the seven questions that bear on the purchase and gives the contact details. Three of them decide items in this part — **question 4** (CB and external antennas), **question 5** (the restricted-areas list) and the possession question at **question 1**. Two further questions arise from this part and neither is urgent:

1. **Whether a 406 MHz personal locator beacon carried by a private individual requires authorisation**, and if so under what instrument and by what procedure (§8.7.6).
2. **Which instrument governs a 2.4 GHz mesh intercom** — G.S.R. 45(E) or G.S.R. 1047(E) Table IX — and whether the 10 mW e.i.r.p. figure in the latter constrains Bluetooth-class devices (§8.3.1). Low practical risk; untidy drafting.

8.7.9 THE SHORT VERSION OF PART 8

1. **What Indian riders actually buy is a 2.4 GHz helmet intercom, and what actually holds their groups together is a phone, a WhatsApp group and a stop-and-regroup rule.** The intercom is the dominant device; the discipline is the dominant system. **[REPORTED]**
2. **Radios are a minority practice**, clustered in commercial operators, off-road event crews and a small pocket of licensed amateurs — and where operators do use them, they put them on **crew roles, not on every person**. Copy that. **[REPORTED]**
3. **Citizen Band remains the best licence-free multi-kilometre option in law and physics, and it is effectively unbuyable and unproven in broken terrain.** Both statements are true. Which one governs your purchase is decided by your route and your patience, not by the radio. §8.5.
4. **The mislabelling problem is quantified and large.** On one platform, 65,931 units were sold with the frequency range blank or outside the exemption. Assume a marketplace "licence-free walkie talkie" is unlawful until the ETA number says otherwise. **[VERIFIED]** — Part 6 §6.4.7.

5. **Community advice is stale in a predictable direction — too pessimistic on law, too credulous on hardware — and excellent on field craft.** Use it for the second and check it against this report for the first. §8.6.
6. **The written convoy SOP outranks every piece of hardware in this part,** and four out of five Indian commercial operators sell it instead of a radio.

8.8 Disclaimer for this part

This part is a research summary of **community practice**, compiled on 2026-08-23. **It is not legal advice, and it is not an endorsement of any practice described.** The legal statements in it are carried from Parts 2, 3, 6 and 7, where the underlying gazette texts were fetched and quoted; the practice statements are overwhelmingly **[REPORTED]** from forums, retailers, tour operators and product reviews, and several are one person's impression rather than any kind of measurement.

Specific limits on the evidence, restated because they matter:

- **Every Team-BHP quotation in this part was read from an Internet Archive capture**, the site having returned HTTP 403 to direct fetching. The most recent Team-BHP material reachable that way is from 2018 to 2021.
- **No Reddit, Quora or YouTube evidence exists in this part at all**, those sources being inaccessible to this research. The community picture is correspondingly thinner and older than it should be, particularly on the motorcycle side.
- **No ETA number quoted anywhere in this part has been verified against WPC's own register**, which is CAPTCHA-gated. Every "ETA approved" claim rests on a vendor's word.
- **No Indian source measured mesh intercom range in Himalayan terrain.** The switchback figures in §8.3.5 are **[ENGINEERING ESTIMATE]**, not measurement.
- **No published list of restricted areas could be found**, and that remains the single biggest practical risk to an overland expedition carrying any radio equipment.

Prices, stock, product line-ups, coverage and enforcement practice all change quickly, and several findings here are expressly tagged **[UNCERTAIN]** because they could not be confirmed against a primary source. Some widely followed practices described in this part are unlawful and are identified as such; **do not read a description as a recommendation.**

Confirm the questions in Part 7 §7.7.2 and §8.7.8 with the WPC Wing of DoT in writing, verify any ETA number yourself on saralsanchar.gov.in, and speak to an established amateur radio club such as ARSI or NIAR, before you buy equipment, before you plan a trip around any of this, and before you transmit.

Part 9 — Listening, receiving, and the Citizen Band verdict

Parts 1 to 6 set out the law. Part 7 applied it to one purchase. Part 8 described what Indian expedition groups actually do. This part answers the two questions that were left open, and it corrects Parts 7 and 8 on one of them.

The two questions are:

1. **"Which radio covers all possible radio frequencies, and may I own it and listen on it in India?"** Answered at §9.2 to §9.5.
2. **"Is a Citizen Band set the convoy answer, as Parts 7 and 8 said it was?"** Answered at §9.6, and the correction is at §9.7.

They turn out to be the same question seen from two ends, because both are decided by the single most important structural fact in radio law: **receiving and transmitting are not the same act, and India does not regulate them the same way**. §9.4 explains why that asymmetry exists. Everything else in this part follows from it.

Position as at 2026-08-25. All hardware specifications, prices, retailer listings and manufacturer pages described here were read on **2026-08-24** or **2026-08-25**; each table says which. Prices in this market move weekly and several Indian retailer pages do not state whether the figure includes Goods and Services Tax (**GST**). Treat every rupee figure as a snapshot, not a quotation.

Evidence tags carry the same meaning as in Part 1 §1.1, Part 7 and Part 8: **[VERIFIED]** = a primary or official source actually fetched, with URL and document date; **[REPORTED]** = a named credible secondary source, or an official document obtained only from a mirror; **[UNCERTAIN]** = could not be confirmed, sources disagree, or the source is undated; **[ENGINEERING ESTIMATE]** = a figure derived from physics or circuit fundamentals, which is a technical judgement and never a legal claim. One additional tag appears in this part where the underlying research used it: **[ARITHMETIC]** = a calculation performed on figures that are themselves tagged above it. Check the numbers rather than trusting the label.

A warning about tag distribution specific to this part. The hardware sections (§9.2, §9.3) are mostly **[VERIFIED]** because manufacturers publish specifications and retailers publish prices. The legal sections (§9.5, §9.6) contain the **highest concentration of [UNCERTAIN] in this entire report**, and that is not laziness — it is the finding. Two of the load-bearing questions here (does India's only receiver exemption cover a scanner? does "built-in antenna" mean what it says?) have never been answered by any Indian court, notification, circular or published clarification, so far as this research can establish. **Where this part hedges, the hedge is the answer, and it must not be read as a soft yes.**

Abbreviations expanded on first use in this part. **SDR** = software-defined radio, a receiver or transceiver in which the demodulation is done by software on a computer rather than by fixed circuitry. **ADC** = analogue-to-digital converter. **ENOB** = effective number of bits, the real resolution of an ADC after noise. **Msps / ksps** = mega- and kilo-samples per second. **IQ** = the in-phase and quadrature pair of signals an SDR produces, from which any modulation can be recovered. **ITU** = International Telecommunication Union; **RR** = its Radio Regulations. **VLF / LF /**

MF / HF / VHF / UHF / SHF / EHF = the ITU frequency band names, tabulated at §9.2.1. **ISM** = industrial, scientific and medical, a class of bands set aside for equipment that uses radio energy for something other than communication. **CEPT** = the European Conference of Postal and Telecommunications Administrations, whose 40-channel plan is the world's default Citizen Band channel plan. **ERP / e.r.p.** = effective radiated power, referenced to a half-wave dipole and including antenna gain; **e.i.r.p.** is the same idea referenced to an isotropic radiator. **dBd / dBW / dBm** = decibels relative to a dipole, to one watt, and to one milliwatt. **PEP** = peak envelope power, the way single-sideband transmitter output is measured. **SO-239 / PL-259 / TNC / BNC / SMA** = standard coaxial antenna connectors; SO-239 is the chunky socket on the back of a mobile radio, TNC and SMA are the small threaded ones on hand-helds. **SWR** = standing wave ratio, the usual measure of how well an antenna matches its feeder. **VNA** = vector network analyser, the instrument that measures it. **ADS-B** = Automatic Dependent Surveillance–Broadcast, the unencrypted 1090 MHz position broadcast every airliner makes. **AIS** = Automatic Identification System, the marine equivalent. **APRS** = Automatic Packet Reporting System, an amateur position-reporting network. **POCSAG** = the common paging protocol. **GNSS** = global navigation satellite system (GPS, GLONASS, Galileo). **DTH** = direct-to-home satellite television. **OSA** = the Official Secrets Act, 1923. **SWL** = Short Wave Listener, the receive-only amateur licence India abolished in 2010. **BRL** = Broadcast Receiver Licence, the domestic radio licence India abolished in 1984. **REPA** = the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026. **WMO** = the Wireless Monitoring Organisation, WPC's field enforcement arm. **DRI** = Directorate of Revenue Intelligence. **CBIC** = Central Board of Indirect Taxes and Customs. **DGFT** = Directorate General of Foreign Trade; **ITC(HS)** = the Indian trade classification of goods, under which import policy conditions are stated. Carried from earlier parts without re-explanation: **WPC, DoT, ASOC, ETA, CB, PMR446, G.S.R.**

9.1 If you only read one page

9.1.1 THE ORGANISING PRINCIPLE: RECEIVE AND TRANSMIT ARE DIFFERENT LEGAL ACTS

A receiver cannot interfere with anybody. A transmitter can deny a frequency to an air traffic controller, a coastguard station or a paramilitary net. That single engineering fact is why every radio regulator on earth, India included, treats width on the receive side as cheap and width on the transmit side as expensive. §9.4 sets out the evidence, including the sentence in which Baofeng's own United Kingdom arm concedes the point.

Everything practical in this part is a consequence:

	Receive side	Transmit side
What width costs in money	₹5,336 buys 500 kHz to 1.766 GHz [VERIFIED]	₹116,500 buys 17.44 MHz of lawful transmit spectrum [VERIFIED spec; ARITHMETIC on the total]

	Receive side	Transmit side
What width costs in legal exposure	Almost nothing, subject to §9.5	Everything. Part 7 §7.3.1 quantified it: about 97 % of a Baofeng UV-32's transmit range is spectrum an Indian amateur may never touch [VERIFIED]
Which Indian gate engages	Possession (2023 Act s.3(1)(c)), and only that	Type approval, import, possession and operation — all four gates of Part 2 §2.2 [VERIFIED]
Worst realistic outcome for the equipment alone	Fine of ₹100 to ₹250 plus discretionary confiscation, or a civil penalty up to ₹50,000 [VERIFIED] — §9.5.6	Seizure, civil penalty up to ₹2,00,000 per instance, and — if you imported it yourself — prosecution under the Customs Act 1962 [VERIFIED / REPORTED] — Part 6 §6.2
The rule to take away	Put the width on the receive side.	Buy the narrowest transmitter that does the job.

That is the inversion at the heart of this part. The reader's original question — "which radio has all possible radio frequencies" — has a hardware answer (§9.3) and a legal answer, and they point in opposite directions. The device that covers the most is the device you should least want to be found holding.

9.1.2 FOUR QUESTIONS THAT GET FOUR DIFFERENT ANSWERS

The internet routinely blends these into one question called "is scanning legal in India". They are four separate questions with four separate governing provisions and four wildly different penalty scales. **Keep them apart or you will get the risk assessment exactly backwards.**

#	The act	Governing provision	Honest answer	Maximum exposure	Tag
1	Owning a receive-only apparatus	Indian Wireless Telegraphy Act 1933 s.3; Telecommunications Act 2023 s.3(1)(c); the exemption in G.S.R. 393(E) of 1997-07-16; REPA rule 10	Probably lawful, and certainly lawful if you hold an ASOC. But a non-ASOC owner is relying on an exemption written in 1997 for domestic broadcast sets, and nobody has ever said whether it reaches a scanner or an SDR	₹100–₹250 fine plus confiscation (1933 Act s.6(1), s.6(3)), or a civil penalty up to ₹50,000 (2023 Act Third Schedule item 1(a)). Not a criminal offence under s.42	Provisions [VERIFIED] ; scope of the exemption [UNCERTAIN]

#	The act	Governing provision	Honest answer	Maximum exposure	Tag
2	Listening to what it hears	No offence under the Indian Telegraph Act 1885; Telecommunications Act 2023 s.42(2)(b) , <i>"intercepts a message unlawfully"</i>	No Indian offence of listening exists. But India, unlike the United Kingdom, wrote no carve-out for transmissions intended for general reception , and whether passive reception is "interception" is unresolved	3 years and/or ₹2 crore ; all s.42 offences are cognizable and non-bailable	Texts [VERIFIED] ; application to a listener [UNCERTAIN]
3	Recording it	Official Secrets Act 1923 s.3(1)(c) , with the s.3(2) reversed burden where the material relates to a "prohibited place"	Materially worse than listening. The verbs in s.3(1)(c) are <i>"obtains, collects, records or publishes"</i> . A monitoring log is a record. A government wireless or signal station is a statutory prohibited place	14 years where the matter relates to naval, military or air force affairs or a secret official code; 3 years otherwise	[VERIFIED]
4	Disclosing or repeating it	OSA s.3(1)(c) <i>"publishes or communicates"</i> ; for a certificate holder, Annexure A para 8 of the Amateur Services Rules 2024	Worse again. Disclosure converts a monitoring hobby into publication. An ASOC holder additionally owes a positive duty of secrecy over anything they were not entitled to receive	As (3), plus suspension or cancellation of the ASOC, plus whatever the substantive act amounts to	[VERIFIED]

The proportion is the point, and it is not close. Owning ≈ ₹100 to ₹50,000. Listening ≈ untested and never prosecuted against a listener in India. **Recording official traffic ≈ up to fourteen years, with a statutory presumption running against you.** §9.5 works each question through, and §9.5.5 states the one real-world risk plainly.

9.1.3 THE CITIZEN BAND VERDICT, IN ONE TABLE

Part 7 §7.6.1 and Part 8 §8.7.4 recommended Citizen Band as the licence-free convoy answer, subject to an open question about antennas. **That open question now has an answer, and the answer is bad.**

G.S.R. 533(E) condition	A President CB set — the only brand that has engineered an Indian band plan	Verdict
1. Band 26.957–27.283 MHz	Every model is a 40-channel 26.965–27.405 MHz radio with a user-switchable country configuration. Its In setting gives 27 channels, 26.965–27.275 MHz, which fits India's window with 8 kHz to spare at each end — genuinely correct engineering. But a five-second key sequence reaches 26.565–27.99125 MHz	Passes on use while parked in In ; fails on capability
2. Not more than 5 W ERP	4 W carrier, AM and FM, across the whole range. Under 5 W on any reading — but only 0.97 dB under , and rule 2(b) folds antenna gain into ERP, so a 5/8-wave or co-phased pair takes you over	Passes on AM/FM. Fails on single sideband , where two models make 12 W PEP
3. Built-in antenna	Hand-helds: a screw-on flexible whip on a 50-ohm TNC female socket , with a mobile adapter <i>in the box</i> for fitting a vehicle antenna instead. Mobiles: an SO-239 socket and no antenna supplied at all	FAILS, on every model surveyed, and the buyer cannot cure it
4. WPC Equipment Type Approval for the model	Claimed by the manufacturer for "its CB portfolio in India"; no ETA number published anywhere ; the India catalogue does not list a Randy III at all	UNVERIFIED

[VERIFIED] on every specification cell from the manufacturer's own specification tables and owner's manual, fetched 2026-08-24/25; **[UNCERTAIN]** on condition 4 and on the interpretation of condition 3. Full working at §9.6.

The one-line answer: a CB radio with an antenna socket is not a CB radio with a built-in antenna, and no manufacturer anywhere was found selling one with a non-detachable antenna. On the face of the gazette that appears to exclude essentially every practical vehicle CB installation — which is a stronger objection than Part 8's market-availability finding, and it is why §9.7 exists.

And a second finding that reframes the whole comparison: **the 5 W ceiling is unreachable by any equipment that satisfies condition 3.** A built-in antenna at 27 MHz is necessarily a stub — a quarter wave is 2.76 metres — so a genuinely compliant Indian CB set would radiate on the order of **0.04 to 0.2 W ERP**, which is the same as or less than a compliant PMR446 handset, on a

noisier band with ISM interference it must accept without protection. **[ENGINEERING ESTIMATE]** The report should stop describing Citizen Band as the high-power licence-free option. §9.6.4.

9.1.4 WHAT TO BUY INSTEAD: THE TWO-DEVICE SPLIT, WITH RUPEE COSTS

Split the job in two. A wide receiver for listening and learning; a narrow, band-limited transmitter for lawful operation. Never one box that does both widely.

Role	Device	Coverage	Price	Tag
RECEIVE — wide, buy today	RTL-SDR Blog V3 dongle plus the multipurpose dipole antenna kit , Fab.to.Lab	500 kHz – 1.766 GHz continuous receive , about 2.4 MHz visible at once, zero transmit capability	₹5,712	[VERIFIED — product page fetched 2026-08-24]
— cheaper, dongle only	RTL-SDR Blog V4, Fab.to.Lab	same coverage	₹5,336	[VERIFIED]
TRANSMIT — narrow, buy after the ASOC	Alinco DJ-VX50E (the European version), Radio Brewery, Chennai	TX 144–146 / 430–440 MHz only , 12 MHz total, 5 / 2 / 1 W	₹9,676 (MRP ₹11,800)	price [VERIFIED] ; the 144–146 / 430–440 figure [VERIFIED from a European retailer's specification, not from the Indian listing — confirm the market version in writing, §9.8.5]
— alternative	Yaesu FT-65E — insist on the European version, not the Asian FT-65	144–146 / 430–440 MHz, 5 / 2.5 / 0.5 W	₹15,000 at RFCharge for an FT-65; version not stated	price [VERIFIED] ; version caveat [VERIFIED]
Antenna — the best rupee in this part	Nagoya VHF/UHF ground-plane kit, Radio Brewery	2 m / 70 cm	₹1,239 (MRP ₹3,540)	[VERIFIED]
The licence	ASOC (Restricted): examination plus certificate	—	₹100 + ₹1,000 (20 years) or ₹100 + ₹2,000 (lifetime)	[VERIFIED] — Part 3 §3.7

Total for the complete starting position: ₹17,727, or ₹18,727 with the lifetime certificate. Budget **₹20,000**, because at least one Indian retailer quotes prices excluding GST and neither Fab.to.Lab nor Radio Brewery states whether GST is included. **[VERIFIED]** that ElectroPi quotes ex-GST; **[UNCERTAIN]** for the other two.

What to spend today: ₹5,812 — the ₹5,712 receiver kit and the ₹100 examination fee. Nothing else should be bought before the certificate exists, because until then a compliant Indian dealer should not sell you a licensed-band radio at all (REPA rule 8(1), Part 2 §2.9.2). **[VERIFIED]**

For comparison, and it is the sharpest way to see the whole argument: the Baofeng UV-32 that Part 7 examined is **₹14,449** on its Indian storefront. **[VERIFIED — Part 7 §7.4]** For **₹939 more than that one radio**, the split above buys a far better receiver, a lawful transmitter, a proper antenna and the licence that makes any of it legal. **[ARITHMETIC]**

9.1.5 THE FIVE THINGS MOST WORTH CARRYING OUT OF THIS PART

1. **There is no radio that covers all radio frequencies, and there cannot be.** The spectrum runs to **3,000 GHz** on the face of India's own 1933 Act. The widest device found anywhere in this research reaches **7.2 GHz** — **0.24 %** of it. Four independent physical reasons each make the idea impossible, and understanding them is most of Section A of the ASOC syllabus. §9.2.
2. **Receive width is cheap; transmit width is legal exposure.** A ₹5,336 dongle already covers about **47 % of the radio spectrum measured in decades**, which is the only unit that makes physical sense. The reader is not far from "everything" — they were measuring it in the wrong unit. §9.2.1, §9.3.
3. **Owning a receiver in India is low-risk but not squarely lawful, and the weak link is import, not ownership.** Buy domestically. A GST invoice from an Indian retailer is the single most useful document you can own. §9.5.7.
4. **The real risk is not the radio. It is recording or repeating aviation, police, military or security traffic, especially near an airport or a defence installation.** That fact pattern produced India's one clear enforcement case, in Delhi in February 2010. §9.5.5.
5. **Citizen Band is out.** Not on availability — on the face of the rule. What replaces it is what Part 8 §8.7 already recommended for every other layer: a written convoy standard operating procedure, 2.4 GHz mesh intercoms, and two to four ETA-verified PMR446 handsets on crew roles. §9.7.

9.2 Why no radio covers all frequencies

This section is deliberately the most technical in the report, because the reader wants to learn radio and because every concept in it is examinable. Where a paragraph maps onto a named line of the ASOC (Restricted) syllabus, the syllabus item is given. The syllabus text is **[VERIFIED]** from Annexure II of WPC Office Memorandum No. L-14011/255/2004-AMT dated 2010-08-13; see Part 4, and note the **[UNCERTAIN]** at Part 1 §1.8 as to whether that 2010 memorandum is still the operative specification.

9.2.1 HOW WIDE "ALL RADIO" ACTUALLY IS

Start with the legal definitions, because they are wider than most people assume.

Boundary	Figure	Source	Tag
Upper limit in India's own statute	3,000 gigacycles per second — i.e. 3,000 GHz, or 3 THz	Indian Wireless Telegraphy Act 1933, s.2(1) Explanation: <i>"Radio waves' or 'Hertzian waves' means electromagnetic waves of frequencies lower than 3,000 gigacycles per second propagated in space without artificial guide"</i>	[VERIFIED]
Upper limit in the Telecommunications Act 2023	None at all	s.2(m): <i>"radio waves' means electromagnetic waves of frequencies propagated in space without any artificial guide"</i> — the 1933 Act's numerical ceiling was simply dropped	[VERIFIED]
Upper limit of India's own allocation table	3,000 GHz	National Frequency Allocation Plan 2022, hosted by DoT	[VERIFIED] — Part 1 §1.3
Upper limit of the European legal definition, for cross-check	3,000 GHz	Directive 2014/53/EU, Art. 2(1) (4): <i>"electromagnetic waves of frequencies lower than 3 000 GHz, propagated in space without artificial guide"</i>	[VERIFIED]
Upper limit of the ITU definition	3,000 GHz — RR No. 1.5 uses the same wording, <i>"arbitrarily lower than 3 000 GHz"</i>	ITU seminar materials on itu.int ; the Radio Regulations text itself could not be extracted (the PDF fetched as binary)	[REPORTED]
Practical lower limit of allocated spectrum	about 8.3 to 9 kHz	Commonly cited as the first row of the ITU allocation table; that row was not fetched	[UNCERTAIN] — treat "about 9 kHz" as approximate

So "all possible radio frequencies" means roughly **9 kHz to 3,000 GHz**. As a ratio that is **1 : 333,000,000** — about **8.5 orders of magnitude**. **[ARITHMETIC]** Taking the more conventional 300 GHz top of the managed spectrum instead gives **1 : 33,000,000**, about **7.5 decades**. **[ARITHMETIC]** Either way, the number is not one a piece of electronics deals with.

The ITU band table, which is a named syllabus item. *"Nomenclature of the Frequency & Wavelength — Article 2"* appears verbatim in the Restricted-grade Section B syllabus, so this table is worth learning rather than merely reading. **[VERIFIED — Annexure II]**

Band number	Symbols	Frequency range (lower limit exclusive, upper inclusive)	Metric subdivision
4	VLF	3 to 30 kHz	Myriametric waves
5	LF	30 to 300 kHz	Kilometric waves

Band number	Symbols	Frequency range (lower limit exclusive, upper inclusive)	Metric subdivision
6	MF	300 to 3 000 kHz	Hectometric waves
7	HF	3 to 30 MHz	Decametric waves
8	VHF	30 to 300 MHz	Metric waves
9	UHF	300 to 3 000 MHz	Decimetric waves
10	SHF	3 to 30 GHz	Centimetric waves
11	EHF	30 to 300 GHz	Millimetric waves
12	—	300 to 3 000 GHz	Decimillimetric waves

[VERIFIED — ITU Radio Regulations 2024, Vol. 1, Article 2.1] Note 1 to the table gives the rule that generates it: *"Band N' (N = band number) extends from 0.3×10^N Hz to 3×10^N Hz."*

[VERIFIED]

Read that table's structure, because it teaches the right mental model. Each row is one decade. The spectrum is **logarithmic, not linear**. A radio that "covers 0 to 300 GHz" would have to work across seven and a half orders of magnitude, and nothing in electronics does that — not antennas, not amplifiers, not mixers, not filters, not converters.

So measure coverage in decades, not megahertz. Measured that way the picture changes completely, and it is much more flattering to the cheap end of the market:

Device	Coverage	Decades	Share of 9 kHz – 300 GHz	Price in India
Baofeng UV-32, receive	six discontinuous blocks, 65–520 MHz	~0.9, full of holes	~12 %, with gaps	₹14,449 [VERIFIED — Part 7 \$7.4]
RTL-SDR Blog V3 / V4	500 kHz – 1.766 GHz continuous	3.55	~47 %	₹4,802 – ₹5,712 [VERIFIED]
HackRF One	1 MHz – 6 GHz continuous, transmits too	3.78	~50 %	₹19,502 – ₹23,159 [VERIFIED]
SDRplay RSP1B	1 kHz – 2 GHz continuous	6.30	~84 %	₹18,007 [VERIFIED]
Keysight N9040B UXA signal analyser	2 Hz – 50 GHz	~6.7 above 9 kHz	~90 %	from USD 80,094.75 used ≈ ₹76.7 lakh [VERIFIED on the USD figure]
A hypothetical 9 kHz – 300 GHz radio	—	7.5	100 %	does not exist at any price

[VERIFIED] on every device's stated coverage and on the Keysight used price, read 2026-08-24; the decade arithmetic is **[ARITHMETIC]**; the rupee conversion uses USD 1 = ₹95.75, the rate reported for 2026-08-21 — **[REPORTED]**, and it moves.

The lesson, and it is the useful answer to the reader's question: a ₹5,712 dongle already covers about half the radio spectrum in the only unit that makes physical sense, and an ₹18,007 one covers about five sixths of it. Nothing at any price closes the last stretch, because the last stretch is where the physics stops cooperating.

9.2.2 REASON ONE: ANTENNAS, AND THE 666,000 : 1 PROBLEM

An antenna is not a wire that picks up radio. It is a **resonant structure**, and its size is set by the wavelength it is meant to work at. Wavelength is fixed by

$$\lambda = c / f \quad \text{where } c \approx 300,000,000 \text{ metres per second}$$

A quarter-wave vertical — the simplest useful antenna, and the "*common type of transmitting and receiving aerial*" the syllabus asks about at item **A(vii) Aerials** — is $\lambda/4$ long:

Frequency	Wavelength λ	Quarter wave ($\lambda/4$)	What that physically is
9 kHz	33.3 km	8.33 km	Taller than any structure ever built. Real VLF stations use kilometres of top-loaded wire and still radiate at single-digit efficiency
500 kHz	600 m	150 m	A broadcast tower
3.5 MHz (Indian 80 m band)	85.7 m	21.4 m	A long wire between two trees
7.1 MHz (Indian 40 m band)	42.2 m	10.6 m	A wire, or a tall vertical with radials
27.12 MHz (Citizen Band centre)	11.05 m	2.76 m	Remember this number — it decides \$9.6
145 MHz (Indian 2 m band)	2.07 m	51.7 cm	The whip on a hand-held
435 MHz (Indian 70 cm band)	69 cm	17.2 cm	A stubby whip
446 MHz (PMR446)	67.3 cm	16.8 cm	An integral antenna that genuinely works
2.4 GHz (a helmet intercom)	12.5 cm	3.1 cm	A trace on a circuit board
6 GHz	50 mm	12.5 mm	A stub
300 GHz	1.0 mm	0.25 mm	A horn or lens fabricated on the chip itself

[ARITHMETIC — $\lambda = c/f$] The Indian amateur band edges are **[VERIFIED]** from Annexure-I to the 2010 memorandum, transcribed in Part 3 §3.4.

The quarter-wave antenna for the bottom of the range is 666,000 times longer than the one for 6 GHz. [ARITHMETIC] No physical object is simultaneously 8 kilometres and 12 millimetres long. This is not an engineering limitation waiting to be overcome; it is what resonance means.

The practical consequence, and it is the single most common disappointment among people who buy a wideband SDR: a receiver that tunes 1 kHz to 2 GHz is not one radio. It is five or six radios sharing a tuning knob, and each of them needs a different antenna. A long wire for HF, a discone for VHF and UHF scanning, a loop for medium wave, a dish for microwave. **A device sold with a 15 cm telescopic whip will hear almost nothing at 500 kHz no matter what its specification sheet says.**

This is also why the RTL-SDR Blog dipole kit ships with **two pairs** of telescopic elements — 23 cm to 1 m and 5 cm to 13 cm — which you extend or collapse when you change band. **That kit is syllabus item A(vii) in a box**, and it is the reason to buy the kit rather than the bare dongle for ₹910 less.

Three further antenna-side consequences, each of which is also examinable:

- **Feedline loss scales with frequency.** The same coaxial cable that is nearly lossless at 145 MHz throws away most of your signal at 10 GHz. That is why systems above a few gigahertz put the electronics *at* the antenna rather than at the operator. It is also why, at 27 MHz, cable loss is so small that **you cannot rely on it to bring you under a power limit** — a point that becomes decisive at §9.6.3.
- **Propagation is different in every band.** 7 MHz bends around the world off the ionosphere at night and dies at midday; 145 MHz goes line of sight; 300 GHz is stopped by rain and by the air itself. A single "all frequencies" radio would still need a different antenna, a different mode and a different time of day for each band. That is syllabus item **A(vi) Radio Wave Propagation** — ground, sky and space waves, skip distance, skip zone and fading.
- **An antenna is only useful across a few per cent of its centre frequency.** Which is why "resonance in series and parallel circuits", syllabus item **A(ii)**, is on the paper at all.

9.2.3 REASON TWO: TUNING RANGE IS NOT INSTANTANEOUS BANDWIDTH

This is the distinction most beginners miss, and it is the one that finally kills the idea of hearing everything at once. Every SDR quotes two completely different numbers, and vendors are content for buyers to confuse them.

- **Tuning range** is where the local oscillator can be pointed. This is the big number in the advertising.
- **Instantaneous bandwidth**, set by the sample rate of the ADC, is how much spectrum the device can actually see and digitise **at one moment**. This is the number that decides what you can do.

Device	Tuning range	Instantaneous bandwidth	Bandwidth as % of range
HackRF One	5 999 MHz	20 MHz (20 Msps)	0.33 %
SDRplay RSPdx-R2	1 999.999 MHz	10 MHz	0.50 %
RTL-SDR Blog V4	1 765.5 MHz	~2.4 Msps stable (3.2 Msps maximum, drops samples)	0.14 %
Airspy R2	1 676 MHz	10 MHz	0.60 %
Airspy HF+ Discovery	30.9995 MHz	0.66 MHz alias-free	2.1 %
USRP B210	5 930 MHz	56 MHz	0.94 %
Baofeng UV-32 (for contrast)	six blocks, 65–520 MHz	one channel, 12.5 or 25 kHz	~0.004 %

[ARITHMETIC, from the specifications cited at §9.3]; the RTL-SDR sample-rate figure is **[VERIFIED — rtl-sdr.com "About RTL-SDR", fetched 2026-08-24];** the UV-32 channel bandwidth follows from its 16K0F3E and 11K0F3E emission designators **[VERIFIED — Part 7 §7.3.1]**.

To sweep 1 MHz to 6 GHz on a HackRF at 20 MHz per look takes about **300 sequential looks**. The radio is therefore blind to **99.67 %** of its advertised range at any given instant, and a signal that appears for two seconds somewhere in that 6 GHz will almost certainly be missed.

[ARITHMETIC]

When a specification says "this radio covers 6 GHz", the honest translation is: "this radio can be told to look at any 20 MHz slice inside a 6 GHz span, one slice at a time."

And here is the part that matters to a learner rather than a collector. A scanning radio like the UV-32 shows you one 25 kHz channel at a time and you hunt for signals by luck. An SDR shows you **2.4 MHz of spectrum as a picture** — a waterfall display in which time runs down the screen and frequency runs across it. That difference, being able to see the band rather than hunt through it, is the entire reason the SDR is the learning instrument and the hand-held is not. It is also why almost every item in syllabus section **A(iv) Radio Receivers** — sensitivity, selectivity, fidelity, adjacent channel and image interference, automatic volume control, squelch, signal-to-noise ratio — becomes something you can watch happening rather than something you memorise.

9.2.4 REASON THREE: CONVERTER AND FRONT-END PHYSICS

Three separate limits, each of which bites before you reach the top of the advertised range.

(a) Dynamic range is bounded by bit depth. An ADC's theoretical dynamic range is about **6.02 dB per bit**. So:

Converter	Theoretical dynamic range	Found in
8-bit, ENOB ≈ 7	~ 48 dB theoretical, less in practice	RTL-SDR, HackRF One [VERIFIED — rtl-sdr.com estimates ENOB ≈ 7]
12-bit, 10.4 ENOB	~ 72 dB	Airspy R2, USRP B210, ADALM-PLUTO [VERIFIED / REPORTED]
14-bit	~ 84 dB	SDRplay RSP1B / RSPdx-R2, Icom IC-R8600 [VERIFIED]
16-bit at 130 Msps	higher again	RX-888 MkII [REPORTED]
18-bit embedded digital down-converter	110 dB blocking dynamic range on HF	Airspy HF+ Discovery [VERIFIED — airspy.com]

In plain terms: the cheap wideband devices are deafened by strong local signals. Width and sensitivity trade against each other, always. That is not a defect in any particular product; it is what it costs to open the front door wide.

(b) A front end that must accept everything cannot be sharply filtered. A receiver open from 500 kHz to 1.766 GHz sees every broadcast FM transmitter and every cellular base station in the city at once, and those strong signals mix inside the receiver to produce signals that are not there. This is exactly the syllabus's *"adjacent channel and image interference"* and *"selectivity"* under item **A(iv)**. It is also why the RTL-SDR Blog V4 added a triplexed input filter and why the SDRplay RSP1B has built-in broadcast-FM and DAB notch filters. [VERIFIED — manufacturer descriptions on the Fab.to.Lab and Engineer Store listings, fetched 2026-08-24]

The clean demonstration of the trade: the **narrow** Airspy HF+ Discovery (0.5 kHz–31 MHz) comfortably outperforms the **wide** HackRF One (1 MHz–6 GHz) on shortwave, at similar money, because 110 dB of blocking dynamic range on a crowded band beats 6 GHz of tuning range you will never use. **Wider is not automatically better; wider without filtering is usually worse.**

(c) Nothing digitises millimetre waves directly. Above roughly 6 GHz, commercial SDRs and radios work by bolting on external down-converters and transverters — Icom's CX-10G, which takes the IC-905 to 10 GHz, is exactly this. [VERIFIED — Icom material] Each one is a separate box, a separate antenna and separate money, and above about 50 GHz you change hardware per waveguide band. **This is the reason the coverage ladder ends rather than tapering.**

9.2.5 REASON FOUR: TRANSMITTING WIDE IS A DIFFERENT PROBLEM FROM RECEIVING WIDE

Receiving badly is a personal disappointment. **Transmitting badly sprays energy across other people's spectrum.**

A transmitter that covers 1 MHz to 6 GHz cannot have a clean output filter at every frequency, so it emits harmonics and spurious products. HackRF's own documentation tells users to add *"an external bandpass filter for your operating frequency"* when amplifying. [REPORTED — greatscottgadgets/hackrf documentation, surfaced 2026-08-24]

Two Indian consequences follow directly:

- **Annexure A para 4(1) of the Amateur Services Rules 2024 requires the apparatus to be "tuned as accurately as possible to ensure that no energy is radiant on any frequency outside the limits of the authorised frequency bands". [VERIFIED — Part 7 §7.3.2]** A harmonic-rich wideband transmitter is in tension with that condition even when the fundamental is inside the band.
- **India's enforcement posture is aimed at wideband transmit hardware specifically.** The reported reason for DoT's action against the Baofeng BF-888S was *"operating beyond permissible frequencies"* — a description of the hardware, not of the operator's intentions. **[REPORTED — Part 2 §2.10, Part 6 §6.4.8]**

9.2.6 REASON FIVE, WHICH IS NOT PHYSICS: COST DOES NOT CLIMB SMOOTHLY

Coverage does not get expensive gradually. It gets expensive in steps, at each point where the physics changes and a new technology has to be bolted on.

Step	Roughly what it costs	Why the jump happens
24 MHz – 1.7 GHz	₹1,500–2,500 for a generic dongle	Mass-market silicon built for digital television
Add HF below 24 MHz	~₹5,000 (RTL-SDR Blog V3/V4, direct sampling or an internal 28.8 MHz upconverter)	Needs a different sampling scheme entirely
Add true VLF/LF from 1 kHz, 14-bit, real filtering, metal case	~₹18,000 (SDRplay RSP1B)	Better converter, real front-end filters
Add 6 GHz and transmit	₹19,500–23,200 (HackRF One) — but 8-bit only	Wideband mixers. You lose receive quality to gain range
Add 6–50 GHz with laboratory dynamic range	from ~USD 80,000 used ≈ ₹77 lakh (Keysight N9040B)	Microwave mixers, YIG preselectors, calibrated front ends
Add 50–300 GHz	External harmonic mixers and frequency extenders, each a separate instrument	No single front end works across millimetre wave

[VERIFIED] on the Keysight page ("*2 Hz to 50 GHz*", "*Starting at USD 80,094.75*", fetched 2026-08-24) and on the Indian retail prices at \$9.3; the characterisation of millimetre-wave extension is engineering background and **[UNCERTAIN]** in the sense that extenders were not priced.

Note the shape of that table, because it is the single most useful generalisation in radio buying. Past a certain point, paying more buys you *range* and costs you *quality*. The HackRF covers 6 GHz and transmits, and is a distinctly worse receiver than a ₹5,712 RTL-SDR on every band this reader will actually use. **Specialised beats general, at every price point.**

9.2.7 SO WHAT DOES "COVERAGE" MEAN? THREE NUMBERS, ALWAYS

For the rest of this part, every device is described with three figures, and a device that scores well on the first and badly on the others is a specification rather than a radio:

1. **Continuous tuning span** — lowest to highest, and whether there are gaps.
2. **Instantaneous bandwidth** — what it can see at once.
3. **Whether the coverage is *usable*** — bit depth, dynamic range, and whether one antenna can plausibly serve it.

9.3 The widest hardware that exists

Everything in this section was read from manufacturer specification pages and Indian retailer pages on **2026-08-24**, except where a later date is given. **Prices and stock change constantly; re-check before buying anything.**

9.3.1 RECEIVE-ONLY SOFTWARE-DEFINED RADIOS

Device	Continuous tuning span	Gaps?	Bandwidth	ADC	India price	Tag
SDRplay RSPdx-R2	1 kHz – 2 000 MHz (span 1 999.999 MHz)	None — gapless	up to 10 MHz	14-bit	₹30,607.37 (was ₹34,224.20), Fab.to.Lab	span/spec [VERIFIED via reseller listings and datasheet material; sdrplay.com itself returned HTTP 403 to every fetch]; price [VERIFIED — page fetched]
SDRplay RSPduo	1 kHz – 2 000 MHz, two tuners	None	10 MHz single-tuner, or 2 MHz per tuner simultaneously	14-bit	₹37,445.26, Fab.to.Lab	[VERIFIED]
SDRplay RSP1B	1 kHz – 2 000 MHz	None	up to 10 MHz	14-bit	₹18,007.37, Fab.to.Lab	[VERIFIED]
SDRplay nRSP-ST (networked)	1 kHz – 2 000 MHz	None	up to 10 MHz	14-bit	₹65,677.89, Fab.to.Lab	[VERIFIED]

Device	Continuous tuning span	Gaps?	Bandwidth	ADC	India price	Tag
SDRplay RSP1A	1 kHz – 2 000 MHz	None	up to 10 MHz	14-bit	discontinued, production ended 2023; ₹17,475.48 still listed at The Engineer Store	[REPORTED] on the discontinuation; price [VERIFIED]
RTL-SDR Blog V4	500 kHz – 1 766 MHz (span 1 765.5 MHz)	HF reached through a built-in 28.8 MHz upconverter; then the tuner above 24 MHz; effectively continuous	3.2 Msps maximum, ~2.4 Msps stable	8-bit, ENOB $\approx$ 7	₹5,336 dongle only, Fab.to.Lab; ₹5,750–₹7,450, RFCharge; ₹4,840 + 18 % GST , ElectroPi	coverage [VERIFIED — RTL-SDR Blog V4 Datasheet v1.0, December 2024]; prices [VERIFIED]
RTL-SDR Blog V3	500 kHz – 1 766 MHz	as V4	~2.4 Msps	8-bit	₹5,712 with the dipole antenna kit; ₹4,802.04 dongle only, Fab.to.Lab	[VERIFIED]
RX-888 MkII	1 kHz – 64 MHz direct sampling plus 64 MHz – 1 700 MHz via tuner	Two architectures in one box; the join at 64 MHz is a mode change, not a gap	64 MHz real-time bandwidth on HF — by far the widest instantaneous bandwidth at this price	16-bit at 130 Msps on HF; 8-bit path above	no Indian retailer found; about USD 200 from Chinese sellers	[REPORTED] — no manufacturer datasheet fetched
Airspy R2	24 – 1 700 MHz native; DC – 1 700 MHz with the SpyVerter upconverter	None within each range	10 Msps (80 with custom firmware)	12-bit, 10.4 ENOB	no Indian retailer found	[VERIFIED — airspy.com]
Airspy Mini	24 – 1 700 MHz native	None	10 / 6 / 3 Msps	12-bit	none found	[VERIFIED]

Device	Continuous tuning span	Gaps?	Bandwidth	ADC	India price	Tag
Airspy HF+ Discovery	0.5 kHz – 31 MHz and 60/64 – 260 MHz	Yes — a large hole from 31 to 60 MHz	768 ksps, 660 kHz alias-free	18-bit DDC; 110 dB blocking dynamic range on HF	none found	[VERIFIED — airspy.com]

How to read that table. The **RSPdx-R2** is the **sweet spot** among devices actually stocked in India: gapless from below the longwave broadcast band to 2 GHz, 14 bits so it survives a strong signal next to a weak one, and on an Indian shelf with a GST invoice. The **Airspy HF+ Discovery** is the **best listener's receiver on the page despite covering the least**, and if the reader's interest turns out to be shortwave rather than scanning, it is the one to want.

A source conflict worth disclosing, because it affects a purchase. One survey in this research concluded that **no functioning Indian SDRplay dealer could be found**, noting that the Indian page search engines still return for the RSP1A (radiojitter.com) now resolves to an unrelated gambling site — the domain has evidently lapsed. **[VERIFIED that the domain now resolves elsewhere]** A second survey, on the same date, **fetches live Fab.to.Lab product pages with the rupee prices in the table above. [VERIFIED]** The fetched listing is the better evidence and the table follows it, but **check stock by telephone before counting on it.**

9.3.2 TRANSMIT-CAPABLE SDRS, LISTED HERE BECAUSE THEIR RECEIVE SPANS ARE THE WIDEST NUMBERS ANYWHERE

Read §9.4 and §9.5 before reading this table, and read Part 2 §2.10 before buying anything in it. Every device here transmits, and in India transmitting requires an authorisation under s.4 of the Telecommunications Act 2023 exactly as much as a walkie-talkie does. **[VERIFIED — Part 2 §2.2]**

Device	Tuning span, receive and transmit	Gaps?	Duplex	Transmit power	Bandwidth	India price	Tag
Ettus USRP X410	1 MHz – 7.2 GHz (specified tunable to 8 GHz), span 7 199 MHz	none claimed	full, 4 TX + 4 RX	not stated on the pages read	400 MHz per channel	not retailed in India at consumer level; quotation only	[REPORTED — radiojitter.com]

Device	Tuning span, receive and transmit	Gaps?	Duplex	Transmit power	Bandwidth	India price	Tag
HackRF One	1 MHz – 6 GHz , span 5 999 MHz	None	half-duplex	5 to 15 dBm — about 3 to 32 milliwatts	2–20 Msps	₹19,502.22 Fab.to.Lab; ₹23,158.99 The Engineer Store; ₹19,500 / ₹24,500 for PortaPack clones at RFCharge; ₹22,000 + 18 % GST at compoindia	all specs greatsc official [VERIFI figure is search s
bladeRF 2.0 micro	47 MHz – 6 GHz, span 5 953 MHz	None	full, 2×2	~8 dBm claimed by resellers [UNCERTAIN]	up to 61.44 MHz	none found	[REPOR
USRP B210	70 MHz – 6 GHz, span 5 930 MHz	None	full, 2×2	~10 dBm class [UNCERTAIN]	56 MHz	₹279,271.50 (The Engineer Store) / ₹430,000 (IndiaMART)	[REPOR snippets
ADALM-PLUTO , stock	325 MHz – 3 800 MHz, span 3 475 MHz	None	full	~7 dBm class [UNCERTAIN]	20 MHz	₹32,568 incl. GST (DNA Technology) / ₹39,990 (IndiaMART)	spec [VE wiki.and [REPOR
ADALM-PLUTO, reconfigured	70 MHz – 6 000 MHz , span 5 930 MHz	None	full, but 1 TX / 1 RX in the wide mode	as above	56 MHz	as above	[VERIFI Devices the reco [UNCER at the ed it
LimeSDR USB	100 kHz – 3.8 GHz, span 3 799.9 MHz	None	full, 2T2R	~10 dBm class [UNCERTAIN]	up to 61.44 MHz	none found	[REPOR
LimeSDR Mini 2.0	10 MHz – 3.5 GHz	None	full	—	up to 40 MHz	₹64,446.99, The Engineer Store	[VERIFI

9.3.3 THE HACKRF CAVEATS, STATED PLAINLY BECAUSE IT IS THE ANSWER PEOPLE FIND FIRST

Search "widest coverage SDR" and the HackRF One is what comes back within ten minutes. It is genuinely the widest device a hobbyist realistically buys, on both receive and transmit. **It is also the wrong purchase, and for five reasons that have nothing to do with Indian law.**

1. **It transmits milliwatts.** Output is **5 to 15 dBm** — about **3 to 32 mW** — and it varies enormously across the range: rising 5 → 15 dBm between 1 and 10 MHz, falling 15 → 5 dBm from 10 MHz to 2.17 GHz, best at 13–15 dBm between 2.17 and 2.74 GHz, 0–5 dBm from 2.74 to 4 GHz, and **–10 to 0 dBm from 4 to 6 GHz**. **[VERIFIED — Great Scott Gadgets' own documentation]** For scale, India's licence-exempt PMR446 allowance is **500 mW e.r.p.**, which is **15 to 160 times more power**. **[VERIFIED — G.S.R. 1047(E) Table V, Part 2 §2.5.1; ARITHMETIC on the ratio]**
2. **It is half-duplex.** It cannot receive while transmitting. That is fine for a bench instrument and disqualifying for a radio you hold a conversation on.
3. **It is harmonic-rich.** Its own documentation tells you to add an external bandpass filter for your operating frequency when amplifying. **[REPORTED]** §9.2.5.
4. **It receives with 8 bits.** About 48 dB of theoretical dynamic range, less in practice. A ₹5,336 dongle is no better; but a ₹18,007 RSP1B is 36 dB better; and on a crowded band that is the difference between hearing something and hearing mush.
5. **Most units sold in India are not Great Scott Gadgets hardware.** Fab.to.Lab describes its ₹19,502 unit as the "*Clifford Heath Version HackRF One*" — a third-party variant — and RFCharge sells "*HackRF R10 PortaPack H2M / H4M*" at ₹19,500 and ₹24,500, which are Chinese clone boards with a screen bolted on. **[VERIFIED — retailer pages fetched 2026-08-24]** **Clone RF performance is not characterised by anybody.**

The honest summary: a HackRF One is a signal generator with a receiver attached. It is a laboratory instrument, not a radio you talk on, and its transmit function exists so that engineers can test things across a bench.

And in India it is the worst object on this page to be found holding. Part 7 §7.3.1 established that a UV-32 exposes you across roughly 198 MHz of transmit capability of which 6 MHz is lawful for a Restricted-grade amateur — about 3 %. A HackRF exposes you across **5 999 MHz**, of which the same 6 MHz is lawful — about **0.1 %**. **[ARITHMETIC on the percentages; the 6 MHz and the emission list are VERIFIED from Annexure-I, Part 3 §3.4]**

	Baofeng UV-32	Quansheng UV-K5 with modified firmware	HackRF One
Continuous transmit span	≈198 MHz	1 282 MHz claimed (18–1 300 MHz)	5 999 MHz
Of which lawful for an Indian Restricted-grade amateur	6 MHz	6 MHz	6 MHz

	Baofeng UV-32	Quansheng UV-K5 with modified firmware	HackRF One
Share of transmit range that is lawful	≈3 %	≈0.5 %	≈0.1 %
Authorised emissions on Indian VHF/UHF	F1B, F2B, F3E, F3C only — FM and digital; no single sideband, no AM	same	same, and an SDR can generate anything at all
Front-panel band lock	none	none, deliberately removed	not applicable

There are three further traps specific to a wideband SDR transmitter that do not arise with a walkie-talkie, and the third is the least appreciated:

- **Import.** Rule 7(3) of the Amateur Services Rules 2024 says *"An ASOC holder shall not require a separate authorisation for the import of amateur radio equipment"*. **[VERIFIED — Part 2 §2.8.3]** Part 7 §7.5.1 already flagged that whether a 198 MHz-wide consumer hand-held is "amateur radio equipment" is exactly what a customs officer may ask. **A 1 MHz–6 GHz laboratory transmitter is a far harder case to argue, and it is not obviously amateur equipment on any reading.**
- **Possession.** REPA rule 10(f) exempts a person who *"holds an Amateur Station Operator Certificate"*. **[VERIFIED]** That is a status test about the person, not about the equipment, so on the plain text an ASOC holder's possession of a HackRF is exempt. **[UNCERTAIN as a matter of DoT practice — no source addresses it, and Part 1 §1.5 Group B carries a further hedge on rule 10(f) itself.]**
- **Emission type.** An ASOC on VHF/UHF authorises **F1B, F2B, F3E and F3C only**. **[VERIFIED — Part 3 §3.4]** A software-defined transmitter emitting an arbitrary waveform is outside the authorised emission set **even when the frequency is inside 144–146 MHz. The licence constrains what you transmit, not only where.** **[UNCERTAIN]** — no source addresses SDRs, and this is a good question to put to WPC in writing (§9.8.5).

The one clean route for an engineer who does want to radiate from an SDR lawfully is not a purchase but a licence category. WPC Office Memorandum No. **R-14016/01/2019-NT(Pt.) dated 2019-07-23** creates **Experimental / Technology Trial / Manufacturing and Testing / Demonstration** licences with import permission built in, at a **₹5,000** licence fee, whose terms expressly cover *"importing wireless products / assemblies / modules and associated accessories ... operating in both licensed and license-exempt bands"*, provided *"such imported products shall not be for commercial use / sale"*. **[VERIFIED — Part 2 §2.8.2]** Realistically this is an **organisational** route rather than an individual one, but it is the correct door to knock on and it is worth knowing it exists. Note also that WPC's compendium reproduces an **Experimental Wireless Telegraphy Station licence (Non-Radiating)** category — a station operating into a dummy load or under shielding, with an application form asking for *"Particulars of dummy load (non-*

Radiating)". **[VERIFIED]** It is **not** a receive licence and should not be presented as one; it is noted only because it is the closest structural analogue in Indian law to a "monitoring station" authorisation, and a WPC officer answering §9.8.5 question 1 may reach for it. **[UNCERTAIN]**

9.3.4 TRADITIONAL STANDALONE WIDEBAND RECEIVERS — AND WHY THE CATEGORY IS NEARLY DEAD

These need no computer. That is their whole advantage, and it is no longer enough.

Device	Continuous coverage	Gaps?	Modes and architecture	Price	Tag
AOR AR5700D	9 kHz – 3.7 GHz, span 3 699.99 MHz — the widest standalone receiver ever sold	none claimed	analogue plus TETRA, NXDN, DCR, D-STAR, P25; IQ output to a computer	USD 4,475.95 sale / 5,035.95 list — listed as discontinued	coverage [VERIFIED — aorja.com] ; price [REPORTED]
Icom IC-R8600	EXP version: 10 kHz – 3 000 MHz, span 2 999.99 MHz, guaranteed 100 kHz–3 GHz. USA version has two blocked cellular gaps	EXP none	USB, LSB, CW, FSK, AM, FM, WFM plus D-STAR, P25 Phase 1, NXDN, dPMR, DCR. 14-bit ADC at 122.88 MHz, IQ output over USB	USD 2,799.95 at GigaParts; roughly ₹2.4–2.6 lakh landed before duty [UNCERTAIN]	coverage and architecture [VERIFIED — icomjapan.com] ; price [REPORTED]
AOR AR5001D	40 kHz – 3.15 GHz, span 3 149.96 MHz	none claimed	14-bit / 65 Msps direct sampling below 25 MHz; P25 optional	USD 3,370.95, discontinued	coverage [VERIFIED] ; rest [REPORTED]
Icom IC-R30 (hand-held)	0.1 – 3 304.999 MHz, span 3 304.9 MHz	version-dependent cellular blocks	all-mode plus P25, NXDN, dPMR, D-STAR, DCR; dual watch, dual recording, GPS	discontinued	[REPORTED]

Device	Continuous coverage	Gaps?	Modes and architecture	Price	Tag
AOR AR-DV1	100 kHz – 1 300 MHz	none claimed	digital voice receiver; TETRA with optional group filtering	discontinued	[VERIFIED for the coverage figure only]
Uniden SDS100 (scanner)	25–512, 758–824, 849–869, 894–960, 1240–1300 MHz	Yes, four large gaps. Actual coverage ≈ 699 MHz	P25 Phase 1/2, X2-TDMA, Motorola, EDACS, LTR, analogue; true IQ front end	USD 699.99	coverage [REPORTED — RadioReference wiki; uniden.com publishes no frequency table] ; price [VERIFIED] ; the gap arithmetic is [ARITHMETIC]

The point of that table is the price and status column. Every genuinely wide standalone receiver is either **discontinued** (AR5700D, AR5001D, IC-R30, AR-DV1) or **costs more than a motorcycle** (IC-R8600). **The market for standalone wideband receivers has been eaten by SDRs.** A ₹30,607 RSPdx-R2 plus a laptop beats a USD 3,370 AR5001D on almost every measure except "works without a computer".

9.3.5 PORTABLE RECEIVE-ONLY SETS — THE CLOSEST THING TO WHAT MOST PEOPLE IMAGINE

Device	Coverage	Gaps?	Modes	India price	Tag
Malahit / Malachite DSP2	10 kHz – 380 MHz and 404 MHz – 2 GHz; about 1 976 MHz of actual coverage	Yes — 380 to 404 MHz is missing	AM, SSB, DSB, CW, NFM, WFM, RDS; 3.5-inch touchscreen; battery; fully standalone	₹37,500 incl. GST, RFCharge — sold out on 2026-08-24	[VERIFIED that RFCharge states 10 kHz–2 GHz with a 380–404 MHz gap]
Malahit DSP3	50 kHz – 380 MHz and 404 MHz – 2 GHz	same gap	as above, plus USB-C	not found in India	[REPORTED — malahiteam.com]
Tecsun PL-990x	LW 100–519 kHz, MW 522–1620 kHz, SW 1621–29 999 kHz, FM 64–108 MHz	Yes — HF, then a jump to broadcast FM. No air band	AM, FM, SSB, Bluetooth, MP3	₹44,997 (was ₹55,120), The Engineer Store	Indian listing states only <i>"100 kHz to 30 MHz (LW, MW, SW, FM bands)"</i> [VERIFIED] ; wider figures [REPORTED] ; price [VERIFIED]

Device	Coverage	Gaps?	Modes	India price	Tag
XHDATA D-808	FM / MW / LW / SW plus AM air band 118–137 MHz , SSB	Yes, several	AM, FM, SSB, RDS	₹8,090 at an IndiaMART seller	[REPORTED] — amazon.in returned HTTP 500 so no marketplace price is first-hand

The Malahit is the closest thing that exists to the device the reader imagined: one battery-powered handheld box, longwave to 2 GHz, touchscreen, no computer. It is also ₹37,500, has a 24 MHz hole in the middle of the UHF band, was out of stock, and — like everything else on this page — will hear almost nothing at 10 kHz through the whip it ships with.

9.3.6 THE LEADERBOARD, AND THE TWO HEADLINE NUMBERS

Widest continuous receive tuning span:

Rank	Device	Span	Figures
1	Ettus USRP X410	7 199 MHz	1 MHz – 7.2 GHz [REPORTED]
2	HackRF One	5 999 MHz	1 MHz – 6 GHz [VERIFIED]
3	bladeRF 2.0 micro	5 953 MHz	47 MHz – 6 GHz [REPORTED]
4=	USRP B210 / PlutoSDR reconfigured	5 930 MHz	70 MHz – 6 GHz [REPORTED / VERIFIED]
5	LimeSDR USB	3 799.9 MHz	100 kHz – 3.8 GHz [REPORTED]
6	AOR AR5700D	3 699.99 MHz	9 kHz – 3.7 GHz — the widest standalone receiver ever sold [VERIFIED]
7	Icom IC-R30	3 304.9 MHz	0.1 – 3 305 MHz [REPORTED]
8	AOR AR5001D	3 149.96 MHz	40 kHz – 3.15 GHz [VERIFIED]
9	Icom IC-R8600 (EXP)	2 999.99 MHz	10 kHz – 3 GHz — the widest good receiver still in production [VERIFIED]
10	SDRplay RSPdx-R2 / RSPduo / RSP1B	1 999.999 MHz	1 kHz – 2 GHz, gapless, 14-bit [VERIFIED]
11	Malahit DSP2	1 976 MHz of actual coverage	10 kHz – 2 GHz minus the 380–404 MHz gap [VERIFIED]
12	RTL-SDR Blog V3 / V4	1 765.5 MHz	500 kHz – 1.766 GHz [VERIFIED]

Widest continuous transmit tuning span: the same list, headed by the X410 at 7 199 MHz and the **HackRF One at 5 999 MHz, half-duplex, at 3 to 32 milliwatts**. For contrast, the widest transmit *reach* in a purpose-built radio is the **Icom IC-905** — 144 MHz to 10 GHz across five bands plus the CX-10G transverter, **₹350,000 at RFCharge and sold out on 2026-08-24 [price VERIFIED; band set REPORTED]** — and even that is five discrete bands, not a continuous span.

The two headline numbers, stated plainly:

Widest receive: HackRF One — 1 MHz to 6 GHz, one continuous span of 5 999 MHz, 8-bit, 20 MHz visible at a time, about ₹19,500 to ₹23,200 in India.

Widest transmit: HackRF One — the same 1 MHz to 6 GHz, half-duplex, at 3 to 32 milliwatts.

Above that and out of hobbyist reach: **Ettus USRP X410, 1 MHz to 7.2 GHz, four channels, 400 MHz per channel** — a research instrument sold on quotation.

And for context on the premise: the radio spectrum runs to 3,000 GHz. The X410 covers 0.24 % of it. There is no device, at any price, that covers all radio frequencies.

9.3.7 A NOTE ON THE QUANSHENG UV-K5 FIRMWARE MODIFICATIONS, BECAUSE THE READER WILL FIND THEM

The Quansheng UV-K5 (₹4,250 at RFCharge for the UV-K5(8) **[VERIFIED]**) has an unusually open architecture and a large open-source firmware ecosystem around it. In stock form it transmits **136–173.9975 and 400–469.9975 MHz** — about 108 MHz in two blocks, of which **6 MHz** is lawful for an Indian amateur — and receives 50–600 MHz in seven segments plus broadcast FM. **[REPORTED — manufacturer specification page and dealer listings]**

The popular community firmware builds do four things, in ascending order of how much they matter:

1. **Widen the receive range** from 50–600 MHz to roughly **18–850 MHz**, or **18–1300 MHz** with an extended build option, and widen AM reception correspondingly. **[REPORTED — the egzumer [uv-k5-firmware-custom](#) README on GitHub]**
2. **Add a spectrum-analyser display**, fast scanning and on-radio channel naming — genuinely useful learning features.
3. **Fix AM reception**, which is broken in stock firmware.
4. **Remove the transmit band restrictions**. The firmware documentation itself describes the radio as usable "*as an all-range receiver and transmitter in the frequency range of 18...1300 MHz*". **[REPORTED — same source]**

What the modifications do not do is change the hardware, and the firmware authors say so. A community write-up of the modified firmware states: "*the hardware is not designed for these frequency range extensions and the reception and transmission powers decrease dramatically and, in the worst case, can destroy the device. In addition, the antenna used is not suitable for such frequency ranges.*" **[REPORTED — [open-boat-projects.org](#), surfaced**

2026-08-24] There is no power amplifier, no filter and no antenna behind the number 1300. The radio will *display* 1250.000 and key up, and almost all of that energy goes into heat and harmonics.

So does a modified UV-K5 "cover 18–1300 MHz on transmit"? As a specification sheet, yes. As a radio, no. As a legal exposure in India, absolutely yes. A front-panel-unlocked 18–1300 MHz transmitter is a strictly worse object to be found holding at a checkpoint than the UV-32 that Part 7 already recommended against — and Part 7's reasoning applies with about six times the force.

9.3.8 THE HONEST GAP: NOBODY IN INDIA STOCKS A DEDICATED WIDEBAND RECEIVER

This has to be said clearly, because it changes what the reader can actually do.

No Indian retailer was found stocking any dedicated standalone wideband communications receiver or scanner. Specifically: no Icom IC-R8600, no IC-R30, no AOR receiver of any model, no Airspy of any model, and no Uniden scanner. RFCharge — the most complete Indian amateur and SDR dealer located in this research, listing 21 Yaesu and 13 Icom models in stock — **returned zero results for Tecsun and stocks no dedicated receive-only scanner at all. [VERIFIED that the searches returned nothing; UNCERTAIN as a negative, because a dealer's website is not their whole inventory]**

What is stocked in India, and it is a narrower list than the leaderboard suggests:

Available in India	Not found in India
RTL-SDR Blog V3 and V4 dongles (multiple retailers)	Icom IC-R8600, IC-R30
SDRplay RSP1B, RSPdx-R2, RSPduo, nRSP-ST (Fab.to.Lab); RSPdx and RSP1A (The Engineer Store)	Any AOR receiver
HackRF One and clone PortaPack variants — a transmitter, see §9.3.3	Any Airspy model, including the HF+ Discovery
Malahit DSP2 (RFCharge, sold out 2026-08-24)	Malahit DSP3
Tecsun PL-990x (The Engineer Store); XHDATA D-808 (IndiaMART)	Any Uniden scanner; RX-888 MkII

[VERIFIED] for every left-column entry from a fetched product page on 2026-08-24; **[UNCERTAIN]** for the right column, which is a negative from searching.

Two consequences, and they pull in the same direction.

First, if the reader wants a high-end receiver, the practical move is to **ask an Indian dealer to special-order one** rather than to import it personally — because a domestic import sidesteps the whole 1953 Form II question at §9.5.7 and puts the import gate on the seller. That is the single largest risk reduction available in this part.

Second, and more usefully: **the devices that are actually stocked in India are the ones this part recommends anyway.** The RTL-SDR at ₹5,336 and the RSP1B at ₹18,007 are both on Indian shelves with GST invoices, and between them they cover 47 % and 84 % of the radio spectrum measured in decades. **The gap in the Indian market is at the top of the price list, where the reader should not be shopping.**

One further note on model choice, because it saves money for nothing: **buy the RTL-SDR Blog V3, not the V4.** RTL-SDR Blog have announced that the V4 is end of line — *"no more Blog V4 productions will be possible"*, the Rafael R828D tuner stockpile being exhausted — and that *"Going forward, the Blog V3 will remain in stable production."* A limited-edition V4L using the R828S is the interim successor. **[VERIFIED — rtl-sdr.com, "RTL-SDR Blog V4 End Of Line"; the post carries no date on its face and the comments suggest roughly three months old as at 2026-08-24, so the date is UNCERTAIN]** Buying a discontinued model to save nothing is a poor trade, and the V3 with the dipole kit at ₹5,712 is a better package than the V4 dongle alone at ₹5,336.

A documented internal inconsistency, disclosed because it is instructive. The official RTL-SDR Blog V4 datasheet names the tuner chip as **R820T2**, while rtl-sdr.com's own "About RTL-SDR" page describes the V4 as **R828D**-based. Both are the manufacturer's own documents. **[UNCERTAIN]** It does not affect the coverage figure, but it is a reminder that even primary sources on this hardware contradict each other — exactly as Part 7 §7.3.1 found for Baofeng's own paperwork. **Get specifications in writing from the seller, for the exact unit they will ship.**

9.4 The legal asymmetry, and why it exists

9.4.1 THE REASON, IN ONE SENTENCE

A receiver cannot interfere with anybody.

That is the whole of it. Radio regulation exists because spectrum is a shared resource that one user's emissions can deny to another. A transmitter can block an air traffic control channel, a coastguard net or a paramilitary link. A receiver observes and emits nothing, so there is no interference to prevent, no coordination to perform and no assignment to make. It follows that the regulatory machinery built to prevent interference has almost nothing to say about listening — and that where a State does regulate listening, it is doing so for a *different* reason, which is secrecy and content control rather than interference. **§9.5.2 shows that this is exactly the reason India has historically regulated reception, and it is why the analysis there is less comfortable than the engineering logic would suggest.**

9.4.2 THE ASYMMETRY VISIBLE IN THE SOURCES

Three independent sources, from three different kinds of author, all say the same thing.

(a) The equipment vendor. Baofeng's own United Kingdom arm states the position in one sentence, on the page from which it sells wideband hand-helds:

"To transmit, you require a valid UK Amateur Radio (Ham) Licence or Business Radio Licence. ...
Reception (listening) is permitted without a licence."

[VERIFIED — baofengradio.co.uk; see Part 7 §7.3.4 for the same page's warning that transmission on PMR446 frequencies is not permitted with the device]

(b) The equipment's own manual, in the middle of a warning against illegal transmission.

This is the more interesting of the two, because it is a concession made against the manufacturer's commercial interest, printed immediately after the manual explains how to key up on an arbitrary frequency from the numeric keypad:

"WARNING! Just because you can program in a channel does not mean you're automatically authorized to use that frequency. Transmitting on frequencies you're not authorized to operate on is illegal, and in most jurisdictions a serious offence. **However, it is legal in most jurisdictions to listen.** Contact your local regulatory body for further information on what laws, rules and regulations apply to your area."

[VERIFIED — Baofeng UV-32 user manual §5.5, p.10; quoted at Part 7 §7.3.2 for the separate point that the radio enforces no band limits]

Note what that sentence is doing. In the same breath in which Baofeng tells the buyer that transmitting out of band is "*a serious offence*", it tells them that **listening is not**. Two capabilities, one box, opposite legal treatment — and the manufacturer draws the line in the same place the regulators do.

(c) India's own amateur rules, which forbid *divulging* rather than *receiving*. This is the most authoritative of the three for present purposes, because it is Indian subordinate legislation rather than vendor copy. Annexure A, paragraph 8 of the Telecommunications (Amateur Services) Rules, 2024, **verbatim and in full**:

"8. Secrecy of correspondence. - If any message which a ASOC holder is not entitled to receive, is received, such ASOC holder shall not make known or allow to be made known its contents, its origin or destination, its existence or the fact of its receipt to any person (other than to an officer duly authorised for this purpose by the Central Government or a court of law) and such ASOC holder **shall not reproduce in writing, copy or make any use of such message or allow the same to be reproduced in writing, copied or made use of.**"

[VERIFIED — G.S.R. 675(E), Annexure A, para 8, gazette text]

Read the structure of that clause carefully, because it is the clearest statement of the asymmetry anywhere in Indian law. It opens by *assuming* that an amateur will receive things they were not entitled to receive — "*If any message which a ASOC holder is not entitled to receive, is received*" — and it does not prohibit that at all. What it prohibits is everything you might do afterwards: disclosing the contents, disclosing the origin or destination, disclosing the existence or the fact of receipt, and reproducing, copying or making any use of it. **Four prohibitions on downstream conduct and none on the reception itself.**

That clause is not new drafting. It is materially identical to paragraph VIII of Annexure V to the 1978 Rules, which WPC's own compendium reproduces — so this obligation has been continuous in Indian amateur licensing for at least **forty-eight years**. [VERIFIED — WPC compendium text]

(d) One more place the concept appears, and it is the most tantalising. WPC's own import and licence application forms carry a declaration in these terms:

"I/We hereby declare that ... **(iv) I/We will not, without authorisation, divulge to any person the purport of message, other than a message intended for the general use of the public, which I/We may transmit or receive by wireless equipment operated by me or which may come to my knowledge with the operation of the said equipment.**"

[VERIFIED — WPC Compendium of Orders related to import licence, File No. R-11018/02/2017-PP dated 2022-07-06, Annexure form]

That phrase — "a message intended for the general use of the public" — is the single most important five words in this part, and its location is the problem. It is the concept the United Kingdom put into statute and India did not (§9.5.3). It shows that WPC understands perfectly well the difference between listening to a broadcast and eavesdropping on a private link. **But the distinction lives in a declaration on an application form, not in the Act.** That is a point worth making to WPC in writing when asking the questions at §9.8.5.

9.4.3 WHERE THE ASYMMETRY DOES NOT HELP, STATED BEFORE ANYONE RELIES ON IT

The asymmetry is real and it is the organising principle of this part. It is not a blanket permission, and three specific limits must be understood before §9.5:

1. **It is about interference, and India's receiver rules are about content.** The interference logic explains why no *operating* licence is needed to receive. It says nothing about **possession**, which is governed by a 1933 statute whose entire original purpose was to prosecute people who owned receivers without paying a licence fee. §9.5.1.
2. **"Radio equipment" in Indian law is capability-based, and a receiver is caught by the words.** Section 2(l) of the 2023 Act defines it as "*telecommunication equipment used or capable of use for telecommunication by means of Hertzian or radio waves*", and s.2(p) defines "telecommunication" to include **reception**. [VERIFIED] There is no interference-based exception written anywhere into that chain. §9.5.1.
3. **The asymmetry protects the act of hearing, not the acts that follow it.** Annexure A para 8 is the model: reception is assumed, disclosure and reproduction are forbidden. **The Official Secrets Act 1923 takes the same shape and adds fourteen years.** §9.5.4.

So the correct statement of the principle, for India specifically, is narrower than the international slogan: *receiving is not an interference offence and there is no receive-side operating licence left to obtain; but possession of the receiver still needs an exemption, and what you do with what you hear is where the serious criminal exposure lives.*

9.5 Receive legality in India, question by question

This section answers the four questions of §9.1.2 in order, then deals with import, then with what an ASOC changes. **It is the most heavily hedged section in this report and the hedges are load bearing.**

9.5.1 QUESTION 1 — OWNING A RECEIVER

The statutory position: caught by both Acts, on their face

There is no receiver exclusion in either Indian wireless statute, and both of them affirmatively include reception in their core definitions.

The Indian Wireless Telegraphy Act, 1933. Its long title is *"An act to regulate the possession of wireless telegraphy apparatus"*, and its Statement of Objects and Reasons is unusually candid about why it exists: **[VERIFIED]**

"An important source of revenue to the Indian State Broadcasting Service is the fees on licenses for wireless apparatus. These licenses are issued under the Indian Telegraph Act which, however, only gives power to control establishment, maintenance and working of such apparatus in British India. The detection of unlicensed apparatus and the successful prosecution of offenders is therefore difficult in practice... It is now proposed to remedy this state of affairs by legislation to prohibit the possession without license of wireless apparatus as distinct from the establishment, maintenance and working of such apparatus."

The 1933 Act was created to prosecute people who owned radio receivers without paying the licence fee. That is its entire original purpose. Any reading of the Act that assumes receivers fall outside it is reading against the Act's own stated object.

The operative definitions and provisions, all **[VERIFIED]** from the bare text:

Provision	Text, abridged where marked	Why it matters here
s.2(1) "wireless communication"	<i>"any transmission, emission or reception of signs, signals, writing, images and sounds, or intelligence of any nature by means of electricity, magnetism, or Radio waves or Hertzian waves..."</i>	Reception is in the definition. It always was
s.2(2) "wireless telegraphy apparatus"	<i>"...instrument or material used or capable of use in wireless communication... but does not include any such apparatus... commonly used for other electrical purposes, unless it has been specially designed or adapted for wireless communication..."</i>	Only two exclusions, and a receiver is in neither . A communications receiver is not "commonly used for other electrical purposes"

Provision	Text, abridged where marked	Why it matters here
s.2(2A) "wireless transmitter" (inserted 1949)	<i>"any apparatus... used or capable of use for transmission or emission of wireless communication"</i>	The key structural point. Parliament created a <i>narrower sub-category</i> for transmitters inside the wider category. That drafting only makes sense if the wider category includes non-transmitters
s.3	<i>"Save as provided by section 4, no person shall possess wireless telegraphy apparatus except under and in accordance with a license issued under this Act."</i>	The prohibition
s.4	<i>"The Central Government may by rules made under this Act exempt any person or any class of persons..."</i>	The Act itself exempts nothing. Every exemption is a rule. This is why the whole question turns on rules rather than on the statute
s.6(1)	fine up to ₹100 first offence, ₹250 subsequent, for possessing apparatus <i>"other than a wireless transmitter"</i>	This is, in terms, the receive-only possession offence. It exists precisely so that a person holding a receiver without a licence can be fined
s.6(1A)	up to 3 years for possessing a wireless transmitter	The asymmetry, written into the penalty scale itself
s.6(2)	a court <i>"may presume"</i> possession where apparatus <i>"is under his ostensible charge, or is located in any premises or place over which he has effective control"</i>	The prosecution need not prove ownership, only effective control of the premises
s.6(3)	on conviction the court decides whether apparatus <i>"should be confiscated"</i>	₹100 is derisory in 2026 money. Confiscation is not
s.7	an empowered officer <i>"may search any building, vessel or place... and take possession thereof"</i>	No warrant is mentioned
s.10(2)(i)	power to make rules <i>"determining that any article or class of article shall be or shall not be wireless telegraphy apparatus"</i>	No rule declaring receivers not to be wireless telegraphy apparatus was found. That is the missing instrument that would have settled this cleanly, and it does not appear to exist

The Telecommunications Act, 2023. The chain is even shorter, and every link is express statutory text: **[VERIFIED]**

"(l) 'radio equipment' means telecommunication equipment used or capable of use for telecommunication by means of Hertzian or radio waves; (p) 'telecommunication' means transmission, emission or **reception** of any messages...; (q) 'telecommunication equipment' means any equipment, appliance, instrument, device, radio station, radio equipment, material, apparatus, or user equipment, that may be or is being used for telecommunication... **and excludes such equipment as may be notified by the Central Government;**"

Chain it: a scanner is *equipment* → *used for telecommunication*, because "telecommunication" includes **reception** → therefore "telecommunication equipment" → and it does so *by means of radio waves* → therefore "**radio equipment**". **There is no gap to argue through.**

Then **s.3(1): [VERIFIED]**

"Any person intending to— (a) provide telecommunication services; (b) establish, operate, maintain or expand telecommunication network; or **(c) possess radio equipment**, shall obtain an authorisation from the Central Government..."

Two further observations, both of which matter:

- **s.2(q) contains an exclusion power that has not been used for receivers.** "*excludes such equipment as may be notified by the Central Government*" — and **no notification excluding receivers, receive-only devices, SDRs or scanners from "telecommunication equipment" was found.** [VERIFIED as a search-limited negative] That is the second missing instrument.
- **s.4(2) — the spectrum-assignment limb — almost certainly does not bite on a receiver.** "*Any person intending to use spectrum shall require an assignment from the Central Government.*" A receiver does not use spectrum in any ordinary sense; it observes it. **That is a meaningful difference from a transmitter, and it is the statutory home of the §9.4 asymmetry.** But it is a reading, not WPC's words. [UNCERTAIN]

Commencement, because it decides whether any of this is live. Section 3 was **not** in either 2024 commencement tranche — S.O. 2408(E) of 2024-06-26 or S.O. 2623(E) of 2024-07-05.

[VERIFIED — the commencement footnotes to s.1] Sections 3(1) and 3(6) were notified into force on 2026-06-23. [REPORTED — a law-firm regulatory update; no S.O. number was given] So as at 2026-08-25 the possession-authorisation obligation in s.3(1)(c) is live, and has been for about two months. Part 1 §1.9 carries the same timeline.

India did licence receivers, for about fifty years

This history is not decoration. It is the only way to see what the surviving exemption actually covers.

WPC publishes its own consolidated list of subordinate legislation, and item 05 in its table of contents is "*Receiver exemption from license- related*". The whole of India's receiver law is **four instruments: [VERIFIED — the compendium's own index page, verbatim]**

Sl.	Rules	Principal Act	Notification
01	Licensing of Wireless Receiving Apparatus Rules, 1965	Indian Telegraph Act 1885	G.S.R. 1319 dated 1965-08-28 , in force 1965-11-01
02	Commercial Broadcast Receiver Licensing (Dealers) Rules, 1965	Indian Telegraph Act 1885	G.S.R. 1320 dated 1965-11-01
03	Radio, Television and Video Cassette Recorder Sets (Exemption from Licensing Requirement) Rules, 1997	Both the 1885 and 1933 Acts	G.S.R. 393(E) dated 1997-07-16
04	The same, Amendment Rules, 2001	Both	G.S.R. 18(E) dated 2001-01-09

And the licences themselves were abolished on these dates:

- **Broadcast Receiver Licence (radio): abolished 1984.** Licences were renewed only to 1984-12-31; the fee had been ₹10 a year in the 1960s and ₹15 later, renewable annually at the post office. **[REPORTED — secondary sources, one reproducing an actual BRL document]** **Corroborating primary evidence:** G.S.R. 393(E) of 1997 supersedes *"the Radio, Television and video Cassette Recorder Sets (Exemption from Licensing Requirements) Rules, 1985"* — so an exemption rule-set of that name existed from 1985, which is exactly what you would expect if the licence died at the end of 1984. **[VERIFIED that the 1985 Rules existed and bore that title]**
- **Television licence: folded into the same 1985/1997 exemption.** The 1961 amendment to the 1933 Act had been made specifically to bring television within it — the Statement of Objects records that *"With the introduction of television in India, it has become necessary to licence the possession and working of television apparatus"*. **[VERIFIED]**
- **Short Wave Listener's licence: abolished 2010-04-01.** It appears in the "Existing" column and not the "As per Amendment" column of the table in WPC OM L-14011/255/2004-AMT dated 2010-08-13. **[VERIFIED]** — Part 3 §3.3 records the same. It was a **receive-only amateur licence**, obtainable from age 12 with no examination. **[REPORTED]**

The instrument that actually matters, and what it does not say

G.S.R. 393(E) dated 1997-07-16. Rule 3, verbatim from the clean typeset text in WPC's own compendium: **[VERIFIED]**

"3. License not required for certain broadcasting apparatus. Notwithstanding any thing contained in the Licensing of Wireless Receiving Apparatus Rules, 1965, **the Indian Wireless Telegraphy (Possession) Rules, 1965** or the Commercial broadcast Receiver Licensing (Dealers) rules, 1965, **no licence shall be required by any person to establish, maintain, work, possess or deal in – (i) radio, television and video cassette recorder sets; (ii) special antenna, including dish antenna, satellite decoder and associated front-end converter, used or capable of being used for reception of transient images of fixed and moving objects direct from satellites operating in**

Broadcasting Satellite Service or Fixed Satellite Service...: 'Provided that the exemption referred to in clause (ii) shall not be available for persons in the States of Arunachal Pradesh, Assam, Jammu and Kashmir, Manipur, Meghalaya, Mizoram, Nagaland and Tripura, and in the Union territories of Andaman and Nicobar islands and Lakshdweep and Minicoy Islands'."

Amended by **G.S.R. 18(E) dated 2001-01-09**, which narrowed clause (ii) to television signals, removed its 4800 MHz ceiling, and **omitted rule 4** — the provision that had prohibited Ku-band satellite reception. **[VERIFIED]** That 2001 amendment is the moment India legalised direct-to-home reception.

What G.S.R. 393(E) does:

- Removes the need for a licence to **establish, maintain, work, possess or deal in** the named articles. **All five verbs**. So it covers operating *and* possessing, and it covers shops as well as households. **[VERIFIED]**
- **Expressly disapplies the Indian Wireless Telegraphy (Possession) Rules, 1965** as well as the two Telegraph Act rule-sets — so it is a genuine exemption under 1933 Act s.4, not merely a Telegraph Act one. **[VERIFIED]** That is what lets it flow forward through 2023 Act s.3(4) (*"Any exemption granted prior to the appointed day under the Indian Telegraph Act, 1885 or the Indian Wireless Telegraphy Act, 1933 shall continue under this Act"*) and s.61 (old rules continue *"unless and until they are superseded"*) into rule 10(d) of REPA. **[VERIFIED]**

What it does not do — and this is the whole problem:

- It **never uses the words "receiver", "receiving apparatus" or "receive-only"**, or anything general. It exempts **three named product classes: radio sets, television sets and video cassette recorder sets**, plus a satellite-television antenna clause. **[VERIFIED]**
- It says nothing about communications receivers, scanners, SDRs or air-band receivers.

The central gap, stated plainly

Is an Icom IC-R8600, or a Tecsun PL-990x, or an RTL-SDR dongle, a "radio ... set"?

Arguments that it is:

- **Ordinary language**. An SDR receiving shortwave broadcasts *is* a radio set, and a Tecsun is sold in India as "a radio". The 1997 Rules use the plain commercial term, not a technical one.
- **Purpose**. The mischief the 1997 Rules removed was the licensing of domestic listening. A shortwave portable that happens to tune 150 kHz–30 MHz plus FM is the paradigm case of what those rules meant to free.
- **Twenty-nine years of practice**. No Indian prosecution for owning a shortwave or scanning receiver could be found, and multi-band shortwave portables are sold openly by mainstream Indian retailers.

Arguments that it is not:

- The Rules are drafted **by product class**, in a notification whose entire subject-matter is **broadcast** reception. Rule 2 defines only satellite-broadcasting terms, and the Explanatory Memorandum is entirely about broadcasting.
- WPC's own compendium files them under "*operation of wireless receiving apparatus and exemption of receivers*" — **but the rules themselves say "sets"**, and a regulator reading them strictly could say that a 3 GHz all-mode digital-decoding scanner is not a "radio set" in 1997 terms.
- **If "radio set" meant "any receiver", rule 3 would have been drafted that way.** The 1965 Rules it disappplies *were* drafted in general terms — "*wireless receiving apparatus*" — and the 1997 drafter chose narrower words.

Honest verdict: this is genuinely unresolved. [UNCERTAIN] For a broadcast-band portable the exemption almost certainly applies. For an IC-R8600 or an RTL-SDR it is arguable both ways and has never been tested. **This single question decides whether a non-ASOC individual has any possession exemption at all**, and it is question 1 at §9.8.5.

One piece of context that should shape expectations about how a regulator would answer it. The 1997 Explanatory Memorandum, which created the rule 4 that the 2001 amendment later deleted, is worth reading for its tone: **[VERIFIED]**

"...the new technology would enable any broadcaster within or outside India to telecast or beam any programme of his choice directly to the houses and bedrooms of general public. It is apprehended that any broadcaster who decides to start the DTH service for reception in India could include a foreign entity unlinking his signals from outside the India soil without any control over either the nature or content of the programme."

That is the mindset in which India's only receiver-exemption rule was drafted: reception is regulated because of content control, not because of interference. Keep it in mind when assessing how a regulator might react to a scanner — and note that it cuts directly against the comfortable international logic of §9.4.1.

Do the 2026 Possession Rules reach receivers? Yes, and there is no carve-out

REPA — G.S.R. 592(E), Gazette of India Extraordinary Part II §3(i) No. 534, dated **2026-07-08**, made under s.56 read with s.3(1)(c) of the 2023 Act, commencing on publication. **[VERIFIED — full gazette PDF]** Part 2 §2.9.2 covers the dealer-facing detail; only the receiver-relevant findings are given here, and they come from searching the entire extracted text of the notification:

- **The word "receive" appears nowhere in any operative rule.** Its only occurrences are procedural — "*the comments and suggestions received in that period*", "*On receipt of application under rule 4*", "*on receipt of a written request*". **[VERIFIED — search of the gazette text]**

- **The word "transmit" appears nowhere at all.** The Rules make **no distinction whatsoever between transmitting and receiving equipment.** [VERIFIED]
- **There is no repeal or savings clause**, and no reference to superseding the Indian Wireless Telegraphy (Possession) Rules, 1965. The only mention of the 1965 Rules is rule 4(2)(d), which makes a person holding a licence under them *eligible to apply* — **which presupposes they still exist.** [VERIFIED]
- Rule 2 defines only "Act" and "portal"; rule 2(2) imports everything else from the Act. So "radio equipment" here means what s.2(1) says, which includes receive-only devices. [VERIFIED]

Conclusion: REPA applies to receive-only equipment on its face and contains no receive-only exemption. Any exemption must come from rule 10. Here is how a receive-only owner gets home, and which routes actually work:

Route	Reasoning	Tag
10(f) — holds an ASOC	<i>"holds an Amateur Station Operator Certificate in accordance with the Telecommunications (Amateur Services) Rules, 2024".</i> Person-based, not equipment-based. No frequency qualification, no transmit/receive qualification, no "used as part of the amateur station" qualification. On the natural reading an ASOC holder needs no possession authorisation for any radio equipment	[VERIFIED] as to the text. But carry the Part 1 §1.5 Group B hedge: the report's own fact-check could not re-fetch the final 2026 Rules from a gov.in host, and the February 2025 draft numbered the exemption differently and contained no express ASOC gateway. [UNCERTAIN]
10(d) — a pre-existing exemption carried forward	G.S.R. 393(E) of 1997 is an exemption in respect of possession, granted under the 1885 and 1933 Acts before the appointed day; s.3(4) continues it; rule 10(d) then dispenses with authorisation. This is the route for an ordinary radio set, and the only route for a non-ASOC individual	Chain [VERIFIED] ; [UNCERTAIN] whether a scanner or SDR is within G.S.R. 393(E) at all
10(c) — licence-exempt-band devices	Requires both a s.3(3) exemption and a s.4(6) assignment exemption. Designed for PMR446-type transmitters; awkward for a receiver, which arguably needs no assignment exemption because it never uses spectrum	[UNCERTAIN] — the drafting does not contemplate receivers
10(a) — existing licence holders	Only if you hold a live Telegraph or 1933 Act licence. Not available to a new entrant ; those application routes closed on 2026-07-03	[VERIFIED] — Part 2 §2.9.1

And rule 4 offers an individual nothing. Rule 4(1) permits an application only by someone intending to *"(a) deal with such radio equipment through manufacture, purchase or import for sale, let for hire, repair or testing or demonstration thereof; or (b) import or purchase or hire radio equipment only for the purposes of testing or demonstration"*, and rule 4(2) restricts eligibility to companies, limited liability partnerships, holders of a trade licence, governments and 1965-Rules licensees. **[VERIFIED] There is no "I want to own a receiver" category, and a private individual is not eligible for any of them.**

Two further REPA provisions bear on a receiver owner:

- **Rule 8(1) is the practical bite:** *"An authorised entity that holds authorisation for possession of radio equipment as referred to in clause (a) of sub-rule (1) of rule 4 shall give possession thereof only to a person as referred to in rule 10 or who holds an authorisation granted under these rules."* **[VERIFIED]** So a compliant Indian dealer may hand a receiver only to a rule-10 person. **If a scanner is not within G.S.R. 393(E), a compliant dealer cannot lawfully sell one to a customer without an ASOC.** In practice no Indian retailer of SDR dongles or shortwave portables appears to apply any such check — Fab.to.Lab, The Engineer Store, ElectroPi, Robu and RFCharge all list them openly with GST invoices and no compliance notice **[VERIFIED — pages fetched 2026-08-24]** — but the rule is there, and it is the *same rule* that Baofeng's Indian storefront displays (without enforcing) on licence-band walkie-talkies at Part 7 §7.4.4.
- **Rule 12 requires a rule-10 person who ceases to be such a person to dispose of the equipment within two months.** **[VERIFIED]** On the unqualified reading of 10(f), **everything you own that is "radio equipment" must be disposed of two months after an ASOC lapses.** That is an odd consequence and another reason to put question 3 at §9.8.5 to WPC. **Do not let the certificate lapse while owning equipment.**

Are the 1965 Possession Rules still alive? It matters, because they are stricter

The **Indian Wireless Telegraphy (Possession) Rules, 1965 — G.S.R. 1318 dated 1965-08-28**, in force from 1965-11-01. **[VERIFIED]** as to citation. Content **[REPORTED]** from a reproduction:

- **Rule 2** defines *"complete wireless set"* as apparatus capable of **transmitting or receiving** wireless communication, with or without power supply, aerials or valves.
- **Rule 3:** no person may possess wireless apparatus without a licence except as provided in rule 5; a dealer must satisfy himself that the buyer holds a valid licence.
- **Rule 4:** an importer has 30 days from customs clearance to obtain a licence.
- **Rule 5** exempts *"(a) such apparatus as is reasonably required for the purpose specified in a current licence issued to him under section 4 of the Indian Telegraph Act, 1885...; (b) wireless telegraph apparatus other than complete wireless set; (c) wireless receiving apparatus established in any motor vehicle which is exempted from registration in India, provided the said wireless apparatus is not used for the reception of wireless signals while in India; (d) crystal wireless sets."*

Note the trap in rule 5(b). Because "complete wireless set" is defined to include apparatus capable of **receiving**, a communications receiver **is** a complete wireless set and is therefore **not** exempt under 5(b). **Rule 5(b) exempts components, not receivers.** And 5(c) is the opposite of helpful — it exempts a receiver in a foreign-registered vehicle **only if it is not used to receive in India.**

So the pre-2010 scheme was internally coherent, and seeing it whole explains what was lost:

Want to listen to shortwave or amateur traffic on a communications receiver? Take an SWL licence — no examination, nominal fee — and rule 5(a) then made your possession exempt. Want a domestic broadcast radio? Covered by the 1985/1997 exemption.

When the SWL licence was abolished on 2010-04-01, the first route disappeared and nothing replaced it. The comfortable reading of that abolition is "receiving needs no licence". **The careful reading is that the route by which a listener used to be licensed was removed, without a corresponding exemption being created.** Whether that leaves a hole or an implied liberty is exactly the question above. **It is not safe to treat the abolition of a licence as the grant of an exemption.**

Are the 1965 Rules in force in 2026? Arguments both ways:

- **For:** REPA does not repeal them, and rule 4(2)(d) treats 1965-Rules licensees as a live category. 2023 Act s.61 continues old rules *"unless and until they are superseded by any rules made under this Act"* — and REPA covers the same ground, which is an argument for supersession, but REPA does not say so.
- **Against:** the 1933 Act's own repeal has not commenced (Part 1 §1.8), so the parent Act survives; DoT nevertheless stopped accepting **all** applications under the 1933 and 1885 Acts from 2026-07-03 [**VERIFIED — Part 2 §2.9.1**], so even if the Rules survive, **the licence they require cannot be obtained.** A prohibition with no available licence is a bad position to be in.

[UNCERTAIN] — and it is question 2 at §9.8.5.

A correction to Part 3 §3.11

Part 3 §3.11's table of common claims answers *"India still issues a Short Wave Listener licence"* with *"Abolished on 2010-04-01. Receiving needs no licence today. ARSI's public page is out of date"*, tagged [**VERIFIED**].

The first and third sentences are right. The middle one is an inference, not a verified fact, and it should be read as [UNCERTAIN]. Nothing in the 2010 memorandum, the 2024 Amateur Services Rules or REPA says that receiving needs no licence. What is accurately true is narrower, and it is worth stating in the form that survives scrutiny:

No *operating* licence is needed to receive, because there is no receive-side operating licence left to obtain, and the Telegraph Act's exclusive privilege concerns *establishing, maintaining and working* a telegraph, which a receiver arguably does not do to anyone else's telegraph. [VERIFIED as to the absence of any such licence]

But *possession* of a receiver is still governed by 1933 Act s.3 and 2023 Act s.3(1)(c), and depends on an exemption whose scope is unclear. [UNCERTAIN]

Part 1 §1.5's grouping of hedged findings should carry this alongside the rule 10(f) hedge.

9.5.2 QUESTION 2 — LISTENING

The Indian Telegraph Act 1885 contains no offence of listening

This is a real and useful finding, and it disposes of most of what the internet asserts. Every interception offence in the 1885 Act is built around telegraph offices, physical plant and officials — the technology of 1885 — and **none of them can catch a person listening on a receiver at home. [REPORTED as to the text, from a state-government hosted copy of the Act; section numbering and substance corroborated against the TRAI-hosted text quoted in Part 6 §6.3]**

Section	What it prohibits	Why it cannot catch a listener
s.23	Entering a signal-room or trespassing in a telegraph office. Fine up to ₹500	Requires physical trespass
s.24	<i>"Unlawfully attempting to learn the contents of messages" — doing any s.23 act "with the intention of unlawfully learning the contents of any message". Up to 1 year</i>	The offence is parasitic on s.23. A person listening at home commits no s.23 act and therefore cannot commit s.24
s.25	Damaging or tampering with telegraph plant with intent <i>"(b) to intercept or to acquaint himself with the contents of any message"</i> . Up to 3 years	Requires damaging, removing, tampering with or touching physical plant. A receiver touches nothing
s.26	Unlawful interception or disclosure by <i>"any telegraph officer, or any person... having official duties connected with any office which is used as a telegraph office"</i> . Up to 3 years	Applies only to officials. Not to a private listener
s.30	Fraudulently retaining or secreting a message. Up to 2 years	Concerns physical custody of a message entrusted for delivery
s.21	Using an unauthorised telegraph. Fine of ₹50	Catches the person who talks on an unlicensed set, not the person who listens

The Telecommunications Act 2023 s.42(2)(b) is the live provision, and it has no listener carve-out

Verbatim: **[VERIFIED — gazette text]**

"42. ... (2) Whoever directly or indirectly or through personation— (a) gains or attempts to gain unauthorised access to a telecommunication network or to data of an authorised entity or transfers data of an authorised entity; or **(b) intercepts a message unlawfully**, shall be punishable with imprisonment for a term which may extend to **three years**, or with fine which may extend up to **two crore rupees**, or with both."

Section 42 commenced **2024-06-26** (S.O. 2408(E)) [VERIFIED], and **s.42(7) makes every s.42 offence cognizable and non-bailable [VERIFIED — Part 6 §6.2].**

This is a standalone, general, criminal offence of unlawful interception, applying to any person. India did not have one before. It replaces a scheme in which interception offences required trespass, tampering or official status with one that requires none of them.

And "message" is defined extremely broadly — s.2(g), *"any sign, signal, writing, text, image, sound, video, data stream, intelligence or information sent through telecommunication"*.

[VERIFIED] A voice transmission on 145.500 MHz is a message. So is an ADS-B position report. So is a pager burst.

What makes interception lawful? Only the State doing it under an order. Section 20(2) is the authorising provision — on grounds of *"the sovereignty and integrity of India, defence and security of the State, friendly relations with foreign States, public order, or for preventing incitement to the commission of any offence"*, an authorised officer may direct that a message *"shall not be transmitted, or shall be intercepted or detained, or shall be disclosed in intelligible format"* — with procedure in the Telecommunications (Procedures and Safeguards for Lawful Interception of Messages) Rules, 2024, notified 2024-12-06. [VERIFIED] **Nothing else, anywhere in the Act, is described as lawful interception.**

Does passive reception count as "interception"? The real uncertainty, not papered over

This is the single most consequential unresolved question in this part. Both cases are set out because the reader deserves the argument rather than a verdict dressed up as one.

The case that reception is not interception:

- **"Intercept" in ordinary and legal usage implies coming *between* a sender and an intended recipient** — diverting, tapping, capturing something on its way. A radio transmission is broadcast omnidirectionally into public space; a receiver that detects it comes between nothing.
- The opening words *"directly or indirectly or through personation"* colour both limbs and suggest **active, deceptive conduct** rather than passive observation.
- s.42(2)(b) sits alongside s.42(2)(a) — unauthorised access to a network — in a sub-section evidently aimed at **hacking and phone-tapping**, and it carries a ₹2 crore fine, which is absurd as a hobbyist penalty.
- **No Indian prosecution of a listener under it could be found.**

The case that it might:

- **The Act supplies no definition of "intercept", and no carve-out for transmissions intended for general reception.** Compare the United Kingdom: Wireless Telegraphy Act 2006 s.48 makes it an offence to use apparatus with intent to obtain information about the contents of a message *"of which neither he nor a person on whose behalf he is acting is an intended recipient"* — and s.48(2) expressly excludes messages **"sent by means of wireless telegraphy for reception by the general public"**. India has the prohibition-shaped concept without the protective exception.
- **The word "unlawfully" is the only limiter, and it is circular:** interception is unlawful unless authorised, and only the State can be authorised.
- **India's regulatory instinct on reception is content control, not interference** — see the 1997 Explanatory Memorandum at §9.5.1. A regulator with that instinct is not obviously going to read a listener-friendly exception into s.42(2)(b).
- **Section 26 of the 1885 Act — the old official interception offence — was drafted narrowly, and the 2023 Act deliberately widened the concept to "whoever". Widening was the point.**

Best available reading, stated as a reading and not as law. [UNCERTAIN]

Listening to a transmission that is in fact intended for general reception — a broadcast station, a beacon, ADS-B, AIS, a weather satellite, an amateur contact, a time signal — is very unlikely to be prosecuted as unlawful interception, and a court would probably decline to read "intercept" that far.

Listening deliberately and repeatedly to a point-to-point transmission between identified parties who plainly did not intend you to hear it — a police net, a private business channel, a security detail — is a materially different case, and there is no statutory shelter for it in India.

The Information Technology Act 2000 is mostly a red herring

Many pages assert that "intercepting communications is illegal under the IT Act". Checked, and it is worth correcting because the claim circulates widely:

- **s.43** creates civil liability for a person who, *without permission of the person in charge of a computer, computer system or computer resource*, accesses it or downloads, copies or extracts data from it; **s.66** criminalises s.43 conduct done dishonestly or fraudulently, up to 3 years and/or ₹5,00,000. **[REPORTED — standard transcription and commentary; the bare Act was not fetched]**
- **The gating concept is "computer resource"**. Detecting a radio wave in free space is not accessing anyone's computer resource, and a voice transmission is not data held in a computer system.

- **Where it could bite:** capturing and decoding a **digital data** transmission and thereby extracting data that resides in someone's system — defeating encryption on a private digital network, or lifting readable records off a data link. There, *"downloads, copies or extracts any data ... from a computer, computer system or computer network"* becomes arguable. **[UNCERTAIN]** — no Indian authority either way.
- **s.72** (breach of confidentiality and privacy) applies to persons who obtain information *"in pursuance of any of the powers conferred under this Act or rules"* — officials, not listeners.

Do not repeat the common claim that the IT Act makes scanning illegal. It does not say that. But do not assume it is irrelevant to **decoding private digital traffic**, which is a genuine question mark.

And there is no Indian scanner law at all

Searched for any Indian instrument, circular, FAQ, gazette notification or DoT/MHA position addressing "scanner", "scanning receiver", "monitoring receiver" or "wideband receiver".

Results:

- **The word "scanner" appears in Indian wireless instruments only in the document-scanner sense.** WPC O.M. No. ETA-WPC/Policy/2018-19 dated 2019-02-26 lists ETA self-declaration categories as *"mobile handsets, smart phones, electronic notepads, laptops, Smart watches, Short Range Devices including accessories, microphones, speakers, headphones, earphones, printers, scanners, cameras etc."* — office scanners, in a list of consumer peripherals. **[VERIFIED]** — Part 2 §2.7.2 quotes the same list.
- A search of the entire text extraction of WPC's compendium of subordinate legislation for "scanner", "scanning receiver" and "monitoring receiver" produced **no hits in any operative provision.** **[VERIFIED — search of the extracted text]**
- **No Indian equivalent** of the several United States states' scanner-in-vehicle statutes, Australia's state-level police-scanner offences, or the United Kingdom's s.48 regime. Nothing.

So India regulates scanners only through the general possession law and the general interception law. There is no scanner-specific prohibition to breach — and equally no scanner-specific permission to rely on. **[UNCERTAIN as a search-limited negative]**

One institutional fact that cuts the other way, and is worth knowing. India's own monitoring capability is institutional: the **Wireless Monitoring Organisation**, established 1952, is WPC's field and enforcement arm and operates International Monitoring Stations. **[VERIFIED — Press Information Bureau release, Ministry of Communications, 2023-12-19]** Part 2 §2.6.3 records the WMO physically going looking for over-powered PMR446 radios. **The State does the monitoring, and a private person doing the same thing is doing something the State treats as its own function.** That is not a legal argument. It is a description of the institutional temperature.

9.5.3 WHAT COULD NOT BE FOUND, AND IT MATTERS

Three negatives, all search-limited, all pointing the same way:

- **No Indian customs seizure of an SDR, a scanner or a receiver.** The March 2026 DRI seizure at Nhava Sheva that features in Part 6 §6.4.8 concerned **Baofeng BF-888S transmitters** in a ₹30 crore consignment of illegal electronics — **not receivers**. **[REPORTED]**
- **No reported Indian judgment** on possession of a receive-only apparatus, under 1933 Act s.6(1) or otherwise. **[UNCERTAIN — absence of evidence, not evidence of absence]**
- **No DoT or MHA published position on scanners** — nothing on dot.gov.in, eservices.dot.gov.in, saralsanchar.gov.in or in the WPC compendia. **[UNCERTAIN]**

And one warning about the sources the reader will find instead. The confident pages asserting that "receiving is legal in India" are, on inspection, of very poor quality. The top search result for "*Is RTL-SDR legal in India?*" is an undated, unattributed page citing no primary sources, which asserts that interception is illegal "*under the Indian Penal Code*" without naming a section. A claim circulating that police-scanner applications have been removed from Indian app stores traces to the same site, and **no app-store or MeitY blocking order could be verified**. **[REPORTED, and expressly unreliable]** Treat every confident "receiving is legal in India" page with the same suspicion; most are of that quality, and this part is deliberately more cautious than the internet consensus because the internet consensus has no sources.

9.5.4 QUESTIONS 3 AND 4 — RECORDING, AND DISCLOSING

This is where the serious exposure lives, and the order of magnitude is not close.

The Official Secrets Act 1923 is the real hazard

s.2(8), "prohibited place", clause (a), verbatim: **[VERIFIED]**

"any work of defence, arsenal, naval, military or air force establishment or station, mine, minefield, camp, ship or aircraft belonging to, or occupied by or on behalf of Government, **any military telegraph or telephone so belonging or occupied, any wireless or signal station or office so belonging or occupied** and any factory, dockyard or other place so belonging or occupied and used for the purpose of building, repairing, making or storing any munitions of war..."

A government wireless station or signal office is a statutory prohibited place. So is a military telegraph or telephone. And clauses (c) and (d) let the Central Government declare further prohibited places by notification, including "*any railway, road way or channel, or other means of communication by land or water*" — which, once notified, could include civil aviation and port infrastructure. **[VERIFIED]**

s.3(1), verbatim in the part that matters: **[VERIFIED]**

"If any person for any purpose prejudicial to the safety or interests of the State ... **(c) obtains, collects, records or publishes or communicates to other person any secret official code or pass word, or any sketch, plan, model, article or note or other document or information which is calculated to be or might be or is intended to be, directly or indirectly, useful to an enemy** [or which relates to a matter the disclosure of which is likely to affect the sovereignty and integrity of India, the security of the State or friendly relations with foreign States], he shall be punishable with imprisonment for a term which may extend, where the offence is committed in relation to any work of defence, arsenal, naval, military or air force establishment or station... or otherwise in relation to the naval, military or air force affairs of Government or in relation to any secret official code, to **fourteen years** and in other cases to **three years**."

The verbs are the important part: "obtains, collects, records or publishes or communicates". A monitoring log is a record. A recorded audio file is a record. A screenshot of decoded traffic is a record. **A frequency list built from observation is arguably a collection.**

s.3(2) — the reversed burden, verbatim: [VERIFIED]

"On a prosecution for an offence punishable under this section it shall not be necessary to show that the accused person was guilty of any particular act tending to show a purpose prejudicial to the safety or interests of the State, and, notwithstanding that no such act is proved against him, he may be convicted if, from the circumstances of the case or his conduct or his known character as proved, it appears that his purpose was a purpose prejudicial to the safety or interests of the State; and **if any sketch, plan, model, article, note, document or information relating to or used in any prohibited place, or relating to anything in such a place, or any secret official code or pass word is made, obtained, collected, recorded, published or communicated by any person other than a person acting under lawful authority**, and from the circumstances of the case or his conduct or his known character as proved it appears that his purpose was a purpose prejudicial to the safety or interests of the State, such sketch, plan, model, article, note, document, information, code or pass word **shall be presumed to have been made, obtained, collected, recorded, published or communicated for a purpose prejudicial to the safety or interests of the State.**"

Read that against a realistic fact pattern. A person is found near a defence establishment or an airport with a wideband receiver, a laptop, a frequency list and recordings of official traffic. The material *"relat[es] to ... any prohibited place"*. They were not *"acting under lawful authority"*. **The prosecution does not have to prove any particular prejudicial act.** Whether the presumption bites turns on *"the circumstances of the case or his conduct or his known character"* — which is to say, **on how it looks.**

s.5 — wrongful communication of information is principally directed at persons who obtained information by virtue of office, contract or entrustment, with a three-year maximum. **[REPORTED — standard commentary; the bare text of s.5 could not be retrieved]** For a private listener s.3(1)(c) is much the more likely charging provision.

Disclosure: a secrecy obligation an ASOC holder may not know they have

For a certificate holder, disclosure breaches a licence condition **independently of the OSA**. Annexure A para 8, quoted in full at §9.4.2, imposes **four separate prohibitions** on an ASOC holder who happens to hear something they were not entitled to hear: do not disclose the **contents**; do not disclose the **origin or destination**; do not disclose **its existence or the fact of its receipt**; and do not **reproduce, copy or make any use of it**. [VERIFIED]

In practice that means an ASOC holder who tunes across a police channel, hears something, and mentions it to a friend has breached a licence condition. One who posts a recording online has breached it flagrantly — and has also handed a prosecutor the *"publishes or communicates"* limb of OSA s.3(1)(c). Breach of a certificate condition is a ground for suspension or cancellation under rule 9(a) of the 2024 Rules, after a hearing. [VERIFIED — Part 3 §3.8]

The proportion, stated once

Act	Provision	Maximum exposure
Owning a receiver without an exemption	1933 Act s.6(1) + s.6(3), or 2023 Act Third Schedule item 1(a)	₹100–₹250 plus confiscation , or a ₹50,000 civil penalty
Listening	2023 Act s.42(2)(b)	3 years / ₹2 crore in principle; no known Indian prosecution of a listener
Recording material relating to a government wireless station	OSA s.3(1)(c) with the s.3(2) presumption	Fourteen years, with the burden reversed

Everything a cautious Indian listener should do follows from that table: do not record official traffic; do not build logs of it; do not carry a receiver and a notebook near defence or airport property; and do not publish anything you hear on official frequencies.

9.5.5 THE BIGGEST REAL RISK, STATED PLAINLY, AND THE ONE INDIAN CASE

The biggest legal risk is not owning the receiver. It is not listening on it at home. It is recording or repeating aviation, police, military or security-agency traffic — and doing it anywhere near an airport or a defence installation.

That is not theoretical. **The one clear Indian enforcement event on point is about recording, not owning.** Facts, from converging contemporary reports, all [REPORTED] and well corroborated:

Who	Stephen Hampton (46) and Steven Ayres (56), British nationals from Bristol
When and where	Arrested 2010-02-15 near Indira Gandhi International Airport, Delhi. They were detained in a fourth-floor room of a hotel at Mahipalpur, directly opposite runway 11-29
Context	Two days after a bomb blast in Pune, during a nationwide security crackdown

Equipment	"a scanner, laptop, binoculars and cameras", plus an AirNav RadarBox — an ADS-B virtual-radar receiver
Conduct charged	"recording the conversation between pilots and air traffic control" / "illegally monitoring aircraft"
Statute used	The Indian Telegraph Act . They initially faced potential spying charges carrying up to ten years
Outcome	Pleaded guilty to a Telegraph Act breach carrying up to three years; fined about USD 550 (roughly ₹25,000 at 2010 rates) and released without jail time on 2010-03-05 ; passports returned and permitted to leave India

What this case establishes, and what it does not.

It **does** establish, as a matter of practice, that: Indian police will arrest for monitoring and recording air traffic control; the charge will be framed under wireless legislation; **the initial framing may be espionage**; proximity to an airport is aggravating; foreign nationality during a security alert is aggravating; and the eventual penalty may be modest **but the process is weeks of detention and passport seizure. That process is the real cost, not the fine.**

It does **not** establish which section of the Telegraph Act was used — and on the analysis at §9.5.2, it is not obvious that any section fitted a passive listener, which suggests either s.20 (working an unauthorised telegraph, if the RadarBox was characterised as apparatus) or a negotiated plea to something approximate. **[UNCERTAIN]** No charge sheet or judgment could be retrieved.

And it is a pre-2023-Act case. Today the natural charge would be **s.42(2)(b) — three years and ₹2 crore, cognizable and non-bailable — and possession would additionally be in issue.**

Aviation, marine, police and military: where the sensitivity actually is

- **Aviation. There is no Indian rule that says "aviation monitoring is prohibited."** No DGCA civil aviation requirement, no Aircraft Rules provision, no WPC notification. **[UNCERTAIN as a search-limited negative]** But the 2010 case happened. The widely repeated online claim that *"in India, reception/monitoring of any aviation related transmissions is deemed illegal, unless you have a formal permission/license"* comes from **aviation-hobbyist forum discussion, not from any instrument [REPORTED]** — treat it as an accurate description of *police attitude*, not of statute. **ADS-B is the underrated trap:** it is an unencrypted broadcast on 1090 MHz designed to be received by anyone, and feeding it to a flight-tracking site from India is done by many people — but in 2010 the police treated an ADS-B box as part of the offence. **[REPORTED]** No current Indian instrument restricting ADS-B reception could be found. **[UNCERTAIN]**

- **Marine.** India's marine VHF regime governs **ship stations** — GMDSS rules, coast radio stations and watchkeeping circulars. **Nothing addressing shore-based civilian monitoring of marine VHF could be found: no prohibition, no permission.** [UNCERTAIN] Coastal security is nonetheless a live sensitivity post-2008, and OSA s.2(8)(d) allows the government to notify *"any railway, road way or channel, or other means of communication by land or water"* as a prohibited place. [VERIFIED]
- **Police, security agencies, military.** **This is where §9.5.4 does all the work, and where the reader should simply not go.** A government wireless or signal station or office is a statutory prohibited place; **recording** anything from such a source engages OSA s.3(1)(c) directly with the s.3(2) presumption available and a fourteen-year maximum. [VERIFIED] Much of it is in any event unlistenable — Indian police and paramilitary forces have moved substantially to encrypted TETRA and DMR networks, so a scanner mostly yields control-channel noise. [UNCERTAIN — **this could not be verified from a primary source**]

The rule to take away, in one line:

Tune wherever your receiver goes, but never record and never repeat anything that came from a police, military, security-agency or air-traffic channel — and do not do your listening within sight of a defence installation or an airport perimeter.

The practical difference is not legal, and that is the point. **Air-band listening at home, on a portable, without recording, is not something anyone in India has ever been troubled for. Air-band listening at an airport fence with a camera and a notebook is the exact fact pattern that produced an arrest.** It is how it looks to a police officer during a security alert.

And note the Part 8 §8.4.6 finding from the other direction: **sensitive areas are policed on the ground, not by spectrum law.** Indian travellers report being asked to declare or deposit communication devices at checkpoints, including a **licensed** satellite phone surrendered at an army post near Leh. [REPORTED] Part 7 §7.2.1 and Part 7 §7.7.2 question 5 record that **no published list of restricted areas could be found. Plan for a checkpoint to take an interest in anything with an antenna, carry paperwork, and declare rather than conceal.**

9.5.6 WHAT THE PENALTY FOR OWNING ACTUALLY IS

Two parallel possibilities, and it matters which applies:

Regime	Sanction for unauthorised possession of a non-transmitting apparatus	Enforceable now?
Indian Wireless Telegraphy Act 1933, s.6(1) with s.6(3)	Fine up to ₹100 first offence, ₹250 subsequent, plus confiscation at the court's discretion; presumption of possession from effective control of the premises (s.6(2)); search without a warrant (s.7)	Yes — the Act is unrepealed. [VERIFIED]; Part 1 §1.8 carries the repeal hedge

Regime	Sanction for unauthorised possession of a non-transmitting apparatus	Enforceable now?
Telecommunications Act 2023, Third Schedule item 1(a)	<i>"Possessing radio equipment without an authorisation or an exemption... First Offence: Civil penalty up to fifty thousand rupees. Each subsequent offence: Civil penalty up to two lakh rupees for each such instance."</i>	[UNCERTAIN] — the Third Schedule is engaged through ss.33 and 34, and the adjudication chapter (ss.31–41) was in neither 2024 commencement tranche and no later notification commencing it could be found. REPA rule 14 also routes breaches to s.32, in the same uncommenced block. Part 6 §6.1 carries the same hedge

And note what is *not* available to a prosecutor. The criminal provisions in 2023 Act s.42(3) do **not** cover simple unauthorised possession: s.42(3)(d) is confined to radio equipment *"that can accommodate more than specified number of subscriber identity modules"*, and s.42(3)(f) to equipment using *"unauthorised or tampered telecommunication identifiers"*. **[VERIFIED] So possessing a receiver, even without an exemption, is not a s.42 crime.** It is a civil-penalty matter under the 2023 Act, or a ₹100 fine under the 1933 Act.

This is the single most reassuring fact in this part, and it is properly sourced. The downside exposure for owning a receiver is small. The downside exposure for what you do with it is not.

9.5.7 IMPORT — THE WEAKEST LINK IN THE CHAIN

The 1953 notification, which names receivers and distinguishes them properly

The controlling instrument is **Notification No. 71 of 1953** (S.R.O. 1815 dated **1953-09-25**, effective **1953-10-15**), issued under s.19 of the Sea Customs Act 1878, superseding Notification No. 55-Customs of 1933-12-09. Part 2 §2.8.1 introduces it; here is the text that matters, verbatim: **[VERIFIED — reproduced as Annex-1 to WPC's 2022 import compendium]**

"...the Central Government hereby restricts with effect from the 15th October 1953, the bringing into India of apparatus for wireless telegraphs as follows: — **(a) in the case of transmitting apparatus and of receiving apparatus incorporated in a single unit with transmitting apparatus, to persons who have been licensed to import such apparatus by the Ministry of Communications in Form I, annexed hereto; (b) in other cases, to persons who apply to a customs officer to import the same in Form II, annexed hereto:** Provided that in the case of wireless receiving apparatus imported by post the customs officer may dispense with the application referred to in clause (b)."

Read the structure, because it is the best news in this section. The 1953 notification draws exactly the distinction the reader cares about:

- **Transmitters and transceivers → clause (a) → a WPC import licence in Form I.**

- **Everything else, i.e. receive-only apparatus → clause (b) → not a licence at all, but an application to the customs officer in Form II** — a one-page declaration of owner, importer, description, value and "*Number of complete wireless receivers*", signed by the importer and countersigned by the customs officer. **[VERIFIED — Form II reproduced in the compendium]**
- **And for a receiver arriving by post, even Form II may be dispensed with.**

Form II carries a printed notice that is a nice period artefact of the licensing era: "*NOTICE.— Under the Indian Wireless Telegraphy Act, 1933, and the Indian Telegraph Act, 1885, a license is required for the possession or for the establishment, maintenance and working of wireless apparatus in India.*" **[VERIFIED]**

WPC's own 2022 restatement is broader than the notification it cites

WPC's Compendium of Orders... in regard to Import licensing requirement, File No. R-11018/02/2017-PP dated 2022-07-06, §1.1, verbatim: **[VERIFIED]**

"Under Notification No. 71 of 1953 issued under the Sea Custom Act 1878, a license is required from Ministry of Communications for import of any wireless **transmitter/receiver/transceiver.**"

And §2.4, verbatim, signed by the Deputy Wireless Adviser: **[VERIFIED]**

"**2.4. Non-requirement of Import license for standalone Antenna and other RF accessories-** As already stated, import license from WPC Wing is required only for RF **transmitter/receiver/transceiver**; No import license / NOC is required from WPC wing for import of any standalone RF antenna / other RF accessories such as cables connector, battery etc.; however, for import of RF Amplifier import licence from WPC Wing is required."

WPC's own summary says a receiver needs an import licence. The notification it cites says a receiver needs a Form II customs application, not a licence. Those are different requirements, and WPC's restatement has flattened clauses (a) and (b) into a single sentence.

This is not academic, because the summary — not the notification — is what Customs was told. CBIC Instruction No. 16/2022-Customs dated 2022-07-21 circulated the compendium to all Customs zones, modified to the extent of the ETA self-declaration change by CBIC Instruction No. 24/2024-Customs dated 2024-10-22. **[VERIFIED as to the 2024 instruction — Part 2 §2.7.2; [REPORTED] for the 2022 instruction, from a reproduction quoting the same sentence]**

Practical effect: a customs officer working from the current instruction has been told that importing a receiver requires a WPC licence.

The DGFT layer, and one thing that could not be verified

- **ITC(HS) heading 8527** is "*Reception apparatus for radio-broadcasting...*"; **8525** is "*Transmission apparatus...*"; **8526** is radar and radio navigational aid apparatus.

- For **8525 60 00** — transmission apparatus incorporating reception apparatus, i.e. walkie-talkies, repeaters and boosters — the policy is "Free" with the condition *"Not permitted to be imported except against a license to be issued by the WPC wing"*. **[REPORTED] This is the condition that bites on a Baofeng**, and Part 2 §2.8.5 records the same.
- **8526 is "Restricted"**. **[REPORTED]**
- **The current policy condition attached to 8527 could not be verified.** The ITC(HS) Schedule text reachable in this research was the 2012 edition. **[UNCERTAIN] Check this on the DGFT site before ordering anything expensive.**
- In practice, RTL-SDR-class dongles are usually declared as **DVB-T television tuners**, which is why they appear on Indian marketplaces alongside TV tuner sticks with no wireless-compliance flag at all. **That is a classification accident, not a legal permission.**

The 2026 gap — and it has no successor category for an individual

Part 2 §2.8.4 identifies a gap in the import chain for transmitters. **The same gap exists for receivers, and it is if anything cleaner.** The dated facts, all **[VERIFIED]** from the DoT notices recorded at Part 2 §2.9.1 and the eServices import-licence page fetched 2026-08-24:

1. DoT stopped accepting new applications under the 1885 Act for **all** licences, registrations and permissions, directing applicants to the Authorisation Portal, **from 2026-06-25**.
2. DoT stopped accepting **all** applications, including renewals, for Dealer and Non-Dealer Possession Licences and Manufacturing and Testing Licences under the 1885 Act **from 2026-07-03**.
3. Radio Equipment Possession Authorisation applications opened on the Authorisation Portal **from 2026-08-06**.
4. The eServices **Import Licence** page still describes the WPC import licence — *"Rs. 500 per import license"*, grantable to *"any entity registered in India and holding valid permission granted by DoT"* — while carrying the same migration notice. **It says nothing about receive-only equipment.**

So if WPC's §2.4 reading is right and a receiver import needs a WPC licence, then as at 2026-08-25 a private individual (a) is not "an entity registered in India holding valid permission granted by DoT", (b) cannot apply under the old regime because it is closed, and (c) cannot apply under the new regime because REPA rule 4 has no category for them and rule 4(2) makes them ineligible.

There may currently be no lawful route for a private individual to import a receive-only wireless apparatus into India. [UNCERTAIN], and held tentatively — the more likely truth is that WPC has never thought about receivers and Customs has never stopped one. **But the paperwork, read literally, produces that conclusion, and this report will not pretend otherwise.**

The mitigation is straightforward, and it is the practical advice of this whole section

Buy inside India.

A domestic purchase engages the dealer's compliance rather than yours, and it avoids the import question entirely. It also avoids the **Customs Act 1962** exposure that Part 2 §2.8.6 and Part 6 §6.4.8 identify as the thing which actually produced *arrests* in the Baofeng cases: **importing yourself moves you from a civil-penalty regime into a confiscation-and-prosecution regime.**

And it gives you the single most useful document you can own: **a GST invoice naming the model, serial number and date.** Part 7 §7.4.9 made the same point about a domestic Baofeng purchase, for the same reason.

Two things not to do:

- **Do not personally import a transmitter of any kind**, including a HackRF. §9.3.3.
- **Do not import an RF amplifier, brick or booster.** It needs its own WPC import licence [VERIFIED — Compendium §2.4], and it takes you straight past both amateur power ceilings. Standalone antennas, coaxial cable, connectors and batteries need none. [VERIFIED]

Is an ETA needed for a receive-only device? Probably not, but nobody official has said so

The structural argument is strong, and it is my reasoning rather than WPC's words:

- ETA exists to certify compliance with the technical parameters of a **licence-exempt band** notification. WPC's compendium §2.1: *"Central government ... allows license exempt use of certain frequency bands subject to compliance with the technical parameters mentioned in the respective notifications; however, equipment complying with such notifications need to be type approved and accordingly Equipment Type approval (ETA) certificate is issued to such compliant RF devices."* [VERIFIED] — Part 2 §2.7.1.
- **Licence-exemption notifications exempt use of spectrum, i.e. radiation. A receiver does not use spectrum.** There is therefore no notification for a receive-only device to comply with, and nothing for an ETA to certify.
- Consistently, **the ETA application format is built around a transmitter**: the annexure has *"Section-B (Details of Transmitter: frequency range, number of preset switchable channels...)"*, and where receivers appear at all it is as *"Section-C – Details of Receivers"* — a sub-part asking for the **spurious response of receiver** inside a transmitter application. [VERIFIED — WPC compendium, multiple ETA annexures]
- DoT's ETA page lists only four excluded product types — radar, jammers, drones, satellite — and otherwise says all licence-exempt-band products may self-declare. **Receivers are not mentioned either way.** [VERIFIED — Part 2 §2.7.2]

Weak secondary support: several compliance consultancies state that "only pure receivers do not need approval, for example GPS products". [REPORTED — none cites an instrument]

So treat "no ETA needed for receive-only" as the probable position, and note that it is consultants' practice rather than published law. [UNCERTAIN] — question 4 at §9.8.5.

One consumer-protection consequence worth stating: Saral Sanchar's ETA equipment-category dropdown does contain a "Receivers" entry, though what it is used for is not stated. **[VERIFIED that the category exists — Part 2 §2.11.1; UNCERTAIN what falls in it]** If a seller waves an "ETA certificate" at you for a receive-only device, that is a warning sign about the seller, not a reassurance about the product — and either way, verify the number by hand at https://saralsanchar.gov.in/wpc_eta_report.php, which is CAPTCHA-gated and cannot be scripted. **[VERIFIED]**

9.5.8 WHAT AN ASOC CHANGES — AND IT IS THE CLEANEST ANSWER AVAILABLE

Possession: solved, decisively

REPA rule 10(f) exempts a person who *"holds an Amateur Station Operator Certificate in accordance with the Telecommunications (Amateur Services) Rules, 2024"* from the possession-authorisation requirement, **with no qualification as to frequency, equipment type, or transmit versus receive capability.** **[VERIFIED as to the text; and see the Part 1 §1.5 Group B hedge, which this part does not upgrade]**

On the natural reading this is the cleanest possession route available to any private individual in India, and it disposes of the §9.5.1 uncertainty entirely — you no longer need to argue whether your receiver is a "radio set", because you are exempt as a person.

That is a strong practical reason to sit the examination **early rather than late**, independent of any wish to transmit. It costs ₹100 to sit and ₹1,000 or ₹2,000 to hold, there is no Morse test for the Restricted grade, and it converts the possession question from "arguable" to "settled".

Three caveats, none of which defeats the point:

- **The Explanation to rule 10** bars a rule-10 person from manufacturing, selling or letting for hire the equipment in their possession. **[VERIFIED]** So an ASOC does not let you become a dealer. Part 2 §2.9.3 flags the open question whether "manufacture" in that Explanation bars an ASOC holder from homebrewing — which would be an odd result for a service statutorily defined around self-training and technical investigation. **[UNCERTAIN]**
- **Rule 12** requires disposal within two months if a rule-10 person ceases to be one. **[VERIFIED]** §9.5.1.
- **The 1965 Possession Rules, if still in force**, exempt possession by reference to a *"licence issued ... under section 4 of the Indian Telegraph Act, 1885"*. An ASOC under the 2024 Rules is granted under s.47 of the 2023 Act, not s.4 of the 1885 Act, so it does not literally fit 1965 rule 5(a). **This is a drafting-overlap artefact rather than a real risk — DoT is plainly administering the 2023 Act regime — but it is another reason the 1965 Rules' status needs clarifying.** **[UNCERTAIN]**

Listening: no change, and in one respect it makes things stricter

An ASOC authorises operation of an amateur station for amateur services. It is not a general monitoring licence. Two provisions bear on this.

Annexure A para 1, verbatim: [VERIFIED]

"1. **Use of the amateur station.** - The amateur station may be used for the purpose of receiving transmissions in the Standard Frequency and Time Signal Service to facilitate operation of the amateur station within the authorised frequency bands."

The only out-of-band reception the Rules positively contemplate for an amateur station is standard frequency and time signals, and even that is justified instrumentally. This clause is inherited verbatim from para I(2) of Annexure V to the 1978 Rules. **[VERIFIED]**

Two readings are available:

- **Narrow:** an ASOC holder using their amateur station to monitor the air band is using the station for a purpose outside para 1, and is therefore in breach of a licence condition — **a worse position than an unlicensed person with a separate receiver, who has no licence conditions to breach.**
- **Wide:** para 1 addresses the *amateur station*, defined by s.47 of the Act as "*a radio station operated by an amateur for amateur services*". A standalone scanner or SDR that is not part of the amateur station and is never used to transmit **is not an "amateur station" at all**, so para 1 does not reach it.

The wide reading is more persuasive and is what amateurs worldwide assume. But the narrow reading is available, and it produces a specific piece of practical advice: keep general monitoring physically and functionally separate from the amateur station. Use a distinct receiver rather than the transceiver's wideband receive function, and **make no monitoring entries in the amateur log.** **[UNCERTAIN]**

And **Annexure A para 8** then imposes a positive secrecy duty on anything received that the holder was not entitled to receive (§9.4.2). **An unlicensed listener has no such duty; an ASOC holder does. So on the disclosure axis, the certificate makes you more constrained, not less.**

Net effect

Axis	Effect of holding an ASOC
Owning any radio equipment	Solved. REPA rule 10(f) [VERIFIED as to the text; Part 1 §1.5 Group B hedge applies]
Buying from an Indian dealer	Solved. REPA rule 8(1) permits a dealer to give possession to a rule-10 person [VERIFIED]
Importing amateur equipment	Improved: rule 7(3) of the 2024 Rules means no separate import authorisation for amateur equipment [VERIFIED] . Whether that extends to a receiver, which is not amateur equipment, is unclear [UNCERTAIN]
Listening within the amateur bands	Squarely lawful — it is the point of the certificate

Axis	Effect of holding an ASOC
General out-of-band monitoring	Not authorised. Arguably restricted if done <i>with the amateur station</i> [UNCERTAIN]
Recording and disclosing	Stricter. Annexure A para 8 adds a positive duty, breach of which is a licence matter on top of any OSA exposure [VERIFIED]
Official Secrets Act exposure	Unchanged. An ASOC is not a defence to anything at §9.5.4

The advice that follows: get the ASOC, and get it early. But do not treat it as a licence to listen to whatever you like — it is not, and it brings a secrecy obligation with it.

9.6 The Citizen Band verdict

Part 7 §7.6.1 recommended Citizen Band as the best licence-free convoy option, subject to one open question about antennas. Part 8 §8.5 held that recommendation alongside a market finding that CB is effectively dead in India, and Part 8 §8.7.4 made it a conditional Layer 3. **This section closes the open question. §9.7 states the consequences for those recommendations.**

The subject is the **President** range, for three reasons: it is the only manufacturer that has engineered an Indian band plan into its firmware; it is the brand Indian retailers actually stock; and a President set is buyable in India today. It is also, to be fair to the company before taking it apart, **the most conscientious actor in this market.**

9.6.1 THE RULE BEING APPLIED, AND WHERE THE BAND COMES FROM

G.S.R. 533(E), New Delhi, 2005-08-12, *Use of low power equipment in the Citizen band 26.957 – 27.283 MHz (Exemption from Licensing Requirement) Rules, 2005*, signed "[No. R-11014/04/2005-LR] **ASHOK KUMAR, Jt. Wireless Advisor**". Rule 3, verbatim:

"3. Use of wireless equipment in the band 26.957 – 27.283 MHz.- Notwithstanding anything contained in any law for the time being in force, no licence shall be required by any person to establish, maintain, work, possess or deal in any wireless equipment **intended to be used while in motion or during halts, on non-interference, non-protection and shared (non-exclusive) basis, in the frequency band 26.957 – 27.283 MHz with 5 Watt Effective Radiated Power and built-in antenna**"

and rule 2(b), verbatim:

"(b) 'Effective Radiated Power' includes the gain of the antenna, if any;"

[VERIFIED] — retrieved independently three times and recorded at Part 2 §2.5.2. Amended as to the interference remedy only by **G.S.R. 35(E) of 2007-01-10; the 5 W figure and the built-in-antenna condition are unchanged. [VERIFIED]**

Four conditions, all of which must hold simultaneously, plus type approval as a separate gate (Part 2 §2.2, §2.7.1):

1. Band **26.957–27.283 MHz** and nothing outside it
2. At most **5 W Effective Radiated Power**
3. A **built-in** antenna
4. **WPC Equipment Type Approval** for the model

Why India's band stops at 27.283 MHz — the ITU footnote 5.150 finding

This is new, it is genuinely illuminating, and it explains two things at once.

26.957–27.283 MHz, centre 27.120 MHz, 326 kHz wide, is the ITU Radio Regulations footnote 5.150 ISM band — designated for industrial, scientific and medical applications. **[REPORTED — ITU RR 5.150 as summarised across ITU-R Report SM.2255-0 and ITU short-range-device workshop material, searched 2026-08-24; the Radio Regulations text itself could not be extracted]**

Two consequences the reader should absorb:

(a) It explains why India's band stops where it does. India did not truncate the international CB band arbitrarily, and it did not choose 27.283 MHz to be awkward. **It de-licensed the ISM band, and nothing more.** The CEPT plan's top thirteen channels run above 27.283 MHz because the CEPT plan is not bounded by the ISM allocation; India's exemption is. That reframes the whole "thirteen missing channels" problem: they are not missing, they were never inside the thing India de-licensed.

(b) It explains the "non-interference, non-protection" wording, and this has real operational consequences. ISM applications — dielectric heaters, RF welders, 27 MHz diathermy machines, industrial plasma equipment — **have priority in that band.** A CB user in India must accept interference from them and has **no protection** against it.

That is not a formality. Near an industrial estate with RF welders or plastic-sealing machines, or near a hospital or physiotherapy clinic running 27 MHz diathermy equipment, a CB channel can be unusable and **you have no remedy and no standing to complain.** A convoy that stops for fuel beside a light-industrial area may find the radios simply do not work there, and the rule says that is your problem. Combine that with 27 MHz's existing susceptibility to vehicle ignition noise, mains electrical noise and long-distance skip interference — the reason you will sometimes hear a station 3,000 km away and nothing from the vehicle 2 km ahead — and the band's practical quality is worse than its power figure suggests.

Part 2 §2.5.2 caveat 3 flags a further, still-unresolved conflict in the same band: G.S.R. 1047(E) Table IX row 2 permits only **10 mW e.r.p.** across 26957–27283 kHz for a non-specific short range device, the 2018 rules contain **no supersession clause**, and both instruments remain on DoT's index as live. **[VERIFIED] If Table IX governed a 27 MHz voice transceiver, a 4 W set would be four hundred times over the limit. [UNCERTAIN]** — nothing in this part resolves it, and it is question 6 at §9.8.5.

9.6.2 GATE 1 — BAND

What President actually builds

President Electronics is a French manufacturer whose models are named after people. **Every current model is "European Multi-Norms"** — the firmware carries several national band configurations and the user selects one. President's own configuration table, published on both its European and its India-facing sites: **[VERIFIED — fetched 2026-08-25 and cross-checked against the European original]**

Configuration code	FM channels	AM channels	Countries
d	80 Ch (4 W)	40 Ch (4 W)	DE
EC	40 Ch (4 W)	—	AT, HU, LT, MT, NO, SK
EU	40 Ch (4 W)	40 Ch (4 W)	BE, BG, CH, CY, CZ, DK, EE, ES, FI, FR, GR, IE, IS, IT, LU, LV, NL, PT, RO, SE, SI
In	27 Ch (4 W)	27 Ch (4 W)	IN
PL	–5 kHz 40 Ch (4 W)	–5 kHz 40 Ch (4 W)	PL
U	CEPT 40 Ch (4 W) + ENG 40 Ch (4 W)	—	GB

And the channel-by-channel frequency tables behind those codes: **[VERIFIED — President's own frequency-table page, read from the page source]**

Band code	Channels	Frequency span	Relative to India's 26.957–27.283 MHz
BAND IN	1 – 27	26.965 – 27.275 MHz	entirely inside
BAND EU / E / EC / U (CEPT)	1 – 40	26.965 – 27.405 MHz	channels 28–40 = 27.285–27.405 MHz, outside
BAND PL	1 – 40	26.960 – 27.400 MHz	channels 29–40 = 27.290–27.400 MHz, outside
BAND D	1–40 as CEPT plus 41–80	adds 26.565 – 26.955 MHz	all forty of those channels outside, below
BAND U (ENG)	1 – 40	27.60125 – 27.99125 MHz	all forty outside, far above

Credit where it is due, because it is deserved. President's **In** configuration is not a marketing gesture. It is engineered to fit G.S.R. 533(E) exactly:

- **In** channel 1 = **26.965 MHz** — 8 kHz inside India's lower edge.

- **In** channel 27 = **27.275 MHz** — 8 kHz inside India's upper edge.
- The next channel up in the CEPT plan, channel 28, is **27.285 MHz** — **2 kHz above India's ceiling**. That is precisely why the India table stops at 27.
- The German plan's channel 80 is **26.955 MHz** — **2 kHz below India's floor**. The same knife edge, mirrored.
- Accounting for sidebands, an FM CB signal occupies roughly ± 5 kHz, so channel 1 reaches down to about 26.960 MHz and channel 27 up to about 27.280 MHz — **about 3 kHz inside each edge. [ENGINEERING ESTIMATE] The 27-channel plan is not a rough approximation of the ISM band; it is engineered to fit inside it. Somebody at President did this arithmetic properly.**

[ARITHMETIC on the margins, against the gazette figures at §9.6.1]

For completeness, the CEPT channels inside India's window — the ones a compliant set would use — are 1 to 27: 26.965, 26.975, 26.985, 27.005, 27.015, 27.025, 27.035, 27.055, 27.065, 27.075, 27.085, 27.105, 27.115, 27.125, 27.135, 27.155, 27.165, 27.175, 27.185, 27.205, 27.215, 27.225, 27.255, 27.235, 27.245, 27.265, 27.275 MHz. **[VERIFIED]** (Channels 23, 24 and 25 are out of numerical order — 27.255, 27.235, 27.245 — which is a historical artefact of the CEPT plan, not an error. All three are inside.) **A caution for anyone re-checking this: President's own website prints channel 14 of the India table as "27.25 MHz", which is impossible; every other President table, including the identical one in the Randy III owner's manual, gives 27.125 MHz. [VERIFIED — both documents read side by side.] Quote the manual, not the website.**

The bad news: the **In** setting is a menu item, not a hardware limit

The Randy III owner's manual gives the procedure in plain language: *"Switch off the transceiver — Turn on the power while pressing the F (7) key... In order to change the configuration, use the ▲ or ▼ key... long press on the F (7) key... confirm the selection by switching off then switching on again."* **[VERIFIED — owner's manual §F(1)]**

That is a five-second, no-tools, no-password operation, and the manufacturer prints it in the manual. So the same radio, unmodified, will transmit anywhere from **26.565 MHz to 27.99125 MHz** depending on which letter is on the display.

Does capability outside the band defeat the exemption? Two readings

Rule 3 exempts *"any wireless equipment intended to be used while in motion or during halts... in the frequency band 26.957 – 27.283 MHz..."*.

- **The use-based reading.** "Intended to be used... in the frequency band" describes the *use*, not the hardware envelope. A set left in **In** is being used inside the band and is therefore within the exemption; the fact that it *could* be reconfigured is no more disqualifying than the fact that a car *could* be driven at 200 km/h.

- **The capability-based reading.** The words attach to *"equipment ... in the frequency band"*, so equipment *capable* of transmitting from 26.565 to 27.99125 MHz is not equipment "in the band" at all, and the exemption never applies to it. The type-approval machinery points the same way: the parallel wording in the 2018 short-range-device rules requires equipment *"designed and constructed in such a manner that the bandwidth of emission and other parameters shall conform to the limits specified"* — **constrained hardware, not a menu.** [VERIFIED — Part 2 §2.7.1]

Which would be applied? [UNCERTAIN] — no WPC guidance, ETA condition, test method, circular or CB enforcement precedent addressing multi-norm sets could be found. But the only Indian enforcement evidence available points toward a **parameter-conformity** test at least as strict as the first reading and arguably supporting the second. DoT's 2021-12-28 public notice on PMR446 misuse says of type-approved devices found operating outside their notified parameters: *"the certificate will not be valid in case any change in the above parameters and not conforming to the relevant Gazette Notification; and ... use of such equipment in case not conforming to above notification will require a specific wireless operating license"*, with non-compliance leading to *"revocation of the ETA certificate issued and initiation of other legal proceedings"*. [REPORTED — Part 2 §2.6.3]

Practical conclusion, stated plainly. An intention to stay on channels 1–27 is not a legal answer to the fact that the set is a 40-channel radio with a documented, user-accessible key sequence for changing configuration — **a sequence President publishes on its own India website.** If you ever transmit on channels 28–40, you are outside the exemption **at that moment** and need "a specific wireless operating license" that a private individual will not have. And if the capability reading prevails, you were never inside it.

One consequence regardless of which reading is right, and it is worth planning around. If a Wireless Monitoring Organisation officer inspects the set, **the display shows the configuration letter.** If that letter is **In** you have a story. **If it is EU — the factory default on a European-market unit, and the setting that gives you 40 channels instead of 27 — you are transmitting outside the exempted band and the argument is over.** [ENGINEERING ESTIMATE] as to enforcement behaviour; the factual premise, that the display shows the letter, is [VERIFIED] from the manual.

And a hard fail for US-market sets, which is what a US retailer will ship you

The **RANDY III FCC** datasheet gives *"Frequency ranges: from 26.965MHz to 27.405MHz"*, 40 channels, and **lists no multiconfiguration row whatsoever.** [VERIFIED — datasheet dated 08-2025, text-extracted 2026-08-25] The same is true of **BILL III FCC**: SO-239 socket, 26.965–27.405 MHz, 4 W AM / 4 W FM, no multi-norm. [VERIFIED — product page] The same applies to JFK III FCC and JOHNNY III USA.

A US-market President cannot be set to India's band. Its channels 28 to 40 lie outside 26.957–27.283 MHz and cannot be switched off. [VERIFIED] If the reader buys from a US retailer, **that is what arrives.** (These sets also contain a 162.400–162.550 MHz NOAA weather receiver — receive-only, useless in India, and subject to \$9.5 like any other receiver. [VERIFIED])

Gate 1 verdict: passes on *use* only while the set is left in **In , and fails on *capability*.** President did the honest engineering work here and then sold it as a switch position on a 40-channel radio rather than as India-only hardware.

9.6.3 GATE 2 — POWER, AND WHY RULE 2(B) IS THE PROVISION THAT BITES

What the sets actually put out

4 W carrier, AM and FM, across the entire President range. [VERIFIED] from the manufacturer's own specification tables for eleven models. Two single-sideband models — **MC KINLEY (TXPR600)** and **GEORGE II** — add **12 W PEP on USB/LSB. [VERIFIED]** The Randy III + also has a HI/LO transmitting power key, but **the manual gives no figure for the low setting and none could be found anywhere. [UNCERTAIN]**

Several of the DIN-mount mobiles President markets to India are **1 W AM / 4 W FM** rather than 4/4 — Taylor III ASC, JFK II ASC, Truman ASC, Walker ASC, Johnson II. **[VERIFIED]** **So if the reader intends to use AM, which is the mode Indian truckers historically used, several President mobiles give only 1 W carrier.**

Carrier output power is not Effective Radiated Power, and this is the whole of gate 2

- **Carrier or conducted output power** is measured at the antenna socket, into a 50 Ω dummy load. It is what a specification sheet quotes. President: 4 W.
- **Effective Radiated Power** is what actually leaves the antenna, referenced to a half-wave dipole. **G.S.R. 533(E) rule 2(b) is explicit that it "includes the gain of the antenna, if any". [VERIFIED]**

The relationship:

$$\text{ERP (dBW)} = \text{transmitter output (dBW)} - \text{feeder loss (dB)} + \text{antenna gain (dBd)}$$

Now the numbers, **[ENGINEERING ESTIMATE]** throughout:

- **4 W = 6.02 dBW. 5 W = 6.99 dBW. The entire legal headroom is 0.97 dB.**
- **Feeder loss at 27 MHz is almost nothing.** RG-58 runs about 0.03 dB/m at 27 MHz, so a 5 m mobile run costs roughly **0.16 dB. Unlike UHF, you cannot rely on cable loss to bring you under the limit** — at 446 MHz the same cable would cost about ten times as much. \$9.2.2 flagged this in advance.
- **Therefore any net antenna gain above roughly +1.1 dBd puts a 4 W President set over 5 W ERP.**

Can a 4 W set exceed 5 W ERP with a gain antenna? Yes, easily — and with exactly the antennas a convoy would choose

Antenna on a 4 W set	Approximate gain	Approximate ERP	Inside 5 W?
Short loaded mobile whip, about 1 m, on a car roof	−8 to −13 dBd	0.2–0.6 W	Yes, comfortably
Full 2.59 m (102-inch) quarter-wave steel whip, good ground plane	−1 to 0 dBd	3–4 W	Yes, but close
5/8-wave mobile or base vertical, good ground plane	+1 to +3 dBd	5–8 W	No
Co-phased twin whips — the classic truck mirror-mount pair	~+3 dBd	~8 W	No
Base-station collinear	+3 dBd or more	8 W+	No

[ENGINEERING ESTIMATE] throughout — these are figures from antenna fundamentals, not vendor claims and not law.

So the answer is unambiguous: a 4 W set with a gain antenna can and does exceed 5 W ERP, and the antennas that do it are exactly the ones a serious convoy installation would fit. The margin is under 1 dB and ordinary antenna choices blow straight through it.

One unresolved point that cuts in the buyer's favour

The Wireless Monitoring Organisation appears to police the conducted figure, not a computed ERP. In the one documented enforcement drive, DoT recorded the WMO finding devices operated *"with higher (than permissible) transmit power of around 5 Watt"*. **[REPORTED — Part 2 §2.6.3]** If that is the practice for CB too, a 4 W set measures as compliant regardless of what antenna is on it. **[UNCERTAIN]** — no CB test method was found, and this is question 5 at §9.8.5.

And President's own India site misstates the Indian power rule

President's India FAQ says *"In India the maximum effective power for CB radios is 5 watts"* and, in its restrictions list, *"They have up to 5w AM/FM output power, license free"*. **[VERIFIED — fetched 2026-08-24/25]**

The gazette does not say 5 W output power. It says 5 W Effective Radiated Power, and rule 2(b) expressly folds antenna gain into it. Output power and ERP are different quantities, and President's formulation is **the more permissive of the two** for a set with a gain antenna. This is the same error Part 2 §2.5.2 and Part 8 §8.4.3 catalogue for PMR446 vendors — *"Maximum Permissible RF Output Power: 0.5 Watts"* against a 500 mW **e.r.p.** limit.

A manufacturer's restatement of an Indian rule is not the rule. Do not rely on the FAQ; rely on the gazette.

Gate 2 verdict: passes on AM/FM carrier power, with under 1 dB of headroom and no margin for a gain antenna. Fails on single sideband — 12 W PEP is 2.4 times the ceiling before any antenna gain at all. Note that President's own **In** configuration lists **AM and FM only, with no SSB entry**, and its India FAQ claims licence-free status for "5w AM/FM" only. **Treat MC KINLEY and GEORGE II as outside the exemption whenever they are in SSB mode.**

9.6.4 GATE 3 — BUILT-IN ANTENNA: THE DECISIVE CONDITION

The finding

Every President CB model surveyed has an external antenna connector. [VERIFIED] — read from the manufacturer's own specification tables for eleven European models, fourteen US models and twelve India-catalogue models:

- **Mobile sets** (BILL II +, BARRY II +, HARRISON II +, HENRY II, JFK III, JIMMY III, JOHNSON III, MC KINLEY, GEORGE II, and the whole India catalogue of DIN-mount sets): **SO-239 socket, 50 Ω**. Their accessory lists contain a microphone, a mounting bracket and a power cord — **no antenna at all**. The buyer supplies one.
- **Hand-helds** (RANDY III + TXPR613, RANDY III TXPR611, JERRY TXPR612, and the Randy II generation TXSE211 / TXSE311): **TNC female socket, 50 Ω**, with the flexible antenna listed as a supplied **accessory** and a **mobile adapter in the box** for connecting an external vehicle antenna.

The specimen that matters most is the Randy III +, because a hand-held is the only form factor that could conceivably have a built-in antenna. Its manufacturer specification: **[VERIFIED — product page fetched 2026-08-24]**

Field	Value
Article number	TXPR613
Channels / modes	40 / AM–FM
Frequency ranges	from 26.965 MHz to 27.405 MHz
Multiconfiguration	EU / U / D / PL / IN
Antenna socket	TNC
Antenna impedance	50 ohms
Carrier power	4 W (AM/FM)
Frequency allowance	± 200 Hz
Power supply	7.4 V handheld / 13.2 V mobile

Field	Value
Dimensions	152 × 66.5 × 37 mm
Weight	305 g with battery, <i>without antenna</i>
Ingress protection	IP54
Accessories supplied	Microphone support, 1 flexible antenna , belt clip, hand strap, Li-Ion battery, 1 mobile adapter , home charger, fuse, user manual

Three details on that page are individually fatal to a "built-in antenna" reading, and they are the manufacturer's own words:

1. **The weight is specified "without antenna".** The manufacturer treats the antenna as a separable item, not as part of the radio.
2. **The antenna is listed under "Accessories supplied", alongside the belt clip and the fuse.**
3. **A "mobile adapter" is in the box**, whose stated purpose is to replace the flexible antenna with an external vehicle CB antenna.

And the owner's manual makes it explicit in both directions. **Portable configuration, first line:**

"Screw the flexible antenna (C2) on the device."

Mobile configuration, antenna installation:

"- Remove the flexible antenna (C2) from the device. - Connect the CB antenna to the mobile charge module (C8). - Clip the mobile charge module (C8) to the bottom of the unit in any side."

and, on the warnings page:

"Never use two antennas together, otherwise the device may be destroyed."

[VERIFIED — Randy III + owner's manual, pages 5, 6 and 8, text-extracted 2026-08-25] That last warning **only makes sense for a radio that can be connected to an antenna other than the one in the box.**

The **EU Declaration of Conformity** treats the antenna as a **separately declared component** — *"Antenna (aerial) — Model: Flexible antena S-20"* **[VERIFIED — DoC for TXPR613 / TXPR611 / TXPR612, dated 2021-01-23]** — and President's accessory catalogue sells it separately, listing *"Rubber antenna (30 cm) (ACMS301)"* as an orderable option alongside an *"adapter with CB antenna and cigarette lighter plug (ACMS304)"*. **[VERIFIED]**

The connector is independently confirmed as **TNC female** by two European retailers **[REPORTED]**, and the aftermarket sells TNC antennas described as compatible with "President Randy, Randy II and Randy III" together with **TNC-to-PL-259 adapters** as the standard way to drive an external CB whip. **[REPORTED]**

And an Indian retailer sells the adapter. Fab.to.Lab lists a "TNC/PL Adapter" at ₹470.00, in stock. [REPORTED — fetched 2026-08-25] That is the exact part that converts the Randy III's TNC socket to the PL-259 world of roof-mount CB whips, and its existence on an Indian shelf next to the radio is the clearest possible demonstration that this antenna is not built in.

One variant deserves separate mention because it fails harder. The RANDY II M (TXSE311) — one of only two hand-helds on President's India-facing catalogue — **does not ship with a rubber antenna at all.** Its supplied accessory is the **external-antenna adapter.** [VERIFIED — India catalogue product sheet] President sells, to India, a hand-held CB whose supplied configuration *requires* an external vehicle antenna.

The strongest argument the other way, and why it does not survive

The reader deserves the best available argument for "built-in", not a straw man.

For "built-in": the S-20 is the antenna the set is supplied with; it is physically attached during all normal hand-held use; the radio is type-tested and its radio-frequency exposure assessed *with that antenna* (the Declaration of Conformity cites EN 62209-2 and EN 50566, the body-worn exposure standards, which is precisely why the antenna is a declared component); the set will not transmit safely without it, and the manual warns against transmitting with no antenna. On that view the removability is an engineering convenience for the mobile mode, and the "built-in antenna" condition was aimed at stopping people bolting a gain antenna or an amplifier onto a hand-held to exceed 5 W ERP — which the S-20 plainly does not do.

Against "built-in", and this is where the analysis lands. Four points, and the fourth is the one that closes it:

- 1. The rule says what it says, and it is conjunctive.** "...with 5 Watt Effective Radiated Power **and** built-in antenna". Part 2 §2.5.2 already recorded the conclusion in terms: *"Built-in antenna' is a hard condition. A CB set with an SO-239 socket or a mag-mount whip falls outside the exemption on its face."* [VERIFIED]
- 2. The socket is a general-purpose 50-ohm RF connector, not a captive fitting.** TNC is a threaded coaxial connector with a defined impedance and an entire ecosystem of adapters — one of which is on an Indian shelf at ₹470.
- 3. The manufacturer documents, supplies and sells the external-antenna path.** A mobile adapter is *in the box*. The manual's very first operating section is how to fit an external CB antenna and adjust its SWR. That is not an incidental capability; **the manual's own cover calls the product a "Portable or Mobile CB Radio".** [VERIFIED]
- 4. Rule 2(b) destroys the purposive argument.** G.S.R. 533(E) already defines ERP to *"include the gain of the antenna, if any"* — **so the drafters handled antenna gain through the ERP definition. If "built-in" were merely a proxy for gain control it would be redundant.** Read as a separate condition it does separate work: **it confines the exemption to self-contained sets.**

The decisive new evidence: India said *why* it requires integral antennas — thirteen years later

Parts 7 and 8 both left this as market-practice-versus-gazette-wording, tagged [UNCERTAIN]. **It can now be pushed further, and it goes against the loose reading.**

India applied the **identical architecture** to PMR446 in 2018 and, this time, **stated the purpose in the instrument**. G.S.R. 1047(E), Table V note, verbatim:

"**Note:** For the purpose of this Table, personal mobile radio 446 MHz device means hand portable radio with no base station or repeater use and **uses integral antennas only in order to maximise sharing and minimise interference**, and which operates in short range peer-to-peer mode and shall be used neither as a part of infrastructure network nor as a repeater;"

[VERIFIED — Part 2 §2.5.1]

That note states the purpose of the integral-antenna requirement in terms: to maximise sharing and minimise interference. It is the same regulatory logic, on the same non-interference, non-protection, shared basis, as the 2005 CB rule.

So "built-in antenna" in G.S.R. 533(E) is not a drafting slip and not an accident of 2005. It is a deliberate and consistent instrument of Indian licence-exempt policy across at least thirteen years, whose function is to cap real radiated power and stop users bolting on gain antennas and masts in a shared band.

And "while in motion or during halts" does not mean what the vehicle-installation reading needs

The obvious objection to all of this is that rule 3 expressly contemplates equipment *"intended to be used while in motion or during halts"* — wording that plainly means vehicles — while demanding a built-in antenna, which at 27 MHz forecloses every practical vehicle installation. **On its face the rule looks drafted for the exact use case it then makes impossible.** Part 7 §7.6.1 and Part 8 §8.5.6 both rested on that objection.

It does not hold, once you see what the phrase is doing. Indian station licensing is **address-bound**: rule 7(2) of the 2024 Amateur Services Rules ties a station to a specified location, and Part 7 §7.2.1 traces the whole architecture. **The function of "while in motion or during halts" in rule 3 is to distinguish mobile and portable use from a fixed station at premises, so that someone using a radio on the move does not need a station licence tied to an address. It describes the manner of use. It is not a permission to drill a roof mount.**

Read that way, the two conditions are entirely coherent and the rule is not self-defeating at all:

G.S.R. 533(E) permits a self-contained hand-held CB set, with a non-detachable antenna, operating within 26.957–27.283 MHz at not more than 5 W ERP, carried and used in or around a vehicle while travelling or stopped. It does not appear to permit a vehicle-installed CB with an external whip and a coaxial feed.

That reading excludes essentially every serious CB installation — including the entire Indian retail CB proposition and everything President's own India FAQ describes. And it is worth noting how directly it contradicts what President tells Indian buyers. On the same site that asserts Indian ETAs, its FAQ gives installation guidance: *"The best place to mount the antenna is in the center of the roof of the vehicle"; "There are a number of different types and styles of antennas for mobile or stationary installations, including magnetic mounts, gutter lip mounts, trunk lip mounts, mirror mounts and permanent mounts"; "Most mobile antennas are actually only half of the effective antenna, the other half is the body of the car or truck."* **[VERIFIED] Every one of those is an external antenna. President's India site tells Indian buyers, in detail, how to do the one thing rule 3 appears to forbid — a few paragraphs from its assertion that its portfolio is WPC-approved.**

The 5 W ceiling is unreachable by compliant hardware — the finding that reframes everything

Now run the arithmetic in the other direction, for a set that actually satisfies condition 3.

A built-in antenna at 27 MHz cannot be resonant. λ at 27.12 MHz is **11.05 m**; a quarter wave is **2.76 m** (§9.2.2). Nothing built into a 152 mm hand-held is a meaningful fraction of that. A short helical of a few centimetres, working against a hand-held's tiny ground and the operator's body, is realistically **–13 to –20 dBd**. **[ENGINEERING ESTIMATE]**

With 4 W into it, that is 0.04 to 0.2 W ERP — against a 5 W ceiling. The permitted maximum is between 25 and 125 times what the compliant hardware can actually radiate.

Compare PMR446: **500 mW e.r.p.** with an integral antenna at 446 MHz, where a quarter wave is 16.8 cm and a hand-held whip is a genuinely efficient radiator — perhaps **–3 to –6 dBd** after body effects, so **0.12 to 0.25 W actually radiated**. **[ENGINEERING ESTIMATE]**

A fully compliant Indian CB set radiates about the same as, or less than, a compliant PMR446 handset — on a band with far worse man-made noise, ISM interference it must accept without protection, and skip.

Part 2 §2.5.2, Part 2 §2.6.1, Part 7 §7.6.1 and Part 8 §8.5.2 all rank CB first among licence-free options at "5 W ERP, ten times PMR446". That ranking compares a ceiling nobody can lawfully reach against a limit that real hardware actually meets. The 5 W figure is only attainable with an external gain antenna — **which is precisely what condition 3 forbids**. §9.7 states what that does to the earlier recommendation.

And there is a bitter irony worth stating once. India's licence-exempt corpus offers a choice between **PMR446**, where the integral-antenna condition matches what manufacturers already build, so compliant hardware genuinely exists and is genuinely licence-free — at 500 mW; and

Citizen Band, nominally ten times the power, where the equivalent condition matches nothing anyone builds, so compliant hardware appears not to exist — **and where a compliant set would radiate no more than the PMR446 handset anyway. The higher-power allocation is the one you cannot lawfully equip, and it would not actually be higher-power if you could.**

Gate 3 verdict: FAILS, for every President model surveyed, and most obviously for the mobiles. It is not curable by the buyer: there is no way to make a radio with an antenna socket into a radio with a built-in antenna.

What can fairly be said on the other side

This will not be overstated. Four points cut the other way and none is worthless:

1. **No authoritative interpretation exists.** G.S.R. 533(E) contains no definition of "built-in antenna" — rules 1 to 4 are short title, definitions, exemption and interference. **No WPC clarification, test method, guidance note, ETA condition or CB enforcement precedent construing the phrase could be found. [UNCERTAIN]**
2. **President asserts it holds Indian ETAs for a portfolio in which every model has an external antenna socket (§9.6.5).** If WPC genuinely type-approved SO-239 mobiles for the Indian CB band, then **WPC itself may read the condition loosely** — or the approvals may be scoped in some way not visible from outside. Either way it is an unverified assertion by an interested party and it does not bind an enforcement officer. **[REPORTED]**
3. **Indian retailers sell mobile CB sets openly** as "licence free european make". **[REPORTED]** — Part 8 §8.5.5.
4. **Field practice records no enforcement.** Part 8 §8.5.2 records a Team-BHP user running a vehicle CB across Jammu and Kashmir and Rajasthan who declared it at a Border Security Force post near Tanot and was told only not to use it beyond the post. **[REPORTED]** And **no Indian prosecution or seizure on the built-in-antenna point could be found anywhere. [UNCERTAIN]**

The gazette is primary. Points 2 to 4 are market practice. Where they conflict, the gazette wins, and the reader carries the risk. And note the asymmetry of that risk: **if the reader is wrong about this, the consequence is seizure of equipment and exposure under the Telecommunications Act 2023. If the reader is right, the gain is thirteen extra channels and a slightly better antenna. That is a bad trade to take on an untested reading.**

9.6.5 GATE 4 — EQUIPMENT TYPE APPROVAL

That ETA is required

The authoritative statement is WPC's Compendium, File No. R-11018/02/2017-PP dated 2022-07-06, §2.1: licence-exempt band use is allowed subject to the technical parameters, *"however, equipment complying with such notifications need to be type approved and accordingly Equipment Type approval (ETA) certificate is issued to such compliant RF devices". [VERIFIED — Part 2 §2.7.1]*

Licence-exempt is not approval-exempt. Part 1 §1.5 and Part 8 §8.6.1 both record this as the single most common Indian confusion in the area — "*CB... Comes with licence paper from WPC is what i understand*" is wrong twice over: there is no user licence to issue, and what a compliant set needs is an **ETA**, which certifies the **equipment**, not the user.

One wrinkle specific to CB, already flagged at Part 2 §2.5.2 caveat 4 and still open. Unlike G.S.R. 1047(E), which has an express type-approval rule at rule 5(2), **G.S.R. 533(E) contains no type-approval rule of its own** — rules 1 to 4 are short title, definitions, exemption and interference. The obligation therefore arrives via the general regime rather than from the CB rules themselves, **which makes the chain less tidy for CB than for PMR446. [UNCERTAIN]** — question 7 at §9.8.5.

What President claims, and what could not be verified

President's India FAQ, verbatim: **[VERIFIED as to the text on the page]**

"Do I need a licence to use a CB radio? No. In the Indian Market usage of CB Radios that comply with the applicable WPC requirements is allowed without a license. The importer or manufacturer however must have an ETA from the WPC in order to sell CB radios in India. **President has received ETAs for its CB portfolio in India and would be pleased to provide this information if needed.**"

and:

"What are the CB radio power and frequency requirements in India? In India the maximum effective power for CB radios is 5 watts and the frequency band for CB Radio ranges from: **26.957 to 27.283 MHz.**"

and:

"Is there any license I can get or buy that will allow extra CB channels or higher power? No. CB radios in India have the following restrictions: - Operate only on the 26-27 Mhz CB channels, - **They must be WPC type approved for CB use**, - They have up to 5w AM/FM output power, license free For usage outside the above parameters the WPC must be consulted."

Give President credit: that is a materially accurate statement of Indian law, it names the band correctly, and it states the ETA duty plainly — which is more than any Indian seller in this research manages. The one error is the "output power" formulation at §9.6.3.

What could not be established, and it is all of gate 4:

1. **No ETA number appears anywhere.** The India site's download library offers manuals, photographs and EU declarations of conformity only — **no Indian ETA certificate for any model. [VERIFIED — download index fetched 2026-08-25]**

2. **The India catalogue contains no Randy III at all.** President's India-facing AM/FM catalogue is **twelve products** — RANDY II M, RANDY II P, HENRY ASC Classic, TEDDY ASC, TRUMAN ASC, ANDY, HARRY III ASC, TAYLOR III ASC Classic, WALKER ASC Classic, WALKER ASC Classic Chrome, JOHNSON II ASC, JFK II ASC — and its FAQ says in terms: *"President offers the Randy II model in handheld/portable format for outdoor use."* **[VERIFIED]** An ETA is **granted per model**, at ₹10,000 per application, to an Indian entity or authorised representative (Part 2 §2.7.3). **An ETA for a RANDY II P (TXSE211) is not an ETA for a RANDY III + (TXPR613)** — different model, different article number, different declaration of conformity, different generation.
3. **The India site is stale, which weakens the claim further.** Its newest general catalogue is *"General catalogue 2015"*, and the two "III" models it does carry are 2012–2015 vintage. **[VERIFIED]** Part 8 §8.5.5 recorded the same staleness from a different direction.
4. **WPC's public register could not be queried.** Saral Sanchar's *"List of Issued ETA (Self-Declaration)"* responded HTTP 200 on 2026-08-25 and its form carries **CAPTCHA fields**. **[VERIFIED]** It cannot be scripted and must be searched by hand. Part 2 §2.11.1 also notes the equipment-category dropdown has **no walkie-talkie, two-way radio, PMR or CB category**, so a CB set would be filed under "Other" — **which makes a by-category search useless and a by-model search essential**.
5. **One number on President's site could be mistaken for an Indian approval and is not.** The only certification number the company publishes is *"Authorized Economic Operator FR00159672"* in its site footer — **an EU customs authorisation, nothing to do with Indian type approval**. **[VERIFIED]**

Note also the **scope** problem, which survives even if an ETA exists: "ETAs for its CB portfolio" says nothing about *which* models, *which* configurations, or whether an approval covers a 40-channel multi-norm set or only the **In** configuration. **An ETA is granted against declared parameters, and DoT's 2021 notice says the certificate "will not be valid in case any change in the above parameters".** **[REPORTED]**

Gate 4 verdict: UNVERIFIED, and leaning negative for a Randy III specifically. President plausibly holds ETAs for the older models it markets to India. **There is no evidence it holds one for a Randy III or Randy III +, and its own India catalogue's silence is meaningful evidence that it does not.** **[UNCERTAIN]**

What the reader must actually do, if they pursue this at all:

1. Ask the seller, **in writing**, for the ETA number of the **exact model and article number** and the **exact configuration** being supplied, plus the name of the ETA holder.
2. Look that number up **by hand** on the Saral Sanchar register.
3. Check that the model string on the certificate matches the box.
4. Keep the printout, and carry it with a printed copy of G.S.R. 533(E) — Part 8 §8.6.3 records that this is the one piece of community field craft everyone agrees on: *"Its the document that makes the difference really."*

An unverifiable "we have ETAs" is marketing, not compliance evidence.

9.6.6 WHAT IS ACTUALLY ON SALE IN INDIA, AND ONE LISTING TO AVOID

All fetched 2026-08-24/25, all **[REPORTED]** — these are vendor pages, not official sources.

Seller	Listing	Price	Stock	Licence claim on the page
Fab.to.Lab	"CB Radio - President RANDY III", code Pres-RANDY-III	₹18,100	In Stock	"President RANDY III Citizen's Band Radio now in India..No license required !"
Fab.to.Lab	"CB Radio - President Jimmy III"	₹5,602	In stock	—
Fab.to.Lab	"President TOMMY ASC"	₹4,330	In stock	"No license required"
Fab.to.Lab	"CB Radio - BILL II ASC"	₹7,969	In stock	"No license required"
Fab.to.Lab	"TNC/PL Adapter"	₹470	In stock	—
The Engineer Store	"CB Radio - President RANDY III", SKU TES-EV02356	₹10,224.76 (list ₹10,754.76)	3–5 working days	none stated
The Engineer Store via IndiaMART	"Cb Radio - President Randy Iii", Bengaluru	₹9,646	out of stock	none stated
NPC Wireless, 27 MHz CB collection	President Bill II	₹9,500 (from ₹12,000)	Sold out	"licence free european make"
NPC Wireless	President Tommy ASC	₹5,799 (from ₹8,000)	Sold out	"27 Mhz licence free"
NPC Wireless	Luiton 27 MHz CB base station	₹6,999 (from ₹12,999)	Sold out	"licence Free"
Manas Rekha, Mumbai, self-described President importer	"Randy Walky Talky"	none published	unknown	"CB radios do not require any license for purchase or usage in India"
Passion Radio, France, for comparison	"Walkie-Talkie 27Mhz President RANDY-3 4W AM FM"	€182.50 excl. tax	—	"Can be used without a licence" — that statement is about France, not India

So yes: Indian listings do claim licence-free status, and yes, a Randy III is buyable in India today. Four warnings before any money moves.

Warning 1 — The Engineer Store's "President RANDY III" listing is not a Randy III. Its description text opens "*President Electronics WALKER II FCC AM Transceiver*" and its entire specification block is that of a **DIN-mount mobile set**: "*Frequency ranges: 26.965 MHz to 27.405 MHz / Weather channels: 162.400 MHz to 162.550 MHz / Antenna impedance: 50 ohms / Power Supply: 13.8 V / Size (WxDxH): 170 × 160 × 52 mm / DIN Size*", with a six-pin front microphone plug, a channel rotary switch and public address. **[VERIFIED — read from the fetched page source]** A Randy III + is 152 × 66.5 × 37 mm, 305 g, battery-powered, with no rotary switch and no PA. The IndiaMART mirror of the same seller carries the same wrong specification. **Two platforms, one seller, one copy-paste error, and the cheapest "Randy III" price in India. Do not order from a listing whose specification describes a different product.** Part 8 §8.5.5 flagged the same listing for a different defect — a US band plan and no Indian compliance statement.

Warning 2 — the price spread is a signal. €182.50 ex-VAT in France, ₹18,100 at Fab.to.Lab, ₹9,646–₹10,225 at The Engineer Store. At any plausible 2026 rate the French ex-VAT price and Fab.to.Lab's are in the same region, while The Engineer Store's is **roughly half — and that is the listing carrying the wrong specification.** **[UNCERTAIN]** as to why; no intent is inferred. **Ask what you are actually being sold.**

Warning 3 — no Indian listing found states an ETA number, a band configuration, or the antenna connector. Not one. The "No license required" claims are bare assertions with no supporting document, and none of them mentions the built-in-antenna condition.

Warning 4 — the category is still dead, and this re-check confirms Part 8 §8.5.5 has not improved. NPC Wireless, the Indian retailer that promotes 27 MHz CB hardest, showed **four products in its 27 MHz CB collection on 2026-08-24, every one of them "Sold out"**, at heavy discounts off list. **[VERIFIED — counted from the fetched collection page]** No Randy of any generation appears there, and its collection blurb still carries the "(3 KMS TO 160 KMS)" claim Part 8 §8.6.1 told the reader to ignore.

One more identification point, for the reader who came to this asking about a "President 3". There is no product called "President 3". President names sets after people and appends generation numerals, so a "President 3" is a third-generation *something* — and **the only third-generation hand-held is the RANDY III, most likely the current RANDY III + (TXPR613).** Two European retailers brand the URL slug "randy-3", so "Randy 3" is ordinary trade shorthand. **[VERIFIED]** For completeness: **there is no BARRY III** — the Barry line stops at BARRY II + (TXPR122) **[VERIFIED as absent from the catalogue]** — and **BILL III exists only in the US FCC line**, which has no India setting (§9.6.2). And the single most important thing to ask Fab.to.Lab is **whether their ₹18,100 unit is the European multi-norm Randy III / Randy III + or a US-market Randy III FCC**, because the listing gives no article number and **that one fact decides gate 1.** **[UNCERTAIN]**

9.6.7 THE FOUR GATES, AND THE HONEST CONCLUSION

"Legal" and "licence-free" are not the same question

They are routinely conflated, including by sellers. Keep them apart:

- **Licence-free** asks: do I need an operator or station licence to transmit? Under G.S.R. 533(E) the answer is no — **but only if all four of rule 3's conditions hold**. Fall outside any of them and DoT's position is that you need *"a specific wireless operating license"*. **[REPORTED — Part 2 §2.6.3] There is no middle ground between "inside the exemption" and "needs a licence"**.
- **Legal** asks a bigger question, across the four independent gates of Part 2 §2.2: type approval, import, possession, operation. A device can be licence-free to operate and still unlawful to import, or approved and still unlawful to use in a given configuration.

The four gates for a President CB set

Gate	Governing instrument	Answer
1. Buying	The ETA obligation binds the manufacturer, importer and seller — an individual cannot apply for an ETA (Part 2 §2.7.3)	Not an offence by the buyer. The ETA and dealer duties sit upstream. But you should refuse to buy without a verifiable ETA number — and in practice the stock mostly is not there anyway (§9.6.6)
2. Possessing	REPA, G.S.R. 592(E), 2026-07-08, rules 4 and 10	Follows the operation gate. A set genuinely within G.S.R. 533(E) is covered by the rule 10 exemption chain. A set outside the exemption — because of the antenna, the band capability or the power — is not obviously covered, and rules 4(1) and 4(2) provide no category a private individual can apply under. Instrument [VERIFIED] ; the consequence is my reading, [UNCERTAIN]
3. Importing	Notification 71-Customs of 1953-09-25; DGFT policy	No clean route for a personal import. ETA'd de-licensed-band devices move on a system-generated import undertaking obtained by the manufacturer or Indian agent, not by an individual. Buying a President set in Europe and carrying it in leaves you with no ETA in your name and no undertaking. Framework [VERIFIED] ; outcome [UNCERTAIN] . Treat as high seizure risk — Part 2 §2.8.4
4. Operating	Telecommunications Act 2023 s.4; G.S.R. 533(E) as amended	Licence-free only inside rule 3. With an external antenna — every model sold — you are outside it. Also outside it on channels 28–40, on the German 41–80, and on SSB at 12 W PEP. Outside the exemption means transmitting without authorisation, exposed to the Third Schedule penalties (Part 6 §6.1) and to the interference kill-switch in rule 4 as amended

The verdict on President specifically

A President CB radio is not demonstrably legal for a private individual in India, and it is not safely licence-free.

- **Condition 3, built-in antenna, fails outright on every model surveyed** — SO-239 on the mobiles, TNC on the hand-helds. **This is the decisive failure, and it is not curable by the buyer.**
- **Condition 1, band, fails on capability.** Every model is a 40-channel 26.965–27.405 MHz radio with a user-accessible configuration switch. It passes on *use* only while parked in **In**, and an intention to stay there is not a legal answer.
- **Condition 2, power, passes on AM/FM** at 4 W carrier, with less than 1 dB of headroom to the 5 W ERP ceiling and no room for a gain antenna — **and fails on SSB** at 12 W PEP.
- **Condition 4, ETA, is unverified.** Claimed by the manufacturer, no number published, register CAPTCHA-gated, model absent from the India catalogue.

President is the most conscientious actor in this market and it deserves saying plainly: it engineered a correct 27-channel Indian configuration when it did not have to, it publishes the channel plan, it tells buyers not to use the wrong configuration, and it states Indian law and the ETA obligation more accurately than any Indian seller in this research. None of that fixes the antenna socket.

Is there ANY CB set a private individual can lawfully buy and use licence-free in India today?

Honest answer: none could be identified, and the built-in-antenna condition appears to make compliant CB practically unobtainable.

A compliant set would have to be:

1. A self-contained **hand-held** — a DIN-mount mobile cannot have a built-in antenna by construction;
2. With a **permanently attached, non-detachable** antenna — no TNC, no BNC, no SMA, no SO-239;
3. Transmitting **only** within 26.957–27.283 MHz — no channels 28–40, no German 41–80, ideally no configuration switch at all;
4. At **≤ 5 W ERP**, which a built-in 27 MHz antenna guarantees by a wide margin anyway (§9.6.4);
5. Holding a current **Indian WPC ETA** for that exact model.

Why nobody builds it. The world CB industry builds to the CEPT and FCC 40-channel plans with standard antenna connectors, because that is what every significant market requires. **India's 27-channel ISM-bounded plan with a sealed antenna is unique, and the Indian CB market is tiny.** There is no commercial reason for any manufacturer to tool a 27-channel,

sealed-antenna, India-only hand-held. President came closer than anyone — it built the **In** configuration into the firmware — **but it did not build India-only hardware. It built a 40-channel multi-norm radio with a switch and a TNC socket.**

No CB radio, from any manufacturer, advertised as having a non-detachable antenna could be found. Every CB hand-held examined uses a removable whip on an RF connector, because CB users universally want to connect a vehicle antenna — **which is the entire point of the form factor and the entire problem under Indian law. [UNCERTAIN — this is a negative finding from searching, not proof]**

And it explains Part 8's market finding rather than contradicting it

Part 8 §8.5.5 concluded that CB in India is *"a legal allocation with a channel plan, an encyclopedia article, a handful of out-of-stock SKUs, and no community"*, and Part 8 §8.5.3 recorded the Bangalore off-road club that tested CB and abandoned it on performance grounds in wooded, quarried terrain.

Those two findings and this one are the same phenomenon seen from three sides, and together they explain it better than any of them does alone:

- **Part 8 found the demand side dead:** 27 MHz performed badly in the hilly, wooded terrain Indian off-roaders actually drive, so the clubs that tried it walked away.
- **Part 8 found the retail side dormant:** every Indian SKU out of stock, at discounts, on the site of the retailer that promotes the category hardest.
- **This part finds the supply side legally foreclosed:** the only form factor the rule as written permits — a sealed-antenna 27 MHz hand-held — is **the worst possible implementation of an already marginal band**, radiating a fraction of a watt from a stub antenna, and nobody manufactures it. **The sets that would actually work are the ones with roof-mount whips, and those appear to fall outside the exemption.**

A rule that permits only the version of a technology that does not work will not produce a market. That is a more complete explanation of India's dead CB category than "no demand", and it predicts that the category will stay dead unless WPC either clarifies the antenna condition permissively or amends it.

The honest caveat, stated once and not buried

This is statutory interpretation, not a finding of fact. No Indian authority, court, ETA condition or published clarification construing "built-in antenna" could be found. **[UNCERTAIN]** No Indian buyer has been found prosecuted or had CB equipment seized on the point. **[UNCERTAIN]** **And if WPC reads the condition loosely — which its apparent grant of ETAs to SO-239 mobiles might suggest — the entire market position in this section changes.**

Which is exactly why §9.7 does not say "Parts 7 and 8 were wrong". It says the question they left open now looks very likely to resolve against them, and it identifies the one body that can settle it.

9.7 The correction to Parts 7 and 8

This section exists because **Part 7 and Part 8 both recommend Citizen Band as the convoy answer, and the research in §9.6 shows that recommendation should not stand in the form they gave it.** A report that quietly changes its mind between parts is worse than useless to someone spending money, so the change is set out here squarely: what was said, what is now known, what survives, what does not, and what it turns on.

9.7.1 WHAT PARTS 7 AND 8 ACTUALLY SAID

Part 7 §7.6, option A, in the table of every lawful way to get voice between vehicles:

"**Citizen Band, 26.957–27.283 MHz** | Licence needed: **None** | Power ceiling: **5 W ERP** | ₹5,800–₹12,000 per set plus antenna | **Best licence-free option.** Highest legal licence-free power in India, and the rule was written for vehicles"

Part 7 §7.6.1 developed that at length: *"Why this is the right instrument for a convoy... This exemption was drafted for vehicles... It is also five watts — ten times PMR446 — and it is the highest-power licence-free two-way voice allocation in the entire Indian de-licensed corpus."* It then flagged the antenna problem honestly — *"The problem that must not be glossed: 'built-in antenna'"* — set out the arguments both ways, and tagged the whole thing **[UNCERTAIN]**, directing the reader to **§7.7.2 question 4** before spending money.

Part 7 §7.1.2, step 2 of the recommended plan, put money behind it:

"**Buy licence-free radios for the convoy now,** in parallel with step 1. **Citizen Band at 5 W ERP** if the antenna question at §7.6.1 resolves acceptably; ETA-approved PMR446 at 500 mW otherwise or as well."

Part 7 §7.7.3 repeated it in the closing summary: *"The lawful convoy answer exists today and costs less than the UV-32. CB at 5 W ERP, subject to the built-in-antenna question..."*

Part 8 §8.5 then reconciled Part 7's legal finding with a field finding that Indian CB is effectively dead, and — importantly — **kept Part 7's legal conclusion intact:**

"**Is it still true? Yes, entirely. Nothing in this part disturbs it.**"

Part 8 §8.5.2 restated it: *"It is the highest-power licence-free two-way voice allocation in India, ten times PMR446... The physics still favours it for this job."* Part 8 §8.5.6 carried the antenna question forward as **Trap 2**, still **[UNCERTAIN]**, still pointing at Part 7 §7.7.2 question 4. Part 8 §8.5.7 built a **seven-point test** for whether to pursue CB, and Part 8 §8.7.4 made CB **Layer 3** of the recommended architecture — conditional, four-wheelers only, ₹21,000 for two vehicles. Part 8 §8.7.9 point 3 summarised the position as: *"Citizen Band remains the best licence-free multi-kilometre option in law and physics, and it is effectively unbuyable and unproven in broken terrain. Both statements are true."*

So the position across both parts was: CB is legally the best licence-free option; it is practically hard to buy; and there is one open legal question, which the reader should put to WPC.

9.7.2 WHAT THE NEW RESEARCH CHANGES

Four findings, and the first is the one that matters.

(a) The built-in-antenna condition appears to exclude essentially every practical vehicle CB installation — and it is not a drafting accident. Parts 7 and 8 treated the condition as a puzzle: the rule contemplates use "while in motion", which is hard to reconcile with a built-in antenna at 27 MHz, so market practice and gazette wording "do not line up". **§9.6.4 closes that puzzle, and it closes it against the loose reading, on four grounds:**

- 1. Every CB radio surveyed, from every manufacturer, has an external antenna connector — SO-239 on mobiles, TNC on hand-helds — and no CB radio advertised with a non-detachable antenna could be found anywhere.** The manufacturer specifies the radio's weight "*without antenna*", lists the antenna under "*Accessories supplied*", sells it separately under its own order number, puts an external-antenna adapter **in the box**, and an Indian retailer stocks the **TNC-to-PL adapter at ₹470. [VERIFIED]**
- 2. Rule 2(b) already handles antenna gain by folding it into ERP. If "built-in" were merely a proxy for gain control it would be redundant.** Read as a separate condition, it confines the exemption to self-contained sets. **[VERIFIED as to the texts; the inference is reasoning]**
- 3. India stated the purpose of the identical condition thirteen years later.** G.S.R. 1047(E) Table V's note says PMR446 devices use "*integral antennas only in order to maximise sharing and minimise interference*". **[VERIFIED] That makes the integral-antenna requirement a deliberate and consistent instrument of Indian licence-exempt policy, not a 2005 slip.**
- 4. "While in motion or during halts" does not do the work Parts 7 and 8 asked of it.** Indian station licensing is address-bound; the phrase distinguishes *mobile and portable use* from a *fixed station at premises*. **It describes the manner of use. It is not a permission to drill a roof mount.**

(b) This is a stronger objection than Part 8's market-availability finding, and the two must not be confused. Part 8's objection was *you cannot buy one and it does not work in your terrain* — a statement about stock and physics, which better stock or flatter terrain would answer. **This objection is that the standard vehicle installation appears to fall outside the exemption on the face of the rule — which no amount of shopping, and no change of route, fixes.** Part 8 §8.5.7's "Do not pursue CB if..." list has five entries, all of them practical. **A sixth now sits above them and is categorical.**

(c) The 5 W ceiling is unreachable by compliant hardware, so the headline comparison was wrong. A built-in antenna at 27 MHz radiates on the order of **0.04 to 0.2 W ERP** from 4 W of carrier — **the same as or less than a compliant PMR446 handset**, on a noisier band with ISM interference it must accept without protection. **[ENGINEERING ESTIMATE] §9.6.4. "Five watts, ten times PMR446" compares a ceiling nobody can lawfully reach against a limit that real hardware actually meets.**

(d) Band capability is a second, independent problem that Parts 7 and 8 under-weighted.

Part 8 §8.5.6 Trap 1 correctly identified that channels 28–40 fall outside the Indian band and told the reader to ask for a set that can be band-limited. §9.6.2 shows that what is actually on sale is a 40-channel radio with a five-second, manual-documented key sequence for changing configuration — and that on the capability reading of rule 3, which India's only relevant enforcement evidence tends to support, that may defeat the exemption regardless of which setting the radio is in.

9.7.3 WHAT SURVIVES OF THE EARLIER RECOMMENDATION, AND WHAT DOES NOT

Stated as two lists, because a reader deciding what to buy needs to know which sentences to keep.

Survives, unchanged

Finding	Where it was stated	Status
G.S.R. 533(E) is real, in force, and de-licenses 26.957–27.283 MHz at 5 W ERP	Part 2 §2.5.2, Part 7 §7.6.1	Unchanged. [VERIFIED]
5 W ERP is the highest <i>nominal</i> licence-free two-way voice figure in the entire Indian de-licensed corpus	Part 2 §2.6.1	Unchanged as a statement about the notification. §9.6.4 changes what it means in practice, not what the ladder says
There is no licence-free 10 W anywhere in India, in any band, under any notification in force	Part 2 §2.6.2, Part 8 §8.5.7	Unchanged, and now reinforced. [VERIFIED]
The rule's wording contemplates mobile and portable use — <i>"while in motion or during halts"</i>	Part 7 §7.6.1	Unchanged as text. §9.6.4 corrects what it means
The physics of 27 MHz genuinely favours the job — longer antenna, better diffraction at 11 m wavelength than at 67 cm	Part 7 §7.6.1, Part 8 §8.5.2	Unchanged — but only reachable with an external antenna, which is the problem
Part 7 §7.6.1's range estimates were not optimistic, and the Bangalore club's field results land inside them	Part 8 §8.5.4	Unchanged. [ENGINEERING ESTIMATE / REPORTED]
Part 8's market finding: CB is effectively dead in India	Part 8 §8.5.5	Unchanged, re-confirmed on 2026-08-24, and now explained (§9.6.7)

Finding	Where it was stated	Status
Part 2 §2.5.2 caveat 2 was right all along — <i>"Built-in antenna' is a hard condition. A CB set with an SO-239 socket or a mag-mount whip falls outside the exemption on its face."</i>	Part 2 §2.5.2	Unchanged, and vindicated. Part 2 stated it flatly; Parts 7 and 8 softened it into an open question; §9.6.4 resolves the softening back in Part 2's direction

Does not survive

Statement	Where	What replaces it
"Best licence-free option" for a convoy	Part 7 §7.6, option A	PMR446 is the best licence-free option that can actually be equipped lawfully. CB is the best licence-free option <i>on paper only</i> , and the paper does not permit the installation that would deliver it
"Buy licence-free radios for the convoy now... Citizen Band at 5 W ERP if the antenna question at §7.6.1 resolves acceptably"	Part 7 §7.1.2 step 2, and repeated at Part 7 §7.7.1 step 1 and §7.7.3 point 4	The condition should now be read as very unlikely to resolve acceptably. Buy ETA-verified PMR446 now; do not buy CB pending a WPC answer
"Five watts — ten times PMR446" as a practical power comparison	Part 7 §7.6.1, Part 8 §8.5.2, and implicitly Part 2 §2.6.1's ladder	A compliant CB set radiates about the same as, or less than, a compliant PMR446 handset. §9.6.4. The report should stop describing CB as the high-power licence-free option
CB as Layer 3 of the recommended architecture, ₹21,000 for two vehicles	Part 8 §8.7.4, and the Layer 3 row in Part 8 §8.1.4 and §8.7.8	Drop the layer. The baseline total at Part 8 §8.7.8 was already ≈₹43,500 without it , and that baseline is unaffected
The seven-point "pursue CB if..." test	Part 8 §8.5.7	Points 5, 6 and 7 are now effectively unpassable in the market as it stands — no Indian listing publishes an ETA number; no set can be hardware-limited to channels 1–27, and question 7 is the very question at issue. And point 3 — "willing to do antenna work" — is the step that takes you outside the exemption
CB as the answer to the bike-to-car and multi-kilometre vehicle-link gap	Part 8 §8.3.11, §8.7.7	That gap is unsolved. It was unsolved before, and removing CB does not make it worse — it removes an answer that would not have been lawful anyway. §9.7.5

One thing that is explicitly *not* being said. Parts 7 and 8 were not careless. Both flagged the antenna condition, both tagged it [UNCERTAIN], both told the reader to ask WPC before spending money, and Part 7 put the exact question in writing at §7.7.2 question 4. **The correction is that the open question now has a strong provisional answer, and the recommendation was written on the assumption that it would go the other way.**

9.7.4 IT TURNS ON A QUESTION ONLY WPC CAN ANSWER, AND THAT QUESTION IS ALREADY DRAFTED

This entire correction rests on the interpretation of two words in a 2005 notification that has never, so far as this research can establish, been construed by any Indian court, WPC clarification, ETA condition, test method or enforcement precedent. [UNCERTAIN]

The question is Part 7 §7.7.2, question 4, and it is reproduced here verbatim because it is now the single most consequential item on that list:

"Rule 3 of G.S.R. 533(E) dated 12.08.2005 exempts equipment at 5 W Effective Radiated Power *with built-in antenna*. Does a Citizen Band transceiver installed in a road vehicle with an externally mounted antenna remain within the exemption, provided ERP does not exceed 5 W? If not, on what basis has ETA been granted to vehicle-mounted CB equipment?"

Send it to the WPC COP Section — **Engineer (COP), 011-23036181, awacop-dot@gov.in** — and to the nodal officer on DoT's ASOC service page — **Senior Deputy Wireless Adviser, 011-23036540, umang.srivastava@nic.in**. [VERIFIED — Part 7 §7.7.2] **Keep the reply.**

Three sharper follow-ups now available, which Part 7 could not have drafted, and which should go with it: the concrete TNC hand-held question, the multi-norm question, and the measurement question. All three are set out at §9.8.5 as questions 8, 9 and 10.

What each possible answer means:

If WPC answers...	Then
"A detachable antenna is outside the exemption"	§9.6 and this section stand. CB is out for a vehicle installation, permanently, and only a sealed-antenna hand-held — which nobody makes — would qualify. Buy PMR446
"A supplied antenna fitted to the equipment counts as built-in"	Part 7 §7.6.1 and Part 8 §8.7.4 revive in full , subject only to the stock problem and the terrain test at Part 8 §8.5.7. The hand-helds come back; the SO-239 mobiles with no antenna in the box still fail, because they have no antenna to be built in
"Vehicle-mounted CB with an external whip is within the exemption"	The earlier recommendation revives entirely, and §9.6.4's ERP arithmetic becomes the live constraint instead — because a 4 W set with a 5/8-wave whip is over 5 W ERP, and the whole headroom is 0.97 dB
No answer, or a non-answer	Treat as the first row. A non-answer on file is itself useful (§9.8.5), and an untested reading is not a basis for an installation that can be seized

Until an answer comes back, the operative advice is the conservative one, and the reason is the asymmetry of the bet: if the reader is wrong, the cost is seizure of equipment and exposure under the Telecommunications Act 2023; if the reader is right, the gain is thirteen extra channels and a slightly better antenna.

9.7.5 WHAT THE CONVOY PLAN LOOKS LIKE WITH CB REMOVED

Almost exactly what Part 8 §8.7 already recommended, which is the reassuring part. CB was Layer 3 of a six-layer architecture, it was already conditional, and the baseline total already excluded it.

Layer	What	Status after this correction
0	Written convoy standard operating procedure — named lead and sweep, maximum interval, regroup at every junction and fuel stop, stop-and-wait rule, rally points in everyone's offline map, an agreed hand-signal set	Unchanged, ₹0, and still the highest-return item in the entire report. Part 8 §8.7.1
1	2.4 GHz mesh helmet intercoms , one per motorcycle plus one off-helmet unit for the lead four-wheeler, all the same brand	Unchanged. ₹6,000–₹11,000 each, licence-free under G.S.R. 45(E). Part 8 §8.7.2
2	Two to four ETA-verified PMR446 handhelds on crew roles — lead, sweep, backup vehicle, spare — with earpieces and, for a rider, a handlebar push-to-talk	Now the primary radio layer rather than the fallback. ₹7,400 a pair. Part 8 §8.7.3
3	~~Citizen Band~~	Removed pending a WPC answer. §9.7.4
4	Your own Restricted ASOC	Unchanged, and more valuable than before — see below. ₹1,100 or ₹2,100. Part 8 §8.7.5
5	A scheduled check-in plan. No satellite messenger.	Unchanged. ₹0. Part 8 §8.7.6, Part 7 §7.6.4

Baseline total for a five-person mixed group: ₹43,500, exactly as Part 8 §8.7.8 already gave it, because that figure never included CB. **[VERIFIED against Part 8's own table]**

Two honest consequences of removing Layer 3, stated rather than glossed:

- 1. The multi-kilometre vehicle-to-vehicle link is now unsolved for an unlicensed group, and there is no lawful substitute.** Part 8 §8.5.7 said it plainly and it is still true: *"There is no licence-free 10 W anywhere in India, in any band, under any notification in force."* PMR446 gives 0.5–2 km between vehicles on an open highway and 100–500 m in mountain switchbacks **[ENGINEERING ESTIMATE — Part 7 §7.6.2]**. **A convoy strung out over 5 km of Himalayan switchbacks cannot be held together by any licence-free radio in India.** The answer to that is not a bigger radio; it is Layer 0. **A disciplined convoy with no radios loses nobody; an undisciplined convoy with radios loses people the moment the radios fail.**

2. **The one lawful route to a genuine multi-kilometre link is that every vehicle carries its own ASOC holder.** Part 7 §7.2.2 established the four independent bars that make amateur frequencies unlawful for unlicensed companions, and none of them has moved. But **a four-vehicle convoy of licensed amateurs costs under ₹9,000 in government fees and is entirely lawful** at 10 W DC input on 2 m and 70 cm. [VERIFIED — Part 7 §7.2.2] With CB gone, that route is now the *only* one that reaches beyond a couple of kilometres, and it is worth taking more seriously than Part 7 presented it.

And one thing this correction does not change at all. Part 8's central finding stands untouched: **what actually holds Indian expedition groups together is a phone, a WhatsApp group and a stop-and-regroup rule, with an intercom layered on top.** Part 8 §8.4.2 found that roughly four out of five Indian commercial operators sell the human version — road captain, mechanic, sweeper, backup vehicle — and no radio at all, at ₹50,000 a head. **Removing a radio layer that was legally doubtful and commercially unbuyable brings this report's recommendation closer to what the professionals already do, not further from it.**

9.8 The recommended progression

One path, in three stages, with real Indian prices. Not a survey.

The sequencing rule that governs all three stages, and it is not negotiable: you cannot lawfully transmit on any amateur frequency, at any power, until you hold an ASOC [VERIFIED — 2023 Act s.4(2); Part 1 §1.4], and you cannot lawfully be *sold* a licence-band handset either, because REPA rule 8(1) says a dealer *"shall give possession thereof only to a person as referred to in rule 10 or who holds an authorisation granted under these rules"*, and rule 10(f) is what makes you such a person. [VERIFIED] **So stage 0 is receive-only — and that is not a compromise. It is where all the learning is.**

9.8.1 STAGE 0 — BEFORE THE LICENCE: BUY A RECEIVER, NOT A TRANSMITTER

Item	Model	Price	Where	Tag
The purchase — buy this	RTL-SDR Blog V3, dongle + multipurpose dipole antenna kit	₹5,712, in stock	Fab.to.Lab, Bengaluru	[VERIFIED — page fetched 2026-08-24]
— dongle only	RTL-SDR Blog V3	₹4,802.04	Fab.to.Lab	[VERIFIED]
— second source	RTL-SDR Blog V4, dongle only	₹5,336 in stock (Fab.to.Lab); ₹4,840 + 18 % GST (ElectroPi); ₹5,750–₹7,450 (RFCharge)	various	[VERIFIED] — but see the end-of-line note at §9.3.8

Item	Model	Price	Where	Tag
The antenna	Twenty metres of wire for HF, plus the dipole kit's two element pairs	~₹200	any hardware shop	—
The software	SDR++, SDRangel, GQRX, CubicSDR, SDRuno, dump1090, WSJT-X, multimon-ng	₹0	all free and open source	—
The examination	ASOC (Restricted) examination fee	₹100 per grade	saralsanchar.gov.in	[VERIFIED — Rule 5(2), 2024 Rules; Part 3 §3.7]
Study material	NIAR and ARSI question banks; ITU RR Articles 2, 5, 15, 19, 25, 30–33 and Appendices 1, 14, 42	₹0 – ₹1,500	Part 4	[per Part 4]
If you need to talk to unlicensed companions now	ETA-verified PMR446 — BRUTFORCE BFR-002 pair ₹7,400, or TalkPRO LF446 ₹9,000 + tax quoting ETA SD 20250100112	as listed	Part 8 §8.4.4	[REPORTED] — verify the ETA number by hand before paying, and read Part 8 §8.4.4's coincidence warning

Spend today: ₹5,812. The receiver kit and the examination fee.

Do not buy at stage 0: a HackRF or any transmit-capable SDR (§9.3.3); a UV-32 or any Baofeng-class hand-held (Part 7); anything described as a "scanner" that also transmits; and — per §9.7 — a CB set.

What stage 0 teaches, mapped to the ASOC syllabus

This is the argument for buying a ₹5,712 receiver rather than treating it as a toy: it is the study aid. Six of the eight Section A headings and most of the named Section B items have real content here, before a rupee has been spent on a transmitter. **The syllabus text is [VERIFIED]** from Annexure II to the 2010 memorandum, transcribed in Part 4; **the mapping is pedagogical judgement, not an official study guide.**

Syllabus item	What the SDR does for it
A(iii) Semiconductor devices; amplifiers, oscillators, detectors, frequency changers	The dongle is a tuner, a mixer and an ADC. Reading its block diagram teaches the superheterodyne chain

Syllabus item	What the SDR does for it
A(iv) Radio receivers — TRF and superheterodyne; CW reception; sensitivity, selectivity, fidelity, adjacent channel and image interference, AVC and squelch, signal-to-noise ratio	The single best match in the whole syllabus. Every one of those is a slider or a visible artefact in SDR software. You <i>watch</i> an image response appear, <i>watch</i> selectivity change as you narrow the filter, and <i>watch</i> signal-to-noise degrade as you detune
A(v) Transmitters; low-power transmitters; crystal oscillators; stability of oscillators	Watch other people's transmitters drift across the waterfall; understand why the dongle's 1 ppm temperature-compensated oscillator matters
A(vi) Radio wave propagation — EM spectrum; wavelength, frequency, frequency bands; ground, sky and space waves; skip distance, skip zone, fading	The waterfall is ITU Article 2 made visible. Watch 7 MHz open at dusk and close at midday; watch fading happen in real time; watch 145 MHz stay stubbornly line-of-sight
A(vii) Aerials	The dipole kit's two element pairs force you to retune per band. §9.2.2 is the theory; the kit is the practical
A(viii) Frequency measurement; simple frequency meters	Read frequencies to the hertz off the waterfall; calibrate against a known broadcast carrier
B — ITU Appendix 1 , designation of emissions (16K0F3E, J3E, A3E, F1B...)	You can see the difference between an FM, SSB, AM and FSK signal on a waterfall and name it. Nothing else teaches this faster
B — ITU Appendix 14 , phonetic alphabet and figure code	Hear it used correctly by real operators, for hours, free
B — Article 2 , nomenclature of frequency bands	§9.2.1, watched rather than memorised
B — Article 5 , amateur allocations	Tune to 144–146 and 434–438 MHz and hear what is actually there; then tune to 430–434 and hear why it is not yours
B — Article 15 , interference and measures against it	Watch your own laptop and phone charger wreck your noise floor; find and kill them. This is the most useful hour you will spend
B — Article 19 and Appendix 42 , identification and Indian callsign series	Log every callsign you hear and look up its prefix; you will learn the VU series and a hundred others
B — Articles 30–33 , distress, urgency and safety signals	Listen on the maritime and aeronautical bands, receive only , and hear real procedure
B — the examined 30 Q-codes and 17 abbreviations	Hear them used in real contacts. Part 4 lists the exact examined set

Note that Section A is exempt for applicants holding an engineering or science degree, or a diploma in engineering with electronics or telecommunications [VERIFIED — Part 4]. If the reader qualifies, **Section B is the whole examination — and the SDR is still the best study aid for it**, because Section B is mostly about emissions, band nomenclature, identification and procedure, all of which you can watch happening.

And one legal reminder that belongs in this stage, not later: everything at \$9.5 applies from the day the dongle arrives. **Broadcast, amateur, ADS-B, AIS, weather satellites, time signals and open aeronautical and marine voice are what the device is for. Do not record official traffic, do not build logs of it, do not use it near a defence installation or an airport perimeter, and do not install software whose purpose is to decode somebody's private traffic.** §9.5.5.

9.8.2 STAGE 1 — AFTER PASSING ASOC (RESTRICTED)

Sequence: pass the examination → pay for and claim the certificate → *then* buy the transmitter. Payment must be made **within two years of the result being declared** or the entitlement lapses and you re-sit. **[VERIFIED — Rule 6(2), 2024 Rules; Part 3 §3.7]**

Item	Model / spec	Price	Where	Tag
The certificate	ASOC, 20-year validity	₹1,000	saralsanchar.gov.in	[VERIFIED — Rule 6(1)]
— or	Lifetime, to age 80	₹2,000	"	[VERIFIED]
Band-limited handheld — buy this	Alinco DJ-VX50E, TX 144–146 / 430–440 MHz, 5 / 2 / 1 W, IP67, F3E	₹9,676 (MRP ₹11,800)	Radio Brewery, brand of Mahesho Technologies LLP, <i>"holding Dealer Possession License from W.P.C. SRHQ Chennai"</i>	price [VERIFIED] ; the TX range is [VERIFIED from a European retailer's specification, not from the Indian listing — §9.8.5]
— alternative	Yaesu FT-65E, European version, 144–146 / 430–440 MHz, 5 / 2.5 / 0.5 W	₹15,000 for an "FT-65"	RFCharge	price [VERIFIED] ; insist on the E version — the Asian FT-65 is 136–174 / 400–480 MHz [VERIFIED, Part 7 §7.5.2]
Antenna — buy this	Nagoya VHF/UHF ground-plane kit, 2 m / 70 cm	₹1,239 (MRP ₹3,540)	Radio Brewery	[VERIFIED]
— portable	RH-775 retractable whip, 144/430 MHz	₹1,239	Radio Brewery	[VERIFIED]
— cheaper whip	Nagoya NA-771, SMA-female, 2.15 dBi	~₹799	IndiaMART	[REPORTED]

Item	Model / spec	Price	Where	Tag
— vehicle	Nagoya NL-770H mobile antenna ₹2,285.74 + B-400 lip mount ₹1,599.01	₹3,884.75	Radio Brewery	[VERIFIED]
Measurement — the real learning tool	NanoVNA H-4, 50 kHz – 1.5 GHz, 4-inch touchscreen	₹8,732 (MRP ₹18,880)	Radio Brewery	[VERIFIED]
— or, simpler	Nissei RS-50 MAX digital SWR and power meter	₹11,210	Radio Brewery	[VERIFIED]
HF — the sleeper pick, and made in India	HF Signals (Hyderabad) uBITX v6 , 3–30 MHz, 10 W, SSB/ CW	basic kit USD 160 , full kit with case USD 209 ≈ ₹15,300 / ₹20,000	hfsignals.com	[VERIFIED on the USD figures]; the rupee conversion is [REPORTED] and moves; no INR pricing or India-shipping statement is published
— newer from the same maker	zBitx (80 m–10 m, 10 W) USD 169; sBitx v3 (25 W) USD 399	≈ ₹16,200 / ₹38,200	hfsignals.com	[VERIFIED on the USD figures]

Why the narrowest radio, not the widest — the one-line version of this whole report. In India, **transmit width is legal exposure**. A UV-32 puts roughly **192 MHz** of transmit capability outside Indian amateur spectrum; a European FT-65E or DJ-VX50E puts **6 MHz** outside it. **[VERIFIED — Part 7 §7.5.1]** That single choice removes almost all of the equipment risk in Part 7.

An update to Part 7 §7.5.2, and it is good news. Part 7 recorded the FT-65E as *"imported; higher than ₹14,449"* because no Indian dealer stocking it could be found. **An Indian dealer holding a WPC dealer licence stocks the equivalent Alinco at ₹9,676 — ₹4,773 less than the UV-32 Part 7 examined, for a radio that transmits over 12 MHz instead of 198. [VERIFIED — Radio Brewery page fetched 2026-08-24]** Buying domestically also moves the entire import gate onto the seller, which is the single largest risk reduction available anywhere in this report.

Why HF belongs at stage 1, not stage 2. Most guides treat the Restricted grade as a VHF/UHF licence. **It is not.** Annexure-I gives ASOC (Restricted) **50 watts DC input on ten HF bands** — 1820–1860, 3500–3700, 3890–3900, 7000–7100, 7100–7200, 14000–14350, 18068–18168, 21000–21450, 24890–24990 and 28000–29700 kHz — in voice emissions **A3E, H3E, J3E, R3E. [VERIFIED**

— **Part 3 §3.4] That is five times the VHF power allowance, on the bands that actually go somewhere.** An Indian-made uBITX at 10 W RF out draws perhaps 20–25 W DC, comfortably inside 50 W, and **sidesteps the import question entirely.** Part 7 §7.2.3 and Part 8 §8.7.5 both identify HF as the only capability in this report that lawfully reaches out of a Himalayan valley. **If the reader's real goal is "learn radio properly", HF is the better second radio.**

Two cautions specific to this stage:

- **The DC-input trap.** India's ceilings are **DC input to the final stage**, not RF output — Annexure-I's column heading is *"Max DC Input Power unless Otherwise specified"* and note (iv) defines it as *"the total direct current power input to the final stage of the transmitter"*. **[VERIFIED]** Consumer handhelds are advertised by **RF output**. The one manufacturer-published data point is Yaesu's FT-65 transmit current — *"1.5 A (5 W Tx, 144 MHz) 7.4 V DC / 1.7 A (5 W Tx, 430 MHz) 7.4 V DC"* **[VERIFIED — FT-65R/FT-65E manual p.46]** — which is **11.1 W of DC for 5 W out on 2 m and 12.6 W for 5 W out on 70 cm**, for the whole radio. **Practical rule for a new VU3 licensee: operate on the low or mid power step.** Not because 5 W is obviously unlawful, but because nobody — including you — can demonstrate compliance at the high step from published data, and rule 14 obliges you to produce the equipment for inspection. **[The ceilings are VERIFIED; the RF-to-DC mapping is inference and UNCERTAIN — Part 7 §7.3.3, Part 1 §1.8]**
- **Emissions on HF for the Restricted grade.** Annexure-I lists **A3E, H3E, J3E, R3E** — all voice. **A1A (Morse) is authorised only to "old Grade II licensees"**, and the digital designators F1B and F2B appear only on VHF/UHF for Restricted. **[VERIFIED — Part 3 §3.4]** So a newly licensed VU3 buying an SSB/CW HF rig should use it on **SSB (J3E)** and **confirm with WPC before operating CW or digital modes on HF.** **[UNCERTAIN]** — question 11 at §9.8.5.

9.8.3 STAGE 2 — AFTER UPGRADING TO ASOC (GENERAL)

The General grade requires the **8 words-per-minute Morse test, sending and receiving** (Rule 5(4), 2024 Rules) **[VERIFIED — Part 3 §3.7]**, and it raises the ceiling to **25 W DC input on VHF/UHF and 400 W DC input on HF**, adds 18068–18168 and 24890–24990 kHz, adds **5725–5840 MHz**, permits **A1A, A2A, A3C, A3F**, and opens the **amateur satellite service** at 30 dBW e.i.r.p. **[VERIFIED — Part 3 §3.4]**

Item	Model / spec	Price	Where	Tag
Wider, better receiver	SDRplay RSP1B, 1 kHz – 2 GHz, 14-bit, 10 MHz visible, FM/DAB notch filters, steel case	₹18,007.37, in stock	Fab.to.Lab	[VERIFIED]
— or, the best wide-receive buy in India	SDRplay RSPdx-R2, 1 kHz – 2 GHz gapless, 14-bit	₹30,607.37	Fab.to.Lab	[VERIFIED]

Item	Model / spec	Price	Where	Tag
— standalone, no computer	Malahit DSP2, 10 kHz – 2 GHz with a 380–404 MHz gap	₹37,500 incl. GST	RFCharge — sold out 2026-08-24	[VERIFIED]
HF base station	Icom IC-7300 , HF + 50 MHz	₹103,500; IC-7300 MK2 ₹165,000; MK2 EUR ₹1,91,160 at Radio Brewery, whose listing states <i>"Valid License required to purchase this radio"</i>	RFCharge / Radio Brewery	[VERIFIED]
— the all-band portable	Icom IC-705 , TX 1.810–440 MHz across 12 amateur bands, RX 30 kHz–199.999 and 400–470 MHz	₹116,500	RFCharge	[VERIFIED]
— alternative	Yaesu FT-991A, 1.8–450 MHz amateur bands	₹138,500	RFCharge	price [VERIFIED] ; spec [REPORTED]
VHF/UHF mobile	Icom IC-2730A dual band	₹41,890 (MRP ₹48,380)	Radio Brewery	[VERIFIED]
— cheaper	Alinco DR-638 dual band	₹31,270	Radio Brewery	[VERIFIED]
Fixed antenna	Diamond X-50, dual band, 136–174 / 400–470 MHz	₹7,080	Radio Brewery	[VERIFIED]
— 2 m only, high gain	Diamond F-22 ₹10,030 / F-23 ₹11,800		Radio Brewery	[VERIFIED]
Analyser	NanoVNA v2 (S-A-A-2), 50 kHz – 3 GHz	₹14,750	Radio Brewery	[VERIFIED]

The same hard warning about mobiles that Part 7 §7.5.3 gave, restated because it catches people. Every 2 m / 70 cm mobile on the market starts at 20–50 W and steps down. India's Restricted ceiling is 10 W DC input, roughly 4–6 W RF out, so **a 50 W mobile is lawful for a VU3 licensee only on its lowest power step, and one button press puts you outside your authorisation.** A General-grade licensee has room to roughly 10–16 W RF out. **[The ceilings are VERIFIED; the RF-out equivalents are inference and UNCERTAIN.]** **Buy the power you are licensed for, not the power the shop is selling.**

An honest note on scale, for perspective. A ₹116,500 IC-705 transmits on **17.44 MHz of spectrum in total** across twelve amateur bands, which is **less than 4 % as much spectrum as a ₹14,449 Baofeng UV-32. [VERIFIED spec; ARITHMETIC on the totals]** That is not a defect. That fence is exactly what makes the expensive radio defensible to own in India, and its absence is exactly what makes the cheap one a liability. §9.1.1.

9.8.4 THE PRINCIPLE THAT OUTRANKS EVERY PURCHASE ON THESE THREE LISTS

An antenna is almost always a better investment than a bigger radio.

Some arithmetic to make that concrete rather than folkloric, **[ENGINEERING ESTIMATE]** throughout:

- **Doubling transmit power is +3 dB.** Going from a handheld's 1 W low step to a 10 W high step is **+10 dB** — and takes a VU3 licensee outside the Restricted grade's DC-input ceiling (§9.8.2).
- **Replacing a handheld's stock rubber whip with a resonant half-wave or a ground-plane vertical commonly buys +3 to +6 dB, on transmit *and* receive, with no legal consequence whatsoever.** There is no power ceiling on an antenna, only on what you feed it.
- **Getting the antenna outside and five metres higher, clear of the building, is frequently worth more than either,** on VHF and UHF, where propagation is essentially line of sight.

So at stage 1 the ₹1,239 ground-plane kit and, later, the ₹7,080 Diamond X-50 will each do more for the reader's range than any radio upgrade on these lists — and the ₹8,732 NanoVNA will teach more about *why* than any amount of reading. It also directly serves syllabus items **A(ii) resonance in series and parallel circuits, A(vii) Aerials and A(viii) Frequency measurement.**

The receive-side version of the same principle is even starker, and it is the sentence to remember from §9.2.2: a wideband receiver with the wrong antenna hears nothing. **Add a decent HF wire before adding anything else; the antenna will improve results more than the radio did.**

9.8.5 WHAT TO CONFIRM BEFORE SPENDING MONEY

Ask WPC, in writing. Part 7 §7.7.2 lists seven questions and gives the contact details; Part 8 §8.7.8 adds two more. **These are the ones this part adds or sharpens.** Address them to the Wireless Adviser / WPC Wing, DoT, Sanchar Bhawan, 20 Ashoka Road, New Delhi 110001, or to the COP Section contacts at §9.7.4. **Number them, keep the replies, and file each answer against its question.**

#	Question	Why it decides something	Tag today
1	"Is a receive-only wideband communications receiver (for example an Icom IC-R8600, 10 kHz–3 GHz, with no transmit capability) covered by the exemption in rule 3 of the Radio, Television and Video Cassette Recorder Sets (Exemption from Licensing Requirements) Rules, 1997 (G.S.R. 393(E) dated 16.07.1997)? If not, under what provision may a private individual lawfully possess one?"	The single question that decides whether a non-ASOC individual has any possession exemption at all	[UNCERTAIN] — §9.5.1
2	"Are the Indian Wireless Telegraphy (Possession) Rules, 1965 (G.S.R. 1318 dated 28.08.1965) still in force, given that the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026 neither repeal them nor supersede them expressly, while rule 4(2)(d) of the 2026 Rules presupposes their continued existence?"	If they survive, rule 5(b)'s "complete wireless set" definition makes a communications receiver licensable — and the licence cannot be obtained	[UNCERTAIN] — §9.5.1
3	"Does an Amateur Station Operator Certificate exempt the holder, under rule 10(f) of the 2026 Rules, from possession authorisation in respect of all radio equipment, or only equipment used as part of the amateur station? And does rule 12 require an ASOC holder whose certificate lapses to dispose of all radio equipment within two months?"	Decides the value of the cleanest possession route available, and an odd consequence of it	[UNCERTAIN] — §9.5.8; compare Part 7 §7.7.2 question 1
4	"Is an Equipment Type Approval required for a receive-only device? If yes, under which rule; and what is the 'Receivers' category in the Saral Sanchar ETA dropdown used for?"	Decides whether a seller's "ETA" on a receiver means anything	[UNCERTAIN] — §9.5.7
5	"Is an import licence from the WPC Wing required for a receive-only wireless apparatus imported by a private individual for personal use, given paragraph 2.4 of the Compendium in File No. R-11018/02/2017-PP dated 06.07.2022 and clause (b) of Notification No. 71 of 1953? If yes, how may such a licence be applied for after 03.07.2026?"	The 1953 notification and WPC's own summary of it say different things, and there may currently be no route at all	[UNCERTAIN] — §9.5.7
6	"Is listening to a transmission not addressed to the listener 'intercepting a message unlawfully' within section 42(2)(b) of the Telecommunications Act, 2023?"	The one criminal provision that could in principle reach a listener	[UNCERTAIN] — §9.5.2. This will probably not get a straight answer. Ask it anyway; a non-answer on file is itself useful

#	Question	Why it decides something	Tag today
7	"Does G.S.R. 533(E) (5 W ERP) or G.S.R. 1047(E) Table IX serial 2 (10 mW e.r.p. in 26957–27283 kHz) govern a 27 MHz Citizen Band voice transceiver ? The 2018 Rules contain no supersession clause and DoT's index lists both."	A four-hundred-fold difference in the operative ceiling	[UNCERTAIN] — §9.6.1; also Part 2 §2.12
8	"A hand-held Citizens Band transceiver is supplied with a short flexible antenna that screws onto a 50-ohm TNC female socket , and is also supplied with an adapter enabling that antenna to be removed and an external vehicle-mounted antenna connected instead. (a) Does such equipment have a 'built-in antenna' within rule 3 of G.S.R. 533(E) when the supplied antenna is fitted? (b) Does the answer change if the adapter is never used? (c) Is any Citizens Band equipment with a detachable antenna eligible for ETA for licence-exempt CB use?"	The concrete version of Part 7 §7.7.2 question 4, and the question §9.7 turns on	[UNCERTAIN] — §9.6.4
9	"Does a CB transceiver whose firmware allows the user to select among several national band configurations — one of which restricts transmission to 27 channels between 26.965 and 27.275 MHz, and others of which permit transmission between 26.565 and 27.99125 MHz — qualify as 'wireless equipment intended to be used... in the frequency band 26.957 – 27.283 MHz'? If yes, is the exemption conditional on the equipment being kept in the India configuration, and must the user be able to demonstrate that?"	Decides whether every multi-norm set on the market is inside or outside the exemption	[UNCERTAIN] — §9.6.2
10	"For the 5 W ERP limit in rule 3 of G.S.R. 533(E) read with rule 2(b), how is compliance measured — as transmitter conducted output power, or as radiated power computed with the antenna actually fitted? What test method and standard does WPC apply?"	With 0.97 dB of headroom, the measurement basis decides whether a 4 W set with any gain antenna is lawful	[UNCERTAIN] — §9.6.3
11	"May an ASOC (Restricted) holder use A1A (Morse) or digital modes on the HF bands, given that Annexure-I to OM No. L-14011/255/2004-AMT lists only A3E, H3E, J3E and R3E for Restricted on HF and reserves A1A to 'old Grade II licensees'? And can a software-defined transmitter be operated lawfully under an ASOC given the F1B/F2B/F3E/F3C restriction on VHF/UHF?"	Decides what a new VU3 may actually do with an HF rig or an SDR	[UNCERTAIN] — §9.3.3, §9.8.2

Ask the seller, in writing, before paying, and keep the replies:

1. **For a receiver:** the exact model and, if imported, whether it was imported against an Indian ETA or as a personal import. **Insist on a GST invoice naming the model and serial number.**
2. **For the Alinco DJ-VX50:** *"Which market version is this — E, T or H — and what exactly is its transmit frequency range?"* The Indian listing does not print it, the European DJ-VX-50E is 144–146 / 430–440 MHz and the North American DJ-VX50T is 144–148 / 420–450 MHz. **[VERIFIED for the E version from a European retailer; REPORTED for the T version]** **This is the same class of document-versus-marketing conflict Part 7 §7.3.1 found inside Baofeng's own paperwork.**
3. **Also for the Alinco:** its power steps. One European listing says "5 / 2 / 1 W"; Radio Brewery says "switchable 1 W, 2.5 W, 4 W, 5 W". **A four-step versus three-step conflict worth resolving before you rely on a power figure. [VERIFIED that the two pages differ]**
4. **For any dealer:** *"You state you hold a Dealer Possession Licence. Have you migrated to an authorisation under the Telecommunications (Radio Equipment Possession Authorisation) Rules, 2026?"* DoT stopped accepting DPL applications and renewals from **2026-07-03** and the new regime went live on **2026-08-06. [VERIFIED — Part 2 §2.9.1]** The transition mechanics are **[UNCERTAIN]**, and the answer tells you a great deal about the dealer.
5. **For any dealer selling you a licensed-band radio:** *"Will you record this sale against my ASOC number and callsign?"* Rule 8(1) requires them to give possession only to a rule-10 person; getting that on the invoice documents your side of it. **[VERIFIED]**
6. **For any CB set, if the reader pursues it despite §9.7:** the model name and President article number; the **WPC ETA number, its date and the name of the ETA holder;** which band configuration the unit ships in; the antenna connector type; and whether it is the European multi-norm version or the US FCC version. **Ask for a copy of the ETA certificate, and verify the number by hand.**

And verify every claimed ETA number yourself at https://saralsanchar.gov.in/wpc_eta_report.php. It is CAPTCHA-gated so it must be done by hand, but it takes two minutes and it is

the best consumer protection available. **[VERIFIED that the lookup exists — Part 2 §2.11.1]**

Note that it is relevant to licence-exempt equipment only: **amateur-band gear has no ETA and cannot have one, and any dealer offering an "ETA-approved ham radio" is selling something that does not exist. [VERIFIED — Part 2 §2.7.1, Part 7 §7.5.4]**

9.8.6 THE SHORT VERSION OF PART 9

1. **There is no radio that covers all radio frequencies, and the reasons why are the syllabus.** The spectrum runs to 3,000 GHz on the face of India's own 1933 Act; a quarter-wave antenna at 9 kHz is 8.33 km long and at 6 GHz is 12.5 mm; a HackRF that tunes across 5,999 MHz sees 20 MHz at a time; and past a certain point paying more buys range and costs quality. §9.2.

2. **Receive width is cheap and mostly harmless; transmit width is what the law regulates.** ₹5,336 buys 500 kHz to 1.766 GHz of receive. ₹116,500 buys 17.44 MHz of lawful transmit. **Put the width on the receive side and buy the narrowest transmitter that does the job.** §9.1.1, §9.3.
3. **Owning a receiver in India is low-risk but rests on an exemption written in 1997 for domestic broadcast sets. Listening is not an offence, but India wrote no general-reception carve-out. Recording official traffic is where fourteen years and a reversed burden of proof begin.** Buy domestically; get the ASOC, which settles possession cleanly; and never record or repeat anything from a police, military, security or air-traffic channel. §9.5.
4. **Citizen Band fails the built-in-antenna condition on every set on the market, and the failure is not curable by the buyer.** A compliant CB set would also radiate no more than a PMR446 handset. **Parts 7 and 8 recommended CB subject to this question; the question now looks very likely to resolve against them, and only WPC can settle it.** §9.6, §9.7.
5. **The plan is unchanged in shape and cheaper than the radio Part 7 talked you out of.** ₹5,712 of receiver plus ₹100 of examination fee today; ₹9,676 of band-limited transmitter and ₹1,239 of antenna after the certificate; HF when the expedition case justifies it. **₹17,727 all in — ₹939 more than the single radio that started this.** §9.1.4, §9.8.
6. **An antenna is almost always a better investment than a bigger radio,** and that is true on receive as much as on transmit. §9.8.4.

9.9 Disclaimer for this part

This part is a research summary compiled on 2026-08-25, from hardware, retailer and manufacturer sources read on 2026-08-24 and 2026-08-25. **It is not legal advice, it is not a buying recommendation from a dealer, and nothing in it authorises the purchase, import, possession or operation of any radio equipment.**

It is deliberately more cautious than the internet consensus on receiver legality, because that consensus consists almost entirely of undated, uncited pages asserting that "receiving is legal in India" — including the page that is currently the top search result for the question, which names no author, no date and no statutory provision. §9.5.3.

Specific limits on the evidence, restated because they matter:

- **The central legal question at §9.5.1 — whether a wideband communications receiver falls within the 1997 broadcast-receiver exemption — is unresolved and, so far as this research can establish, has never been addressed by any Indian court, notification, circular or FAQ.** Several supporting questions are equally open: whether the 1965 Possession Rules survive; whether the adjudication chapter of the 2023 Act has commenced; whether passive reception is "interception" for s.42(2)(b); and whether a private individual can currently obtain any import authorisation for a receiver. **§9.8.5 gives the exact wording to put to WPC. Those hedges must not be read as soft yeses.**

- **The central question at §9.6.4 — whether "built-in antenna" in G.S.R. 533(E) means a non-detachable antenna — has never been defined, interpreted or litigated in India so far as this research can find.** The conclusion at §9.6 and the correction at §9.7 are **statutory interpretation, not findings of fact**, and if WPC reads the condition loosely the position changes substantially. §9.7.4 sets out what each possible answer would mean.
- **The 5 W ERP arithmetic at §9.6.3 and §9.6.4, the antenna-efficiency figures, and the range and gain estimates throughout are [ENGINEERING ESTIMATE] — physics, not measurement and not law.**
- **Rule 10(f) of the 2026 Possession Rules, on which the cleanest possession answer in this part depends, is carried at [UNCERTAIN] per Part 1 §1.5 Group B,** because the report's own fact-check could not re-fetch the final 2026 Rules from a government host. This part does not upgrade that hedge.
- **Every band, emission and power figure for the amateur service rests on a WPC Office Memorandum of 2010-08-13** whose continued validity under the 2024 Amateur Services Rules could not be independently confirmed, because wpc.dot.gov.in was unreachable throughout this research. Part 1 §1.8.
- **No ETA number anywhere in this part has been verified against WPC's own register,** which is CAPTCHA-gated and cannot be queried programmatically. Where a vendor quotes an ETA number, this part records the number and the fact that it was quoted — nothing more.
- **Several manufacturer documents contradict each other,** and where they do, both readings are shown: the RTL-SDR Blog V4's tuner chip, President's own India frequency table, the Alinco power steps, and the Randy III's multiconfiguration list. **Get specifications in writing from the seller for the exact unit they will ship.**
- **Every price is a snapshot from a retailer's own page on one day, and several of those pages do not state whether GST is included.** Prices in this market move weekly.
- **Sources that could not be reached,** and whose absence shapes the findings: sdrplay.com (HTTP 403 to every fetch), yaesu.com product pages (HTTP 500), amazon.in and flipkart.com (HTTP 500 / 403), robu.in (HTTP 403), the ITU Radio Regulations text itself, indiacode.nic.in and mha.gov.in for several bare Acts, and the Saral Sanchar ETA register's query results.

Indian telecom regulation changed substantially in June, July and August 2026 and is still moving. Anything written before mid-2026 about possession licences, dealer licences or import licences describes a dead regime. Re-verify every notification against the live gazette before acting.

Before buying any transmitter, confirm your own position with the WPC Wing of DoT in writing, using the questions at §9.8.5, Part 7 §7.7.2 and Part 8 §8.7.8; verify any claimed ETA number yourself on saralsanchar.gov.in ; and speak to an established amateur radio club such as ARSI or NIAR. Before buying any imported radio, **confirm the market version** — as Part 7 §7.5.2 shows for the Yaesu FT-65, the same model number can be band-limited in Europe and wide open in Asia, and in India that difference is the whole ballgame.

And getting the receive side wrong exposes you to equipment seizure; recording or publishing official traffic exposes you to prosecution under the Official Secrets Act 1923, with a statutory presumption running against you and a fourteen-year maximum. §9.5.4. Behave accordingly.

Disclaimer

This document is a research summary compiled from public sources on the date shown below. It is **not legal advice**. Indian spectrum regulation changes — fees are revised, portals are migrated, bands are reallocated, and the Telecommunications Act 2023 continues to bring new rules into force. Any claim tagged [REPORTED] or [UNCERTAIN] should be confirmed before you rely on it.

Before spending money or transmitting, confirm the current position with the Wireless Planning and Coordination Wing of the Department of Telecommunications, or with an established amateur radio club or institute that processes these applications regularly.

Compiled 2026-08-23 for a private individual in India. Sources are listed per claim throughout and collected in the appendices.